



National Institutes
of Health

National Institutes of Health

Bldg 31B Landscape
Water Intrusion

BETHESDA, MARYLAND

SPECIFICATION

LSY No. 24024
WR# C116635

IFC SUBMISSION
August 26, 2025



LSY

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Laboratory Planners

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SECTION 01 1000 - SUMMARY

PART 1 - GENERAL

1.1 PROJECT IDENTIFICATION

- A. Project Name and Location: Design Services – Bldg. 31B Exterior Landscape and Grade / Excavate to Fix Water Intrusion Issues in National Institutes of Health (NIH) Bethesda Campus.
- B. Phasing:
 - 1. Architectural systems infrastructure and related construction. Phasing will occur in this sequence; Refer to phasing drawings and SOW attached in Basis of Design.
 - 2. Engineering systems infrastructure construction including as documented in the drawings will follow the phasing noted above.
- C. Architect: The term Architect refers to the project designer. The Architect's status relative to the construction will be delineated in writing by the Contracting Officer prior to the pre-construction conference. The project was designed by:

Louviere, Stratton and Yokel, LLC
8484 Georgia Avenue, Suite 650
Silver Spring, Maryland 20910
Contact: Giwon Lee 301.588.1500
- D. All correspondence from the Contractor to the Architect will be through the Project Officer.
- E. Engineer: The term Engineer refers to the various project engineering systems designers, unless noted otherwise on drawings, all working as sub-consultants to the Architect. The Engineers' status relating to the construction will be delineated in writing by the Contracting Officer prior to the pre-construction conference. The project was designed by the companies indicated on the drawings.
 - 1. All correspondence from the Contractor to the Engineer will be through the Project Architect via the Project Officer.
- F. Project Officer: The Project Officer has been assigned for this Project to serve as the representative of the Owner and to provide assistance in administering the Contract for Construction between Government and Contractor. The Project Officer is:

Contact: Mohammad Biswas, AIA
NIH
9000 Rockville Pike,
Building 13, Suite 2S75
Bethesda, MD 20982
202.934.2138

1.2 WORK UNDER OTHER CONTRACTS

- A. The Contractor shall cooperate with other contractors performing related work, including providing labor, materials and other costs necessary to satisfactorily coordinate the Contract work with work performed under other contracts.

- B. Preceding Work: Separate contract work may be awarded by NIH for performance of certain construction operations that are expected to be substantially complete before the Work of this Contract begins. Follow-on activities under this Contract shall be satisfactorily conducted to protect and, if indicated on the drawings, to complete the related preceding separate Contract work.
- C. Concurrent Work: Separate contract work may be awarded by NIH for performance of certain construction operations that will be conducted simultaneously with the work of this contract. Interface work under this contract shall be satisfactorily conducted in compliance with the schedule requirements of the related concurrent separate contract work.
 - 1. A separate contract will be awarded for installation of telecommunication and data systems (IT Systems).

1.3 SCHEDULE OF NIH-FURNISHED PROPERTY

- A. If shown on the Project drawings as Government Furnished, Contractor Installed (GFCI), the project includes the following NIH furnished property to be installed by the Contractor: biological safety cabinets, ice machines, and other miscellaneous equipment.
 - 1. The Contractor shall perform a physical survey of GFCI equipment and prepare BIM Coordination drawings for each item of equipment.
 - 2. The Contractor shall notify the Project Officer in writing at least 14 calendar days in advance of the date on which the materials and equipment are required. Pick up the materials and equipment no later than 3 calendar days after the indicated required date.
 - 3. The Contractor shall verify quantities of NIH-furnished property upon delivery and shall submit an acknowledging receipt thereof in writing to the Project Officer. Property found damaged or quantity short shall be reported to the Project Officer in writing within 24 hours.
 - 4. The Contractor shall accept delivery at the time and location specified when delivery is made. The Contractor shall provide all handling and transportation necessary from the receiving location to the point of incorporation into the work.
 - 5. If the materials and equipment are not picked up by the date as scheduled, the Contractor will be charged for storage at the rate of \$100.00 per 45 Kg per month or fraction thereof.
 - 6. The materials and equipment are located within 10 miles of the job site. The contractor shall provide all personnel required and means of safe transportation from the storage area to the job site.
- B. The Contractor shall be responsible for disconnecting, loading, transporting, unloading, uncrating, assembly, installation, connection, and testing new materials and equipment. Schedule a joint inspection with the Project Officer to inspect the materials and equipment to determine their existing condition. Uncrate the equipment in the presence of the Project Officer and inspect for possible damaged or missing parts. The Contractor shall be responsible for all damage sustained by the materials and equipment after this inspection.

1.4 MISCELLANEOUS PROVISIONS

- A. Work in the extension of existing conditions shall correspond in all respects with the existing conditions to which it connects, or to similar existing conditions, in materials, workmanship and finish.
- B. Alterations to Existing Conditions: Existing conditions shall be cut, drilled, removed, temporarily removed, or removed and replaced, as necessary for performance of work under the contract.

1. Replacements of existing conditions that are removed shall match similar existing conditions.
 2. Unless otherwise indicated, existing structural members shall not be cut or altered without authorization by the Project Officer.
 3. Conditions remaining in place, which are damaged or defaced during the work, shall be restored to the condition existing at time of award of contract.
 4. Discolored or unfinished surfaces exposed by removal of existing conditions, that are indicated to be final exposed surfaces, shall be refinished or replaced as necessary to produce uniform and harmonious contiguous surfaces.
- C. Existing structures will remain in place unless otherwise indicated in project drawings.
- D. Existing utility services shall be disconnected and removed to the extent indicated.
- E. Removed items indicated to remain the property of NIH or the government and shall be stored on site where directed by the Project Officer.
- F. Work includes comprehensive Building Information Modeling (BIM) services related to production of the requisite Coordination BIM for the project and the Record BIM.

1.5 SPECIFICATION FORMATS AND CONVENTIONS

- A. Specification Format: The Specifications are organized into Divisions and Sections using the 33-division format and CSI/CSC's "Master Format" numbering system.
- B. Section Identification: The Specifications use section numbers and titles to help cross-referencing in the Contract Documents. Sections in the specifications are in numeric sequence; however, the sequence is incomplete. Consult the table of contents at the beginning of the specifications to determine numbers and names of sections included in the Contract Documents.
- C. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:
1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be inferred as the sense requires. Singular words shall be interpreted as plural, and plural words shall be interpreted as singular where applicable as the context of the Contract Documents indicates.
 2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood are to be performed by Contractor. Occasionally, the indicative or subjunctive mood may be used in the Section Text for clarity to describe responsibilities that must be fulfilled indirectly by Contractor or by others when so noted.
 - a. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 1200 – GENERAL SITEWORK REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SITEWORK LAYOUT

- A. Monuments and Benchmarks

- 1. Maintain all monuments, property corners, bench marks and other reference points.
 - 2. If these are disturbed or destroyed during construction operations, have them replaced by a surveyor licensed in the State of Maryland. This replacement shall be at no additional expense to the Contract.

- B. Laying out the Work.

- 1. Locate all existing bench marks and other reference points.
 - 2. Protect these points throughout construction.
 - 3. Layout work utilizing these reference points.

- C. Record Drawings

- 1. Maintain a record of the horizontal and vertical locations of all underground utilities and piping, both newly installed and existing encountered.
 - 2. Maintain a record of any variations of the work.
 - 3. Record Drawings shall be certified by a Land Surveyor registered in the State of Maryland.
 - 4. Maintain record drawings throughout construction, recording information as work is performed. Record Drawings shall be available for periodic inspection during the course of the project.
 - 5. Submit final, complete record drawings at Substantial Completion.

1.3 EASEMENTS

- A. Verify the acquisition of all off-site easements and Rights-of-Way prior to the start of off-site construction. This may be done by contacting the Architect.
- B. Restore all off-site easements to the condition existing prior to the start of work.

1.4 MAINTENANCE OF TRAFFIC

- A. Maintain vehicular and pedestrian traffic across the frontage of this project. Comply with all applicable safety requirements.

1.5 CORRELATION OF CONSTRUCTION DOCUMENTS

- A. Review construction documents thoroughly prior to the start of construction.
- B. Report any conflict or discrepancy discovered in the Construction Documents to the Architect prior to the start of construction.

- C. Report any conflict or discrepancy discovered between the Construction Documents and state and local governmental regulations to the Architect prior to the start of construction.

1.6 PROJECT CONDITIONS

- A. The conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to natural occurrences prior to the start of clearing work.
- B. The location of existing underground utilities indicated is approximate only. Field locate all existing underground utilities within the limits of disturbance, regardless of whether or not they are indicated. The Contractor is required by NIH to use a private utility locator for locating all existing utilities. It's required by state law to Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities.
- C. Should charted, uncharted or incorrectly charted utilities be encountered during demolition, contact the Architect immediately for instructions. Cooperate with Owner and utility companies to keep services and facilities in operation.

1.7 SCHEDULING

- A. The Contractor shall submit a detailed construction schedule for review and approval by the Owner / Architect.

PART 2 - PRODUCTS

Not Applicable

PART 3 – EXECUTION

3.1 PROJECT CLEAN UP

- A. Clean site as construction progresses. Do not allow trash or other waste materials to accumulate.
- B. Prior to requesting the punch-list inspection, clean the site to the following requirements:
 - 1. Power wash all walks and pavements.
 - 2. The remainder of the site shall be broom clean.
 - 3. Remove all trash and debris.

3.2 EXISTING FACILITIES

- A. Preserve existing signs, markers, guardrails, fences and other amenities in their original condition unless written permission is obtained for their removal and replacement.
- B. Replace damaged items at no additional cost to the Contract.

END OF SECTION 01 0200

SECTION 01 1400 - WORK RESTRICTIONS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section documents conditions and procedures on the NIH campus that may impact the performance of work by the contractor including the following:
 - 1. Contractor use of premises
 - 2. NIH occupancy of adjacent premises
 - 3. NIH occupancy prior to substantial completion
 - 4. Working hours
 - 5. Use of existing elevators
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to work restriction documentation including but not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Project Management and Coordination"
 - 3. Division 01 Section "Construction Progress Documentation"
 - 4. Division 01 Section "Submittal Procedures"
 - 5. Division 01 Section "Quality Requirements"
 - 6. Division 01 Section "Construction Quality Control"
 - 7. Division 01 Section "Temporary Facilities and Controls"
 - 8. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 9. Division 01 Section "Safety and Health"
 - 10. Division 01 Section "Execution Requirements"
 - 11. Division 01 Section "Cutting and Patching"
 - 12. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 13. Division 01 Section "Closeout Procedures"
 - 14. Division 01 Section "Selective Demolition"

1.3 CONTRACTOR USE OF PREMISES

- A. The Project Officer will conduct a pre-construction survey with the Contractor to review and document the existing conditions surrounding the project premises prior to the beginning of any construction activity.
- B. All contractor personnel shall abide by the NIH campus wide no smoking policy.
- C. All corridors and stairs required for emergency egress shall remain clear and unobstructed at all times.

- D. Access to emergency services and for fire, police, and other emergency forces shall remain free and unobstructed at all times.
- E. When exit access or exit components need to be changed or modified in any way, this action shall be done only with prior written approval by the authority having jurisdiction (AHJ). The AHJ will keep the NIH Fire Department and relevant occupants notified of all route changes.
- F. Existing fire alarm, detection, and suppression systems shall remain in good working order. All modifications or planned shutdowns of the fire protection systems must be scheduled and approved by Maintenance Engineering. It is the responsibility of Maintenance Engineering to notify the AHJ and the NIH Fire Department of all modifications in these systems. Contractor shall ensure that temporary, but equivalent, fire safety measures are in place when the operation of any fire system is impaired. Temporary and equivalent systems must be approved by the AHJ and tested monthly.
- G. Restrict combustible storage on the jobsite to the minimum level acceptable to the AHJ for daily operations.
- H. Penetrations in fire and smoke walls contiguous with occupied areas will be properly sealed at the end of each work shift.
- I. Remove wastes from the worksite at the end of each work shift or more frequently if directed by Project Officer.
- J. The Contractor shall limit use of the site and premises to the work in areas indicated in the contract documents, to allow for NIH occupancy and public use of areas surrounding project site.
- K. The Contractor shall schedule his work to minimize the amount of interference with NIH building and campus operations. All work schedules shall be approved by the Project Officer.
- L. Permission to interrupt any building services and/or utility services shall be requested in writing a minimum of eighteen (18) working days prior to the desired date of interruption. NIH reserves the right to refuse any request and to schedule such interruption on a later or earlier date and time which is mutually agreeable to NIH and the Contractor.
- M. The Contractor, his employees and all subcontractors shall become familiar with and comply with all NIH regulations, including fire, traffic, safety, and security regulations.
- N. All personnel employed by the Contractor or subcontractors and working on the NIH campus shall keep within the limits of the work and avenues of ingress and egress. Entry to any restricted area is forbidden unless entry is required and is approved in writing by the Project Officer.
- O. The Contractor's equipment shall be conspicuously marked for identification purposes.
- P. Confine operations at the site to areas indicated. Do not disturb portions of the site beyond the areas in which work is indicated.
- Q. Keep driveways and entrances serving the premises clear and available at all times to NIH employees and visitors. Do not use these areas for parking or storage of materials.
- R. Lock automotive type vehicles, such as trucks or other mechanized or motorized construction equipment, when parked and unattended so as to prevent unauthorized use. Do not leave such vehicles or equipment unattended with the motor running or the ignition key in place.

- S. Schedule deliveries to minimize space and time requirements for storage of material and equipment on site.
- T. Maintain existing building in a safe and weather tight condition throughout the construction period. Provide temporary heating and cooling as required to keep indoor temperatures between 65 degrees F and 80 degrees F. A heating and cooling plan shall be submitted to the Project Officer within 14 calendar days of the Notice to Proceed for approval.
- U. Repair damage caused by construction operations. Take precautions to protect the building, its occupants, and the public during the construction period.
- V. Keep public areas, such as hallways, stairs, lobbies and toilet rooms, free from accumulation of waste material, rubbish, construction debris, and construction materials and remove such daily.
- W. All demolition and construction debris shall be taken to a construction dumpster at the B1 level East loading dock or as directed by the Project Officer.
- X. Prior to bid, become familiar with all work restrictions applicable to the building in which work is to be performed and develop work plan and project cost accordingly. Such work restrictions include but are not limited to:
 - 1. Restricted use of loading zones, spaces, docks and other delivery facilities,
 - 2. Restricted use of elevators and stairs,
 - 3. Restrictions regarding removal of exterior windows,
 - 4. Restrictions regarding use of VOC or other volatile chemicals.
- Y. Use of the existing loading dock facilities will be shared with NIH activities on a first come-first-served, wait-your-turn basis, unless otherwise directed.
 - 1. Access to the building for material delivery shall be limited to the use of the B1 level east loading dock, unless otherwise directed.
- Z. Existing materials and equipment that are removed as part of the construction operations, and that are not reused or designated to be salvaged as NIH or other's property, shall be removed from the site. Storage or sale of excess salvageable materials and equipment is not permitted on site.
- AA. Pollution producing equipment shall not be located near air intakes where airborne smoke or fumes could be drawn into the building.
- BB. For parking restrictions in effect under this contract see Contract for Construction and applicable Office of Research Facilities contractor parking policy.
- CC. Electric arc welding equipment shall not be connected to the building power supply system.
- DD. Where available, the contractor will be allowed limited use of existing toilet facilities.
- EE. Existing Stair Usage: Use of existing stairs will be permitted, as long as stairs are cleaned and maintained in a condition acceptable to the NIH. Prior to Substantial Completion, restore stairs to condition existing before initial use and maintain in their original condition until Substantial Completion.
 - 1. Provide protective coverings, barriers, devices, signs, or other procedures to protect stairs and to maintain means of egress. If, despite such protection, stairs become damaged, restore damaged areas so no evidence remains of correction work.

1.4 NIH OCCUPANCY OF ADJACENT PREMISES

- A. NIH will occupy the site and the existing building areas immediately adjacent to, above and below the construction site during the entire period of construction, unless otherwise indicated on project drawings.
 - 1. Cooperate with NIH representatives during construction operations to minimize conflicts and facilitate NIH usage. Perform the work in a manner that does not interfere with NIH operations. Delays may be incurred. If the Project Officer determines that the safety and health of building occupants is in jeopardy, the Contractor must modify his construction plan and procedures to avoid the potential hazard before continuing with construction.
 - 2. Some areas within the limits of the contract may be occupied during performance of work under this Contract. In addition, NIH reserves the right to complete the work not specifically in this contract, but within the physical limits of the contract, by NIH or NIH contract personnel.

1.5 NIH OCCUPANCY PRIOR TO SUBSTANTIAL COMPLETION

- A. NIH reserves the right to occupy and to place and install equipment in completed areas of the building prior to Substantial Completion, provided such occupancy does not interfere with completion of the work. NIH installation of equipment and partial occupancy shall not constitute acceptance of the total work.
 - 1. Prior to partial NIH occupancy, mechanical and electrical systems for the space shall be fully operational, and required inspections and tests shall be successfully completed. Upon occupancy, NIH will operate and maintain mechanical and electrical systems serving the occupied portions of the building.
 - 2. Upon occupancy, NIH will assume responsibility for maintenance and custodial service for the occupied portions of the building.
 - 3. The warranty period for the occupied portion of the building only will commence on the date of occupancy by NIH.

1.6 UNSCHEDULED OPERATIONS

- A. NIH may restrict work in or around hospital, laboratory, or animal holding areas on an unscheduled basis. NIH will provide not less than 24 hour notification each time work restrictions due to unscheduled NIH operations are invoked, except for emergencies.

1.7 WORKING HOURS

- A. Contractor's General Working Hours: The normal work day is between the hours of 6:30am to 4:30pm Monday through Friday, except for the holidays and other times as listed in the Contract for Construction.
- B. The Contractor shall not encumber peak traffic flows during morning rush hour (6:30 am to 8:30 am) and evening rush hour (4:00pm to 5:30pm).
- C. Noise Hours shall be between the hours of 7:00pm and 7:30am.
- D. Work shall not be performed during other than General Working Hours except when such timeliness of performance is required to safeguard life or property. Other deviations may be authorized by the Project Officer.

- E. Requests for deviations from General Working Hours, including work on holidays, shall be submitted in writing to the Project Officer not less than fifteen (15) calendar days in advance of the proposed work period. Once approved, the schedule must be delivered to the NIH Police prior to the start of work.
- F. NIH Unoccupied Hours will be defined, if applicable during the Preconstruction Meeting or on the project drawings. The Contractor shall submit a proposed schedule and gain the Project Officer's approval at least 72 hours before proceeding with any work during NIH Unoccupied Hours.
- G. Contractor's Required Working Hours: The following work shall be performed during NIH Unoccupied Hours:
 - 1. See General Notes on Project Drawings.

1.8 USE OF EXISTING ELEVATORS

- A. The Contractor will be permitted to use an existing freight elevator for freight service and transportation of construction personnel during the construction period.
 - 1. The designated freight elevator is indicated on the drawings or, if not indicated will be designated by the Project Officer.
- B. Elevators will be used by both contractor and NIH and must be available to NIH at all times. Coordinate usage with the Project Officer. Use of any elevator is on a first come first use basis. Delays may be incurred.
- C. Protection of existing elevators: Provide protective pads for the cab and other appropriate protective measures for the car and entrance doors and frames. The Contractor shall not load the elevator to more than 75% of the rated capacity or the capacity indicated by the elevator operator, whichever is less. The Contractor is responsible for making repairs to any damage to the elevators during his use. If the Contractor fails to promptly repair any damage to the elevator or if the elevator is required to discontinue service, NIH may make any necessary repairs it deems required and deduct all associated costs directly from this construction contract. At the end of construction, restore the elevator to its original condition, replace worn cables, guide shoes and similar items of limited life.
- D. Oversized or overweight items: Where items cannot be fabricated and delivered in dimensions or weight to fit in freight elevator, but can fit in elevator shafts, provide all necessary dunnage, wenchers and other appurtenances to hoist items within shaft. Coordinate work with Project Officer.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 2900 - PAYMENT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section specifies administrative and procedural requirements necessary to prepare and process Applications for Payment.
- B. Related Sections:
 - 1. Division 01 Section "Construction Progress Documentation" for administrative requirements governing the preparation and submittal of the Contractor's construction schedule.
 - 2. Division 01 Section "Submittal Procedures" for administrative requirements governing the preparation and submittal of the submittal schedule.

1.3 SCHEDULE OF VALUES

- A. Coordination: Coordinate preparation of the schedule of values with preparation of Contractor's construction schedule. Cost-loaded Critical Path Method Schedule shall be used to satisfy requirements for the schedule of values.
 - 1. Correlate line items in the schedule of values with other required administrative forms and schedules, including the following:
 - a. Application for Payment forms with continuation sheets.
 - b. Submittal schedule.
 - c. Items required to be indicated as separate activities in Contractor's construction schedule.
 - 2. Submit the schedule of values to Project Officer at earliest possible date but no later than seven days before the date scheduled for submittal of initial Applications for Payment.
 - 3. Sub-schedules for Phased Work: Where the Work is separated into phases requiring separately phased payments, provide sub-schedules showing values correlated with each phase of payment.
 - 4. Sub-schedules for Separate Elements of Work: Where the Contractor's construction schedule defines separate elements of the Work, provide sub-schedules showing values correlated with each element.
- B. Format and Content: Use the Project Manual table of contents as a guide to establish line items for the schedule of values. Provide at least one line item for each Specification Section.
 - 1. Identification: Include the following Project identification on the schedule of values:

- a. Project name and location.
 - b. Name of Project Officer
 - c. Government's project number
 - d. Name of Architect.
 - e. Architect's project number.
 - f. Contractor's name and address.
 - g. Date of submittal.
2. Arrange schedule of values consistent with format of AIA Document G703.
3. Arrange the schedule of values in tabular form with separate columns to indicate the following for each item listed:
 - a. Related Specification Section or Division.
 - b. Description of the Work.
 - c. Name of subcontractor.
 - d. Name of manufacturer or fabricator.
 - e. Name of supplier.
 - f. Change Orders (numbers) that affect value.
 - g. Dollar value of the following, as a percentage of the Contract Sum to nearest one-hundredth percent, adjusted to total 100 percent.
 - 1) Labor.
 - 2) Materials.
 - 3) Equipment.
4. Provide a breakdown of the Contract Sum in enough detail to facilitate continued evaluation of Applications for Payment and progress reports. Coordinate with the Project Manual table of contents. Provide multiple line items for principal subcontract amounts in excess of five percent of Contract Sum.
5. Round amounts to nearest whole dollar; total shall equal the Contract Sum.
6. Provide a separate line item in the schedule of values for each part of the Work where Applications for Payment may include materials or equipment purchased or fabricated and stored, but not yet installed.
 - a. Differentiate between items stored on-site and items stored off-site. If required, include evidence of insurance.
7. Provide separate line items in the schedule of values for initial cost of materials, for each subsequent stage of completion, and for total installed value of that part of the Work.
8. Each item in the schedule of values and Applications for Payment shall be complete. Include total cost and proportionate share of general overhead and profit for each item.
 - a. Temporary facilities and other major cost items that are not direct cost of actual work-in-place may be shown either as separate line items in the schedule of values or distributed as general overhead expense, at Contractor's option.
9. Schedule Updating: Update and resubmit the schedule of values before the next Applications for Payment when Change Orders or Construction Change Directives result in a change in the Contract Sum.

1.4 APPLICATIONS FOR PAYMENT

- A. Each Application for Payment shall be consistent with previous applications and payments as certified by Architect and paid for by Owner.
 - 1. Initial Application for Payment, Application for Payment at time of Substantial Completion, and final Application for Payment involve additional requirements.
- B. Payment Application Times: Progress payments shall be submitted to Project Officer by the fifteenth day of the month. The period covered by each Application for Payment is one month, ending on the last day of the month.
 - 1. Submit draft copy of Application for Payment seven days prior to due date for review by Project Officer.
- C. Application for Payment Forms: Use AIA Document G702 and AIA Document G703 as form for Applications for Payment.
- D. Application Preparation: Complete every entry on form. Notarize and execute by a person authorized to sign legal documents on behalf of Contractor. Project Officer will return incomplete applications without action.
 - 1. Entries shall match data on the schedule of values and Contractor's construction schedule. Use updated schedules if revisions were made.
 - 2. Include amounts for work completed following previous Application for Payment, whether or not payment has been received. Include only amounts for work completed at time of Application for Payment.
 - 3. Include amounts of Change Orders and Construction Change Directives issued before last day of construction period covered by application.
 - 4. Indicate separate amounts for work being carried out under Owner-requested project acceleration.
- E. Stored Materials: Include in Application for Payment amounts applied for materials or equipment purchased or fabricated and stored, but not yet installed. Differentiate between items stored on-site and items stored off-site.
 - 1. Provide certificate of insurance, evidence of transfer of title to Owner, and consent of surety to payment, for stored materials.
 - 2. Provide supporting documentation that verifies amount requested, such as paid invoices. Match amount requested with amounts indicated on documentation; do not include overhead and profit on stored materials.
 - 3. Provide summary documentation for stored materials indicating the following:
 - a. Materials previously stored and included in previous Applications for Payment.
 - b. Work completed for this Application utilizing previously stored materials.
 - c. Additional materials stored with this Application.
 - d. Total materials remaining stored, including materials with this Application.
- F. Transmittal: Submit three signed and notarized original copies of each Application for Payment to Project Officer by a method ensuring receipt within 24 hours. One copy shall include waivers of lien and similar attachments if required.
 - 1. Transmit each copy with a transmittal form listing attachments and recording appropriate information about application.

- G. Waivers of Mechanic's Lien: With each Application for Payment, submit waivers of mechanic's lien from entities lawfully entitled to file a mechanic's lien arising out of the Contract and related to the Work covered by the payment.
1. Submit partial waivers on each item for amount requested in previous application, after deduction for retainage, on each item.
 2. When an application shows completion of an item, submit conditional final or full waivers.
 3. Owner reserves the right to designate which entities involved in the Work must submit waivers.
 4. Waiver Forms: Submit waivers of lien on forms, executed in a manner acceptable to Owner.
- H. Initial Application for Payment: Administrative actions and submittals that must precede or coincide with submittal of first Application for Payment include the following:
1. List of subcontractors.
 2. Schedule of values including proposed earned value curve for project.
 3. Contractor's construction schedule (preliminary if not final).
 4. Products list (preliminary if not final).
 5. Schedule of unit prices.
 6. Submittal schedule (preliminary if not final).
 7. List of Contractor's staff assignments.
 8. List of Contractor's principal consultants.
 9. Initial progress report.
 10. Report of preconstruction conference.
 11. Certificates of insurance and insurance policies.
 12. Performance and payment bonds.
 13. Data needed to acquire Owner's insurance.
- I. Application for Payment at Substantial Completion: After issuing the Certificate of Substantial Completion, submit an Application for Payment showing 100 percent completion for portion of the Work claimed as substantially complete.
1. Include documentation supporting claim that the Work is substantially complete and a statement showing an accounting of changes to the Contract Sum.
 2. This application shall reflect Certificates of Partial Substantial Completion issued previously for Owner occupancy of designated portions of the Work.
- J. Final Payment Application: Submit final Application for Payment with releases and supporting documentation not previously submitted and accepted, including, but not limited, to the following:
1. Evidence of completion of Project closeout requirements.
 2. Insurance certificates for products and completed operations where required and proof that taxes, fees, and similar obligations were paid.
 3. Updated final statement, accounting for final changes to the Contract Sum.
 4. AIA Document G706, "Contractor's Affidavit of Payment of Debts and Claims."
 5. AIA Document G706A, "Contractor's Affidavit of Release of Liens."
 6. AIA Document G707, "Consent of Surety to Final Payment."
 7. Evidence that claims have been settled.
 8. Final meter readings for utilities, a measured record of stored fuel, and similar data as of date of Substantial Completion or when Owner took possession of and assumed responsibility for corresponding elements of the Work.
 9. Final liquidated damages settlement statement.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION

SECTION 01 3100 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS.

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative provisions for managing and coordinating construction operations including, but not limited to, the following:

1. General project coordination.
2. Contractor Security Plan
3. Document Control
4. Communication
5. Release of Information
6. Utility service interruptions.
7. Secure, managed, internet-accessible collaboration website
8. Administrative and supervisory personnel.
9. Conferences and meetings.
10. Cleaning and protection.

- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to project management and coordination. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:

1. Division 01 Section "Summary"
2. Division 01 Section "Work Restrictions"
3. Division 01 Section "Construction Progress Documentation"
4. Division 01 Section "Photographic Documentation"
5. Division 01 Section "Submittal Procedures"
6. Division 01 Section "Quality Requirements"
7. Division 01 Section "Construction Quality Control"
8. Division 01 Section "Temporary Facilities and Controls"
9. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
10. Division 01 Section "Safety and Health"
11. Division 01 Section "Product Requirements"
12. Division 01 Section "Execution Requirements"
13. Division 01 Section "Cutting and Patching"
14. Division 01 Section "Fire Prevention Precautions for Hot Work"
15. Division 01 Section "Closeout Procedures"
16. Division 01 Section "Operation and Maintenance Documentation"
17. Division 01 Section "Project Record Documentation"
18. Division 01 Section "Demonstration and Training"

19. Division 01 Section "Selective Demolition"

1.3 GENERAL PROJECT COORDINATION

- A. Contractor Security Plan: The Contractor Security Plan shall include provisions to address an approach to overall security that is consistent with the goals contained in existing NIH Security Policies, Guidelines and Regulations. Security goals are as follows:
1. Screen A/E contractor workforce consistent with NIH policies and procedures.
 2. Safeguard NIH employees and assets from events or persons who could cause harm.
 3. Limit Project information distribution and ensure compliance with the National Institutes of Health Confidentiality Non-Disclosure Certification to authorized persons and entities to the greatest extent that is practical.
- B. Document Control
1. The Contractor shall assume responsibility for and maintain records of distribution of all drawings, specifications and other project information. Such information shall be distributed only to those specifically with a need for them to accomplish the Work.
 2. Certain documents may be marked as "Sensitive Unclassified - For Official Use Only" or "Law Enforcement Sensitive", hereinafter called "sensitive" information which may include drawings, sketches, digital photographs, exposed negatives, video, written descriptions of work requirements, and similar information. Return such information promptly to Project Officer upon request, including any copies.
 - a. Store all Project Information in file cabinets locked during non-business hours. Store "Sensitive" documents in separate containers with access restricted to those with a specific need for use of the information.
 - b. Reproduction of documents shall be kept at the minimum needed to accomplish the Work, and all copies shall be numbered and recipients recorded. "Sensitive" documents shall be returned to the NIH Project Officer when no longer needed, and the NIH will arrange for their proper disposal.
 - c. Limit dissemination of sensitive information on need-to-know basis.
 - d. Any proposed deviations from these requirements must be submitted to the Project Officer for review. Approval may or may not be granted.
 - e. Do not allow removal/transmission of such information from Project Site without prior written approval by Project Officer.
 - f. "Sensitive" documents shall be transmitted using U.S. Postal Service or any commercial services that permit delivery only after receiving signature.
 - g. All paper waste and electronic media such as diskettes and CDs of "Sensitive" documents shall be shredded or otherwise destroyed in a manner acceptable to the NIH.
 3. Notification and Reporting: Notify both the Project Officer and DPSM immediately when known or suspected loss/compromise of "sensitive" information or other documents, notes, drawings, sketches, reports, photographs, exposed films, or similar information has occurred which may affect the security interests of the NIH. Extend this requirement to employees and other personnel working on behalf of Contractor, and expand the responsibility to include prompt reporting of security issues, including observed efforts by unauthorized persons to gain access to sensitive information.
 4. The Contractor and each sub-contractor or other entity involved in the Project shall submit an electronic security memorandum describing the approach to meeting the above goals and maintaining confidentiality of Project files. The memorandum shall describe security within the Project, including any intrusion prevention and detection methodology.
 - a. The Contractor shall not proceed without NIH's written approval of the memorandum.

- b. Any requests for exceptions from these requirements shall be submitted in writing to the Project Officer for review. No entity or person shall proceed with such exception without written approval by the NIH.
- 5. Electronic Document Security
 - a. All Electronic files shall be stored in specified location following NIH standards and procedures using an Engineering Document Management Software (EDMS).
 - b. Each Project shall have a registry file, listing the project team with their contact information, drawing list, and Revision index stored in the same location of the other project documents.
 - c. Security, access and maintenance of all project engineering drawings and related documents, both scanned and electronic, shall be performed and tracked through the EDMS system.
 - d. All documents shall be distributed among team members, including A/E, DM, CM, and their sub-contractors with approval and knowledge of the NIH Project Officer.
 - e. Three principles shall always be followed in providing information:
 - 1) Only give the information to those that have a need to know.
 - 2) Keep records of who received the information.
 - 3) Safeguard the information during use.
- C. Communication
 - 1. Use of telephone or facsimile for communications will not be restricted except that these means shall not be used for "Sensitive" information.
 - 2. "Sensitive" information, including drawings and other documents. May be attached to electronic mail provided a commercially available encryption software compatible with Microsoft Outlook approved in advance for use by The NIH.
 - 3. If computer area networks are used for performing administrative or technical work, electronic partitions must be installed to limit access by unauthorized personnel to electronic files. Electronic files shall be organized to allow complete purging of files at the conclusion of the Work.
- D. Release of Information: No information pertaining to the Work shall be released for publication, including, but not limited to, press releases, internet posting, advertisement, marketing or solicitation without prior written approval of the NIH.
- E. Coordination of Trades: Coordinate construction operations included in the various sections of the Specifications to provide an efficient and orderly installation of each part of the Work. Coordinate construction operations included under different sections of the Specifications that depend on each other for proper installation, connection or operation.
 - 1. Schedule construction operations in the sequence required to obtain the best results where the installation of one part of the work depends on installation of other components before or after that part.
 - 2. Coordinate installation of different components to provide maximum accessibility for required maintenance, service, and removal/installation of component parts, testing and repair.
 - 3. Accommodate items scheduled for later installation.
 - 4. Provide for coordinated incorporation of Contractor's accepted Value Engineering proposals and Change Orders.
 - 5. Locate pipes, conduits, ducts, equipment, and their related supports so that they do not interfere with the intended use of lifting devices for adjacent equipment and components.
- F. Notification: Where required by this section and others, prepare and distribute memoranda to each party performing work at the project site, outlining special procedures required for

coordination, including required notices, reports, attendance at meetings and meeting minutes as part of the memoranda.

- G. Administrative Procedures: Coordinate scheduling and timing of administrative procedures with other construction activities to avoid conflicts and promote orderly progress of the work. Administrative procedures include but are not limited to the following:

1. Preparation of schedules and management plans.
2. Installation and removal of temporary facilities and controls.
3. Delivery and processing of submittals.
4. Progress meetings.
5. Pre-installation conferences

- H. Utility Service Interruption Plan

1. Contractor shall submit a Utility Service Interruption Plan for the project. Plan shall include proposed dates and times of each scheduled interruption based on the construction schedule, with estimated period of outage, list of existing equipment that will be affected by the interruption, proposed sequence of equipment shut-down and start-up, and responsible personnel. The plan should incorporate the following guidance:
 - a. Frequency and duration of interruption should be minimized.
 - b. Interruptions should be scheduled during times when the facility is unoccupied.
 - c. Requirements contained in the NIH ORF Manual "Policy and Procedure Utility Shutdown", latest edition, available on the NIH Website.
2. Plan must be approved in writing by the Contracting Officer and the Project Officer. If not approved, consult with the Project Officer, and revise and resubmit the plan until approved.
3. Requests for Utility Outages and Connections shall be made in writing to the Project Officer at least 18 calendar days in advance of the date and time required. Each request shall state the system involved, area involved, approximate duration of outage, and the nature of work involved.
4. Utility outages and connections required during the performance of the work that affect existing systems and are deemed by the Project Officer to jeopardize normal NIH operations shall be made during other than normal duty hours.
5. Coordination and notification to NIH tenants and activities will be accomplished by the Contracting Officer following notification by the contractor.
6. Prior to each interruption, the Contractor will confirm with the Contracting Officer that any necessary NIH personnel are prepared to execute the shut-down and start-up of affected existing equipment.

- I. Hazardous Materials Waste Management Plan

1. Contractor shall establish and submit for approval a Hazardous Materials Waste Management Plan within 14 calendar days of the Notice to Proceed or at least 14 calendar days prior to introducing hazardous materials on the site.
2. The Hazardous Materials Waste Management Plan shall be included (in draft form if approved copy is not available) with the submission of the Contractor's Safety and Health Program.
3. Requirements for the Hazardous Material Waste Management Plan are located in Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material."

- J. Heating and Cooling Plan

1. Contractor shall submit a Heating and Cooling Plan for the project when the work will impact the conditioned spaces of existing NIH facilities. Plan shall include written description of how the contractor intends to maintain the interior temperature between 65 degrees F and 80 degrees F or at other environmental conditions specified in the project documents. Include in the plan the heating and cooling of temporary facilities which enclose existing NIH spaces. Where plan requires use of permanent equipment to provide this conditioning, refer to 015000 for submission of the Temporary Operation and Conditioning Plan.

1.4 SUBMITTALS

- A. Contractor Security Plan: Develop and submit a Contractor Security Plan to the Contracting Officer for Approval within 14 calendar days after Notice to Proceed.
- B. Utility Service Interruption Plans
 1. Develop and submit a Utility Service Interruption Plan to the Contracting Officer for approval within 14 calendar days after the Notice to Proceed and at least 18 calendar days in advance of the first requested interruption to coincide with the request for utility outage.
 2. Update the plan and re-submit when conditions change at least 15 calendar days in advance of the first requested change to a utility interruption.
 3. Include a copy as an appendix to the Quality Control Plan.
- C. Hazardous Waste/Waste Management Plan
 1. Develop and submit to the Contracting Officer for approval within 14 calendar days after Notice to Proceed a Hazardous Waste/Waste Management Plan that conforms to the requirements contained in Division 1 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials."
 2. Update the plan and re-submit for information when conditions change.
 3. Include a copy as an appendix to the Quality Control Plan.
- D. Heating and Cooling Plan
 1. Develop and submit to the Contracting Officer for approval within 14 calendar days after Notice to Proceed a Heating and Cooling Plan for occupied NIH spaces effected by the project work.
 2. Update the plan and re-submit for approval when conditions change.
 3. Include a copy as an appendix to the Quality Control Plan.
- E. Integrated Pest Management Service Plan
 1. Develop and submit to the Contracting Officer for approval within 14 calendar days after Notice to Proceed a Integrated Pest Management Service Plan conforming to the requirements identified in Division 01 Section "Temporary Facilities and Controls."
 2. Update the plan and re-submit for approval when conditions change.
 3. Include a copy as an appendix to the Quality Control Plan.
- F. Demolition Plan
 1. Develop and submit to the Contracting Officer for review within 14 calendar days after Notice to Proceed and at least 14 calendar days prior to the planned commencement of demolition activities a Demolition Plan conforming to the requirements identified in Division 01 Section "Selective Demolition" or other demolition related sections.

2. Update program and re-submit for approval when conditions change.
3. Include a copy as an appendix to the Quality Control Plan.

G. Photographic Documentation Plan

1. Develop and submit to the Contracting Officer for review within 14 calendar days after Notice to Proceed a Photographic Documentation Plan conforming to the requirements identified in Division 01 Section "Photographic Documentation."
2. Update program and re-submit for approval when conditions change.
3. Include a copy as an appendix to the Quality Control Plan.

H. Quality Control Plan

1. Develop and submit to the Contracting Officer for approval within 30 calendar days after Notice to Proceed a Quality Control Plan conforming to the requirements identified in Division 01 Section "Construction Quality Control."
2. Update program and re-submit for approval when conditions change.

I. Safety and Health Program

1. Develop and submit to the Contracting Officer for review within 14 calendar days after Notice to Proceed the Contractor's Safety and Health Program as required by Division 01 Section "Safety and Health."
2. Update program and re-submit for approval when conditions change.

J. Cutting and Patching Plan

1. Develop and submit to the Contracting Officer for approval a cutting and patching plan for approval at least 21 calendar days in advance of the time cutting and patching will initially be performed as required by Division 01 Section "Cutting and Patching."
2. Update program and re-submit for approval when conditions change.
3. Include a copy as an appendix to the Quality Control Plan.

K. Project Closeout and Work Plan

1. Develop and submit to the Contracting Officer for approval a project closeout and work plan for approval at least 15 calendar days prior to the Substantial Completion Inspection as required by Division 01 Section "Closeout Procedures."
2. Update program and re-submit for approval when conditions change.
3. Coordinate contents of this plan with Division 01 Section "Construction Quality Control" and the contents of the Quality Control Plan.

L. Staff Names: Within 14 calendar days of the start of site operations, submit a list of principal staff assignments, including superintendent and other personnel in attendance at Project site. Identify individuals and their duties and responsibilities; list addresses and telephone numbers, including home and office telephone numbers. Provide names, addresses, and telephone numbers of individuals assigned as standbys in the absence of individuals assigned to Project.

1. Post copies of list in Project meeting room, in temporary field office, and by each temporary telephone.
2. Provide updated lists when individuals assigned to positions change or positions are added to the contractor's staff team.

M. Meeting minutes: The contractor shall provide meeting minutes from the following meeting as required by this section. Additional meeting minutes requirements may be located in other sections.

1. Preconstruction Conference.
2. Preinstallation Conferences.
3. Progress meetings.
4. Coordination Meetings
5. Coordination BIM Meetings
6. Health and Safety Meetings

1.5 ADMINISTRATIVE AND SUPERVISORY PERSONNEL

- A. In addition to the Project Superintendent, the Contractor shall provide other administrative and supervisory personnel as required for proper performance of the work.
- B. Project Coordinator: Provide a full-time project coordinator, experienced in the administration and supervision of building construction, including mechanical and electrical work. The project coordinator shall at minimum have a Bachelors degree from an accredited university or college in the field of construction management, engineering, architectural engineering or architecture, and shall be authorized to act as the coordinator of construction activities, including but not limited to the following:
 1. Scheduling and sequencing of work.
 2. Sharing access to work spaces.
 3. Installations.
 4. Protection of work.
 5. Cutting and patching.
 6. Selections for compatibility.
 7. Preparation of coordination of drawings.
 8. Inspection and tests.
 9. Temporary services and facilities.
 10. Conduct Project Coordination Meetings
- C. Mechanical and Electrical Coordinator: Provide a full-time mechanical and electrical coordinator, with a Bachelor of Science degree in mechanical, electrical or architectural engineering from an accredited college or university, a minimum of five years experience in the coordination of mechanical and electrical construction of the types and complexity required for the project, and experienced in coordination of mechanical and electrical construction with other operations. The mechanical and electrical coordinator shall be authorized to act as the coordinator for mechanical and electrical activities, including but not limited to the following:
 1. Scheduling and sequencing of mechanical and electrical activities.
 2. Sharing access to mechanical and electrical work spaces.
 3. Integration of mechanical and electrical work into limited spaces available for mechanical and electrical installations.
 4. Protection of mechanical and electrical work.
 5. Cutting and patching for mechanical and electrical work.
 6. Tolerances for mechanical and electrical work.
 7. Preparation of mechanical and electrical coordination drawings.
 8. Application of BIM-based coordination techniques to identify and resolve spatial conflicts
 9. Mechanical and electrical inspections and tests.
 10. Utilization of mechanical and electrical temporary services and facilities.
 11. Commissioning
- D. Safety and Health Officer: Provide a full time safety and health officer whose duties shall consist of developing and implementing safety and health programs specified in Division 01 Section "Safety and Health."

1. Other Duties: Safety and Health Officer shall not have other duties.
2. Additional Requirements: See Division 01 Section "Safety and Health" for additional requirements and information regarding the Safety and Health Officer.
3. Additional Safety and Health Officers:
 - a. For each additional 75 employees (including Subcontractor employees) above 40 on any shift, provide one additional full time Safety and Health Officer.
 - 1) Upon review by the Project Officer, requirement for additional Safety and Health Officer(s) may be wholly or partially waived at Project Officer's sole discretion.
- E. Environmental Safety Officer/Industrial Hygienist: Provide the services of an Industrial Hygienist with duties as indicated in Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material."
- F. Energy Conservation / Waste Management Officer: Provide a full time energy conservation / waste management officer whose duties shall consist of:
 1. Developing and implementing a program for minimizing use of energy on the site.
 2. Developing and implementing a program for maximizing recycling of waste.
- G. Integrated Pest Management Quality Control Supervisor: Provide an Integrated Pest Management Quality Control Supervisor with duties as indicated in Division 01 Section "Temporary Facilities and Controls."

1.6 CONFERENCES AND MEETINGS

- A. Pre-construction Conference:
 1. The Contractor shall attend a pre-construction conference scheduled by the Project Officer at a time and place convenient to both parties. Work shall not commence prior to the conference. Conference shall review responsibilities and personnel assignments.
 2. Attendees: Participants at the conference shall be familiar with the project, shall be authorized to conclude matters relating to the Work, and at a minimum include representatives of the following parties or their designated representatives:
 - a. Contracting Officer.
 - b. NIH Project Officer
 - c. Architect.
 - d. Major design consultants.
 - e. Contractor
 - f. Major subcontractors
 - g. Major Suppliers
 - h. NIH Police
 - i. NIH Division of Safety, including representatives from the Pest Management Section and the Occupational Safety and Health Branch
 - j. Other concerned parties.
 3. Agenda: Subjects for discussion shall include items of significance that could affect progress, including but not limited to the following:
 - a. Tentative construction schedule.
 - b. Critical work sequencing.
 - c. Designation of responsible personnel.
 - d. Procedures for processing field decisions and Change Orders.
 - e. Procedures for processing Applications for Payment.

- f. Distribution of Contract Documents.
 - g. Submittal of Shop Drawings, Product Data, and Samples.
 - h. Preparation of Record Documents.
 - i. Use of the premises.
 - j. Parking availability.
 - k. Office, work, and storage areas.
 - l. Equipment deliveries and priorities.
 - m. Safety and Health procedures
 - n. First Aid
 - o. Security
 - p. Housekeeping and progress cleaning
 - q. Working hours
 - r. Permits
 - s. Integrated Pest Management procedures.
- 4. Reporting: No later than 3 calendar days after the conference, the Contractor shall distribute minutes of the conference to each party present and to other concerned parties, including the Contracting Officer.
 - 5. This meeting may satisfy the requirement for a Pre-construction Safety Meeting in Division 01 Section "Safety and Health."
- B. Pre-installation Conferences: The Contractor shall conduct a pre-installation conference at the Project Site before each construction activity that requires coordination with other construction. Pre-installation conferences may be combined with Preparatory Phase meetings described in Division 01 Section "Construction Quality Control."
- 1. Attendees: In addition to the Contractor's representative, the installer and representatives of manufacturers and fabricators involved in or affected by the installation, and its coordination or integration with other materials and installations that have preceded or will follow, shall attend. Prior to conferences, advise the Contracting Officer of scheduled conference dates.
 - 2. Agenda: Review the progress of other construction activities and preparations for the particular activity under consideration at each pre-installation conference, including requirements for the following.
 - a. Contract Documents
 - b. Options
 - c. Related Change Orders, including accepted Value Engineering proposals
 - d. Purchases
 - e. Deliveries
 - f. Submittals
 - g. Review of mockups
 - h. Possible conflicts
 - i. Compatibility problems
 - j. Time schedules
 - k. Weather limitations
 - l. Manufacturer's recommendations
 - m. Warranty requirements
 - n. Compatibility of materials
 - o. Acceptability of substrates
 - p. Quality requirements
 - q. Temporary facilities and controls
 - r. Space and access limitations
 - s. Governing regulations
 - t. Safety
 - u. Testing and inspection requirements

- v. Required performance results
 - w. Protection.
3. Reporting: Record significant discussions and agreements and disagreements of each conference. No later than 3 calendar days after each conference, the Contractor shall distribute minutes of the conference to each party present and to other concerned parties, including the Contracting Officer.
 4. Do not proceed with the installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to the performance of the work, and reconvene the conference at the earliest feasible date.
- C. Pre-Demolition Conference: The Contractor shall conduct conference at project site. Review methods and procedures related to selective demolition including, but not limited to, the following:
1. Inspect and discuss condition of construction to be selectively demolished.
 2. Review structural load limitations of existing structure.
 3. Review and finalize demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
 4. Review requirements of work performed by other trades that rely on substrates exposed by demolition operations.
 5. Discussion of any outstanding issues from NIH review of Contractor's Demolition Plan.
- D. Progress Meetings: The Project Officer shall conduct progress meetings at the Project Site at frequency stipulated by the Project Officer at the Pre-construction Meeting. Dates of meetings shall be coordinated with preparation of the payment request.
1. Attendees: In addition to the Contractor and Project Officer, each subcontractor, supplier, or other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented. All participants at the conference shall be familiar with the Project and authorized to conclude matters relating to the Work. The Project Officer shall include the Contracting Officer and/or the project designer in these meetings as required. The contractor is responsible for requesting the Contracting Officer and/or designer attend and provide justification to the Project Officer.
 2. Agenda: Review and correct or approve minutes of the previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to the status of the Project.
 3. Contractor's Construction Schedule: Review progress since the last meeting. Determine where each activity is in relation to the Contractor's Construction Schedule, whether on time or ahead or behind schedule. Determine the required contractor response to construction behind schedule; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to insure that current and subsequent activities will be completed within the Contract Time.
 - a. At each progress meeting, the Contractor will provide a list of construction activities completed since the previous Progress Meeting, activities in progress with percent completed, activities scheduled to start in the next 14 calendar days, and activities to start in the next thirty calendar days as derived from the accepted project schedule.
 - b. At each progress meeting, the Contractor will provide a list of construction activities whose start date or anticipated completion date deviate from the approved Construction Schedule by 7 or more calendar days. For each such identified activity, the Contractor shall:
 - 1) Indicate whether or not the activity is on the critical path of the project.

- 2) Identify action items necessary to minimize delay of the activity start or finish.
 - 3) Identify action items necessary to recover lost time.
- c. Schedule Updating: The Contractor shall revise the Contractor's Construction Schedule to reflect project progress after each progress meeting and include revisions approved by the Contracting Officer.
- 1) Schedule updates shall retain the Approved Construction Schedule in the form of a visual baseline format.
 - 2) Schedule updates shall include links to Coordination BIM.
- d. The revised schedule shall be issued monthly to correspond with the submission of the Contractor's application for payment.
4. Review the present and future needs of each entity present, including but not limited to the following:
- a. Interface requirements
 - b. Time
 - c. Sequence of operations
 - d. Status of submittals
 - e. Deliveries
 - f. Off-Site fabrication
 - g. Access
 - h. Site Utilization
 - i. Temporary facilities and controls
 - j. Hours of work
 - k. Hazards and risks
 - l. Proposed or scheduled utility service interruptions
 - m. Housekeeping and progress cleaning
 - n. Quality and work standards
 - o. Deficiencies
 - p. Change Orders
 - q. Documentation of information for payment requests
 - r. Status of Record BIM Activities and Updates
 - s. Status of Coordination BIM Activities
5. Reporting: The Project Officer may stipulate minutes of Progress Minutes to be prepared by the Architect of CQM consultant. If not so stipulated, no later than 3 calendar days after each meeting, the Contractor shall prepare and distribute minutes of the meeting to each party present and to other concerned parties who should have been present, including the Contracting Officer. Include a brief summary, in narrative form, of progress since the previous meeting and report. Minutes shall contain a minimum of the following:
- a. Summary of "old business" items including date item was first discussed, discussion at subsequent meetings, and description of resolution of item and resolution date. Old business items will not be removed from minutes until fully resolved.
 - b. Summary of "new business" items including assignment of action items necessary for resolution.
 - c. Summary of safety issues.
 - d. Summary of project activity related to Contractor Requests for Information (RFIs), Architect Supplemental Informations (ASIs), and submittals and submittal reviews. Include:

- 1) List of any items affecting or likely to affect Project cost or schedule.
6. The Contractor shall present Record Drawings at progress meetings for review by the NIH, if so requested.
7. The Progress meeting may be in conjunction with the Quality Control (QC) Meeting required per Division 01 Section "Construction Quality Control."
- E. Coordination Meetings: Conduct weekly project coordination meetings as required, to verify detailed coordination procedures for the upcoming construction operations in order to avoid potential problems and misunderstandings.
 1. Attendees: In addition to the Contractor and Project Officer, each subcontractor, supplier or other entity involved in coordination or planning construction activities shall be represented. All participants shall be authorized to conclude matters relating to the work.
 2. Agenda: Review the plans and requirements of each entity present, including but not limited to the subjects listed for Progress Meetings.
 3. Reporting: No later than 3 calendar days after each meeting, the Contractor shall distribute minutes of the meeting to each party present and to other concerned parties who should have been present, and including the Contracting Officer. Include a brief summary, in narrative form, of progress since the previous meeting and report.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION

3.1 GENERAL COORDINATION PROVISIONS

- A. Inspection of Conditions: Prior to installations, require the installer of each component to inspect both the substrate and conditions under which work is to be performed.
 1. Do not proceed until unsatisfactory conditions have been corrected in an acceptable manner.
 2. Coordinate temporary enclosures with required inspections and tests to minimize the necessity of uncovering completed construction for that purpose.
- B. Construction in Progress: Keep construction in progress, adjoining materials in place, and adjoining materials clean during handling and installation. Apply protective coverings where required for protection from damage or deterioration.
- C. Completed Construction: Clean completed construction, and provide maintenance, as frequently as necessary to prevent damage or soiling or other deterioration through the remainder of the construction period. Adjust and lubricate operable components as necessary to assure operability without damage.
- D. Limiting Exposures: Supervise construction operations to prevent exposure of any part of construction, completed or in progress, to harmful, dangerous, damaging or otherwise deleterious conditions during the construction period. Such conditions include but are not limited to the following:
 1. Excessive static or dynamic loading.
 2. Excessive internal or external pressures.
 3. Excessively high or low temperatures.
 4. Thermal shock.

5. Excessively high or low humidity.
6. Pollution and air contamination.
7. Water or ice.
8. Chemicals and solvents.
9. Light
10. Radiation
11. Puncture
12. Abrasion
13. Heavy Traffic
14. Soiling, staining, and corrosion
15. Bacteria
16. Rodent and insect infestation
17. Combustion
18. Electrical Current
19. High-speed operation
20. Improper lubrication
21. Unusual wear or other misuse
22. Contact between incompatible materials
23. Destructive testing
24. Misalignment
25. Excessive weathering
26. Unprotected storage
27. Improper shipping or handling
28. Theft or vandalism

END OF SECTION

SECTION 01 3200 - CONSTRUCTION PROGRESS DOCUMENTATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes certain schedules and reports required for documenting the progress of construction during the performance of the work, including the following:
- B. This section includes the entire spectrum of construction schedule requirements from simple hand-written bar charts to complex CPM diagrams. Edit this section after determining and confirming with the Project Officer the level of schedule complexity required.
- C. Select the following sub-paragraphs which apply to this project. Selection should be confirmed with the Project Officer when confirming the scope of the proposed contractor's quality control effort being required for the project. Coordinate selection with schedules included in Submittals article and Part 2 of this section.
 - 1. Preliminary Construction Schedule
 - 2. Contractor's Construction Schedule
 - 3. Material and Equipment Delivery Schedule
 - 4. Submittal Schedule
 - 5. Contractor's Production Report
 - 6. Material Location Reports
 - 7. Field Condition reports
 - 8. Special reports
- D. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to construction progress documentation. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Photographic Documentation"
 - 4. Division 01 Section "Project Management and Coordination"
 - 5. Division 01 Section "Submittal Procedures"
 - 6. Division 01 Section "Quality Requirements"
 - 7. Division 01 Section "References"
 - 8. Division 01 Section "Construction Quality Control"
 - 9. Division 01 Section "Temporary Facilities and Controls"
 - 10. Division 01 Section "Safety and Health"
 - 11. Division 01 Section "Execution Requirements"
 - 12. Division 01 Section "Cutting and Patching"
 - 13. Division 01 Section "Closeout Procedures"
 - 14. Division 01 Section "Demonstration and Training"
 - 15. Division 01 Section "Selective Demolition"

1.3 DEFINITIONS

- A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.
 - 1. Critical activities are activities on the critical path. They must start and finish on the planned early start and finish times.
 - 2. Predecessor activity is an activity that must be completed before a given activity can be started.
- B. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.
- C. Critical Path: The longest continuous chain of activities through the network schedule that establishes the minimum overall Project duration and contains no float.
- D. Event: The starting or ending point of an activity.
- E. Float: The measure of leeway in starting and completing an activity.
 - 1. Float time is not for the exclusive use or benefit of either the NIH or the Contractor.
 - 2. Free float is the amount of time an activity can be delayed without adversely affecting the early start of the following activity.
 - 3. Total float is the measure of leeway in starting or completing an activity without adversely affecting the planned Project completion date.
- F. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.
- G. Milestone: A key or critical point in time for reference or measurement.
- H. Network Diagram: A graphic diagram of a network schedule, showing activities and activity relationships.

1.4 SUBMITTALS

- A. Preliminary Construction Schedule: Submit the Preliminary Construction Schedule to the Contracting Officer within 10 calendar days following receipt of the Notice to Proceed.
- B. Contractor's Construction Schedule: Submit the Contractor's Construction Schedule to the Contracting Officer within 30 calendar days after Notice to Proceed.
- C. Submittal Schedule: Prepare and submit a complete schedule of submittals within 10 calendar days after receipt of Notice to Proceed.
- D. Contractor's Production Report: Reports are required for each day that work is performed and shall be attached to the Contractor Quality Control Report specified in Division 01 Section "Construction Quality Control" prepared for the same day.
- E. Material Location Reports: At not more than weekly intervals, prepare and submit a comprehensive list of materials delivered to and stored at the site.
- F. Field Condition Reports: When the need to take corrective action requires a departure from the Contract Documents, prepare a detailed report describing the problem and recommended changes and submit a copy to the Project Officer.

G. Special reports

1. Unusual Event Reports: When an event of an unusual and significant nature occurs at the site, prepare a detailed report. List the chain of events, persons participating, response by the Contractor's personnel, evaluation of the results or effects, and similar pertinent information. Submit a copy to the Project Officer immediately. Advise the Project Officer in advance when such events are known or anticipated.

PART 2 - PRODUCTS

2.1 MATERIAL AND EQUIPMENT DELIVERY SCHEDULE.

- A. Submit for Contracting Officer approval a schedule showing procurement plans for materials, plant, and equipment. Submit in the format and content as directed by the Contracting Officer and include as a minimum the following information:
 1. Description
 2. Date of purchase order
 3. Promised shipping date
 4. Name of manufacturer or supplier
 5. Date delivery is expected
 6. Date the material or equipment is required according to the current construction schedule
- B. Update the equipment delivery schedule at monthly intervals indicating material deliveries and purchases. Reflect any changes occurring since the last update. Submit copies of the purchase orders and confirmation of the delivery dates as directed.
 1. Revisions to the approved schedule which will change completion dates require Contracting Officer approval.

2.2 PRELIMINARY CONSTRUCTION SCHEDULE:

- A. Prepare and submit for Contracting Officer approval a preliminary time scaled arrow diagram for the project comprised of an outline of activities for the first 30 calendar days of construction. Include each significant construction activity. Coordinate each activity in the network with other activities. Schedule each activity in proper sequence and show the critical path graphically. Include the following minimum information for each activity:
 1. Beginning event number
 2. Description
 3. Duration estimate
 4. Scheduled start or days after commencement of the contract
 5. Scheduled finish date or days after commencement of the contract.
- B. Provide a separate fragnet for each Definable Feature of Work included in the Draft Contractor QC Plan required by Division 01 Section "Construction Quality Control."
- C. Schedule shall be time scaled in not more than weekly increments with dates of the first day of each week (Sunday) indicated.
- D. Completion of the work shall be in advance of the date for completion of the contract.
- E. Provide a skeleton diagram for the work beyond the preliminary time scaled arrow diagram.
- F. Cash Requirements Prediction: With the submission of the Preliminary Construction Schedule, include a preliminary monthly cash requirement prediction based on the scheduled progress of the activities.
- G. Distribution: Submit 3 copies of the preliminary construction schedule to the Contracting Officer.

- H. Within 10 calendar days after receipt of the preliminary construction schedule, the Contracting Officer will meet with the Contractor for a joint review, correction and adjustment of the preliminary schedule.
- I. Within 5 calendar days after the joint review, the Contractor shall revise the preliminary construction schedule in accordance with agreements at the joint review, and submit 3 copies of the revised Preliminary Construction Schedule to the Contracting Officer. The resubmission will be reviewed by the Contracting Officer and, if found to be adequate, one approved copy will be returned to the Contractor as the basis for the Contractor's construction schedule.
- J. Review comments made by NIH on the Preliminary Construction Schedule shall not relieve the Contractor from compliance with the requirements of the Contract Documents. The Contractor remains responsible for scheduling, sequencing and prosecuting the work to comply with the requirements of the Contract documents. NIH acceptance extends only to the activities of the contractor's schedule that NIH has been assigned responsibility for and agrees it is responsible. NIH will also review for contract imposed schedule constraints and conformance, and cost loading of the CPM activities. Comments offered on other parts of the schedule which the Contractor is assigned responsibility are offered as a courtesy and are not conditions of NIH acceptance; but are for the general conformance with established industry schedule concepts.

2.3 CONTRACTOR'S CONSTRUCTION SCHEDULE

- A. Prepare and periodically update a comprehensive, fully developed, CPM-program based type construction schedule.
- B. Basis for developing the Contractor's Construction Schedule shall be the approved Preliminary Construction Schedule including updates and feedback received since the start of construction. All requirements for the Contractor's Construction Schedule are in addition to those of the Preliminary Construction Schedule as appropriate.
- C. Include each significant construction activity. Provide a separate fragnet for each Definable Feature of Work included in the Draft Contractor QC Plan required by Division 01 Section "Construction Quality Control." Coordinate each activity in the network with other activities. Schedule each activity in proper sequence. Show critical path graphically. Include the following minimum data for each activity:
 - 1. Beginning event number.
 - 2. Description.
 - 3. Duration estimate.
 - 4. Cost estimate.
 - 5. Scheduled start date.
 - 6. Scheduled finish date.
 - 7. Actual start date, when applicable.
 - 8. Actual finish date, when applicable.
 - 9. Percent complete.
- D. Schedule shall be produced by a computer program that has been developed specifically to manage construction schedules and capable of accepting revised completion dates as modified by approved time adjustments and re-computation of all tabulation dates and float accordingly. Computer program shall be capable of displaying the schedule in a variety of graphical formats including Gantt Charts and Pert Charts.
- E. Cost Correlation: Provide a cost correlation line at the head of the schedule to indicate planned and actual cost. As work progresses, show dollar volume of work performed as of the dates used for applications for payment. The total of all activity costs shall equal the contract price.

- F. Within each construction activity's time fragnet, indicate estimated completion percentages in not more than 5 percent increments and, as work progresses, place marks to indicate actual completion percentages.
- G. Within 10 calendar days after receipt of the preliminary construction schedule, the Contracting Officer will meet with the Contractor for a joint review, correction and adjustment of the preliminary schedule.
- H. Within 5 calendar days after the joint review, the Contractor shall revise the preliminary construction schedule in accordance with agreements at the joint review, and submit 3 copies of the revised Preliminary Construction Schedule to the Contracting Officer. The resubmission will be reviewed by the Contracting Officer and, if found to be adequate, one approved copy will be returned to the Contractor as the basis for the Contractor's construction.
- I. Review comments made by NIH on the Preliminary Construction Schedule shall not relieve the Contractor from compliance with the requirements of the Contract Documents. The Contractor remains responsible for scheduling, sequencing and prosecuting the work to comply with the requirements of the Contract documents. NIH acceptance extends only to the activities of the contractor's schedule that NIH has been assigned responsibility for and agrees it is responsible. NIH will also review for contract imposed schedule constraints and conformance, and cost loading of the CPM activities. Comments offered on other parts of the schedule which the Contractor is assigned responsibility are offered as a courtesy and are not conditions of NIH acceptance; but are for the general conformance with established industry schedule concepts.
- J. Each activity will be assigned a Schedule Activity Number.
- K. Activity Duration: Define activities so no activity is longer than 21 days, unless specifically allowed by Project Officer.
- L. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 days, as separate activities in the schedule. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.
- M. Submittal Review Time: Include review and resubmittal times indicated in Division 01 Section "Submittal Procedures" in the schedule. Coordinate submittal review times in the Contractor's Construction Schedule with the Submittals Schedule.
- N. Include a separate activity for each contract if work is under more than one contract.
- O. Include a separate activity for each portion of the work performed by NIH or other NIH contractors that directly impacts the work of this contract.
- P. Include a separate activity for each product. Include any delivery dates indicated in Division 01, Section "Summary." Delivery dates indicated stipulate the latest possible delivery date.
- Q. Work Restrictions: Show the effect of the following items on the schedule:
 - 1. Coordination with existing construction.
 - 2. Limitations of continued occupancies.
 - 3. Uninterruptible services.
 - 4. Partial occupancy before Substantial Completion.
 - 5. Use of premises restrictions.
 - 6. Provisions for future construction.
 - 7. Seasonal variations.
 - 8. Environmental control.

- R. Work Stages: Indicate important stages of construction for each major portion of the work, including, but not limited to, the following:
1. Subcontract awards.
 2. Submittals.
 3. Purchases.
 4. Mockups.
 5. Fabrication.
 6. Sample testing.
 7. Deliveries.
 8. Installation.
 9. Tests and inspections.
 10. Adjusting.
 11. Curing.
 12. Startup and placement into final use and operation.
 - 13.
- S. Area Separations: Identify each major area of construction for each major portion of the work. Indicate where each construction activity within a major area must be sequenced or integrated with other construction activities to provide for the following:
1. Structural and demolition completion.
 2. Permanent space enclosure.
 3. Completion of West Tunnel construction.
 4. Completion of mechanical installation.
 5. Completion of electrical installation.
 6. Substantial Completion.
- T. Other Constraints: None.
- U. Milestones: Include milestones indicated in the Contract Documents in the schedule including, but not limited to, the Notice to Proceed, Substantial Completion, and Final Acceptance. Contract Modifications: For each proposed contract modification and concurrent with its submission, prepare a time-impact analysis using fragnets to demonstrate the effect of the proposed change on the overall project schedule.
- V. Updating: Provide an updated schedule in conjunction with project meeting requirements specified in Division 01 Section "Project Management and Coordination." In addition, should actual time lines for events or activities suggest a significant change in the planned progress of the work, the contractor shall submit for Contracting Officer approval a revised schedule with justification for the change. Highlight or otherwise emphasize revisions for ease of identification.
- W. Reports: As a minimum be prepared to furnish the following computer produced reports from sorts of the basic activity data. Actual reports required and report frequency will be determined by the Project Officer.
1. Activity sort by the preceding event number from lowest to highest and then in the order of the following event number.
 2. Activity sort by amount of total float, then in order of preceding event number.
 3. Activity sort in order of latest allowable start dates, then in order of preceding event numbers.
 4. Activity sort in order of latest allowable finish dates, then in order of preceding event numbers.
 5. Activity sort by trade code, early start date and total float.
 6. Contractor's monthly payment request sorted by responsibility.
- X. Float is not for the exclusive use or benefit of either NIH or the Contractor, but must be used in the best interest of completing the project on time. Extensions of time for performance required

under the Contract Clauses entitled "Changes", "Differing Site Conditions", or other applicable clauses, will be granted only to the extent that the equitable time adjustment for the activity or activities affected exceeds the total float along the activities involved.

- Y. Distribution: Print and distribute copies of the initial Contractor's Construction Schedule, and each updated or revised schedule, to the Contracting Officer, as well as to subcontractors and other parties required to comply with upcoming scheduled dates. Post copies in the project meeting room and temporary field office.

2.4 Submittal Schedule:

- A. Prepare and submit a complete schedule of submittals to the Contracting Officer for approval. The submittal schedule shall list all submittals required by the specifications, listed in order by the specification section in which they are required. Coordinate the Submittal Schedule with the Contractor's Construction Schedule and other related documents. The Submittal Schedule shall include the following information:
 - 1. Submittal Number.
 - 2. Submittal Description
 - 3. Related specification section and paragraph numbers
 - 4. Category of Submittal (Sample, Shop Drawing, Certified Test Report, etc.)
 - 5. Name of Subcontractor, if applicable.
 - 6. Description of the part of the work covered by the submittal.
 - 7. Corresponding activity or event number on the Contractor's Construction Schedule, if applicable.
 - 8. Scheduled date for first submittal (planned and actual dates).
 - 9. Schedule dates for resubmittal of disapproved submittals (planned and actual dates).
 - 10. Reviewing authority (NIH, Contractor QC Manager)
 - 11. Type of action by Contracting Officer's Technical Representative (approved; approved as noted; not approved, revise and resubmit).
 - 12. Date of action by Contracting Officer's Technical Representative (planned and actual dates).
- B. Submittals should be scheduled using the review timeframes indicated in Division 01 Section "Submittal Procedures" to maintain orderly progress of the work.
- C. Submittal schedule is to be compatible with the same software system used to generate the contractor's preliminary construction schedule and contractor's construction schedule.
- D. Schedule a re-submittal for each submittal using the re-submittal review timeframes indicated in Division 01 Section "Submittal Procedures."
- E. Distribution: Initially submit 3 paper copies and one electronic copy of the Submittal Schedule to the Contracting Officer for review. After the Contracting Officer's response, print and distribute revised copies to the Contracting Officer and entities designated by the Contracting Officer, and to subcontractors and other parties required to comply with upcoming submittal dates.
 - 1. Post copies in the project meeting room and temporary field office.
 - 2. When the submittal schedule is updated, repeat distribution and posting.
- F. Updating: The Submittal Schedule shall be kept up-to-date by the Contractor. A current updated copy of the Submittal Schedule shall be submitted with each Contractor payment request, and at other times as requested by the Contracting Officer, until all submittals are approved. Failure to provide the requested information, or delay in submitting required submittals may result in the payment request being returned to the Contractor until the required schedule or submittals are received.

2.5 CONTRACTOR PRODUCTION REPORT:

- A. Reports are required for each day that work is performed and shall be attached to the Contractor Quality Control Report specified in Division 01 Section "Construction Quality Control" prepared for the same day. Account for each calendar day throughout the life of the Contract. The reporting of work shall be identified by terminology consistent with the construction schedule. Contractor Production Reports are to be prepared, signed and dated by the project superintendent and shall contain the following information:
1. Date of report, report number, name of contractor, Contract number, title and location of Contract and superintendent present.
 2. Weather conditions in the morning and in the afternoon including maximum and minimum temperatures.
 3. Identify work performed by corresponding Schedule Activity No., PC#, Modification No., etc.
 4. A list of Contractor and subcontractor personnel on the work site, their trades, employer, work location, description of work performed, hours worked by trade, daily total work hours on work site this date (including hours on continuation sheets), and total work hours from start of construction.
 5. A list of job safety actions taken and safety inspections conducted. Indicate that safety requirements have been met including the results on the following:
 - a. Was a job safety meeting held this date? (If YES, attach a copy of the meeting minutes.)
 - b. Were there any lost time accidents this date? (If YES, attach a copy of the completed OSHA report.)
 - c. Was crane/manlift/trenching/scaffold/HV electrical/high work/hazmat work done? (If YES, attach a statement or checklist showing inspection performed.)
 - d. Was hazardous material/ waste released into the environment? (If YES, attach a description of incident and proposed action.)
 6. Identify Schedule Activity # related to safety action and list safety actions taken today and safety inspections conducted.
 7. Identify Schedule Activity Number, Submittal Number and list equipment/ material received each day that is incorporated into the job.
 8. Identify Schedule Activity Number, Owner and list construction and plant equipment on the work site including the number of hours used.
 9. Include any meter readings.
 10. Describe any emergency procedures implemented.
 11. Identify significant meetings and unusual events conducted during the day.
 12. Include a "remarks" section in this report which will contain pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, equipment started or tested, partial completion or Use of Occupancy granted, delays encountered and a record of visitors to the work site. For each remark given, identify the Schedule Activity Number that is associated with the remark.
 13. Attach additional pages if necessary to complete the report.

2.6 MATERIAL LOCATION REPORTS:

- A. At not more than weekly intervals, prepare and submit a comprehensive list of materials delivered to and stored at the site. The list shall be cumulative, showing materials previously reported plus items recently delivered. Include a statement of progress on and delivery dates for materials or items or equipment fabricated or stored away from the site. Submit an electronic copy through e-mail to the Project Officer.

2.7 FIELD CONDITION REPORTS:

- A. When the need to take corrective action requires a departure from the Contract Documents, the Contractor will prepare a detailed report. Include a statement describing the problem and recommended changes. Indicate reasons the Contract Documents cannot be followed. Submit a copy to the Project Officer via e-mail.

2.8 UNUSUAL EVENT REPORTS:

- A. When an event of an unusual and significant nature occurs at the site, the Contractor will prepare a detailed report. List the chain of events, persons participating, response by the Contractor's personnel, evaluation of the results or effects, and similar pertinent information. Submit a copy to the Project Officer immediately. Advise the Project Officer in advance when such events are known or predictable.

PART 3 - EXECUTION

3.1 CONTRACTOR'S CONSTRUCTION SCHEDULE

- A. Scheduling Consultant: Engage a consultant to provide planning, evaluation, and reporting using CPM scheduling.
 - 1. In-House Option: Owner may waive the requirement to retain a consultant if Contractor employs skilled personnel with experience in CPM scheduling and reporting techniques. Submit qualifications.
 - 2. Meetings: Scheduling consultant shall attend all meetings related to Project progress, alleged delays, and time impact.
- B. Contractor's Construction Schedule Updating: At bi-weekly intervals, update schedule to reflect actual construction progress and activities. Issue schedule two days before progress meeting occurring during week of schedule update.
 - 1. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.
 - 2. Include a report with updated schedule that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.
 - 3. As the work progresses, indicate Actual Completion percentage for each activity.
 - 4. Retain original approved Construction Schedule baseline and include baseline indicators on each schedule update.
- C. Distribution: Distribute copies of approved schedule to all parties identified by the Project Officer and the Contractor with a need-to-know schedule responsibility.
 - 1. Post copies in project meeting rooms and temporary field offices.
 - 2. When revisions are made, distribute updated schedules to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the work and are no longer involved in performance of construction activities.

3.2 COORDINATION

- A. Coordinate Contractor's Construction Schedule with subcontractors schedules, submittals schedule, progress reports, payment requests, and other required schedules and reports.

END OF SECTION

SECTION 01 3220 - PHOTOGRAPHIC DOCUMENTATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for the following:

1. Pre-construction and pre-demolition photographs.
2. Periodic construction photographs.
3. Record construction photographs.
4. Demonstration and training videos

- 1.3 RELATED SECTIONS: This specification section is related to any and all specification sections with explicit or implicit reference to photographic documentation. Specific submittal requirements of these related specification sections are not included in this section.

1.4 QUALIFICATIONS

- A. Pre-construction / pre-demolition and periodic construction and record photographs may be taken by any individual who demonstrates competency with camera.

1.5 SUBMITTALS

- A. Qualification Data: Submit photographer's list of completed projects with project names and addresses, and names and addresses of owners and architects.
- B. Photographic Documentation Plan: Include a photographic documentation plan section or appendix in the Quality Control Plan specified in Division 01 Section "Construction Quality Control". The plan should include but not be limited to:
 1. Proposed photographic media and equipment
 2. Vantage Points
 3. Photo frequency
 4. Storage plan
 5. Submission schedule
- C. Pre-construction and Pre-demolition Photographs: Submit one digital copy of all required photographs on CD ROM within 7 calendar days of Notice to Proceed.
- D. Periodic Construction Photographs: Submit one digital copy of all required photographs on CD ROM within 7 calendar days following each required set of progress construction photographs.

- E. Record Photographs: Submit 3 prints of each view and one digital copy containing all images on CD ROM within 7 calendar days of Substantial Completion.
- F. Digital Images: Electronically name each digital file with logical naming convention containing name of view and date of photo. Submit un-cropped images that have the same aspect ratio as the sensor. Submit un-compressed electronic files.

PART 2 - PRODUCTS

2.1 PHOTOGRAPHIC MEDIA

- A. Digital Images: JPEG format with minimum sensor size of 6.0 megapixels
- B. Demonstration and Training Video Format: High quality digital color video on DVD media.
- C. Video Identification: Each copy shall be labeled with an applied label containing the following information:
 - 1. Project name.
 - 2. Name of Contractor.
 - 3. Date video was recorded.
 - 4. Description of training or demonstration session and equipment or system name.

2.2 RECORDS

- A. A paper copy of all photographs is to be maintained on the project site by the contractor's quality control organization and available to NIH Project Officer during normal business hours.

PART 3 - EXECUTION

3.1 COOPERATION WITH PHOTOGRAPHER

- A. Contractor shall cooperate with the photographer's work, including providing auxiliary services as requested, access to the project site, and use of temporary lighting and other facilities.

3.2 CONSTRUCTION PHOTOGRAPHS

- A. Pre-construction and Pre-Demolition Photographs: Before construction is started, take photographs of the site and surrounding properties from different points of view selected by the Project Officer.
 - 1. Take not less than 8 photographs to show existing conditions adjacent to and within the project limits of construction, in sufficient detail to accurately record the physical conditions at the start of construction.
- B. Periodic Construction Photographs: Periodically take photographs of the project's progress from different points of view. Vantage points shall be selected by the photographer unless otherwise directed by the Project Officer.

1. Frequency: Take photographs weekly, with timing each month adjusted to coincide with the cutoff date associated with each application for payment.
 2. Number: Take not less than eight photographs each time, to best show the status of construction and progress since taking previous photographs.
- C. Demonstration and Training Videos: Record instruction of Owner's personnel in the operation and maintenance of equipment and systems. Edit video to remove non-instructional conversation. Photographer shall select vantage points to best show equipment, systems, and procedures demonstrated.

END OF SECTION

SECTION 01 3300 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes certain administrative and procedural requirements for shop drawings, coordination drawings, fire protection working plans, schedules, samples and certain other quality assurance submittals.
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to submittals. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Project Management and Coordination"
 - 3. Division 01 Section "Construction Progress Documentation"
 - 4. Division 01 Section "Photographic Documentation"
 - 5. Division 01 Section "Quality Requirements"
 - 6. Division 01 Section "References"
 - 7. Division 01 Section "Construction Quality Control"
 - 8. Division 01 Section "Temporary Facilities and Controls"
 - 9. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 10. Division 01 Section "Safety and Health"
 - 11. Division 01 Section "Product Requirements"
 - 12. Division 01 Section "Execution Requirements"
 - 13. Division 01 Section "Cutting and Patching"
 - 14. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 15. Division 01 Section "Closeout Procedures"
 - 16. Division 01 Section "Demonstration and Training"
 - 17. Division 01 Section "Selective Demolition"

1.3 DEFINITIONS

- A. Action Submittals: Unless specifically indicated as an "informational submittal" or as not requiring NIH approval or other action, submittals are action submittals and require NIH approval or other indicated action.
- B. Informational Submittals: Written and graphic information that is indicated as not requiring NIH approval or other NIH action but rather is indicated as an informational submittal in the specifications. Informational submittals may be rejected by NIH for not complying with requirements.

- C. Shop drawings: Drawings and schedules specifically prepared for the project, except for coordination drawings.
- D. Coordination Building Information Model (Coordination BIM) and Coordination Drawings: See Division 01 Section "Project Management and Coordination" for definition and contract requirements.
- E. Product data: Manufacturer's standard catalogs, pamphlets and other printed materials, and includes but is not limited to the following:
 - 1. Product specifications
 - 2. Installation instructions
 - 3. Color charts
 - 4. Catalog cuts
 - 5. Rough-in diagrams and templates
 - 6. Wiring diagrams
 - 7. Performance curves
 - 8. Operational range diagrams
 - 9. Mill reports
- F. Samples: Product samples of such scale to allow delivery for review, as well as field samples or mock-ups of full-size physical examples erected on-site or elsewhere, or establish a true-scale standard by which the corresponding work will be judged or a standard for compliance testing.
- G. Other quality assurance submittals include materials specifically prepared for the project, except drawings and schedules, and include but are not limited to the following:
 - 1. Design data and calculations
 - 2. Certifications of compliance or conformance
 - 3. Manufacturer's instructions and field reports

1.4 GENERAL SUBMITTAL REQUIREMENTS

- A. Format: Each required submittal shall be submitted in both paper and electronic format.
 - 1. Electronic format of each submittal item to be compatible with Coordination BIM and Record BIM requirements identified in Division 01 Sections "Project Management and Coordination" and "Record Project Documentation" respectively.
 - 2. Post electronic submittals to project server.
 - 3. Submit final electronic submittals for inclusion in Project Record BIM. All submittal files to be fully indexed and hyper-linked within Record BIM.
- B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities and with the Submittal Schedule specified in Division 01 Section "Construction Progress Documentation."
- C. All submittals will be certified by the Contractor as conforming to the requirements of the contract documents prior to being forwarded to the Contracting Officer.
 - 1. Submittals not identified as informational submittals are action submittals and require NIH approval prior to the product or item submitted being incorporated into the project.
 - 2. Submittals identified as information submittals are informational submittals and must be submitted for information not later than 10 working days prior to the product or item submitted being shipped to site or incorporated into the work.

- D. All submittals shall be transmitted to the Contracting Officer within 30 calendar days after receipt of Notice to Proceed, unless the approved Submittal Schedule specifically provides for an earlier or later submission. Transmit each submittal sufficiently in advance of the scheduled performance of related construction activities to avoid delaying the work, allowing for the review times specified in this section.
- E. Coordinate each submittal with other submittals and related activities that require sequential scheduling, to allow for testing, purchase, fabrication and product delivery in a timely manner.
- F. Schedule transmittal of different categories of submittals for the same Definable Feature of Work (DFOW) and for different elements of related parts of the work at the same time so as to minimize delay because of the need to review submittals concurrently for coordination. See Division 01 Section "Construction Quality Control" for definition of DFOW
 - 1. NIH reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.
- G. Allow sufficient time for submittal review, corrections following the initial review and re-submittal review before activities scheduled after the submittal approval.
 - 1. Initial submittal review: Allow 14 calendar days from receipt by NIH of the submittals for the initial review. If Architect's consultants are included in the review process add an additional 7 calendar days for review.
 - 2. Any resubmission required after NIH review shall be made within 14 calendar days after return of the submittal, unless specifically authorized otherwise by the Project Officer.
 - 3. Re-submittal review: Allow 14 calendar days from receipt by NIH of the submittals for the initial review. If consultants are included in the review process add an additional 7 calendar days for review.
 - 4. Special Review Timelines:
 - a. Provide not less than 30 calendar days for review of each fire protection submittal and re-submittal.
- H. Construction will generally not be allowed to proceed without approved submittals. The Contracting Officer may, as requested in writing on a case-by-case basis, allow construction to proceed with submittal approval pending. Failure by the Contractor to provide the required submittals in a timely manner will not result in an extension to the Contractor's Construction Schedule.
- I. Failure by the Contractor to provide the required submittals in a timely manner may result in progress payment requests being returned to the Contractor until submittals are up-to-date.
- J. Submittal Preparation: Identify and prepare drawings and samples as specified in the Construction Contract Clauses and other specification sections. Provide a permanent label on each submittal with the following information:
 - 1. Project name, contract number and work request number.
 - 2. Date of submission
 - 3. Name, address and telephone number of firm or entity that prepared the submittal.
 - 4. Name, address and telephone number of the Contractor.
 - 5. Name, address and telephone number of the subcontractor, supplier or manufacturer.
 - 6. Number and title of appropriate specification section.
 - 7. Drawing number and detail references, as appropriate.
 - 8. Space to record Contractor's review and approval markings approximately 5 by 5 inches by 5 inches by 5 inches (250 by 250 mm).

9. A blank page with a listing of all pages in the submittal shall be attached to the front of the submittal package with at least a 4 inches by 6 inches space for the NIH.
- K. Submittal Transmittal: Package each submittal for transmission and handling. Transmit each submittal from the Contractor by use of a transmittal form. The following minimum information shall be included on the transmittal form.
1. Project name and number.
 2. Date
 3. Destination (To:)
 4. Source (From:)
 5. Names of subcontractor, manufacturer and supplier, as applicable.
 6. Category of submittal
 7. Description of submittal
 8. Number and title of appropriate specification section.
 9. Submittal number, including means to separately identify initial submittal and each resubmittal.
 10. Certification by Contractor stating that submittal complies with the Contract Documents, or statement of deviations from the requirements of the Contract Documents including minor variations and limitations. Deviations may be listed on an attached sheet referenced on the transmittal form.
 11. Signature of transmitter.
- L. Direct Transmittal to Architects and Engineers: Unless prohibited by the Project Officer, submittals may be transmitted directly to design Architect and, where specifically stipulated for certain submittals, to Architect's Engineering consultants. Provide duplicate copy of transmittal to Project Officer, and in the case of direct transmittal to Architect's consultants, to the Architect.
1. Unless indicated otherwise in above or following paragraphs, transmit all submittals to Architect,
 2. Transmit all submittals requiring structural, mechanical, electrical, plumbing and fire protection review directly to the cognizant engineering consultant. Provide copy of transmittal to Architect.
 3. Transmit light fixture and other electrical device submittals to Electrical Engineer with concurrent single copy of submittal to Architect. Provide copy of transmittal to Architect.
 4. Transmit the following submittals to Architect or Architect's engineering consultant (as applicable) with concurrent copy to Project Officer or CQM consultant for NIH review:
 - a. Safing and firestopping
 - b. Fire protection system
 - c. Door hardware
 - d. Electronic security system
 - e. Elevator
 - f. Other submittals identified by Project Officer
- M. Certifications: Submit notarized certifications from the party certifying compliance of the submittal with specified requirements. Certifications shall be signed by an officer or other individual authorized to sign documents on behalf of the company certifying compliance. Certifications shall be as described in Division 01 Section "Construction Quality Control."
- N. Delegated-Design Submittal: Comply with requirements in Division 01 Section "Quality Requirements."
- O. Contractor's Construction Schedule: Comply with requirements in Division 01 Section "Construction Progress Documentation."

- P. Submittal Schedule: Comply with requirements in Division 01 Section "Construction Progress Documentation."

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

- A. General: Prepare and submit Action Submittals required by individual Specification Sections.
- B. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
 2. Mark each copy of each submittal to show which products and options are applicable to the project.
 3. Include the following information, as applicable:
 - a. Manufacturer's written recommendations.
 - b. Manufacturer's product specifications.
 - c. Manufacturer's installation instructions.
 - d. Standard color charts.
 - e. Manufacturer's catalog cuts.
 - f. Wiring diagrams showing factory-installed wiring.
 - g. Printed performance curves.
 - h. Operational range diagrams.
 - i. Mill reports.
 - j. Standard product operating and maintenance manuals.
 - k. Compliance with recognized trade association standards.
 - l. Compliance with recognized testing agency standards.
 - m. Application of testing agency labels and seals.
 - n. Notation of coordination requirements.
 - o. Notation of dimensions verified for fit by field measurements.
 4. Number of Copies: Submit five copies of each submittal, unless otherwise indicated or concurrent NIH review is required. Submit an additional copy of MEP submittals. NIH will return three copies. Mark up and retain one returned copy as a Project Record Document. Additional copies, if submitted, will not be returned to Contractor.
- C. Shop Drawings: Prepare and submit originally prepared information, drawn accurately to scale. Do not reproduce Contract Documents or copy standard printed materials as the basis for Shop Drawings and Coordination Drawings.
1. Preparation: Include the following information, as applicable:
 - a. Dimensions.
 - b. Identification of products and materials.
 - c. Fabrication and installation drawings.
 - d. Roughing-in and setting diagrams.
 - e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring.
 - f. Shopwork manufacturing instructions.
 - g. Templates and patterns.
 - h. Schedules.

- i. Design calculations.
 - j. Compliance with specified standards.
 - k. Notation of coordination requirements.
 - l. Notation of dimensions established by any field measurement.
 - m. Highlighted or encircled deviations from the Contract Documents, if any.
- 2. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring.
- 3. Sheet Size: Except for templates, patterns, and similar full-size drawings, submit Shop Drawings on sheets at least 8-1/2 by 11 inches (215 by 280 mm) but no larger than 30 by 42 inches (750 by 1000 mm).
- 4. For products and installations required to be included in Coordination BIM, prepare shop drawings using specified Building Information Modeling software and coordinated properly for inclusion in the Coordination BIM.
 - a. Time submittal of shop drawings as required for maximum effectiveness of Coordination BIM.
- D. Number of Copies: Submit five copies of each drawing, unless otherwise indicated or concurrent NIH review is required. Submit an additional copy of MEP submittals. NIH will return three copies. Mark up and retain one returned copy as a Project Record Document. Additional copies, if submitted, will not be returned to Contractor. Contractor shall be responsible for reproduction of marked up shop drawings required for operation and maintenance manuals or other construction management purpose. Coordination Drawings: Comply with requirements in Division 01 Section "Project Management and Coordination."
- E. Fire Protection Working Plan Submittals: Submit fire protection working plans, prepared in full compliance with all requirements stated in NFPA-13. Sheets shall be no smaller than 30 by 42 inches.
- F. Samples: Prepare physical units of materials or products, including the following:
 - 1. Comply with requirements in Division 01 Section "Quality Requirements" for mockups.
 - 2. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
 - 3. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from the same material to be used for the work, cured and finished in manner specified, and physically identical with the product proposed for use, and that show full range of color and texture variations expected. Samples include, but are not limited to, the following: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.
 - 4. Preparation: Mount, display, or package Samples in manner specified to facilitate review of qualities indicated. Prepare Samples to match Architect's sample where so indicated. Attach label on unexposed side that includes the following:
 - a. Generic description of Sample.
 - b. Product name or name of manufacturer.
 - c. Sample source.
 - 5. Additional Information: On an attached separate sheet, prepared on Contractor's letterhead, provide the following:
 - a. Size limitations.
 - b. Compliance with recognized standards.
 - c. Availability.

- d. Delivery time.
- 6. Submit Samples for review of kind, color, pattern, and texture for a final check of these characteristics with other elements and for a comparison of these characteristics between final submittal and actual component as delivered and installed.
 - a. If variation in color, pattern, texture, or other characteristic is inherent in the product represented by a Sample, submit at least three sets of paired units that show approximate limits of the variations.
 - b. Refer to individual Specification Sections for requirements for Samples that illustrate workmanship, fabrication techniques, details of assembly, connections, operation, and similar construction characteristics.
- 7. Number of Samples for Initial Selection: Submit three full sets of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. One copy of the submittal will be returned with options selected.
- 8. Number of Samples for Verification: Submit four sets of Samples. Two sets will be returned.
 - a. Submit a single Sample where assembly details, workmanship, fabrication techniques, connections, operation, and other similar characteristics are to be demonstrated.
- 9. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
 - a. Samples that may be incorporated into the work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
 - b. Samples not incorporated into the work, or otherwise designated as Owner's property, are the property of Contractor.
- G. Product Schedule or List: Submit a written summary indicating types of products required for the work and their intended location. Include the following information in tabular form:
 - 1. Type of product. Include unique identifier for each product.
 - 2. Number and name of room or space.
 - 3. Location within room or space.

2.2 INFORMATIONAL SUBMITTALS

- A. General: Prepare and submit Informational Submittals required by other Specification Sections. General qualitative requirements are as specified for Action Submittals elsewhere in this section unless otherwise indicated in this Article.
 - 1. Number of Copies: Submit two copies of each non-MEP submittal, unless otherwise indicated. Submit three copies of each MEP submittal. NIH will not return copies.
 - 2. Certificates and Certifications: Provide a notarized statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity.
 - 3. Test and Inspection Reports: Comply with requirements in Division 01 Section "Construction Quality Control."

- B. Contractor's Construction Schedule: Comply with requirements in Division 01 Section "Construction Progress Documentation."
- C. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person as required in the technical specifications and in Division 01 Section "Construction Quality Control." Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.
- D. Product Certificates: Prepare written statements on manufacturer's letterhead certifying that product complies with requirements.
- E. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on AWS forms. Include names of firms and personnel certified.
- F. Installer Certificates: Prepare written statements on manufacturer's letterhead certifying that Installer complies with requirements and, where required, is authorized for this specific Project.
- G. Manufacturer Certificates: Prepare written statements on manufacturer's letterhead certifying that manufacturer complies with requirements. Include evidence of manufacturing experience where required.
- H. Material Certificates: Prepare written statements on manufacturer's letterhead certifying that material complies with requirements.
- I. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements.
- J. Pre-construction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements.
- K. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.
- L. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements.
- M. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.
- N. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project. Include the following information:
 - 1. Name of evaluation organization.
 - 2. Date of evaluation.
 - 3. Time period when report is in effect.

4. Product and manufacturers' names.
 5. Description of product.
 6. Test procedures and results.
 7. Limitations of use.
- O. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements in Division 01 Section "Closeout Procedures."
- P. Design Data: Prepare written and graphic information, including, but not limited to, performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.
- Q. Manufacturer's Instructions: Prepare written or published information that documents manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer. Include the following, as applicable:
1. Preparation of substrates.
 2. Required substrate tolerances.
 3. Sequence of installation or erection.
 4. Required installation tolerances.
 5. Required adjustments.
 6. Recommendations for cleaning and protection.
- R. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. Include the following, as applicable:
1. Name, address, and telephone number of factory-authorized service representative making report.
 2. Statement on condition of substrates and their acceptability for installation of product.
 3. Statement that products at Project site comply with requirements.
 4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
 5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
 6. Statement whether conditions, products, and installation will affect warranty.
 7. Other required items indicated in individual Specification Sections.
- S. Insurance Certificates and Bonds: Prepare written information indicating current status of insurance or bonding coverage. Include name of entity covered by insurance or bond, limits of coverage, amounts of deductibles, if any, and term of the coverage.
- T. Construction Photographs: Comply with requirements in Division 01 Section "Photographic Documentation."

- 2.3 Material Safety Data Sheets: Submit information directly to NIH. MSDS sheets submitted directly to Architect will be returned.

PART 3 - EXECUTION

3.1 CONTRACTOR'S REVIEW

- A. Review each submittal prior to transmission to NIH and check for compliance with the Contract Documents. Note corrections and field dimensions. Mark with certification stamp before submitting to NIH
- B. Certification Stamp: Stamp each submittal with a uniform certification stamp. Include Project name and location, submittal number, Specification Section title and number, name of reviewer, date of Contractor's certification, and statement certifying that submittal has been reviewed, checked, and complies with the Contract Documents.

3.2 NIH ACTION

- A. Except for submittals for record or for information or for another purpose where no action and return is required, the Project Officer will review submittals and mark returned copies to indicate action taken.
- B. Compliance with specified characteristics is the Contractor's responsibility, and is not part of the Contracting Officer's review and indication of action taken. No matter what review action is taken, final acceptance will depend on the contractor's full compliance with the Contract Documents.
- C. Submittals that do not contain the required marking of approval by the Project Officer, as indicated in the specifications, shall not be used for construction.
- D. Action Stamp: Each submittal will be stamped with a uniform action stamp. The stamp shall be marked to indicate one of the following actions taken:
 - 1. "Approved" or "Approved as Submitted": The work covered by the submittal may proceed, provided it complies with the notations or corrections on the submittal and with the requirements of the Contract Documents.
 - 2. "Approved as noted, Resubmission not required": The contractor is authorized to proceed with work as noted provided the contractor takes no exception to the notations.
 - 3. "Approved as noted, Resubmission Required": The contractor is authorized to proceed with portions of the work as noted. The contractor must resubmit those items/components so noted with additional information or requirements, for approval, before work may proceed on that portion of the submittal.
 - 4. "Disapproved": The submittal is incomplete or does not comply with design concept or requirements of the contract documents. No work shall proceed for this item until re-submittal with appropriate changes is approved.
 - 5. "Not Reviewed": A submittal marked "not reviewed" will indicate submittal does not have evidence of being reviewed and certified by the Construction Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals which are returned for lack of review by the contractor or for being incomplete, with appropriate action, coordination, or change.
 - 6. "Receipt Acknowledged" or no action: Submittals which are for the record or for information only or for another purpose not requiring review action.

END OF SECTION

SECTION 01 3550 - CONSTRUCTION SECURITY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Other general provisions of the Construction Contract, including Contract Clauses and Section I, Provisions and other Division I sections apply to requirements of this Section.
 - 1. Refer to Section, "Work Restrictions" for Contractor use of the premises
 - 2. Refer to Section, "Temporary Facilities and Controls" for construction requiring temporary security specified herein.

1.2 SUBMITTALS

- A. Provide a list of Contractor and Subcontractor personnel proposed to work on the Project Site as required in Article, "CONTRACTOR PERSONNEL".
- B. Vehicle Authorization Requests as required in Article, "Motor Vehicle Restrictions".
- C. Electronic Security Memorandum as required in Article, "Document Control".
- D. Weekly Schedule as required in Article, "Schedule".

1.3 GENERAL

- A. The Contractor agrees on behalf of himself and all subcontractors that the following security regulations will be observed by Contractor and subcontractor personnel on the property. The Contractor shall make it a specific provision of his subcontracts that these regulations be accepted.
- B. The Contractor shall be accountable for compliance with the provisions of this Section by all individuals and entities employed by or under contract to the Contractor.
- C. Notification: Notify the Project Officer, or a designated representative, not less than 48 hours prior to performing work in a restricted-access area as defined by the NIH. Include the following:
 - 1. Companies: Name of each company performing the work.
 - 2. Personnel: Name, social security number, date and place of birth, citizenship and, where applicable, visa status of each individual who is to work.
 - 3. Time: The exact time, date, and hours of work.
 - 4. Areas: Specific areas of the building in which work is to be performed.

1.4 CONTRACTOR PERSONNEL

- A. Upon commencement of the work under this Contract, the following security procedures will apply:

1. The Contractor shall provide information about all Contractor and subcontractor personnel and others who require continuing access to the site, before access is required and when access ceases.
2. No interviews of prospective Contractor employees shall be conducted within the project site or NIH property. The Contractor and subcontractors will be required to maintain a field office, outside the project site or NIH property, for all public contacts. Applicants for employment and other persons not entitled to access to the project site shall be required to contact the Contractor or subcontractor at these offices.
3. Within 10 calendar days after the award of the Contract, the Contractor shall submit a list on the Contractor's letterhead stationary of all employees, subcontractors and their employees, and others who will perform work or otherwise require access to the site. Personnel shall be listed in alphabetical order by company. The list shall include the full name, social security number and date and place of birth, citizenship and, where applicable, visa status for each individual.
4. Name of any employee added later to the original list shall be submitted with the same information on the Contractor's letterhead stationary at least 8 calendar days in advance of the date of access by the employee.
5. The Contractor shall notify the Government in writing when personnel are no longer employed by the Contractor or a subcontractor. Individual's name, social security number and date and place of birth, citizenship, visa status where applicable and company who employed the individual, shall be included.
6. Each company or entity with personnel assigned to this Project shall ensure that all such personnel undergo a personnel security screening to determine their suitability for access to NIH facilities, information and data.
 - a. The Security Plan establishes two levels of contractor personnel involvement: those involved in sensitive duties; and those not involved in sensitive duties. The NIH will use two different screening processes accordingly.
 - b. Completion of a background questionnaire and assorted forms ("long form screening") as well as a credit check is required for:
 - 1) Contractor personnel with direct management responsibilities on the Project.
 - c. A police check (single, "short form" screening) is required for all other contractor personnel working on the Project.
 - d. In addition to other disqualifying factors as determined by the NIH, the following shall apply:
 - 1) Conviction for tax evasion may disqualify contractor personnel being considered for sensitive positions.
 - 2) A history of acts of violence, arrests for firearms or explosives violations, illegal alien status, or any felony convictions will disqualify contractor personnel from working on this Project.
7. The Contractor shall notify the DPS Police Branch of any change in personnel assigned to the project site, including changes in employee status such as disciplinary actions.
8. The NIH reserves the right to require the removal of any Contractor employee from the project site if the employee is deemed a security risk by the Division of Public Safety.
9. In order to permit the NIH Division of Police to supply badges for on-site personnel, the Contractor shall cause each individual to complete a personnel identification form. These forms will be provided by the NIH to the Contractor at the pre-construction conference. Processing of the forms will be performed by the NIH at NIH expense.
10. At a time designated by the Contracting Officer or when an individual reports to the site for work the first time, a period of at least 2 hours will be required for security processing, including review of identification forms and fabrication of a permanent badge. Personnel will then be permitted to go to work without further processing of identification forms by

the Government, but at least 15 minutes should be allowed each day for signing in with security to obtain access to the site.

11. The permanent badge furnished by the NIH to each Contractor employee or other person granted access to the site will serve to authorize the wearer to enter and leave the project area. The badge must be worn so as to be clearly visible at all times when on the work site.
12. The badge will be retained by the individual as long as he requires continued admittance to the site, but the Contractor will arrange for its immediate return to the Government when such need ceases. Temporary or visitor badges will be provided for persons who are identified as having an infrequent or temporary legitimate business need for access to the site.

1.5 SECURITY PROCEDURES

- A. Upon commencement of the work under this contract, the following security procedures shall apply to the Contractor and all subcontractors.
 1. Contractor personnel shall not enter the NIH campus without appropriate badge.
 2. For work outside "work hours" as defined in the Contract, the Contractor shall give the Contracting Officer or his designated representative at least 3 calendar days notice. This notice is required so that security arrangements may be provided and is separate and distinct from any notices required for utility shutdown or other outages. Police after hours permit is required.
 3. The NIH reserves the right to restrict photography of the Project or other areas of the NIH premises.
 - a. Cameras are not permitted without written permission from the Contracting Officer or his designated representative. If approved, permission will be granted in writing and will provide additional guidelines.
 4. Personnel may be subject to inspection of their personal effects when entering and leaving the Project Site. In addition, unscheduled inspections of personnel may be made while on site.
 5. The NIH reserves the right to close down the project site and order Contractor personnel off the premises in the event of a national emergency or a shut-down, for as long as the national emergency or shut down exists. The Contractor may only return to the site with written approval from the Contracting Officer or his authorized representative.

1.6 MOTOR VEHICLE RESTRICTIONS

- A. For motor vehicle restrictions in effect under this contract, see Section H of the Contract.
- B. Vehicle authorization requests shall be required for any vehicle entering the site, and shall be submitted 24 hours before the date and time of access. Access shall be restricted to picking up and dropping off materials and supplies and, with a permit issued by the NIH, to parking in designated areas.
- C. For vehicle inspection procedures, see Section H of the Contract.

1.7 PARKING

- A. For parking restrictions in effect under this contract, see Section H of the Contract.

1.8 KEY CONTROL

- A. Control of keys and lock combinations is essential for NIH security. Do not allow keys to be duplicated or removed from Project Site, nor allow lock combinations to be divulged without specific written advanced authorization from Project Officer. Such loss of control, observed or suspected by Project Officer, will result in requirement to change locks involved at Contractor's expense. Provide duplicate keys and lock combinations to Project Officer, as requested for the purpose of security inspections and emergency actions; include keys/combinations needed for unrestricted access to every area and element of Project, tool boxes, parked machines, storage units, and similar items.

1.9 DOCUMENT CONTROL

- A. The Contractor shall assume responsibility for and maintain records of distribution of all drawings, specifications and other project information. Such information shall be distributed only to those specifically with a need for them to accomplish the Work.
- B. Certain documents may be marked as "Sensitive Unclassified - For Official Use Only" or "Law Enforcement Sensitive", hereinafter called "sensitive" information which may include drawings, sketches, digital photographs, exposed negatives, video, written descriptions of work requirements, and similar information. Return such information promptly to Project Officer upon request, including any copies.
 - 1. Store all Project Information in file cabinets locked during non-business hours. Store "Sensitive" documents in separate containers with access restricted to those with a specific need for use of the information.
 - 2. Reproduction of documents shall be kept at the minimum needed to accomplish the Work, and all copies shall be numbered and recipients recorded. "Sensitive" documents shall be returned to the NIH Project Officer when no longer needed, and the NIH will arrange for their proper disposal.
 - 3. Limit dissemination of sensitive information on need-to-know basis.
 - 4. Any proposed deviations from these requirements must be submitted to the Project Officer for review. Approval may or may not be granted.
 - 5. Do not allow removal/transmission of such information from Project Site without prior written approval by Project Officer.
 - 6. "Sensitive" documents shall be transmitted using U.S. Postal Service or any commercial services that permit delivery only after receiving signature.
 - 7. All paper waste and electronic media such as diskettes and CDS of "Sensitive" documents shall be shredded or otherwise destroyed in a manner acceptable to the NIH.
- C. The Contractor and each sub-contractor or other entity involved in the Project shall submit an electronic security memorandum describing the approach to meeting the above goals and maintaining confidentiality of Project files. The memorandum shall describe security within the Project, including any intrusion prevention and detection methodology.
 - 1. The Contractor shall not proceed without written approval of the memorandum by the NIH.
 - 2. Any requests for exceptions from these requirements shall be submitted in writing to the Project Officer for review. No entity or person shall proceed with such exception without written approval by the NIH.
- D. Electronic Document Security

1. All Electronic files shall be stored in specified location following NIH standards and procedures using an Engineering Document Management Software (EDMS).
2. Each Project shall have a registry file, listing the project team with their contact information, drawing list, and Revision index stored in the same location of the other project documents.
3. Security, access and maintenance of all project engineering drawings and related documents, both scanned and electronic, shall be performed and tracked through the EDMS system.
4. All documents shall be distributed among team members, including A/E, DM, CM, and their sub-contractors with approval and knowledge of the NIH Project Officer. Three principles shall always be followed in providing information:
 - a. Only give the information to those that have a need to know
 - b. Keep records of who received the information
 - c. Safeguard the information during use.

1.10 COMMUNICATION

- A. Use of telephone or facsimile for communications will not be restricted except that these means shall not be used for "Sensitive" information.
- B. "Sensitive" information, including drawings and other documents. May be attached to electronic mail provided a commercially available encryption software compatible with Microsoft Outlook approved in advance by the NIH is used.
- C. If computer area networks are used for performing administrative or technical work, electronic partitions must be installed to limit access by unauthorized personnel Project to electronic files. Electronic files shall be organized to allow complete purging of Project files at the conclusion of the Work.

1.11 RELEASE OF INFORMATION

- A. No information pertaining to the Project shall be released for publication, including, but not limited to, press release, internet posting, advertisement or solicitation without prior written approval of the NIH.

1.12 SCHEDULE

- A. Before the close of business at the end of each week, the Contractor shall prepare a schedule for the following week's work, including the following:
 1. Specific location of work for each trade.
 2. Description of work for each trade.
 3. Number of persons who will be on site for each location and trade.
 4. Specific impacts required, such as equipment or utility shutdowns.
 5. Hours of Work.

PART 2 - PRODUCTS (Not used)

PART 3 - EXECUTION (Not used)

END OF SECTION

SECTION 01 4000 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes general administrative and procedural requirements for the contractor's quality control program. Requirements for the Contractor to provide quality control services required by the Contracting Officer are not limited by provisions of this Section. The specific technical quality control program required for the work is defined by the combination of this section and Division 01 Section "Construction Quality Control."
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to quality requirements. Specific submittal requirements of these related specification sections are not included in this section. Product quality requirements are contained in Divisions 02 through 33. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Photographic Documentation"
 - 6. Division 01 Section "References"
 - 7. Division 01 Section "Construction Quality Control"
 - 8. Division 01 Section "Temporary Facilities and Controls"
 - 9. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 10. Division 01 Section "Safety and Health"
 - 11. Division 01 Section "Product Requirements"
 - 12. Division 01 Section "Execution Requirements"
 - 13. Division 01 Section "Cutting and Patching"
 - 14. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 15. Division 01 Section "Closeout Procedures"
 - 16. Division 01 Section "Demonstration and Training"
 - 17. Division 01 Section "Selective Demolition"

1.3 DEFINITIONS

- A. Delegated Design: Contractual requirement for Contractor to provide professional services where indicated by Contract Documents.

- B. Quality Assurance: Activities, actions, and procedures performed by NIH or their designated representatives, before and during execution of the work to verify that the contractor's quality control program is producing the quality of work required by the contract documents and ensure that the construction complies with all requirements.
- C. Quality Control: Tests, inspections, procedures, and related actions performed by the contractor during and after execution of the work to evaluate that completed construction complies with contract requirements. Services do not include contract enforcement activities performed by the NIH or their designated representative.
- D. Mockups: Full-size, physical example assemblies to illustrate finishes and materials. Mockups are used to verify selections made under Sample submittals, to demonstrate aesthetic effects and, where indicated, qualities of materials and execution, and to review construction, coordination, testing, or operation; they are not Samples. Mockups establish the standard by which the remaining work will be judged.
- E. Testing Agency: An entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

1.4 DELEGATED DESIGN

- A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of the Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to the Contracting Officer.

1.5 SUBMITTALS

- A. Delegated-Design Submittal: When project includes requirement for Delegated Design, in addition to Shop Drawings, Product Data, and other required submittals, submit a statement, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional, indicating that the products and systems are in compliance with performance and design criteria indicated. Include list of codes, loads, and other factors used in performing these services.
- B. Permits, Licenses, and Certificates: For NIH's records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of the Work.

1.6 QUALITY CONTROL QUALIFICATIONS

- A. Qualification requirements for the specified contractor's quality control organization are included in Division 01 Section "Construction Quality Control."
- B. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.
- C. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of

manufacturer's products that are similar in material, design, and extent to those indicated for this project.

- D. **Installer Qualifications:** A firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent to that indicated for this project, whose work has resulted in construction with a record of successful in-service performance.
- E. **Manufacturer Qualifications:** A firm experienced in manufacturing products or systems similar to those indicated for this project and with a record of successful in-service performance.
- F. **Professional Engineer Qualifications:** A professional engineer who is legally qualified to practice in jurisdiction where the project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of the system, assembly, or product that are similar to those indicated for this project in material, design, and extent.
- G. **Specialists:** Certain sections of the specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for the activities indicated.
 - 1. Requirement for specialists shall not supersede building codes and similar regulations governing the work, nor interfere with local trade-union jurisdictional settlements and similar conventions.

1.7 PRE-CONSTRUCTION TESTING

- A. The testing agency shall perform pre-construction testing for compliance with specified requirements for performance and test methods.
 - 1. Contractor responsibilities include the following:
 - a. Provide test specimens and assemblies representative of proposed materials and construction. Provide sizes and configurations of assemblies to adequately demonstrate capability of product to comply with performance requirements.
 - b. Submit specimens in a timely manner with sufficient time for testing and analyzing results to prevent delaying the Work.
 - c. Fabricate and install test assemblies using installers who will perform the same tasks for project.
 - d. When testing is complete, remove assemblies; do not reuse materials on the project.
 - 2. **Testing Agency Responsibilities:** Submit a certified written report of each test, inspection, and similar quality control service to the Project Officer or designated representative with copy to Contractor. Interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from the Contract Documents.

1.8 MOCKUPS

- A. Before installing portions of the work requiring mockups, build mockups for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed work:
 - 1. Build mockups in location and of size indicated or, if not indicated, as directed by the Project Officer.

2. Notify Project Officer 7 calendar days in advance of dates and times when mockups will be constructed.
3. Demonstrate the proposed range of aesthetic effects and workmanship using products identical to those proposed for the project.
4. Obtain Project Officer's approval of mockups before starting fabrication, or construction of remaining work represented by the mockup.
5. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
6. Demolish and remove mockups when directed, unless otherwise indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 REPAIR AND PROTECTION

- A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
 1. Comply with the Contract Document requirements for Division 01 Section "Cutting and Patching."
- B. Protect construction exposed by or for quality control service activities.
- C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality control services.

END OF SECTION

SECTION 01 4200 -REFERENCES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 DEFINITIONS

- A. General Explanation: Specification language often includes terms that are defined elsewhere in the Contract Documents, including the Construction Contract Clauses. Certain terms are defined in this section. These definitions or explanations are not necessarily complete or exclusive, but are general for the work and may be explained more explicitly in other Sections.
- B. "General Conditions" refer collectively to the Construction Contract Clauses, Labor Standards and the U.S. Department of Labor Wage Decision and Special Contract Requirements bound into the specifications.
- C. "Indicated" refers to graphic representations, notes or schedules on the Drawings, or to requirements elsewhere in the Specifications or other Contract Documents. Terms such as "shown", "noted", "scheduled" and "specified" have the same meaning as "indicated" and are used to further help locate the reference, but no limitation on location is intended except as specifically stated.
- D. Where "directed", "authorized", "selected", "approved", or a similar term is used in conjunction with the Contractor's submittals, applications, requests and other activities, and the specifications state that an individual other than the Contracting Officer, such as the Project Officer, Architect or Construction Engineer, shall provide this action, it is understood that only the Contracting Officer has this authority unless the individual stated is so authorized in writing by the Contracting Officer.
 - 1. When the individual is so authorized by the Contracting Officer, the Contractor may still appeal the action to the Contracting Officer.
 - 2. The Contracting Officer's decision will be final.
 - 3. In no case shall the Contracting Officer's action be interpreted as releasing the Contractor from responsibility to fulfill the requirements of the Contract Documents.
- E. "Regulations" include laws, ordinances, statutes and lawful orders issued by authorities having jurisdiction, as well as rules, conventions and agreements within the construction industry that control performance of the Work.
- F. "Project site" refers to the space available to the Contractor for performance of the Work, either exclusively or in conjunction with others performing other work.
- G. "Furnish" means to supply and deliver to the Project site, ready for unloading, unpacking, assembling, installation and similar operations.

- H. "Government Furnished" means the Government will supply the items so referenced, with the Contractor having the responsibility of pick-up, storage, delivery to the work site and final installation.
- I. "Install" describes operations at the Project site, including unloading, temporary storage, unpacking, assembling, erecting, placing, anchoring, applying, working to dimension, finishing, curing, protecting, cleaning and similar operations.
- J. "Provide" means to furnish and install, complete in place and ready for full use.
- K. The requirement for packaging, packing, marking, and preparation for shipment or delivery included in the referenced specifications will apply only to those materials and equipment that are furnished directly to the Government and not to materials and equipment that are furnished and installed by the Contractor.
- L. "Installer" is the Contractor or another entity engaged by the Contractor, either directly or indirectly through subcontracting, to perform a particular construction operation at the Project site, including installation, erection, application and similar operations. Installers shall be skilled in the operations they perform. Where indicated, installers shall also be Specialists as defined in the Construction Contract Clauses.
 - 1. Using a term such as "carpentry" does not imply that certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of the corresponding generic name.
- M. "Owner" refers to the National Institutes of Health (NIH).
- N. "Government" refers to the National Institutes of Health (NIH).
- O. "Project Officer" refers to the NIH Contracting Officer's Technical Representative (COTR).
- P. "Building Manager/Facility Manager" is the Government employee responsible for the administration, operation and maintenance of the building.
- Q. "Construction Quality Manager" (CQM) is the individual or entity, under Contract to the Government, responsible for performing the day-to-day coordination and administration of the construction Contract, including performing field inspections, recommending approval or rejection of material and workmanship, monitoring labor and safety provisions, maintaining inspection logs and records of defects, and similar activities.
- R. "Notice to Proceed" is the Contracting Officer's notification in writing to the Contractor to proceed with the individual task orders, activating the time period for construction and establishing the completion date.

1.3 DRAWING SYMBOLS

- A. Except as otherwise indicated, symbols used on the Drawings are those symbols recognized in the construction industry for the purposes.
 - 1. These include graphic symbols defined by "Architectural Graphic Standards", published by John Wiley & Sons, Inc., latest edition, as well as graphic symbols recommended by ASHRAE, ASME, ASPE, CSI, IEEE and similar technical organizations for the mechanical and electrical Drawings.

2. Refer uncertainties as to meaning of symbols to the Contracting Officer for clarification before proceeding.

1.4 INDUSTRY STANDARDS

- A. **Applicability of Standards:** Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect, to the extent referenced, as if bound or copied directly into the Contract Documents. Such standards are made a part of the Contract Documents by reference.
- B. **Conflicting Requirements.** Where compliance with two or more standards is specified and the standards establish different or conflicting requirements for minimum quantity or quality, comply with the most stringent requirement. Immediately refer uncertainties, and requirements that are different but apparently equal, to the Contracting Officer in writing for a decision before proceeding.
- C. **Minimum Quantity and Quality:** The quantity or quality indicated shall be the minimum provided. The actual installation may comply exactly with the minimum quantity or quality indicated, or it may exceed the minimum levels within reasonable limits.
 1. Indicated numeric values are minimum or maximum as appropriate for the context of the requirements.
 2. Refer uncertainties to the Contracting Officer for a decision before proceeding.

1.5 ABBREVIATIONS AND ACRONYMS

- A. **Industry Organizations:** Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. Names, telephone numbers, and Web site addresses are subject to change and are believed to be accurate and up-to-date as of the date of the Contract Documents.

B. CBHF	State of California, Department of Consumer Affairs	(800) 952-5210
	Bureau of Home Furnishings and Thermal Insulation	(916) 445-1254
	www.dca.ca.gov/bhfti	
	www.cpsc.gov	(301) 504-0990
DOC	Department of Commerce www.doc.gov	(202) 482-2000
EPA	Environmental Protection Agency www.epa.gov	(202) 260-2090
FAA	Federal Aviation Administration www.faa.gov	(202) 366-4000
FCC	Federal Communications Commission www.fcc.gov	(202) 418-0190
FDA	Food and Drug Administration www.fda.gov	(888) 463-6332
GSA	General Services Administration	(202) 708-5082

	www.gsa.gov	
HUD	Department of Housing and Urban Development www.hud.gov	(202) 708-1112
LBL	Lawrence Berkeley Laboratory (See LBNL)	
LBNL	Lawrence Berkeley National Laboratory www.lbl.gov	(510) 486-5605
NCHRP	National Cooperative Highway Research Program (See TRB)	
NIST	National Institute of Standards and Technology www.nist.gov	(301) 975-6478
OSHA	Occupational Safety & Health Administration www.osha.gov	(202) 693-1999
RUS	Rural Utilities Service (See USDA)	(202) 720-9540
TRB	Transportation Research Board www.nas.edu/trb	(202) 334-2934
USDA	Department of Agriculture www.usda.gov	(202) 720-2791
USPS	Postal Service www.usps.com www.aham.org	(202) 268-2000
AI	Asphalt Institute www.asphaltinstitute.org	(859) 288-4960
AIA	American Institute of Architects (The) www.e-architect.com	(202) 626-7300
AISC	American Institute of Steel Construction www.aisc.org	(800) 644-2400 (312) 670-2400
AISI	American Iron and Steel Institute www.steel.org	(202) 452-7100
AITC	American Institute of Timber Construction www.aitc-glulam.org	(303) 792-9559
ALA	American Laminators Association (See LMA)	
ALCA	Associated Landscape Contractors of America www.alca.org	(800) 395-2522 (703) 736-9666
ALSC	American Lumber Standard Committee	(301) 972-1700

AMCA	Air Movement and Control Association International, Inc. www.amca.org	(847) 394-0150
ANLA	American Nursery & Landscape Association (Formerly: AAN - American Association of Nurserymen) www.anla.org	(202) 789-2900
ANSI	American National Standards Institute www.ansi.org	(202) 293-8020
AOSA	Association of Official Seed Analysts www.aosaseed.com	(402) 476-3852
APA	APA - The Engineered Wood Association www.apawood.org	(253) 565-6600
APA	Architectural Precast Association www.archprecast.org	(941) 454-6989
API	American Petroleum Institute www.api.org	(202) 682-8000
ARI	Air-Conditioning & Refrigeration Institute www.ari.org	(703) 524-8800
ASCA	Architectural Spray Coaters Association www.ascassoc.com	(609) 848-6120
ASCE	American Society of Civil Engineers www.asce.org	(800) 548-2723 (703) 295-6300
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers www.ashrae.org	(800) 527-4723 (404) 636-8400
ASME	ASME International (The American Society of Mechanical Engineers International) www.asme.org	(800) 843-2763 (212) 591-7722
ASSE	American Society of Sanitary Engineering www.asse-plumbing.org	(440) 835-3040
ASTM	American Society for Testing and Materials www.astm.org	(610) 832-9585
AWCI	AWCI International (Association of the Wall and Ceiling Industries International) www.awci.org	(703) 534-8300
AWCMA	American Window Covering Manufacturers Association (See WCMA)	
AWI	Architectural Woodwork Institute	(800) 449-8811

	www.awinet.org	(703) 733-0600
AWPA	American Wood-Preservers' Association www.awpa.com	(817) 326-6300
AWS	American Welding Society www.aws.org	(800) 443-9353 (305) 443-9353
AWWA	American Water Works Association www.awwa.org	(800) 926-7337 (303) 794-7711
BHMA	Builders Hardware Manufacturers Association www.buildershardware.com	(212) 297-2122
BIA	Brick Industry Association (The) www.bia.org	(703) 620-0010
BIFMA	BIFMA International (Business and Institutional Furniture Manufacturer's Association International) www.bifma.com	(616) 285-3963
CCC	Carpet Cushion Council www.carpetcushion.org	(203) 637-1312
CCFSS	Center for Cold-Formed Steel Structures www.umn.edu/~ccfss	(573) 341-4471
CDA	Copper Development Association Inc. www.copper.org	(800) 232-3282 (212) 251-7200
CEA	Canadian Electricity Association www.canelect.ca	(613) 230-9263
CFFA	Chemical Fabrics & Film Association, Inc. www.chemicalfabricsandfilm.com	(216) 241-7333
CGA	Compressed Gas Association www.cganet.com	(703) 412-0900
CGSB	Canadian General Standards Board www.pwgsc.gc.ca/cgsb	(819) 956-0425
CIMA	Cellulose Insulation Manufacturers Association www.cellulose.org	(888) 881-2462 (937) 222-2462
CISCA	Ceilings & Interior Systems Construction Association www.cisca.org	(630) 584-1919
CISPI	Cast Iron Soil Pipe Institute www.cispi.org	(423) 892-0137
CLFMI	Chain Link Fence Manufacturers Institute www.chainlinkinfo.org	(301) 596-2583

CPA	Composite Panel Association (Formerly: National Particleboard Association) www.pbmdf.com	(301) 670-0604
CPPA	Corrugated Polyethylene Pipe Association www.cppa-info.org	(800) 510-2772 (202) 462-9607
CRI	Carpet & Rug Institute (The) www.carpet-rug.com	(800) 882-8846 (706) 278-3176
CRSI	Concrete Reinforcing Steel Institute www.crsi.org	(847) 517-1200
CSA	CSA International (Formerly: IAS - International Approval Services) www.csa-international.org	(800) 463-6727 (416) 747-4000
CSI	Construction Specifications Institute (The) www.csinet.org	(800) 689-2900 (703) 684-0300
CSSB	Cedar Shake & Shingle Bureau www.cedarbureau.org	(604) 820-7700
CTI	Cooling Technology Institute (Formerly: Cooling Tower Institute) www.cti.org	(281) 583-4087
DHI	Door and Hardware Institute www.dhi.org	(703) 222-2010
EIA/TIA	Electronic Industries Alliance/Telecommunications Industry Association www.eia.org	(703) 907-7500
EIMA	EIFS Industry Members Association www.eifsfacts.com	(800) 294-3462 (770) 968-7945
EJMA	Expansion Joint Manufacturers Association, Inc. www.ejma.org	(914) 332-0040
FCI	Fluid Controls Institute www.fluidcontrolsintitute.org	(216) 241-7333
FGMA	Flat Glass Marketing Association (See GANA)	
FM	Factory Mutual System (See FMG)	
FMG	FM Global (Formerly: FM - Factory Mutual System) www.fmglobal.com	(401) 275-3000
GA	Gypsum Association www.gypsum.org	(202) 289-5440

GANA	Glass Association of North America (Formerly: FGMA - Flat Glass Marketing Association) www.glasswebsite.com/gana	(785) 271-0208
GRI	Geosynthetic Research Institute www.drexel.edu/gri	(215) 895-2343
GTA	Glass Tempering Division of Glass Association of North America (See GANA)	
HI	Hydraulic Institute www.pumps.org	(888) 786-7744 (973) 267-9700
HI	Hydronics Institute www.gamanet.org	(908) 464-8200
HMMA	Hollow Metal Manufacturers Association (See NAAMM)	
HPVA	Hardwood Plywood & Veneer Association www.hpva.org	(703) 435-2900
HPW	H. P. White Laboratory, Inc. www.hpwhite.com	(410) 838-6550
IAS	International Approval Services (See CSA International)	
ICEA	Insulated Cable Engineers Association, Inc. www.icea.net	(508) 394-4424
ICRI	International Concrete Repair Institute (The) www.icri.org	(703) 450-0116
IEC	International Electrotechnical Commission www.iec.ch	41 22 919 02 11
IEEE	Institute of Electrical and Electronics Engineers, Inc. (The) www.ieee.org	(212) 419-7900
IESNA	Illuminating Engineering Society of North America www.iesna.org	(212) 248-5000
IGCC	Insulating Glass Certification Council www.igcc.org	(315) 646-2234
ILI	Indiana Limestone Institute of America, Inc. www.iliai.com	(812) 275-4426
IRI	Industrial Risk Insurers www.industrialrisk.com	(800) 243-8308 (860) 520-7300
ITS	Intertek Testing Services	(800) 345-3851

	www.itsglobal.com	(607) 753-6711
IWS	Insect Screening Weavers Association (Now defunct)	
KCMA	Kitchen Cabinet Manufacturers Association www.kcma.org	(703) 264-1690
LEED	LEED-NC, U.S. Green Building Council www.usgbc.org	(202) 223-7299
LGSi	Light Gage Structural Institute www.loseke.com	(972) 370-0967
LMA	Laminating Materials Association (Formerly: ALA - American Laminators Association) www.lma.org	(201) 664-2700
LPI	Lightning Protection Institute www.lightning.org	(800) 488-6864 (847) 577-7200
LSGA	Laminated Safety Glass Association (See GANA)	
MBMA	Metal Building Manufacturers Association www.mbma.com	(216) 241-7333
MCA	Metal Construction Association www.metalconstruction.org	(312) 201-0193
MFMA	Maple Flooring Manufacturers Association www.maplefloor.org	(847) 480-9138
MFMA	Metal Framing Manufacturers Association	(312) 644-6610
MGPHO	Medical Gas Professional Healthcare Organization, Inc. www.mgpho.org	(877) 238-5157 (913) 681-6548
MHIA	Material Handling Industry of America www.mhia.org	(800) 345-1815 (704) 676-1190
MIA	Marble Institute of America www.marble-institute.com	(614) 228-6194
ML/SFA	Metal Lath/Steel Framing Association (See SSMA)	
MSS	Manufacturers Standardization Society of The Valve and Fittings Industry Inc. www.mss-hq.com	(703) 281-6613
NAAMM	National Association of Architectural Metal Manufacturers www.naamm.org	(312) 332-0405
NAAMM	North American Association of Mirror Manufacturers	

(See GANA)

NACE	NACE International (National Association of Corrosion Engineers International) www.nace.org	(281) 228-6200
NAIMA	North American Insulation Manufacturers Association (The) www.naima.org	(703) 684-0084
NAMI	National Accreditation and Management Institute, Inc.	(304) 258-5100
NAPM	National Association of Photographic Manufacturers (See PIMA)	
NBGQA	National Building Granite Quarries Association, Inc. www.nbgqa.com	(800) 557-2848
NCMA	National Concrete Masonry Association www.ncma.org	(703) 713-1900
NCPI	National Clay Pipe Institute www.ncpi.org	(414) 248-9094
NCTA	National Cable Television Association www.ncta.com	(202) 775-3669
NEBB	National Environmental Balancing Bureau www.nebb.org	(301) 977-3698
NECA	National Electrical Contractors Association www.necanet.org	(301) 657-3110
NeLMA	Northeastern Lumber Manufacturers' Association www.nelma.org	(207) 829-6901
NEMA	National Electrical Manufacturers Association www.nema.org	(703) 841-3200
NETA	InterNational Electrical Testing Association www.netaworld.org	(303) 697-8441
NFPA	National Fire Protection Association www.nfpa.org	(800) 344-3555 (617) 770-3000
NFRC	National Fenestration Rating Council www.nfrc.org	(301) 589-6372
NGA	National Glass Association www.glass.org	(703) 442-4890
NHLA	National Hardwood Lumber Association www.natlhardwood.org	(800) 933-0318 (901) 377-1818
NLGA	National Lumber Grades Authority	(604) 524-2393

	www.nlga.org	
NOFMA	National Oak Flooring Manufacturers Association www.nofma.org	(901) 526-5016
NPA	National Particleboard Association (See CPA)	
NRCA	National Roofing Contractors Association www.nrca.net	(800) 323-9545 (847) 299-9070
NRMCA	National Ready Mixed Concrete Association www.nrmca.org	(888) 846-7622 (301) 587-1400
NSA	National Stone Association www.aggregates.org	(800) 342-1415 (703) 525-8788
NSF	NSF International (National Sanitation Foundation International) www.nsf.org	(800) 673-6275 (734) 769-8010
NTMA	National Terrazzo and Mosaic Association, Inc. www.ntma.com	(800) 323-9736 (703) 779-1022
NWWDA	National Wood Window and Door Association (See WDMA)	
PCI	Precast/Prestressed Concrete Institute www.pci.org	(312) 786-0300
PDCA	Painting and Decorating Contractors of America www.pdca.com	(800) 332-7322 (703) 359-0826
PDI	Plumbing & Drainage Institute www.pdionline.org	(800) 589-8956 (508) 230-3516
PGI	PVC Geomembrane Institute //pgi-tp.ce.uiuc.edu	(217) 333-3929
PIMA	Photographic & Imaging Manufacturers Association (Formerly: NAPM - National Association of Photographic Manufacturers) www.pima.net	(914) 698-7603
RCSC	Research Council on Structural Connections www.boltcouncil.org	(800) 644-2400 (312) 670-2400
RFCI	Resilient Floor Covering Institute (Contact by mail only)	
RIS	Redwood Inspection Service www.calredwood.org	(888) 225-7339 (415) 382-0662
RMA	Rubber Manufacturers Association www.rma.org	(800) 220-7620 (202) 682-4800

SAE	SAE International www.sae.org	(724) 776-4841
SDI	Steel Deck Institute www.sdi.org	(847) 462-1930
SDI	Steel Door Institute www.steeldoor.org	(440) 899-0010
SEFA	Scientific Equipment and Furniture Association www.sefalabfurn.com	(843) 689-6878
SGCC	Safety Glazing Certification Council www.sgcc.org	(315) 646-2234
SIGMA	Sealed Insulating Glass Manufacturers Association www.sigmaonline.org/sigma	(312) 644-6610
SJI	Steel Joist Institute www.steeljoist.org	(843) 626-1995
SMA	Screen Manufacturers Association	(561) 533-0991
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association www.smacna.org	(703) 803-2980
SPFA	Spray Polyurethane Foam Alliance (Formerly: SPI/SPFD - The Society of the Plastics Industry, Inc.; Spray Polyurethane Foam Division) www.sprayfoam.org	(800) 523-6154
SPI	The Society of the Plastics Industry www.plasticsindustry.org	(202) 974-5200
SPIB	Southern Pine Inspection Bureau (The) www.spib.org	(850) 434-2611
SPI/SPFD	The Society of the Plastics Industry Spray Polyurethane Foam Division (See SPFA)	
SPRI	SPRI (Single Ply Roofing Institute) www.spri.org	(781) 444-0242
SSINA	Specialty Steel Industry of North America www.ssina.com	(800) 982-0355 (202) 342-8630
SSMA	Steel Stud Manufacturers Association (Formerly: ML/SFA - Metal Lath/Steel Framing Association) www.ssma.com	(312) 456-5590
SSPC	SSPC: The Society for Protective Coatings	(800) 837-8303

	www.sspc.org	(412) 281-2331
STI	Steel Tank Institute www.steeltank.com	(847) 438-8265
SWI	Steel Window Institute www.steelwindows.com	(216) 241-7333
SWRI	Sealant, Waterproofing, and Restoration Institute www.swrionline.org	(816) 472-7974
TCNA	Tile Council of North America, Inc. www.tileusa.com	(864) 646-8453
TIA/EIA	Telecommunications Industry Association/Electronic Industries Alliance www.tiaonline.org	(703) 907-7700
TPI	Truss Plate Institute	(608) 833-5900
TPI	Turfgrass Producers International www.turfgrasssod.org	(800) 405-8873 (847) 705-9898
UFAC	Upholstered Furniture Action Council www.ufac.org	(336) 885-5065
UL	Underwriters Laboratories Inc. www.ul.com	(800) 704-4050 (847) 272-8800
UNI	Uni-Bell PVC Pipe Association www.uni-bell.org	(972) 243-3902
USITT	United States Institute for Theatre Technology, Inc. www.culturenet.ca/usitt	(800) 938-7488 (315) 463-6463
USGBC	LEED-NC, U.S. Green Building Council www.usgbc.org	(202) 223-7299
USP	U.S. Pharmacopeia www.usp.org	(800) 822-8772 (301) 881-0666
WASTEC	Waste Equipment Technology Association www.wastec.org	(800) 424-2869 (202) 244-4700
WCLIB	West Coast Lumber Inspection Bureau www.wclib.org	(800) 283-1486 (503) 639-0651
WCMA	Window Covering Manufacturers Association (Formerly: AWCMA - American Window Covering Manufacturers Association) www.windowcoverings.org	(800) 506-4653 (212) 661-4261
WDMA	Window & Door Manufacturers Association (Formerly: NWWDA - National Wood Window and Door Association)	(800) 223-2301 (847) 299-5200

www.wdma.com

WIC	Woodwork Institute of California www.wicnet.org	(916) 372-9943
WMMPA	Wood Moulding & Millwork Producers Association www.wmmpa.com	(800) 550-7889 (530) 661-9591
WWPA	Western Wood Products Association www.wwpa.org	(503) 224-3930

- C. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. Names, telephone numbers, and Web site addresses are subject to change and are believed to be accurate and up-to-date as of the date of the Contract Documents.

CBHF	State of California, Department of Consumer Affairs Bureau of Home Furnishings and Thermal Insulation www.dca.ca.gov/bhfti www.cpssc.gov	(800) 952-5210 (916) 445-1254 (301) 504-0990
DOC	Department of Commerce www.doc.gov	(202) 482-2000
EPA	Environmental Protection Agency www.epa.gov	(202) 260-2090
FAA	Federal Aviation Administration www.faa.gov	(202) 366-4000
FCC	Federal Communications Commission www.fcc.gov	(202) 418-0190
FDA	Food and Drug Administration	(888) 463-6332

- D. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. Names, telephone numbers, and Web site addresses are subject to change and are believed to be accurate and up-to-date as of the date of the Contract Documents.

CBHF	State of California, Department of Consumer Affairs Bureau of Home Furnishings and Thermal Insulation www.dca.ca.gov/bhfti www.cpssc.gov	(800) 952-5210 (916) 445-1254 (301) 504-0990
DOC	Department of Commerce www.doc.gov	(202) 482-2000
EPA	Environmental Protection Agency www.epa.gov	(202) 260-2090
FAA	Federal Aviation Administration www.faa.gov	(202) 366-4000
FCC	Federal Communications Commission	(202) 418-0190

	www.fcc.gov	
FDA	Food and Drug Administration www.fda.gov	(888) 463-6332
GSA	General Services Administration www.gsa.gov	(202) 708-5082
HUD	Department of Housing and Urban Development www.hud.gov	(202) 708-1112
LBL	Lawrence Berkeley Laboratory (See LBNL)	
LBNL	Lawrence Berkeley National Laboratory www.lbl.gov	(510) 486-5605
NCHRP	National Cooperative Highway Research Program (See TRB)	
NIST	National Institute of Standards and Technology www.nist.gov	(301) 975-6478
OSHA	Occupational Safety & Health Administration www.osha.gov	(202) 693-1999
RUS	Rural Utilities Service (See USDA)	(202) 720-9540
TRB	Transportation Research Board www.nas.edu/trb	(202) 334-2934
USDA	Department of Agriculture www.usda.gov	(202) 720-2791
USPS	Postal Service www.usps.com	(202) 268-2000

- E. State Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized or reasonably inferrable name of the entities having jurisdiction in the State in which the project is constructed.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 4500 - CONSTRUCTION QUALITY CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for the contractor's Quality Control (QC) Program. The QC Program required for the work is defined by the combination of this section and Division 1 Section "Quality Requirements."
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to quality control. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Photographic Documentation"
 - 6. Division 01 Section "Submittal Procedures"
 - 7. Division 01 Section "Quality Requirements"
 - 8. Division 01 Section "References"
 - 9. Division 01 Section "Temporary Facilities and Controls"
 - 10. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 11. Division 01 Section "Safety and Health"
 - 12. Division 01 Section "Product Requirements"
 - 13. Division 01 Section "Execution Requirements"
 - 14. Division 01 Section "Cutting and Patching"
 - 15. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 16. Division 01 Section "Closeout Procedures"
 - 17. Division 01 Section "Project Record Documents"
 - 18. Division 01 Section "Demonstration and Training"
 - 19. Division 01 Section "Selective Demolition"
- C. The contractor's QC Program includes tests, inspections, procedures, and related actions performed by the contractor or other contractually designated party during and after execution of the work to verify that completed construction complies with contract requirements. Services do not include contract enforcement activities performed by the NIH or their designated representative.
- D. Specific QC requirements for individual construction activities are included in the technical sections that specify those construction activities.

- E. The intent of this section is to describe the duties and responsibilities of the contractor's QC Program. The Contractor has the option of assembling a QC Program to match that described in this specification section or proposing an alternate program that meets the same intent. The Contractor Officer can waive all or portions of this specification if the contractor has an established proven alternative QC Program which can be documented as meeting the intent of the provisions in this specification section. The contractor is required to submit any alternate QC Program to meet the same milestones described herein and have the alternate QC Program accepted by the Contracting Officer prior to the start of work. No additional time will be allowed for the review process of an alternate program. Any costs savings resulting from an alternate QC Program will require a change order to be incorporated into the contract. No additional costs will be authorized for an alternate QC Program.
- F. This specification section does not relieve the Contractor of responsibility for compliance with Contract Document requirements, and does not limit the Contractor's QC procedures that facilitate compliance with Contract Document requirements.
- G. The term Contracting Officer and Project Officer shall also mean his/her representative who has been so designated in writing.

1.3 GOVERNING REGULATIONS AND AUTHORITIES

- A. The latest edition of the publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.
 1. AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) ASTM A880 Criteria for Use in Evaluation of Testing Laboratories and Organization for Examination and Inspection of Steel, Stainless Steel, and Related Alloys
 2. ASTM C1077 Laboratories Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Laboratory Evaluation
 3. ASTM D3666 Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
 4. ASTM D3740 Minimum Requirements for Agencies Engaged in the Testing and/ or Inspection of Soil and Rock as Used in Engineering Design and Construction
 5. ASTM E329 Agencies Engaged in the Testing and/ or Inspection of Materials Used in Construction
 6. ASTM E543 Agencies Performing Nondestructive Testing

1.4 QUALITY CONTROL RESPONSIBILITIES

- A. Work is to be performed under the general direction of the Contracting Officer and shall be subject to inspection by the Contracting Officer's Technical Representative. No representative of the Contracting Officer is permitted to change specifications or drawings without the written authorization of the Contracting Officer.
- B. The contractor shall be familiar with the latest provisions of the inspection requirements of local jurisdictions and the NIH and shall include compliance with those requirements in the work of this contract. These local provisions shall be enforced when they exceed the provisions as outlined in the specifications. The provisions of the specifications shall be considered a minimum.

- C. Unless specifically indicated otherwise, the Contractor shall provide a contractor QC (QC) Program specified or required by this contract. Costs for these services are included in the Contract price.
- D. The contractor's QC Program shall include but not be limited to a QC Plan, a QC Staff, specified QC meetings, a three phased QC system described in this section, submittal review and certification, specified testing, completion inspections, QC certifications and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations which comply with the requirements of this Contract.
 - 1. The QC Program shall cover on-site and off-site work and shall be keyed to the work sequence identified in the contractor's construction schedule.
 - 2. No work or testing may be performed unless the contractor's QC Manager is on the work site.
 - 3. The contractor's QC Manager shall report to an officer of the firm and shall not be subordinate to the Project Superintendent or the Project Manager.
 - 4. The contractor's QC Manager, Project Superintendent, and the contractor's management team must work together effectively. Although the QC Manager is the primary individual responsible for QC, the Contractor will ultimately be held responsible for the quality of work on the job. The project superintendent will be held responsible for the quality of production.
- E. Testing and inspecting services may be required to verify compliance with requirements specified or indicated in the contract documents. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.
- F. Where individual technical specification sections specifically indicate that certain tests, inspections or other QC services are to be provided by a testing agency, the Contractor shall employ and pay for a qualified independent testing agency to perform the QC services.
- G. Specified tests, inspections, and related actions do not limit Contractor's quality control procedures that facilitate compliance with the Contract Document requirements.
- H. Where specific quality control tests or services are indicated as a NIH responsibility, the NIH will engage a qualified agency to perform these services.
 - 1. NIH will furnish Contractor with names, addresses, and telephone numbers of testing agencies engaged and a description of the types of testing and inspecting they are engaged to perform.
 - 2. Costs for retesting and re-inspecting construction that replaces or is necessitated by work that failed to comply with the Contract Documents will be charged to Contractor and the Contract Sum will be adjusted by Change Order.
- I. The Contractor shall cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. The contractor will provide the following:
 - 1. Access to the Work.
 - 2. Incidental labor and facilities necessary to facilitate tests and inspections.
 - 3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
 - 4. Facilities for storage and field-curing of test samples.
 - 5. Delivery of samples to testing agencies.

- 6. Preliminary design mix proposed for use for material mixes that require control by testing agency.
 - 7. Security and protection for samples and for testing and inspecting equipment at Project site.
- J. Coordination: Coordinate sequence of activities to accommodate required quality assurance and quality control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspecting.
- 1. Schedule times for tests, inspections, obtaining samples, and similar activities.

1.5 SUBMITTALS

- A. Submit a preliminary Definable Features of Work listing within 14 calendar days of receipt of the notice to Proceed.
- B. Submit a draft Quality Control Plan within 30 calendar days following receipt the Notice to Proceed.
- C. The only work that is authorized to proceed prior to the approval of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying.
- D. Contractors should plan on a NIH review period of 14 calendar days prior to receiving comments on the draft QC Plan submission. The planned duration of the NIH review is provided as a planning figure and will vary dependent on the complexity and accuracy of the submission. No additional time will be allowed the Contractor for a NIH review longer than the above duration.
- E. Submit revised Quality Control Plan within 14 calendar days following receipt of NIH comments.

1.6 CONTRACTOR'S QUALITY CONTROL ORGANIZATION

- A. Quality Control Manager Duties and Qualifications
 - 1. Provide a full time QC Manager at the work site to implement and manage the QC Program.
 - a. The only duties and responsibilities of the QC Manager are to manage and implement the QC Program on this contract. The QC Manager shall not be designated as the Project Superintendent or the safety competent person as defined OSHA Regulations (Standards) 29 CFR 1926.
 - b. The QC Manager is required to attend the QC Plan Meeting, attend the Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of QC "except for those phases of control designated to be performed by QC specialists" if QC Specialists are required, perform submittal review and certification, ensure testing is performed and provide QC certifications and documentation required in this contract. The QC Manager is responsible for managing and coordinating the three phases of QC and documentation of work performed by "the QC specialists," if QC Specialists are required Testing Laboratory personnel and any other inspection and testing personnel required by this Contract.

- 1) Specific duties of the QC Manager include but are not limited to:
 - a) Maintains access to quality references called for in the specifications
 - b) Ensures all submittals are prepared, certified, and submitted as required in a timely manner to avoid project delays.
 - c) Coordinates changes or substitution requests made by the contractor to the Project Officer, however, he does not have the authority to approve them.
 - d) Inspects all work for compliance and maintains a Rework Items list on all non-conforming work.
 - e) Coordinates all testing requirements to maintain the production schedule.
 - f) Ensures that As-Built Drawings and Specifications and As-Built Record of Material documents are current and on site.
 - g) Coordinates submission of all miscellaneous plans required by various Division 01 specification sections to the Contracting Officer.
- 2) The QC Manager reports to an officer of the firm and not to the site superintendent.
- 3) The QC Manager has the authority and responsibility to stop the work if the work does not conform to the contract requirements. Failure to exercise this responsibility is cause for NIH directing replacement of the QC Manager.
- 4) Regardless who is designated as the Safety Competent Person as defined in OSHA Regulations (Standards) 29 CFR 1926.32(f) and assigned other safety responsibilities as described in Division 01 Section "Safety and Health." The QC Manager is always responsible for observing the work and monitoring safe work practices during the normal course of his jobsite duties.

2. QC Manager Qualifications.

- a. A graduate of a four year accredited college program in one of the following disciplines: Engineering, Architecture, Construction Management, Engineering Technology, Building Construction, or Building Science, with a minimum of 10 years experience as a superintendent, inspector, QC Manager, project manager, or construction manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must be familiar with the requirements of OSHA Regulations (Standards) 29 CFR 1926, and have experience in the areas of hazard identification and safety compliance.

B. Additional QC Management Staff Requirements. The QC requirements of this project require the QC staff to also include the following individuals:

1. Alternate QC Manager. The contractor will provide an Alternate QC Manager with similar qualifications as those of the QC Manager to serve in the event of the QC Manager's absence. The period of absence of the QC Manager shall not exceed two weeks at any one time, and not more than 30 workdays in the aggregate during a calendar year. The alternate QC Manager shall be designated in the QC Plan and should be familiar with the workings and status of the QC Program. The Alternate QC Manager may be a member of the QC staff performing other QC duties when not acting as the QC Manager.
2. QC Specialists. Provide a separate QC Specialist at the work site for each of the areas of responsibilities, indicated in the following table, who shall assist and report to the QC Manager and who shall have no duties other than their assigned QC duties. QC specialists are required to attend the Coordination and Mutual Understanding Meeting,

QC meetings, and be physically present at the construction site to perform the three phases of quality control and prepare documentation for each definable feature of work in their area of responsibility at the frequency specified below.

Area of Responsibility	Qualifications	Frequency
HVAC, Piping and Plumbing Systems QC Specialist	A graduate of a four year accredited college program in mechanical engineering with a minimum of 10 years experience as a superintendent, inspector, QC Manager, project manager, or construction manager on similar size and complexity mechanical systems installations that are part of this Contract	Daily during installation, startup and testing of systems included in area of responsibility and two days per week during pre-installation period of project.

3. Submittal Reviewer[s]. Provide Submittal Reviewer[s], other than the QC Manager, qualified in the discipline[s] being reviewed, to review and certify that the submittals meet the requirements of this Contract prior to certification by the QC Manager.
4. Submittal Reviewers with specific qualifications are required for the following specification sections. Qualifications are contained in the technical specifications. The following listing is intended to be representative and may not include the complete requirement. Technical specification sections should be reviewed for additional requirements.
5. The technical specification sections may also require the presence of manufacturer or factory technical or quality representatives or engineers to be on-site prior to observe or conduct all or portions of the installation. See Division 01 Section "Quality Requirements" for individual qualifications. These representatives shall be considered an extension of the QC staff.
6. The QC Supervisor for the Pest Management Contractor (PMC) required by Division 01 Section "Temporary Facilities and Controls" shall be considered a member of the QC staff. Copies of all PMC reports will be maintained by the QC Organization and PMC activities will be included in QC Reports

1.7 QUALITY CONTROL PLAN

- A. QC Plan Meeting: Conduct a QC Plan meeting with the Project Officer within 10 calendar days of receipt of the Notice To Proceed with the purpose of developing a mutual understanding of the QC Plan requirements prior to plan development and submission.
- B. Submit a QC Plan to the Contracting Officer for approval conforming to the requirements specified elsewhere in this section within 30 calendar days after receipt of the Notice to Proceed. Coordinate submission of the QC Plan with the preparation and submission of the contractor's preliminary and final construction schedules as described in Division 01 Section "Construction Progress Documentation." The QC Plan submission should be concurrent or follow the submission of the schedule.
- C. The final submission of the QC Plan should be coordinated with the submission of the contractor's construction schedule as described in Division 01 Section "Construction Progress Documentation." If the completed list of DFOWs and accepted contractor's schedule is not received within the time indicated in Division 01 Section "Construction Progress Documentation," work beyond that authorized under the "approved as noted" will not

commence. The contractor will not be entitled to a contract time extension for lost time due to failure to submit a conforming QC Plan or Construction Schedule.

- D. The contractor's QC Program is subject to continuous evaluation, review, and verification by the Project Officer and the Contracting Officer. Acceptance of the Contractor's QC Plan is conditional and will be predicated on satisfactory performance during the construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time in order to verify the submitted qualifications. All QC organization personnel shall be subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the contract.
- E. The Contractor will notify the Contracting Officer, in writing, of any proposed change to the submitted QC Plan, including pending changes in the QC organization personnel, a minimum of seven calendar days prior to the effective date of the proposed change. Proposed changes shall be subject to acceptance by the Contracting Officer prior to implementation.
- F. QC Plan Requirements. Provide, for approval by the Contracting Officer, a QC Plan with pages numbered sequentially that covers both on-site and off-site work and includes the following:
 - 1. A table of contents listing the major sections identified with tabs in the following order:
 - a. QC Organization
 - b. Names And Qualifications
 - c. Duties, Responsibility And Authority Of QC Personnel
 - d. Outside Organizations
 - e. Appointment Letters
 - f. Submittal Procedures And Initial Submittal Register
 - g. Testing Laboratory Information
 - h. Testing Plan And Log
 - i. Procedures To Complete Rework Items
 - j. Documentation Procedures
 - k. List Of Definable Features Of Work (DFOW)
 - l. Procedures For Performing The Three Phases Of Control
 - m. QC Personnel Matrix
 - n. Procedures For Completion Inspection
 - o. Appendix
 - 2. A chart showing the contractor's QC organizational structure.
 - 3. Names and qualifications, in resume format, for each person in the QC organization.
 - 4. Duties, responsibilities and authorities of each person in the QC organization.
 - 5. A listing of outside organizations such as, architectural and consulting engineering firms that will be employed by the Contractor and a description of the services these firms will provide.
 - 6. Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager and stating that they are responsible for implementing and managing the QC Program as described in this contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of quality control, and their authority to stop work which is not in compliance with the contract.
 - 7. Copies of letters of direction signed by the QC Manager to all other QC staff outlining their duties, authorities, and responsibilities.

8. Procedures for reviewing, approving and managing submittals. Provide the names of the persons in the QC organization authorized to review and certify submittals prior to submission to the Contracting Officer.
9. Include a copy of the initial submittal of the Submittal Register per the requirements of Division 01 Section "Submittal Procedures."
10. Testing laboratory information required by the paragraphs entitled "Testing Agencies and Reports" of this section
11. A Testing Plan and Log that includes the tests required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
12. Procedures to identify, record, track and complete rework items.
13. Documentation procedures, including proposed report formats.
14. List of definable features of work.
 - a. A definable feature of work (DFOW) is a task which is separate and distinct from other tasks, has the same control requirements and work crews.
 - b. The list shall be cross-referenced to the contractor's Construction Schedule and the specification sections.
 - c. For projects requiring a CPM Schedule, the list of definable features of work shall include but not be limited to all critical path activities.
 - d. All activities for which this specification requires QC Specialists or special inspection personnel should also be included as separate DFOWs.
 - e. Cutting and Patching activities as described in Division 01 Section "Cutting and Patching" shall be treated as separate DFOWs.
 - f. All activities that can be expected to impact NIH operations will be treated as separate DFOWs including but not limited to utility outages and temporary traffic provisions.
15. Procedures for performing the three phases of quality control. For each DFOW, provide copies of the DFOWs Preparatory and Initial Phase Checklists. Each list shall include a breakdown of quality checks that will be used when performing the QC functions, inspections, and tests required by the contract documents. The preparatory and initial phases and meetings shall be conducted with a view towards obtaining quality construction by planning ahead and identifying potential problems for each definable feature of work. The three phases of quality control are further defined in a following section of this specification.
16. Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
17. Appendix to include the anticipated miscellaneous project management and coordination plans being submitted for the project including but not limited to the following as applicable for project:
 - a. Utility Service Interruption Plan
 - b. Energy Conservation Plan
 - c. Waste Management Plan
 - d. Hazardous Waste/Waste Management Plan
 - e. Heating and Cooling Plan
 - f. Storm Water Management Plan (draft until State approved plan available)
 - g. Integrated Pest Management Plan
 - h. Severe Storm Plan
 - i. Demolition Plan
 - j. Photographic Documentation Plan
 - k. LEED Submittals

1.8 COORDINATION AND MUTUAL UNDERSTANDING MEETING

- A. After submission of the QC Plan, and prior to the start of construction, meet with the Contracting Officer to present the QC Program proposed by the contractor. The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, and the coordination of the Contractor's management, production and QC personnel.
- B. At the meeting, the Contractor will be required to explain in detail how the three phases of quality control will be implemented for each definable feature of work.
- C. As a minimum, the Contractor's personnel required to attend shall include an officer of the firm, the Project Manager, Project Superintendent, QC Manager and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities shall have a principal of the firm at the meeting.
- D. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, and the Contracting Officer (or their designated representative). A copy of the signed minutes shall be provided by the Contractor to all attendees. Repeat the coordination and mutual understanding meeting when a new QC Manager is appointed.

1.9 QC MEETINGS.

- A. After the start of construction, the QC Manager shall conduct bi-weekly QC meetings at the work site with the Contractor's project manager, project superintendent and sub-contractors' project managers. The QC Manager shall prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer or their designated representative may attend these meetings. The QC Manager shall notify the Contracting Officer or their designated representative at least 48 hours in advance of each meeting. As a minimum, the following shall be accomplished at each meeting:
- B. Review the minutes of the previous meeting;
- C. Review the schedule and the status of work:
 - 1. Work or testing accomplished since last meeting
 - 2. Rework items identified since last meeting
 - 3. Rework items completed since last meeting;
- D. Review the status of submittals:
 - 1. Submittals reviewed and approved since last meeting
 - 2. Submittals required in the near future;
- E. Review the work to be accomplished in the next 4 weeks and documentation required:
 - 1. Establish completion dates for outstanding rework items
 - 2. Update the schedule showing planned and actual dates of the preparatory, initial and follow-up phases, including testing and any other inspection required by this contract
 - 3. Discuss construction methods and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each definable feature of work.
 - 4. Discuss status of off-site work or testing

5. Discuss documentation required for the scheduled tasks;
6. Discuss safety requirements for upcoming activities:

F. Resolve QC and production problems:

1. Assist in resolving Request for Information issues; and

G. Address items that may require revising the QC Plan:

1. Changes in QC organization personnel
2. Changes in QC procedures.
3. Review health and safety plan
4. Other issues or topics as requested by the Project Officer.

1.10 THREE PHASES OF QUALITY CONTROL

A. The three phases of quality control, describes the process that forms the backbone of the required QC system. The three phases, Preparatory, Initial, and Follow up shall adequately cover both on-site and off-site work and shall include the following for each definable feature of work.

1. Preparatory Phase

- a. Notify the Contracting Officer at least 2 workdays in advance of the beginning of each Preparatory Phase. This phase shall include a meeting conducted by the QC Manager and attended by Contractor's Project Manager, the Superintendent, and the Foreman responsible for the DFW. Document the results of the Preparatory Phase actions in the daily Contractor Quality Control Report and in the Preparatory Phase Checklist. The Preparatory Phase Meeting can be combined with the Pre-installation Conference if held.
- b. Perform the following prior to beginning work on each DFW:
 - 1) Review each paragraph of the applicable specification sections
 - 2) Review the Contract drawings
 - 3) Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required
 - 4) Review the testing plan and cutting and patching plan and other miscellaneous project management and coordination plans as appropriate and ensure that provisions have been made to provide the required QC testing
 - 5) Examine the work area to ensure that the required preliminary work has been completed. If manufacturer's field services are specified, verify that inspections have been accomplished and results noted. See Division 01 Section "Execution Requirements" for additional requirements.
 - 6) Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data
 - 7) Discuss construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFW

- 8) Review the Safety Plan and appropriate activity hazard analysis to ensure that applicable safety requirements are met, and that required Material Safety Data Sheets (MSDS) are submitted.
- 9) Review any hazardous material processes required for materials involved in the DFW.

2. Initial Phase

- a. Notify the Contracting Officer at least 2 workdays in advance of the beginning of each Initial Phase. When construction crews are ready to start work on a DFW, conduct an Initial Phase meeting with the superintendent, and the foreman responsible for that DFW. Observe the initial segment of the definable feature of work to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily Contractor Quality Control Report and in the Initial Phase Checklist. Repeat the Initial Phase for each new crew to work on-site, or when acceptable levels of specified quality are not being met.
- b. Perform the following for each DFW:
 - 1) Establish the quality of workmanship required
 - 2) Resolve conflicts
 - 3) Ensure that testing is performed as specified and as incorporated into the testing plan.
 - 4) Check work procedures for compliance with the Safety Plan and the appropriate activity hazard analysis to ensure that applicable safety requirements are met.
 - 5) Verify that all the requirements agreed to as a result of the Preparatory Phase have been or are being accomplished.

3. Follow-Up Phase

- a. Perform the following for on-going work daily, or more frequently as necessary until the completion of each DFW and document in the daily Contractor Quality Control Report:
 - 1) Ensure the work is in compliance with Contract requirements;
 - 2) Maintain the quality of workmanship required
 - 3) Ensure that testing is performed by the specified or approved source;
 - 4) Ensure that rework items are being corrected
 - 5) Perform safety inspections.
 - 6) Ensure Preparatory and Initial Phase requirements are being met.

4. Additional Preparatory and Initial Phases. Additional Preparatory and Initial Phases shall be conducted on the same DFWs for the following conditions:

- a. If the quality of on-going work is unacceptable, if there are changes in the applicable QC organization,
- b. If there are changes in the on-site production supervision or work crew,
- c. If cutting or patching is required following completion of the DFW,
- d. If work on a definable feature is resumed after substantial period of inactivity, or
- e. If other problems develop.

- B. Notify the Contracting Officer at least two weeks prior to the start of the preparatory and initial phases if off site QC will be required.

1.11 SUBMITTAL REVIEW AND APPROVAL

- A. The QC organization shall be responsible for reviewing and certifying that all submittals are in compliance with the contract requirements. Those submittals requiring additional NIH approval shall be certified prior to being forwarded to NIH for approval.
- B. Those submittals requiring specific NIH approval are specifically identified in the technical specifications and will be forwarded for approval as described in Division 01 Section "Submittal Procedures."
- C. Submittal documentation requirements for contractor QC certified submittals are identical to those of NIH approved submittals and are described in Division 01 Section "Submittal Procedures."
- D. Procedures for submission, review and approval of technical submittals are described in Division 01 Section "Submittal Procedures."

1.12 TESTING AGENCIES AND REPORTS

- A. Qualification Data: Contractor shall submit for Contracting Officer Approval each testing agency's firm name, and credentials to perform the specified services, to the NIH for the Contracting Officer's approval at least 15 calendar days before scheduled inspections or tests.
 - 1. A qualified independent testing agency shall be an accredited entity engaged to perform tests or inspections, either at the Project site or elsewhere, and to report on and, if required, to interpret results of those tests or inspections.
 - 2. Testing agencies shall be acceptable to the Contracting Officer.
 - 3. Contractor shall not employ the same testing agency engaged by NIH, unless agreed to in writing by NIH.
 - 4. Unless other accreditation is specified in the applicable individual technical specification section, each testing agency shall be an agency pre-qualified with the experience and capability to conduct testing and inspections indicated, as documented by ASTM E 548 that specializes in types of tests and inspections to be performed or shall be recognized by the Occupational Safety and Health Administration (OSHA) in accordance with 29 CFR Part 1910.7 to test and approve equipment or materials for their safe intended use.
 - 5. Testing agencies shall be authorized by authorities having jurisdiction to operate in the geographic location of the project.
 - 6. The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract.
 - 7. Testing and inspecting requested by the Contractor not required by the Contract Documents are Contractor's responsibility.
 - 8. Submit additional copies of each written test report directly to authorities having jurisdiction, when they so direct.
 - 9. Testing agencies may not release, revoke, alter, or increase requirements of the Contract Documents or approve or accept any portion of the work.
- B. Schedule of Tests and Inspections: The Contractor shall prepare and submit a comprehensive Schedule of Tests and Inspections required by the contract documents.
 - 1. Coordinate the Schedule of Tests and Inspections with the Contractor's Construction Schedule and other related documents. Prepare in tabular form to include the following:

- a. Specification Section number and title.
- b. Description of test and inspection.
- c. Identification of applicable standards.
- d. Identification of test and inspection methods.
- e. Number of tests and inspections required.
- f. Time schedule or time span for tests and inspections.
- g. Entity responsible for performing tests and inspections.
- h. Requirements for obtaining samples.
- i. Unique characteristics of each quality-control service.

C. Test Results and Reports

1. Provide actual results in a written format in the number of copies required by the QC Manager and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify Contracting Officer directly through the Project Officer immediately. Written test reports will include the following information:
 - a. Date of issue.
 - b. Project title and number.
 - c. Name, address, and telephone number of testing agency.
 - d. Dates and locations of samples and tests or inspections.
 - e. Names of individuals making the test or inspection.
 - f. Designation of the work and test method.
 - g. Identifications of product and specification section.
 - h. Complete test or inspection data.
 - i. Test results and an interpretation of test results.
 - j. Ambient conditions at the time of sample taking and testing.
 - k. Comments or professional opinion on whether tested or inspected Work complies with Contract Document requirements.
 - l. Name and signature of laboratory inspector.
 - m. Recommendations on retesting.
2. Provide three copies to the Project Officer of each test and inspection report.
3. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results shall be signed by a testing laboratory representative authorized to sign certified test reports.
4. Furnish the signed reports, certifications, and other documentation to the Contracting Officer or his designated representative via the QC Manager. Furnish a summary report of field tests at the end of each month. Attach a copy of the summary report to the last daily Contractor Quality Control Report of each month.

1.13 QUALITY CONTROL CERTIFICATIONS AND COMPLETION INSPECTIONS

A. QC Certifications

1. Contractor Quality Control Report Certification. Each Contractor QC Report shall contain the following statement:
2. "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used and work performed during this reporting period is in compliance with the contract drawings and specifications to the best of my knowledge, except as noted in this report." (signed by the QC Manager)

3. Invoice Certification. Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that Project Record Documents are current and attesting that the work for which payment is requested, including stored material, is in compliance with Contract requirements.
4. Completion Certification. Upon completion of work under this Contract, the QC Manager shall furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract."

B. Completion Inspections

1. Individual DFOW Punch-Out Inspections.
 - a. Near the completion of each DFOW, the QC Manager shall conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings and specifications.
 - b. Include in the punch list any remaining items on the "Rework Items List" which were not corrected prior to the Punch-Out Inspection.
 - c. The punch lists shall include the estimated date by which the deficiencies will be corrected. A copy of the punch lists shall be provided to the Contracting Officer.
 - d. The QC Manager or staff shall make follow-on inspections to ascertain that all deficiencies have been corrected.
 - e. The Contractor shall notify the Contracting Officer or his designated representative when each DFOW has had all deficiencies corrected.
2. Punch-Out Inspection (prior to Pre-Final Inspection).
 - a. Near the completion of all work or any increment thereof established by a contract clause or stated elsewhere in the specifications, the QC Manager shall conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings and specifications.
 - b. Include in the punch lists any remaining items on the "Rework Items List" which were not corrected prior to the Punch-Out Inspection.
 - c. The punch lists shall include the estimated date by which the deficiencies will be corrected. A copy of the punch lists shall be provided to the Contracting Officer.
 - d. The QC Manager or staff shall make follow-on inspections to ascertain that all deficiencies have been corrected.
 - e. The Contractor shall notify the Contracting Officer or his designated representative that the facility is ready for the NIH Pre-Final Inspection.
3. Pre-Final Inspection
 - a. The NIH will perform this inspection to verify that the facility is complete and ready to be occupied. A NIH Pre-Final Punch List may be developed as a result of this inspection.
 - b. Any items noted on the Pre-Final inspection shall be corrected in a timely manner and shall be accomplished before the contract completion date or phase completion date if the project is divided into phases with separate completion dates.
 - c. The QC Manager shall ensure that all items on the Pre-Final Punch List are corrected prior to notifying the NIH that a Final inspection with the customer can be scheduled.
4. Final Acceptance Inspection

- a. The QC Manager, the superintendent and other primary contractor management personnel, and the Contracting Officer's representative will be in attendance at this inspection. Additional NIH personnel may be in attendance.
- b. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-Final Inspection. Notice shall be given to the Contracting Officer at least 14 days prior to the Final Inspection stating that all specific items previously identified to the Contractor as being unacceptable, along with all the remaining work performed under the contract, will be complete and acceptable by the date scheduled for the final acceptance inspection.
- c. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the additional NIH inspection costs in accordance with the Contract Clause entitled "Inspection of Construction."
- d. When the Contracting Officer takes possession of partially completed work, it will be in accordance with Contract Clause "Use and Possession Prior to Completion".

1.14 QUALITY CONTROL DOCUMENTATION

- A. Maintain current and complete records of on-site and off-site QC Program operations and activities. QC Documentation is to be coordinated with Division 01 Sections "Construction Progress Documentation," "Photographic Documentation," and "Closeout Procedures," if included in the project specifications.
- B. The Contractor shall have an identification and data retrieval system. Records, reports, drawings, submittals, and equipment shall be identified to reference the following:
 - 1. Contract Number
 - 2. Contract Specification Number
 - 3. Contract Drawing Number
 - 4. Submittal Document Number
 - 5. Contract Change Number
- C. Quality Control Site Records. Establish and maintain on the jobsite the following documentation readily available to the NIH Project Officer during all business hours.
 - 1. All completed Preparatory and Initial Phase Checklists, arranged by specification section.
 - 2. All milestone inspections, arranged by Activity/ Event Number.
 - 3. Photographic Documentation
 - 4. A current up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section.
 - 5. Copies of all contract modifications, arranged in numerical order. Also include documentation that the modified work was accomplished.
 - 6. A current up-to-date copy of the Rework Items List.
 - 7. Current up-to-date copies of all punch lists issued by the QC Staff of the Contractor and Sub-Contractors and all punch lists issued by the NIH.
- D. Contractor Production Report. Reports are required for each day that work is performed and shall be attached to the Contractor Quality Control Report prepared for the same day. Account for each calendar day throughout the life of the Contract. The reporting of work shall be identified by terminology consistent with the construction schedule as described in Division 01 Section "Construction Progress Documentation."

- E. Contractor Quality Control Report. Reports are required for each day that work is performed and for every seven consecutive calendar days of no work and on the last day of a no-work period. Account for each calendar day throughout the life of the Contract. The reporting of work shall be identified by terminology consistent with the construction schedule.
1. Contractor Quality Control Reports are to be prepared, signed and dated by the QC Manager and shall contain the following information:
 - a. Date of report, report number, Contract Number, and Contract Title.
 - b. Indicate if Preparatory Phase work was performed (Yes/No checkboxes).
 - c. If Preparatory Phase work was performed (including on-site and off-site work), identify its Schedule Activity Number and DFOW. The Index # is a cross reference to the Preparatory Phase Checklist. An example of the Index # is: 0025-P01, where "0025" is the Contractor Quality Control Report Number, "P" indicates Preparatory Phase, and "01" is the Preparatory Phase Checklist number(s) for this date. Each entry in this section must be accompanied with a corresponding copy of the Preparatory Phase Checklist.
 - d. Indicate if Initial Phase work was performed (Yes/No checkboxes).
 - e. If Initial Phase work was performed today (including on-site and off-site work), identify its Schedule Activity No. and DFOW. The Index # is a cross reference to the Initial Phase Checklist. An example of the Index # is: 0025-I01, where "0025" is the Contractor Quality Control Report Number, "I" indicates Initial Phase, and "01" is the Initial Phase Checklist number(s) for this date. Each entry in this section must be accompanied with a copy of the corresponding Initial Phase Checklist.
 - f. Results of the Follow-up Phase inspections held that day (including on-site and off-site work), including Schedule Activity No., the location of the DFOW, Specification Sections, etc. Indicate in the report for this definable feature of work that the work complies with the Contract as approved in the Initial Phase, work complies with safety requirements, and that required testing has been performed and include a list of who performed the tests.
 - g. List the rework items identified, but not corrected by close of business; along with its associated Schedule Activity Number.
 - h. List the rework items corrected from the rework items list along with the corrective action taken and its associated Schedule Activity Number.
 - i. Include a "remarks" section in this report which will contain pertinent information including directions received, QC problem areas, deviations from the QC Plan, construction deficiencies encountered, photographic documentation accomplished, QC meetings held, acknowledgement that as-built drawings have been updated, corrective direction given by the QC Organization and corrective action taken by the Contractor. For each remark given, identify the Schedule Activity Number that is associated with the remark.
 - j. Contractor Quality Control Report certification, signature and date.
 2. Attach a summary report to the last daily Contractor Quality Control Report of each month.
 - a. Include reference to submission of Monthly Integrated Pest Management Report.
- F. Preparatory Phase Checklist. Each DFOW that is in the Preparatory Phase shall have this checklist filled out for it. The checklist shall be identified by terminology consistent with the construction schedule. Attach a copy of the completed checklist to the Contractor Quality Control Report of the same date.
1. The checklist shall contain the following information:

- a. Specification Section, date of report, and Contract number shall be filled out. Duplicate this information in the header of the second page of the report.
- b. DFOW, Schedule Activity Number and Index # entry and format will match entry in the Preparatory Phase section of the Contractor Quality Control Report. Duplicate this information in the header of the second page of the report.
- c. Personnel Present: Indicate the number of hours of advance notice that was given to the NIH Representative and indicate (Yes/No checkboxes) whether or not the NIH Rep was notified. Indicate the Names of Preparatory Phase Meeting attendees, their position and company/NIH Activity they are with.
- d. Submittals: Indicate if submittals have been approved (Yes/No checkboxes), if no indicate what has not been submitted. Are materials on hand (Yes/No checkboxes) and if not, what items are missing. Check delivered material/equipment against approved submittals and comment as required.
- e. Material Storage: Indicate if materials/equipment is stored properly (Yes/No checkboxes) and if not, what action is/was taken.
- f. Specifications: Review and comment on specification paragraphs that describe the material/equipment, procedure for accomplishing the work and clarify any differences.
- g. Preliminary Work & Permits: Ensure preliminary work is in accordance with the contract documents and any necessary permits are on file, if not, describe the action taken.
- h. Testing: Identify who performs tests, the frequency, and where tests are to occur. Review the testing plan, report abnormalities, and if the test facilities have been approved.
- i. Safety: Indicate if an activity hazard analysis has been reviewed (Yes/No checkboxes) and comment on the review of the applicable portions of OSHA Regulations (Standards) 29 CFR 1926.
- j. Meeting Comments: Note comments and remarks during the Preparatory Phase Meeting that were not addressed in previous sections of this checklist.
- k. Other Items or Remarks: Note any other remarks or items that were a result of the Preparatory Phase.

2. QC Manager will sign and date the checklist.

G. Initial Phase Checklist. Each DFOW that is in the Initial Phase shall have this checklist filled out for it. The checklist shall be identified by terminology consistent with the construction schedule. Attach this checklist to the Contractor Quality Control Report of the same date.

1. The checklist shall contain the following information:

- a. Specification Section, date of report, and Contract number shall be entered.
- b. DFOW, Schedule Activity Number and Index Number entry and format will match entry in the Initial Phase section of the Contractor Quality Control Report.
- c. Personnel Present: Indicate the number of hours of advance notice that was given to the NIH Representative and indicate (Yes/No checkboxes) whether or not the NIH Rep was notified. Indicate the Names of Initial Phase Meeting attendees, their position and company/NIH Activity they are with.
- d. Procedure Compliance: Comment on compliance with procedures identified at Preparatory Phase of Control and assurance that work is in accordance with plans, specifications and submittals.
- e. Preliminary Work: Ensure preliminary work being placed is in compliance and if not, what action is/was taken.

- f. Workmanship: Identify where initial work is located; if a sample panel is required (Yes/No checkboxes); is the initial work the sample (Yes/No checkboxes); and if Yes, describe the panel location and precautions taken to preserve the sample.
- g. Resolution: Comment on any differences and the resolutions reached.
- h. Check Safety: Comment on the safety review of the job conditions.
- i. Other: Note any other remarks or items that were a result of the Initial Phase.

2. The QC Manager will sign and date the checklist.

H. Testing Plan and Log.

- 1. As tests are performed, the QC Manager shall record on the "Testing Plan and Log" the date the test was conducted, the date the test results were forwarded to the Contracting Officer, remarks and acknowledgement that an accredited or Contracting Officer approved testing laboratory was used.
- 2. Attach a copy of the updated "Testing Plan and Log" to the last daily Contractor Quality Control Report of each month.
- 3. Testing Plan and Log may be an expanded version of the Testing and Inspection Schedule.

I. Rework Items List.

- 1. The QC Manager shall maintain a list of work that does not comply with the Contract, identifying what items need to be reworked, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.
- 2. There is no requirement to report a rework item that is corrected the same day it is discovered. Attach a copy of the "Rework Items List" to the last daily Contractor Quality Control Report of each month.
- 3. The Contractor shall be responsible for including on this list items needing rework including those identified by the Contracting Officer.

J. Record Documents.

- 1. The QC Manager is required to ensure the record documents including Record Drawings and Record Specifications required by Division 01 Section "Closeout Procedures" are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings.
- 2. Ensure each deviation has been identified with the appropriate modifying documentation (e. g. PC No., Modification No., Request for Information No., etc.).
- 3. The QC Manager, shall initial each deviation and each revision.
- 4. Upon completion of work, the QC Manager shall furnish a certificate attesting to the accuracy of the Record Documents prior to submission to the Contracting Officer.

K. Report Forms. The Project Officer will make available sample formats for the various reports required by this contract which will meet the requirements of this specification. While use of these specific formats are not required, any other format used shall contain the same information.

L. Notification of Non-Compliance. The Contracting Officer will notify the Contractor of any detected non-compliance with the foregoing requirements. The Contractor shall take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, shall be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such

stop orders shall be made the subject of claim for extension of time for excess costs or damages by the Contractor.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 5000 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes requirements for construction facilities and temporary controls, including temporary utilities, support facilities and security and protection. All costs associated with such work and subsequent removal shall be the sole burden of the Contractor. The Contractor is responsible for any repairs required to restore NIH property to original or better condition if damaged by temporary construction.
- B. Temporary utilities include but are not limited to the following:
 - 1. Temporary water service and distribution.
 - 2. Temporary electric power and lighting.
 - 3. Temporary heat and ventilation.
 - 4. Temporary telephone service.
 - 5. Temporary sanitary facilities, including drinking water.
- C. Support facilities include but are not limited to the following:
 - 1. Field offices, storage sheds and fabrication shops.
 - 2. Temporary roads and paving.
 - 3. Dewatering facilities and drains.
 - 4. Temporary enclosures.
 - 5. Temporary lifts and hoists.
 - 6. Construction Signage.
 - 7. Waste disposal services.
 - 8. Other construction aids and miscellaneous services and facilities.
- D. Security and protection facilities include but are not limited to the following:
 - 1. Environmental protection.
 - 2. Stormwater control.
 - 3. Site enclosure fence.
 - 4. Security enclosure and lockup.
 - 5. Barricades, warning signs, and lights.
 - 6. Temporary enclosures.
 - 7. Temporary partitions.
 - 8. Temporary Fire protection.
- E. Integrated pest management services.
- F. Provide temporary facilities and controls required for construction activities except, if any, for facilities and controls indicated as existing or provided by NIH or others.
- G. NIH will not be responsible for any cost or use charges for temporary facilities or utilities as a basis of claims for Change Orders.

- H. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to temporary facilities and controls. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:

1. Division 01 Section "Summary"
2. Division 01 Section "Work Restrictions"
3. Division 01 Section "Project Management and Coordination"
4. Division 01 Section "Construction Progress Documentation"
5. Division 01 Section "Photographic Documentation"
6. Division 01 Section "Quality Requirements"
7. Division 01 Section "References"
8. Division 01 Section "Construction Quality Control"
9. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
10. Division 01 Section "Safety and Health"
11. Division 01 Section "Execution Requirements"
12. Division 01 Section "Cutting and Patching"
13. Division 01 Section "Fire Prevention Precautions for Hot Work"
14. Division 01 Section "Closeout Procedures"
15. Division 01 Section "Operation and Maintenance Documentation"
16. Division 01 Section "Demonstration and Training"
17. Division 01 Section "Selective Demolition"

1.3 DEFINITIONS

- A. Permanent Enclosure: As determined by the Project Officer, permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and all openings are closed with permanent construction or substantial temporary closures.

1.4 UTILITY USE CHARGES

- A. The Contractor shall provide all temporary utilities which are used and required by all entities engaged in construction activities at the Project site.
- B. The point at which the NIH will deliver such utilities and the quantity available are as directed by the Project Officer. The Contractor shall pay all costs incurred in connecting, converting and transferring NIH utilities to the work. The Contractor shall make connections, including providing backflow preventing devices on connections to domestic water lines, shall provide transformers, and shall make all disconnections.
- C. Notwithstanding any other provision in the contract documents, Contractor use of NIH utilities is contingent on the Contracting Officers approval of the following contractor developed documents:
1. Waste Management Plan
 2. Energy Conservation Plan
 3. Heating and Cooling Plan
 4. Utility Service Interruption Plan
- D. Water Service: The Contractor may use reasonable amounts of water from the existing NIH water system, without metering and without payment of use charges.
- E. Sewer Service: The Contractor may use the existing NIH sewer system, without payment of use charges.

- F. Electric Power Service: The Contractor may use reasonable amounts of electric power from the existing NIH electric power distribution system, without metering and without payment of use charges.

1.5 SUBMITTALS

- A. Reports: Submit reports of tests, inspections, meter readings and similar procedures for temporary utilities.
- B. Implementation and Termination Schedule: Within 18 calendar days after the date established for the submittal of the Contractor's Construction Schedule, submit a schedule indicating implementation and termination of each temporary utility. If implementation or termination will interrupt utility service outside the limit of construction include milestones for submission of outage requests in the schedule. Coordinate schedule with requirements for the Utility Service Interruption Plan contained in Division 01 Section "Project Management and Coordination."
- C. Shop Drawings: Submit layout indicating location of all fencing, gate locations, and size and type of fencing and gates.
- D. Integrated Pest Management (IPM) Program. Within 14 calendar days after the date of Notice to Proceed, submit the following for approval:
 - 1. IPM Service Plan. The plan shall include a description of the implemented pest management program during all phases of construction. Include at a minimum, the following:
 - a. Grounds and ground cover for task orders which involve any exterior work.
 - b. Solid waste management.
 - c. Site (interior and exterior) sanitation.
 - d. Other factors that contribute to pest infestation.
 - e. Regularly scheduled monitoring and survey program for the identification and control of insect and rodent pests.
 - 2. Identification of the IPM Quality Control supervisor. Include a copy of the qualification certificate and resume.
 - 3. Copy of the Commercial Pesticide Applicators Business License.
 - 4. Copy of Pesticide Applicators Certificates and resume for all personnel assigned to each task order.

1.6 QUALITY REQUIREMENTS

- A. Standards and Regulations: Comply with industry standards and with applicable laws and regulations of authorities having jurisdiction, including but not limited to the following:
 - 1. Building code requirements.
 - 2. Health and Safety regulations.
 - 3. Utility company regulations.
 - 4. Police, fire department and rescue squad rules.
 - 5. Environmental protection regulations.
 - 6. NFPA 241 "Standards for Safeguarding Construction, Alterations and Demolition Operations".
 - 7. ANSI-A10 Series standards for "Safety Requirements for Construction and Demolition".
 - 8. NECA Electrical Design Library "Temporary Electrical Facilities", and NEMA, NECA and UL standards and regulations for temporary electric service. Install service in accordance with NFPA 70, "National Electric Code."
 - 9. Maryland Pesticide Applicators Laws and Regulations.

- B. Inspections: Arrange for authorities having jurisdiction to inspect and test each temporary utility before use. Obtain required certifications and permits.

1.7 PROJECT CONDITIONS

- A. Install, operate, maintain and protect temporary facilities and controls.
 - 1. Keep temporary services and facilities clean and neat in appearance.
 - 2. Operate temporary services in a safe and efficient manner.
 - 3. Relocate temporary services and facilities as needed as work progresses.
 - 4. Do not overload temporary services and facilities or permit them to interfere with progress.
 - 5. Provide necessary fire prevention measures.
 - 6. Do not allow hazardous, dangerous or unsanitary conditions, or public nuisances to develop or persist on-site.
- B. Prepare a schedule indicating dates for implementation and termination of each temporary utility and incorporate into the project schedule. At the earliest feasible time, when acceptable to the Project Officer, change over from temporary services to use of permanent services and remove temporary facilities when no longer needed.
- C. Temporary Use of Permanent Facilities: Contractor shall assume responsibility for the operation, maintenance and protection of each permanent service during its use as a construction facility prior to acceptance by NIH.
 - 1. Protect permanently installed mechanical ducts and equipment to remain from dust and odors in accordance with SMACNA IAQ Guidelines for Occupied Buildings under Construction.
 - a. Where possible, do not use permanently installed HVAC systems during construction. Seal all duct and equipment openings with plastic.
 - b. If permanently installed HVAC system must be operated to protect finished work or to serve occupied areas of building, seal all non-essential openings with plastic and install and maintain temporary filters to protect openings required to operate.
 - 1) Temporary filters shall be identical in quantity, arrangement and MERV rating to specified permanent filters for each air handling and exhaust system.
- D. Existing Equipment and Items: Cover or otherwise protect and provide security for existing equipment and other items that are to remain in place, to prevent soiling, damage and loss.
 - 1. Temporarily move equipment and other items that interfere with the performance of required work. Upon completion of the work, return the equipment and items to their original location and installation condition.
 - 2. Store equipment and other items that have been temporarily removed. Upon reinstallation, clean and, if damaged, repair or replace equipment and items to match their condition prior to removal.
 - 3. Protect existing return air plenums above construction sites to remain by installing ceiling panels or otherwise separating plenum from construction area.
 - 4. Protect existing mechanical ducts and equipment to remain from dust and odors in accordance with SMACNA IAQ Guidelines for Occupied Buildings under Construction.
 - a. Where possible, do not use existing HVAC systems during construction. Seal all duct and equipment openings with plastic.
 - b. If existing HVAC system must be operated to protect finished work or to serve occupied areas of building, seal all non-essential openings with plastic and install and maintain temporary filters to protect openings required to operate.

- 1) Temporary filters shall be identical in quantity, arrangement and MERV rating to specified permanent filters for each air handling and exhaust system.

1.8 INTERIOR CONSTRUCTION BARRIERS

- A. All work within existing NIH buildings shall comply with the following requirements for Construction Barriers.
 1. All work areas not completely enclosed by permanent walls shall be separated from the remainder of the building with full smoke tight dust barrier protection. The work area shall be made negative in air pressure with respect to all surrounding areas, unless otherwise indicated on the contract drawings.
 2. All material used for the erection of temporary dust barriers shall consist of only noncombustible or fire retardant materials, as defined by NFPA 241. All polyethylene plastic sheeting shall be fire retardant treated, as defined by NFPA 241.
 3. All dust partitions (other than partitions required around asbestos abatement areas) shall be constructed of metal studs faced with gypsum wallboard, installed from the floor to underside of slab above in areas where the contract documents indicate the ceiling to be removed. If ceiling removal is not part of the contract, extend dust partition to the finished ceiling.
 4. Provide full construction barrier protection with locking doors around the construction. Provide 12 keys for each lock to the Project Officer. Where possible, entrances shall be constructed of a manufactured door and include a manufactured hydraulic door closer.
 5. Carpet runners shall be installed and extend 3000 mm beyond construction entrance doors. Carpet shall be vacuumed nightly. Sweeping is not acceptable.
 6. Where construction areas include food service or dumb waiters, these areas shall be separated again from the general construction area with additional full smoke tight dust barrier protection, including installation of carpet runners outside of the elevator/dumb waiter temporary enclosure.
 7. The contractor shall maintain a clear safe means of emergency egress through the construction site at all times to appropriate emergency exits.

1.9 ACCESS ROAD DUST CONTROL

- A. The Contractor's dust control methods and procedures shall be approved by the Project Officer. Dust abatement on access roads shall be with applications of gravel, water sprinklers, or similar methods or treatments approved by the Project Officer.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. General: Provide new or undamaged, previously used materials in serviceable condition. Provide materials suitable for use intended.
- B. Tarpaulins: Waterproof, fire-resistant UL labeled with flame spread rating of 15 or less. For temporary enclosures, provide translucent, nylon-reinforced, laminated polyethylene or polyvinyl chloride, fire-retardant tarpaulins.
- C. Water: Potable and approved by local health authorities.

- D. Wood: Lumber complying with DOC PS 20 and applicable grading rules of an inspection agency certified by ALSC's Board of Review for specific use. Provide preservative treated lumber where partially or fully in contact with the earth, concrete or masonry.
- E. Sign, Directory and Other Graphic Panel Materials: Unless otherwise indicated, products shall comply as follows:
 - 1. Panels: Exterior type Grade B-B high density concrete-form-overlay plywood.
 - 2. Paint: Exterior primer and exterior grade alkyd gloss enamel top coat.
- F. Open-Mesh Fencing: Minimum 0.12 inch (3 mm) thick galvanized 2 inch (50 mm) chainlink fabric fencing with galvanized steel pipe posts, 1-1/2 inches (38 mm) inside diameter for line posts and 2-1/2 inches (64 mm) inside diameter for corner posts. At the discretion of the Project Officer, steel posts and vinyl "snow fencing", or removable chain link fencing, may be provided on a temporary basis for work areas adjacent to the project site.
 - 1. Fence height: Minimum 8 feet (2.5 m).
 - 2. Top Protection: Galvanized barbed-wire top strand.
- G. Gypsum Board: Minimum 1/2 inch (12.7 mm) thick by 48 inches (1219 mm) wide by maximum available lengths; regular-type panels with tapered edges. Comply with ASTM C 36.
- H. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indices of 25 and 50, respectively.

2.2 EQUIPMENT

- A. General: Provide new or undamaged, previously used equipment in serviceable condition. Provide equipment suitable for use intended. All equipment must meet applicable local codes governing its use.
- B. Water Hoses: 3/4 inch (19 mm) heavy duty abrasion-resistant flexible rubber hoses, 100 feet (30 m) long with pressure rating greater than the maximum pressure of the water distribution system. Provide adjustable shutoff nozzles at hose discharge.
- C. Electric Outlets: Properly configured NEMA-polarized outlets to prevent insertion of 110 to 120 Volt plugs into higher voltage outlets. Provide receptacle outlets equipped with ground fault circuit interrupters, reset button and pilot light for connection of power tools and equipment.
- D. Electric Power Cords: Grounded extension cords.
 - 1. Provide hard-service cords where exposed to abrasion or traffic.
 - 2. Provide waterproof connectors to connect separate lengths of electric cords where single lengths will not reach areas of construction activity.
 - 3. Do not exceed safe length-voltage ratio.
- E. Lamps and Light Fixtures: General service incandescent lamps of wattage required for adequate illumination.
 - 1. Provide guard cages or tempered glass enclosures where exposed to breakage.
 - 2. Provide exterior fixtures where exposed to moisture.
- F. Unless the Project Officer authorizes the use of permanent heating system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.
 - 1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.

2. Heating Units: Temporary heating units that have been tested and labeled by UL, FM or another recognized trade association related to the type of fuel consumed as appropriate for the space being heated.
- G. Temporary Offices: Space is allocated for temporary contractors' offices on the 14th floor of Building 10.
- H. Fire Extinguishers: Hand-carried portable UL-rated fire extinguishers.
1. Class A extinguishers for temporary offices and similar spaces.
 2. Class ABC dry chemical extinguishers or a combination of extinguishers of NFPA recommended classes for the exposures in other locations.
 3. Comply with NFPA 10 and NFPA 241 for classification, extinguishing agent and size required by location and class of fire exposure.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Use qualified personnel for installation of temporary facilities.
- B. Locate facilities where they will serve the project adequately and result in minimum interference with performance of construction activities. Maintain, relocate and modify facilities as required during the construction period. Contractor is responsible for scheduling in order to provide each facility ready for use when needed to avoid delay. Temporary facilities shall remain in place until no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

- A. General: The Contractor shall connect to existing services.
1. Notification to interrupt any building service and/or utility service shall be requested in writing to the Project Officer a minimum of 15 working days prior to the desired date of interruption. NIH reserves the right to refuse any request and to schedule such interruption on a later or earlier date and time which is mutually agreeable to NIH and the Contractor.
 2. Provide adequate capacity at each stage of construction. Prior to temporary utility availability, provide trucked-in services.
 3. Obtain easements to bring in temporary utilities to the site where NIH easements are not available for that purpose.
- B. Water Service: Install temporary water service and distribution piping of sizes and pressures adequate for construction needs until permanent water service is in use. Sterilize water piping prior to use.
1. Provide rubber hoses as necessary to serve Project site.
 2. As soon as water is required at each level, extend service to form a temporary water- and fire-protection standpipe. Provide distribution piping. Space outlets so water can be reached with a 100-foot (30-m) hose. Provide one hose at each outlet.
 3. Where installations below an outlet might be damaged by spillage or leakage, provide a drip pan of suitable size to minimize water damage. Drain accumulated water promptly from pans.
 4. Provide backflow preventors at connections to existing water service.

5. Provide pumps to supply a minimum of 30-psi (200-kPa) static pressure at highest point. Equip pumps with surge and storage tanks and automatic controls to supply water uniformly at reasonable pressures.
- C. Electric Power Service: Provide weatherproof, grounded, electric power service and distribution system of sufficient size, capacity and power characteristics for construction needs. Include meters, transformers, overload-protected disconnects, automatic ground-fault interrupters and main distribution switch gear.
1. Make connections at location(s) designated by the Project Officer.
 2. Feeder and branch wiring with area distribution boxes shall be located so that power is available throughout the project site by use of power cords.
 3. Provide waterproof connectors to connect separate lengths of electrical power cords if single lengths will not reach areas where construction activities are in progress. Do not exceed safe length-voltage ratio.
 4. Install all electrical devices, both temporary and permanent, in accordance with the National Electric Code.
 5. Provide 4-gang outlets, spaced so 100-foot (30-m) extension cord can reach each area for power hand tools and task lighting. Provide a separate 125-V ac, 20-A circuit for each outlet.
- D. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations and traffic conditions at all times.
1. Install and operate temporary lighting that will fulfill security and protection requirements without operating the entire system.
 2. Install temporary lighting in accordance with the National Electric Code.
- E. Heating and Cooling: Provide temporary heat and cooling required for the construction activities in new and existing spaces, including but not limited to curing or drying completed installations and protecting construction from adverse effects of low temperatures and/or high humidity. Select equipment from that specified that will not have a harmful effect on completed installations or elements being installed.
1. Maintain a minimum temperature of 50 degrees F (10 degrees C) in permanently enclosed portions of building for normal construction activities, and 65 degrees F (18.3 degrees C) for finishing activities and areas where finished Work has been installed.
 2. Provide temporary heating and cooling for occupied NIH spaces as required to maintain specified temperature and product manufacturer recommended humidity levels without need to rely on existing building systems.
- F. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment from that specified that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.
- G. Heating Facilities: Except where NIH authorizes use of the permanent system, provide vented self-contained, LP-gas or fuel oil heaters with individual space thermostatic control. Do not use gasoline-burning space heaters, open flame or salamander-type heating units. All temporary heating and cooling systems shall be operated in accordance with the manufacturer's instructions.
- H. Telephone Service: The Contractor shall provide temporary telephone service within the contract area only if requested in writing by the Project Officer. If required, telephone service shall be removed prior to contract completion. The Contractor shall pay all costs of service. Installation and removal of service shall be subject to the approval of the Project Officer.

- I. Use of Existing NIH Toilets: Use of existing NIH toilet facilities designated by the Project Officer will be permitted. Clean and maintain facilities in a condition acceptable to the Contracting Officer and, at completion of construction, restore facilities to condition prevalent at the time of initial use.
 - 1. Drinking -Water Facilities: Provide bottled-water, drinking-water units.
 - a. Provide electric water coolers to maintain dispensed water temperature at 45 to 55 degrees F (7.2 to 12.7 degrees C).
- J. Sewers and Drainage: Where sewers are available, provide temporary connections to remove effluent that can be discharged lawfully. If sewers are not available, provide containers to remove and dispose of effluent off-site in a lawful manner.
 - 1. Filter out excessive amounts of soil, construction debris, chemicals, oils and similar contaminants that might clog sewers or pollute waterways.
 - 2. Connect temporary sewers as directed by the Project Officer and sewer utility officials.
 - 3. Maintain temporary sewers and facilities in a clean, sanitary condition. Following use, promptly restore sewers and facilities to normal conditions.
- K. Erosion and Sediment Control: The Contractor shall comply with all provisions of the Maryland Department of the Environment requirements for Storm Water Management, including all required submittals to the state.

3.3 SUPPORT FACILITIES INSTALLATION

- A. General: Comply with the following:
 - 1. Locate storage sheds and other temporary construction and support facilities for easy access and as directed by the Project Officer.
 - 2. Provide incombustible construction for offices, shops and sheds located within the construction area or within 30 feet (9 m) of building lines. Comply with NFPA 241.
 - 3. Maintain support facilities until near Substantial Completion. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to NIH.
- B. Construction Signage and Other Temporary Signs: Locate signs where indicated or directed by the Project Officer to inform the public and instruct persons seeking entrance to the project. Support exterior signs on posts or framing of steel or preservative-treated wood. All signage shall conform to the standards set forth in the DCAB Guide for Construction Site Signage.
 - 1. Exterior Construction Site Sign: All exterior construction sites shall have at least one and not more than three site signs. Signs will be provided by the Project Officer for installation by the Contractor. The Contractor shall provide a written request to the Project Officer 21 calendar days before work starts of his need for exterior construction site signs. The signs for smaller projects and those with a construction duration of less than six months will be 3 feet by 4 feet (0.9 by 1.2 m).
 - 2. Interior Construction Sign:
 - a. All interior construction areas shall have a minimum of 2 interior construction information signs. The signs will be provided by the Project Officer for installation by the Contractor. The Contractor shall provide a written request to the Project Officer 21 calendar days before work starts of his need for the Interior Construction Signs. The signs shall be placed on site not less than 14 calendar days prior to the start of construction. The signs will be either 8 1/2 by 11 inches (216 by 279 mm) or 11 by 17 inches (279 by 432 mm).
 - b. Construction in Progress Door Hanger Signs: Door hanger signs shall be hung on the door knobs on the outside of the entrance door(s) of the room(s) in which construction is being performed. The signs will be provided by the Project Officer for installation by the Contractor. The Contractor shall provide a written request to

the Project Officer 21 calendar days before work starts of his need for the Construction in Progress signs. Include the number of signs needed. Door hanger signs shall be in place not less than 7 calendar days prior to construction. The signs are 100 x 225 mm paper.

- c. Construction in Your Neighborhood Door Hanger Signs: Door hanger signs shall be hung on the door knobs on the outside of the adjacent corridors entrance door(s) on floors above and below the room(s) that may be affected by construction. The signs will be provided by the Project Officer for installation by the Contractor. The Contractor shall provide a written request to the Project Officer 21 calendar days before work starts of his need for the Construction in Progress signs. Include the number of signs needed. Door hanger signs shall be in place not less than 7 calendar days prior to construction. The signs are 100 x 225 mm paper.
 3. Hard Hat Area Sign: The entire work area under this contract is designated as a Hard Hat Area. The Contractor shall assure that all contractor personnel, vendors, and visitors utilize hard hats within the project area. The Contractor shall provide hard hat area signs at each entrance to the Project site for all construction work performed under this contract. Sign shall be approximately 450 x 600 mm, with minimum 50 mm tall lettering as follows:

HARD HAT AREA
AUTHORIZED PERSONNEL ONLY
ALL PERSONNEL ENTERING THIS
CONSTRUCTION SITE SHALL BE EQUIPPED
WITH PROPER SAFETY ATTIRE
 4. Other Temporary Signs: The Contractor shall provide all signage for temporary construction, closing of roads, parking lots, sidewalks or other areas. Signage shall conform to the standards set forth in the DCAB Guide for Construction Site Signage. A copy of the guide may be obtained from the Project Officer.
 5. Do not permit installation of unauthorized signs.
- C. Waste Disposal Facilities:
1. See Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material" for specific requirements related to waste disposal.
 2. Provide waste-collection containers in sizes adequate to handle waste from construction operations.
 3. Comply with Division 01 Section "Execution Requirements" for progress cleaning requirements.
- D. Janitorial Services: Provide janitorial services on a daily basis for temporary offices, first-aid stations, toilets, wash facilities, lunchrooms, and similar areas.
- E. Temporary Lifts and Hoists: When existing elevators are not available or otherwise restricted from use by construction personnel, provide facilities for hoisting materials and personnel as needed to efficiently execute the project. Truck cranes and similar devices for hoisting materials are considered "tools and equipment" and not temporary facilities.
- F. Existing Elevator Usage: See Division 1 Section "Work Restrictions" for additional requirements on use of existing elevators.
1. Provide protective coverings, barriers, devices, signs, or other procedures to protect elevator car and entrance doors and frame. If, despite such protection, elevators become damaged, engage elevator Installer to restore damaged work so no evidence remains of correction work. Return items that cannot be refinished in field to the shop, make required repairs and refinish entire unit, or provide new units as required.

- G. Existing Stair Usage: Use of existing stairs will be permitted, as long as stairs are cleaned and maintained in a condition acceptable to NIH. At Substantial Completion, restore stairs to condition existing before initial use.
 - 1. Provide protective coverings, barriers, devices, signs, or other procedures to protect stairs and to maintain means of egress. If, despite such protection, stairs become damaged, restore damaged areas so no evidence remains of correction work.
- H. Collection and Disposal of Waste: Collect waste from construction areas and elsewhere daily. Enforce requirements strictly and dispose of material lawfully.
 - 1. Comply with NFPA 241 for removal of combustible waste material and debris.
 - 2. Do not hold waste materials more than 7 days during periods when the ambient temperature remains continuously less than 80 deg F (27 deg C), or more than 3 days when the temperature exceeds or is expected to rise above 80 deg F (27 deg C).
 - 3. Handle and properly containerize hazardous, dangerous or unsanitary waste materials separately from other waste.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

- A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and that minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects. Avoid using tools and equipment that produce harmful noise. Restrict use of noisemaking tools and equipment to hours that will minimize complaints from persons or firms near Project site.
- B. Storm Water Control: Provide earthen embankments and similar barriers in and around excavations and subgrade construction, sufficient to prevent flooding by runoff of storm water from heavy rains. Comply with State of Maryland Department of the Environment requirements and approved Storm Water Management permit drawings.
- C. Site Enclosure Fence: When excavation begins, install an enclosure fence with lockable entrance gates. Install in a manner that will prevent people, dogs and other animals from easily entering the site, except by the entrance gates.
 - 1. Locate where indicated or, if not indicated, enclose the entire site or the portion determined sufficient to accommodate construction operations.
 - 2. Provide either open-mesh chainlink fencing with posts set in a compacted mixture of gravel and earth, or provide 8 feet (2.5 m) high plywood fence framed with four nominal 2-by-4 inch (50-by-100 mm) wood rails and preservative-treated wood posts spaced not more than 8 feet (2.5 m) apart.
 - 3. The Contractor shall provide a locking system, approved by the Project Officer, and the enclosure fence shall be locked at the end of every work day and at all times whenever the Contractor is not present on the site.
- D. Barricades, Warning Signs, and Lights: Comply with standards and code requirements for erecting structurally adequate barricades. Paint with appropriate colors, graphics, and warning signs to inform personnel and the public of the hazard involved. Where appropriate and needed, provide lighting, including flashing red or amber lights.
- E. Security Enclosure and Lockup: Install substantial temporary enclosure of partially completed areas of construction. Provide locking entrances to prevent unauthorized entrance, vandalism, theft and similar violations of security.
 - 1. Storage: Provide a secure lockup for valuable stored materials and equipment.
 - 2. Enforce discipline in connection with the installation and release of material to minimize the opportunity for theft and vandalism.

- F. Temporary Fire Protection: Until fire protection needs are supplied by permanent facilities, install and maintain temporary fire protection facilities of types needed to protect against reasonably predictable and controlled fire losses. Comply with NFPA 241 "Standard for Safeguarding Construction, Alterations, and Demolition Operations".
1. Comply with NFPA 10 "Standard for Portable Fire Extinguishers". Provide fire extinguishers, installed on walls on mounting brackets, visible and accessible from space being served, with sign mounted above.
 - a. Field Offices: Class A stored-pressure water-type extinguishers.
 - b. Other Locations: Class ABC dry-chemical extinguishers or a combination of extinguishers of NFPA-recommended classes for exposures.
 2. Locate fire extinguishers where convenient and effective for their intended purpose, but not less than one extinguisher on each floor at or near each access route exit or entrance.
 3. Store combustible materials in containers in fire-safe locations.
 4. Maintain unobstructed access to fire extinguishers, fire hydrants, temporary fire protection facilities and access routes. Prohibit smoking in hazardous fire-exposure areas.
 5. Provide supervision of welding operations, combustion-type temporary heating units and other sources of fire ignition.
 6. All required standpipe systems and sprinkler systems shall be maintained in conformity with the progress of building activity in such a manner that they are always in working order.
 7. Develop and supervise an overall fire-prevention and first-aid fire-protection program for personnel at Project site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
 8. Provide temporary standpipes and hoses for fire protection. Hang hoses with a warning sign stating that hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.
 9. Notification to interrupt any fire protection system shall be requested in writing to the Project Officer a minimum of fifteen working days prior to the desired date of interruption. NIH reserves the right to refuse any request and to schedule such interruption on a later or earlier date and time which is mutually agreeable to NIH and the Contractor.
- G. Permanent Fire Protection: At the earliest feasible date in each area of the Project, complete installation of the permanent fire protection facilities, including connected services, and place into operation and use. Instruct key personnel in the use of the facilities.
- H. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress or completed, from exposure, inclement weather, other construction operations and similar conditions.
1. Where heat is needed and the permanent building enclosure is not complete, provide temporary enclosures where there is no other provision for containment of heat. Coordinate enclosure with ventilating and material drying or curing requirements to avoid dangerous conditions or unacceptable effects.
 2. Install tarpaulins securely using fire-retardant-treated wood framing and other materials.
 3. Vertical Openings: Close openings of 25 sq. ft. (2.3 sq. m.) or less with plywood or similar materials.
 4. Horizontal Openings: Close openings through floor or roof decks and other horizontal surfaces with load-bearing wood-framed construction.
 5. Where enclosure exceeds 100 sq. ft. (9.2 sq. m) in area, use UL labeled fire-retardant-treated wood and plywood for framing and sheathing.
- I. Safety: The contractor shall protect the integrity of any installed safety systems or personnel safety devices. If entrance into systems serving safety devices is required, the Contractor shall

obtain prior approval from the Project Officer. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish contract requirements, the Contractor shall provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and shall obtain prior written approval from the Project Officer.

3.5 INTEGRATED PEST MANAGEMENT (IPM)

- A. Integrated Pest Management (IPM) services for the control of pests are required on construction sites during all phases of construction. These services are to provide exclusion and control of pests while minimizing the use of pesticides. The Construction Contractor shall retain the services of a licensed and certified pest management contractor (PMC) to provide proactive IPM services, identify inadequate site sanitation and facility deficiencies during the duration of the contract. The PMC and all pest management personnel assigned to the contract must understand the principles and practices of urban IPM. The PMC must recognize that the facility in which they are working may support sensitive biomedical research which can not be compromised through pest infestation or the improper use of toxicants, such as pesticides.
 - 1. IPM services shall include all invertebrate and vertebrate pests including, but not limited to, cockroaches, ants (including carpenter ants), fleas, stinging insects and nests accessible from the ground or windows, rats and mice, flies, silverfish, stored product pests, raccoons, opossums, pigeons, and incidental invaders, (i.e., crickets, earwigs, midges, millipedes, centipedes, ground beetles, clover mites, birds, and bats).
 - 2. Preventive caulking and sealing of pest harborage and pathways is the preferred method of correcting or controlling an infestation. The PMC shall review the construction contract plans and specifications and shall document whether the application of sealants and other exclusion materials under sinks, around cabinets, pipe chases, windows and doors, and the exterior, etc. have been indicated on the contract documents. The project-specific recommendations shall be submitted in writing concurrently to the Construction Contractor, Contracting Officer, and Project Officer for their review and action.
- B. Personnel
 - 1. The PMC shall designate a Quality Control Supervisor who is a manager/supervisor in the PMC firm and who will have primary responsibility for the performance of the IPM program.
 - a. The Quality Control Supervisor shall perform a thorough review of all pertinent construction plans and specifications, survey the site and the surrounding grounds, survey all portions of a building involved, and meet with the Project Officer, Construction Contractor, and contractor's QC Manager to discuss the IPM service requirements.
 - b. The PMC QC Supervisor shall participate in preconstruction meetings and QC meetings as appropriate.
 - c. The PMC QC Supervisor shall be considered a member of the contractor's QC Staff and coordinate activities with the contractor's QC Manager. See Division 01 Section "Construction Quality Control" for additional coordination information.
 - 2. Pest management services shall be performed only by technicians Certified by the Maryland Department of Agriculture in Category VII-General and Rodent Control. In addition, the PMC shall designate an alternate technician who is Certified by the Maryland Department of Agriculture in Category VII-General and Rodent Control. The alternate shall be familiar with all aspects of this program. Copies of certificates and the PMC's Maryland Department of Agriculture Pesticide Business License shall be kept in the PMC logbook at the work site.
- C. The PMC shall develop a written IPM service plan that describes the pest management program in effect during all phases of construction. The service plan shall include recommendations concerning the grounds and existing ground cover, solid waste management,

and site sanitation, and other factors that contribute to pest infestation. This plan shall also designate a day and time that the PMC shall perform pest management services. A copy of this report shall be kept with the PMC logbook at the work site. IPM services shall consist of the following:

1. Bi-weekly survey of the grounds, trailers, and building site for pest activity, structured so that all interior space and the surrounding grounds are surveyed at least once each month. The PMC shall survey the site using visual inspections, and by placing traps and other pest monitoring devices in areas where pests may live or forage. Traps shall be identified with the technician's initials, date, and location on the site. Sticky traps shall be maintained in all areas, where feasible, throughout the construction process.
2. Reports: Sanitation and poor housekeeping practices that may contribute to pest infestations shall be reported, in writing, to the Project Officer and the Construction Contractor at the completion of each service. Submission of these reports shall be coordinated through the contractor's QC Manager.
3. Pest Management Logbook: A copy of all reports shall be kept in a pest management logbook in the Construction Contractors QC office. The logbook shall contain clear and concise records reflecting the following:
 - a. Common name(s) of pests. Codes for pests shall not be permitted.
 - b. Location(s) of pest activity.
 - c. Common names of pesticides used (as required by the Maryland Pesticide Applicators Law and Regulations). Codes for pesticides shall not be permitted.
 - d. Structural and sanitation deficiencies.
 - e. Recommendations regarding pest management.
 - f. Non-pesticidal control measures applied.
 - g. MSDS and labels for all products that may be applied.
 - h. A site map indicating the location of rodent activity and rodent control devices used on the site.
 - i. A section of the Pest Management Logbook shall be allocated for construction personnel to record pest sightings.
4. Monthly Reports: At the end of each month the PMC Quality Control Supervisor shall prepare a written report summarizing pest management activities at the site. This report shall be submitted to the Contracting Officer, with copies sent to the Project Officer and Construction Contractor. The report shall be referenced in the monthly QC report prepared by the construction contractor's QC Manager. The report shall contain information on the following:
 - a. Types of pests encountered, pest numbers and locations.
 - b. Pesticides applied, amounts and locations.
 - c. Sanitation and structural deficiencies.
 - d. Non-chemical methods applied.
 - e. Recommendations regarding pest management issues.
5. Progress Meetings: The PMC technician and Quality Control Supervisor shall be available, as needed, to attend progress meetings as specified in Division 01 Section "Project Management and Coordination" and "Construction Quality Control" to discuss program activities.
6. Special Service and Emergency Requests: The PMC shall respond to special service requests within one (1) working day of the request from NIH or Construction Contractor. The PMC shall provide emergency service on the same day of the request and be capable of performing emergency services when construction work is on-going. A list of telephone and pager numbers (along with instructions) shall be included in the logbook. The date and time of emergency service and the pest(s) requiring emergency service shall be listed in the logbook and monthly summary report.

- D. Pesticide Use. Pesticide applications shall be made only to areas of known pest infestation or activity and where alternative control measures such as traps, caulking, sealing, cleaning and disposal of goods were not successful or are not feasible. Because of the odor associated with

some pesticides, indoor treatments shall only be performed after notification is given to Project Officer. If necessary, the PMC shall schedule pesticidal treatments for a later time and/or date in order to accommodate the needs of the site personnel and management. Insecticides shall be applied using crack and crevice injection devices, such as Whitmire System III, or other aerosols with injection tips. Granular formulations and compressed air sprayers shall only be used for treatment of utility areas or areas outside the building to correct pest problems. Pesticides permitted for use are listed below. Any products proposed for use that are not on the list require the written approval of the NIH Pest Management Unit through the Contracting Officer.

1. chlorpyrifos
2. hydramethylnon
3. avermectin
4. boric acid
5. gencor
6. methoprene
7. cholecalciferol
8. silica gel
9. Diazinon
10. pyrethrins and pyrethroids
11. brodifacoum (place packs)
12. sulfuramid
13. Chlorophacinone-Liphadione or Diphacinone Tracking Powder
14. Fipronil
15. The use of zinc phosphide is prohibited.

- E. Rodent Control: Rodent control shall be on-going throughout the entire construction period. The primary focus of this program is preventive rodent management with emphasis placed on site sanitation, exclusion, and continuous surveillance of the entire site. At a minimum it shall consist of the following:

1. Thorough monitoring of all construction areas including the building interior, perimeter and grounds of the construction site. Monitoring shall consist of using traps, visual inspections for burrows and other signs of activity, and interviewing site personnel. The PMC shall collect and dispose of all dead rodents immediately.
2. Rodenticide baits shall only be applied outside the structure using "place-packs" or "bait blocks" in bait stations, labeled with the common name of the rodenticide applied and date of service, or applied directly in the burrow. The location of each bait station shall be identified on the site map located in the logbook. The use of tracking powder rodenticides shall be limited to the exterior areas. Rodent burrows that have been treated with a rodenticide shall be sealed with newspaper or by closing the burrow opening.
3. Known or suspected rodent activity in areas surrounding the construction site shall be reported to the Project Officer immediately.
4. Rodent management inside the structure shall consist of using rodent traps and glueboards. The use of grain based rodenticides is discouraged because of the potential odor problems caused by dead rodents and infestation by stored product pests in walls and other inaccessible places. If special circumstances arise the PMC may request permission from the NIH Pest Management Unit to use rodenticides inside the building.
5. Traps and glueboards shall be placed in areas where they will not be removed or pose a hazard to the construction personnel. Traps and monitoring devices should be checked during each regular service to assure adhesiveness of the surface. If repeating live traps are used, they shall be serviced the next day to ensure quick removal of the trapped animal.

- F. Post-Construction Survey and Clearance: Prior to final acceptance of the facility by NIH, the PMC shall perform a complete and thorough survey of the building(s) and grounds involved with

the individual task order. The PMC will perform all necessary pest control operations inside the facility and on the grounds to control pest activity. Upon completion of this service the PMC shall issue written clearance certification for the building and grounds, with a thirty (30) day guarantee, under which the PMC will return to correct any pest problems without charge. Termites and other Wood Destroying Organisms are excluded from this requirement.

3.6 OPERATION, TERMINATION, AND REMOVAL

- A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.
- B. Maintenance: Maintain facilities in good operating condition until removal. Protect from damage by freezing temperatures and similar elements.
 - 1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.
 - 2. Protection: Prevent water-filled piping from freezing. Maintain markers for underground lines. Protect underground lines from damage during excavation operations.
- C. Temporary Facility Changeover: Except for using permanent fire protection facilities as soon as available, do not change over from temporary protection facilities until authorized by the Project Officer.
- D. Termination and Removal: Unless the Project Officer requests that a temporary facility be maintained longer, each temporary facility shall be removed when the need for its service has ended, when it can be replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with the temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
 - 1. Materials and facilities that constitute temporary facilities are the property of the Contractor, except NIH reserves the right to take possession of project identification signs.
 - 2. Remove temporary paving not intended for or acceptable for integration into permanent paving. Where the area is intended for landscape development, remove soil and aggregate fill that do not comply with requirements for subsoil or fill in the area. Remove materials contaminated with road oil, asphalt and other petrochemical compounds, and other substances that might impair growth of plant materials or lawns. Repair or replace street paving, curbs, and sidewalks at temporary entrances, in accordance with the requirements of the governing authority.

3.7 TEMPORARY CONDITIONING AND OPERATION USING PERMANENT EQUIPMENT

- A. Contractor shall only use building permanent equipment to provide temporary conditioning on the approval of the AE, and NIH). Approval for such will only be given upon acceptance of a detailed plan provided by the individually involved subcontractors and compiled by the CM. The temporary conditioning plan shall consider/address the following at a minimum.
- B. Indicate that the full start up protocol as required by the specification for final acceptance will be performed for the temporary start up. Temporary conditioning plan shall include the start up forms to be used which will be the same as those that will be used for final start up.
- C. Contractor shall address how equipment will be maintained in good, clean condition. Specifically address.

1. Temporary filtering of air: Temporary filters used during construction shall be identical in quantity, arrangement and MERV rating to specified permanent filters for each air handling and exhaust system. Contractor shall remove construction filters and replace with new filters at substantial completion. Filters shall be maintained and replaced at the specified final pressure drop. Contractor shall install a magnehelic for visual indication of pressure drop as well as set up the loaded filter DP switch for monitoring on the BAS.
 2. Temporary Filtering of Water and Condensate: Construction strainers shall be used while circulating fluid during construction. Strainer shall be finer than specified for final strainers.
 3. Sealing/Filtering of Open Ducts: Address that all open ducts shall be either sealed or protected with filter media. Generally return or exhaust systems shall not be used during construction unless otherwise approved. Seal ducts in accordance with SMACNA IAQ Guidelines for Occupied Buildings under Construction.
 4. Lubrication and Maintenance: Contractor shall maintain the systems and equipment in accordance with the manufacturer's instructions. Contractor shall coordinate lubricants used with NIH's operators. Frequency of lubrication and inspection shall be as recommended by manufacturer's literature. Applicable maintenance lubrication schedules shall be included in the plan. Draft maintenance logs shall be submitted with plan and completed as maintenance is performed.
 5. Operation outside of Normal ranges: Systems and equipment shall not be operated outside the range of specified conditions. Plan shall address how the contractor will ensure that operation will not harm the equipment.
 6. Emergency Condition Identification and Response protocols: Plan shall address protocols for responding to equipment malfunctions and or harmful operation. Automatic safeties and remote enunciation shall be in place to protect people and property. Temporary operation shall not be allowed until there is a automatic communication/enunciation medium such as a phone connection or an internet connection. At a minimum, an alarm on the equipment used for temporary service shall be automatically sent to the contractor's 24 hour monitoring service and to the Owner's help desk. The contractor shall respond to and be responsible for securing conditions within the building. Owner shall assess the situation and as necessary secure utilities feeding the building from isolation points outside of the building.
- D. Campus Utility Impact: The plan shall address the expected impact on the campus utilities involved in the temporary conditioning equipment. Specifically address:
1. How the systems will be controlled to both ensure they are operating in range, and to avoid energy waste or inefficient conditions.
 2. Project the range of loads and flows to be imposed on the campus systems. For cooling, indicated how you will ensure a temperature split of at least 8C.
 3. For campus chilled water connections, the bridge connection and automatic control of the bridge related sequences shall be installed, functional and tested.
- E. Building Protection: Address how the system will be controlled to avoid humidity conditions that will either promote mold growth or cause corrosion.
- F. Equipment Reconditioning: Address with specific means and methods how the equipment used for temporary conditioning will be re-conditioned to new condition. Belts, seals, bearings, couplings, or other parts that wear more than 3% of their expected life shall be replaced.
- G. Cleaning: Address how ducts, pipes, coils, converters, air handling equipment, terminal units, etc. shall be cleaned at final turn over.
- H. Operations Log: Contractor responsible for operating the equipment shall maintain a log of all activities associated with operating and maintaining equipment. Log shall be submitted to Owner on a frequency specified by them.

- I. Operating System Alterations: Plan shall address specific protocol for doing work the systems.
- J. Any material, device, component, equipment, etc. that is assessed as damaged or as having a substantially shortened life as a result of temporary conditioning operation shall be replaced by the contractor at no cost to the contract.
- K. Segregation: Where only portions of a system are to be used, contractor shall specifically indicate how the used portion will be isolated from the unused portion. Plan shall address how to ensure that the reduced operation condition will be maintained within acceptable ranges, and/or how capacity will be throttled to keep all operating parameters in recommended ranges.

END OF SECTION

SECTION 015480 - USE, HANDLING, STORAGE, TRANSPORTING, ACCUMULATION, AND DISPOSAL OF NIH CONTROLLED MATERIAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section covers the demolition, use, handling, storage, transporting, accumulation and disposal of hazardous or recyclable materials/substances/chemicals that may be encountered or that may be used by the Contractor during the course of the work. The Contractor is made aware by this specification that hazardous or recyclable materials/substances/chemicals are regulated by a multitude of statutes and regulations and require special care.
- B. Only hazardous and recyclable materials and wastes generated or removed during this construction contract at NIH will be disposed of by the Contractor.
- C. Hazardous wastes belonging to NIH will be disposed of by NIH. The Contractor shall notify the Contracting Officer when NIH owned hazardous wastes are discovered and/or become necessary to be removed from the project site.
- D. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to project management and coordination. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Photographic Documentation"
 - 6. Division 01 Section "Submittal Procedures"
 - 7. Division 01 Section "Quality Requirements"
 - 8. Division 01 Section "References"
 - 9. Division 01 Section "Construction Quality Control"
 - 10. Division 01 Section "Temporary Facilities and Controls"
 - 11. Division 01 Section "Safety and Health"
 - 12. Division 01 Section "Product Requirements"
 - 13. Division 01 Section "Execution Requirements"
 - 14. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 15. Division 01 Section "Closeout Procedures"
 - 16. Division 01 Section "Demonstration and Training"
- E. Additional Requirements: Refer to the individual technical specification sections for additional requirements for the use, handling, storage, transporting, accumulation and disposal of NIH controlled material.

1.3 REFERENCES

- A. In addition to publications referenced in the Construction Contract Clauses, the following publications designate and define hazardous materials and conditions, and establish procedures for handling these materials and conditions. Omission of any publication in this section does not remove any obligation or legal requirement on the part of the contractor to comply with all legal requirements for the location of the work.
1. Code of Federal Regulations (CFR):
 - a. 29 CFR, Part 1910: Occupational Safety and Health Administration (OSHA) General Industry and Health Standards.
 - b. 29 CFR, Part 1910.134: Respiratory Protection
 - c. 29 CFR, Part 1910.145: Specifications for Accident Prevention Signs and Tags
 - d. 29 CFR, Part 1910.20, Sub-part C: General Safety and Health Provisions
 - e. 40 CFR, Part 61: National Emissions Standards for Hazardous Air Pollutants
 - f. 40 CFR, Parts 9 and 82: Protection of Stratospheric zone (CFCs), Clear Air Act Amendments of 1990
 - g. 40 CFR, Part 122 and 125: National Pollutant Discharge Elimination System Clean, Water Act
 - h. 40 CFR, Parts 260 - 272: Solid and Hazardous Wastes, Resource Conservation and Recovery Act
 - i. 40 CFR, Part 165: Disposal and Storage of Pesticides and Pesticide Containers
 - j. 40 CFR, Subchapter J, Parts 300 - 373, Superfund Emergency Planning, and Community Right-to-Know Programs
 - k. 40 CFR, Parts 700 - 799: Toxic Substances Control Act (TSCA)
 - l. 49 CFR, Parts 171 - 179: Department of Transportation (DOT)
 - m. Federal Standard 313A: Material Safety Data Sheets, Preparation and Submission of
 2. American National Standards Institute (ANSI)
 - a. Z288.2: Standard for Respiratory Protection
 3. American Society for Testing and Materials (ASTM)
 - a. E 849-82: Safety and Health Requirements Relating to Occupational Exposure to Asbestos
 4. State and Local Regulations: The Contractor shall comply with all current
 - a. State of Maryland and Montgomery County Regulations.
 - b. Title 26, Code of Maryland Regulation (COMAR)
 - c. Washington Suburban Sanitary Commission (WSSC) discharge standards
 5. NIH Policies and Procedures
 - a. NIH Manual Issuance: 3032 Waste Management at NIH
 - b. NIH Waste Calendar.
- B. Acquisition of Publications: Referenced CFR publications may be purchased from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

1.4 DEFINITIONS

- A. "NIH Controlled Material" is defined as any material that poses a threat to human health or to the environment; that can be recycled or reused; for which disposal in municipal landfills is regulated or restricted; for which unregulated introduction into groundwater, land, or the atmosphere is irresponsible; and that should be designated as "hazardous waste."
1. It is imperative that the use, handling, storage, transporting and disposal of hazardous and recyclable materials and solid waste in NIH facilities and on NIH property be disciplined and consistent -- to both ensure the safety of NIH personnel and visitors and to avoid incurring liabilities or penalties to the NIH as a consequence of reckless or

improper disposal or recycling of waste generated in, deposited on, brought to, or transported from NIH facilities.

2. A number of statutes and regulations define the term "hazardous" in a variety of ways depending on the nature, condition, and intended use or disposal of the particular material, substance, and chemical. To simplify the manner in which these materials are addressed, the term "hazardous material" is used to identify all materials, substances, and chemicals that exhibit the properties defined by this specification:
 - a. Any material, substance, and chemical that because of its quantity, concentration, or physical, chemical or infectious characteristics is toxic, lethal, corrosive, flammable or combustible, reactive, an irritant, a strong sensitizer, or generates pressure by decomposition, heat or other means and is injurious to human beings, animal life and/or the environment shall be considered "hazardous."
 - b. Exhaustive but not inclusive lists of these hazardous materials are detailed in the following regulations: RCRA 40 CFR 261; CERCLA 40 CFR 302; SARA 40 CFR 355, 370 & 372; FIFRA 40 CFR 152; TSCA 40 CFR 700-799; COMAR Title 26.13.02.15.19.
- B. A "recyclable material" is defined as any material that must be recycled or reused under federal, state, or local regulations, or can be recycled and reused in an economically feasible manner.

1.5 NIH CONTROLLED MATERIAL STANDARDS AND CONTRACTUAL RESPONSIBILITIES

- A. Hazardous materials defined by paragraph 1.4 may not be dumped into storm drains, sewage lines, or dumpsters, nor are they to be introduced into the environment in an uncontrolled manner.
- B. Regulatory Compliance: The Contractor shall comply with all federal, state, and local regulations and the conditions of NIH facility operating permits and licenses applicable to activities and services performed under the contract. A partial but not an inclusive list of such laws and regulations are identified in paragraph 1.3 of this specification.
- C. All contractors will be informed of hazardous materials and potential hazards present in the areas in which the contractor's employees will be working.
- D. Upon request by the Contracting Officer, the Contractor shall provide documentation that personnel are properly trained in generating, handling and storing hazardous wastes.
- E. Contractors shall become familiar with and abide by the requirements of the DES Standard for the Use, Handling, Storage, Transporting, and Disposal of Hazardous Materials. Consult with the NIH Division of Safety, Environmental Protection Branch as required or if so directed by the Project Officer.
- F. The Contractor shall dispose of hazardous materials belonging to him or which is generated as a result of this contract, at no additional expense to the NIH. Waste shall be disposed of on a daily basis.
- G. Hazardous wastes belonging to the NIH will be disposed of by the NIH. The Contractor shall notify the Contracting Officer when NIH-owned hazardous wastes are discovered and/or become necessary to be removed from the project site.
- H. Contractor Liability and Responsibility

1. The Contractor shall assume full responsibility and liability for compliance with all applicable regulations affecting the health and safety of Contractor personnel and others at the NIH during work operations.
 2. In the performance of the contract, the Contractor shall provide for the protection of the health and safety of NIH employees, guest workers, and the public who may be located in areas contiguous, adjacent, or near the work site, and prevent damage to NIH property, materials, supplies, and equipment.
 3. The Contractor shall assume full liability and responsibility for all hazardous materials brought to the contract site. The use, handling, storage, transporting and disposal of such materials shall comply with all regulatory conditions, the manufacturer's recommendations, and the Material Safety Data Sheet (MSDS) for that substance. Copies of the MSDS's will be maintained on-site. Quantities of materials stored and used at the work site shall be limited to the minimum amount required to accomplish the prescribed task or activity.
 4. NIH operates as a treatment, storage and disposal facility as well as an owner and generator. The Contractor will be responsible for site storage, shipment and disposal of hazardous wastes and substances. The Contractor will be responsible for proper shipment and disposal at Contractor-proposed, NIH-approved disposal facilities.
- I. Provisions for and Conditions of Temporary Storage of Hazardous Materials on NIH Property: Hazardous waste for which the Contractor is responsible for transportation and disposal may be accumulated at NIH facilities for up to 30 days, except for extremely hazardous waste materials, provided that:
1. The containers holding the hazardous waste are in good condition, are labeled with all contents, and are not leaking.
 2. The containers are made of or lined with materials that will not react with, and are otherwise compatible with, the hazardous waste to be stored, so that the ability of the container to contain the waste is not impaired.
 3. The containers are closed during storage, except when it is necessary to add or remove waste, and the containers are not opened, handled, or stored in a manner which may rupture the container or cause it to leak.
 4. The containers are inspected at least weekly, to check for leaks and for deterioration caused by corrosion or other factors.
 5. The containers are securely stored away from high traffic areas in a locked area to prevent deliberate or accidental release of hazardous materials.
 6. The location selected for storage of the waste containers provides secondary containment of accidental discharge from the containers.
 7. The containers are stored at a location on the job-site pre-approved by the Contracting Officer.
 8. If a container is leaking, the waste must be immediately transferred to a sound container by the Contractor and treated as a hazardous material spill.
- J. Disposal Method
1. The Contractor shall maximize the use of treatment facilities that employ valid, best available technologies to permanently destroy or render the hazardous material nonhazardous.
 2. Land disposal of hazardous waste shall be minimized. Landfills shall only be employed by the Contractor when no other satisfactory method of treatment for a particular waste is available.
 3. Recycling and reuse of materials are encouraged to the extent economically feasible.
- K. Hazardous Material Spill: Hazardous material spills on-campus shall be reported immediately to the NIH Fire Department (Dial 911), or if off-campus and the telephone is on the NIH exchange, dial 9-911. The Contractor shall take immediate measures to contain the spill,

prevent its spread, and ensure proper cleanup. If there is a threat to health or safety, or danger of further contamination, NIH may elect to take charge of the cleanup, perform the required cleanup, and charge the Contractor for expenses incurred. Hazardous material spill response is explained in paragraph 1.8.

- L. Hazardous Materials Log: The Contractor shall maintain a hazardous materials log on the job site that identifies the type and amounts of materials generated on the site, type and amounts of materials brought to the site and the type and amounts of materials transported from the site. Documentation shall be in chronological order to identify the sequence of activities.
- M. The contractor shall maintain an Material Safety Data Sheet (MSDS) binder on the construction site that shall include all hazardous materials used on the construction site and shall be clearly identified and easily accessible to the NIH Fire Department and Safety and Health Branch during and after work hours. The NIH Fire Department shall be notified in writing, through the Project Officer, of the binder location.
- N. Alternative Materials: Alternative products and materials that are less hazardous and will produce comparable results must be used.

1.6 DELEGATION OF AUTHORITY FOR NIH APPROVAL

- A. For work related to NIH Controlled Material, including use, handling, storage, transporting, accumulation and disposal, the Contracting Officer may designate a representative of the Division of Safety, hereinafter referred to as the NIH EPB Compliance Officer, to act in his behalf in hazardous waste issues only. Such delegation of approval authority shall be presented to the Contractor in writing prior to the beginning of efforts by the Contractor in those elements of the contract that relate directly to the work either requiring the use, handling, storage, transporting, accumulation or disposal of NIH Controlled Material.

1.7 SUBMITTALS

- A. Hazardous Materials Waste Management Plan: Any hazardous materials (i.e. solvents, floor mastic strippers, adhesives, etc.) brought onto the job site by the Contractor or generated as a result of project work shall require the Contractor to prepare a Hazardous Materials Waste Management Plan. This plan shall be prepared and submitted to the Contracting Officer and will be subject to review by the Division of Safety (DS), Occupational Health and Safety Branch (OSHB) and Environmental Protection Branch (EPB). No work shall begin until the plan has been approved by the Contracting Officer, OSHB, and the EPB Compliance Officer. The Hazardous Materials Waste Management Plan shall:
 1. Include a list of all hazardous materials to be used on the job site, including how the materials will be used, handled, stored, packaged, manifested, transported and disposed.
 2. Ensure that any chemical agents, such as floor-mastic strippers, sealants, adhesives, methylene chloride, or other agents containing strong solvents which might create odor problems, headaches, or other physiological symptoms among NIH or Contractor personnel, will be cleared for use through the Occupational Safety and Health Branch (OSHB), Division of Safety (496-2346).
 3. Include the Material Safety Data Sheets (MSDS) for all hazardous materials to be used or generated as a result of the project. This requirement shall be coordinated with Section H of the Contract.
 4. Delineate the process, procedures, and methods to be used by the Contractor to contain or evacuate fumes, vapors, dusts, aerosols, etc. that may be released or generated so as to protect NIH personnel when hazardous materials are used or generated.

5. Delineate the safety procedures, personnel protection measures, and precautions to be taken covering all operations involving hazardous materials, including handling, use, loading, transporting, and first aid.
6. Include a hazardous materials spill control plan for work on federal facilities.
7. Include a hazardous materials spill control plan for materials in transit.
8. Include the identification of the State/EPA permit number and Certification of Insurance of any proposed transporter.
9. Include the identification, description, location, and the State/EPA permit number of any proposed disposal site(s).
10. Include the plan to supply and post OSHA Form 20, Safety Data Sheets, during the use of all solvents, liquids, fluids, and substances brought onto Government property.
11. Include a list of all contract employees assigned to the project along with their titles, duties, education, training, and experience.
12. Include a scheme for minimization of the waste volume to the maximum extent possible either through recycling, alternate methods or processes, or conservation of waste materials.

B. Hazardous Waste Manifest

1. The Contractor is responsible for all Hazardous Waste Manifest documentation to include properly filling out the required documentation, maintaining the records, tracking the manifest, and distribution. The Hazardous Waste Manifest will be approved and signed by the EPB Environmental Compliance Officer.
2. The Contractor shall provide the Hazardous Waste Manifest and any other required regulatory shipment documents to the Contracting Officer for review and approval at least one day preceding a planned shipment of hazardous materials or waste. When the manifest is approved and signed by the Contracting Officer, the Contractor shall leave a copy with the Contracting Officer and ensure that appropriate copies accompany the shipment.
3. While the Contractor may arrange off-site transportation activities at his convenience, the Contracting Officer will be available for signing manifests only during normal business hours. The Contractor's transporter shall sign the manifest and provide copies of all documents required for shipment to the Project Officer prior to commencement of shipment.
4. The receiving or disposal facility official shall sign the manifest certifying receipt of the shipment and forward a copy from that facility to the Project Officer within seven (7) days after receipt. The Project Officer shall receive this return manifest within ten (10) days of the shipping order.

C. Approval of Transporter

1. All transporters the Contractor proposes to use must be approved by the Contracting Officer in writing prior to the generation, storage, or removal of hazardous waste materials on the job site. The Contractor shall furnish the Project Officer with a compliance summary of the proposed transporter which minimally contains the following:
 - a. Current copies of the transporter's operating permits and licenses.
 - b. Copies of environmental impairment insurance.
2. Should there be a need for an alternate transporter in the case of an accident or release in transit, the alternate transporter must be approved by the Contracting Officer.
3. If remedial action is required to contain and cleanup a spill, the Contractor shall work under the direction of the local regulatory authorities responding to the incident. The Contractor shall utilize the most efficient resources available to expedite the control and cleanup. If such an event should occur in transit, the Contractor shall immediately notify the Project Officer.

D. Approval of Treatment, Storage and Disposal Facilities (TSDF)

1. All transfer facilities or treatment, storage and disposal facilities the Contractor proposes to use must be approved in writing by the Contracting Officer prior to removal, generation, or storage of hazardous materials on the job site.
 2. NIH reserves the right to prohibit the Contractor from employing the services of a subcontractor that does not possess the ability to satisfactorily perform in conformance with the provisions of the contract. The use of a subcontractor(s) does not relieve the Contractor of any requirements or responsibilities set forth in the contract. The Contractor is responsible to ensure that any subcontractor(s) perform in accordance with the terms and conditions of the contract.
 3. If, during the period of the contract, the Contractor requests the approval of additional TSDFs, NIH must be allowed reasonable time to evaluate such requests.
- E. PCB Plan of Action: Provide a PCB Plan of Action in the Hazardous Materials Waste Management Plan for approval by the Contracting Officer addressing all requirements set forth in paragraph 3.3.D of this section, including the location of EPA approved recycling and incineration sites, qualifications of transporter, methods and transport, and a description of the methods to be employed to prevent release to the environment. The Contractor shall explain the method for documenting proper PCB disposal (incineration) to the Contracting Officer and include the written procedures in the Hazardous Materials Waste Management Plan.
- F. Certification of Final Disposition
1. It is the Contractor's responsibility to obtain all necessary documentation to prove the final treatment/disposal of all wastes has been accomplished. This documentation shall be included in the Certificate of Final Disposition and submitted with, or prior to, the invoice for payment. A Certificate of Final Disposition shall itemize each container received at the treatment/disposal facility by its NIH Transportation Container number, shall document the date and method of treatment or disposal, and shall be forwarded to the Contracting Officer within four (4) months of the date the material was received at the disposal facility.
 2. If the material is disposed of in a secured chemical landfill facility, then the Certificate of Final Disposition shall also indicate the location within the landfill where the waste container was buried through the use of cell numbers, coordinates, or other appropriate identifiers.

1.8 EMERGENCY RESPONSE AND REMEDIATION OF SPILLS AND RELEASES

- A. Notify Contracting Officer: The Contractor shall be responsible, in the event of a spill of hazardous material resulting from the execution of this contract, to immediately notify the Project Officer. The following information will be supplied to the Contracting Officer in writing as part of the notification process:
1. Type of material.
 2. Quantity of material spilled.
 3. Date and time spill occurred.
 4. Environmental media released; i.e., air, water, soil, etc.
 5. Location, type and extent of spill.
 6. Cause of spill.
 7. Methods used or proposed to control or cleanup the spill.
- B. Spill Kits: The Contractor is responsible for providing spill kits at the work site for all hazardous materials brought into the workspace by the contractor.
- C. Reporting of Spills, Leaks, and Releases: The Contractor is not authorized to issue press releases concerning incidents on NIH installations or to report releases on or from NIH

installations to regulatory authorities. Such news releases and reporting shall be performed only by NIH personnel designated by the Contracting Officer.

- D. Compliance with Contingency and Spill Prevention Control and Countermeasure Plans and Participation Drills: In compliance with Title 40 CFR, some NIH installations have internal Contingency and Spill Prevention Control and Countermeasure plans. Upon award of this contract, the Contractor shall ensure that all on-site contractor personnel are thoroughly familiar with these plans and comply with them as they apply to contract operations.

1.9 PERSONNEL QUALIFICATIONS AND REQUIREMENTS

- A. The Contractor must ensure that every person is trained pursuant to the directions in 49 CFR 172.202 and HM-126F if they are responsible for any of the following:
 - 1. Prepares hazardous materials for use, storage or disposal.
 - 2. Handles, loads, unloads or moves hazardous materials.
 - 3. Fills out forms for the transportation of hazardous materials.
 - 4. Is in any way responsible or accountable for any hazardous materials in NIH facilities.
- B. Qualified personnel are essential and shall be provided by the Contractor for the performance of this contract to assure that public health, safety, and protection of the environment are assured during the use, handling, packaging, transporting, treatment and/or disposal of hazardous materials and that property of the NIH is protected.
- C. Contractor personnel must possess all personal licenses, permits, and certifications required to perform their duties. For example, truck drivers hauling hazardous wastes in Maryland must hold current CHS Driver Certifications.
- D. The Contractor's personnel shall perform in a professional and courteous manner and shall abide by applicable NIH rules, regulations and procedures; and shall present a neat appearance when working at NIH facilities. The Contractor personnel shall not wear protective equipment or clothing when in lunch rooms, cafeterias, and other public areas where such protective items are unnecessary.

1.10 ENVIRONMENTAL SAFETY OFFICER

- A. The Contractor shall retain the part time services of an Industrial Hygienist certified by the American Board of Industrial Hygiene (ABIH), and possessing previous experience in hazardous materials work. The Industrial Hygienist shall interpret the published rules and regulations governing hazardous materials and will provide direction in the implementation of said regulations and safety requirements. This work shall include but not be limited to:
 - 1. Monitoring the Contractor's work with hazardous material for compliance with published rules and regulations by EPA, OSHA, and the State of Maryland including use, handling, storage, packaging, transporting and disposal.
 - 2. Monitoring the generation of hazardous vapors, fumes, dusts, aerosols, etc.
 - 3. Providing the Project Officer with all required documentation pertaining to hazardous materials use, handling, removal, cleanup, transport, and disposal.
 - 4. The Industrial Hygienist is required to be present on site for the following activities/phases of this project. The contractor may elect to employ the Industrial Hygienist for additional times as required in the Contractor's Safety and Health Program as necessary to meet established regulations.
 - a. All construction activities involving the use, handling, transporting and disposal of NIH controlled material.

PART 2 - PRODUCTS

2.1 EQUIPMENT

- A. Equipment, including disposable protective clothing, used in the execution of this contract and/or provided to visitors to the site, shall comply with applicable Federal, State, and local regulations.

2.2 TOOLS

- A. Tools used in the execution of this contract and/or provided to visitors to the site, shall comply with applicable Federal, State, and local regulations. Tools shall be used in strict compliance with manufacturers written instructions for their intended use.

2.3 MATERIALS

- A. Materials used in the execution of this contract and/or provided to visitors to the site, shall comply with applicable Federal, State, and local regulations. Materials shall be used in strict compliance with manufacturers written instructions for their intended use.
- B. Product Material Safety Data Sheets (MSDS): MSDS's will be submitted as part of the Contractor's Hazardous Materials Waste Management Plan and maintained on-site along with all product data from the manufacturer for all applicable substances to be used. All products must comply with NIH Fire Safety Requirements.
- C. Container requirements: All transportation containers purchased and used by the Contractor shall meet DOT Title 49 CFR requirements and regulations for shipment of hazardous materials. Only new containers, free from damage which could compromise the integrity of the container, shall be used. Any damaged containers shall be rejected by the Project Officer and replaced by the Contractor at no cost to NIH. The Contractor shall provide all documents and container labels as required.
- D. All Product Data and corresponding manufacturer's information, shop drawings and manuals must be submitted and approved prior to initiation of work. All product data must be maintained on site at all times during work.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Isolate the Hazardous Material area for the duration of the work to prevent unauthorized access by designating the area off limits to all but authorized personnel. Maintain a log of all persons visiting the Hazardous Material work site.
- B. Post warning signs and labels as required by this contract, 29 CFR 1910, 40 CFR 761, and as directed by the Contracting Officer.

3.2 WORK PROCEDURES

- A. General Procedures: Perform all Hazardous Materials related work in strict compliance with the general safety and health provisions of the referenced requirements. If a conflict arises in the regulations, the more stringent application/requirements for overall safety shall apply until a determination is made otherwise by the Contracting Officer.
- B. Coordination of Work of all Trades: Coordinate the work of all trades to assure their work is performed in accordance with the applicable regulations and the hazardous materials control area remains separated from the remaining work areas.
- C. All contractors will be informed of hazardous materials and potential hazards present in the areas in which the contractor's employees will be working.

3.3 REQUIREMENTS FOR SELECTED NIH CONTROLLED MATERIALS

- A. Lead
 - 1. SPECIAL PERMITS REQUIRED: The State of Maryland requires that lead abatement and lead removal be done only by those with a specific license. Verification of a current license will be submitted to the Contracting Officer.
 - 2. ITEMS USUALLY CONTAINING LEAD: paint, soft solder, radiation shielding, pipe, terne roofing.
 - 3. OPERATIONS INVOLVED: chipping, sanding, grinding, soldering, welding, flame torch cutting abrasive blasting, and spray painting.
 - 4. DEMOLITION PROCEDURES: Demolition procedures shall meet OSHA Guidelines "Working with lead in the construction industry," OSHA Publication 3126", latest edition. This standard lists hazard-reducing techniques for lead demolition such as using vacuum dust collection systems to capture lead dust and fumes at the point of generation. Paint removal shall not employ an open flame, or dry sanding or grinding. Also, the use and selection of respiratory protection and other protective equipment shall be as specified.
 - 5. RECYCLING: Large quantities of lead such as lead sheeting and lead blocks must be recycled.
 - 6. DISPOSAL: Disposal of lead shall be in accordance with paragraph 1.5.
- B. Mercury
 - 1. ITEMS CONTAINING MERCURY: manometers, thermometers, pumps, switches, relays, thermostats, fluorescent tubes.
 - 2. REMOVAL: Segregate all materials containing mercury and prevent mercury from being combined with other liquid or solid hazardous or nonhazardous materials.
 - 3. STORAGE: Mercury and mercury-containing materials shall be stored in appropriate containers that are clearly labeled to identify the contents. Appropriate containers are those that will not deteriorate or react with mercury or allow mercury to leak into the environment during normal use handling, and disposal procedures. Regulations for containing and labeling mercury and mercury-containing materials can be found in 49 CR 172.101.
 - 4. DISPOSAL: Disposal of mercury shall be in accordance with paragraph 1.5.
- C. Batteries: Batteries are not to be discarded in waste dumpsters that are sent to the Montgomery County Solid Waste Transfer Station.
 - 1. USES OF BATTERIES: Emergency lightning, uninterruptible power supplies (UPS), emergency power generation.
 - 2. DISPOSAL: All batteries, regardless of size or material, are to be properly disposed of or recycled by the Contractor in accordance with paragraph 1.5.

- D. Items with Liquid Dielectric Suspected of Containing PCBs (Polychlorinated BiPhenyls)
1. TYPES OF ITEMS: Fluorescent light fixture ballasts, oil circuit breakers, capacitors, liquid transformers, electrical ballasts. Items manufactured after 1978, should carry a label "NO PCBs". Items not clearly labeled, or without labels, or having unknown dielectric shall be considered to contain PCBs under this specification and shall be treated, handled and disposed as hazardous PCB waste.
 2. REMOVAL: The following is a basic outline of the minimum steps that shall be taken during the removal of PCB containing ballasts from buildings. The plan of action shall be submitted by the Contractor for approval by the Contracting Officer, shall include the procedures he intends to follow and shall address these points:
 - a. A minimum of two layers of 6-mil plastic shall be placed on the floor beneath the disposal drums, the actual work area, and beneath stored disposal drums.
 - b. Pour a minimum three inch layer of oil absorbent in a DOT 17-H drum labeled as containing PCB.
 - c. Remove the light ballast from the light fixture. Wires exiting from the ballast shall be cut to less than three (3) inches long. Place the ballast in the drum. If the ballast is clearly labeled as "Non-PCB" or "No PCB", then the ballast can remain with the fixture and the fixture can be removed and disposed in accordance with standard procedures established in the contract for demolition debris.
 - d. If an unmarked ballast shows any sign of leaking, PCB resistant gloves shall be worn when handling the fixture. If the PCB status cannot be clearly determined from the label, the ballast shall be removed from the fixture and treated as hazardous waste.
 - e. Dispose of any light fixture which held an unlabeled leaking ballast as PCB contaminated material.
 - f. Alternately layer the ballasts and 2-inch layers of absorbent until the drum is full, carefully checking to assure that the last layer in the drum will be absorbent.
 - g. Place all used disposable protective clothing and plastic in the drum.
 - h. Label the drum with a DOT-OREM label with the following minimum information:
 - 1) Date materials were placed in the drum.
 - 2) Materials in the drum, i.e. "75 two-tube light ballasts.
 - 3) Name, address and phone number of the manufacturer of the generator, or owner of the light ballasts.
 - 4) Container identification number (ex; DC0093ZZ-001BAL)
 - 5) The weight of each container is to be recorded (in kilograms) and entered on the manifest.
 - i. Make arrangements, i.e. obtain EPA Generator Identification Number, make EPA notifications, fill out hazardous waste manifest, and incinerate or recycle the material in the drum(s) as PCB contaminated solid waste.
 - j. Transport the drum(s) and any other PCB materials to an EPA approved incinerator or ballast recycling facility.
 - k. The Contractor may transport the drum(s) and other PCB materials to an EPA approved ballast recycling facility which dismantles the ballast, segregates, and packages the PCB components of a ballast for incineration and then reclaims non-contaminated materials remaining after recycling are to be destroyed by incineration.
 3. STORAGE: Short term storage shall be in accordance with paragraph 1.5 herein.
 4. DISPOSAL: Immediately arrange for disposal of these items. Collect and dispose of all other PCB contaminated waste, rags, scrap, debris, bags, containers, equipment, and PCB-contaminated clothing in properly labeled PCB disposal drums. Waste PCB containing material shall be transported to an incineration site or ballast recycling facility. Incinerate all PCB-contaminated materials remaining after recycling. The Contractor shall provide the Project Officer with a copy of all manifests and continuation sheets resulting from the incineration of the PCB-containing waste. In some instances, a

temporary holding area can be established upon approval by the Contracting Officer for properly packaged PCB waste.

5. EMERGENCIES: Should the Contractor find a ballast that is leaking, the NIH Fire Department shall be called immediately (dial 911 on the main NIH campus) to assess the danger and to decontaminate any areas that may have been contaminated by the leakage.
- E. Oils - Confirmed NON-PCB Containing
1. ITEMS CONTAINING OILS: Internal combustion engines, bearings, refrigeration and heating units, all industrial machinery. Waste oil may also be in oily rags used for maintenance and possibly in soil contaminated by leaks or spills.
 2. RECYCLING: The Contractor shall make every effort to recycle the waste oil. If there is an economically feasible method to recycle the waste oil, the Contractor shall do so. This is required under the provisions of Executive Order 12780, dated October 31, 1991.
 3. STORAGE: The Contractor shall not mix waste oils with general waste. Waste oils shall be stored in appropriate containers that are clearly labeled to identify contents. Appropriate containers are those that will not deteriorate or react with the oil or allow the oil to leak into the environment during normal use, handling, and disposal procedures. Regulations for containing and labeling petroleum oil waste can be found in 49 CFR 172.101.
 4. DISPOSAL: All materials are to be properly disposed of or recycled by the Contractor in accordance with paragraph 1.5 herein.
- F. Solvents. The Contractor shall use non-toxic and non-flammable solvents whenever these alternative solvents are feasible.
1. KINDS OF SOLVENTS: Paint solvents, gasoline, and other toxic and ignitable solvents.
 2. STORAGE: The Contractor shall not mix solvents with general waste. Solvents shall be stored in NFPA-approved flammable liquid containers that are clearly labeled to identify contents. Appropriate containers are those that will not deteriorate or react with the solvents or allow the solvents to leak into the environment during normal use, handling, and disposal procedures. Regulations for containing and labeling solvents can be found in 49 CFR 172.101.
 3. DISPOSAL: All materials are to be properly disposed of by the Contractor in accordance with paragraph 1.5.
- G. Chlorofluorocarbons (CFCs)
1. USES OF CHLOROFLUOROCARBONS: Refrigerants, propellants in some aerosol cans.
 2. FEDERAL REGULATIONS FOR CFCs: As of July 1, 1992, the Clean Air Act makes it unlawful to vent CFCs to the atmosphere. During the repair, serving, or replacement of items such as cold rooms and chillers, CFCs shall be captured before they would otherwise be released to the atmosphere. When refrigerant systems are to be purged of air, venting of CFCs must be kept to a minimum.
 3. STORAGE: The Contractor shall store used refrigerant in containers meeting DOT regulations for refill services. Drums and other storage containers shall be checked for leaks prior to use. When known, these containers must be labeled with the specific CFC they contain.
 4. DISPOSAL OF AEROSOL CANS:
 - a. Aerosol cans that are empty in accordance with 40 CFR 261.7 provisions may be disposed of as conventional waste.
 - b. Aerosol cans that are not empty in accordance with 40 CFR 261.7 shall be considered hazardous material and shall be disposed of in conformance with hazardous material regulations.

- c. Aerosol cans having an unknown propellant shall be considered to contain CFCs under this specification and shall be treated as such.
- d. Disposal shall be in accordance with paragraph 1.4
- e. CFC CAPTURE: CFCs will be captured for reuse or recycling.

H. Gas Cylinders

- 1. KINDS OF GAS CYLINDERS: All compressed gases.
- 2. INSPECTION: Gas cylinders shall be visually checked for damage and shall be handled and moved in a manner that will not damage the valves or rupture the cylinders.
- 3. DISPOSAL: Gas cylinders shall be returned to the gas vendor either for refilling or proper disposal depending on the condition of the cylinder. In no case shall the Contractor discard these cylinders in waste dumpsters.

I. Scrap metal and containers (contaminated)

- 1. KINDS OF HAZARDOUS SCRAP METAL: Metal as well as plastic drums and containers that once held NIH regulated waste (including solvents, oils, paint); cadmium; duct systems that contain residue of hazardous materials, etc.
- 2. STORAGE: Spent metal or plastic drums and containers over 5-gallon capacity shall not be included with general waste. These items shall be stored and disposed of separately. The containers shall be clearly labeled to identify they are empty but also the label shall clearly identify the original contents to facilitate disposal. Other hazardous scrap metal shall be stored in containers that will not react with the material or allow the hazardous material to disperse into the environment.
- 3. DISPOSAL: All materials are to be properly disposed of by the Contractor in accordance with paragraph 1.5.

J. Scrap Metal (non-contaminated).

- 1. SOURCES OF NON-HAZARDOUS SCRAP METAL: Lab casework, metal wall partitions, sheet metal, nuts, bolts, and other metal construction waste.
- 2. RECYCLING: Every attempt shall be made to recycle these materials by the Contractor to conform with Executive Order 12780 dated October 31, 1991.
- 3. DISPOSAL: If material cannot be recycled, non-hazardous scrap metal shall be disposed of as construction debris in accordance with the contract.

K. Paint: The Contractor shall use non-toxic and non-flammable paints whenever these alternative paints are feasible. Use of lead containing paint is prohibited. For removal or disposal of lead-containing paint, see paragraph 3.3.A.

- 1. KINDS OF PAINTS: Toxic and/or flammable oil based paint.
- 2. STORAGE: The Contractor shall not mix hazardous paints with general waste. Toxic and/or flammable paints shall be stored in NFPA-approved flammable liquid containers/cabinets that are clearly labeled to identify contents. Appropriate containers are those that will not deteriorate or react with the paints or allow the paints to leak into the environment during normal use, handling, and disposal procedures. Regulations for containing and labeling paints/solvents can be found in 49 CFR 172.101. All paint containers shall be sealed securely to prevent spilling or dispersal during transport.
- 3. DISPOSAL: All materials are to be properly disposed of by the Contractor in accordance with paragraph 1.5.

3.4 PACKAGING, LABELING, AND MARKING OF HAZARDOUS WASTE MATERIALS FOR SHIPMENT OFF-SITE

A. General Shipping Requirements: Contractor shipments of hazardous and mixed wastes or materials shall be performed in accordance with the latest revision of all applicable EPA Title 40

CFR, NRC Title 10 CFR, and DOT Title 49 CFR requirements. As new packaging, labeling, and shipping regulations are promulgated and approved, the Contractor shall take the necessary measures to comply. The Contractor shall provide all documents and labels required for shipping wastes and materials.

3.5 SHIPPING TO TREATMENT, STORAGE, AND DISPOSAL FACILITIES AND OTHER OFF-SITE DESTINATIONS.

- A. Use of Hazardous Waste Manifest: Uniform Hazardous Waste Manifests (EPA Form 8700-22 or latest revision thereof) shall be properly completed by the Contractor for each waste shipment and shall list each transportation container including any non-hazardous waste or hazardous materials shipped. The Contractor shall use the manifest of the receiving state unless that state does not have one, in which case the Contractor shall use the generator state manifest. All manifest information shall be neatly typed and contain all information required by applicable federal, state, and local hazardous waste or materials regulations. The Contractor shall provide all data required for waste transportation, treatment, or disposal, and for completion of hazardous waste or material generator report as required by the regulatory agency of jurisdiction.
- B. DOT Emergency Response Information Requirements: The Contractor and the transporter must comply with the DOT Emergency Response Communication Standards applicable to the shipment of hazardous materials.
- C. Use of Permitted Hazardous Waste Facilities with Full RCRA Permit Status.
 - 1. Storage, treatment or disposal of hazardous wastes and materials shall be carried out only at facilities that have been issued final operating permits pursuant to RCRA and implemented by the hazardous waste regulatory authority(ies) of jurisdiction. Facilities that have applied for formal permits, and are in interim status, may be used by the Contractor only upon written certification by the Contractor that no fully permitted facilities are available and upon written approval of the interim status facility by the Contracting Officer. All Facility Permits must explicitly state that they are permitted for each particular waste and material that is to be treated, stored, or disposed of under this contract.
 - 2. All facilities used for interim treatment or final treatment and disposal of items under this contract shall have as a minimum an EPA and state approved interim status permit showing EPA hazardous waste numbers for each waste the facility is permitted to handle, as described in 40 CFR 261, Subparts C and D. Mere acceptance of the waste or material at a properly permitted TSDF does not meet the definition of final treatment and disposal under this contract.

3.6 FIELD QUALITY CONTROL

- A. Site Inspection and Stop Work Orders: While performing this work, the Contractor shall be subject to on-site inspection by NIH officials or NIH contracted inspection services. Work shall also be subject to inspection by OSHA and EPA inspectors and/or local building or health officials. If found to be in violation by any of these officials, the Contractor shall cease all work immediately. Until the violation is resolved, standby time required to resolve the violation shall be at the Contractor's expense.

3.7 CLEANUP AND DISPOSAL

- A. Permits and Notifications: Secure necessary permits in conjunction with hazardous material removal, hauling and disposition and provide timely notification of such actions, as may be required by federal, state, regional and local authorities. Notify the Regional Office of the United States Environmental Protection Agency and provide copies of the notification to the Contracting Officer 10 calendar days prior to the commencement of the work.
- B. Housekeeping: Essential elements of Hazardous Materials control include housekeeping and cleanup procedures. Maintain all surfaces within the work area free of accumulations of debris to prevent further dispersion and contamination. Given meticulous attention to restricting the spread of debris, keep waste from being distributed over the general area or to other areas in the building. The blowing down of the work area with compressed air is forbidden. Post appropriate hazard warning signs. In all possible instances workmen shall cleanup their own areas. Equip personnel engaged in cleaning up scrap and waste with appropriate personal protective clothing.
- C. Approval of Final Cleanup: The Contracting Officer will inspect the work area in concert with the NIH industrial hygienist for approval of the Hazardous Material cleanup. Visual observation of potential contamination, dust or debris is not permitted on any surface in or around the work area. The Contracting Officer will approve final cleaning and restoration of the work area.

END OF SECTION

SECTION 01 5639 - TEMPORARY TREE AND PLANT PROTECTION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes general protection and pruning of existing trees and plants that are affected by execution of the Work, whether temporary or permanent construction.
- B. Related Requirements:
 - 1. Section 312500 "Erosion Control" for temporary site fencing.

1.3 DEFINITIONS

- A. Caliper: Diameter of a trunk measured by a diameter tape at a height 6 inches above the ground for trees up to and including 4-inch size at this height and as measured at a height of 12 inches above the ground for trees larger than 4-inch size.
- B. Plant-Protection Zone: Area surrounding individual trees, groups of trees, shrubs, or other vegetation to be protected during construction and indicated on Drawings.
- C. Tree-Protection Zone: Area surrounding individual trees or groups of trees to be protected during construction and indicated on Drawings.
- D. Vegetation: Trees, shrubs, groundcovers, grass, and other plants.

1.4 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.
 - 1. Review methods and procedures related to temporary tree and plant protection including, but not limited to, the following:
 - a. Tree-service firm's personnel, and equipment needed to make progress and avoid delays.
 - b. Arborist's responsibilities.
 - c. Quality-control program.
 - d. Coordination of Work and equipment movement with the locations of protection zones.
 - e. Management and alterations to vegetated-roof systems.
 - f. Trenching by hand or with air spade within protection zones.

- g. Field quality control.

1.5 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Shop Drawings:
 - 1. Include plans, elevations, sections, and locations of protection-zone fencing and signage, showing relation of equipment-movement routes and material storage locations with protection zones.
 - 2. Detail fabrication and assembly of protection-zone fencing and signage.
 - 3. Indicate extent of trenching by hand or with air spade within protection zones.
- C. Samples: For each type of the following:
 - 1. Organic Mulch: 1-quart volume of organic mulch; in sealed plastic bags labeled with composition of materials by percentage of weight and source of mulch.
 - 2. Protection-Zone Fencing: Assembled Samples of manufacturer's standard size made from full-size components.
 - 3. Protection-Zone Signage: Full-size Samples of each size and text, ready for installation.
- D. Tree Pruning Schedule: Written schedule detailing scope and extent of pruning of trees to remain that interfere with or are affected by construction.
 - 1. Species and size of tree.
 - 2. Location on site plan. Include unique identifier for each.
 - 3. Reason for pruning.
 - 4. Description of pruning to be performed.
 - 5. Description of maintenance following pruning.

1.6 INFORMATIONAL SUBMITTALS

- A. Qualification Data: For arborist and tree service firm.
- B. Certification: From arborist, certifying that trees indicated to remain have been protected during construction according to recognized standards and that trees were promptly and properly treated and repaired when damaged.
- C. Maintenance Recommendations: From arborist, for care and protection of trees affected by construction during and after completing the Work.

- D. Existing Conditions: Documentation of existing trees and plantings indicated to remain, which establishes preconstruction conditions that might be misconstrued as damage caused by construction activities.
 - 1. Use sufficiently detailed photographs or video recordings.
 - 2. Include plans and notations to indicate specific wounds and damage conditions of each tree or other plants designated to remain.
- E. Quality-control program.

1.7 QUALITY ASSURANCE

- A. Arborist Qualifications: Certified Arborist-Municipal Specialist as certified by ISA and Registered Consulting Arborist as designated by ASCA.
- B. Tree Service Firm Qualifications: An experienced tree service firm that has successfully completed temporary tree and plant protection work similar to that required for this Project and that will assign an experienced, qualified arborist to Project site during execution of the Work.
- C. Quality-Control Program: Prepare a written program to systematically demonstrate the ability of personnel to properly follow procedures and handle materials and equipment during the Work without damaging trees and plantings. Include dimensioned diagrams for placement of protection zone fencing and signage, the arborist's and tree-service firm's responsibilities, instructions given to workers on the use and care of protection zones, and enforcement of requirements for protection zones.

1.8 FIELD CONDITIONS

- A. The following practices are prohibited within protection zones:
 - 1. Storage of construction materials, debris, or excavated material.
 - 2. Moving or parking vehicles or equipment.
 - 3. Foot traffic.
 - 4. Erection of sheds or structures.
 - 5. Impoundment of water.
 - 6. Excavation or other digging unless otherwise indicated.
 - 7. Attachment of signs to or wrapping materials around trees or plants unless otherwise indicated.
- B. Do not direct vehicle or equipment exhaust toward protection zones.
- C. Prohibit heat sources, flames, ignition sources, and smoking within or near protection zones and organic mulch.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Backfill Soil: Planting soil of suitable moisture content and granular texture for placing around tree; free of stones, roots, plants, sod, clods, clay lumps, pockets of coarse sand, concrete slurry, concrete layers or chunks, cement, plaster, building debris, and other extraneous materials harmful to plant growth.
 - 1. Planting Soil: Planting soil as specified in Section 329115 "Soil Preparation (Performance Specification)."
- B. Organic Mulch: Free from deleterious materials and suitable as a top dressing for trees and shrubs, consisting of one of the following:
 - 1. Type: Shredded hardwood.
 - 2. Size Range: 3 inches maximum, 1/2 inch minimum.
 - 3. Color: Natural.
- C. Protection-Zone Fencing: Fencing fixed in position and meeting the following requirements: Previously used materials may be used when approved by Architect.
 - 1. Chain-Link Protection-Zone Fencing: Galvanized-steel fencing fabricated from minimum 2-inch opening, 0.148-inch- diameter wire chain-link fabric; with pipe posts, minimum 2-3/8-inch- OD line posts, and 2-7/8-inch- OD corner and pull posts; with 1-5/8-inch- OD top rails and 0.177-inch- diameter bottom tension wire; with tie wires, hog ring ties, and other accessories for a complete fence system.
 - a. Height: 72 inches.
 - 2. Gates: Double-swing access gates matching material and appearance of fencing, to allow for maintenance activities within protection zones; leaf width 36 inches.
- D. Protection-Zone Signage: Shop-fabricated, rigid plastic or metal sheet with attachment holes prepunched and reinforced; legibly printed with nonfading lettering and as follows:
 - 1. Size and Text: Signs 24-inch high by 36-inch wide, unless otherwise indicated on Drawings.
 - a. Message: TREE PROTECTION ZONE. DO NOT ENTER.
 - 2. Lettering: 3-inch high minimum, black characters on white background.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Erosion and Sedimentation Control: Examine the site to verify that temporary erosion- and sedimentation-control measures are in place. Verify that flows of water redirected from construction areas or generated by construction activity do not enter or cross protection zones.
- B. Prepare written report, endorsed by arborist, listing conditions detrimental to tree and plant protection.

3.2 PREPARATION

- A. Locate and clearly identify trees, shrubs, and other vegetation to remain. Flag each tree trunk at 54 inches above the ground.
- B. Protect tree root systems from damage caused by runoff or spillage of noxious materials while mixing, placing, or storing construction materials. Protect root systems from ponding, eroding, or excessive wetting caused by dewatering operations.
- C. Tree-Protection Zones: Mulch areas inside tree-protection zones and other areas indicated. Do not exceed indicated thickness of mulch.
 - 1. Apply 2-inch uniform thickness of organic mulch unless otherwise indicated. Do not place mulch within 6 inches of tree trunks.

3.3 PROTECTION ZONES

- A. Protection-Zone Fencing: Install protection-zone fencing along edges of protection zones before materials or equipment are brought on the site and construction operations begin in a manner that will prevent people and animals from easily entering protected areas except by entrance gates. Construct fencing so as not to obstruct safe passage or visibility at vehicle intersections where fencing is located adjacent to pedestrian walkways or in close proximity to street intersections, drives, or other vehicular circulation.
 - 1. Chain-Link Fencing: Install to comply with ASTM F567 and with manufacturer's written instructions.
 - 2. Posts: Set or drive posts into ground one-third the total height of the fence without concrete footings. Where a post is located on existing paving or concrete to remain, provide appropriate means of post support acceptable to Architect.
 - 3. Access Gates: Install where indicated; adjust to operate smoothly, easily, and quietly; free of binding, warp, excessive deflection, distortion, nonalignment, misplacement, disruption, or malfunction throughout entire operational range. Confirm that latches and locks engage accurately and securely without forcing or binding.

- B. Protection-Zone Signage: Install protection-zone signage in visibly prominent locations in a manner approved by Architect. Install one sign spaced approximately every 50 feet on protection-zone fencing, but no fewer than four signs with each facing a different direction.
- C. Maintain protection zones free of weeds and trash.
- D. Maintain protection-zone fencing and signage in good condition as acceptable to Architect and remove when construction operations are complete and equipment has been removed from the site.
 - 1. Do not remove protection-zone fencing, even temporarily, to allow deliveries or equipment access through the protection zone.
 - 2. Temporary access is permitted subject to preapproval in writing by arborist if a root buffer effective against soil compaction is constructed as directed by arborist. Maintain root buffer so long as access is permitted.

3.4 EXCAVATION

- A. General: Excavate at edge of protection zones and for trenches indicated within protection zones according to requirements in Section 312000 "Earth Moving" unless otherwise indicated.
- B. Trenching within Protection Zones: Where utility trenches are required within protection zones, excavate under or around tree roots by hand or with air spade, or tunnel under the roots by drilling, auger boring, or pipe jacking. Do not cut main lateral tree roots or taproots; cut only smaller roots that interfere with installation of utilities. Cut roots as required for root pruning. If excavating by hand, use narrow-tine spading forks to comb soil and expose roots.
- C. Redirect roots in backfill areas where possible. If encountering large, main lateral roots, expose roots beyond excavation limits as required to bend and redirect them without breaking. If encountered immediately adjacent to location of new construction and redirection is not practical, cut roots approximately 3 inches back from new construction and as required for root pruning.
- D. Do not allow exposed roots to dry out before placing permanent backfill. Provide temporary earth cover or pack with peat moss and wrap with burlap. Water and maintain in a moist condition. Temporarily support and protect roots from damage until they are permanently relocated and covered with soil.

3.5 ROOT PRUNING

- A. Prune tree roots that are affected by temporary and permanent construction. Prune roots as follows:
 - 1. Cut roots manually by digging a trench and cutting exposed roots with sharp pruning instruments; do not break, tear, chop, or slant the cuts. Do not use a backhoe or other equipment that rips, tears, or pulls roots.
 - 2. Cut Ends: Coat cut ends of roots more than 1-1/2 inches in diameter with an emulsified asphalt or other coating formulated for use on damaged plant tissues and that is acceptable to arborist.
 - 3. Temporarily support and protect roots from damage until they are permanently redirected

- and covered with soil.
- 4. Cover exposed roots with burlap and water regularly.
- 5. Backfill as soon as possible according to requirements in Section 312000 "Earth Moving."
- B. Root Pruning at Edge of Protection Zone: Prune tree roots 12 inches outside of the protection zone by cleanly cutting all roots to the depth of the required excavation.
- C. Root Pruning within Protection Zone: Clear and excavate by hand or with air spade to the depth of the required excavation to minimize damage to tree root systems. If excavating by hand, use narrow-tine spading forks to comb soil to expose roots. Cleanly cut roots as close to excavation as possible.

3.6 CROWN PRUNING

- A. Prune branches that are affected by temporary and permanent construction. Prune branches as directed by arborist.
 - 1. Prune to remove only injured, broken, dying, or dead branches unless otherwise indicated. Do not prune for shape unless otherwise indicated.
 - 2. Do not remove or reduce living branches to compensate for root loss caused by damaging or cutting root system.
 - 3. Pruning Standards: Prune trees according to ANSI A300 (Part 1) and as indicated on Drawings.
 - a. Type of Pruning: Cleaning and thinning where indicated.
- B. Unless otherwise directed by arborist and acceptable to Architect, do not cut tree leaders.
- C. Cut branches with sharp pruning instruments; do not break or chop.
- D. Do not paint or apply sealants to wounds.
- E. Provide subsequent maintenance pruning during Contract period as recommended by arborist.
- F. Chip removed branches and dispose of off-site.

3.7 REGRADING

- A. Lowering Grade: Where new finish grade is indicated below existing grade around trees, slope grade beyond the protection zone. Maintain existing grades within the protection zone.
- B. Lowering Grade within Protection Zone: Where new finish grade is indicated below existing grade around trees, slope grade away from trees as recommended by arborist unless otherwise indicated.
 - 1. Root Pruning: Prune tree roots exposed by lowering the grade. Do not cut main lateral roots or taproots; cut only smaller roots. Cut roots as required for root pruning.
- C. Raising Grade: Where new finish grade is indicated above existing grade around trees, slope

grade beyond the protection zone. Maintain existing grades within the protection zone.

- D. Minor Fill within Protection Zone: Where existing grade is 2 inches or less below elevation of finish grade, fill with backfill soil. Place backfill soil in a single uncompacted layer and hand grade to required finish elevations.

3.8 FIELD QUALITY CONTROL

- A. Inspections: Engage a qualified arborist to direct plant-protection measures in the vicinity of trees, shrubs, and other vegetation indicated to remain and to prepare inspection reports.

3.9 REPAIR AND REPLACEMENT

- A. General: Repair or replace trees, shrubs, and other vegetation indicated to remain or to be relocated that are damaged by construction operations, in a manner approved by Architect.

1. Submit details of proposed pruning and repairs.
2. Perform repairs of damaged trunks, branches, and roots within 24 hours according to arborist's written instructions.
3. Replace trees and other plants that cannot be repaired and restored to full-growth status, as determined by Architect.

- B. Trees: Remove and replace trees indicated to remain that are more than 25 percent dead or in an unhealthy condition before the end of the corrections period or are damaged during construction operations that Architect determines are incapable of restoring to normal growth pattern.

1. Small Trees: Provide new trees of same size and species as those being replaced for each tree that measures 6 inches or smaller in caliper size.
2. Large Trees: Provide two new tree(s) of 6-inch caliper size for each tree being replaced that measures more than 6 inches in caliper size.

- a. Species: As selected by Architect.

3. Plant and maintain new trees as specified in Section 329300 "Plants."

- C. Excess Mulch: Rake mulched area within protection zones, being careful not to injure roots. Rake to loosen and remove mulch that exceeds a 4-inch uniform thickness to remain.

- D. Soil Aeration: Where directed by Architect, aerate surface soil compacted during construction. Aerate 10 feet beyond drip line and no closer than 36 inches to tree trunk. Drill 2-inch-diameter holes a minimum of 12 inches deep at 24 inches o.c. Backfill holes with an equal mix of augered soil and sand.

3.10 DISPOSAL OF SURPLUS AND WASTE MATERIALS

- A. Disposal: Remove excess excavated material, displaced trees, trash, and debris and legally dispose of them off Owner's property.

END OF SECTION 01 5639

SECTION 01 5950 - SAFETY AND HEALTH

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Inclusive Nature of the Requirements of this Section: All Work performed under this contract is subject to the requirements of this Section.
- B. Safety and Health Program:
 - 1. The Contractor shall develop and implement an effective safety and health program in accordance with Occupational Safety and Health Administration (OSHA) Regulations and the US Army Corps of Engineers Manual 385-1-1 (EM 385-1-1).
 - a. EM-385-1-1 Waiver: The Contractor may request a waiver from the EM 385-1-1 requirements prior to commencing work activities. The following process shall be used when requesting a waiver or a variance:
 - 1) Specify the exact requirement from which relief is being sought, providing the exact citation and quoting the requirement in question. A separate request must be made for each requirement;
 - 2) Provide statement as to whether a waiver (total elimination of the requirement) or a variance (retaining the basic requirement, but doing it differently) is being sought;
 - 3) Provide details as to why it is not possible or practical to comply with the requirement;
 - 4) Explain method the Contractor suggests to use in lieu of the existing requirement and how it provides protection equal to or greater than the requirement under waiver review. The burden of proof rests with the requesting Contractor.
 - 2. Each contract employee is responsible for complying with applicable safety and occupational health requirements, wearing prescribed safety and health equipment, reporting unsafe conditions/activities, and avoid actions and conditions that may result in an accident.
 - 3. The Contractor will not begin work authorized under this contract, except for authorized preliminary activities (i.e. mobilization), without first submitting for review a Site Specific Accident Prevention Plan to the NIH Project Officer and ORF Safety Officer (safety@nih.gov) that meets the requirements of EM 385-1-1.
- C. Contractor's Responsibility:
 - 1. The Contractor assumes full and sole responsibility for and shall comply with all laws, regulations ordinances, and governmental orders pertaining to employee worksite safety in the performance of this Contract. The Contractor shall be responsible for employing appropriate safety measures and taking all other actions reasonably necessary to protect the life, health, and safety of the public and to protect adjacent and NIH-owned property in connection with the performance of the scope of work.
 - 2. The Contractor shall be solely and completely responsible for the safety of all workers, NIH employees, public and NIH property in performance of the scope of work. This

requirement shall apply at all times while work is being performed under this contract, and is not limited to only normal working hours. Nothing the NIH may do, or fail to do, with respect to safety in the performance of the scope of work shall relieve Contractor of this responsibility.

D. General Contractor Obligations:

1. The Contractor is responsible for accident prevention and worksite safety. This responsibility cannot be delegated to Subcontractors, suppliers, the NIH, or other persons. To this end, the Contractor shall:
 - a. Promote a safe and healthy work environment.
 - b. Provide a Site-Specific Accident Prevention Program.
 - c. Ensure employees are adequately trained in occupational safety and health topics relevant to the activities to be performed under this contract. Maintain documentation of the employee training, to include the date and subject taught.
 - d. Instruct all employees of safe work methods and practices when assigning work.
 - e. Ensure that employees have and use the proper protective equipment and tools for the job.
 - f. Ensure that all heavy equipment operators (i.e. cranes, loaders and forklifts) are properly qualified and trained on the specific piece of equipment in use.
 - g. Plan and execute all work to comply with the stated objectives and safety requirements contained in the contract provisions, Federal, State, local laws and regulations, and industry standards.
 - h. Cooperate fully with the NIH, and its Representatives in connection with all matters pertaining to safety.
 - i. Conduct an orientation training program for new employees that includes at a minimum, a review of:
 - 1) Potential hazards in the work areas
 - 2) Required personal protective equipment and apparel
 - 3) Methods to mitigate hazards as appropriate
 - 4) Emergency response relevant to the area
2. Site Inspections:
 - a. The Contractor Site Safety and Health Officer shall perform and document daily inspections of the project site, assess and immediately correct any substandard safety conditions or practices, including those of any Subcontractor.
 - b. The Contractor shall specifically respond in writing to any substandard safety conditions or practices identified by the NIH.
 - c. Daily inspection records shall be maintained at the project site and be made available upon request by the NIH Project Officer or ORF Safety Officer.
3. Safety meetings shall be conducted to review past activities, plan for new or changed operations, review pertinent aspects of appropriate Activity Hazard Analysis, establish safe working procedures for anticipated hazards, and provide pertinent safety and health training and motivation.
 - a. Meetings shall be conducted at least once a month for all supervisors on the project location and at least once a week for all workers by supervisors or foremen.
 - b. Meetings shall be documented, including the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Documentation shall be maintained and copies furnished to the NIH on request.
4. Injuries and Illnesses:
 - a. Ensure prompt medical treatment is administered to any injured or ill employee.
 - b. During business hours (7:30 am -5:00 pm, M-F) contractor injuries and illnesses should be treated by the Occupational Medical Service (Building 10, 6th Room 6C306, phone 301-496-4411).
 - c. Injuries and illnesses other than first-aid treatment should be administered by third party provider.
 - d. In all cases, an injury or illness resulting from work under this contract shall be

reported to the Occupational Medical Service, the NIH Project Officer and ORF Safety Officer within 24-hours of the incident

- e. Undertake a complete investigation of all accidents, injuries, illnesses and implement corrective action to prevent a recurrence.
- 5. Communication of Requirements to Subcontractors and Others: Ensure that all of its Subcontractors, suppliers delivering materials or services to the worksite, etc., are provided with a copy of this specification and are informed of their obligations regarding worksite safety under this requirement.
- 6. Compliance with Regulations and Applicable Reference Standards: Comply with all applicable OSHA regulations and the US Army Corps of Engineers Manual (EM 385-1-1).

E. NIH Rights

- 1. Inspections and Investigations:
 - a. The NIH Project Officer may, in any reasonable manner, observe and inspect the Contractor's safety and accident prevention procedures for all activities and personnel working at the construction sites, including the Contractor, Subcontractors, visitors, and materials or equipment suppliers. This specifically includes, but is not limited to, the right to attend all safety meetings.
 - b. Upon request, the NIH Project Officer shall receive copies of any daily Construction Safety Inspection Reports completed by the Contractor or anyone performing work for, on behalf of or under the Contractor.
 - c. The NIH Project Officer shall receive written copies of accident or incident reports completed by the Contractor within 24-hours of occurrence. This report shall include but not be limited to those reports prepared pursuant to OSHA regulations.
 - d. The NIH Project Officer may, in any reasonable manner, observe or participate in any accident investigation conducted by the Contractor or anyone performing work for, on behalf of or under the Contractor. The NIH may also, at its sole discretion and in any reasonable manner, undertake its own accident investigation.
- 2. Corrective Actions / Stop Work
 - a. The NIH Project Officer shall have the right to direct the Contractor to correct unsafe working conditions, including taking corrective action when unsafe working conditions are observed (i.e. lack of good housekeeping practices, use of equipment in obviously poor condition, failure to adhere to statutory OSHA regulations, etc.).
 - b. The NIH Project Officer shall have the right to require the removal, from the work site, any person, property or equipment that, in the NIH's opinion, is deemed unsafe.
 - c. The NIH Project Officer shall have the right to instruct the Contractor to immediately cease any action and/or stop work (or any action thereof) when any conditions exist that, in the NIH's opinion, constitutes and imminent danger or could result in serious harm.
 - d. The NIH Project Officer shall have the right to suspend the work pending the completion of any accident/incident investigation, whether undertaken by Contractor, the NIH or others.
 - e. The Contractor is responsible for costs, expenses and other obligations paid or incurred, as a result of the Contractor or Subcontractor's noncompliance with Federal, State, or Local safety regulations; or failure to comply with terms and conditions of this contract.
- 3. NIH Action / Inaction
 - a. Nothing the NIH may do, or fail to do, with respect to safety in the performance of the work shall relieve the Contractor of its responsibility to comply strictly with this Contract and all standards referenced in this document.

- F. References: In addition to publications referenced in the Construction Contract Clauses, the following Code of Federal Regulations (CFR) publications designate and define hazardous

materials and conditions, and establish procedures for handling these materials and conditions. Omission of any publication in this section does not remove any obligation or legal requirement on the part of the contractor to comply with all legal requirements for the location of the work.

1. 29 CFR, Part 1910: Occupational Safety and Health Administration (OSHA) General Industry and Health Standards.
2. 29 CFR, Part 1926: OSHA Construction Industry Standards.
3. 40 CFR, Part 61: National Emission Standards for Hazardous Air Pollutants.
4. 40 CFR, Part 261: Environmental Protection Agency (EPA) Characteristics of Hazardous Waste.
5. 40 CFR, Part 761, EPA Polychlorinated Biphenyls (PCBs), Manufacturing, Processing, Distribution in Commerce and Use Prohibitions.
6. 40 CFR, Part 763: EPA Asbestos.
7. Federal Standard 313A: Material Safety Data Sheets, Preparation and the Submission of.
8. NIH DES Instruction 1340-7: Precautions and Procedures for Entering Manholes or Other Below Grade Confined Spaces.
9. NIH DCAB publication "Standards for Temporary Construction," March 1988.
10. U.S. Army Corps of Engineers Safety and Health Requirements Manual, (EM 385-1-1, 11/3/03)
11. National Fire Protection Association (NFPA) publication NFPA 70E
12. National Fire Protection Association (NFPA) publication NFPA 101

G. **Related Sections:** This specification section is related to any and all specification sections with explicit or implicit reference to cutting and patching. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:

1. Division 01 Section "Summary"
2. Division 01 Section "Work Restrictions"
3. Division 01 Section "Project Management and Coordination"
4. Division 01 Section "Construction Progress Documentation"
5. Division 01 Section "Photographic Documentation"
6. Division 01 Section "Submittal Procedures"
7. Division 01 Section "Quality Requirements"
8. Division 01 Section "References"
9. Division 01 Section "Construction Quality Control"
10. Division 01 Section "Temporary Facilities and Controls"
11. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation, and Disposal of NIH Controlled Material"
12. Division 01 Section "Safety and Health"
13. Division 01 Section "Product Requirements"
14. Division 01 Section "Execution Requirements"
15. Division 01 Section "Cutting and Patching"
16. Division 01 Section "Fire Prevention Precautions for Hot Work"
17. Division 01 Section "Closeout Procedures"
18. Division 01 Section "Demonstration and Training"
19. Division 01 Section "Selective Demolition"

H. **Hazardous Materials:** Some hazardous and toxic materials and substances are included in 29 CFR Part 1910, subparts H and Z, and in 29 CFR Part 1926 and others additionally defined in Federal Standard 313A. Commonly encountered hazardous materials include but are not limited to asbestos, PCBs, explosives and radioactive material.

1. PCBs may be contained in transformers, capacitors, voltage regulators, oil switches, mechanical insulation and other materials.

I. **Acquisition of Publications:** Referenced CFR publications may be purchased from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

1.3 DEFINITIONS

- A. ORF Safety Officer: An employee of the NIH, or designated representative who is responsible for the day-to-day management of the ORF Safety Program.

1.4 SUBMITTALS

- A. General: All submittals required under this Section must be provided to the NIH Project Officer.
 - 1. Copies: Provide copy of the submittals via email to safety@nih.gov.
 - 2. Software: Submittals shall be in either MS Word or Adobe Acrobat.
 - 3. Delays: Delays caused by failure to submit the required documentation shall not be considered a reason for extension of contract time or increase in costs to the Government.
- B. Contractor's Safety and Health Program: The contractor shall submit a Site Specific Safety and Health Program which includes a Site Specific Safety and Health Plan and a copy of the Company Safety and Health Plan to the Contracting Officer within 14 calendar days of the Notice to Proceed or before work commences on the project site, whichever is earlier.
 - 1. NIH will review the submitted Safety and Health Plan for general conformance to the requirements of the Contract Documents.
 - 2. NIH will not make any judgment as to the adequacy of the submitted plans with respect to federal, state, and local safety laws and safety regulations. If the plans are found to meet the requirements of the Contract Documents, NIH will acknowledge that the Contractor has provided copies of the plans and accept them as meeting the contract requirement for the Contractor to provide copies of his safety program.
 - 3. NIH will not approve any contractor Safety and Health Plan. Safety is the Contractor's responsibility under the current laws and governing regulation.
 - 4. NIH's review of, or comment on, the Contractor's Safety and Health Plan, including the Site Specific Accident Prevention Plan or safety program shall not, in any way, relieve the Contractor of any responsibility or liability for compliance with applicable regulations, NIH policies, or other requirements stipulated in the Contract.
 - 5. Contents of Site Specific Accident Prevention Plan:
 - a. In accordance with Appendix A, EM 3985-1-1.
 - 6. Additional content requirements for the Contractor's Safety and Health Program are included in this section.
 - 7. Safety Health Officer Qualifications: Submit the qualifications of the Safety and Health Officer as part of the Safety and Health Plan.
 - a. The NIH Project Officer will review the qualifications and a personal interview may be required. The NIH Project Officer may reject anyone deemed "Not Qualified".
- C. Hazardous Materials/Waste Management Plan: The contractor will coordinate the Safety and Health Program with the Hazardous Material/Waste Management Plan required by Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials." The Hazardous Material/Waste Management Plan is included in this section for reference only.
- D. HAZMAT disposal Certificates: The contractor will provide copies of all permits and certificates for any hazardous materials disposed of off site as required by Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials."
- E. Accident Reports. The Contractor must submit to the Contracting Officer and Project Officer a written report within three calendar days of any accident, fire, emergency, theft or incident in which any personal or property damage took place, regardless of any other notifications performed. Include a copy of each accident report that is submitted by the Contractor or

Subcontractors to their insurance carriers, within seven calendar days after the date of the accident.

- F. Gas Tests: Copies of the Daily Log of gas tests shall be provided to the Contracting Officer.
- G. Severe Storm Plan. The Contractor shall develop and submit to the Contracting Officer within 14 calendar days of receipt of the notice to proceed a plan describing contractor procedures for responding to threats of Severe Weather.

1.5 SAFETY AND HEALTH OFFICER REQUIREMENTS

- A. The Contractor's Site Safety and Health Officer(s) shall be categorized as a Full-time Safety Professional.
- B. Full-time Safety Professional qualifications include:
 - 1. Other Duties: He/She shall have no other duties.
 - 2. Experience: Minimum of five years progressive experience managing safety programs on large projects comparable to this contract in scope and complexity.
 - 3. Working Knowledge: Knowledgeable concerning all Federal, State, and Local regulations applicable to construction and industrial safety.
 - 4. Certification: "Competent Person" certification in safety disciplines related to the work performed and possess verifiable training.
 - a. Identify "Competent Persons" required by State and Federal safety standards for which they are not certified.
 - 5. Training:
 - a. Successfully completed the OSHA 500 Safety & Health Course.
 - 1) This requirement may be waived if individual possesses a safety and health degree or professional safety or industrial hygiene certification (i.e. CSP or CIH).
 - b. Trained in, and possess current certification for CPR and First Aid.
 - 6. Skills
 - a. Capable of performing accident investigations and developing a concise report.
 - b. Proficient in the development and presentation of "tool box" meetings and safety training.
- C. Responsibilities:
 - 1. The responsibility for safety lies with the Contractor. Each Contractor shall appoint an individual(s) responsible for safety on each contract. This individual(s) must be employed in a supervisory position, empowered by their employer to take corrective action; be present on the project while work is being performed; and spend the amount of time necessary to ensure the Contractor's compliance with safety requirements..
 - 2. A safety inspection shall be performed and documented for each shift worked by the Contractor's Site Safety and Health Officer.
 - 3. The Contractor Site Safety and Health Safety Officer shall be primarily responsible for ensuring Contractor's compliance with the safety requirements including but not limited to those provided in this Section. Without limiting the generality of the foregoing, the Contractor Site Safety and Health Officer shall:
 - a. Review all Subcontractor and sub-tier contractor's Site-Specific Accident Prevention Plan and Activity Hazard Analysis for compliance with applicable safety standards.
 - b. Perform or ensure that all Contractor, Subcontractors and sub-tier contractors' employees have received a site specific safety orientation prior to beginning work. Training will include discussion of the site specific Activity Hazards Analysis reports. This site specific orientation is in addition to the NIH's safety orientation.

- c. Perform daily safety inspections of the Contractor and Subcontractor's worksite to evaluate the project and to eliminate unsafe conditions and/or practices, and take the appropriate corrective action(s) as required.
- d. Immediately report all injuries to personnel, vehicle accidents, near miss incidents and property damage and insure immediate corrective action is taken. Assist in the preparation of all accident investigations and reporting procedures and ensure reports are submitted within 24-hours to both the NIH Project Officer and ORF Safety Officer (safety@nih.gov).
- e. Ensure appropriate, weekly safety meetings are held for all onsite employees, to include Subcontractors. Maintain documentation of training dates and times, topics discussed and attendees, with copies of records kept onsite.
- f. Be responsible for the control, availability, and use of necessary safety equipment, including personal protective equipment and apparel for the employees.

D. Substitution and Replacement:

- 1. Replacement Directed by NIH: If NIH, in its sole discretion, determines the Contractor's Site Safety and Health Safety Officer is not performing duties in accordance with this Section, the Contractor shall replace the Officer with an individual meeting the requirements of this Section.
- 2. Replacement Requested by Contractor: The Site Safety and Health Safety Officer may not be removed from this contract or replaced without the NIH's advanced written approval.
- 3. Substitution: Contractor shall notify the NIH Project Officer whenever a Safety and Health Safety Officer cannot be on duty while work is being performed and shall submit the name(s) and qualifications of the individual assigned to perform said duties during the Officer's absence.

1.6 PRECONSTRUCTION SAFETY MEETING

- A. Prior to commencing construction, representatives of the Contractor, including the Contractor's safety officer, general superintendent and one or more safety representatives, shall meet with the Contracting Officer for the purpose of reviewing Contract safety and health requirements.
 - 1. The Contractor's Safety and Health Program shall be reviewed, and implementation of safety and health provisions pertinent to the Work shall be discussed.
 - 2. The Contractor shall be prepared to discuss, in detail, the Contractor's site specific Safety and Health Plan including measures intended to control any unsafe or unhealthy conditions associated with the work to be performed under the contract.
 - 3. This meeting may be held in conjunction with the preconstruction conference, if so directed by the Contracting Officer. The conduct of this meeting is not contingent upon a general preconstruction meeting.
 - 4. The level of detail for the safety meeting is dependent upon the nature of the work and the potential inherent hazards.
 - 5. The Contractor shall advise the Contracting Officer of any special safety restrictions he has established so that NIH personnel can be notified of these restrictions.
 - 6. No later than 3 calendar days after the Preconstruction Safety Meeting, the Contractor shall distribute minutes of the meeting to each party present and to other concerned parties, including the Contracting Officer.

1.7 CONTRACTOR ORIENTATION

- A. NIH Safety Orientation: Prior to commencing construction, ensure that all Contractor and Subcontractor Supervisors at any tier have completed the NIH Safety Orientation to be held by the NIH at a time and location to be specified by the NIH, prior to commencing work. The time

expended and any associated costs such as travel time, parking, and other expenses are to be borne by the Contractor.

1. Orientation Registration: It is the responsibility of the Contractor and Subcontractor to contact the ORF Safety Officer to register each supervisor for orientation. Class registration must be scheduled at least 14-working days prior to the project start date, or the date that the supervisor is assigned to NIH
2. Orientation Schedule: Contact the ORF Safety Officer at: (301) 496-3353 for the orientation schedule

1.8 COMPLIANCE WITH REGULATIONS

- A. The work, including contact with or handling of hazardous materials, disturbance or dismantling of structures containing hazardous materials, and disposal of hazardous materials, shall comply with the applicable requirements of 29 CFR Parts 1910 and 1926, and 40 CFR Parts 61, 261, 761 and 763 and any other requirements contained in Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials."
- B. Work shall additionally comply with all applicable state and local safety and health regulations.
- C. In case of a conflict between applicable regulations, the more stringent requirements shall apply.
- D. Contractor Responsibility: The Contractor shall obtain all required permits for work to be performed. The Contractor shall assume full responsibility and liability for compliance with all applicable codes, standards and regulations pertaining to the health and safety of personnel during execution of the Work, and shall hold the Government harmless for any action on the Contractor's part, or that of the Contractor's employees or subcontractors, that results in illness, injury or death.
- E. Safety and Health Programs: The Contractor shall have written safety and health programs that, at a minimum, address and are in compliance with the following:
 1. Contractor's Construction Risk Assessment (CRA) and Interim Life Safety Measures (ISLM) plan developed for the project
 2. 29 CFR Parts 1910 and 1926.
 3. U.S. Army Corps of Engineers Safety and Health Requirements Manual, (EM 385-1-1, 11/3/03)
 4. Other Reference Standards as applicable.
 5. Permits, Inspections, Tests, and Reports: The required permits inspections, tests, and reports made by the Contractor, subcontractors, specially trained technicians, equipment manufacturers, and others as required, shall be at the Contractor's expense.

1.9 ACCIDENT PREVENTION

- A. The Contractor shall be responsible for correcting hazardous conditions and practices. When more than one Contractor is working within a job site, any project management personnel shall have the authority to prevent physical harm or significant property damage.
 1. Imminent Danger: If it is determined there is "Imminent Danger" the Contractor shall:
 - a. Take immediate action to remove/safeguard workers from the hazard and stabilize or stop work until corrective actions can be implemented to eliminate the hazard.
 - b. Immediately notify the NIH Project Officer and then the ORF Safety Officer.
 - c. Each worker shall immediately report any condition known, or suspected to be unsafe or unhealthy to the Contractor Site Safety Officer. If there is no resolution of

the concern at that level, the employee shall report the concern to the NIH Project Officer.

1.10 ONSITE FIRST AID

- A. At least one person shall be available at the work site at all times to render first aid. This person must have a valid certificate in first-aid training from the U.S. Bureau of Mines, the American Red Cross, or equivalent verifiable training program. A minimum ratio of one such qualified person for every 25 employees shall be maintained throughout the project. Additionally, the Contractor shall:
 - 1. Post emergency procedures which shall include telephone numbers and locations of facilities including, but not limited to, hospitals, physicians, police, fire and emergency medical services, in conspicuous locations at the job site and at all telephone locations.
 - 2. Provide, in a readily accessible location, first-aid supplies of sufficient size and number to handle common first-aid incidents.
 - 3. Regularly discuss actions to be taken during emergencies with the Contractor's supervisory personnel and at "tool box" safety meetings.

1.11 USE OF EXPLOSIVES

- A. Use of explosives shall be prohibited. Explosive actuated tools (ex: nailguns, etc) shall not be used or brought to the project site without prior written approval from the Project Officer. Such approval shall not relieve the Contractor of responsibility for injury to persons or for damage to property due to the use of such explosives.

1.12 FIRE PREVENTION

- A. The Contractor shall ensure that the fire prevention measures on-site are in accordance with OSHA, NIH Fire Department policies, and NFPA standards. Approved safety cans (approved or listed by a nationally recognized testing laboratory) shall be used for flammable and combustible liquids. Fire extinguishers shall be provided where required.
 - 1. Open Flame Devices: Prohibit the use of unapproved fuel-burning types of lanterns, torches, flares or other open-flame devices on NIH property.
 - 2. Hot Work Permit: Open flame welding and spark producing equipment and tasks require the Contractor to secure a "Hot Work Permit" from the NIH Fire Department. This can be obtained by calling the NIH Fire Marshall at (301) 496-0414

1.13 ELECTRICAL

- A. The Contractor shall appoint an individual responsible for the electrical safety of each work team to restrict entry to dangerous locations to those authorized by him jointly with NIH.
- B. Electrical – Safe Clearance Procedures
 - 1. Entry into High Voltage Areas: Work under this contract may require entry into High Voltage Areas.
 - 2. In the event entry is required, the Contractor is obligated to identify any High Voltage areas that may be involved in work under this contract. Before entry into a High Voltage work area the Contractor shall notify the NIH Project Officer.
 - 3. To prevent employee exposure or damage to electrical systems the Contractor shall exhaust all options and means to de-energize live electrical parts.

- C. Electrical arc welding equipment shall not be connected to the building power supply.

1.14 GAS PROTECTION

- A. The Contractor shall have one or more employees properly trained in operation of gas testing equipment and formally qualified as gas inspectors who shall be on duty during times workmen are in confined spaces. Their primary functions shall be to test for gas and operate testing equipment. Unless equipment of constant supervisory type with automatic alarm is employed, gas tests shall be made at least every 2 hours or more often when character of ground or experience indicates gas may be encountered. A gas test shall be made before workmen are permitted to enter the excavation after an idle period exceeding one-half hour.
- B. Readings shall be permanently recorded daily, indicating the concentration of gas, number and location of drilled piers, point of test, date, and time of test.
- C. Special requirements, coordination, and precautions will apply to areas that contain a hazardous atmosphere or, by virtue of their use or physical character, may be oxygen deficient. A check by NIH is required prior to entering areas that contain hazardous or oxygen-deficient atmospheres. Surveillance and monitoring shall be required in these types of work spaces by both Contractor and Government personnel.

1.15 MATERIAL DELIVERIES

- A. Whenever practicable, deliveries shall be made during regular NIH working hours and only when the Contractor's representative is available to receive them. Deliver material in approved containers and with properly licensed vehicles and operators. Open delivery vehicles are not permitted. Deliver materials in fully closed vehicles or tarp covered vehicles. All dump trucks shall be fully covered while in transport to and from the unloading site. All loads shall be securely fastened until unloading. Engines shall not be left running while vehicles are loading, unloading, waiting or parked. Do not block roads, walks, building entrances/exits, fire hydrants and standpipes, exterior tanks or building gas connections.

1.16 HAZARDOUS MATERIALS

- A. The Contractor shall bring to the attention of the Contracting Officer, or the Contracting Officer's authorized representative, any materials encountered during execution of the Work that the Contractor suspects are hazardous. The Contracting Officer shall determine whether the Contractor shall perform tests to determine if the material is hazardous. If the suspected material is leaking or spilling from its location the contractor is to contact the NIH Fire Department immediately.
- B. If the Contracting Officer directs the Contractor to perform tests on suspected hazardous materials and the material is found to be hazardous, or if the material is found to be hazardous without Contractor testing, and/or if additional protective measures are required, a change to the Contract price may be provided, subject to the applicable provisions of the Contract.
- C. Asbestos and Contractor Personnel Asbestos Training: Ensure that all workers have received and documented initial and annual Asbestos Awareness training prior to the start of work.

1.17 ADDITIONAL NIH SAFETY REQUIREMENTS

- A. Excavating & Trenching: Coordination with the NIH Project Officer shall be required for work performed.
- B. Construction activities that pose a potential risk of exposure to contaminated soil (such as excavations) shall be supervised by personnel who have a current 40-hour Hazardous Waste Supervisor's certification. These individuals shall be able to identify the potential need for upgrading the level of health and safety protection. All personnel working in direct contact with contaminated soil shall have a current 40-hour Hazardous Waste Certification and medical monitoring, in accordance with OSHA regulations. The Contractor is responsible for soil sampling and air monitoring to determine hazards and exposures to their employees.
- C. Individuals who operate hoisting equipment shall possess the appropriate certification (e.g. National Commission for the Certification of Crane Operators). A copy of the certification(s) shall be submitted to the NIH Project Officer prior to commencing work involving hoisting activities.
- D. Work Glove Policy: To reduce the possibility of hand injuries the Contractor shall implement a policy that requires the use of gloves for all personnel under their control. In certain circumstances, it shall be the responsibility of the Contractor to supply the proper gloves to perform the task.
- E. Chemical Exposure Plan: The Contractor shall submit a Chemical Exposure Plan for any products containing isocyanates, methylene chloride, lead, silica, hydrofluoric acid and processes involving floor sealers, traffic coatings, terrazzo sealers, specialty paints or any other chemical which can produce nuisance odors. The plan shall include employee exposure control methods, isolation methods to prevent spread of chemicals and odors outside the work area and safeguarding of the NIH employees and public. Material Safety Data Sheets for each chemical must be maintained on site.
- F. Protection of the Public: The Contractor shall submit a plan for the protection of the public on or adjacent to construction and demolition operations.
- G. The Contractor shall comply with all established NIH Standards, Codes and Regulations and obtain appropriate approvals from the NIH Divisions of Safety, Health, Security and Fire Protection
- H. The Contractor shall comply with all established NIH Standards, Codes and Regulations and obtain appropriate approvals from the NIH Divisions of Safety, Health, Security and Fire Protection.
- I. The Contractor shall additionally comply with all requirements contained in the U.S. Army Corps of Engineers Safety and Health Requirements Manual, (EM 385-1-1, 11/3/03)
- J. No work shall be performed in any area occupied by the public or NIH employees unless approved by the Project Officer.
- K. In the event of an emergency in a construction site that has been secured with a chain, the NIH Fire Department will cut the chain to gain entry. The Contractor shall be responsible for a new chain.
- L. Accident Treatment and Records: The Contractor shall post emergency first aid information and NIH emergency Telephone Numbers at the work site.

- M. NIH Emergency Telephone Numbers: Emergency response for medical and other emergencies will be provided by the Government. The Division of Public Safety has implemented 9-1-1 service on the NIH Campus. Calls requesting the assistance of the NIH Police or Fire Department (fires, ambulance, rescue or hazardous materials incidents) are received at a state of the art Emergency Communications Center where trained and certified Emergency Services Dispatchers ensure that the appropriate NIH Police and/or Fire Department personnel and equipment are promptly dispatched. Contractors shall instruct their employees to immediately report any fire even if the fire has been extinguished as well as any hazardous material spill, ambulance or rescue emergency.

ON-CAMPUS Emergency (Police/Fire/Rescue)	Dial 9-1-1
TTY Telephone Line	496-0063
NIH Police Non-Emergency	496-5685
NIH Fire Department Non-Emergency	496-2372
NIH Fire Prevention Section	496-0487
NIH Occupational Safety and Health Branch	496-2960
OFF-CAMPUS Emergency POLICE & FIRE/RESCUE	Dial 9-9-1-1
(For off-campus locations when using a telephone with an NIH exchange, e.g. 496, 402,435, 480 or 594)	

- N. Safety Clearance Procedures (Lockout/Tagout Tag System). The following Lockout/Tagout procedures will be followed unless more stringent procedures are required by current laws and regulations. The following procedures do not relieve the contractor from any more stringent requirements of the applicable codes and regulations. The safety of contractor and NIH personnel in areas impacted by the project remains the contractor's responsibility during construction.
1. Lockout/Tagout procedures shall be in accordance with OSHA 29 CFR 1910 and OSHA 29 CFR 1926. Lockout/Tagout procedures shall be included in the Contractor's Safety and Health Plan.
 2. Contractor shall ensure that each employee is familiar with and complies with these procedures.
 3. The Project Officer will, at the Contractor's request, apply lockout tagout tags and take other actions that, because of experience and knowledge, are known to be necessary to make the particular NIH equipment safe to work on.
 4. No person, regardless of position or authority, shall operate any switch, valve, or equipment that has an official lockout/tag out tag attached to it, nor shall such tag be removed except as provided in this section.
 5. No person shall work on any equipment that requires a lockout tagout tag unless he, his immediate supervisor, project leader, or a subordinate has in his possession the keys to the required lockout/tagout tags.
 6. When work is to be performed on electrical circuits, the work shall be performed only by qualified personnel following the required safety procedures.
 7. A Contractor's supervisor who is required to enter an area protected by a lockout/tagout tag will be considered a member of the protected group provided he notifies the holder of the tag stub each time he enters and departs from the protected area.
 8. Identification markings on building light and power distribution circuit breakers shall not be relied on for establishing safe work conditions.
 9. Before clearance will be given on any equipment other than electrical (generally referred to as mechanical apparatus), the apparatus, valves, or systems shall be secured in a passive condition with the appropriate vents, pins, and locks.
 10. Pressurized or vacuum systems shall be vented to relieve differential pressure completely.

11. Vent valves shall be lockout/tagout tagged open during the course of the work.
12. Where dangerous gas or fluid systems are involved, or in areas where the environment may be oxygen deficient, system or areas shall be purged, ventilated, or otherwise made safe prior to entry. See paragraph "Gas Protection."
13. Tag Placement
 - a. Lockout/tagout tags shall be completed in accordance with the regulations printed on the back thereof and attached to any device which, if operated, could cause an unsafe condition to exist.
 - b. If more than one group is to work on any circuit or equipment, the employee in charge of each group shall have a separate set of lockout/tagout tags completed and properly attached and locked.
 - c. When it is required that certain NIH owned and operated equipment be tagged, the NIH will review the characteristics of the various systems involved that affect the safety of the operations and the work to be done; take the necessary actions, including voltage and pressure checks, grounding, and venting, to make the system and equipment safe to work on; and apply such lockout/tagout tags to those switches, valves, vents, or other mechanical devices needed to preserve the safety provided. This operation is referred to as "Providing Safety Clearance."
14. Tag Removal
 - a. Lockout/tagout tags shall be removed only by those persons who initiated the lockout/tagout tag and who retain possession of the keys. Otherwise, lockout/tagout tags may be removed only with the authorization of the Project Officer.

1.18 PERSONNEL PROTECTIVE EQUIPMENT

- A. Special facilities, devices, equipment and similar items used by the Contractor in execution of the work shall comply with 29 CFR, Part 1910, Subpart I and other applicable regulations.

1.19 SAFETY PERFORMANCE

- A. If the Contractor experiences ongoing safety concerns such as a Lost Work Day Case or Recordable Incident Rate greater than the Bureau of Labor Statistics National Average for their respective industry, experiences repeated violations of safety and health rules and regulations or "Imminent Danger" situations; or fails to abate violations in a timely manner, the Contractor shall be subject to the following action at the NIH Project Officer's discretion:
 1. Removal and replacement of management personnel.
 2. Submitting a written Safety Recovery plan to the NIH Project Officer detailing what changes will be made to their safety program and a timeline as to when the changes will be implemented.
 3. Hiring an independent safety consultant who shall audit the Contractor's procedures and operations. The consultant shall compile a plan detailing what changes the Contractor shall implement. This report shall be submitted to the NIH Project Officer.
 4. Conduct a "Safety Stand Down" (suspend all work or any action thereof). Suspended work shall not be allowed to resume until the Contractor has completed the following actions for review and acceptance by the NIH Project Officer.
 - a. Hazardous conditions leading up to the Safety Stand Down shall be abated.
 - b. Training of such type and duration shall be conducted to educate personnel on the awareness of, identification of, and correction of hazards leading up to the stand down.

PART 2 - PRODUCTS

- 2.1 Safety and Health Programs: The Contractor shall submit copies of the written site specific project safety and health plan and emergency action procedures, as applicable to the work scope, as required as a result of the safety meeting, or as required by OSHA 29 CFR, Part 1926 including but not necessarily limited to the procedures and programs that support the requirements of the following:
- A. Designation of Safety Competent Person.
 - B. Occupational Noise Exposure.
 - C. Fall Protection.
 - D. Personnel Protective Equipment.
 - E. Control of Hazardous Energy.
 - F. Hazardous Materials Waste Management Plan (draft if final plan has not been accepted).
 - G. Electrical Safety Related Work Practices.
 - H. Lead.
 - I. Asbestos.
 - J. Respirator Protection.
 - K. Confined spaces.
 - L. Emergency evacuation and reporting.
- 2.2 Contractor's Safety and Health Plan: In addition to specific safety and health programs applicable to the project, Contractor shall submit to the Project Officer a copy of the firm's general Safety and Health Plan listing emergency procedures and contact persons with home addresses and telephone numbers.
- 2.3 Permits: If hazardous materials are disposed of off-site, submit copies of shipping manifests and permits from applicable federal, state or local authorities and disposal facilities, and submit certificates that the material has been disposed of in accordance with regulations as required in Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material".
- 2.4 Accident Reporting: The Contractor must submit to the Contracting Officer a written report within three calendar days of any accident, fire, emergency, theft or incident in which any personal or property damage took place, regardless of any other notifications performed. Include a copy of each accident report that is submitted by the Contractor or Subcontractors to their insurance carriers, within seven calendar days after the date of the accident.
- A. Gas Test Reports: Submit copies of daily log of gas tests

PART 3 - EXECUTION

3.1 EMERGENCY SUSPENSION OF WORK

- A. When the Contractor is notified by the Contracting Officer, or the Project Officer, of non-compliance with the safety or health provisions of the Contract, the Contractor shall immediately, unless otherwise instructed, correct the unsafe or unhealthy condition.
 - 1. If the Contractor fails to comply promptly, all or part of the work will be stopped by notice from the Contracting Officer.
 - 2. When, in the opinion of and by notice given by the Contracting Officer, satisfactory corrective action has been taken by the Contractor, work shall resume.
 - 3. The Contractor shall not be allowed any extension of time or compensation for damages in connection with a work stoppage for an unsafe or unhealthy condition.

3.2 PROTECTION OF PERSONNEL

- A. The Contractor shall take all necessary precautions to prevent injury to the public, occupants, or damage to property of others. The public and occupants includes all persons not employed by the Contractor or a subcontractor.
- B. Wherever practical, the work area shall be fenced, barricaded or otherwise blocked off from the public or occupants to prevent unauthorized entry into the work area.
 - 1. Provide traffic barricades and traffic control signage where construction activities occur in vehicular areas.
 - 2. Corridors, aisles, stairways, doors and exit ways shall not be obstructed or used in a manner to encroach upon routes of ingress or egress utilized by the public or occupants, or to present an unsafe or unhealthy condition to the public or occupants.
 - 3. Store, position and use equipment, tools, materials, scraps and trash in a manner that does not present a hazard to the public or occupants by accidental shifting, ignition or other hazardous activity.
 - 4. Store and transport refuse and debris in a manner to prevent unsafe and unhealthy conditions for the public and occupants. Cover refuse containers, and remove refuse on a frequent regular basis acceptable to the Contracting Officer. Use tarpaulins or other means to prevent loose transported materials from dropping from trucks.
- C. Alternate Precautions: When the nature of the work prevents isolation of the work area and the public or building occupants may be in or pass through, under or over the work area, alternate precautions such as the posting of signs, the use of signal persons, the erection of barricades or similar protection around particularly hazardous operations shall be used as appropriate.
- D. Public Thoroughfare: When work is to be performed over a public thoroughfare such as a sidewalk, roadway or other site access way, the thoroughfare shall be closed, if possible, or other precautions taken such as the installation of screens or barricades. When the exposure to heavy falling objects exists, as during the erection of building walls or during demolition, special protection of the type detailed in 29 CFR, Parts 1910 and 1926 shall be provided.

3.3 ENVIRONMENTAL PROTECTION

- A. Dispose of solid, liquid and gaseous contaminants in accordance with local codes, laws, ordinances and regulations.

- B. Comply with applicable federal, state and local noise control laws, ordinances and regulations, including but not limited to 29 CFR, Part 1910.95 and 29 CFR, Part 1926.52.

3.4 HAZARDOUS MATERIALS

- A. Requirements for hazardous Materials are contained in Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials."

END OF SECTION

SECTION 01 6000 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements governing the Contractor's selection of products for use in the Project, including manufacturers' standard warranties on products and special warranties.
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to product requirements and warranties. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Options"
 - 3. Division 01 Section "Work Restrictions"
 - 4. Division 01 Section "Project Management and Coordination"
 - 5. Division 01 Section "Construction Progress Documentation"
 - 6. Division 01 Section "Submittal Procedures"
 - 7. Division 01 Section "Quality Requirements"
 - 8. Division 01 Section "References"
 - 9. Division 01 Section "Construction Quality Control"
 - 10. Division 01 Section "Temporary Facilities and Controls"
 - 11. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation, and Disposal of NIH Controlled Material"
 - 12. Division 01 Section "Safety and Health"
 - 13. Division 01 Section "Execution Requirements"
 - 14. Division 01 Section "Cutting and Patching"
 - 15. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 16. Division 01 Section "Closeout Procedures"
 - 17. Division 01 Section "Selective Demolition"

1.3 DEFINITIONS

- A. The following definitions are not intended to change the meaning of other terms used in the Contract Documents, such as "specialties," "systems," "structure," "finishes," "accessories," and similar terms that are self-explanatory and have well-recognized meanings in the construction industry.
- B. "Products" are items purchased for incorporation in the work, whether purchased for the Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and other terms of similar intent.

1. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.

1.4 QUALITY REQUIREMENTS

- A. Source Limitations: To the fullest extent possible, provide products of the same kind from a single source. Equipment of the same function shall be manufactured by the same entity, unless otherwise indicated.
- B. Compatibility of Options: When the contractor is given the option of selecting between 2 or more products for use on the project, the product selected shall be compatible with products previously selected, even if previously selected products were also options. Total compatibility among options is not assured by limitations within the Contract Documents, but must be provided by the contractor. Compatibility is a basic general requirement of product/material selections.
- C. Manufacturers: Specific manufacturers and models of equipment and materials sited throughout the contract documents establish the desired performance and minimum quality of equipment and materials. The contractor may propose substitute manufacturers of equipment and materials unless specifically indicated otherwise. The contractor shall bear any cost related to the proposed substitution in all respects including but not limited to cost for establishing equality of the specified product and any coincidental construction costs directly related to the substitution. Rejection by the Project Officer does not allow for additional compensation.
 1. The terms desired performance and minimum quality shall be interpreted in the broadest sense possible and are intended to imply attributes including but not limited to size limitations, aesthetic intent, compatibility with utility services shown, availability for project, technical performance characteristics, durability, and ease of maintenance.
- D. Labels and nameplates: Except for required labels and operating data, do not attach or imprint manufacturer's or producer's nameplates or trademarks on surfaces of products that will be exposed to view in occupied spaces or on the exterior.
 1. Labels: Locate required product labels and stamps on concealed surfaces or, where required for observation after installation, on accessible surfaces that are not conspicuous.
 2. Equipment Nameplates: Provide a permanent nameplate on each item of service-connected or power-operated equipment. Locate nameplate on an easily accessible surface that is inconspicuous in occupied spaces. The nameplate shall contain the following information:
 - a. Name of product and manufacturer.
 - b. Model and serial numbers.
 - c. Operating data such as capacity, speed and ratings and similar essential operating data.
 3. Protection: Labels and nameplates shall be protected from defacement and other damage during the remainder of the work.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

- A. Deliver, store and handle products according to the manufacturer's recommendations, using means and methods that will prevent damage, deterioration and loss, including theft.
 - 1. Schedule product delivery to minimize long-term storage at the site and to prevent overcrowding of construction spaces.
 - 2. Deliveries shall be addressed to the contractor and be scheduled to arrive at the worksite during normal working hours, unless otherwise authorized in writing by the Contracting Officer. The contractor shall take receipt of all deliveries. NIH will not accept delivery of materials.
 - 3. Coordinate delivery with installation time to provide minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft and other losses.
 - 4. Deliver products to the site in an undamaged condition, in the manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting and installing.
 - 5. Inspect products upon delivery to ensure compliance with the Contract Documents and to ensure that products are undamaged and properly protected.
 - 6. Store products at the site in a manner that will facilitate inspection and measurement of quantity or counting of units.
 - 7. Store heavy materials away from the project structure in a manner that will not endanger the supporting construction.
 - 8. Store products subject to damage by the elements above ground, under cover in a weather-tight enclosure, with ventilation adequate to prevent condensation. Maintain temperature and humidity within range required by manufacturer's instructions.
 - 9. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
 - 10. Protect stored products from damage.
- B. If, in the opinion of the Contracting Officer, delivered or stored items are considered damaged or defective, they shall be replaced at no additional cost to NIH.
- C. Deliveries which require cranes shall be scheduled with the Project Officer a minimum of 15 working days prior to expected delivery.

PART 2 - PRODUCTS

2.1 PRODUCT COMPLIANCE AND REQUIREMENTS

- A. General: The compliance requirements, for individual products as indicated in the Contract Documents, are multiple in nature and may include generic, descriptive, performance, prescriptive, compliance with standards, conformance with graphic details and other similar forms and methods of indicating requirements, all of which must be complied with.
- B. Provide products complete with accessories, blocking, anchors, trim, finish, safety guards, devices and other items needed for a complete installation that provides the intended use and effect. Provide all ancillary materials and work shown, implied or required to fully integrate products into surrounding construction with fully finished appearance.
- C. Standard Products: Products shall be essentially the standard catalogued products of manufacturers regularly engaged in production of such products and shall be the manufacturer's latest standard design that complies with the specification requirements.

Equipment shall essentially duplicate items that have been in satisfactory commercial and industrial use at least two years, or more if otherwise specified, prior to award of the contract or in lieu thereof shall have been used and operated in a test installation which, in the opinion of the Project Officer, duplicate its field performance for the same period of time. The Project Officer reserves the right to require the Contractor to submit evidence to this effect for his approval. When two or more units of the same class of equipment are required, these units shall be the product of a single manufacturer; however, the component parts of the system need not be the products of the same manufacturer.

- D. Continued Availability: Products which, by nature of their application, are likely to be needed at a later date for maintenance and repair or replacement work, shall be current models for which replacement parts are available. Where specified and available, provide standard products of types that have been produced and used successfully in similar situations on other projects.
- E. Product Selection Procedures: Contractor's options for selecting products are limited by the Contract Document requirements including the Construction Contract Clauses, and governing regulations, and are not controlled by industry traditions or procedures experienced by the Contractor on previous construction projects. Required procedures include, but are not necessarily limited to, the following for various indicated methods of specifying:
 - 1. Where the Specifications or Drawings list manufacturers' names or product designations, the Contractor may provide any product that complies with the requirements, subject to the following conditions:
 - a. Manufacturers: Where a Specification or Drawings paragraph, subparagraph or note titled "Manufacturers" lists manufacturers' names, provide a compliant product by one of the manufacturers named, or request a Substitution of another compliant product by another manufacturer.
 - b. Available Manufacturers: Where a Specification or Drawings paragraph, subparagraph or note titled "Available Manufacturers" lists manufacturers' names, provide a compliant product by one of the manufacturers named or by another manufacturer.
 - c. Products: Where a Specification or Drawings paragraph, subparagraph or note titled "Products" lists product designations, provide one of the products designated, or request a Substitution of another compliant product.
 - d. Available Products: Where a Specification or Drawings paragraph, subparagraph or note titled "Available Products" lists product designations, provide one of the products designated or another compliant product.
 - e. Basis of Design Product: Where a Specification or Drawings paragraph, subparagraph or note titled "Basis of Design Product" includes a product designation, provide the product designated, or request a substitution of another compliant product by one of the other manufacturers named, if any, or by another manufacturer.
 - 2. Descriptive Requirements: Where Specifications or Drawings describe a product or assembly, listing exact characteristics required, provide a product or assembly that provides the characteristics and otherwise complies with Contract requirements.
 - 3. Performance Requirements: Where Specifications or Drawings require compliance with performance requirements, provide products that comply with these requirements and are recommended by the manufacturer for the application indicated. Manufacturer's recommendations may be contained in published product literature or by the manufacturer's certification of performance.
 - 4. Prescriptive Requirements: Where Specifications or Drawings require products that are produced using specified ingredients and components, including specific requirements for mixing, fabricating, curing, finishing, testing and similar operations in the manufacturing

process, provide products produced in accordance with the prescriptive requirements that otherwise comply with Contract requirements.

5. Codes, Standards and Regulations: Where Specifications or Drawings require compliance with an imposed code, standard or regulation, select a product that complies with the codes, standards or regulations specified.
6. Visual Matching: Where Specifications or Drawings require matching an established sample, or matching existing conditions, the Project Officer decision will be final on whether a proposed product matches satisfactorily.
7. Visual Selection: Where specified product requirements include the phrase "as selected from manufacturer's standard colors, patterns, textures" or a similar phrase, select a product and manufacturer that complies with other specified requirements. The Project Officer will select the color, pattern and texture from the manufacturer's product line.

PART 3 - EXECUTION (Not Applicable)

END OF SECTION

SECTION 01 7000 - EXECUTION REQUIREMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes certain general procedural requirements governing the Contractor's execution of the work, including, but not limited to the following:
 - 1. Laying out of the work.
 - 2. General installation of products.
 - 3. Correction of Defective Work
 - 4. Progress cleaning.
 - 5. Starting and adjusting.
 - 6. Protection of installed construction.
- B. Substitutions: Changes in methods of construction required by the Contract Documents proposed by the Contractor after award of the Contract shall comply with the procedures and conditions specified for Substitutions in the Construction Contract Clauses.
- C. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to execution requirements. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Photographic Documentation"
 - 6. Division 01 Section "Submittal Procedures"
 - 7. Division 01 Section "Quality Requirements"
 - 8. Division 01 Section "References"
 - 9. Division 01 Section "Construction Quality Control"
 - 10. Division 01 Section "Temporary Facilities and Controls"
 - 11. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 12. Division 01 Section "Safety and Health"
 - 13. Division 01 Section "Product Requirements"
 - 14. Division 01 Section "Cutting and Patching"
 - 15. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 16. Division 01 Section "Closeout Procedures"
 - 17. Division 01 Section "Demonstration and Training"
 - 18. Division 01 Section "Selective Demolition"

- D. Additional Requirements: Refer to the individual technical specification sections for additional execution requirements.

1.3 SUBMITTALS

- A. Land Surveying Submittals: Where requirement for laying out work with land surveyor is indicated on drawings, provide the following submittals:
 - 1. Surveyor Qualification Statement: Submit for review a statement attesting to previous experience from the land surveyor or professional engineer engaged to lay out the work. Include list of completed projects, with project names and addresses, and names and addresses of architects and owners.
 - 2. Certificates: Submit a certificate signed by the land surveyor or professional engineer certifying that the location and elevation of improvements comply with requirements indicated.
- B. Contractor Requests for Information (RFI): The Contractor shall submit Requests for Information (RFI) to the Project Officer. Upon receipt of an RFI, the Project Officer will review and coordinate a response. The response will be provided within 14 calendar days.
- C. Manufacturer's Field Services Submissions: Where product manufacturers are required by the individual sections of the Specifications to provide qualified personnel to observe conditions of surfaces or other project conditions, installation or workmanship, start up or adjustment of equipment, tests or other activities, and to initiate instructions when necessary, the following shall be submitted to the Project Officer:
 - 1. Qualifications: For approval, submit qualifications of observer at least 30 calendar days in advance of scheduled activities.
 - 2. Report: For information, submit report of activities and findings within 15 calendar days after the successful execution of the specified work. Include logs and other documented data where applicable.

1.4 QUALITY REQUIREMENTS

- A. Workmanship Standards: Initiate and maintain procedures to ensure personnel performing the work are skilled and knowledgeable in the methods and craftsmanship needed to produce the required levels of workmanship in the completed work. Remove and replace work that does not comply with workmanship specified and standards recognized in the construction industry for the applications indicated. Remove and replace work damaged or deteriorated by faulty workmanship or replacement of other work.
- B. Manufacturer's Instructions: Where installations include manufactured products, comply with manufacturer's applicable installation instructions and recommendations to the extent that those instructions and recommendations are more explicit or stringent than requirements contained in the Contract Documents.
- C. Specialists: Where the individual sections of the specifications require specialists to perform the work, comply with the requirements specified in the Construction Contract Clauses. The assignment of a specialist shall not relieve the contractor from complying with applicable regulations, union jurisdictional settlements or similar conventions, and the final responsibility for fulfillment of the entire requirements remains with the Contractor.

- D. Minimum Quality and Quantity: The quality level or quantity shown or specified shall be the minimum required for the work. Except as otherwise indicated, the actual work shall comply exactly with that minimum or may be superior to that minimum within limits acceptable to the Contracting Officer. Specified numeric values are either minimums or maximums as indicated or as appropriate for the context of the requirements.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION

3.1 LAYING OUT THE WORK

- A. Projects Requiring Land Survey Work (when indicated on drawings):
1. Identification: NIH will identify existing benchmarks, control points and property line corner stakes.
 2. Reference Points: Locate existing permanent benchmarks, control points and similar reference points before beginning work. Preserve and protect permanent benchmarks and control points during construction operations.
 - a. Do not change or relocate existing benchmarks or control points without prior written approval of the Project Officer. Report lost or destroyed permanent benchmarks or control points promptly. Report the need to relocate benchmarks or control points to the Project Officer before proceeding.
 - b. Replace lost or destroyed permanent benchmarks and control points promptly. Base replacements on the original control points.
 3. Before proceeding to lay out the work, verify layout information shown on the drawings, in relation to existing benchmarks. If discrepancies are discovered notify the Project Officer promptly.
 4. Establish and maintain a minimum of 2 permanent benchmarks on the site, referenced to data established by survey control points.
 - a. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.
 - b. Where the actual location or elevation of layout points cannot be marked, provide temporary reference points sufficient to locate the work.
 - c. Remove temporary reference points when no longer needed. Restore marked construction to its original condition.
 5. Record Log: Maintain a log of layout control work. Record deviations from required lines and levels. Make the log available for reference by the Project Officer.
- B. Existing Utilities and Equipment: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Additional underground utilities, not indicated on drawings, may be present and interfere with new work. Before beginning sitework, investigate and verify the existence and location of underground utilities and other construction including but not limited to the following:
1. Employ services of independent "Utility Finder" consultant to survey all proposed excavation areas.

- a. Use ground penetrating radar, x-ray, metal detection, and video-scoping as appropriate and in combination as required to identify all utilities in excavation area.
 - b. Mark locations of utilities using temporary markers durable enough to remain in place until excavation activities commence. Survey location of located utilities and add survey information to project Record Documents.
- 2. Hand excavate test pits to determine absolute location of all utilities that may be damaged by excavation activities.
- 3. Prior to construction, verify the location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping.
- 4. Furnish location data for work related to the project which must be performed by public utilities serving the Project site.
- C. Layout the work using accepted surveying practices.
 - 1. Establish benchmarks and markers to set lines and levels as needed to locate each element of the project.
 - 2. Establish dimensions within tolerances indicated.
 - 3. Inform installers of lines and levels to which they must conform.
 - 4. Check the location, level and plumb of every major element as the work progresses.
 - 5. Notify the Project Officer when deviations from required lines and levels exceed allowable tolerances.

3.2 EXAMINATION

- A. Examine applicable substrates and conditions under which the work will be performed before starting construction operations.
- B. If unsafe or otherwise unsatisfactory conditions are encountered take corrective action before proceeding.
- C. Require installer of each major unit of work to inspect substrate to receive work, and conditions under which work will be performed, and report in writing to the Contractor any unsuitable conditions. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner acceptable to the installer. Absence of written notice of unsuitable conditions will be relied upon to assure the Government that condition of substrate is acceptable.
- D. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.

3.3 PREPARATION

- A. Site Improvements: Locate and lay out site improvements, including pavements, stakes for grading, fill and topsoil placement, utility slopes, and invert elevations.
- B. Take field measurements as required to fit the work properly. Indicate field measurements on shop drawings. Recheck measurements before installing each product.
 - 1. Dimensions indicated on project drawings are approximate and intended to show general design intent. Contractor is responsible for all field measurements required to properly design, fabricate, fit and install products.

2. Survey existing floor slabs to determine actual deviation from "level". Identify conditions that exceed tolerance specified for new concrete work and propose corrective action plan for approval by Project Officer.
- C. Verify space requirements of items shown diagrammatically on drawings.
- D. Drawing Verification: Check all drawings furnished immediately upon their receipt and promptly notify the Project Officer of any discrepancies. Figures marked on drawings shall in general be followed in preference to scale measurements. Large scale drawings shall in general govern small scale drawings. The Contractor shall compare all drawings and verify the figures before laying out the work or ordering custom furniture, equipment or material and will be responsible for any errors which might have been avoided thereby. Dimensions on drawings shall be checked for accuracy by the Contractor.
- E. Expansion and Control Joints: Layout expansion and control joints at sufficient intervals as required or recommended by manufacturer for each product.

3.4 INSTALLATION

- A. Locate the work and components of the work accurately.
 1. Make vertical work plumb and horizontal work level.
 2. Where space is limited, install components to maximize space available for maintenance and to maximize ease of removal for replacement.
 3. Conceal pipes, ducts, and wiring in finished areas.
 4. Coordinate, layout and install pipes, conduits and other components as tight to underside of floor slabs above as possible except where this requirement compromises a zone system of above ceiling utility installation or compromises requirement to install components to maximize ease of maintenance.
- B. Install products at the time and under conditions that will produce satisfactory results.
 1. Maintain temperature, humidity and other weather controls for best performance.
 2. Isolate units of non-compatible work to prevent deterioration.
- C. Conduct construction operations so that no part of the work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.
- D. Tools and Equipment: Do not use tools or equipment that produce harmful levels of noise.
- E. Anchors and Fasteners: Provide anchors and fasteners as required to withstand stresses, vibration and physical distortion. Anchor each component securely in place, accurately located and aligned with other work.
 1. Allow for building movement, including thermal expansion and contraction.
- F. Mounting Heights: In no case shall exposed items be mounted at a height which violates the provisions of the Americans with Disabilities Act Accessibility Guidelines (ADAAG)/Uniform Federal Accessibility Standards (UFAS) for the application indicated. Refer questionable component mounting heights to the Project Officer for final decision.
- G. Joints: Make like joints of uniform width within contiguous surfaces. Where joint locations in exposed work are not indicated, arrange joints for a uniform and balanced visual effect.

- H. Control and Expansion Joints: Install control and expansion joints where shown on drawings, and where required or recommended by manufacturer for each product to accommodate thermal or other movement without damaging product.
- I. Adjust operating components for proper operation without binding.

3.5 CORRECTION OF DEFECTIVE WORK

- A. Repair or remove and replace defective construction. Restore damaged substrates and finishes.
- B. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and proper adjustment of operating equipment.
- C. Restore permanent facilities used during construction to their specified condition.
- D. Remove and replace damaged surfaces that are exposed to view if the surfaces cannot be repaired without visible evidence of repair.
- E. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired to operate properly.
- F. Remove and replace chipped, scratched or broken surfaces.
- G. Re-install equipment, piping, conduits and other devices where Project Officer determines installation tighter to slab above is feasible and such re-installation is warranted to provide improved future construction flexibility or improved personnel circulation.
- H. Relocate equipment, piping, conduits and other devices where Project Officer determines adjusted location would provide better access for maintenance.

3.6 PROGRESS CLEANING

- A. Maintain the project work areas free of waste material and debris.
- B. Effect daily cleaning activities designed to control contaminants in building spaces during construction and prior to occupancy.
 - 1. Protect porous materials from exposure to moisture, dust or other contaminants.
 - 2. Remove construction debris daily from construction site and render construction site "broom clean" at end of each work day.
 - 3. During construction activities with high dust generation, augment daily construction cleaning with twice weekly vacuuming using wet/dry vacuum cleaner. High dust generation activities include but are not limited to demolition, cutting and patching, masonry cutting and installation, and installation and finishing gypsum board.
- C. Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the work.
 - 1. Remove liquid spills promptly.

- D. Keep installed work clean. Clean installed surfaces in accordance with the recommendations of the manufacturer or fabricator of the product installed, using only the cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and will not damage exposed surfaces.
- E. Remove debris from concealed spaces prior to enclosing the space.
- F. Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at the time of project completion.
- G. Collection and Disposal of Waste: Collect waste from construction areas and elsewhere daily. Enforce requirements strictly and dispose of material lawfully.
 - 1. Comply with NFPA 241 for removal of combustible waste material and debris.
 - 2. Do not hold waste materials more than 7 days during periods when the ambient temperature remains continuously less than 80 deg F or for more than 3 days when the temperature exceeds or is expected to rise above 80 deg F.
 - 3. Handle and properly containerize hazardous, dangerous or unsanitary waste materials separately from other waste. Coordinate with Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material".

3.7 STARTING AND ADJUSTING

- A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.
- B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.
- C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.
- D. Manufacturer's Field Service: If a factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Division 01 Section "Quality Requirements."

3.8 PROTECTION OF INSTALLED CONSTRUCTION

- A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.
- B. Comply with manufacturer's written instructions for temperature and relative humidity.

END OF SECTION

SECTION 01 7310 - CUTTING AND PATCHING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes procedural requirements for cutting and patching.
- B. Definition: Cutting and patching includes cutting into existing construction to provide for the installation or performance of other work and subsequent fitting and repair required to restore surfaces to their original condition.
- C. Refer to other sections for other requirements and limitations applicable to cutting and patching individual parts of the work.
- D. Coordinate cutting and patching with demolition requirements specified in Division 02 Section "Selective Structure Demolition."
- E. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to cutting and patching. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Photographic Documentation"
 - 6. Division 01 Section "Submittal Procedures"
 - 7. Division 01 Section "Quality Requirements"
 - 8. Division 01 Section "References"
 - 9. Division 01 Section "Construction Quality Control"
 - 10. Division 01 Section "Temporary Facilities and Controls"
 - 11. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 12. Division 01 Section "Safety and Health"
 - 13. Division 01 Section "Product Requirements"
 - 14. Division 01 Section "Execution Requirements"
 - 15. Division 01 Section "Fire Prevention Precautions for Hot Work"
 - 16. Division 01 Section "Closeout Procedures"
 - 17. Division 01 Section "Selective Demolition"

1.3 SUBMITTALS

- A. Cutting and Patching Plan: Submit a written plan to the Project Officer through the Quality Control manager describing procedures at least 21 calendar days in advance of the time cutting and patching will initially be performed.
1. Include the following information, as applicable:
 - a. Description of the extent of cutting and patching required. Show how it will be performed and indicate why it cannot be avoided.
 - b. Description of the anticipated results in terms of changes to existing construction. Include changes to structural elements and operating components as well as changes in appearance and other significant visual elements.
 - c. List of products to be used and entities that will perform work.
 - d. Dates and hours of operation when cutting and patching will be performed.
 - e. List of utilities that cutting and patching procedures will disturb or affect. List utilities that will be relocated and those that will be temporarily out-of-service. Indicate how long service will be disrupted.
 - f. Compatibility and cohesion characteristics of patching compounds with adjacent materials.
 - g. Details and engineering calculations showing integration of reinforcement with the original structure, where cutting and patching involves adding reinforcement to structural elements.
 2. Approval by the Project Officer to proceed with cutting and patching does not waive the right to later require complete removal and replacement of unsatisfactory work.
 3. Samples: Provide cutting and patching samples for the following items within 14 calendar days after notice to proceed in order that special reviews and coordination can be arranged with approval authorities.
 - a. Where cutting and patching samples are indicated as a requirement on Drawings.

1.4 QUALITY REQUIREMENTS

- A. Requirements for Structural Work: Do not cut and patch structural elements in a manner that would change their load-carrying capacity or load-deflection ratio.
1. The cutting and patching plan shall include but not be necessarily limited to work required at the following structural elements:
 - a. Foundation construction.
 - b. Bearing and retaining walls.
 - c. Structural concrete.
 - d. Structural steel.
 - e. Lintels.
 - f. Timber and primary wood framing.
 - g. Structural decking.
 - h. Stair systems.
 - i. Miscellaneous structural metals.
 - j. Exterior curtain-wall construction.
 - k. Equipment supports.
 - l. Piping, ductwork, vessels, and equipment.
 - m. Structural systems of other construction.
- B. Operational Limitations: Do not cut and patch operating elements, safety related systems, or related components in a manner that would result in reducing their capacity to perform as intended. Do not cut and patch operating elements, safety related systems or related components in a manner that would result in increased maintenance or decreased operational life or safety. Operating elements or safety related systems include but are not limited to the following:

1. Primary operational systems and equipment.
2. Air or smoke barriers.
3. Water, moisture, or vapor barriers.
4. Membranes and flashings.
5. Fire protection systems.
6. Noise and vibration control elements and systems.
7. Control systems.
8. Communication systems.
9. Conveying systems.
10. Electrical wiring systems.
11. Operating systems of other construction.

- C. Visual Requirements: Do not cut and patch construction exposed on the exterior or in occupied spaces in a manner that would, in the Project Officer's opinion, reduce the building's aesthetic qualities. Do not cut and patch construction in a manner that would result in visual evidence of cutting and patching. Remove and replace construction that was previously cut and patched in a visually unsatisfactory manner. When requested by the Contracting Officer, engage a Specialist who is specifically experienced in the work. Visual elements include but are not limited to the following:

1. Processed concrete finishes.
2. Stonework and stone masonry.
3. Ornamental metal.
4. Matched-veneer woodwork.
5. Preformed metal panels.
6. Firestopping.
7. Window wall system.
8. Stucco and ornamental plaster.
9. Acoustical ceilings.
10. Terrazzo.
11. Finished wood flooring.
12. Fluid-applied flooring.
13. Carpeting.
14. Aggregate wall coating.
15. Wall covering.
16. Mechanical system enclosures, cabinets, or covers.

1.5 EXISTING WARRANTIES

- A. Remove, replace, patch, and repair material and surfaces cut or damaged during cutting and patching operations by methods and with materials in such a manner as not to void any existing warranties.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Comply with requirements specified in other Sections of these Specifications.
- B. Use materials identical to existing materials. For exposed surfaces, use materials that visually match existing adjacent surfaces to the fullest extent possible.
1. If identical materials are unavailable or cannot be used, use materials that, when installed, will match the visual and functional performance of existing materials.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Before cutting, examine surfaces to be cut and patched and conditions under which cutting and patching is to be performed. If unsafe or unsatisfactory conditions are encountered, take corrective action before proceeding.
- B. Before proceeding with cutting and patching involving two or more trades, meet at the project site with the Project Officer and the entities providing or affected by the cutting and patching. Site meeting should be incorporated into the Three Phases of Quality Control process managed by the Contractor's Quality Control Manager. Review areas of potential interference and conflict. Coordinate procedures and resolve potential conflicts before proceeding.
- C. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.

3.2 PREPARATION

- A. Provide temporary support of work to be cut.
- B. Protect existing conditions during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of the Project that might be exposed during cutting and patching operations.
- C. Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.
- D. Where existing services are required to be removed, relocated, abandoned, bypass such services before cutting to avoid interruption of services to occupied areas.

3.3 PERFORMANCE

- A. Employ skilled workmen to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time and complete without delay.
- B. Cutting: Cut existing construction using methods least likely to damage elements retained and adjoining construction. Where possible, review proposed procedures with the original installer and comply with the original installer's recommendations.
 - 1. In general, use hand or small power tools designed for sawing or grinding, not for hammering and chopping.
 - 2. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
 - 3. To avoid marring existing finished surfaces, cut or drill from the exposed or finished side into concealed surfaces.
 - 4. Cut through concrete and masonry using a cutting machine, such as a Carborundum saw or a diamond-core drill.
 - 5. After utility services are bypassed, cut-off pipe or conduit in walls or partitions to be removed. Cap, valve or plug and seal the remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after by-passing and cutting.

- C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other work. Patch with durable seams that are as invisible as possible. Provide materials and comply with installation requirements specified in other Sections of these Specifications.
 - 1. Where feasible, inspect and test patched areas to demonstrate integrity of the installation.
 - 2. Restore exposed finishes of patched areas and extend finish restoration into adjoining construction in a manner that will eliminate evidence of patching and refinishing.
 - 3. Where removed walls or partitions extends one finished area into another finished area, patch and repair floor and wall surfaces to provide an even surface of uniform color, finish, texture, and appearance. Remove existing floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
 - 4. Where patching occurs in a smooth painted surface, extend final paint coat over entire unbroken surface that contains the patch after the area has received primer and other undercoats.
 - 5. Patch, repair or rehang existing ceilings as necessary to provide an even-plane surface of uniform appearance.
 - 6. Maintain integrity of fire barriers, vapor barriers and insulation.
- D. Perform cutting and patching work listed in Division 01 Section "Work Restrictions" during NIH Unoccupied Hours.

3.4 CLEANING

- A. Clean areas and spaces where cutting and patching are performed. Completely remove paint, mortar, oils, putty, and similar items.
- B. Thoroughly clean piping, conduit, and similar features before applying paint, restored pipe coverings, or other finishing materials.

END OF SECTION

SECTION 01 7320 - SELECTIVE DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes the following:
 - 1. Demolition and removal of selected portions of a building.
 - 2. Demolition and removal of selected site elements.
 - 3. Repair procedures for selective demolition operations.
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to selective demolition. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 1 Section "Summary"
 - 2. Division 1 Section "Work Restrictions"
 - 3. Division 1 Section "Project Management and Coordination"
 - 4. Division 1 Section "Construction Progress Documentation"
 - 5. Division 1 Section "Photographic Documentation"
 - 6. Division 1 Section "Submittal Procedures"
 - 7. Division 1 Section "Quality Requirements"
 - 8. Division 1 Section "References"
 - 9. Division 1 Section "Construction Quality Control"
 - 10. Division 1 Section "Temporary Facilities and Controls"
 - 11. Division 1 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Material"
 - 12. Division 1 Section "Temporary Traffic Controls"
 - 13. Division 1 Section "Safety and Health"
 - 14. Division 1 Section "Product Requirements"
 - 15. Division 1 Section "Execution Requirements"
 - 16. Division 1 Section "Cutting and Patching"
 - 17. Division 1 Section "Fire Precautions for Hot Work"
 - 18. Division 1 Section "Closeout Procedures"

1.3 DEFINITIONS

- A. Remove: Detach items from existing construction and legally dispose of them.
- B. Remove and Salvage: Detach items from existing construction and deliver them to NIH.
- C. Remove and Reinstall: Detach items from existing construction, prepare them for reuse, and reinstall them where indicated.
- D. Existing to Remain: Existing items of construction that are not to be removed.

1.4 MATERIALS OWNERSHIP

- A. Except for items or materials indicated to be salvaged, reinstalled or otherwise indicated to remain NIH property, demolished materials shall be removed from the site by the Contractor, with further disposition at Contractor's option.
- B. Historical items, relics, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, antiques, and other items of interest or value to NIH, which may be encountered during selective demolition, shall remain NIH property.
 - 1. Carefully remove and salvage each item or object in a manner to prevent damage, and deliver it promptly to NIH.
 - 2. Cooperate with the NIH archaeologist or historical adviser.

1.5 SUBMITTALS

- A. Qualification Data: Provide qualifications of persons or firms performing demolition which demonstrate their capabilities and experience. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other amplifying information. Qualification data must conform to that required in Division 1 Section "Quality Requirements" for Specialists or Professional Engineers as appropriate to the work.
- B. Proposed noise-control or dust-control measures: Submit statement or drawing to the Contracting Officer for approval at least 14 calendar days prior to the start of demolition that indicates the measures proposed for use, proposed locations, and proposed time frame for their operation. Identify options if proposed measures are later determined to be inadequate.
- C. If requirement is indicated on drawings or on construction solicitation documents, submit a proposed Demolition Plan to the Contracting Officer indicating the following:
 - 1. Detailed sequence of selective demolition work, with starting and ending dates for each activity.
 - 2. Interruption of utility services.
 - 3. Coordination for shutoff, capping, and continuation of utility services.
 - 4. Use of elevator and stairs.
 - 5. Locations of temporary partitions and means of egress.
 - 6. Procedures and safety precautions to be used during demolition.
 - 7. Trees and shrub protection.
 - 8. Silt and erosion control handling.
 - 9. Removal, transportation, and reclamation or disposal of removed materials.
 - 10. Types of demolition activities that can be expected to interfere with NIH operations.
 - 11. Procedures to ensure uninterrupted progress of NIH operations on-site.
 - 12. Coordination of continuing NIH occupancy of portions of existing building and of partial occupancy by NIH of completed work.
- D. Inventory: Items to be removed and salvaged.
- E. Photographs or Videotape: Before work begins, submit sufficiently detailed photographs showing pre-demolition existing conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by selective demolition operations. Requirements for photographs are located in Division 1 Section "Photographic Documentation."
- F. Landfill Records: Indicate receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes. Comply with all requirements of Division 1 Section "Use, Handling, Storage, Transporting, Accumulation and Disposal of NIH Controlled Materials."

1.6 QUALITY REQUIREMENTS

- A. Demolition Firm Qualifications: Firm shall be a specialist in demolition work of similar materials and extent to that indicated for this Project.
- B. Professional Engineer Qualifications: Comply with Division 1 Section "Quality Requirements."
- C. Regulatory Requirements: Comply with governing EPA notification regulations before starting selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction. Comply with all regulatory requirements of those agencies and organizations having jurisdiction.
- D. Standards: Comply with ANSI A10.6 and NFPA 241.
- E. Pre-Demolition Conference: Conduct conference at Project site to comply with requirements in Division 1 section "Project Management and Coordination." Review methods and procedures related to selective demolition including, but not limited to, the following:
 - 1. Inspect and discuss condition of construction to be selectively demolished.
 - 2. Review structural load limitations of existing structure.
 - 3. Review and finalize demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
 - 4. Review requirements of work performed by other trades that rely on substrates exposed by demolition operations.
- F. Demolition Plan: The Contractor shall prepare a detailed plan of the demolition work procedures and safety precautions to be used in the identification, demolition, trees and shrub protection, silt and erosion control handling, removal, transportation, and reclamation or disposal of removed materials. The plan shall be submitted to the Contracting Officer for review within 14 calendar days after receipt of the Notice to Proceed and at least 14 calendar days before the planned commencement of demolition activities.
 - 1. Review and acceptance by NIH of the Contractor's demolition plan will not relieve the Contractor of any responsibility regarding damage from any demolition activity.

1.7 PROJECT CONDITIONS

- A. NIH will occupy portions of the building immediately adjacent to the selective demolition area. Refer to Division 1 Section "Work Restrictions" for additional requirements.
 - 1. Conduct selective demolition so NIH operations will not be disrupted.
 - 2. Provide the Project Officer with not less than 72 hours notice prior to activities that will affect NIH operations. Include in the notification the expected demolition activities and expected level and duration of disturbance to NIH operations.
- B. Safe access to existing walkways, corridors and other adjacent occupied or used facilities must be maintained. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from the Contracting Officer.
- C. NIH will remove the existing items so indicated on the Drawings.
- D. The Contractor shall remove and salvage the existing items so indicated on the Drawings.
- E. The Contractor shall remove and reinstall the existing items so indicated on the Drawings.
- F. Hazardous Materials: Except where noted, hazardous materials are not expected to be encountered in the work. If any material suspected of containing hazardous materials is encountered, do not disturb the material, immediately notify the Contracting Officer and the

Project Officer. If suspected hazardous material is spilling out or leaking, notify the fire department immediately.

- G. On-site storage or sale of removed items or materials will not be permitted.
- H. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.
- I. Fire Protection: Maintain fire-protection services during selective demolition operations.
- J. Use of explosives is prohibited. Explosive actuated tools (ex: nailguns, etc) shall not be used or brought to the project site without prior written approval from the Project Officer. Such approval shall not relieve the Contractor of responsibility for injury to persons or for damage to property due to the use of such explosives.
- K. Contractor dumpsters will be permitted only if a location for the dumpster is shown on the contract drawings.

1.8 WARRANTIES

- A. Existing Special Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during selective demolition, by methods and with materials that do not void existing warranties. Verify existence of warranties with the Project Officer.

PART 2 - PRODUCTS

2.1 REPAIR MATERIALS

- A. Where available and appropriate for use, provide repair materials that are identical to existing materials.
- B. Where identical materials are unavailable or cannot be used for exposed surfaces, use materials that visually match existing adjacent surfaces to the fullest extent possible.
- C. Use materials whose installed performance equals or surpasses that of existing materials.
- D. Use fireproof materials for dust barriers and other temporary enclosures. See Division 1 Section "Temporary Facilities and Controls" for additional barrier requirements.
- E. Comply with material and installation requirements specified in individual Specification Sections.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verify that utilities to be removed have been disconnected and capped.
- B. Survey existing conditions and correlate with requirements indicated to determine extent of selective demolition required.
- C. Inventory and record the condition of items to be removed and reinstalled, and items to be removed and salvaged.

- D. When encountering unanticipated mechanical, electrical or structural elements that conflict with the intended function or design, investigate and measure the nature and extent of the conflict. Promptly submit a written report to the Project Officer.
- E. Engage a professional engineer to survey the condition of the building to determine whether removing any element might result in a structural deficiency or unplanned collapse of any portion of the structure or adjacent structures during selective demolition.
- F. Perform surveys as the selective demolition progresses to detect hazards resulting from the activities.
- G. Do not start any construction activities in an existing laboratory until the NIH Division of Safety has inspected the space and certified it to be safe for construction work by putting a Division of Safety sticker on the door.

3.2 UTILITY SERVICES

- A. Existing Utilities: Maintain services indicated to remain and protect them against damage during selective demolition operations.
- B. Do not interrupt existing utilities serving occupied or operating facilities, except when authorized in writing by the Project Officer. See Division 1 Section "Project Management and Coordination" for additional requirements associated with utility shutdowns.
 - 1. Provide temporary services during interruptions to existing utilities, as acceptable to governing authorities. See Division 1 Section "Temporary Facilities and Controls" for additional temporary utility information.
 - 2. Provide not less than 18 calendar days notice to the Project Officer if shutdown of service is required during changeover.
- C. Utility Requirements: Locate, identify, disconnect, and seal or cap off indicated utility services serving areas to be selectively demolished.
 - 1. NIH will arrange to shut off indicated utilities when requested by Contractor.
 - 2. Where utility services are required to be removed, relocated or abandoned, provide bypass connections to maintain continuity of service to other parts of the building before proceeding with selective demolition.
 - 3. Cut off pipe or conduit in walls or partitions to be removed. Cap, valve or plug and seal the remaining portion of pipe or conduit after bypassing.
 - 4. Do not start selective demolition work until utility disconnection and sealing have been completed and verified.
- D. Smoke Detectors: Request a utility outage for all smoke detectors where demolition work is planned. Do not start work until smoke detectors are shut off.

3.3 PREPARATION

- A. Dangerous Materials: If chemicals, gases, explosives, acids, flammable or other dangerous materials of unknown content or origin are found which are not shown on the drawings, contact the Project Officer immediately before proceeding with demolition.
- B. Pest Control: Employ a certified, licensed exterminator to treat building and to control rodents and vermin before and during selective demolition operations. See Division 1 Section "Temporary Facilities and Controls" for requirements to provide Integrated Pest Management on all NIH construction projects.

3.4 POLLUTION CONTROLS

- A. Dust Control: Use temporary enclosures and other suitable methods complying with governing environmental protection regulations to limit the spread of dust and dirt.
 - 1. Do not use water when it may damage existing construction or create hazardous or objectionable conditions, such as ice, flooding, pollution, or slippery conditions.
 - 2. Wet mop floors to eliminate trackable dirt, and wipe down walls and doors of demolition enclosure.
- B. Disposal: Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
 - 1. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in accordance with applicable safety regulations.
- C. Cleaning: Clean adjacent structures and site improvements of dust, dirt and debris caused by selective demolition operations. Return adjacent areas to condition existing before start of selective demolition.

3.5 SELECTIVE DEMOLITION

- A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete selective demolition within limitations of governing regulations and as follows:
 - 1. Proceed with selective demolition systematically. Conduct work in an order that avoids transporting removed items and debris through areas with completed selective demolition work, and that allows for removal of items before supports for those items are removed in another area.
 - 2. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage adjoining construction to remain. Where possible, use hand or small power tools designed for sawing or grinding, not for hammering and chopping, to minimize disturbance of adjacent surfaces. Contractor is to use the correct tool for the work and operate it in accordance with the manufacturers instructions. Temporarily cover openings to remain.
 - 3. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
 - 4. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations, and maintain adequate ventilation when using cutting torches. See Division 1 Section "Fire Prevention Precautions for Hot Work."
 - 5. Maintain adequate ventilation when using cutting torches.
 - 6. Remove decayed, vermin-infested and other dangerous or unsuitable materials, and promptly dispose of these materials off-site.
 - 7. Lower removed structural framing members to ground by method suitable to avoid free fall and to prevent ground impact or dust generation.
 - 8. Locate selective demolition equipment throughout the structure and remove debris and materials so as not to impose excessive loads on supporting walls, floors or framing.
 - 9. Dispose of demolished items and materials promptly.
 - 10. Return elements of construction and surfaces to remain to condition existing before start of selective demolition operations.
- B. Existing Facilities: Comply with all regulations for using and protecting elevators, stairs, walkways, loading docks, building entries and other building facilities during selective demolition operations. Coordinate with the Project Officer for building-specific requirements.

- C. Removed and Salvaged Items. Comply with the following:
 - 1. Clean salvaged items.
 - 2. Pack or crate items after cleaning. Identify contents of containers.
 - 3. Store items in a secure area until delivery to NIH.
 - 4. Delivery to NIH: Transport the materials and equipment to the area on-site designated by the Contracting Officer or indicated on the drawings.
 - 5. Protect items from damage during transport and storage.
 - D. Removed and Reinstalled Items. Comply with the following:
 - 1. Clean and repair items to functional condition adequate for intended reuse. Paint damaged or deteriorated painted surfaces of equipment to match new equipment.
 - 2. Pack or crate items after cleaning and repairing. Identify contents of containers.
 - 3. Protect items from damage during transport and storage.
 - 4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.
 - E. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by the Contracting Officer, items may be removed to a suitable, protected storage location during selective demolition and then cleaned and reinstalled in their original locations after selective demolition operations are complete.
 - F. Concrete: Demolish in small sections. Cut concrete to a depth of at least 3/4 inch ((19 mm)) at junctures with construction to remain, using power-driven saw. Dislodge concrete from reinforcement at perimeter of areas being demolished, cut reinforcement, and then remove remainder of concrete indicated for selective demolition. Neatly trim openings to dimensions indicated.
 - G. Concrete: Demolish in sections. Cut concrete full depth at junctures with construction to remain and at regular intervals, using power-driven saw, then remove concrete between saw cuts.
 - H. Concrete Slabs on Grade: Break up and remove concrete slabs on grade, unless otherwise shown to remain.
 - I. Masonry: Demolish in small sections. Cut masonry at junctures with construction to remain, using power-driven saw, then remove masonry between saw cuts.
 - J. Resilient Floor Coverings: Remove floor coverings and adhesive, and prepare substrate for new floor covering, according to recommendations of the Resilient Floor Covering Institute (RFCI).
 - K. Roofing: Remove no more existing roofing than can be covered in one day by new roofing. Refer to applicable Division 7 Section for new roofing requirements.
 - L. Air-Conditioning Equipment: Remove equipment without releasing refrigerants.
- 3.6 PATCHING AND REPAIRS
- A. General: Promptly repair damage to adjacent construction caused by selective demolition operations.
 - B. Patching: Comply with Division 1 Section "Cutting and Patching."

- C. Repairs: Where repairs to existing surfaces are required, patch to produce surfaces suitable for new materials.
 - 1. Completely fill holes and depressions in existing masonry walls to remain with an approved masonry patching material, applied according to the manufacturer's written recommendations.
- D. Finishes: Restore exposed finishes of patched areas and extend finish restoration into adjoining construction to remain in a manner that eliminates evidence of patching and refinishing.
- E. Floor and Wall Surfaces: Patch and repair floor and wall surfaces in each space where demolished walls or partitions result in extending one finished area into another. Provide a flush and even surface of uniform color, texture and appearance.
 - 1. Closely match texture and finish of existing adjacent surface.
 - 2. Patch with durable seams that are as invisible as possible. Comply with specified tolerances.
 - 3. Where patching occurs in a painted surface, apply primer and other specified undercoats. Apply specified intermediate paint coats over patch and apply final paint coat over entire unbroken surface containing patch. Provide additional coats until patch blends with adjacent surfaces.
 - 4. Remove existing floor and wall coverings and replace with new materials, if necessary to achieve uniform color and appearance.
 - 5. Where feasible, inspect and test patched areas to demonstrate integrity of the installation.
- F. Ceilings: Patch, repair or rehang existing ceilings as necessary to provide an even-plane surface of uniform appearance.

3.7 DISPOSAL OF DEMOLISHED MATERIALS

- A. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.
- B. Burning of any materials removed during demolition is not permitted on NIH property.
- C. Disposal: Transport demolished materials off NIH property and legally dispose of them.

3.8 SELECTIVE DEMOLITION SCHEDULE

- A. Existing items to remain, be removed and/or be salvaged are indicated on the drawings.

END OF SECTION

SECTION 01 7350 - FIRE PREVENTION PRECAUTIONS FOR HOT WORK

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section applies to safeguards to be observed in performing hot work, including welding, soldering, brazing and similar applications producing a spark, flame or heat.
- B. Very specific requirements exist for performing hot work on the NIH campus. Any deviations from the requirements outlined in this section must be thoroughly coordinated with the Project Officer and the NIH Division of Public Safety.
- C. This specification section is related to any and all specification sections with explicit or implicit reference to fire prevention and requirements for hot work including but not limited to the following sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Work Restrictions"
 - 3. Division 01 Section "Project Management and Coordination"
 - 4. Division 01 Section "Construction Progress Documentation"
 - 5. Division 01 Section "Submittal Procedures"
 - 6. Division 01 Section "Quality Requirements"
 - 7. Division 01 Section "References"
 - 8. Division 01 Section "Construction Quality Control"
 - 9. Division 01 Section "Temporary Facilities and Controls"
 - 10. Division 01 Section "Use, Handling, Storage, Transporting, Accumulation, and Disposal of NIH Controlled Material"
 - 11. Division 01 Section "Safety and Health"
 - 12. Division 01 Section "Product Requirements"
 - 13. Division 01 Section "Execution Requirements"
 - 14. Division 01 Section "Cutting and Patching"
 - 15. Division 01 Section "Closeout Procedures"
 - 16. Division 01 Section "Selective Demolition"
- D. Additional Requirements: Refer to the individual technical specification sections for additional requirements.

1.3 SAFETY PRECAUTIONS

- A. Prior to commencing hot work operations, a positive determination shall be made that it is impractical to conduct the hot work in a shop area or outside of the building. Coordinate suitable locations for hot equipment operations agreeable to the Project Officer and the NIH Division of Public Safety.

B. Hazardous Work Permit

1. The Contractor shall ensure that operations involving the use of open-flame, spark-producing tools, or electrical arc welding equipment are not conducted until a Hazardous Work Permit has been issued by the NIH Division of Public Safety. The Contractor shall call the NIH Division of Public Safety at 301-496-0414 to obtain a permit. The permit shall be conspicuously posted at the work site at all times.

1.4 NOTIFICATION

- A. The Contractor shall notify the Project Officer of the area of hot work operations for each day and of all subsequent changes that occur. A separate Hazardous Work Permit is required for each area in which hot work will be performed and for all subsequent changes that occur.

PART 2 - PRODUCTS - (Not Applicable)

PART 3 - EXECUTION

3.1 GENERAL

- A. The Contractor shall not start work until all precautions noted in the "Hazardous Work Permit" have been implemented.

3.2 SAFEGUARDS

- A. Hot work shall not be done in or near rooms or areas where flammable liquids or explosive atmospheres are present or thought to be present.
- B. Insofar as possible, the Contractor shall remove and keep the area free from all combustibles, including rubbish, paper and waste within a radius of 35 feet (10.66 m) from hot work operations.
 1. If combustible material cannot be removed, the Contractor shall furnish retardant blankets to cover such materials. At the direction of the Project Officer, floors, walls, and ceilings of combustible material shall be wetted thoroughly with water before, during, and after operations sufficiently to afford adequate protection. Where floors have been wet down, personnel operating arc welding or cutting equipment shall be protected from possible shock.
- C. The Contractor shall prevent the spread of sparks and particles of hot metal through openings or cracks in windows, doors, floors, walls, ceilings, roofs and ducts.
- D. Cylinders of gas used in hot work shall be placed a safe distance from the work. The Contractor shall provide hoses and equipment free of deterioration, malfunction and leaks. Suitable supports shall be provided to prevent accidental overturning of cylinders. All cylinder control valves shall be shut off while not in use.
- E. When hot work operations are completed or ended for the day, each location of the day's work shall be inspected by the Contractor at least 30 minutes after completion of operations to detect

for hidden or smoldering fires and to ensure that proper housekeeping is maintained. The Contractor shall cleanup the area of work at the end of each shift or workday.

- F. Where sprinkler protection exists, the sprinkler system shall be maintained without interruption while operations are being performed. If operations are performed close to automatic sprinkler heads, a wet rag may be used to shield the individual heads temporarily. The heads shall be inspected by the Contractor immediately after hot work operations cease, to ensure all materials have been removed from the heads and that the heads have not been damaged.
- G. Fully-charged and operable portable fire extinguishers that are appropriate for the type of fire shall be available at all times during hot work operations.
- H. If any of the above safeguards are not employed, or are violated, the Contracting Officer may, by written notice, stop the work until compliance is obtained. Such stoppage shall not relieve the Contractor from performing his work within the Contract period for the Contract price.

END OF SECTION

SECTION 01 7700 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:
 - 1. Substantial Completion requirements and inspection procedures
 - 2. Final Completion requirements and inspection procedures
 - 3. Warranties.
 - 4. Instruction of NIH personnel.
 - 5. Final cleaning.
- B. Closeout requirements for specific construction activities are included in the individual sections in Divisions 02 through 33.
- C. This specification section is related to any and all specification sections with explicit or implicit reference to closeout procedures. Specific submittal requirements of these related specification sections are not included in this section. Related sections include but are not limited to the following specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Project Management and Coordination"
 - 3. Division 01 Section "Construction Progress Documentation"
 - 4. Division 01 Section "Photographic Documentation"
 - 5. Division 01 Section "Submittal Procedures"
 - 6. Division 01 Section "Quality Requirements"
 - 7. Division 01 Section "Construction Quality Control"
 - 8. Division 01 Section "Temporary Facilities and Controls"
 - 9. Division 01 Section "Safety and Health"
 - 10. Division 01 Section "Product Requirements"
 - 11. Division 01 Section "Project Record Documents"
 - 12. Division 01 Section "Demonstration and Training"
- D. Substantial Completion is defined as that state when the Contractor has complied with the Contract requirements, except for minor deviations, and the project is sufficiently complete and capable of being occupied and used by NIH for the intended purpose. Achievement of Substantial Completion is determined by the Contracting Officer.

1.3 SUBSTANTIAL COMPLETION

- A. Preliminary Procedures: Before requesting inspection for determining the date of Substantial Completion, complete the following.

1. Provide supporting documentation for completion as indicated elsewhere in the Contract Documents and a statement showing an accounting of changes to the Contract Sum.
2. Submit copy of Contractor's punch list for each room or element of work, annotated by Contractor to indicate successful completion and inspection by Contractor of corrected work required by punch list.
3. Submit a list of incomplete items to the Project Officer. Indicate the value of incomplete construction, and reasons the work is not complete.
4. Obtain and submit any necessary releases enabling NIH unrestricted use of the project and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
5. Make final changeover of permanent locks. Use 8 bit Lockwood cylinders in the locks. Set the cylinders for blank keys and transmit two blank keys for each cylinder to the Project Officer. Advise the NIH user personnel of changeover in security provisions.
6. Complete startup testing of systems and instruction of NIH operation and maintenance personnel.
7. Discontinue and remove temporary facilities from the site, along with mockups, construction tools, and similar elements.
8. Submit Operation and Maintenance Manuals, final project photographs, and final surveys as specified.
9. Submit draft or final warranty documents for Contracting Officer review for approval.
10. Complete final clean-up requirements, including touch-up painting. Touch-up and otherwise repair and restore marred exposed finishes.
11. Record Documents (Record Document BIM, Record Prints, drawings, specifications, product data, and miscellaneous record documentation).
 - a. Initial submission shall be made to the Project Officer prior to the Pre-final Inspection.
 - b. NIH will review the submission and provide appropriate comments. If comments are significant the initial submission will be returned to the contractor for correction and re-submission incorporating the comments prior to the Final Inspection.
 - c. See Division 01 Section "Project Record Documentation" for additional requirements.
12. Provide all required submittals to the Contracting Officer.
13. Submit test/adjust/balance records.
14. Submit changeover information related to NIH occupancy, use, operation, and maintenance.

B. Inspection Procedures:

1. Substantial Completion Inspection corresponds to the Pre-Final Completion Inspection described in Division 01 Section "Construction Quality Control."
2. On receipt of a request for inspection, the Project Officer will either schedule the inspection or advise the Contractor of unfilled requirements. The Contracting Officer will prepare the Certificate of Substantial Completion following the inspection or advise the Contractor of construction that must be completed or corrected before the certificate will be issued.
3. The Project Officer will repeat the inspection when requested and when assured that the work is substantially complete.
4. Results of the completed inspection will form the basis of the requirements for final completion.

1.4 SUBMITTALS

- A. The following shall be submitted prior to final payment and closeout.
1. Manufacturer's cleaning instructions
 2. Posted instructions

3. All required submittals
4. Final project warranty documents reflecting changes directed by any comments from the Contracting Officer's review of draft documents.
5. Record BIM Model.

1.5 Project Closeout Work Plan. Contractor shall submit a Project Closeout Work Plan for each phase of occupancy to the Project Officer for approval at least twenty-one calendar days prior to the Substantial Completion Inspection of the phase to be occupied. The plan should include all scheduled inspections, instruction classes, items to be submitted, closeout dates for all functions and the required NIH and Contractor personnel for these functions that will be taking part.

1.6 FINAL COMPLETION

- A. Preliminary Procedures: Before requesting final inspection for Certification of Final Completion and Final Payment, complete the following. Note that the following are to be completed, submitted as appropriate, and approved by NIH as applicable prior to the final inspection and are not to be submitted for approval or otherwise at the final inspection unless specifically indicated. List exceptions in the request.
1. Submit final payment request with releases and supporting documentation not previously submitted and accepted.
 2. Submit an updated final statement, accounting for final additional changes to the Contract price.
 3. Verify that all required submittals have been provided to the Contracting Officer including but not limited to the following:
 - a. Manufacturer's cleaning instructions
 - b. Posted instructions
 - c. Record Documents (Record Document BIM, Record Prints, drawings, specifications, product data, and miscellaneous record documentation) incorporating any changes required by the Contracting Officer as a result of the review of the submission prior to the pre-final inspection. See Division 01 Section "Project Record Documentation" for additional requirements.
 - d. Final project warranty documents reflecting changes directed by any comments from the Contracting Officer's review of draft documents.
 4. Submit a certified copy of the previous Substantial Completion inspection list of items to be completed or corrected. The certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance, and shall be endorsed and dated by the Contractor.
 5. Submit Pest management Post Construction Survey and Certification.
 6. Submit specific warranties, workmanship bonds, maintenance agreements, final certifications and similar documents.
 7. Submit record documents and similar final record information.
 8. Deliver tools, spare parts, extra stock and similar items.
 9. Complete final clean-up requirements including touch-up painting of marred surfaces.
 10. Submit final meter readings for utilities, a measured record of stored fuel, and similar data as of the date when NIH took possession of and assumed responsibility for corresponding elements of the work.
- B. Re-inspection Procedure: The Project Officer will re-inspect the work upon receipt of notice from the Contractor that the project work, including inspection list items from earlier inspections, has been completed, except for items whose completion is delayed under circumstances acceptable to the Contracting Officer.

1. Upon completion of re-inspection, the Project Officer will advise the Contracting Officer, who will prepare a Certificate of Final Completion, or the Contracting Officer will advise the Contractor of work that is incomplete or of obligations that have not been fulfilled and are required for Final Completion.
2. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for any additional NIH inspection costs in accordance with the Contract Clause entitled "Inspection of Construction."

1.7 WARRANTIES

- A. Standard and special warranties required by the individual sections of the Specifications and coincidental warranties shall provide guarantees in terms of time limits or rights of NIH in addition to those contained in the Construction Contract clauses.
 1. Manufacturer's disclaimers and limitations on product warranties do not relieve the contractor of the warranty on the work that incorporates the products.
 2. Manufacturer's disclaimers and limitations on product warranties do not relieve suppliers, manufacturers, and subcontractors required to countersign special warranties with the Contractor.
 3. Standard product warranties shall be preprinted written warranties published by individual manufacturers for particular products, and shall be specifically endorsed to NIH by the manufacturer.
 4. Special project warranties shall be specifically written to incorporate particular requirements of the Contract Documents, and shall be endorsed to NIH by the entities responsible for the work, as stated in the individual section.
 5. Coincidental product warranties shall be provided where available on a product incorporated into the work by virtue of the fact that the manufacturer of the product has published a warranty in connection with purchases and uses of the product without regard for specific applications except as otherwise limited by terms of the warranty.
- B. Reinstatement of Warranty: When work covered by a warranty has failed and been corrected by replacement or rebuilding, reinstate the warranty by written endorsement. The reinstated warranty shall be equal to the original warranty with an equitable adjustment for depreciation.
- C. Replacement Cost: Upon determination that work covered by a warranty has failed, replace or rebuild the work to an acceptable condition complying with requirements of the Contract Documents. The contractor is responsible for the cost of replacing or rebuilding defective work regardless of whether NIH has benefited from the use of the work through a portion of its anticipated useful service life.
- D. Rejection of Warranties: The Contracting Officer reserves the right to reject warranties and to limit selection to products with warranties not in conflict with requirements of the Contract Documents.
- E. Where the Contract Documents require a special warranty, or similar commitment for the work or part of the work, the Contracting Officer reserves the right to refuse to accept the work on behalf of NIH until the contractor presents evidence that entities required to countersign such commitments are willing to do so.
- F. Where a warranty is not specifically required by the Contract Documents but is available on a product incorporated into the work, by virtue of the fact that the manufacturer of the product has published a warranty in connection with purchases and uses of the product without regard for specific applications except as otherwise limited by terms of the warranty, that warranty shall be provided to NIH.

- G. Submit written warranties to the Contracting Officer as required by the contract documents.
1. Draft copies of required warranty documents shall be submitted to the Contracting Officer for review in the specified format prior to warranty execution and prior to the date certified for Substantial Completion, unless an earlier time of submission is specified elsewhere in the contract documents or requested by the Contracting Officer.
 - a. Submit three copies of draft warranty documents. All but one copy of the draft submission shall be returned to the Contractor for corrections and resubmission.
 2. Warranties will comply with the requirements included in the technical specification sections.
 3. Unless indicated otherwise warranties are to take effect on the date of Substantial Completion.
 4. When the contract documents require the Contractor, or the Contractor and a subcontractor, supplier or manufacturer to execute a special warranty, provide a written document that contains the appropriate terms and identification, executed by the required parties.
 - a. Refer to Division 02 through 33 sections for specific content requirements and particular requirements for submitting special warranties.
 5. Following Contracting Officer review, correct draft warranty documents as required and submit three copies of each final warranty document properly executed by the contractor, subcontractor, supplier, or manufacturer at Final Completion.
 6. Organize the warranty documents into an orderly sequence based on the Specification Divisions and Section Numbers.
 - a. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2-by-11-inch (216-by-280-mm) paper.
 - b. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Project name, and name of Contractor.
 - c. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation.
 - d. Provide a typed description of each product or installation being warranted, including the name of the product, and the name, address, and telephone number of the Installer.
 7. When warranted construction requires operation and maintenance manuals, provide additional copies of each required warranty in each required manual. Refer to Division 1 Section "Operation and Maintenance Documentation" for requirements of Operation and Maintenance manuals.
- H. When a designated portion of the work is completed and occupied or used by NIH, by a separate agreement with the contractor during the construction period, submit properly executed warranties within 14 calendar days after completion of that designated portion of the work.

PART 2 - PRODUCTS (Not Applicable)

PART 3 - EXECUTION

3.1 POSTED EQUIPMENT INFORMATION

- A. Post critical information for equipment in location approved by Project Officer.
- B. Critical Information: Critical information includes the following as recommended by equipment manufacturer or required by Project Officer:

1. Equipment Name, Manufacturer, Model Number and Serial Number
 2. Safety precautions
 3. Wiring and control (P&ID) diagrams where appropriate showing system layout
 4. Critical start up and shut down precautions where appropriate
- C. Format: Engraved, framed under glass or approved laminated plastic and/or other durable, permanent format as appropriate for each type of information. Postings exposed to the weather shall be made of weather-resisting materials or shall be suitably enclosed to be weather protected. Operating instructions shall not fade when exposed to sunlight and shall be secured to prevent easy removal or peeling.

3.2 FINAL CLEANING

- A. General cleaning during construction is required by the General Conditions and included in Section H of the Contract.
- B. Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to the condition expected in a normal, commercial cleaning and maintenance program. Use only those cleaning materials which will not create hazards to health or property and which will not damage surfaces. Use only those cleaning materials and methods recommended by the manufacturer of the surface material to be cleaned. Use cleaning materials only on surfaces recommended by cleaning material manufacturer. Comply with manufacturer's instructions.
- C. Complete the following cleaning operations before requesting inspection for Final Completion.
1. Remove labels and stickers that are not permanent from fixtures and equipment. Do not remove permanent nameplates, equipment model numbers and ratings.
 2. Polish glossy surfaces to a clear shine.
 3. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
 4. Sweep exterior paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
 5. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.
 6. Remove tools, construction equipment, machinery, and surplus material from Project site.
 7. Remove snow and ice to provide safe access to building.
 8. Remove grease, mastic, adhesives, dust, dirt, stains, fingerprints, labels, and other foreign materials and substances from sight-exposed interior and exterior surfaces. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
 9. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
 10. Sweep exposed concrete floors broom clean followed by wet mopping in occupied and unoccupied spaces.
 11. Clean resilient floor surfaces and finish with wax or other finish cleaning and buffing in accordance with manufacturer's recommendations.
 12. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo if visible soil or stains remain.
 13. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision-obscuring materials. Replace chipped or broken glass and other damaged transparent materials. Polish mirrors and glass, taking care not to scratch surfaces.
 14. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

- a. Do not paint over "UL" and similar labels, including mechanical and electrical nameplates.
 - 15. Internally clean the entire system of piping and equipment. Open dirt pockets and strainers, completely blowing down as required and clean strainer screens of accumulated debris.
 - 16. Wipe surfaces of mechanical and electrical equipment, elevator equipment, and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
 - 17. Replace parts subject to unusual operating conditions.
 - 18. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.
 - 19. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
 - 20. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out bulbs, and those noticeably dimmed by hours of use, and defective and noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
 - 21. Perform LEED Credit EQ 3.2 HVAC system flush out procedure.
 - a. Remove temporary filters and duct covers installed as part of the Construction IAQ Management Plan.
 - b. Perform flush out as required to introduce minimum of 14,000 cu.ft. outside air per sq. ft. of project area.
 - 22. Leave Project clean and ready for occupancy.
- D. Pest Control: Engage an experienced, licensed exterminator to make a final inspection and rid the project of rodents, insects, and other pests.
- E. Dust Control: Handle materials in a controlled manner with as little dust and over spray as possible.
- F. Removal of Protection: Remove temporary protection and facilities installed for the protection of the Work during construction.
- G. Compliance: Comply with the regulations of authorities having jurisdiction and with safety standards for cleaning and disposal operations. Do not burn or bury rubbish, waste, and/or excess materials on NIH property. Do not discharge volatile, harmful, or dangerous materials into drainage systems. Remove waste materials from the site and dispose of it lawfully.
- H. Remaining Materials: Where extra materials of value are remaining after completion of associated work, they become NIH property. Arrange for disposition of these materials as directed by the Project Officer.

END OF SECTION

SECTION 01 7810 - PROJECT RECORD DOCUMENTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This Section includes administrative and procedural requirements for Project Record Documents
- B. Related Sections: This specification section is related to any and all specification sections with explicit or implicit reference to Project Record Document including but not limited to the following Division 01 specification sections:

1. Division 01 Section "Summary"
2. Division 01 Section "Project Management and Coordination" for:
 - a. General project management and coordination requirements
3. Division 01 Section "Construction Progress Documentation"
4. Division 01 Section "Quality Requirements"
5. Division 01 Section "Construction Quality Control"
6. Division 01 Section "Safety and Health"
7. Division 01 Section "Closeout Procedures"
8. Division 01 Section "Operation and Maintenance Documentation"
9. Division 01 Section "Functional Performance Testing"
10. Division 01 Section "Interactive System Functional Testing"
11. Division 01 Section "Demonstration and Training"
12. Divisions 02 through 33 sections for project record document requirements for products included in those sections.

- C. Project Record Documents and samples are to be stored by the Contractor in the field office apart from the Contract Documents used for construction. Project Record Documents are not to be used for construction purposes. Record Documents shall be maintained in good order and in a clean, dry, legible condition and available at all times for the Contracting Officer's inspection.

1.3 DEFINITIONS

- A. Equipment Inventory Card: Data sheet in Excel format for each NIH-numbered building system and component. See Section 1.5.B for detailed deliverables requirement.
- B. Record Prints: Prints of Contract Drawings and Shop Drawings marked up to show actual as-built conditions. See Section 1.5.A for detailed deliverables requirements.
- C. Record Specifications: As-built specifications in .pdf format marked up to reflect actual as-built conditions. Paper copies where required. See Section 1.5.A for detailed deliverables requirements.

D. Record Documents: Paper copies of project records, including:

1. Record Prints
2. Record Specifications
3. Record Product Data
4. Miscellaneous Record Submittals
5. Administrative Project Record

1.4 SUBMITTALS

A. Record Documents: Comply with the following:

1. Record Prints and Specifications : Submit Record Prints and Record Specifications to the Contracting Officer for review and approval prior to final payment.
2. Hard Copies: During Project Closeout and prior to final payment submit 2 hard copies of Record Documents.

PART 2 - PRODUCTS

2.1 RECORD PRINTS

A. Record Prints: During construction, maintain a set of blue or black-line white prints of Contract Drawings and Shop Drawings for Project Record Document purposes.

1. Mark Contract Drawings or Shop Drawings to show the actual installation where the installation varies from the installation shown originally. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later. Items required to be marked include, but are not limited to, the following:
 - a. Dimensional changes to the Drawings.
 - b. Revisions to details shown on the Drawings.
 - c. Depths of foundations below the first floor.
 - d. Locations and depths of underground utilities.
 - e. Revisions to routing of piping and conduits.
 - f. Revisions to electrical circuitry.
 - g. Actual equipment locations.
 - h. Duct size and routing.
 - i. Locations of concealed internal utilities.
 - j. Changes made by Change Order or Construction Change Directives.
 - k. Changes made following the Contracting Officer's written orders.
 - l. Details not on original Contract Drawings.
2. Mark record prints of Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. Where Shop Drawings are marked, show cross-reference to and mark Contract Drawings location.
3. Mark record sets with red erasable colored pencil. Use other colors to distinguish between changes for different categories of the work at the same location.
4. Mark important additional information that was either shown schematically or omitted from original Drawings.
5. Note alternate numbers, Change Order numbers, and similar identifications.
6. Responsibility for Markup: The individual, installer, subcontractor or other entity who obtained the record data shall prepare the markup on record drawings.
 - a. Accurately record information in an understandable drawing technique.

- b. Record data as soon as possible after obtaining it. Record and check the markup prior to enclosing concealed installations.
- B. Format: Identify and date each Record Print; include the designation "PROJECT RECORD PRINT" in a prominent location.
 - 1. Record Prints: Organize Record Prints and newly prepared Record Drawings into manageable sets. Bind each set with durable paper cover sheets. Include identification on cover sheets.

2.2 RECORD SPECIFICATIONS

- A. During the construction period, maintain three copies of the Project Specifications, including addenda and other modifications issued, for Project Record Document purposes.
 - 1. Mark the Specifications to indicate the actual installation where the installation varies from that indicated in Specifications. Note related project record drawing information, where applicable. Give particular attention to substitutions, selection of product options, and information on concealed installations that would be difficult to identify or measure and record later.
 - a. In each Specification section where products, materials or units of equipment are specified or scheduled, mark the copy with the proprietary name and model number of the product furnished.
 - b. Record the name of the manufacturer, supplier, installer, and other information necessary to provide a record of selections made and to document coordination with record Product Data submittals and maintenance manuals.
 - 2. Upon completion of markup, submit Record Specifications to the Project Officer for NIH records.

2.3 RECORD PRODUCT DATA

- A. During the construction period, maintain one copy of each Product Data submittal for Project Record Document purposes.
 - 1. Mark Product Data to indicate the actual product installation where the installation varies substantially from that indicated in Product Data submitted. Include significant changes in the product delivered to the site and changes in manufacturer's instructions and recommendations for installation.
 - 2. Give particular attention to information about concealed products and installations that cannot be readily identified and recorded later.
 - 3. Note related Change Orders and markup of Record Drawings, where applicable.
 - 4. Upon completion of markup, submit a complete set of record Product Data to the Project Officer.
 - 5. Where record Product Data is required as part of operation and maintenance manuals, submit marked-up Product Data as an insert in the manual instead of submittal as record Product Data.
- B. See Section 1.5 Submittals above for instructions on preparation and delivery of Record Product Data.

2.4 RECORD SAMPLE SUBMITTAL

- A. Immediately prior to the date of Substantial Completion, the Contractor shall meet with the Project Officer at the site to determine which of the Samples maintained during the construction period shall be transmitted to NIH for record purposes.
- B. Comply with the Project Officer's instructions for packaging, identification marking and delivery to the NIH sample storage space. Dispose of other samples in a manner specified for disposing of surplus and waste materials.

2.5 MISCELLANEOUS RECORD SUBMITTALS

- A. Refer to other specification sections for miscellaneous record-keeping requirements and submittals in connection with various construction activities. Prior to Final Acceptance, assemble completed miscellaneous records and place in good order, properly identified and bound or otherwise organized to allow for use and reference.
- B. Submit to the Project Officer for NIH records.
- C. Miscellaneous records include, but are not limited to, the following:
 - 1. Field records on excavations and foundations.
 - 2. Field records on underground construction and similar work.
 - 3. Survey showing locations and elevations of underground lines.
 - 4. Invert elevations of drainage piping.
 - 5. Surveys establishing building lines and levels.
 - 6. Authorized measurements utilizing unit prices or allowances.
 - 7. Records of plant treatment.
 - 8. Ambient and substrate condition tests.
 - 9. Certifications received in lieu of labels on bulk products.
 - 10. Batch mixing and bulk delivery records.
 - 11. Testing and qualification of tradespersons.
 - 12. Documented qualification of installation firms.
 - 13. Load and performance testing.
 - 14. Inspections and certifications by governing authorities.
 - 15. Leakage and water-penetration tests.
 - 16. Fire-resistance and flame-spread test results.
 - 17. Final inspection and correction procedures.

2.6 ADMINISTRATIVE RECORD

- A. Maintain comprehensive electronic file administrative record for the project including copies of the following:
 - 1. Contracts and contract modifications
 - 2. Correspondence
 - 3. Records relating to payment requests and payment
 - 4. Addenda
 - 5. RFIs
 - 6. ASIs
 - 7. Change Orders
 - 8. Meeting Minutes
 - 9. Project Schedule
 - 10. Construction documentation

PART 3 - EXECUTION

3.1 RECORDING AND MAINTENANCE

- A. Recording: Maintain one hard copy of each submittal during the construction period for Project Record Documents purposes. Post changes and modifications to Project Record Documents as they occur; do not wait until the end of the Project.

END OF SECTION

SECTION 01 7823 - OPERATION AND MAINTENANCE DOCUMENTATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. This section includes administrative and procedural requirements for operation and maintenance manuals and instruction, including the following.
 - 1. Preparing and submitting instruction manuals covering the care, preservation and maintenance of materials and finishes.
 - 2. Preparing and submitting operation and maintenance manuals for equipment and building operating systems.
 - 3. Instruction of NIH operating personnel in the operation and maintenance of building systems and equipment.
- B. This specification section is related to any and all specification sections with explicit or implicit reference to operation and maintenance documentation including but not limited to the following Division 01 specification sections:
 - 1. Division 01 Section "Summary"
 - 2. Division 01 Section "Project Management and Coordination"
 - 3. Division 01 Section "Construction Progress Documentation"
 - 4. Division 01 Section "Quality Requirements"
 - 5. Division 01 Section "Construction Quality Control"
 - 6. Division 01 Section "Safety and Health"
 - 7. Division 01 Section "Closeout Procedures"
 - 8. Division 01 Section "Project Record Documents"
 - 9. Division 01 Section "Demonstration and Training"
- C. Additional Requirements: Refer to the individual Division 02-33 technical specification sections for specific operations and maintenance manual requirements for products and systems in those sections and additional requirements for the care and maintenance of materials and finishes, and for the operation and maintenance of the various pieces of equipment and operating systems.

1.3 QUALIFICATIONS

- A. Operation and Maintenance Manual Preparation: In preparation of manuals, use personnel thoroughly trained and experienced in the maintenance of the material or finish involved, or in the operation and maintenance of the equipment or system involved.
 - 1. Where manuals require written instructions, use the personnel skilled in technical writing where necessary for communication of essential data.
 - 2. Where manuals require drawings or diagrams, use draftspersons capable of preparing drawings clearly in an understandable format.

- B. Instructions for NIH Personnel: Use instructors thoroughly trained and experienced in the operation and maintenance of the equipment or system involved to instruct NIH operation and maintenance personnel.

1.4 DEFINITIONS

- A. System: An organized collection of parts, equipment, or subsystems united by regular interaction.
- B. Subsystem: A portion of a system with characteristics similar to a system.

1.5 COORDINATION

- A. Where operation and maintenance documentation includes information on installations by more than one factory-authorized service representative, assemble and coordinate information furnished by representatives and prepare manuals.

1.6 SUBMITTALS

- A. Initial Submittal: Submit to the Contracting Officer 1 draft hard copies and one electronic copy of each manual. Include a complete operation and maintenance directory (with relative hyperlinks to each document in the electronic version). The Contracting Officer will return 1 copy of draft with comments for incorporation into the final submission.
 - 1. Should comments be extensive, the Contracting Officer may require the Initial Submission to be repeated prior to scheduling a Final Inspection.
 - 2. Initial submittal shall occur at the initial occupancy phase
- B. Final Submittal: Submit 1 copy and one electronic copy of each manual in final form at least two working days before the beginning of the Acceptance Phase. If comments on the Initial Submittal are extensive, the Contracting Officer may require additional time in advance of the Acceptance Phase. Contracting Officer will provide return a copy with comments within 14 calendar days after the beginning of the Acceptance Phase.
 - 1. Correct or modify each manual to comply with the Contracting Officer's comments. Submit 1 hard copy and one electronic copy of each corrected manual within 14 calendar days of receipt of Contracting Officer's comments.

PART 2 - PRODUCTS

2.1 OPERATION AND MAINTENANCE DOCUMENTATION DIRECTORY

- A. Provide a comprehensive Operation and Maintenance Documentation Directory which provides a quick reference document for all the manuals provided for the project.
- B. Organization: Include a section in the directory for each of the following:
 - 1. Systems and Subsystems.
 - a. List systems and subsystems alphabetically. Include references to operation and maintenance manuals that contain information about each system and subsystem.

2. Equipment.
 - a. List equipment for each system and subsystem, organized alphabetically by system. For pieces of equipment not part of system or subsystem, list alphabetically in separate list.
3. Tables of contents.
 - a. Include a copy of the table of contents for each emergency, operation, and maintenance manual.
- C. Identification: In the documentation directory and in each operation and maintenance manual, identify each system, subsystem, and piece of equipment with the same designation used in the contract documents. If no designation exists, assign a designation according to ASHRAE Guideline 4, "Preparation of Operating and Maintenance Documentation for Building Systems."

2.2 MANUALS, GENERAL

- A. Organization and Content: Organize each manual into separate sections for each related product or piece of equipment. To the extent applicable, each manual shall contain a title page, table of contents, general information, copies of Product Data, written text, drawings and copies of each warranty and service contract issued.
 1. Title Page: Provide a title page in a transparent, plastic envelope as the first sheet of each manual. As a minimum, provide the following information:
 - a. Subject matter covered by the manual.
 - b. Name and address of the Project.
 - c. Date of submittal.
 - d. Name, address, and telephone number of the Contractor.
 - e. Cross-reference to related products in other operation and maintenance manuals, if applicable.
 2. Table of Contents: After title page, include a typewritten table of contents for each volume, arranged systematically by specification section according to the specifications format. Include a list of each product included, identified by product name or other appropriate identifying symbol and indexed to the content of the volume. Where more than one volume is required to accommodate the data, provide a comprehensive table of contents for all volumes in each volume of the set. The hard copy Table of Contents and Indexes shall be the printed version of the active hyperlink indexes of the electronic version.
 3. General Information: Provide a general information section immediately following table of contents, listing each product included in the manual, identified by product name. Under each product, list the name, address, and telephone number of the subcontractor or installer and the maintenance contractor. Clearly delineate the extent of responsibility for each of these entities. Include a local source for replacement parts for equipment.
 4. Product Data: Where the manuals include manufacturer's standard printed data, include only those sheets that are pertinent to the part or product installed. Mark each sheet to identify each part or product included in the installation. Where the project includes more than one item contained in the product data, identify each item, using appropriate references from the Contract Documents. Identify data that is applicable to the installation, and delete references to information that is not applicable.
 5. Written Text: Prepare written text to provide necessary information where manufacturer's standard printed data is not available, and the information is necessary for proper maintenance of materials or finishes, or for proper operation and maintenance of

equipment or systems. Prepare written text where it is necessary to provide additional information or to supplement data included elsewhere in the manual. Organize text in a consistent format under separate headings for different procedures. Where necessary, provide a logical sequence of instruction for each operation or maintenance procedure.

6. Drawings: Provide specially prepared drawings where necessary to supplement manufacturer's printed data to illustrate the relationship of component parts of equipment or systems or to provide control or flow diagrams. Coordinate these drawings with information contained in Project Record Drawings to assure correct illustration of the completed installation.
 7. Warranties, and Service Contracts: Provide a copy of each warranty or service contract in the appropriate manual for the information of NIH operating personnel. Provide written data outlining procedures to follow in the event of product failure. List circumstances and conditions that would affect the validity of warranty.
 8. Where required for full understanding, include a copy of applicable Project Record Drawings. Do not use original Project Record Documents as part of operation and maintenance manuals.
- B. Hard Copy Format: Prepare operation and maintenance manuals in the form of an instructional manual for use by operating and maintenance personnel. Organize into suitable sets of manageable size. Where possible, assemble instructions for similar products into a single binder.
1. Binders: For each manual, provide heavy-duty, commercial-quality, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to receive 8-1/2-by-11-inch (115-by-280-mm) paper. Provide a clear plastic sleeve on the spine to hold labels describing contents. Provide pockets in the covers to receive folded sheets.
 - a. Where 2 or more binders are necessary to accommodate data, correlate data in each binder into related groupings according to the Specifications table of contents. Cross-reference other binders where necessary to provide essential information for proper operation or maintenance of the product.
 - b. Identify each binder on front and spine, with the printed type of manual Project title or name, and subject matter covered. Indicate volume number for multiple volume sets of manuals.
 2. Dividers: Provide heavy paper dividers with celluloid-covered tabs for each separate section. Mark each tab to indicate contents. Provide a typed description of the product or major parts of equipment included in the section on each divider.
 3. Protective Plastic Jackets: Provide protective, transparent, plastic jackets designed to enclose diagnostic software for computerized electronic equipment.
 4. Text Material: Where maintenance manuals require written material, use the manufacturer's standard printed materials, where available. If manufacturer's standard printed materials are not available, provide specially prepared data, neatly typewritten, on 8-1/2-by-11-inch (115-by-280-mm), 20-lb/sq. ft. (75-g/sq. m) white bond paper.
 5. Drawings: Where manuals require drawings or diagrams, provide reinforced, punched binder tabs on drawings and bind in with text.
 - a. Where oversize drawings are necessary, fold drawings to the same size as text pages and use as a foldout.
 - b. If drawings are too large to be used practically as a foldout, place the drawing, neatly folded, in the front or rear pocket of binder. Insert a typewritten page indicating drawing title, description of contents and reference to the applicable location in the manual.
- C. Electronic Copy: All operations and maintenance data shall be provided in both hard copy (as indicated above) and electronic copy. All operation and maintenance data shall be TEXT

SEARCHABLE (scanned versions are not acceptable).provided in a combination of the following file formats:

1. pdf format
2. Hypertext Mark Up Language format (*.html)
3. Microsoft Office file formats
4. Autocad v2007 or later

- D. The electronic version of the Operation and Maintenance Manual shall be a series of files organized in a logical series of hierarchical subdirectories of the primary OM directory with a summary Index (located in that primary directory) with hyperlinks to the various documents that make up the content. The Index will typically be the electronic version of the printed table of contents for the hard copy with live RELATIVE hyperlinks. Final OM Manual shall be provided on a portable hard drive large enough to hold the files with 50% excess storage capacity. Interim submissions for review may be provided on multiple DVDs as long as directory structure is maintained and clear.

2.3 PRODUCT MAINTENANCE MANUAL

- A. Content and Organization: Organize manual into a separate section for each product, material, and finish. Order manual by specification section of the applicable product. Include the following for each product:
1. Source information. List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list :
 - a. Name, Address, Telephone number of Installer or supplier and maintenance service agent.
 - b. Cross-reference to Specification Section number and title in Project Manual.
 2. Product information, Include the following, as applicable:
 - a. Product name and model number.
 - b. Manufacturer's name.
 - c. Color, pattern, and texture.
 - d. Material and chemical composition.
 - e. Reordering information for specially manufactured products.
 3. Maintenance procedures: Include manufacturer's written recommendations and the following:
 - a. Inspection procedures.
 - b. Types of cleaning agents to be used and methods of cleaning.
 - c. List of cleaning agents and methods of cleaning detrimental to product.
 - d. Schedule for routine cleaning and maintenance.
 - e. Repair instructions.
 4. Repair materials and sources: Include lists of materials and local sources of materials and related services.
 5. Warranties and bonds, Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
 - a. Include procedures to follow and required notifications for warranty claims.

2.4 EQUIPMENT OPERATION AND MAINTENANCE MANUAL

- A. Content and Organization: Provide information needed for daily operations and management of systems and equipment. Organize manual into a separate section for each equipment, component, packaged/preengineered system, etc.. Order manual by specification section of the applicable product. Include the following for each component:
1. Source information. List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list :
 - a. Name, Address, Telephone number of Installer or supplier and maintenance service agent.
 - b. Cross-reference to Specification Section number and title in Project Manual.
 2. Product information, Include the following, as applicable:
 - a. Product name and model number.
 - b. Manufacturer's name.
 - c. Color, pattern, and texture.
 - d. Material and chemical composition.
 - e. Reordering information for specially manufactured products.
 - f. Equipment identification with serial number of each component.
 - g. Equipment function.
 - h. Operating characteristics.
 - i. Limiting conditions.
 - j. Performance curves.
 - k. Engineering data and tests.
 - l. Complete nomenclature and number of replacement parts.
 3. Performance and design criteria if contractor is delegated design responsibility.
 4. Operating standards.
 5. Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:
 - a. Inspection procedures.
 - b. Types of cleaning agents to be used and methods of cleaning.
 - c. List of cleaning agents and methods of cleaning detrimental to product.
 - d. Schedule for routine cleaning and maintenance.
 - e. Repair instructions.
 - f. Troubleshooting guide.
 - g. Standard printed maintenance instructions and bulletins.
 - h. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
 - i. Identification and nomenclature of parts and components.
 - j. List of items recommended to be stocked as spare parts.
 6. Maintenance procedures: Include the following information and items that detail essential maintenance procedures:
 - a. Test and inspection instructions.
 - b. Precautions against improper maintenance.
 - c. Disassembly; component removal, repair, and replacement; and reassembly instructions.
 - d. Aligning, adjusting, and checking instructions.
 - e. Demonstration and training videotape, if available.

7. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
 - a. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
 - b. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.
8. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
9. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.
10. Warranties and bonds, Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
 - a. Include procedures to follow and required notifications for warranty claims.

2.5 SYSTEMS MANUAL

- A. The Systems Manual will be authored by the CxA during the Construction Phase of the project. Contractor shall provide the following content for the System Manual in the template provided by the CxA:
 - a. Emergency Shutdown procedures
 - b. Startup procedures.
 - c. Equipment or system break-in procedures.
 - d. Routine and normal operating instructions.
 - e. Regulation and control procedures.
 - f. Instructions on stopping.
 - g. Normal shutdown instructions.
 - h. Seasonal and weekend operating instructions.
 - i. Required sequences for electric or electronic systems.
 - j. Special operating instructions and procedures.
2. Operating logs.
3. Piped Systems: Diagram piping as installed, and identify color-coding where required for identification. Provide charts of valve-tag numbers, with the location and function of each valve.

PART 3 - EXECUTION

3.1 MANUAL PREPARATION

- A. Operation and Maintenance Documentation Directory: Prepare a separate manual that provides an organized reference to operation, and maintenance manuals.
- B. Product Maintenance Manual: Assemble a complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into the Work.

- C. Operation and Maintenance Manuals: Assemble a complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.
 - 1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
 - 2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by Owner's operating personnel.
- D. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark each sheet to identify each product or component incorporated into the Work. If data include more than one item in a tabular format, identify each item using appropriate references from the Contract Documents. Identify data applicable to the Work and delete references to information not applicable.
 - 1. Prepare supplementary text if manufacturers' standard printed data are not available and where the information is necessary for proper operation and maintenance of equipment or systems.
- E. Drawings: Prepare drawings supplementing manufacturers' printed data to illustrate the relationship of component parts of equipment and systems and to illustrate control sequence and flow diagrams. Coordinate these drawings with information contained in Record Drawings to ensure correct illustration of completed installation.
 - 1. Do not use original Project Record Documents as part of operation and maintenance manuals. If Record drawings must be used, copies are to be included in the manuals.
 - 2. Comply with requirements of newly prepared Record Drawings in Division 01 Section "Project Record Documents."
- F. Comply with Division 01 Section "Closeout Procedures" for the schedule for submitting operation and maintenance documentation.

END OF SECTION

SECTION 02 4113 - SELECTIVE SITE DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Demolition and removal of existing curbs and gutters, and other exterior site items indicated or not indicated which interfere with the Work.
 - 2. Removal and disposal of storm drainage pipe indicated.
 - 3. Removal and disposal of material from dump areas.

1.3 DEFINITIONS

- A. Remove: Remove and legally dispose of items except those indicated to be reinstalled, salvaged, or to remain the Owner's property.
- B. Remove and Salvage: Items indicated to be removed and salvaged remain the Owner's property. Remove, clean, and pack or crate items to protect against damage. Identify contents of containers and deliver to Owner's designated storage area.
- C. Remove and Reinstall: Remove items indicated; clean, service, and otherwise prepare them for reuse; store and protect against damage. Reinstall items in the same locations or in locations indicated.
- D. Existing to Remain: Protect items indicated to remain against damage and soiling. When permitted by the Architect, items may be removed to a suitable, protected storage location and then cleaned and reinstalled in their original locations.

1.4 MATERIALS OWNERSHIP

- A. Except for items or materials indicated to be reused, salvaged, reinstalled, or otherwise indicated to remain the Owner's property, remove demolished materials from the site with further disposition at the Contractor's option.
- B. Storage or sale of removed items or materials on-site will not be permitted.
- C. Historical items, relics, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, antiques, and other items of interest or value to the Owner, which may be encountered, remain the Owner's property. Carefully remove and salvage each item or object in a manner to prevent damage and deliver promptly to the Owner.

1.5 SUBMITTALS

- A. Photographs or videotape, sufficiently detailed, of existing conditions of adjoining construction and site improvements that might be misconstrued as damage caused by the Work.
- B. Record drawings at Project closeout.
 - 1. Provide record drawings that accurately reflect as-built installed site improvements and utilities to include subsurface location of all dry and wet utilities. The information is to be field

certified by a licensed surveyor in the State of Maryland and provided the Owner in CADD format.

- C. Proposed dust-control measures.
- D. Schedule of selective demolition activities indicating the following:
 - 1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity.
 - 2. Interruption of utility services.
 - 3. Coordination for shutoff, capping, and continuation of utility services.
 - 4. Detailed sequence of selective demolition and removal work to ensure uninterrupted progress of Owner's on-site operations.
 - 5. Coordination of Owner's continuing occupancy of portions of existing building and of Owner's partial occupancy of completed Work.
 - 6. Locations of temporary partitions and means of egress.
- E. Inventory of items to be removed and salvaged or turned over to Owner.
- F. Landfill records indicating receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.6 QUALITY ASSURANCE

- A. Regulatory Requirements: All work shall comply with Federal, State and Local laws and regulations concerning hauling and disposal of demolition debris.
- B. Notify the proper agencies prior to the start of work and obtain all necessary permits for this work.

1.7 PROJECT CONDITIONS

- A. Owner assumes no responsibility for actual condition of items or structures to be demolished. Conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to Owner's removal and salvage operations prior to the start of demolition work.
- B. The location of existing underground utilities indicated is approximate only. Field locate all existing underground utilities within the limits of disturbance, regardless of whether or not they are indicated. The Contractor is required by NIH to use a private utility locator for locating all existing utilities. It's required by state law to Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities.
- C. Should charted, uncharted or incorrectly charted utilities be encountered during demolition, contact the Architect immediately for instructions. Cooperate with Owner and utility companies to keep services and facilities in operation.
- D. Do not interrupt existing utilities serving facilities occupied and used by the Owner and others, except when permitted in writing by the Owner. Provide acceptable temporary utility service as required to maintain Owner's operations.

1.8 SCHEDULING

- A. Notify and coordinate any required relocation and/or removal of existing underground utilities, poles, meters or other above ground appurtenances with the appropriate utility company (i.e. power, telephone, cable and natural gas/propane) prior to the start of selective demolition work.

1.9 PAYMENT FOR UTILITY REMOVAL / RELOCATIONS / EXTENSIONS

- A. Electric Service - Contractor's responsibility
- B. Phone Service - Contractor's responsibility
- C. Cable Television - Contractor's responsibility
- D. Gas - Contractor's responsibility
- E. Fiber Optic Lines - Contractor's responsibility
- F. Petroleum Pipelines (Not Applicable to this project)

1.10 USE OF EXPLOSIVES

- A. Do not use explosives to perform selective site demolition work.

PART 2 - PRODUCTS

(Not Applicable)

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Call "Miss Utility" prior to the start of demolition work for assistance in the location of existing underground utilities. Field locate all existing underground utilities in the area of work, regardless of whether or not they are indicated.
- B. Should uncharted or incorrectly charted existing utilities be identified, contact the Architect immediately for instructions. Provide a scale drawing with the location of the uncharted or incorrectly charted utilities for use by the Architect in preparing additional direction.
- C. Verify that utilities indicated as removed, abandoned and/or relocated have been disconnected and capped.
- D. Survey existing conditions and correlate with requirements indicated to determine extent of selective demolition required.
- E. Inventory and record the condition of items to be removed and reinstalled and items to be removed and salvaged and turned over to the Owner.

3.2 PROTECTION OF PERSONS AND PROPERTY

- A. Drain, purge, or otherwise remove, collect, and dispose of chemicals, gases, explosives, acids, flammables, or other dangerous materials before proceeding with selective demolition operations.
- B. Conduct demolition operations and remove debris to ensure minimum interference with roads, streets, walks and other adjacent occupied and used facilities.
 - 1. Do not close or obstruct streets, walks, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by governing regulations.
- C. Conduct demolition operations to prevent injury to people and damage to adjacent buildings and facilities to remain. Ensure safe passage of people around selective demolition area.

1. Erect temporary protection, such as walks, fences, railings, canopies, and covered passageways, where required by authorities having jurisdiction.
 2. Protect existing site improvements, appurtenances, and landscaping to remain.
- D. Barricade areas of demolition occurring as part of this work, and post with warning lights as required by authorities having jurisdiction.
- E. Protect structures, buildings, utilities, walks, pavements, existing vegetation and other facilities to remain from damage caused by settlement, lateral movement, undermining, washout and other hazards created by demolition operations.

3.3 POLLUTION CONTROLS

- A. Perform all work in accordance with the requirements of the latest edition of the MDE Handbook and those of the local Erosion Control official.
- B. Clean adjacent structures and improvements of dust, dirt, and debris caused by the Work. Return adjacent areas to condition existing before start of selective demolition.

3.4 DEMOLITION OF EXISTING FACILITIES

- A. Concrete Pavement Walks and Curbs
1. Remove concrete pavement and walks to the nearest joint. Sawcut concrete if joints are not present adjacent to the area of demolition.
 2. Sawcut concrete along straight lines to a depth of not less than 2 inches. Break out remainder of concrete, provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or sawcut entirely through concrete.

3.5 DISPOSAL OF DEMOLISHED MATERIALS

- A. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site.
- B. Do not burn demolished materials or debris.
- C. Transport and legally dispose of demolished materials off of Owner's property.

3.6 CLEANUP AND REPAIR

- A. Upon completion of demolition work remove all tools, equipment and demolition materials from site. Remove demolition work area protection and leave areas clean.
- B. Repair any demolition performed in excess of that required. Return elements of construction and surfaces to remain to the condition existing prior to the start of construction. Repair adjacent construction or surfaces soiled or damaged by demolition work.

END OF SECTION 02 4113

SECTION 03 3000 - CAST-IN-PLACE CONCRETE

PART 1 – GENERAL

1.01 GENERAL REQUIREMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

- A. This Section specifies cast-in place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes, for the following:
 - 1 Exterior Stairs
 - 2 Concrete Wall

1.03 DEFINITIONS

- A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.

1.04 SUBMITTALS

- A. Product Data: For each type of product indicated.
- B. Design Mixtures: For each concrete mixture. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.
 - 1. Indicate amounts of mixing water to be withheld for later addition at Project site.
- C. Steel Reinforcement Shop Drawings: Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, bent bar diagrams, bar arrangement, splices and laps, mechanical connections, tie spacing, hoop spacing, and supports for concrete reinforcement.
- D. Welding certificates.
- E. Qualification Data: For Installer, manufacturer, testing agency.
- F. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:
 - 1. Aggregates. Include service record data indicating absence of deleterious expansion of concrete due to alkali aggregate reactivity.
- G. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
 2. Admixtures.
 3. Form materials and form-release agents.
 4. Steel reinforcement and accessories.
 5. Curing compounds.
 6. Bonding agents.
 7. Adhesives.
 8. Semirigid joint filler.
 9. Joint-filler strips.
 10. Repair materials.
- H. Floor surface flatness and levelness measurements to determine compliance with specified tolerances.
- I. Field quality-control test and inspection reports.
- J. Minutes of preinstallation conference.
- 1.05 QUALITY ASSURANCE
- A. Installer Qualifications: A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and a supervisor who is an ACI-certified Concrete Flatwork Technician.
- B. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C94 requirements for production facilities and equipment.
1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."
- C. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C1077 and ASTM E329 for testing indicated, as documented according to ASTM E548.
1. Personnel conducting field tests shall be qualified as ACI Concrete Field Testing Technician, Grade 1, according to ACI CP-01 or an equivalent certification program.
 2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician - Grade II.
- D. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from one source, and obtain admixtures through one source from a single manufacturer.
- E. Welding: Qualify procedures and personnel according to AWS D1.4, "Structural Welding Code--Reinforcing Steel."

- F. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:
 - 1 ACI 301, "Specification for Structural Concrete," Sections 1 – 5.
 - 2 ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."
- G. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.
- H. Preinstallation Conference: Conduct conference at Project site to comply with requirements in Division 01 Section "Project Management and Coordination."
 - 1. Before submitting design mixtures, review concrete design mixture and examine procedures for ensuring quality of concrete materials. Require representatives of each entity directly concerned with cast-in-place concrete to attend, including the following:
 - a. Contractor's superintendent.
 - b. Independent testing agency responsible for concrete design mixtures.
 - c. Ready-mix concrete manufacturer.
 - d. Concrete subcontractor.
 - 2. Review special inspection and testing and inspecting agency procedures for field quality control, concrete finishes and finishing, cold-and hot-weather concreting procedures, curing procedures, construction contraction and isolation joints, and joint-filler strips, semirigid joint fillers, forms and form removal limitations, shoring and reshoring procedures, vapor-retarder installation, anchor rod and anchorage device installation tolerances, steel reinforcement installation, floor and slab flatness and levelness measurement, concrete repair procedures, and concrete protection.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Steel Reinforcement: Deliver, store, and handle steel reinforcement to prevent bending and damage.

PART 2 – PRODUCTS

2.01 MANUFACTURERS

- A. In other Part 2 articles where titles below introduce lists, the following requirements apply to product selection:
 - 1. Available Products: Subject to compliance with requirements, products that may be incorporated into the Work include, but are not limited to, products specified.
 - 2. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, manufacturers specified.

2.02 FORM-FACING MATERIALS

- A. Smooth-Formed Finished Concrete: Form-facing panels that will provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

1. Plywood, metal, or other approved panel materials.
 2. Exterior-grade plywood panels, suitable for concrete forms, complying with DOC PS 1, and as follows:
 - a. High-density overlay, Class 1 or better.
 - b. Medium-density overlay, Class 1 or better; mill-release agent treated and edge sealed.
 - c. Structural 1, B-B or better; mill oiled and edge sealed.
 - d. B-B (Concrete Form), Class 1 or better; mill oiled and edge sealed.
 - B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.
 - C. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.
 1. Formulate form-release agent with rust inhibitor for steel form-facing materials.
- 2.03 STEEL REINFORCEMENT
- A. Reinforcing Bars: ASTM A615, Grade 60, deformed.
 - B. Plain-Steel Welded Wire Reinforcement: ASTM A185, plain, fabricated from as-drawn steel wire into flat sheets.
- 2.04 REINFORCEMENT ACCESSORIES
- A. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice," of greater compressive strength than concrete and as follows:
 1. For concrete surfaces exposed to view where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected steel wire or CRSI Class 2 stainless-steel bar supports.
- 2.05 CONCRETE MATERIALS
- A. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:
 1. Portland Cement: ASTM C150, Type II, gray. Supplement with the following:
 - a. Fly Ash: ASTM C618, Class C.
 - b. Ground Granulated Blast-Furnace Slag: ASTM C989, Grade 100 or 120.
 - B. Silica Fume: ASTM C1240, amorphous silica.
 - C. Normal-Weight Aggregates: ASTM C33, Class 3S, coarse aggregate or better, graded. Provide

aggregates from a single source with documented service record data of at least 10 years' satisfactory service in similar applications and service conditions using similar aggregates and cementitious materials.

- 1 Maximum Coarse-Aggregate Size: 1 in. nominal.
- 2 Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.

D. Water: ASTM C94 and potable.

2.06 ADMIXTURES

A. Air-Entraining Admixture: ASTM C260.

B. Chemical Admixtures: Provide admixtures certified by manufacturer to be compatible with other admixtures and that will not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.

- 1 Water-Reducing Admixture: ASTM C494, Type A.
- 2 Retarding Admixture: ASTM C494, Type B.
- 3 Water-Reducing and Retarding Admixture: ASTM C494, Type D.
- 4 High-Range, Water-Reducing Admixture: ASTM C494, Type F.
- 5 High-Range, Water-Reducing and Retarding Admixture: ASTM C494, Type G.
- 6 Plasticizing and Retarding Admixture: ASTM C1017, Type II.

2.07 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

1. Products:
 - a. Axim Concrete Technologies; Cimfilm.
 - b. Burke by Edoco; BurkeFilm.
 - c. ChemMasters; Spray-Film.
 - d. Conspec Marketing & Manufacturing Co., Inc., a Dayton Superior Company; Aquafilm.
 - e. Dayton Superior Corporation; Sure Film.
 - f. Euclid Chemical Company (The); Eucobar.
 - g. Kaufman Products, Inc.; Vapor Aid.
 - h. Lambert Corporation; Lambco Skin.
 - i. L&M Construction Chemicals, Inc.; E-Con.
 - j. MBT Protection and Repair, Div. of ChemRex; Confilm.
 - k. Meadows, W. R., Inc.; Sealtight Evapre.

- l. Metalcrete Industries; Waterhold.
 - m. Nox-Crete Products Group, Kinsman Corporation; Monofilm.
 - n. Sika Corporation, Inc.; SikaFilm.
 - o. Symons Corporation, a Dayton Superior Company; Finishing Aid.
 - p. Unitex; Pro-Film.
 - q. US Mix Products Company; US Spec Monofilm ER.
 - r. Vexcon Chemicals, Inc.; Certi-Vex EnvioAssist.
- B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz /sq yd when dry.
- C. Moisture-Retaining Cover: ASTM C171, polyethylene film or white burlap-polyethylene sheet.
- D. Water: Potable.
- E. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C309, Type 1, Class B, dissipating.
- 1. Products:
 - a. Anti-Hydro International, Inc.; AH Curing Compound #2 DR WB.
 - b. Burke by Edoco; Aqua Resin Cure.
 - c. ChemMasters; Safe-Cure Clear.
 - d. Conspec Marketing & Manufacturing Co., Inc., a Dayton Superior Company; W.B. Resin Cure.
 - e. Dayton Superior Corporation; Day Chem Rez Cure (J-11-W).
 - f. Euclid Chemical Company (The); Kurez DR VOX.
 - g. Kaufman Products, Inc.; Thinfilm 420.
 - h. Lambert Corporation; Aqua Kure-Clear.
 - i. L&M Construction Chemicals, Inc.; L&M Cure R.
 - j. Meadows, W. R., Inc.; 1100 Clear.
 - k. Nox-Crete Products Group, Kinsman Corporation; Resin Cure E.
 - l. Symons Corporation, a Dayton Superior Company; Resi-Chem Clear Cure.
 - m. Tamms Industries, Inc.; Horncure WB 30.
 - n. Unitex; Hydro Cure 309.
 - o. US Mix Products Company; US Spec Maxcure Resin Clear.
 - p. Vexcon Chemicals, Inc.; Certi-Vex Enviocure 100.
- F. Clear, Waterborne, Membrane-Forming Curing and Sealing Compound: ASTM C 1315, Type 1, Class A.
- 1. Products:

- a. Burke by Edoco; Cureseal 1315 WB.
- b. ChemMasters; Polyseal WB.
- c. Conspec Marketing & Manufacturing Co., Inc., a Dayton Superior Company; Sealcure 1315 WB.
- d. Euclid Chemical Company (The); Super Diamond Clear VOX.
- e. Kaufman Products, Inc.; Sure Cure 25 Emulsion.
- f. Lambert Corporation; UV Safe Seal.
- g. L&M Construction Chemicals, Inc.; Lumiseal WB Plus.
- h. Meadows, W. R., Inc.; Vocomp-30.
- i. Metalcrete Industries; Metcure 30.
- j. Symons Corporation, a Dayton Superior Company; Cure & Seal 31 Percent E.
- k. Tamms Industries, Inc.; LusterSeal WB 300.
- l. Unitex; Hydro Seal 25.
- m. US Mix Products Company; US Spec Radiance UV-25.
- n. Vexcon Chemicals, Inc.; Vexcon Starseal 1315.

2.08 RELATED MATERIALS

- A. Expansion-and Isolation-Joint-Filler Strips: ASTM D1751, asphalt-saturated cellulosic fiber.
- B. Semirigid Joint Filler: Two-component, semirigid, 100 percent solids, epoxy resin with a Type A shore durometer hardness of 80 or aromatic polyurea with a Type A shore durometer hardness range of 90 to 95 per ASTM D2240.
- C. Bonding Agent: ASTM C1059, Type II, non-redispersible, acrylic emulsion or styrene butadiene.
- D. Epoxy Bonding Adhesive: ASTM C881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade to suit requirements, and as follows:
 - 1. Types IV and V, load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

2.09 CONCRETE MIXTURES, GENERAL

- A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301.
 - 1. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.
- B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland cement in concrete as follows:
 - 1 Fly Ash: 25 percent.

- 2 Combined Fly Ash and Pozzolan: 25 percent.
 - 3 Ground Granulated Blast-Furnace Slag: 50 percent.
 - 4 Combined Fly Ash or Pozzolan and Ground Granulated Blast-Furnace Slag: 50 percent portland cement minimum, with fly ash or pozzolan not exceeding 25 percent.
 - 5 Silica Fume: 10 percent.
 - 6 Combined Fly Ash, Pozzolans, and Silica Fume: 35 percent with fly ash or pozzolans not exceeding 25 percent and silica fume not exceeding 10 percent.
 - 7 Combined Fly Ash or Pozzolans, Ground Granulated Blast-Furnace Slag, and Silica Fume: 50 percent with fly ash or pozzolans not exceeding 25 percent and silica fume not exceeding 10 percent.
- C. Limit water-soluble, chloride-ion content in hardened concrete to 0.06 percent by weight of cement.
- D. Admixtures: Use admixtures according to manufacturer's written instructions.
- 1 Use water-reducing, high-range water-reducing, or plasticizing admixture in concrete, as required, for placement and workability.
 - 2 Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
 - 3 Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.50.
- 2.10 FABRICATING REINFORCEMENT
- A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."
- 2.11 CONCRETE MIXING
- A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C94 and ASTM C1116, and furnish batch ticket information.
1. When air temperature is between 85 and 90°F, reduce mixing and delivery time from 1-1/2 hrs to 75 min.; when air temperature is above 90°F, reduce mixing and delivery time to 60 min.
- B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete according to ASTM C94. Mix concrete materials in appropriate drum-type batch machine mixer.
- 1 For mixer capacity of 1 cu yd or smaller, continue mixing at least 1-1/2 min., but not more than 5 min. after ingredients are in mixer, before any part of batch is released.
 - 2 For mixer capacity larger than 1 cu yd, increase mixing time by 15 sec for each additional 1 cu yd.
 - 3 Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mixture type, mixture time, quantity, and amount of water added. Record approximate location of final deposit in structure.

2.12 Grout Injection

- A. Use hydrophobic polyurethane grout, non-shrinking, water-reactive, and approved by Engineer.
- B. Contractor to clean cracks/joints, drill ports, and inject under controlled pressure to ensure complete penetration.
- C. Remove ports, patch holes, and restore surfaces after injection.
- D. Contractor shall provide a log of each injection, noting room number, location, pressure, volume, and material used.
- E. Re-inject any areas found leaking until watertight.

PART 3 – EXECUTION

3.01 FORMWORK

- A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.
- B. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.
- C. Limit concrete surface irregularities, designated by ACI 347R as abrupt or gradual, as follows:
 - 1 Class A, 1/8 in. for smooth-formed finished surfaces.
 - 2 Class B, 1/4 in. for rough-formed finished surfaces.
- D. Construct forms tight enough to prevent loss of concrete mortar.
- E. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.
 - 1 Install keyways, reglets, recesses, and the like, for easy removal.
 - 2 Do not use rust-stained steel form-facing material.
- F. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.
- G. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.
- H. Chamfer exterior corners and edges of permanently exposed concrete.
- I. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.
- J. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.
- K. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

- L. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

3.02 REMOVING AND REUSING FORMS

- A. General: Formwork for sides of beams, walls, columns, and similar parts of the Work that does not support weight of concrete may be removed after cumulatively curing at not less than 50°F for 24 hrs after placing concrete, if concrete is hard enough to not be damaged by form-removal operations and curing and protection operations are maintained.
 - 1 Leave formwork for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved at least 70 percent of its 28-day design compressive strength.
 - 2 Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.
- B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.
- C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Architect.

3.03 SHORES AND RESHORES

- A. Comply with ACI 318 and ACI 301 for design, installation, and removal of shoring and reshoring.
 - 1. Do not remove shoring or reshoring until measurement of slab tolerances is complete.
- B. In multistory construction, extend shoring or reshoring over a sufficient number of stories to distribute loads in such a manner that no floor or member will be excessively loaded or will induce tensile stress in concrete members without sufficient steel reinforcement.
- C. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

3.04 STEEL REINFORCEMENT

- A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.
- B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.
- C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars.
- D. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

- E. Install welded wire reinforcement in longest practicable lengths on bar supports spaced to minimize sagging. Lap edges and ends of adjoining sheets at least one mesh spacing. Offset laps of adjoining sheet widths to prevent continuous laps in either direction. Lace overlaps with wire.

3.05 JOINTS

- A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.
- B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Architect.
 - 1 Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints, unless otherwise indicated. Do not continue reinforcement through sides of strip placements of floors and slabs.
 - 2 Form keyed joints as indicated. Embed keys at least 1-1/2 in. into concrete.
 - 3 Locate joints for slabs in the middle third of spans.
 - 4 Use a bonding agent at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.
 - 5 Use epoxy-bonding adhesive at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.
- C. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.
 - 1 Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface, unless otherwise indicated.
 - 2 Terminate full-width joint-filler strips not less than 1/2 in. or more than 1 in. below finished concrete surface where joint sealants, specified in Division 07 Section "Joint Sealants," are indicated.
 - 3 Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.
- D. Doweled Joints: Install dowel bars and support assemblies at joints where indicated. Lubricate or asphalt coat one-half of dowel length to prevent concrete bonding to one side of joint.

3.06 CONCRETE PLACEMENT

- A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.
- B. Do not add water to concrete during delivery, at Project site, or during placement unless approved by Architect.
- C. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.
 - 1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.

- D. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.
- 1 Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
 - 2 Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
 - 3 Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 in. into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.
- E. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
- 1 Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
 - 2 Maintain reinforcement in position on chairs during concrete placement.
 - 3 Screed slab surfaces with a straightedge and strike off to correct elevations.
 - 4 Slope surfaces uniformly to drains where required.
 - 5 Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.
- F. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.
- 1 When average high and low temperature is expected to fall below 40°F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
 - 2 Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
 - 3 Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.
- G. Hot-Weather Placement: Comply with ACI 301 and as follows:
- 1 Maintain concrete temperature below 90°F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
 - 2 Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.07 FINISHING FORMED SURFACES

- A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defects repaired and patched. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
 - 1. Apply to concrete surfaces not exposed to public view.
- B. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
 - 1. Apply to concrete surfaces exposed to public view, to be covered with a coating or covering material applied directly to concrete.
- C. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces, unless otherwise indicated.

3.08 CONCRETE PROTECTING AND CURING

- A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.
- B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.
- C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.
- D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.
- E. Cure concrete according to ACI 308.1, by one or a combination of the following methods:
 - 1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:
 - a. Water.
 - b. Continuous water-fog spray.
 - c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 12 in. lap over adjacent absorptive covers.
 - 2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 in., and sealed by waterproof tape or adhesive. Cure for not less

than 7 days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

- a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
 - b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
 - c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of floor covering used on Project.
3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
- a. After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound will not interfere with bonding of floor covering used on Project.
4. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hrs later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

3.09 JOINT FILLING

- A. Prepare, clean, and install joint filler according to manufacturer's written instructions.
 1. Defer joint filling until concrete has aged at least one month(s). Do not fill joints until construction traffic has permanently ceased.
- B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.
- C. Install semirigid joint filler full depth in saw-cut joints and at least 2 in. deep in formed joints. Overfill joint and trim joint filler flush with top of joint after hardening.

3.10 CONCRETE SURFACE REPAIRS

- A. Defective Concrete: Once B3 slab and B4 and B3 foundation walls are exposed, the contractor should sound all concrete to determine the extent of required concrete repairs. Repair and patch defective areas when approved by Architect. Remove and replace concrete that cannot be repaired and patched to Architect's approval.
- B. Patching Mortar: Mix dry-pack patching mortar, consisting of one part portland cement to two and one-half parts fine aggregate passing a No. 16 sieve, using only enough water for handling and placing. For all patch repairs, normal-weight concrete materials shall be used (i.e., no lightweight concrete).
- C. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and

stains and other discolorations that cannot be removed by cleaning.

1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 in. in any dimension in solid concrete, but not less than 1 in. in depth.
 2. Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent.
 3. Fill and compact with patching mortar before bonding agent has dried.
 4. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
 5. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.
 6. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by Architect.
- D. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template. For bonded overlay installations, ensure a 1/8 in. per ft slope to drain is achieved.
- 1 Repair finished surfaces containing defects. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.01 in. wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
 - 2 After concrete has cured at least 14 days, correct high areas by grinding.
 - 3 Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
 - 4 Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
 - 5 Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 1/4 in. to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
 - 6 Sloped Bonded Overlays:
 - a. Normal-weight concrete shall be used for all sloped bonded overlays (i.e., no lightweight concrete).
 - b. Bonded overlay material shall be normal weight concrete, except where overlay thickness is less than 1.5 inches. Where less than 1.5 inches thick, SikaTop 122 Plus by Sika Corporation shall be used.
 - c. Installation shall achieve the required 1/8 in. per ft slope to drain as noted above. Prepare substrate and apply materials in accordance with manufacturer's written instructions and ACI guidelines.

- 7 Repair defective areas, except random cracks and single holes 1 in. or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least a 3/4 in. clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mixture as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
 - 8 Repair random cracks and single holes 1 in. or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hrs.
- E. Perform structural repairs of concrete, subject to Architect's approval, using epoxy adhesive and patching mortar.
- F. Additional Repair Requirements and Approvals:
1. Repair materials and installation not specified above may be used, subject to Architect's approval.
 2. Pull-off testing of bonded overlay and concrete repair materials in accordance with ASTM C1583 shall be performed. Frequency and acceptance criteria to be approved by the Architect.
- 3.11 FIELD QUALITY CONTROL
- A. Testing and Inspecting: Owner will engage a special inspector or qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.
- B. Testing and Inspecting: Engage a qualified testing and inspecting agency to perform tests and inspections and to submit reports.
- C. Inspections:
- 1 Steel reinforcement placement.
 - 2 Steel reinforcement welding.
 - 3 Verification of use of required design mixture.
 - 4 Concrete placement, including conveying and depositing.
 - 5 Curing procedures and maintenance of curing temperature.
 - 6 Verification of concrete strength before removal of shores and forms from beams and slabs.
- D. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C172 shall be performed according to the following requirements:
1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu yd, but less than 25 cu yd, plus one set for each additional 50 cu yd or fraction thereof.
 2. Testing Frequency: Obtain at least one composite sample for each 100 cu yd or fraction thereof of each concrete mixture placed each day.
 - a. When frequency of testing will provide fewer than five compressive-strength

tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.

3. Slump: ASTM C143; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
4. Air Content: ASTM C231, pressure method, for normal-weight concrete; ASTM C173, volumetric method, for structural lightweight concrete one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
5. Concrete Temperature: ASTM C1064; one test hourly when air temperature is 40°F and below and when 80°F and above, and one test for each composite sample.
6. Unit Weight: ASTM C567, fresh unit weight of structural lightweight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
7. Compression Test Specimens: ASTM C31.
 - a. Cast and laboratory cure two sets of two standard cylinder specimens for each composite sample.
 - b. Cast and field cure two sets of two standard cylinder specimens for each composite sample.
8. Compressive-Strength Tests: ASTM C39; test one set of two laboratory-cured specimens at 7 days and one set of two specimens at 28 days.
 - a. Test one set of two field-cured specimens at 7 days and one set of two specimens at 28 days.
 - b. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.
9. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor shall evaluate operations and provide corrective procedures for protecting and curing in-place concrete.
10. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength and no compressive-strength test value falls below specified compressive strength by more than 500 psi.
11. Test results shall be reported in writing to Architect, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7-and 28-day tests.
12. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by Architect but will not be used as sole basis for approval or rejection of concrete.
13. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Architect. Testing and inspecting

agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C42 or by other methods as directed by Architect.

14. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
15. Correct deficiencies in the Work that test reports and inspections indicate dos not comply with the Contract Documents.

END OF SECTION

SECTION 04 2000 - GENERAL MASONRY

PART 1 – GENERAL

1.01 GENERAL REQUIREMENTS

- A. The General Conditions of the Contract for Construction and General Requirements are hereby made a part of this Section.
- B. Refer to other Divisions of these Specifications to determine the type and extent of work therein affecting the work of this trade, whether or not such work is specifically mentioned in this Section.
- C. Provide a copy of all applicable drawings, including shop drawings, and Specifications at the site during all work covered under this Section.

1.02 SCOPE OF WORK

- A. The scope of work specified herein includes, but is not limited to, the following:
 - 1. Determine a source for, and the availability of, replacement brick to match existing brick.
 - 2. Remove existing masonry, and replace as shown on the Drawings and specified herein. Replicate existing coursing as shown in approved shop drawings.
 - 3. Cut and/or reconstruct existing masonry at vertical expansion joints where shown on the Drawings Provide a min.1/2-in joint, or wider, as required to provide sufficient movement capability.
 - 4. Coordinate masonry work with new sheet metal flashings.
 - 5. Clean the brick masonry, newly pointed mortar and cast stone on all elevations after installation.
- B. The work includes all labor, materials, equipment, and services required for completion of work under this Section, as shown on Drawings and as specified herein.
- C. Refer to Section 01 73 20 – Demolition for information regarding the removal of masonry.

1.03 COORDINATION AND SEQUENCING WITH RELATED WORK

- A. Refer to the other Divisions of these Specifications to determine the type and extent of work therein affecting the work of this trade, whether or not such work is specifically mentioned in this Section.
- B. Work in cooperation with other trades by the timely performance of the masonry work, including temporary protection, as required. Coordinate with other trades to minimize disruption to the building occupants, maintain accessibility of building entrances, and prevent leakage to the interior.
- C. Work in cooperation with other trades on this project, both under this Contract and under separate Contract(s). Other Sections of these Specifications that relate to the work of this Section include the following:

1. Section 01 7320 - NIH Selective Demolition
2. Section 03 30 00 - Cast in Place Concrete
3. Section 07 60 00 – Sheet Metal Flashing
4. Section 07 92 00 – Joint Sealant

1.04 SUBMITTALS

- A. Submit the following items in accordance with Section 01 33 00 – Submittal Procedures.

1. Back Up Brick
2. Face Brick
3. Cast Stone Cap

- B. Submit the following items to the Engineer in time to prevent delay of the project and to allow adequate time for the Engineer's review and resubmittals, if needed. Do not order materials or start work before receiving the Engineer's written approval.

1. Contractor Qualifications:

- a. The Contractor performing the work under this Section must have a minimum of five years' experience in comparable work and must submit a list, with references, of three buildings on which they worked in the last five years employing workers skilled in the restoration processes and operations indicated.
- b. List building name and address, engineer, general contractor, and appropriate subcontractors with phone numbers and contact person.

2. Product Data:

- a. Manufacturer's literature for all materials specified or proposed for use on the project, properly labeled and referenced to the appropriate Specification Section.
- b. Material Safety Data Sheets (MSDS) for each material where appropriate.
- c. Certifications by the producers of the materials that all materials supplied comply with all the requirements of the appropriate referenced standards, that all materials are compatible with adjacent materials, and that all materials are suitable for their intended purpose, except where the material is labeled with such certification by the producers of the materials.

3. Cast-stone concrete mix design from supplier.

4. Mortar mix design.

5. Shop Drawings:

- a. Existing and new brick coursing with dimensions and relative positions shown.

- b. Cast-stone layout with dimensions and relative positions shown.
 - c. Show masonry and any embedded items by related trades to ensure proper coordination of the work.
 - d. Anchor Installation: Include manufacturer specifications.
 - 6. Proposed Construction Procedures: Submit detailed written descriptions to accompany shop drawings showing methods, equipment, and sequencing for the following items:
 - a. Hot- and cold-weather construction per ACI 530.1 – Specification for Masonry Structures.
 - b. Materials, methods, or details where the Contractor proposes to deviate from the Contract plans or Specifications, if any.
 - 7. Samples:
 - a. All non-mortar materials specified or proposed for use on the project, properly labeled and referenced to the appropriate Specification Section.
 - b. Cast stone, brick, and mortar samples for product selection, color matching, mockups, and testing.
 - (1) Samples measuring 1 ft x 1 ft x 2 in. are required, showing full range of color, texture, and finish of proposed stone and brick for comparison with the appearance, color, and finish of the existing exterior masonry. The Engineer's review and acceptance of samples will be for color, texture, and pattern only. Resubmit as many samples as may be required by the Engineer until the Engineer gives final review.
 - (2) Mortar samples for color matching. Provide a range of samples for the Engineer until final review by the Engineer.
 - C. Preconstruction Testing: Prior to construction, test masonry and mortar strength to determine required performance properties for replacement materials.
 - 1. Remove samples of existing masonry and mortar from the building.
 - 2. Provide samples to a certified testing lab for strength and modulus of rupture testing and mortar composition analysis.
- 1.05 PROTECTION, HANDLING, AND STORAGE
- A. Protect mortar and other materials from deterioration by moisture and temperature. Use only dry masonry materials.
 - B. Protect all materials in original unopened labeled containers and packaging and in compliance with manufacturer's directions. Comply with manufacturer's recommendations for minimum and maximum time and temperature limits for storage.
 - C. Handle all materials to prevent damage.

- D. Keep all materials dry while they are transported, stored, and delivered. Deliver materials in manufacturer's unbroken containers. Store all materials on pallets and cover with fireproof canvas tarpaulins completely, top to bottom. Polyethylene covers are not acceptable. Store materials in a secure area designated by the Owner with adequate tie-downs against wind gusts.
- E. Store elastomeric materials, adhesives, solvents, and sealants in their original containers at temperatures recommended by manufacturers.
- F. Materials shall be marked with the date of manufacture and shelf life. Do not use products beyond the expiration of their shelf life. Store flammable materials in a cool, dry, and protected area away from sparks and open flames.

1.06 QUALITY ASSURANCE

- A. Provide full-time monitoring of the progression of the work to ensure that all items are constructed in accordance with the Drawings, Specifications, and all referenced standards. Replace deficient or rejected work at no cost to the Owner and in a manner so as to prevent delay to the project.
- B. Attend a pre-construction conference to be held with representatives of the Owner, the Contractor, the Engineer, the Masonry Foreman, and all other involved trades to discuss the work covered under this Section.
- C. Attend weekly job meetings during the course of the work as required by the Owner or Engineer.
- D. Establish vertical and horizontal controls so that the position of the replaced masonry exactly matches the original. Ensure that all joints are of the right thickness.
- E. Check the progress of masonry work twice each workday to ensure that the replaced masonry is properly installed in accordance with the Drawings, Specifications, and referenced standards.
- F. The Engineer will periodically conduct field sampling and testing of fresh mortar in accordance to ASTM C780, Test Methods A4 and A5, as paid for by the Owner. Provide fresh mortar samples as required by the Engineer at no cost to the Owner.
- G. Masonry Testing:
 - 1. Sampling and testing shall be completed following ASTM C1072 – The Standard Test Method for Measurement of Masonry Flexural Bond Strength, except as modified within.
 - 2. Provide three masonry prisms to the Engineer for testing for each stone, brick, and mortar type to be used.
 - 3. Prisms shall be constructed using the exact materials and proportions to be used during construction. Construct prisms in a protected area to prevent disturbance prior to curing.
 - 4. Each prism shall be one unit wide and six units high, with five mortar joints.
 - 5. The mortar joints shall match existing bed joints. Strike protruding mortar from joints on all faces, but only tool the joint on one face.

6. Prisms shall cure for a minimum of twenty-eight days prior to testing.
7. The Engineer will test each prism and provide recommendations for each brick and mortar type.

H. Cast Stone:

1. Three field-cut cube specimens for each of the samples shall have an average minimum compressive strength of not less than 85%, with no single specimen testing less than 75% of design strength as allowed by ACI 318.
2. Three field-cut cube specimens from each of these samples shall have an average maximum cold-water absorption of 6%.
3. Field specimens shall be tested in accordance with ASTM C1194 and C1195.
4. Test compressive strength and absorption from specimens taken from every 500 cu ft of product produced. Have tests performed by an independent testing laboratory every six months. New and existing mix designs shall be tested for strength and absorption compliance prior to producing units. Retain copies of all test reports for a minimum of two years.

1.07 MOCKUPS

- A. Notify the Engineer of Record (EOR) at least 48 hrs before construction of brick masonry mockup walls begin so that a representative may be present during construction. Mockup must be large enough to show all typical conditions, including flashing at corners.
- B. Coordinate mockup construction with other trades included in the building, such as sheet metal flashing.
- C. The mockup may be constructed on the ground at a location approved by the Owner. The mockup will be used to determine precise mortar mix, and it will be used as a standard for the subsequent masonry construction. The mockup may be constructed concurrently with the submittal and Shop Drawing approval process.
- D. Do not start masonry installation until the Engineer has approved the mockup. Rebuild the mockup as many times as required to meet the Engineer's approval. Reproduce the mockup accurately in construction using identical materials, mixtures, and quality of workmanship.

1.08 WARRANTY

- A. Guarantee all work under this Section in a document stating that if, within two years after the Date of Substantial Completion of the Work, any of the work of this Section is found to be defective or not in accordance with the Contract Documents, the Contractor shall correct it promptly after receipt of a written notice from the Owner to do so, unless the Owner has previously given the Contractor a written acceptance of such condition. State that the obligation of these Guarantees shall run directly to the Owner and may be enforced by the Owner against the Contractor, shall survive the termination of the Contract, and shall not be limited by conditions other than this Contract.

PART 2 – PRODUCTS

2.01 MANUFACTURERS

- A. Manufacturers' products and specifications are generally referred to for identification; the products of other manufacturers meeting the specifications and standards of the specified systems may be submitted for review. The burden of proof for "equal" materials is on the Contractor, who shall bear the costs and delays involved in the Engineer's review of substitutions. Check all specified items upon Contract signing and initiate submittals in time to allow early ordering so that the work is not delayed. All materials are to be new, unless designated otherwise.

2.02 MASONRY MATERIALS

- A. Water: Potable.
- B. Mason's Sand: ASTM C144, fineness modulus 2.0 to 2.5; matching the color of the sand used in the original construction.
- C. Hydrated Lime: ASTM C207, Type "S."
- D. Portland Cement: ASTM C150, Type I or II, low alkali (equivalent alkalies less than 0.6%). Type III may be used for cold-weather construction.
- E. General Mortar: ASTM C270, to match existing. Do not use ground limestone. Use the same brands of cement and lime and the same source of sand throughout the project. Do not use any admixture except those called for herein without written approval by the Engineer.
- F. Mortar Pigment: ASTM C979, do not use mortar coloring material unless specifically requested by the Engineer to obtain color match. If requested, integral coloring material shall consist of inert, non-fading, finely ground, alkali-fast mineral oxides, made especially for cement/lime mortars. Limit coloring additive so as not to exceed 10% of the weight of portland cement. Do not use carbon black as a coloring additive.
- G. Mortar for Pointing: ASTM C270, to match existing. Do not use ground limestone or prepared masonry mortar mixes. Use the same brands of cement and lime and the same source of sand throughout the project for each masonry type.
 - 1. Do not use any admixture except those called for herein, unless specifically reviewed by the Engineer.
 - 2. Mortar colors must match the corresponding final pointing mock-up.
- H. Brick: ASTM C216, Type FBA Grade SW. Brick shall match in color and be the same size as the existing brick. Brick shall have a cold water absorption between 4% and 10%. If the initial rate of absorption (suction) exceeds 20 g per 30 sq in./min., apply to the Engineer for written directions on preparing the brick for installation. All faces shall be free of cracks.
- I. Brick Ties: Stainless steel, 12 ga, DW-10 (Vee Tie) by Hohmann & Barnard, Inc. Space anchors minimum 16 in. o.c. vertically and 16 in. o.c. horizontally. Provide ties with length so that the anchor is located in the middle third of the brick.

- J. Fasteners for Cast-Stone Anchors and Brick Ties: Stainless steel Tapper screws, 3/16 in. dia., by Powers Fasteners or approved equal. Provide lengths to obtain 1-1/2 in. penetration minimum into backup.
- K. Weep Baffles: Dur-O-Wal Cell Vent weep baffles, 3/8 in. thick, 3-5/8 in. deep; height to extend the full height of the mortar joint.
- L. Setting Dowels: ASTM A167, Type 304 stainless-steel threaded rod, 3/8 in. dia.

2.03 CAST STONE COPING

- A. Replace existing cast stone coping and provide new cast stone where indicated on the Drawings.
- B. Tolerances: Maintain new cast-stone face dimensions within 1/8 in. of the face dimensions of the corresponding original piece of cast stone. All faces shall be free of cracks. All surfaces intended to be exposed to view shall have no air voids in excess of 1/32 in. Minor chipping resulting from shipment and delivery shall not be grounds for rejection. Minor chips shall not be obvious under direct daylight illumination from a 20 ft distance.
- C. Cast Stone Coping: Cast stone shall match the existing in color and texture and shall be modified as needed to match existing sizes. Fabricate each new piece of cast stone to match the existing cast stone it is replacing as closely as possible. Provide cast stone to meet the following properties:
 - 1. Compressive Strength (ASTM C1194): 6,500 psi minimum at twenty-eight days.
 - 2. Absorption (ASTM C1195): 6% maximum by the cold-water method or 10% maximum by the boiling method at twenty-eight days.
 - 3. Air Content (ASTM C173 or C231): Wet cast product shall be 4% to 8%.
 - 4. Freeze-Thaw (ASTM C1364): The CPWL shall be less than 5% after 300 cycles of freezing and thawing.
 - 5. Linear Shrinkage (ASTM C426): Shrinkage shall not exceed 0.065%.
- D. Portland Cement: ASTM C150, Type I or II, low alkali (equivalent alkalies less than 0.6%). Type III may be used for cold-weather construction. Grey and/or white to match existing.
- E. Coarse Aggregates: Granite, quartz, or limestone, ASTM C33, except for gradation, and are optional for the VDT casting method.
- F. Fine Aggregates: Manufacturer or natural sands, ASTM C33, except for gradation.
- G. Colors: Inorganic iron oxide pigments, ASTM C979 except that carbon black pigments shall not be used.
- H. Admixtures shall comply with the following:
 - 1. ASTM C260 for air-entraining admixtures.
 - 2. ASTM C494/C405M Types A-G for water-reducing, retarding, accelerating and high-range admixtures.

3. Other Admixtures: Integral water repellents and other chemicals, for which no ASTM standard exists, shall be previously established as suitable for use in concrete by proven field performance or through laboratory testing.
 4. ASTM C618 mineral admixtures of dark and variable colors shall not be used in surfaces intended to be exposed to view.
 5. ASTM C989 granulated blast furnace slag may be used to improve physical properties. Tests are required to verify these features.
- I. Water: Potable
 - J. Curing: Cure units in a warm curing chamber approximately 100°F at 95% relative humidity for approximately 12 hrs, or cure in a 95% moist environment at a minimum of 70°F for 16 hrs after casting. Additional yard curing at 95% relative humidity shall be 350 degree-days prior to shipping. Form cured units shall be protected from moisture evaporation with curing blankets or curing compounds after casting.
 - K. Epoxy Adhesive for Cast Stone Repair: Exterior grade, two-part epoxy adhesive by Bonstone or equivalent.

PART 3 – EXECUTION

3.01 EXAMINATION

- A. Examine conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of work.
 1. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of work.
- B. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of penetrations.
- C. Do not proceed with masonry installation until all associated waterproofing and flashing work has been completed; unsatisfactory conditions have been corrected; and the Engineer has approved the area for installation.

3.02 GENERAL PROCEDURES

- A. Thickness: Build cavity walls and other masonry construction to full thickness shown on the Drawings.
- B. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match the construction immediately adjacent to opening.
- C. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws to obtain true, even, and undamaged edges. Do not use overburned or oil- or grease-marked masonry. Do not use masonry units with cracks or splits in any face, or with chips extending more than 1/8 in. from edges or 1/4 in. from corners. Do not break masonry with a mason's hammer. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed. Do not use units cut to less than one-half size.

- D. Select and arrange units for exposed unit masonry to produce a uniform blend of colors and textures. Mix units from several pallets or cubes as they are placed.
- E. General Masonry Procedures:
1. Masonry workmanship shall comply with all applicable recommendations of the Brick Institute of America and ACI 530.1 – Specifications for Masonry Structures, except as modified below. Report any damage to new flashing or waterproofing within the work area to the Engineer and provide for repairs by appropriately skilled mechanics, at no cost to the Owner.
 2. Masonry construction should match the existing masonry construction, including profiles and colors. Any unspecified variances from the existing construction will be cause for rejection.
 3. Hot Weather:
 - a. Above 80°F: Protect the masonry and mortar from direct sunlight and exposure to wind to avoid rapid evaporation of water in the mortar before, during, and after masonry construction.
 - b. Above 90°F: Do not use mortar when masonry surface temperature is above 90°F. Protect the masonry and mortar from direct sunlight and exposure to wind to avoid rapid evaporation of water in the mortar before, during, and after masonry construction.
 4. Cold Weather (below 40°F): Follow procedures given by ACI 530.1 – Specifications for Masonry Structures. Do not work in average daily temperatures below 40°F without providing cold-weather protection as described in ACI 530 and outlined in the table below. Continue to operate heaters overnight with appropriate supervision. Do not use heaters that produce oily deposits on the masonry. If any oily deposits occur, consult with the Engineer to determine how best to remove oily deposits and remove at the Contractor's expense. Preheat all mortar ingredients, i.e., sand, cement, lime, and water, to 70°F before mixing. Preheat the bricks to a minimum of 40°F before laying.

Temp.	WORK IN PROGRESS			COMPLETED WORK
	Masonry	Mortar	Assemblage	Assemblage
Above 40°F	No requirements.	No requirements.	No requirements.	No requirements.
40°F to 25°F	Remove visible ice using hot-air blowers.	Heat during mixing to between 40°F and 120°F. Maintain above freezing while in use.	No requirements.	Protect masonry with a weather-resistive cover for 24 hrs after construction. Completely cover masonry when temp. is less than 32°F.
25°F to 20°F	Remove visible ice.	Heat during mixing to between 40°F and 120°F. Maintain above	Use heat sources on both sides of wall. Install wind breaks when velocity is over 15 mph.	Completely cover with insulated blanket for 24 hrs after construction.

Temp.	WORK IN PROGRESS			COMPLETED WORK
	Masonry	Mortar	Assemblage	Assemblage
		freezing while in use.		
Below 20°F	Heat to above 20°F; remove visible ice.	Heat during mixing to between 40°F and 120°F. Maintain above freezing while in use.	Provide an enclosure and use heat sources to maintain temp. above 32°F within the enclosure.	Provide an enclosure and use heat sources to maintain temp. above 32°F within the enclosure.

5. Mix mortar using sufficient quantity of water to ensure good workability in accordance with BIA recommendations. Control batching procedures to ensure proper proportions by measuring materials by volume. Measure materials using volumetric device acceptable to the Engineer, not by shovel. Use volumetric dispenser for color admixtures where used. Mix with mechanical mixer only for at least 3 min., but not more than 5 min. Use only workable mortar. Limit to the extent practical, and only retemper mortar on the board or in the tub one time. Completely clean remaining mortar off a board or tub before adding fresh mortar. Use mortar within 2 hrs of initial mixing at temperatures over 80°F.
6. Do not use admixtures of any kind in mortar, unless specifically reviewed by the Engineer.

F. Brick Installation Procedures:

1. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thickness and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less than half size units, particularly at corners and jams.
2. Lay all masonry plumb and true, with uniform and level joints. Coursing and patterns to match the existing masonry construction and the approved shop drawings. Any variances from the existing construction not specified will be cause for rejection.
3. Lay the brick masonry following the original coursing and pattern and within the tolerances defined below:
 - a. External Corners and Other Conspicuous Lines and Levels: Maximum deviation from plumb or level $\pm 1/4$ in. in any 10 ft section with a maximum cumulative amount of $3/8$ in. in any one direction beyond 10 ft.
 - b. Mortar Bed Joint Thickness: Match existing mortar bed joint thickness.
 - c. Mortar Head Joint Thickness: Match existing mortar head joint thickness.
4. Lay solid masonry units with completely filled bed and head joints; butter ends with sufficient mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.
5. Tool exposed joints when thumbprint hard to match existing.
6. Keep cavities clean of mortar droppings and other materials during construction. Bevel beds away from cavity to minimize mortar protrusions into cavity. Do not attempt to trowel or remove mortar fins protruding into cavity.

7. Provide cleanouts in the bottom course of masonry every third brick to remove all mortar from the cavity. Install bricks in cleanouts once construction of the wall is complete and all mortar is removed from the cavity.
8. Coordinate and allow access for sheet metal flashing to be installed in cavity.

G. Cast-Stone Installation Procedures:

1. Kerf the top of each stone and insert.
2. Do not lay bed mortar more than 2 ft ahead of work. Do not furrow the mortar in the bed joint to a depth that results in voids when the units are in their final position. Match existing mortar joint thicknesses on the tower. Completely butter the faces of the masonry units and shove them into place so that mortar squeezes out of the top and sides of the joints requiring mortar to produce well-compacted joints of full width, front to back. Do not slush head joints. Adjust the final position of each masonry unit while the mortar is plastic. To adjust the position of the masonry unit after the mortar has stiffened, remove the unit, replace stiffened mortar with plastic mortar, and replace the unit. At closure units, butter all surfaces of the unit to be placed as well as the abutting surfaces of the existing unit.
3. Strike exterior mortar joints in a manner consistent with the existing joints (as they existed prior to degradation) during laying. When mortar is thumbprint hard on exposed surfaces, tool joints to match existing.
4. Slightly bevel bed joint mortar away from the cavity space before placing the unit to minimize mortar protrusions into any cavity space intended to be free of mortar. "Roll" unit into place to reduce amount of mortar oozing from bed joint into cavity. Back parge or strike mortar extrusions in the cavity space.
5. At all flashings, set masonry on a mortar bed. Provide solid mortar on both sides of flashing end dams that turn up into head joints. Provide specified weep holes within 12 in. to 16 in. of flashing end dams.
6. Do not stain stone face or any surface with mortar drippings or other substances. Dry-brush newly completed masonry and exposed flashing surfaces daily to clean and remove mortar before it can set.
7. Protect uncompleted masonry walls, whenever work is not being performed, by well-fastened waterproof coverings. Protect flashings against any penetration or other damage. Protect base of wall to prevent rain-splashed mud or mortar drippings from contacting masonry.

H. Masonry Pointing Procedures

1. Procedure for removing existing mortar shall be decided through completion of mock-up and review process. Areas of mortar removal will be decided at the completion of the mortar survey.
2. Remove existing mortar by cutting down the center of the joint with power tools and then removing remaining mortar with hand tools. To prevent joint enlargement, do not chip or cut into adjacent masonry. Do not overcut masonry.

3. Remove mortar from joints scheduled for pointing to depths equal to 2-1/2 times their widths, but not less than 5/8 in. Leave no cavities in the existing mortar and remove all deteriorated mortar.
4. Remove mortar from masonry surfaces within raked-out joints to provide reveals with square backs and exposed masonry sides for contact with pointing mortar.
5. Rinse joint surfaces with water to remove dust and mortar particles. Time application of rinsing so that, at time of pointing, excess water has evaporated or run off, and joint surfaces are damp but free of standing water.
6. Apply first later of pointing mortar to areas where existing mortar was removed to depths greater than surrounding areas. Apply mortar in layers of equal depths, a maximum of 3/8 in. thick, until joint is full. Compact each layer thoroughly and allow each layer to become thumbprint-hard before applying next layer. Allow at least 24 hrs to pass between successive stages of mortar application to allow for mortar shrinkage between stages. Where existing masonry may have rounded edges, recess final layer slightly from face. Take care not to spread mortar over edges onto exposed surfaces.
7. When mortar at exterior face is thumbprint hard, tool to form the specified joint type, with edges flush with the face of the masonry. Tool using a flat tool slightly larger than the joint size to compact mortar against masonry joint surface. Remove excess mortar from edge of joint by brushing. When areas of newly pointed mortar are required to blend in with adjacent existing weathered mortar, consult with the Engineer to determine how best to obtain a uniform appearance. Joint geometry to match existing joint geometry.
8. Do not wash the newly pointed mortar.

3.03 INSTALLATION OF BRICK TIES

- A. Install ties in accordance with the manufacturer's current recommendations, except as modified herein.
- B. Locate anchors on the middle third of the stone bed joint, but recess ends of anchors at least 5/8 in. from exterior masonry surfaces. Space anchors 24 in. o.c. vertically and 16 in. o.c. horizontally. Mortar shall be installed to masonry prior to setting of anchor. Place all anchors so that they are completely surrounded by mortar. Direct masonry-to-anchor contact is not permitted. Lateral free movement of the anchor should be limited to less than 0.05 in.

3.04 INSTALLATION OF CAST-STONE ANCHORS

- A. Install anchors in accordance with the manufacturer's current recommendations, except as modified herein. Do not penetrate new flashing with fasteners.
- B. Kerf the top and bottom of each piece of cast stone in the middle third of the depth and at quarter points and insert stone anchors. Place all anchors so that they are completely surrounded by mortar. Direct stone-to-anchor contact is not permitted. Lateral free movement of the anchor is not permitted. Completely fill the kerfs around the anchors with mortar.
- C. Attach stone anchors to the backup using specified fasteners.

3.05 CLEANING PROCEDURES

- A. General Masonry Cleaning

1. Conduct all masonry work in a neat and competent manner to prevent staining any surface with mortar or other spills. Avoid dropping mortar on completed brick work or other elements of the building. If mortar drops or spills on a nonporous material, spot clean immediately using a sponge and clean water. Brush down recently installed exposed masonry units with a soft bristle brush while mortar is hard enough not to smear, but soft enough not to bond tightly to the face of the units.
 2. Clean all masonry work promptly after curing by wetting surfaces and washing with a stiff bristle brush to produce a clean and unmarred appearance. Begin cleaning with hot soapy water first, without chemical cleaners. If hot soapy water alone with a scrub brush is not successful, as determined by the Engineer, use an approved cleaning compound. Dilute the compound with the maximum amount of water that will allow proper cleaning, as approved by the Engineer.
 3. Test the cleaning compound in a small inconspicuous area before beginning full scale cleaning. Cover all existing exposed metal, glass, and other areas as required before cleaning the masonry with any cleaning compounds. Do not use metal scrapers to clean the masonry. Rinse repeatedly with clean water after cleaning to remove all traces of mortar and debris. Protect all exterior finishes, including glass and aluminum, from any damage or staining caused by this work.
 4. Protect all trees and nearby vegetation from wash down water and chemicals used during wash down.
- B. Cleaning for Newly-Pointed Mortar
1. After mortar has fully hardened, and within thirty days, thoroughly clean exposed masonry surfaces of excess mortar and foreign matter using stiff nylon or bristle brushes and clean water that is spray-applied at low pressure.
 2. Do not use metal scrapers or brushes.
 3. Do not use acid or alkali cleaning agents.

3.06 REGLETS

- A. Saw cut reglets in the back-up masonry at mortar joints to accept mechanically attached flashing where shown on the Drawings. Coordinate the exact placement of reglets with flashing installer. Unless noted otherwise on the Drawings, saw cut reglets 1/2 in. wide x 1 in. deep, and slope saw cut down slightly so that open end is lower. Fasten flashing and cover with sealant joint, as shown in Drawings.

END OF SECTION

SECTION 05 5210 - RAILINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section includes the following:
 - 1. Stainless-steel railings.

1.2 PERFORMANCE REQUIREMENTS

- A. Structural Performance: Provide railings capable of withstanding the effects of gravity loads and the following loads and stresses within limits and under conditions indicated:
 - 1. Comply with ASTM E 985, based on testing per ASTM E 894 and ASTM E 935.
 - 2. Capable of withstanding structural loads required by ASCE 7 without exceeding allowable design working stresses of materials involved.
 - 3. Top Rails of Guards:
 - a. Uniform load of 50 lbf/ ft. (0.73 kN/m) applied horizontally and concurrently with uniform load of 100 lbf/ft. (1460 N/m) applied vertically downward..
 - b. Concentrated load of 200 lbf (0.89 kN) applied in any direction.
 - c. Uniform and concentrated loads need not be assumed to act concurrently.
 - 4. Infill of Guards:
 - a. Concentrated load of 200 lbf 890 N) applied horizontally on an area of 1 sq. ft. (0.093 sq. m).
 - b. Uniform load of 25 lbf/sq. ft. (1.2 kN/sq. m) applied horizontally.
 - c. Infill load and other loads need not be assumed to act concurrently.
- B. Control of Corrosion: Prevent galvanic action and other forms of corrosion by insulating metals and other materials from direct contact with incompatible materials.

1.3 SUBMITTALS

- A. Product Data: For mechanically connected railings, grout and anchoring cement products.
- B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
 - 1. For installed products indicated to comply with design loads, include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
- C. Samples: For each exposed finish required.
- D. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, according to ASTM E 894 and ASTM E 935.

1.4 WARRANTY

- A. See Section 01 7700 - Closeout Submittals, for additional warranty requirements.
- B. Correct defective work within a five year period after Date of Substantial Completion.
- C. Special Installer's Warranty: Installer agrees to repair or replace safety railings that do not

comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. Stainless-Steel Railings:
 - a. Blum, Julius & Co., Inc.
 - b. AGS Stainless, Inc
 - c. Sterling Dula Architectural Products, Inc.

2.2 METALS

- A. Brackets, Flanges, and Anchors: Cast or formed metal of same type of material and finish as supported rails, unless otherwise indicated.
- B. Stainless Steel:
 1. Castings: ASTM A 743/A 743M, Grade CF 8 or CF 20 with Type 304; or CF 8M or CF 3M with type 316 or 316L.
 2. Plate and Sheet: ASTM A 666, Type 304 or 316L.
 3. Expanded Metal: ASTM F 1267, Type I (expanded) or II (expanded and flattened), Class 3 (corrosion-resistant steel), ASTM A 666, Type 304 or 316.

2.3 MISCELLANEOUS MATERIALS

- A. Fasteners: Provide concealed fasteners, unless unavoidable or standard for railings indicated; same basic metal as fastened metal.
 1. Stainless-Steel Railings: Type 304 or 316 stainless-steel fasteners.
- B. Anchors: Provide cast-in-place, chemical, or torque-controlled expansion anchors, fabricated from corrosion-resistant materials with capability to sustain, without failure, as determined by testing per ASTM E 488.
- C. Welding Rods and Bare Electrodes: Select according to AWS specifications for metal alloy welded.
- D. Grout and Anchoring Cement: Factory-packaged, nonshrink, nonmetallic grout complying with ASTM C 1107; or water-resistant, nonshrink anchoring cement; recommended by manufacturer for exterior use.

2.4 FABRICATION

- A. General: Fabricate railings to comply with design, dimensions, and details indicated, but not less than that required to support structural loads.
- B. Welded Connections: Cope components at connections to provide close fit, or use fittings designed for this purpose. Weld all around at connections, including at fittings.
- C. Nonwelded Connections: Connect members with concealed mechanical fasteners and fittings.

- D. Form changes in direction by bending or by inserting prefabricated elbow fittings.
- E. Form curves by bending in jigs to produce uniform curvature; maintain cross section of member throughout bend without cracking or otherwise deforming exposed surfaces.
- F. Close exposed ends of railing members with prefabricated end fittings.
- G. Provide wall returns at ends of wall-mounted handrails, unless otherwise indicated.
- H. Brackets, Flanges, Fittings, and Anchors: Provide wall brackets, flanges, miscellaneous fittings, and anchors to interconnect railing members to other work, unless otherwise indicated.

2.5 FINISHES

- A. Stainless Steel:
 - 1. 320-Grit Polished Finish: Oil-ground, uniform, smooth finish with non-slip treatment for enhanced durability, corrosion resistance, and safety in exterior applications.

2.6 INSTALLATION

- A. General: Perform cutting, drilling, and fitting required for installing railings. Set railings accurately in location, alignment, and elevation.
- B. Anchor railing ends to exiting metal door frame with flanges connected to railing ends.
- C. Adjusting and Cleaning:
 - 1. Immediately after erection, clean field welds, bolted connections, and abraded areas of shop paint, and paint exposed areas with the same material as used for shop painting.

END OF SECTION 05 5210

SECTION 07 1113 - SELF-ADHERING SHEET WATERPROOFING

PART 1: GENERAL

1.01 GENERAL REQUIREMENTS

- A. The General Conditions, the Supplementary Conditions, the Instructions to Bidders and Division One General Requirements shall be read in conjunction with and govern this section.
- B. The Specification shall be read as a whole by all parties concerned. Sectioning of the Specification is for convenience. Each Section may contain more or less than the complete work of any trade. The Contractor is solely responsible to make clear to the Subcontractors the extent of their work.

1.02 DESCRIPTION

- A. Supply labor, materials, plant, tools and equipment to complete the Work as shown on the Drawings and as specified herein including, but not limited to the following:
 - 1. Foundation Walls (by others),
 - 2. Primer & Self-Adhered SBS Modified Asphalt Waterproofing Membrane,
 - 3. Drain Board/Protection Board,
 - 4. Specified Backfill.

1.03 RELATED WORK

- A. Division 3 Concrete
- B. Division 4 Masonry
- C. Division 7 THERMAL AND MOISTURE PROTECTION
- D. Division 31 EARTHWORK

1.04 REFERNECES

- A. CAN/CGSB-37.9M: Primer, Asphalt, Unfilled for Asphalt Roofing, Dampproofing and Waterproofing.

1.05 SUBMITALS

- A. Prior to commencing the Work, submit copies of manufacturers current certification to ISO. Membrane, primers, sealants, adhesives and associated auxiliary materials shall be included.
- B. Prior to commencing the Work, submit references clearly indicating that the materials proposed have been installed for not less than fifteen years on projects of similar scope and nature. Submit references for a minimum of ten projects.
- C. Prior to commencing the Work submit manufacturers complete set of standard details for waterproofing systems.

1.06 Quality Assurance

- A. Perform Work in accordance with the printed requirements of the membrane manufacturer and this specification. Advise designer of any discrepancies prior to commencement of the Work.
- B. Maintain one copy of manufacturers literature on site throughout the execution of the Work.
- C. At the beginning of the Work and at all times during the execution of the Work, allow access to site by the waterproofing membrane manufacturers representative.
- D. Materials used in this Section, including, primers, mastics and membranes, asphaltic protection boards, composite drainage boards and expansion joint membranes shall be

- fully compatible and shall be sourced and or produced by one manufacturer.
- E. Submit copies of the membrane manufacturers current ISO certification including the manufacturing of the membrane, primer, mastics, adhesives and protection board.
- 1.07 DELIVERY, STORAGE AND HANDLING
- A. Deliver materials to the job site in undamaged and original packaging indicating the name of the manufacturer and product.
 - B. Cold applied elastomeric membrane should be stored in closed containers outdoors.
 - C. Store membrane at temperature of 40 degrees F and above to facilitate handling.
 - D. Membrane contains petroleum solvents and are flammable. Do not use near open flame.
 - E. Store role materials horizontally in original packaging.
 - F. Store adhesives and primers at temperatures of 40 degrees F and above to facilitate handling.
 - G. Keep solvents away from open flame or excessive heat.
- 1.08 CO-ORDINATION
- A. Ensure continuity of the waterproofing membrane throughout the scope of this section.
 - B. Work shall be so scheduled as to provide a watertight seal at the end of each working day on the areas worked upon during the day.
- 1.09 SITE CONDITONS
- A. Environmental Requirements
 - 1. No installation work shall be performed during rainy or inclement weather and on frost or wet covered surfaces.
 - B. Protection
 - 1. Provide adequate protection of materials and work of this section from damage by weather backfilling operations and other causes.
 - 2. Protect work of other trades from damage resulting from work of this section. Make good such damage at own expense to satisfaction of the consultant.
 - 3. Apply protection board as soon as possible after installation of membrane.
- 1.10 ALTERNATES
- A. Submit requests for alternates.
 - B. Alternate submission format to include:
 - 1. Submit evidence that alternate materials meet or exceed performance characteristics of Product requirements and documentation from an approved independent testing laboratory certifying that the performance of the waterproofing membrane system including drain boards and transition sheets, exceed the requirements of the National Building Code.
 - 2. Submit copies of manufacturers' current ISO certification.
 - 3. Submit references clearly indicating that the membrane manufacturer has successfully completed projects on a annual basis of similar scope and nature for a minimum of fifteen years.
 - 4. Submit manufacturers' complete set of standard details for the waterproofing membrane systems showing a continuous plane of water tightness throughout the building envelope.
 - C. Submit requests for alternate to this specification a minimum of fifteen (10) working days prior to tender closing for evaluation. Include a list of 25 projects executed over the past fifteen years.
 - C. Acceptable alternates will be confirmed by addendum. Substitute materials not approved in writing prior to tender closing shall not be permitted for use on this project.
- 1.11 Warranty
- A. For the Work of this Section, the 12 months warranty period prescribed in subsection GC 32.1 of General Conditions "C" is extended to 60 months.
 - B. Contractor hereby warrants the waterproofing membrane leak coverage in accordance

with GC24, but for two years.

- C. Waterproofing membrane manufacturer hereby warrants that the waterproofing membrane for leak coverage as a result of faulty materials for a period of ten years for single ply without Henry DB Drain Board). Scope of warranty shall include materials required to return the membrane to a watertight condition.

PART 2: PRODUCTS

2.01 MATERIALS

- A. Waterproofing membrane components and accessories must be obtained as a single-source from the membrane manufacturer to ensure total system compatibility and integrity.

1. Acceptable Manufacturer: Henry Company.
909 N Sepulveda Blvd, Suite 650
El Segundo, CA 90245
(800) 598 7663
Web Site: www.Henry.com

2.02 WATERPROOFING MEMBRANE (Basis-of-Design)

- A. Primary sheet applied self-adhered waterproofing membrane shall be Blueskin® WP200 manufactured by Henry, 1.5mm (60 mils) SBS modified bitumen, self-adhering sheet membrane with a cross-laminated polyethylene film, and having the following physical properties:
1. Thickness: 1.5 mm (60 mils) min.,
 2. Flexibility: Pass @ -40 degrees C to ASTM D1970,
 3. Vapour permeance: 2.8 ng/Pa.s.m² (0.05 perms) to ASTM E96,
 4. Tensile strength (membrane): 2.24 MPa to ASTM D412,
 5. Tensile strength (film): 34.5 MPa to ASTM D882,
 6. Elongation: 300% to ASTM D412,
 7. Puncture resistance: 222 N min. to ASTM E154.

2.03 PRIMER

- A. Primer for self-adhering membranes at temperatures above 25 degrees F shall be Aquatac™ Primer manufactured by Henry, a polymer emulsion based adhesive, quick setting, having the following physical properties:
1. Colour: Aqua;
 2. Weight: 8.7 lbs/gal;
 3. Solids by weight: 53%;
 4. Water based, no solvent odours
 5. Drying time (initial set): 30 minutes at 50% RH and 70 degrees F;

2.04 LIQUID MEMBRANE & TERMINATION SEALANT

- A. Termination Sealant shall be HE925 BES Sealant manufactured by Henry; a moisture cure, medium modulus polymer modified sealing compound having the following physical properties:
1. Compatible with sheet air barrier, roofing and waterproofing membranes and substrate,
 2. Complies with Fed. Spec. TT-S-00230C, Type II, Class A
 3. Complies with ASTM C 920, Type S, Grade NS, Class 25
 4. Elongation: 450 – 550%
 5. Remains flexible with aging
 6. Seals construction joints up to 1 inch wide

2.05 PREFABRICATED DRAIN BOARDS: Two part prefabricated geocomposite drain board

consisting of a formed polystyrene or PVC core covered on one side with a woven or non-woven polypropylene filter fabric:

- A. Henry DB 200: For vertical and horizontal installations, shallower depths.
- B. Henry DB 500: For vertical installations requiring high compressive strength and high flow capacity.
- C. Henry DB 650: For horizontal applications requiring high compressive strength, high flow capacity & woven geotextile. Suitable for use under topping slab in split slab applications.

2.06 PREFABRICATED DRAIN BOARD ACCESSORIES

- A. Securement Bars: Continuous 1/4 inch x 3/4 inch HDPE bar for screw attachment.
- B. Moulding Strip: Continuous 3 1/2 inch wide 'Z' flashing strip to fit over exposed top edge of drain board.
- C. Drain Board Plugs & Nails: HDPE pre-moulded washer to fit dimples c/w high strength, corrosion resistant concrete nails, UCAN AFH 37 or equal.

2.07 INSULATION

- A. Minimum thermal resistance (R-Value): Project specific Minimum R-Value
- B. Compressive Strength: 100 psi.
- C. Insulation supplied by Henry a Carlisle Company
 - 1. Accepted Manufacturers:
 - a. Extruded polystyrene (XPS) - 2" minimum thickness
 - 1. DOW®
 - 2. Owens Corning®

PART 3: EXECUTION

3.01 EXAMINATION

- A. Verify that surfaces and conditions are ready to accept the Work of this section. Commencement of the work or any parts thereof shall mean acceptance of the prepared substrate.

3.02 PREPARATION

- A. All surfaces must be sound, dry, clean and free of oil, grease, dirt, excess mortar, frost or other contaminants. Fill spalled areas in substrate to provide an even plane.
- B. New concrete should be cured for a minimum of 7 days and must be dry before waterproofing membranes are applied. Lightweight structural concrete must be cured a minimum of 14 days.
- C. Use appropriate waterproofing membrane primer as recommended by manufacturer based on air and surface temperature at time of application.

3.03 PRIMER

- A. Apply primer for self-adhered membrane by roller or spray at rate recommended by manufacturer.
- B. Allow minimum 30 minute open time. Primed surfaces not covered by waterproofing membrane during the same working day must be re-primed.

3.04 FOOTING/FOUNDATION WALLS, JUNCTURES, CRACKS IN SLAB AND PROTRUSIONS

- A. Coat penetrations, such as brackets, clips, braces, etc. that are set into the concrete with a 90 mil coating of liquid membrane to the height of the wearing course and around projections to ensure a complete seal prior to coating the entire area.
- B. To all cracks and cold joints less than 1/16 inch, apply a coat of liquid membrane at a minimum thickness of 30 mils extending 3 inches on either side of joint, embed a 6 inch wide strip of primary self-adhered waterproofing membrane over.
- C. To all cracks greater than 1/8 inch, fill void with non-shrink cementitious patching material and allow to cure dry. Prime area and install self-adhered waterproofing membrane, extend 3 inches on either side of crack. Overlap end joint of sheet a minimum 3 inches.
- D. At monolithic and non-monolithic wall/slab junctures, prime area, trowel-in fillet bead of liquid membrane to inside corners and install self-adhered waterproofing membrane

sheet to the required height on the wall and at least 4 inches on the slab. Lap primary waterproofing membrane over a minimum of 2 inches.

- E. Horizontal to vertical inside corner transition areas are to be pre-treated with a liquid membrane fillet extending 3/4 inch vertically and horizontally from the corner. Apply a minimum 10 inch strip of self-adhered waterproofing membrane centred at the joint.
- F. All outside corners are to be pre-treated with a minimum 10 inch strip of waterproofing membrane centred at the joint.
- G. Where three or more planes come into contact reinforce with cut sections of waterproofing membrane reinforcing sheet as per manufacturer's instructions.

3.05 PROJECTIONS

- A. Extend waterproofing membrane tight to projection and seal with liquid membrane extending 3 inches along projection and 3 inches onto waterproofing membrane.

3.06 WATERPROOFING MEMBRANE - VERTICAL APPLICATIONS

- A. Apply waterproofing membrane to prepared substrate in lengths of 6 feet or less.
- B. Provide 3 inch laps at both sides and ends. Position for alignment and remove protective film. Press firmly into place. Promptly roll all laps with a counter top roller to effect seal. If more than one length is required on a vertical surface, apply in a shingle fashion.
- C. Terminate membrane using termination mastic or termination bar, reglet or counter flashing as indicated. Refer to manufacturers standard details.
- D. All laps within 12 inches of a 90 degrees change in plane are to be sealed with termination sealant.

3.07 WATERPROOFING MEMBRANE - HORIZONTAL APPLICATIONS

- A. Apply 2 plies of waterproofing membrane to prepared substrate in lengths of 6 feet or less.
- B. Provide 3 inches laps at both sides and ends. Position for alignment and remove protective film. Press firmly into place. Promptly roll all laps with a counter top roller to effect seal. If more than one length is required on a vertical surface, apply in a shingle fashion.
- C. Terminate membrane using termination mastic or termination bar, reglet or counter flashing as indicated. Refer to manufacturers standard details.
- D. All laps within 12 inches of a 90 degrees change in plane are to be sealed with termination sealant.

3.08 Installation of Protection Board

- A. Protection Boards shall be installed over the waterproofing membrane to prevent damage from materials used in backfilling.
- B. Apply protection board adhesive in 1 inch wide strips spaced at 16 inches o/c to cure waterproofing membrane. Immediately embed protection board and press into adhesive to ensure full contact.
- C. Do not backfill until adhesive has cure dried. Do not use excessive levels of adhesive.

3.09 Installation of Drainage Board (Vertical)

- A. Align and hang drainage up to foundation wall. Position bottom edge of drainage board to be in moderate contact with weeping tile system.
- B. Secure drainage board to foundation wall with nails and washers spaced 450 mm o/c horizontally. Install minimum of 2 rows staggered and spaced 6 inches apart and min 6 inches from top edge.
- C. Align and install termination strip along top edge with nails spaced 12 inches o/c and seal with termination sealant.
- D. Align and install moulding strip over completed top edge detail.
- E. Overlap end laps, pull back loose fabric to expose drain core and position core of second panel over the overlap flange of first panel.
- F. Bend drain board to create inside corners and cut board to create outside corners, provide 4 inches of extra fabric to wrap corner.

- G. Stagger or offset joints of drain board sheets.
- H. Place all subsequent sheets in an overlapping single fashion.
- I. Backfill bottom edge in conjunction with weeping tile system.

3.10 Installation of Drainage Board (Horizontal)

- A. The edge of the core flange shall be at the higher edges of the substrate, away from drains.
- B. Overlap in the direction of water flow. Pull back loose fabric to expose drain core and position core of second panel over the overlap flange of first panel.
- C. Bend drain board to create inside corners and cut board to create outside corners, provide 3 inch of extra fabric to wrap corner.
- D. Stagger or offset joints of drain board sheets.
- E. Place all subsequent sheets in an overlapping single fashion.

3.11 CLEAN-UP

- A. Promptly as the work proceeds and on completion clean up and remove from site all rubbish and surplus materials resulting from the foregoing work.

3.12 PROTECTION

- A. Protect waterproofing membrane and drain board work from other trades during construction.
- B. Backfill with specified materials, protect membrane from damage.

END OF SECTION

SECTION 07 1413 - HOT FLUID-APPLIED RUBBERIZED ASPHALT WATERPROOFING

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section Includes
 - 1. Hot rubberized asphalt assembly for the protection and waterproofing of a plaza deck.
- B. Related Requirements
 - 1. Section 07 62 00 – Sheet Metal Flashing and Trim
 - 2. Section 07 92 00 – Joint Sealants

1.2 REFERENCES

- A. Reference Standards
 - 1. American Society for Testing and Materials (ASTM)
 - a. ASTM D41M – Asphalt Primer used in Roofing, Dampproofing, and Waterproofing.
 - b. ASTM D92 – Standard Test Method for Flash and Fire Points by Cleveland Open Cup
 - c. ASTM D3407 – Standard Test Method for Joint Sealants, Hot Poured, for Concrete and Asphalt Pavements.
 - d. ASTM D4263 - Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
 - e. ASTM D5329 – Standard Test Method for Sealants and Fillers, Hot-Applied for Joints and Cracks in Asphaltic and Portland Cement Concrete Pavements.
 - f. ASTM D5957 – Standard Guide for Flood Testing Horizontal Waterproofing Installations
 - g. ASTM D7877 – Standard Guide for Electronic Methods for Detecting and Locating Leaks in Waterproof Membranes
 - h. ASTM E96 – Water Vapor Transmission of Materials
 - 2. Canadian General Standards Board (CGSB)
 - a. CAN/CGSB-37.50-M89 – Standard for Asphalt, Rubberized, Hot Applied, for Roofing and Waterproofing
 - 3. Factory Mutual (FM):
 - a. FM4470 – Single-Ply, Polymer-Modified Bitumen Sheet, Built-Up Roof (BUR) and Liquid Applied Roof Assemblies for use in Class 1 and Noncombustible Roof Deck Construction
 - 4. International Concrete Repair Institute (ICRI)
 - a. ICRI Technical Guidelines number 310-2R-2013
 - 5. Underwriters Laboratories (UL)
 - a. UL 790 – Fire Tests of Roof Coverings

1.3 ADMINISTRATIVE REQUIREMENTS

- A. Coordination
 - 1. Do not allow access to installation areas by other trades during waterproofing installation.
 - 2. Do not use spark producing equipment near waterproofing until vapors dissipate.
- B. Pre-installation Meetings

1. Review conditions, installation procedures, schedules, and coordination with other Work.

1.4 SUBMITTALS

- A. Provide submittals in accordance with Section 01 33 00 Submittal Procedures.
- B. Action Submittals
 1. Product data
 - a. Guide specification
 - b. Safety data sheets
 - c. Standard details
 - d. Technical data sheets
 2. Certificates
 - a. Product certification stating assembly components are supplied and warranted by a single source Manufacturer.
 - b. Statement confirming Installer is authorized by Waterproofing Manufacturer to complete Work as specified.
 3. Site quality control submittals
 - a. Watertightness integrity test results
- C. Closeout Submittals
 1. Warranty Documentation
 - a. Final executed warranty document

1.5 QUALITY ASSURANCE

- A. Qualifications
 1. Manufacturer qualifications
 - a. Minimum of 20 years of experience in the production and sales of waterproofing.
 2. Installer qualifications
 - a. Authorized by Waterproofing Manufacturer to complete Work as specified.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery and Acceptance Requirements
 1. Deliver materials in original, factory-sealed, unopened containers with intact and legible product label and manufacturer name.
- B. Storage and Handling
 1. Store materials as recommended by the Manufacturer in a protected area and out of direct sunlight. Protect materials from rain and physical damage.

1.7 SITE CONDITIONS

- A. Ambient Conditions
 1. Do not perform Work during rain or inclement weather.
 2. Do not perform Work on surfaces covered in frost, snow, or are damp or wet to touch.
 3. Do not perform Work when temperatures exceed product specific limitations. Refer to product specific technical data sheet for minimum application temperature.

1.8 WARRANTY

- A. Manufacturer Warranty
 - 1. Material warranty
 - a. Warrants product against product defect. Provides material only for a period of 20 years from date of purchase.
 - 2. System warranty
 - a. Warrants system. Provides material and labor costs for repair for a period of 10 years from the date of installation completion as a result of any of the following:
 - 1. Manufacturing product defect
 - b. Insulation shall retain a minimum of 80% of its thermal value for the duration of the insulation warranty when supplied by Waterproofing Manufacturer.
 - c. Pavers shall not split, crack, or disintegrate prematurely due to freeze-thaw cycling for the duration of the paver warranty when supplied by Waterproofing Manufacturer.
 - 3. Workmanship warranty
 - a. Warrants system and installation. Provides material and labor costs for repair for a period of 5 years from the date of installation completion as a result of any of the following:
 - 1. Manufacturing product defect
 - 2. Faulty Installer's workmanship
 - b. Insulation shall retain a minimum of 80% of its thermal value for the duration of the insulation warranty when supplied by Waterproofing Manufacturer.
 - c. Pavers shall not split, crack, or disintegrate prematurely due to freeze-thaw cycling for the duration of the paver warranty when supplied by Waterproofing Manufacturer.

PART 2 - PRODUCTS

2.1 ASSEMBLIES

- A. Manufacturers
 - 1. Acceptable manufacturers
 - a. Henry® a Carlisle Company
336 Cold Stream Rd.
Kimberton, PA 19442
(800) 486-1278
www.henry.com
- B. Performance Criteria
 - 1. Complies with CAN/CGSB-37.50-M89
 - 2. Meets ASTM D5329, chemically resistant to water, calcium, chloride, salt, mild acid, alkaline solutions, fertilizer, and animal waste.
 - 3. Fire rating: Classified by UL for use in Class A ballasted systems
 - 4. Meets FM 4470: Class A
- C. Materials
 - 1. Hot Applied Rubberized Asphalt Waterproofing/Roofing Membrane
 - a. Hot applied, rubberized asphalt composed of a specially selected blend of refined asphalts, synthetic rubber, and mineral stabilizers to promote adhesion, improve low temperature flexibility, and provide a monolithic fully

bonded roofing and waterproofing membrane, having the following typical properties:

1. Basis of design: Henry 790-11 Hot Applied Rubberized Asphalt Waterproofing/Roofing Membrane
2. Complies with CAN/CGSB-37.50-M89
3. Meets ASTM D5329, chemically resistant to water, calcium, chloride, salt, mild acid, alkaline solutions, fertilizer, and animal waste.
4. Fire rating: Classified by UL for use in Class A ballasted systems
5. Meets FM 4470: Class A
6. Solids content: 100%
7. Flash point (Open cup): 555 degrees F (291 degrees C)
8. Flow (1/8" film, 75-degree angle, 5 hours @ 140 degrees F): No Flow
9. Toughness (CAN/CGSB-37.50): 16.0J
10. Ratio of toughness to peak load (CAN/CGSB-37.50): 0.05
11. Water absorption (CAN/CGSB-37.50): +0.10 g
12. Low temperature flexibility/adhesion @ Minus 13°F (CAN/CGSB-37.50): Pass
13. Heat stability for 5 hours @ 390°F (CAN/CGSB-37.50): Pass
14. Crack bridging capability: Pass
15. Viscosity @ 390 degrees F (CAN/CGSB-37.50): Pass
16. Resistance to hydrostatic pressure
17. Volatile organic content (VOC): 0 grams/liter
18. Water vapor permeability (ASTM E96): 0.016 perms
- b. Hot applied, rubberized asphalt composed of a specially selected blend of refined asphalts, synthetic rubber, and mineral stabilizers to promote adhesion, improve low temperature flexibility, and provide a monolithic fully bonded roofing and waterproofing membrane, having the following typical properties:
 1. Basis of design: Henry 790-11 EV Environmental Grade Hot Applied Rubberized Asphalt Waterproofing/Roofing Membrane
 2. Complies with CAN/CGSB-37.50-M89
 3. Meets ASTM D5329, chemically resistant to water, calcium, chloride, salt, mild acid, alkaline solutions, fertilizer, and animal waste.
 4. Fire rating: Classified by UL for use in Class A ballasted systems
 5. Meets FM 4470: Class A
 6. Recycled content: 25%
 7. Solids content: 100%
 8. Flash point (Open cup): 555 degrees F (291 degrees C)
 9. Flow (1/8" film, 75-degree angle, 5 hours @ 140 degrees F): No Flow
 10. Toughness (CAN/CGSB-37.50): 16.0J
 11. Ratio of toughness to peak load (CAN/CGSB-37.50): 0.05
 12. Water absorption (CAN/CGSB-37.50): +0.10 g
 13. Low temperature flexibility/adhesion @ Minus 13°F (CAN/CGSB-37.50): Pass
 14. Heat stability for 5 hours @ 390°F (CAN/CGSB-37.50): Pass
 15. Crack bridging capability: Pass
 16. Viscosity @ 390 degrees F (CAN/CGSB-37.50): Pass
 17. Resistance to hydrostatic pressure
 18. Volatile organic content (VOC): 0 grams/liter
 19. Water vapor permeability (ASTM E96): 0.016 perms
2. Adhesives/primers
 - a. Asphalt based primer:
 1. Basis of design: Henry 910-01 Asphalt Primer
 - b. Solvent based synthetic rubber adhesive:

1. Basis of design: Henry 930-18 Polymer Modified Adhesive
3. Polyester fabric reinforcement
 - a. Unsaturated spun bonded polyester reinforcement sheet:
 1. Basis of design: Henry Polyester Fabric
4. Flashing membranes
 - a. Torch grade flashing
 1. SBS modified bitumen non-woven polyester reinforced granulated cap sheet with a thermofusible poly lower surface for indefinite UV exposure:
 - a. Basis of design: ModifiedPLUS® NP180gT4
 2. SBS modified bitumen with a thermofusible poly upper and lower surface for torching to substrate and cap sheets:
 - a. Basis of design: ModifiedPLUS NP180p/p
 3. SBS modified bitumen with a sanded upper surface to receive liquid applied membranes and a thermofusible lower surface for torch applied installation:
 - a. Basis of design: ModifiedPLUS NP180s/p
 - b. Mop grade flashing
 1. SBS modified bitumen non-woven polyester reinforced granulated cap sheet with a sanded lower surface for indefinite UV exposure:
 - a. Basis of design: ModifiedPLUS NP180gM4
 2. SBS modified bitumen non-woven polyester reinforced membrane with a sanded upper and lower surface:
 - a. Basis of design: ModifiedPLUS NP180s/s
 3. Butyl and EPDM polymer combination flexible flashing:
 - a. Basis of design: Henry 990-25 Elastomeric Flashing Sheet Unreinforced
 4. Uncured neoprene flexible flashing membrane:
 - a. Basis of design: Neoflash Uncured Neoprene
 - c. Liquid applied flashing
 1. Polyurethane modified methyl methacrylate (PUMA) reinforced liquid flashing:
 - a. Basis of design: Pumadeq™ System
5. Protection course/separation sheet
 - a. SBS modified bitumen glass reinforced membrane:
 1. Basis of design: ModifiedPLUS G100s/s
6. Sealant
 - a. Moisture cure, medium modulus polymer modified sealing compound:
 1. Basis of design: Henry 925 BES Sealant

2.2 ACCESSORIES

- A. Insulation
 1. Minimum thermal resistance (R-Value): 30
 2. Compressive Strength: 100 psi.
 3. Insulation supplied by Henry® a Carlisle Company:
 - a. Acceptable Manufacturers:
 1. Expanded polystyrene (EPS)
 - a. Insulfoam® a Carlisle Company
 2. Extruded polystyrene (XPS)
 - a. DOW®
 - b. Owens Corning®
- B. Drainage Composite

1. Two-part prefabricated geo-composite drain board consisting of a formed polystyrene core covered on one side with a woven or non-woven polypropylene filter fabric:
 - a. Basis of design: Drainage composite supplied by Henry a Carlisle Company
- C. Filter Fabric
 1. Non-woven biodegradable geotextile fabric:
 - a. Basis of design: Henry Filter Fabric N04
- D. Pavers
 1. Paver and pedestal assembly supplied by Henry a Carlisle Company
- E. Expansion Joints
 1. Expansion joints supplied by Henry a Carlisle Company

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verification of Conditions
 1. Verify substrates are in accordance with Waterproofing Manufacturer requirements and as specified in this Section prior to waterproofing installation. Commencement of the Work indicates installer acceptance of the substrate.
 - a. Verify surfaces are sound, clean, dry, and free of oil, grease, dirt, excess mortar, frost, laitance, loose and flaking particles, or other contaminants.
 - b. Verify substrates are continuous and secure, smooth and without large voids, spalled areas, or sharp protrusions.
 - c. Verify cold joints are on the same plane.
- B. Preinstallation Testing
 1. Moisture detection survey
 - a. Visual inspection
 - b. ASTM D4263 - Plastic Sheet Method
 2. Adhesion/pull test
 - a. Complete a waterproofing adhesion test in accordance with Waterproofing Manufacturer's published literature prior to installation of waterproofing assembly.
- C. Evaluation and Assessment
 1. Concrete surface profiles
 - a. Recommended surface profiles:
 1. ICRI Concrete Surface Profiles: CSP 2 – 5
 2. Broom finish
 3. Wood float

3.2 PREPARATION

- A. Protection of In-Place Conditions
 1. Protect areas and surfaces not included in scope of Work against damage or soiling.
 2. Post "NO SMOKING" signs near waterproofing installation until vapors dissipate.
- B. Surface Preparation

1. Repair voids, rock pockets, and rough surfaces that do not meet required concrete surface profiles with approved non-shrink grout or grind to match un-repaired areas.
2. Remove curing compounds or foreign matter detrimental to the adhesion.

3.3 INSTALLATION

- A. Primer
 1. Apply primer in accordance with product specific technical data sheet and Waterproofing Manufacturer details.
 2. Limit primer installation to areas where waterproofing is installed by end of day.
 3. Do not over apply primer; excessive and/or ponding primer is not recommended.
 4. Allow primer to dry in accordance with product specific technical data sheet prior to waterproofing installation.
- B. Detailing and Flashing
 1. Install detailing and flashings per Waterproofing Manufacturer's details.
- C. Waterproofing
 1. Limit waterproofing installation to areas where waterproofing and protection course are installed by end of day.
 2. Install one layer of waterproofing membrane at 90 mils minimum to form a continuous, monolithic membrane over horizontal and vertical surfaces.
 3. Fully embed polyester fabric into warm and tacky waterproofing membrane.
 4. Coat polyester fabric side and end laps with waterproofing membrane. Overlap of dry polyester fabric is not acceptable.
 5. Overlap polyester fabric 1/4-inch (5 millimeters) minimum.
 6. Apply second layer of waterproofing membrane at 125 mils minimum to form a continuous, monolithic membrane over previously coated areas.
 7. Verify total reinforced waterproofing membrane thickness is 215 mils minimum.
- D. Protection Course
 1. Install protection course in a shingle pattern to promote drainage while waterproofing membrane is partially cured to a warm and tacky consistency.
 2. Overlap protection course adjoining edges 2 inches (50 millimeters) minimum and splice overlap with waterproofing.
- E. Insulation
 1. Refer to Insulation Manufacturer installation guidelines and exposure limitations.
- F. Drainage Composite
 1. Refer to Waterproofing Manufacturer product specific technical data sheet for installation guidelines and exposure limitations.
- G. Filter Fabric
 1. Install filter fabric in a shingle pattern to promote drainage.
 2. Loose lay filter fabric to promote debris obstruction.
 3. Overlap filter fabric adjoining edges approximately 6 inches (150 millimeters).
- H. Pavers
 1. Install paver system in accordance with Paver Manufacturer's published literature.

3.4 SITE QUALITY CONTROL

- A. Site Tests and Inspections

1. Refer to Waterproofing Manufacturer for warranty requirements.
2. Observe waterproofing installation during the following phases:
 - a. Substrate verification
 - b. Waterproofing membrane installation start
 - c. Waterproofing integrity test
 - d. Final inspection of waterproofing assembly prior to overburden installation
3. Waterproofing Integrity Test
 - a. Conduct waterproofing integrity test prior to overburden placement.
 - b. Repair membrane breaches and retest the system.
 - c. Report test results to the architect and Waterproofing Manufacturer.
 - d. Acceptable methods:
 1. Electronic Leak Detection in accordance with ASTM D7877
 2. Flood Test in accordance with ASTM D5957

3.5 CLEANING

- A. Waste Management
 1. As the Work proceeds, and upon completion, promptly clean up and remove from the premises all rubbish and surplus materials resulting from the foregoing Work.
 2. Clean soiled surfaces, spatters, and damage caused by Work of this Section.
 3. Dispose of waterproofing per local code ordinances.

3.6 PROTECTION

- A. Protect waterproofing from damage by other trades.

END OF SECTION

SECTION 07 1416 - COLD FLUID-APPLIED WATERPROOFING

PART 1 - GENERAL

1.1 SUMMARY

- A. This Section Includes
 - 1. Polyurethane modified methyl methacrylate (PUMA) for the protection and waterproofing of a plaza deck
- B. Related Requirements
 - 1. Section 04 20 00 – General Masonry
 - 2. Section 07 62 00 – Sheet Metal Flashing and Trim
 - 3. Section 07 92 00 – Joint Sealants

1.2 REFERENCES

- A. Reference Standards
 - 1. American Society for Testing and Materials (ASTM)
 - a. ASTM C1583 – Standard Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method)
 - b. ASTM C7234 – Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers
 - c. ASTM D4263 – Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
 - d. ASTM D5957 – Standard Guide for Flood Testing Horizontal Waterproofing Installations
 - e. ASTM D7877 – Standard Guide for Electronic Methods for Detecting and Locating Leaks in Waterproof Membranes
 - 2. International Concrete Repair Institute (ICRI)
 - a. ICRI Technical Guidelines number 310-2R-2013
 - 3. The Society for Protective Coatings (SSPC)
 - a. Surface Preparation Standards – Method of surface preparation on steel surfaces with the use of power assisted hand tools

1.3 ADMINISTRATIVE REQUIREMENTS

- A. Coordination
 - 1. Do not allow access to installation areas by other trades during waterproofing installation.
 - 2. Do not use spark producing equipment near waterproofing until vapors dissipate.
- B. Pre-installation Meetings
 - 1. Review conditions, installation procedures, schedules, and coordination with other Work.

1.4 SUBMITTALS

- A. Provide submittals in accordance with Section 01 33 00 Submittal Procedures.
- B. Action Submittals
 - 1. Product data
 - a. Guide specification
 - b. Safety data sheets

- c. Standard details
 - d. Technical data sheets
- 2. Certificates
 - a. Product certification stating assembly components are supplied and warranted by a single source Manufacturer.
 - b. Statement confirming Installer is authorized by Waterproofing Manufacturer to complete Work as specified.
- 3. Site quality control submittals
 - a. Watertightness integrity test results
- C. Closeout Submittals
 - 1. Warranty Documentation
 - a. Final executed warranty document

1.5 QUALITY ASSURANCE

- A. Qualifications
 - 1. Manufacturer qualifications
 - a. Minimum of 20 years of experience in the production and sales of waterproofing
 - 2. Installer qualifications
 - a. Authorized by Waterproofing Manufacturer to complete Work as specified.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Delivery and Acceptance Requirements
 - 1. Deliver materials in original, factory-sealed, unopened containers with intact and legible product label and manufacturer name.
- B. Storage and Handling
 - 1. Store materials as recommended by the Manufacturer in a protected area and out of direct sunlight. Protect materials from rain and physical damage.

1.7 SITE CONDITIONS

- A. Ambient Conditions
 - 1. Do not perform Work during rain or inclement weather.
 - 2. Do not perform Work on surfaces covered in frost, snow, or are damp or wet to touch.
 - 3. Do not perform Work when temperatures exceed product specific limitations. Refer to product specific technical data sheet for minimum application temperature.

1.8 WARRANTY

- A. Manufacturer Warranty
 - 1. Material warranty
 - a. Warrants product against product defect. Provides material only for a period of 20 years from date of purchase.
 - 2. System warranty
 - a. Warrants system. Provides material and labor costs for repair for a period of 10 years from the date of installation completion as a result of any of the following:
 - 1. Manufacturing product defect

- b. Insulation shall retain a minimum of 80% of its thermal value for the duration of the insulation warranty when supplied by Waterproofing Manufacturer.
- c. Pavers shall not split, crack, or disintegrate prematurely due to freeze-thaw cycling for the duration of the paver warranty when supplied by Waterproofing Manufacturer.
- 3. Workmanship warranty
 - a. Warrants system and installation. Provides material and labor costs for repair for a period of 5 years from the date of installation completion as a result of any of the following:
 - 1. Manufacturing product defect
 - 2. Faulty Installer's workmanship
 - b. Insulation shall retain a minimum of 80% of its thermal value for the duration of the insulation warranty when supplied by Waterproofing Manufacturer.
 - c. Pavers shall not split, crack, or disintegrate prematurely due to freeze-thaw cycling for the duration of the paver warranty when supplied by Waterproofing Manufacturer.

PART 2 - PRODUCTS

2.1 ASSEMBLIES

- A. Manufacturers
 - 1. Acceptable manufacturers
 - a. Henry® a Carlisle Company
336 Cold Stream Rd.
Kimberton, PA 19442
(800) 486-1278
www.henry.com
- B. Performance Criteria
 - 1. Waterproofing
 - a. PUMA technology
 - b. Minimum application temperature: 20 degrees F
 - c. Rain ready within 1 hour
 - d. Foot traffic ready within 1 hour
 - e. Abrasion resistance (ASTM C501): 64mg
 - f. Solids content by volume (ASTM D1644): 100 percent
 - g. Adhesion (ASTM C1583): greater than 425 psi, substrate failure
 - h. Tensile strength (ASTM D638): 1680 psi
 - i. Elongation (ASTM D638): 283 percent
 - j. Volatile organic content (VOC) (ASTM C1250): 0 g/l
- C. Materials
 - a. PUMA waterproofing
 - 1. 100 percent solids, cold fluid-applied PUMA, having the following typical properties:
 - a. Basis of design: Pumadeq™ Flex 30SL
 - b. Polyurethane modified methyl methacrylate (PUMA) technology
 - c. Minimum application temperature: 20 degrees F
 - d. Rain ready within 1 hour
 - e. Foot traffic ready within 1 hour
 - f. Abrasion resistance (ASTM C501): 64mg
 - g. Solids content by volume (ASTM D1644): 100 percent
 - h. Adhesion (ASTM C1583): greater than 425 psi, substrate failure

- i. Tensile strength (ASTM D638): 1680 psi
 - j. Elongation (ASTM D638): 283 percent
 - k. Volatile organic content (VOC) (ASTM C1250): 0 g/l
- b. Catalyst
 - 1. Dibenzoyl peroxide catalyst powder (BPO) used to initiate curing of PUMA liquid resins:
 - a. Basis of design: Pumadeq Catalyst
- c. Adhesives/primers
 - 1. Standard primer:
 - a. Basis of design: Pumadeq Primer 20
 - 2. Moisture mitigating epoxy primer:
 - a. Basis of design: Henry GC Epoxy Primer
- d. Fleece reinforcement
 - 1. 100 percent polyester, non-woven, needle punch constructed fleece:
 - a. Basis of design: Pumadeq N-Fleece
- e. Flashing
 - 1. 100 percent solids, medium viscosity, cold fluid-applied PUMA:
 - a. Pumadeq Flex 31MV
- f. Wear coat
 - 1. Liquid applied coating:
 - a. Elastomeric PUMA aggregate holding wear coat for heavy construction traffic areas or overburden bonded directly to waterproofing:
 - 1. Basis of design: Pumadeq Grip 40 and silica sand, sieve size: #20-30, #12 Silica, or NJ0
 - 2. Wear coat filler:
 - a. Finely graded filler powder:
 - 1. Basis of design: Henry Filler
- g. Top coat
 - 1. Poly methyl methacrylate (PMMA) topcoat for areas requiring long term color stability:
 - a. Basis of design: Henry Deqcoat™ 50
 - b. Color: Gray
- h. Substrate repair paste
 - 1. Thixotropic PUMA for repairing concrete substrate deficiencies:
 - a. Basis of design: Pumadeq Flex 32TX
- i. PUMA cleaner
 - 1. Liquid-based Methyl Methacrylate (MMA) PUMA resin surface reactivator:
 - a. Pumadeq Cleaning Fluid

2.2 ACCESSORIES

- A. Insulation
 - 1. Minimum thermal resistance (R-Value): 30
 - 2. Compressive Strength: 100 psi.
 - 3. Insulation supplied by Henry a Carlisle Company
 - a. Acceptable Manufacturers:
 - 1. Extruded polystyrene (XPS)
 - a. DOW®
 - b. Owens Corning®
- B. Drainage Composite
 - 1. Two-part prefabricated geo-composite drain board consisting of a formed polystyrene core covered on one side with a woven or non-woven polypropylene filter fabric:
 - a. Drainage composite supplied by Henry a Carlisle Company

- C. Filter Fabric
 - 1. Non-woven biodegradable geotextile fabric:
 - a. Basis of design: Henry Filter Fabric N04
- D. Pavers
 - 1. Paver and pedestal assembly supplied by Henry a Carlisle Company
- E. Expansion Joints
 - 1. Expansion joints supplied by Henry a Carlisle Company

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verification of Conditions
 - 1. Verify substrates are in accordance with Waterproofing Manufacturer requirements and as specified in this Section prior to waterproofing installation. Commencement of the Work indicates installer acceptance of the substrate.
 - a. Verify surfaces are sound, clean, dry, and free of oil, grease, dirt, excess mortar, frost, laitance, loose and flaking particles, or other contaminants.
 - b. Verify substrates are continuous and secure, smooth and without large voids, spalled areas, or sharp protrusions.
 - c. Verify cold joints are on the same plane.
 - d. Verify penetrations are in place, including coping, railings, flashing, electrical conduit, pipes, pedestals, and curbs.
- B. Preinstallation Testing
 - 1. Moisture detection survey
 - a. Visual inspection
 - b. ASTM D4263 - Plastic Sheet Method
 - c. Electronic hand-held moisture meter with minimum 3/4-inch penetration
 - 2. Adhesion/pull test
 - a. Complete a waterproofing adhesion test in accordance with Waterproofing Manufacturer's published literature prior to installation of waterproofing assembly.
- C. Evaluation and Assessment
 - 1. Concrete surface profiles
 - a. Recommended surface profiles:
 - 1. ICRI Concrete Surface Profiles: CSP 3-5
 - 2. Broom finish
 - 3. Wood float

3.2 PREPARATION

- A. Protection of In-Place Conditions
 - 1. Protect areas and surfaces not included in scope of Work against damage or soiling.
 - 2. Post "NO SMOKING" signs near waterproofing installation until vapors dissipate.
 - 3. Seal air intakes with activated carbon filters, nearby windows, and doors.
 - 4. Ensure a constant supply of "fresh air", required to remove monomers from the resin surface and allow for cure.

- B. Surface Preparation
 - 1. Prepare substrates 2 inches (50 millimeters) minimum beyond area of anticipated waterproofing.
 - 2. Repair voids, rock pockets, and rough surfaces that do not meet required concrete surface profiles with substrate repair paste, non-shrink grout, or grind to match un-repaired areas.
 - 3. Remove curing compounds or foreign matter detrimental to the adhesion.
 - 4. Remove laitance with shot blast or mechanical grinder.
 - 5. Mechanically abrade metal surfaces to meet SSPC-SP3. Clean metal, remove debris and sharp edges, and immediately prime substrate to avoid flash rusting.
 - 6. Lightly sand PVC pipes to a rough surface. Clean pipe and remove debris and sharp edges.

3.3 INSTALLATION

- A. Primer
 - 1. Apply primer in accordance with product specific technical data sheet and Waterproofing Manufacturer details.
 - 2. Limit primer installation to areas where waterproofing is installed by end of day.
 - 3. Do not over apply primer; excessive and/or ponding primer is not recommended.
 - 4. Allow primer to dry in accordance with product specific technical data sheet prior to waterproofing installation.
- B. Detailing and Flashing
 - 1. Install detailing and flashings per Waterproofing Manufacturer's details.
- C. Waterproofing
 - 1. Limit waterproofing installation to areas where waterproofing is installed by end of day.
 - 2. Do not install waterproofing beyond cured primer.
 - 3. Install one layer of waterproofing membrane at 50 mils minimum to form a continuous, monolithic membrane over horizontal surfaces.
 - 4. Fully embed fleece reinforcement into waterproofing membrane and roll or brush fleece to ensure adhesion and remove voids, folds, and wrinkles.
 - 5. Coat side and end laps of embedded fleece reinforcement with waterproofing membrane. Overlap of dry fleece is not acceptable.
 - 6. Overlap fleece reinforcement a minimum of 3 inches (75 millimeters).
 - 7. Apply second layer of waterproofing membrane at 30 mils minimum to form a continuous, monolithic membrane over previously coated areas.
 - 8. Feather second layer of waterproofing at fleece reinforcement edges to eliminate a hard line at waterproofing perimeter.
 - 9. Verify total reinforced waterproofing membrane thickness is 80 mils minimum.
- D. Wear Coat
 - 1. Install wear coat onto cured waterproofing at 40 mils minimum to form a continuous, monolithic membrane at horizontal surfaces or otherwise noted per project specific requirements.
 - 2. Fully broadcast aggregate, to rejection, onto wear coat.
 - a. Average application rate = approximately 1.5 lbs per square foot (7.3 kilograms per square meter)
 - 3. Allow wear coat to cure prior to subsequent installations.
 - 4. Remove excess aggregate, choose from the following:
 - a. Heavy duty brooms
 - b. Mechanical blowing equipment
 - c. Industrial vacuum

- E. Top Coat
 - 1. Install one layer of top coat at 20 mils minimum to form a continuous monolithic membrane over horizontal and vertical surfaces.
 - 2. Allow top coat to cure prior to subsequent installations.
 - 3. Apply second layer of top coat at 12 mils minimum to form a continuous, monolithic membrane over previously coated areas.
- F. Insulation
 - 1. Refer to Insulation Manufacturer installation guidelines and exposure limitations.
- G. Drainage Composite
 - 1. Refer to Waterproofing Manufacturer product specific technical data sheet for installation guidelines and exposure limitations.
- H. Filter Fabric
 - 1. Loose lay filter fabric, extending up vertical surfaces 6 inches (300 millimeters) beyond anticipated height of growing media.
 - 2. Overlap filter fabric adjoining edges approximately 6 inches (150 millimeters).
 - 3. Trim filter fabric to growing media height after growing media compaction.
- I. Pavers
 - 1. Install paver system in accordance with Paver Manufacturer's published literature.

3.4 SITE QUALITY CONTROL

- A. Site Tests and Inspections
 - 1. Refer to Waterproofing Manufacturer for warranty requirements.
 - 2. Observe waterproofing installation during the following phases:
 - a. Substrate verification
 - b. Waterproofing membrane installation start
 - c. Waterproofing integrity test
 - d. Final inspection of waterproofing assembly prior to overburden installation
 - 3. Waterproofing Integrity Test
 - a. Conduct waterproofing integrity test prior to overburden placement.
 - b. Repair membrane breaches and retest the system.
 - c. Report test results to the Architect and Waterproofing Manufacturer.
 - d. Acceptable methods:
 - 1. Electronic Leak Detection in accordance with ASTM D7877
 - 2. Flood Test in accordance with ASTM D5957

3.5 CLEANING

- A. Waste Management
 - 1. As the Work proceeds, and upon completion, promptly clean up and remove from the premises all rubbish and surplus materials resulting from the foregoing Work.
 - 2. Clean soiled surfaces, spatters, and damage caused by Work of this Section.
 - 3. Dispose of waterproofing per local code ordinances.

3.6 PROTECTION

- A. Protect waterproofing from damage by other trades.

END OF SECTION

SECTION 07 2726 - FLUID-APPLIED MEMBRANE AIR BARRIERS

PART 1: GENERAL

1.1. GENERAL REQUIREMENTS

- A. The General Conditions, Supplementary Conditions, Instructions to Bidders, and Division 01- General Requirements shall be read in conjunction with and govern this section.
- B. The Specification shall be read as a whole by all parties concerned. Each Section may contain more or less than the complete Work of any trade. The Contractor is solely responsible to make clear to the Subcontractor the extent of their Work.

1.2. SUMMARY

- A. This Section includes requirements for supplying labor, materials, tools, and equipment to complete the Work as shown on the Drawings Architectural Division as specified herein including, but not limited to, the following:
 - 1. Adhesives/Primers
 - 2. Fluid Applied, Vapor Permeable Air & Water Barrier Membrane
 - 3. Transition Membranes
 - 4. Sealant
 - 5. Thru-wall flashing
 - 6. Insulation Adhesive

1.3. RELATED REQUIREMENTS

- A. DIVISION 03 – Concrete - 03 3000-CAST IN PLACE CONCRETE
- B. DIVISION 04 – Masonry Section 04 20 00 – General Masonry
- C. DIVISION 07 – Thermal and Moisture Protection Section 07 62 00 - Sheet Metal Flashing and Trim
- D. DIVISION 07 – Thermal and Moisture Protection Section 07 92 00 - Joint Sealants

1.4. ALTERNATES

- A. Primary membranes defined as Water Resistive Coatings are only considered acceptable substitutions when installed in conjunction with EIFS in accordance with ICC-ES AC 212 and are not considered acceptable substitutions for wall assemblies with alternate claddings.
- B. Submit requests for alternates as follow:
 - 1. Alternate submission format to include:
 - a. Evidence that alternate materials meet or exceed performance characteristics of product requirements and documentation from an approved independent testing laboratory certifying that the performance of the system including auxiliary components exceed the requirements of the local building code.

- b. References clearly indicating that the Air Barrier Manufacturer has successfully completed projects of similar scope and nature on an annual basis for a minimum of ten (10) years.
 - c. Air Barrier Manufacturer's guide specification.
 - d. Air Barrier Manufacturer's complete set of technical data sheets for assembly.
 - e. Air Barrier Manufacturer's complete set of details for assembly.
 - f. Product certification that the assembly components are supplied and warranted by single source Air Barrier Manufacturer.
 - g. Air Barrier Manufacturer statement that anticipated wall assembly passes NFPA 285.
 - h. Sample warranty as specified.
- C. Submit requests for alternates to this specification a minimum of ten (10) working days prior to bid date. Include a list of twenty-five (25) projects executed over the past five (5) years.
- D. Acceptable alternates will be confirmed by addendum. Substitute materials not approved in writing prior to tender closing shall not be permitted for use on this project.

1.5. REFERENCES

- A. American Architectural Manufacturers Association (AAMA):
 - 1. AMMA 2400-02, Standard Practice for Installation of Windows with a Mounting Flange in Stud Frame Construction
- B. American Society for Testing and Materials (ASTM):
 - 1. ASTM D412, Standard Test Method for Vulcanized Rubber and Thermoplastic Elastomers - Tension
 - 2. ASTM D471, Standard Test Method for Rubber Property - Effect of Liquids
 - 3. ASTM D1970, Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection
 - 4. ASTM D2243, Standard Test Method for Freeze-Thaw Resistance of Water-Borne Coatings
 - 5. ASTM D5590, Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay
 - 6. ASTM E84, Standard Test Method for Surface Burning Characteristics of Building Materials
 - 7. ASTM E96, Standard Test Methods for Water Vapor Transmission of Materials
 - 8. ASTM E283, Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen
 - 9. ASTM E330, Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference
 - 10. ASTM E331, Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
 - 11. ASTM E1354, Standard Test Method for Heat and Visible Smoke Release Rates for Materials and Products Using an Oxygen Consumption Calorimeter
 - 12. ASTM E1677, Standard Specification for Air Barrier (AB) Material or System for Low-Rise Framed Building Walls
 - 13. ASTM E2112, Standard Practice for Installation of Exterior Windows, Doors and Skylights

14. ASTM E2178, Standard Test Method for Air Permeance of Building Materials
15. ASTM E2357, Standard Test Method for Determining Air Leakage of Air Barrier Assemblies

- C. National Fire and Protection Agency (NFPA):
 1. NFPA 285, Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Non-Load-Bearing Wall Assemblies Containing Combustible Components

1.6. ADMINISTRATIVE REQUIREMENTS

- A. Coordination:
 1. Coordinate the Work of this Section with the installation of exterior substrate. Sequence Work so that installation of fluid-applied air barrier coincides with installation of substrate preparation without causing delay to the Work.
- B. Pre-installation meetings:
 1. When required, and with prior notice, an Air Barrier Manufacturer representative will meet with the necessary parties at the jobsite to review and discuss project conditions as it relates to the integrity of the assembly.

1.7. SUBMITTALS

- A. Provide the following requested information in accordance with Section 013300 Submittal Procedures.
- B. Action Submittals:
 1. Product Data:
 - a. Air Barrier Manufacturer's guide specification.
 - b. Air Barrier Manufacturer's complete set of technical data sheets for assembly.
 - c. Air Barrier Manufacturer's complete set of standard detail drawings.
 2. Certificates:
 - a. Product certification that the assembly components are supplied and warranted by single source Air Barrier Manufacturer.
 - b. Statement that installing contractor is authorized by Air Barrier Manufacturer to complete Work as specified.
 3. Tests and Evaluation Reports:
 - a. NFPA 285 wall assembly compliance:
 1. Air Barrier Manufacturer statement that anticipated wall assembly passes NFPA 285.
 4. Warranty:
 - a. Sample warranty as specified.

1.8. QUALITY ASSURANCE

- A. Single Source Responsibility:
 1. Obtain fluid-applied membrane air barrier, transition membranes, air barrier sealants, primers, mastics, and adhesives from a single Air Barrier Manufacturer regularly engaged in the manufacturing and supply of the specified products.
 2. Contactor to verify product compliance with federal, state, and local regulations controlling use of Volatile Organic Compounds (VOC).

- B. Manufacturer Qualifications:
 - 1. Air Barrier Manufacturer shall demonstrate qualifications to supply materials of this section by certifying the following:
 - a. Air Barrier Manufacturer must not issue warranties for terms longer than they have been manufacturing and supplying specified products for similar scope of Work.
- C. Installer Qualifications:
 - 1. Perform Work in accordance with Air Barrier Manufacturer published literature and as specified in this section.
 - 2. Maintain one (1) copy of Air Barrier Manufacturer's instructions on site.
 - 3. At all times during the execution of the Work allow access to site by the Air Barrier Manufacturer representative.
 - 4. If meeting with Air Barrier Manufacturer during project construction, contact Air Barrier Manufacturer a minimum of two weeks prior to schedule meeting.

1.9. DELIVERY, STORAGE, AND HANDLING

- A. Delivery of Materials:
 - 1. Materials shall be delivered to the jobsite in undamaged and clearly marked containers indicating the name of the Air Barrier Manufacturer and product.
- B. Storage of Materials:
 - 1. Store materials as recommended by Air Barrier Manufacturer and conforming to applicable safety regulatory agencies. Refer to all applicable data including but not limited to MSDS sheets, Product Data sheets, product labels, and specific instructions for personal protection.
 - 2. Keep solvents away from open flame or excessive heat.
 - 3. Products should be stored in closed containers.
 - 4. Store rolled materials on end in original packaging.
 - 5. Protect rolls from direct sunlight until ready for use.
 - 6. Refer to Air Barrier Manufacturer published literature.
- C. Handling:
 - 1. Refer to Air Barrier Manufacturer published literature.

1.10. SITE CONDITIONS

- A. Environmental Requirements:
 - 1. No Work shall be performed during rain or inclement weather.
 - 2. No Work shall be performed on frost or wet covered surfaces.
- B. Protection:
 - 1. Cap and protect exposed back-up walls against wet weather conditions during and after application of membrane.
- C. Ensure all preparation Work is completed prior to installing fluid-applied membrane air barrier.

1.11. WARRANTY

- A. Manufacturer Material Warranty:
 - 1. Provide Air Barrier Manufacturer's standard material warranty.

PART 2: PRODUCTS

2.1. MATERIALS MANUFACTURER

- A. Components and auxiliary materials must be obtained as a single-source from the assembly Air Barrier Manufacturer to ensure total system compatibility and integrity.
- B. Acceptable Manufacturers:
 - 1. Henry Company
999 N. Sepulveda Blvd. Suite 800
El Segundo, CA 90245
(800) 486-1278
www.Henry.com

2.2. MATERIALS

- A. Primary Fluid-Applied Membrane Air Barrier (Basis of Design):
 - 1. One-component, water-based, elastomeric emulsion membrane, designed to provide a vapor permeable air and water barrier when applied above-grade wall assemblies, having the following properties:
 - a. Basis of Design Product: Air-Bloc 17MR
 - b. Color: Graphite
 - c. Solids Content:
 - 1. By Weight: 63%
 - 2. By Volume: 53%
 - d. Service Temperature:
 - 1. Low Temperature: -40 degrees F (-40 degrees C)
 - 2. High Temperature: +180 degrees F (+80 degrees C)
 - e. Application Temperature:
 - 1. Low Temperature: +20 degrees F (-6 degrees C)
 - 2. High Temperature: +122 degrees F (+50 degrees C)
 - f. Tensile Strength (ASTM D412): 104 psi (717 kPa)
 - g. Elongation (ASTM D412): 420%
 - h. Low Temperature Flexibility @ -22 degrees F (-30 degrees C) (ASTM D1970): Pass
 - i. Freeze-Thaw Resistance (ASTM D2243): Pass; 10 cycles
 - j. Nail Sealability (ASTM D1970): Pass
 - k. VOC Content: 100 grams/liter max.
 - l. Water Absorption (ASTM D471, modified): 5.6%
 - m. Water Vapor Permeance (ASTM E96 B) @ 40 mils nominal dry film: 14 perms
 - n. Air Permeability:
 - 1. Assembly Air Leakage (ASTM E2357): Pass
 - 2. Building Material (ASTM E2178): 0.0001 cfm/ft2 (0.0005 L/s.m2)
 - o. Chemical Resistance: Resists salt solutions, mild acids and alkalis. Non-resistant to oils, grease or solvents
 - p. Fire Testing (NFPA 285): Complies in various assemblies
 - q. Flame Spread/Smoke Development (ASTM E84): 10/15
 - r. Resistance to Mold, Mildew, and Fungal Growth (ASTM D5590): No growth
- B. Auxiliary Materials
 - 1. Transition Membranes:

- a. Liquid applied flashings:
 - 1. Moisture-curing one component elastomeric liquid applied flashing membrane using a highly advanced STPe (Silyl-Terminated Polyether) polymer, having the following properties:
 - a. Basis of Design Product: Air-Bloc LF
 - b. Color: Blue
 - c. Air Leakage (ASTM E2178): <0.004 L/s/m² @ 75Pa
 - d. Water Vapor Permeance (ASTM E96, Method B): 21.8 perms @25 mils
 - e. Air Leakage of Air Barrier Assemblies (ASTM E2357): Pass
 - f. Water Resistance (AC212/ASTM D2247): Pass
 - g. Nail Sealability (AMMA 711): Pass
 - h. Surface Burning Characteristics (ASTM E84):
 - 1. Class A
 - 2. Flame Spread/Smoke Development (ASTM E84): 20/5
 - i. Tensile Strength (ASTM D412): 132 psi
 - j. Elongation (ASTM D412): 264%
 - b. Self-Adhering flashings:
 - 1. Non-vapor permeable, self-adhered water resistive air and vapor barrier membrane consisting of an SBS rubberized asphalt compound, which is integrally laminated to a blue engineered thermoplastic film, having the following properties:
 - a. Basis of Design Product: Blueskin SA
 - b. Color: Blue
 - c. Water Vapor Permeance (ASTM E96, Method A): .86 perms
 - d. Air Leakage of Air Barrier Assemblies (ASTM E2357): Pass
 - e. Air Leakage (ASTM E2178): <0.0005 L/s/m² @ 75Pa
 - f. Water Tightness (CAN/CGSB-37.58-M86): Pass.
 - g. Nail Sealability (ASTM D1970): Pass.
 - h. Tensile Strength:
 - 1. Membrane (ASTM D412-modified): 500 psi minimum
 - 2. Film (ASTM D828): 5000 psi minimum
 - i. Elongation (ASTM D412-modified): 200% minimum
- 2. Sheathing Joint Membranes:
 - a. Vapor permeable, self-adhered water resistive air barrier membrane consisting of an engineered film and patented, permeable adhesive technology with split-back poly-release film, having the following properties:
 - 1. Basis of Design Product: Blueskin VP160
 - 2. Color: Blue
 - 3. Air Leakage (ASTM E2178): <0.02 L/s/m² @ 75Pa
 - 4. Water Vapor Permeance (ASTM E96, Method A): 29 perms
 - 5. Air Leakage of Air Barrier Assemblies (ASTM E2357): Pass
 - 6. Resistance to Water Penetration (ICC-ES AC 38): Pass.
 - 7. Nail Sealability (ASTM D1970): Pass
 - 8. Surface Burning Characteristics (ASTM E84):
 - a. Class A
 - b. Flame Spread/Smoke Development (ASTM E84): 0/105
 - 9. Tensile Strength (ASTM D828): 182N MD/129N CD
 - 10. Cycling and Elongation (ICC-ES AC48): Pass
 - b. Contact Air Barrier Manufacturer for a complete list of authorized transition membranes.
 - 3. Adhesives and Primers:
 - a. Spray adhesive, and having the following properties:

1. Basis of Design Product: Blueskin Spray Prep
2. Color: Clear amber
3. Solids Content (By Weight): 35%
4. Aerosol
- b. Synthetic rubber based adhesive type, quick setting, having the following properties:
 1. Basis of Design Product: Blueskin Adhesive
 2. Color: Blue.
 3. Solids Content (By Weight): 35%.
 4. Solvent based: Maximum VOC: 450 g/L
- c. Polymer emulsion based adhesive type, quick setting, low VOC content, having the following properties:
 1. Basis of Design Product: Blueskin LVC Adhesive
 2. Color: Blue.
 3. Solids Content (By Weight): 40%.
 4. Solvent based: 240 g/L.
- d. Polymer emulsion based primer for self-adhered membranes, and having the following properties:
 1. Basis of Design Product: Aquatac Primer
 2. Color: Aqua.
 3. Solids Content (By Weight): 58%.
 4. Water based: Maximum VOC: 50 g/l
4. Sealants:
 - a. Building Envelope Sealant:
 1. Moisture cure, medium modulus polymer modified sealing compound, having the following properties:
 - a. Basis of Design Product: HE925 BES Sealant
 - b. Complies with Fed. Spec. TT-S-00230C, Type II, Class A.
 - c. Complies with ASTM C920, Type S, Grade NS, Class 35.
 - d. Elongation: 450 – 550%.
 - e. Remains flexible with aging.
 - b. Sheathing Joint Sealants:
 1. As recommended by Air Barrier Manufacturer
 - c. Contact Air Barrier Manufacturer for a complete list of authorized sealants.
5. Self-Adhesive Thru-Wall Flashing Membrane:
 - a. Non-vapor permeable, self-adhered water resistive air and vapor barrier membrane consisting of an SBS rubberized asphalt compound, which is integrally laminated to a blue engineered thermoplastic film, having the following properties:
 1. Basis of Design Product: Blueskin TWF
 2. Color: Yellow
 3. High Temperature Stability - Flow Resistance (ASTM D5147): Pass
 4. Air leakage (ASTM E283): 0.005 L/s.m² @ 75 Pa
 5. Water vapor permeance (ASTM E96, Method B): 0.03 perms
 6. Low temperature flexibility (CGSB 37-GP-56M): Pass
- C. Insulation Adhesive: (optional)
 1. Synthetic rubber base compound having the following characteristics:
 - a. Basis of Design Product: Air-Bloc 21
 - b. Color: Cream.
 - c. Compatible with air barrier membrane, substrate and insulation materials.
 - d. Long term flexibility (CGSB 71-GP-24M): Pass.
 - e. Chemical resistance: Alkalies, mild acid and salt solutions.

PART 3: EXECUTION

3.1. EXAMINATION

- A. Substrate Conditions:
 - 1. Verify substrates to receive work and surrounding adjacent surfaces are in accordance with Air Barrier Manufacturer published literature prior to installation of fluid applied membrane air barrier assembly.
 - 2. Sheathing panels must be securely fastened and installed flush to ensure a continuous substrate in accordance with Air Barrier Manufacturer published literature.
 - 3. Fastener penetrations must be set flush with sheathing and fastened into solid backing.
 - 4. Mortar joints in concrete block and form tie holes/voids in poured concrete shall be filled, flush, smooth, and allowed to be cured for a minimum of twenty-four (24) hours.
 - 5. New concrete should be cured for a minimum of sixteen (16) hours after forms are removed.
 - 6. Cap and protect exposed back-up walls against wet weather conditions prior to application of fluid applied membrane air barrier assembly.
- B. Notify contractor in writing of any conditions that are not acceptable.
- C. The installing contractor shall examine and determine that surfaces and conditions are ready to accept the Work of this section in accordance with published literature. Commencement of Work or any parts thereof shall mean installer acceptance of the substrate.

3.2. PREPARATION

- A. All surfaces must be sound, dry to touch, clean, and free of oil, grease, dirt, excess mortar, frost, laitance, loose and flaking particles, or other contaminants.
- B. Protect adjacent surfaces not included in scope of Work to prevent spillage and overspray.
- C. Hot weather or direct-sun applications over porous substrates, such as concrete, promote rapid surface drying and can form blisters in the fluid applied membrane air barrier during curing. To aid in blister prevention prepare substrate in accordance with one of the following optional procedures:
 - 1. Prime coat:
 - a. Apply a thin prime coat of fluid applied membrane air barrier to substrate.
 - b. Allow fluid applied membrane air barrier to fully cure prior to subsequent application.
 - c. Install primary fluid applied membrane air barrier to Air Barrier Manufacturer minimum recommended mil thickness.
 - 2. Two coat:
 - a. Apply fluid applied membrane air barrier to achieve one-half (1/2) of Air Barrier Manufacturer minimum recommended mil thickness.
 - b. Allow fluid applied membrane air barrier to fully cure prior to subsequent application.
 - c. Apply fluid applied membrane air barrier to achieve one-half (1/2) of Air Barrier Manufacturer minimum recommended mil thickness.

- d. Overall dry mil thickness shall be in accordance with Air Barrier Air Barrier Manufacturer published literature.

3.3. INSTALLATION

- A. Ensure substrate is ready to receive fluid applied membrane air barrier in accordance with published literature.
- B. If fluid applied membrane air barrier should freeze while in storage, move containers to a controlled environment above 32 degrees F (0 degrees C) until thawed and re-mix using a hand operated power mixer prior to use.
- C. Fluid applied membrane air barrier shall not be applied when ambient (air) and substrate temperatures are below 20 degrees F (-6 degrees C).
- D. Do not proceed with application of air barrier membrane when rain is expected within 16 hours.
- E. Apply sealant at sharp corners, changes in substrate plane, penetrations, and edges to form a smooth transition from one plane to another.
- F. Non-Moving Substrate Joint and Crack Treatment:
 - 1. Gaps equal to or less than 3/8 inch (10 mm) wide:
 - a. Sheathing Joint Sealant:
 - 1. Apply sealant at rate recommended by Air Barrier Manufacturer.
 - 2. Spread sealant at joint extending a minimum one (1) inch beyond gap to ensure a continuous air and watertight assembly.
 - 2. Gaps equal to or less than 1/2 inch (12 mm) wide:
 - a. Building Envelope Sealant:
 - 1. Apply sealant at rate recommended by Air Barrier Manufacturer.
 - 2. Spread sealant at joint extending a minimum one (1) inch on each side of substrate gap.
 - b. Liquid applied flashings:
 - 1. Apply liquid applied flashing at rate recommended by Air Barrier Manufacturer.
 - 2. Apply liquid applied flashing in accordance with Air Barrier Manufacturer published literature extending a minimum of two (2) inches on each side of substrate gap.
 - c. Self-adhering flashings:
 - 1. Apply primer to substrate and allow curing in accordance with published literature prior to installation of self-adhered flashing.
 - 2. Apply self-adhering flashing in accordance with Air Barrier Manufacturer published literature extending a minimum of three (3) inches on each side of substrate gap.
 - 3. Roll membrane with countertop roller to eliminate air pockets between self-adhered flashing and substrate ensuring full adhesion of membrane onto substrate.
 - 4. Seal exposed leading edges of self-adhered membrane with sealant.
 - 3. Gaps greater than 1/2 inch wide:
 - a. Contact Air Barrier Manufacturer.
 - 4. Refer to Air Barrier Manufacturer published literature for a complete list of authorized Non-Moving Substrate Joint and Crack Treatment details.

- G. Moving Joints:
 - 1. Contact Air Barrier Manufacturer.
- H. Refer to Air Barrier Manufacturer detail drawings for installation procedures including, but not limited to, the following:
 - 1. Inside corners
 - 2. Outside corners
 - 3. Crack treatment
 - 4. Penetrations
 - 5. Rough openings
 - 6. Control joints
 - 7. Expansion joints
 - 8. Changes in substrate
- I. Contact Air Barrier Manufacturer to coordinate transition of fluid applied membrane air barrier to adjacent areas including, but not limited to, the following:
 - 1. Roof to air barrier
 - 2. Air barrier to waterproofing
 - 3. Fastener penetrations
- J. Primary Liquid Air Barrier Membrane
 - 1. Install fluid applied membrane air barrier in accordance with Air Barrier Manufacturer published literature to ensure an air and watertight fluid applied membrane air barrier assembly.
 - 2. Fluid applied membrane air barrier assembly must be installed in a monolithic application without sags, runs or voids, and transitioning with auxiliary components to create a uniform drainage plane and air barrier.
 - 3. Install fluid applied membrane air barrier and transition membranes so that subsequent membrane installation laps one (1) inch (2.5 cm) onto existing membrane ensuring an air and watertight fluid applied membrane air barrier assembly.
 - 4. Fluid applied membrane air barrier total dry thickness shall be in accordance with Air Barrier Manufacturer published literature. Refer to Air Barrier Manufacturer Technical Data Sheet.
- K. Insulation Adhesive:
 - 1. Upon curing of the air barrier membrane system apply insulation adhesive in a serpentine pattern.
 - 2. Immediately embed insulation into the adhesive and press firmly into place to ensure full contact. Apply additional adhesive if allowed to skin over.
 - 3. Fully butter all joints of insulation panels with adhesive during installation, with the exception of expansion joints.

3.4. FIELD QUALITY CONTROL

- A. Final Observation and Verification:
 - 1. Final inspection of fluid applied membrane air barrier assembly shall be carried out by the Owner's representative, the contractor, or Air Barrier Manufacturer as required by warranty.
 - 2. Contact Air Barrier Manufacturer for warranty issuance requirements.
- B. Fluid applied membrane air barrier assembly is not designed for permanent UV

exposure. Refer to Air Barrier Manufacturer published literature for product limitations.

3.5. CLEANING

- A. Promptly as the Work proceeds, and upon completion, clean up and remove from the premises all rubbish and surplus materials resulting from the foregoing Work.
- B. Clean soiled surfaces, spatters, and damage caused by Work of this Section.
- C. Check area to ensure cleanliness and remove debris, equipment, and excess material from the site.

END OF SECTION

SECTION 07 6000 - SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014) Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A308/A308M (2010) Standard Specification for Steel Sheet, Terne (Lead-Tin Alloy) Coated by the Hot Dip Process

ASTM A480/A480M (2016) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip

ASTM A653/A653M (2015; E 2016) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM B101 (2012) Standard Specification for Lead-Coated Copper Sheet and Strip for Building Construction

ASTM B209 (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221 (2014) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

ASTM B221M (2013) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

ASTM B32 (2008; R 2014) Standard Specification for Solder Metal

ASTM B370 (2012) Standard Specification for Copper Sheet and Strip for Building Construction

ASTM B69 (2013) Standard Specification for Rolled Zinc

ASTM D1784 (2011) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds

ASTM D226/D226M	(2009) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D4586/D4586M	(2007; E 2012; R 2012) Asphalt Roof Cement, Asbestos-Free
SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)	
SMACNA 1793	(2012) Architectural Sheet Metal Manual, 7th Edition
SINGLE PLY ROOFING INDUSTRY (SPRI)	
ANSI/SPRI RD-1	(2009) Performance Standard for Retrofit Drains

1.2 GENERAL REQUIREMENTS

Finished sheet metalwork will form a weathertight construction without waves, warps, buckles, fastening stresses or distortion, which allows for expansion and contraction. Sheet metal mechanic is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous roofing operations.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval and for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.

SD-02 Shop Drawings

Counterflashing; G

Indicate thicknesses, dimensions, fastenings and anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction.

SD-11 Closeout Submittals

Submit for sheet metal work in accordance with paragraph entitled "Field Quality Control."

1.4 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until immediately before installation.

1.5 WARRANTY

- A. See Section 01 7700 - Closeout Submittals, for additional warranty requirements.
- B. Correct defective work within a five year period after Date of Substantial Completion.
- C. Provide twenty (20) year manufacturer's material and labor warranty to cover failure to prevent penetration of water.

PART 2 PRODUCTS

2.1 MATERIALS

Use any metal listed by SMACNA Arch. Manual for a particular item, unless otherwise specified or indicated. Conform to the requirements specified and to the thicknesses and configurations established in SMACNA Arch. Manual for the materials. Different items need not be of the same metal, except that if copper is selected for any exposed item, all exposed items must be copper.

2.1.1 Exposed Sheet Metal Items

Must be of the same material. Consider the following as exposed sheet metal: parapet flashings and related accessories.

2.1.2 Stainless Steel Sheet, Type 304

2.1.2.1 Finish

2D Finish

2.1.3 Solder

ASTM B32, 95-5 tin-antimony.

2.1.4 Bituminous Plastic Cement

ASTM D4586/D4586M, Type I.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Nailing

Confine nailing of sheet metal generally to sheet metal having a maximum width of 450 mm (18 inch). Confine nailing of flashing to one edge only. Space nails evenly not over 75 mm (3 inch) on center and approximately 13 mm (1/2 inch) from edge unless otherwise specified or indicated. Face nailing will

not be permitted. Where sheet metal is applied to other than wood surfaces, include in shop drawings, the locations for sleepers and nailing strips required to secure the work.

3.1.2 Cleats

Provide cleats for sheet metal 450 mm (18 inch) and over in width. Space cleats evenly not over 300 mm (12 inch) on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 50 mm wide by 75 mm long (2-inch-wide by 3-inch-long) and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam.

3.1.3 Soldering

Where soldering is specified, apply to copper, terne-coated stainless steel, zinc-coated steel, and stainless-steel items. Seal the joints in aluminum sheets of 0.040 inch or less in thickness with specified sealants. Do not solder aluminum.

3.1.3.1 Edges

Scrape or wire-brush the edges of lead-coated material to be soldered to produce a bright surface. Flux brush the seams in before soldering. Treat with soldering acid flux the edges of stainless steel to be pretinned. Seal the joints in aluminum sheets of one mm (0.040 inch) or less in thickness with specified sealants. Do not solder aluminum.

3.1.4 Protection from Contact with Dissimilar Materials

3.1.4.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

3.1.4.2 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

3.1.4.3 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

3.1.5 Expansion and Contraction

Provide expansion and contraction joints at not more than 9750 mm (32 foot) intervals for aluminum and at not more than 12 meter (40 foot) intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascias by expansion and contraction joints spaced not more than 3600 mm (12 feet) apart.

3.1.6 Metal Reglets

Provide factory fabricated caulked type or friction type reglets with a minimum opening of 6 mm (1/4 inch) and a depth of 30 mm (1 1/4 inch), as approved.

3.1.6.1 Caulked Reglets

Provide with rounded edges and metal strap brackets or other anchors for securing to the concrete forms. Provide reglets with a core to protect them from injury during the installation. Provide built-up mitered corner pieces for internal and external angles. Wedge the flashing in the reglets with lead wedges every 450 mm (18 inch), caulked full and solid with an approved compound.

3.1.6.2 Friction Reglets

Provide with flashing receiving slots not less than 16 mm (5/8 inch) deep, 25 mm (one inch) jointing tongues, and upper and lower anchoring flanges installed at 24 inch maximum snaplock receiver. Insert the flashing the full depth of the slot and lock by indentations made with a dull-pointed tool, wedges, and filled with a sealant. For friction reglets, install flashing snaplock receivers at 24 inch on center maximum. When the flashing has been inserted the full depth, caulk the slot and lock [with wedges] and fill with sealant.

3.2 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

3.3 FIELD QUALITY CONTROL

Establish and maintain a Quality Control Plan for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the contract requirements. Remove work that is not in compliance with the contract and replace or correct. Include quality control, but not be limited to, the following:

- a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.
- b. Verification that specified material is provided and installed.
- c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

3.3.1 Procedure

Submit for approval prior to start of roofing work. Include a checklist of points to be observed. Document the actual quality control observations and inspections. Furnish a copy of the documentation to the Contracting Officer at the end of each day.

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES					
Sheet Metal Items	[Copper kilograms per square meter]	[Aluminum, mm]	[Stainless Steel, mm]	[Terne-Coated Stainless Steel, mm]	[Zinc-Coated Steel, mm]
[Flashings:]					

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES					
Sheet Metal Items	[Copper kilograms per square meter]	[Aluminum, mm]	[Stainless Steel, mm]	[Terne-Coated Stainless Steel, mm]	[Zinc-Coated Steel, mm]
[Base]	6.125	1.02	0.46	0.46	0.6
[Cap (Counter-flashing)]	4.9	0.81	0.6	0.38	0.5
[Roof drain]	4.9 (b)				
[Pipe vent sleeve (d)]					
[Coping]	4.9	-	-	-	-
[Gravel stops and fascias:]					
[Extrusions]	-	1.91	-	-	-
[Sheets, corrugated]	4.9	0.81	0.6	0.38	-
[Sheets, smooth]	6.125	1.27	0.46	0.46	0.6
[Edge strip]	7.35	1.27	0.635	-	-

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES					
Sheet Metal Items	[Copper kilograms per square foot]	[Aluminum, inch]	[Stainless Steel, inch]	[Terne-Coated Stainless Steel, inch]	[Zinc-Coated Steel, U.S. Std. Gage]
[Flashings:]					
[Base]	20	.040	.018	.018	24
[Cap (Counter-flashing)]	16	.032	.015	.015	26
[Eave]	16	-	.015	.015	24
[Spandrel beam]	10	-	.010	.010	-
[Bond barrier]	16	-	.015	.015	-

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES

Sheet Metal Items	[Copper kilograms per square foot]	[Aluminum, inch]	[Stainless Steel, inch]	[Terne-Coated Stainless Steel, inch]	[Zinc-Coated Steel, U.S. Std. Gage]
[Stepped]	16	.032	.015	.015	-
[Valley]	16	.032	.015	.015	-
[Roof drain]	16 (b)				
[Pipe vent sleeve (d)]					
[Coping]	16	-	-	-	-
[Gravel stops and fascias:]					
[Extrusions]	-	.075	-	-	-
[Sheets, corrugated]	16	.032	.015	.015	-
[Sheets, smooth]	20	.050	.018	.018	24
[Edge strip]	24	.050	.025	-	-

TABLE II. SHEET METAL JOINTS

TYPE OF JOINT			
Item Designation	Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel	Aluminum	Remarks
Joint cap for building expansion seam, cleated joint at roof	30 mm single lock, standing seam, cleated	30 mm single lock, standing	--
Flashings			

TABLE II. SHEET METAL JOINTS			
TYPE OF JOINT			
Item Designation	Copper, Terne-Coated Stainless Steel, Zinc- Coated Steel and Stainless Steel	Aluminum	Remarks
Base	25 mm 75 mm lap for expansion joint	25 mm flat locked, soldered; sealed; 75 mm lap for expansion joint	Aluminum producer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound compound.
Cap-in reglet	75 mm lap	75 mm lap	Seal groove with joint sealing compound.
Reglets	Butt joint	--	Seal reglet groove with joint sealing compound.
Gravel stops:			
Extrusions	--	Butt with 13 mm space	Use sheet flashing beneath and a cover plate
Sheet, smooth	Butt with 6 mm space	Butt with 6 mm space	Use sheet flashing backup plate.
Sheet, corrugated	Butt with 6 mm space	Butt with 6 mm space	Use sheet flashing beneath and a cover plate or a combination unit

TABLE II. SHEET METAL JOINTS
TYPE OF JOINT

Item Designation	Copper, Terne-Coated Stainless Steel, Zinc- Coated Steel and Stainless Steel	Aluminum	Remarks
Joint cap for building expansion seam, cleated joint at roof	1.25 inch single lock, standing seam, cleated	1.25 inch single lock, standing	--
Flashings			
Base	One inch 3 inch lap for expansion joint	One inch flat locked, soldered; sealed; 3 inch lap for expansion joint	Aluminum producer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound compound.
Cap-in reglet	3 inch lap	3 inch lap	Seal groove with joint sealing compound.
Reglets	Butt joint	--	Seal reglet groove with joint sealing compound.
Gravel stops:			
Extrusions	--	Butt with 1/2 inch space	Use sheet flashing beneath and a cover plate
Sheet, smooth	Butt with 1/4 inch space	Butt with 1/4 inch space	Use sheet flashing backup plate.
Sheet, corrugated	Butt with 1/4 inch space	Butt with 1/4 inch space	Use sheet flashing beneath and a cover plate or a combination unit

END OF SECTION

SECTION 07 9200 - JOINT SEALANTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Silicone joint sealants.
 - 2. Polysulfide joint sealants.

1.3 PRECONSTRUCTION TESTING

- A. Preconstruction Compatibility and Adhesion Testing: Submit to joint-sealant manufacturers, for testing indicated below, samples of materials that will contact or affect joint sealants.
 - 1. Use ASTM C 1087 or manufacturer's standard test method to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of joint sealants to joint substrates.
 - 2. Submit not fewer than eight pieces of each kind of material, including joint substrates, shims, joint-sealant backings, secondary seals, and miscellaneous materials.
 - 3. Schedule sufficient time for testing and analyzing results to prevent delaying the Work.
 - 4. For materials failing tests, obtain joint-sealant manufacturer's written instructions for corrective measures including use of specially formulated primers.
 - 5. Testing will not be required if joint-sealant manufacturers submit joint preparation data that are based on previous testing, not older than 24 months, of sealant products for adhesion to, and compatibility with, joint substrates and other materials matching those submitted.

1.4 SUBMITTALS

- A. Product Data: For each joint-sealant product indicated.
 - 1. Written explanation to decipher code numbers used on material containers to record manufacturing dates.
 - 2. Provide two tubes of sealant and adequate cleaners and primers, along with packaging carton containing coded manufacturing date.

- B. Joint-Sealant Schedule: Include the following information:
 - 1. Joint-sealant application, joint location, and designation.
 - 2. Joint-sealant manufacturer and product name.
 - 3. Joint-sealant formulation.
 - 4. Joint-sealant color.
- C. Qualification Data: For qualified Installer.
- D. Product Certificates: For each kind of joint sealant and accessory, from manufacturer.
- E. Product Test Reports: Based on evaluation of comprehensive tests performed by a qualified testing agency, indicating that sealants comply with requirements.
 - 1. Laboratory test reports from the sealant manufacturer for adhesion-in-peel tests showing adequate adhesion to the various brick, mortar, aluminum, metal flashing, and other substrates used on this project before and after seven days' water immersion, along with the manufacturer's recommendations for cleaning and priming each substrate.
- F. Preconstruction Compatibility and Adhesion Test Reports: From sealant manufacturer, indicating the following:
 - 1. Materials forming joint substrates and joint-sealant backings have been tested for compatibility and adhesion with joint sealants.
 - a. Reports for tests made within three years of the submittal date showing compliance with ASTM C920, and the standards C920 references including C793 and C719, using the standard substrates.
 - 2. Interpretation of test results and written recommendations for primers and substrate preparation needed for adhesion.
- G. Warranties: Sample of special warranties.

1.5 QUALITY CONTROL

- A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.
- B. Source Limitations: Obtain each kind of joint sealant from single source from single manufacturer.
- C. Conduct a quality control program that includes the following as a minimum:
 - 1. Inspect conditions and materials to ensure conformity to the contract requirements.
 - 2. Inspect substrate conditions and coordinate with the Engineer to ensure proper substrate preparation in conformance to the contract requirements.

3. Inspect work in progress and during inclement weather to ensure that the work is in compliance with approved procedures and that no water ponds on the membrane or leaks into the roofing system or the building.
4. Inspect all completed and any corrected work for compliance with the Contract Documents and the sealant manufacturer's recommendations.

D. Preinstallation Conference: Conduct conference at Project site.

E. Protection: Comply with Section 01020, OSHA, and published, approved manufacturer's recommendations.

1.6 PROJECT CONDITIONS

A. Do not proceed with installation of joint sealants under the following conditions:

1. When ambient and substrate temperature conditions are outside limits permitted by joint-sealant manufacturer or are below 40 deg F.
2. When joint substrates are wet.
3. Where joint widths are less than those allowed by joint-sealant manufacturer for applications indicated.
4. Where contaminants capable of interfering with adhesion have not yet been removed from joint substrates.

1.7 MATERIAL HANDLING AND STORAGE

A. All materials shall be delivered to the jobsite in their original, unopened containers with the manufacturer's name, grade, number, batch identification and date of production on the container or packaging. Do not use materials whose shelf life have expired or are more than six months old. Do not allow liquid materials to freeze. Use identical materials from the same manufacturer throughout the entire installation.

1.8 WARRANTY

A. Special Installer's Warranty: Manufacturer's standard form in which Installer agrees to repair or replace joint sealants that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

B. Special Manufacturer's Warranty: Manufacturer's standard form in which joint-sealant manufacturer agrees to furnish joint sealants to repair or replace those that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Provide five year manufacturer's warranty for installed polyurethane sealants and accessories which fail to achieve airtight seal or watertight seal, exhibit loss of adhesion or cohesion, and sealants which do not cure.

2. Provide twenty year manufacturers warranty for installed non-traffic silicone sealants and accessories which fail to achieve airtight seal or watertight seal, exhibit loss of adhesion or cohesion, and sealants which do not cure.
 3. Provide five year manufacturers warranty for installed traffic silicone sealants and accessories which fail to achieve airtight seal or watertight seal, exhibit loss of adhesion or cohesion, and sealants which do not cure.
- C. Special warranties specified in this article exclude deterioration or failure of joint sealants from the following:
1. Movement of the structure caused by structural settlement or errors attributable to design or construction resulting in stresses on the sealant exceeding sealant manufacturer's written specifications for sealant elongation and compression.
 2. Disintegration of joint substrates from natural causes exceeding design specifications.
 3. Mechanical damage caused by individuals, tools, or other outside agents.
 4. Changes in sealant appearance caused by accumulation of dirt or other atmospheric contaminants.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

- A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer, based on testing and field experience.
- B. Stain-Test-Response Characteristics: Where sealants are specified to be nonstaining to porous substrates, provide products that have undergone testing according to ASTM C 1248 and have not stained porous joint substrates indicated for Project.
- C. Colors of Exposed Joint Sealants: As selected by Architect from manufacturer's full range.

2.2 SILICONE JOINT SEALANTS

- A. Silicone Exterior Joints: ASTM C920, Type S, Grade NS, Class 25/50/100; NT, M, G, A and O: single component, neutral or moisture curing, nonstaining, nonbleeding, color as selected:
 1. General Electric; SilPruf LM SCS2700 for EIFS joints or SilPruf NB SCS9000 for other joints.
 2. Dow Corning; 795.
 3. Pecora; 864.
 4. Tremco; Spectrem 2.
 5. Applications: Use for exterior non-traffic bearing joints.

- a. Control and soft joints in masonry.
 - b. Joints between concrete and other materials.
 - c. Joints between metal frames and other materials.
 - d. Other exterior non-traffic bearing joints for which no other sealant is indicated.
 - e. All glazing seals.
- B. Polyurethane Traffic Joints: ASTM C920, Type M, Grade P, Class 25; self leveling, two component, chemical curing, nonstaining, nonbleeding, capable of continuous water immersion, color as selected; manufactured by:
- 1. Tremco, Inc.; THC-900.
 - 2. Pecora; Urexpan NR-200.
 - 3. BASF Building Systems; Sonneborn; Sonolastic SL2.
 - 4. Applications: Use for exterior horizontal traffic bearing joints:
 - a. Joints in concrete floors and paving.
 - b. Soft joints in unit pavers.
 - c. Other traffic bearing joints for which no other sealant is indicated.

2.3 MISCELLANEOUS SEALANTS

- A. Exterior Metal Lap Joint Sealant: ASTM C1311, butyl or polyisobutylene, nondrying, non-skinning, non-curing.
- 1. Tremco, Inc.; Tremco Butyl Sealant.
 - 2. Pecora; BC-158; Butyl Rubber Sealant.
 - 3. Applications: Concealed sealant bead in sheet metal and flashing work.
 - a. Concealed sealant bead in sheet metal work.
 - b. Concealed sealant bead in siding overlaps.
 - c. Concealed sealant bead in vapor barrier.
 - d. Concealed sealant bead in through wall flashing.
 - e. Other concealed sealant joints where specified in other Sections.

2.4 JOINT SEALANT BACKING

- A. General: Provide sealant backings of material that are nonstaining; are compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.
- B. Cylindrical Sealant Backings: ASTM C 1330, Type C (closed-cell material with a surface skin), Type O (open-cell material, unsuitable for horizontal surfaces), Type B (bicellular material with a surface skin) or any of the preceding types, as approved in writing by joint-sealant manufacturer for joint application indicated, and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance. Closed-cell sealant backer rod, nongassing polyethylene, with the diameter of the rod is to be approximately 25 percent in excess of joint width. Surface

skin of rod to be continuous and unbroken and of sufficient thickness to preclude outgassing and formation of voids in the overlying sealant.

- C. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint. Adhesive-backed on one side, width as required.

2.5 MISCELLANEOUS MATERIALS

- A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.
- B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants to joint substrates.
- C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting joint-sealant performance.
- B. Verify all site conditions and dimensions by field measurement in consideration of the special conditions associated with repairs to existing construction prior to development of submittals and to material fabrication or delivery. Notify the Engineer immediately of any inconsistency between the conditions found and those shown in the Drawings.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Surface Preparation General:
 - 1. Mechanical Cleaning: Cut away as much existing sealant from existing substrates as possible. Grind any existing masonry or concrete areas scheduled to receive sealant, and expose virgin surface.
 - 2. Solvent Cleaning: Use two clean, white, soft, lint-free, cotton cloths and clean, fresh solvent as required to clean all surfaces. Wet one cloth with solvent and wipe surface vigorously. Use second cloth to clean surface before solvent evaporates. Pump solvent from cans onto first cloth. Do not dip cloth in solvent, to avoid contamination of solvent. Repeat this two-cloth procedure until surface

does not discolor cloth, and repeat at least once. Do not solvent clean at temperatures below 45°F. Allow solvent to evaporate from non-porous surfaces before continuing.

3. Remove all dirt or other foreign substances, including existing sealant, from surfaces to receive sealant. All surfaces shall be dry before preparation begins. The solvent cleaning preparation is to be done immediately before insertion of the final backer rod or bond breaker and after any temporary rods or seals are removed. Do not prepare or seal over masonry that is less than fourteen days old.
- B. Surface Cleaning of Joints: Grind and clean out joints, as required, immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions and the following requirements:
1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
 2. Clean porous joint substrate surfaces by brushing, grinding, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining after cleaning operations above by vacuuming or blowing out joints with oil-free compressed air. Porous joint substrates include the following:
 - a. Concrete.
 - b. Masonry.
 - c. Exterior insulation and finish systems.
 3. Remove laitance and form-release agents from concrete.
- C. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.
- D. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 INSTALLATION OF JOINT SEALANTS

- A. General: Comply with joint-sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.
1. Provide sealant at reglet-set metal flashings.
 2. Provide miscellaneous other seals where shown on the Drawings.
 3. Provide sealant at all window surrounds, and at all building control joints, shelf angle locations etc.

4. Install Back-Up Material:
 - a. Unless noted otherwise, install clean, dry joint filler/back-up rod or tape into all joints openings against dry substrates. Remove all wet materials from the jobsite. Replace any backer rod not sealed over by the end of each day and solvent clean surfaces again.
 - b. Place the backer rod or bond breaker so the sealant shape will meet the joint shape requirements of this section and as shown on the Drawings.
 - c. Change rod sizes as frequently as required by the variation in the joint width. Do not twist rods together. Butt ends of rods tightly. Provide a full range of rod sizes at the site of all sealant work.
 - d. Do not touch with fingers or otherwise contaminate the substrate surfaces while inserting the backer rod or bond breaker tape.
 - e. Do not rupture the skin of the closed-cell backer rod during installation. Remove any rod containing punctures and solvent-clean the surfaces again.
5. Primer:
 - a. Apply primer to all substrates after backer rod installation. Apply primer to clean, dry substrates at ambient temperatures above 45 degrees F.
 - b. Pour primer into a clean container for use. Do not pour more than a 10 min. supply into container to prevent deterioration.
 - c. Replace cap on primer can immediately after use. Remove from the site any primer that is discolored, contains a precipitate or has thickened.
 - d. Apply primer with a clean brush. Do not apply primer to exposed surfaces beyond sealant. Mask all surfaces before priming, except where surface irregularities will allow the primer to wick beneath the masking tape. Use only one coat of primer. Do not apply primer in a thick layer.
 - e. Allow primer to dry. Do not allow primer to become wet before sealant application.
6. Sealant:
 - a. Inspect each cartridge or container of sealant before use and verify that the production date is within six months of the date of application. Remove from the site all sealant more than six months old. Each applicator shall understand the method of coding the production date on the cartridge.
 - b. Mask all exposed surfaces along joint before applying sealant.
 - c. Recheck correct backer rod and bond breaker tape positioning before applying sealant.
 - d. Apply sealant only to clean, dry, primed surfaces at ambient temperatures above 45°F. Seal joints within 10 hrs of primer application.
 - e. Fill all joints solidly and continuously with sealant, neatly applied with a standard caulking gun in a continuous motion, using a slight pressure. "Push" the sealant bead ahead of the nozzle; do not "drag" the nozzle.
 - f. Within 5 min. of sealant application and before skin develops on sealant, dry tool the joint surface with a concave tool to insure intimate contact with substrate and to eliminate air bubbles. Do not use any liquid for tooling. Provide a smooth, uniform finished surface.
 - g. Remove masking tape within 10 min. of tooling. Avoid contaminating adjacent surfaces with excess sealant. Remove all traces of smears and

droppings on metal or glass surfaces promptly, using a solvent recommended by the sealant manufacturer and that will not damage or discolor the building surfaces. Remove smears and droppings on porous surfaces by mechanical means after the initial cure of the sealant.

- h. Coordinate work with other trades to prevent contamination of fresh sealant by dust or other debris.
- B. Sealant Installation Standard: Comply with recommendations in ASTM C 1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.
- C. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.
 - 1. Do not leave gaps between ends of sealant backings.
 - 2. Do not stretch, twist, puncture, or tear sealant backings.
 - 3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.
- D. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and backs of joints.
- E. Install sealants using proven techniques that comply with the following and at the same time backings are installed:
 - 1. Place sealants so they directly contact and fully wet joint substrates.
 - 2. Completely fill recesses in each joint configuration.
 - 3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.
- F. Acoustical Sealant Installation: At sound-rated assemblies and elsewhere as indicated, seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Comply with ASTM C 919 and with manufacturer's written recommendations.

3.4 CLEANING

- A. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

3.5 PROTECTION

- A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from original work.

END OF SECTION

SECTION 09 2900 - GYPSUM BOARD

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Interior gypsum board.

1.2 ACTION SUBMITTALS

- A. Product data.
- B. Shop Drawings: Show locations and installation of control and expansion joints, including plans, elevations, sections, details of components, and attachments to other work.
- C. Samples: For each texture finish indicated on same backing indicated for Work.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Fire-Resistance-Rated Assemblies: For fire-resistance-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E119 by an independent testing agency.
- B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E90 and classified according to ASTM E413 by an independent testing agency.

2.2 GYPSUM BOARD, GENERAL

- A. General: Complying with ASTM C 36/C 36M or ASTM C 1396/C 1396M, as applicable to type of gypsum board indicated and whichever is more stringent.
1. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. American Gypsum Co.
 - b. BPB America Inc.
 - c. G-P Gypsum.
 - d. Lafarge North America Inc.
 - e. National Gypsum Company.
 - f. PABCO Gypsum.
 - g. Temple.

h. USG Corporation.

- B. Size: Provide maximum lengths and widths available that will minimize joints in each area and that correspond with support system indicated.

2.3 INTERIOR GYPSUM BOARD

1. Thickness: 5/8 inch.
2. Long Edges: Tapered

2.4 TRIM ACCESSORIES

A. Interior Trim: ASTM C1047.

1. Material: Galvanized or aluminum-coated steel sheet, rolled zinc, plastic, or paper-faced galvanized-steel sheet.
2. Shapes:
 - a. Cornerbead.
 - b. Bullnose bead.
 - c. LC-Bead: J-shaped; exposed long flange receives joint compound.
 - d. L-Bead: L-shaped; exposed long flange receives joint compound.
 - e. U-Bead: J-shaped; exposed short flange does not receive joint compound.

2.5 JOINT TREATMENT MATERIALS

A. General: Comply with ASTM C475/C475M.

B. Joint Tape:

1. Interior Gypsum Board: Paper.

C. Joint Compound for Interior Gypsum Board: For each coat, use formulation that is compatible with other compounds applied on previous or for successive coats.

1. Prefilling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting-type taping compound.
2. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use drying-type compound.
 - a. Use setting-type compound for installing paper-faced metal trim accessories.
3. Fill Coat: For second coat, use drying-type, all-purpose compound.
4. Finish Coat: For third coat, use drying-type, all-purpose compound.
5. Skim Coat: For final coat of Level 5 finish, use drying-type, all-purpose compound.

2.6 AUXILIARY MATERIALS

A. Provide auxiliary materials that comply with referenced installation standards and manufacturer's written instructions.

- B. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.
- C. Steel Drill Screws: ASTM C1002 unless otherwise indicated.
 - 1. Use screws complying with ASTM C954 for fastening panels to steel members from 0.033 to 0.112 inch (0.84 to 2.84 mm) thick.
- D. Sound-Attenuation Blankets: ASTM C665, Type I (blankets without membrane facing) produced by combining thermosetting resins with mineral fibers manufactured from glass, slag wool, or rock wool.
- E. Acoustical Sealant: As specified in Section 079219 "Acoustical Joint Sealants."

PART 3 - EXECUTION

3.1 INSTALLATION OF PANELS

- A. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.
- B. Comply with ASTM C840.
- C. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments. Provide 1/4- to 1/2-inch- (6.4- to 12.7-mm-) wide spaces at these locations and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.
- D. For trim with back flanges intended for fasteners, attach to framing with same fasteners used for panels. Otherwise, attach trim according to manufacturer's written instructions.

3.2 FINISHING OF GYPSUM BOARD

- A. Prefill open joints, rounded or beveled edges, and damaged surface areas.
- B. Apply joint tape over gypsum board joints, except for trim products specifically indicated as not intended to receive tape.
- C. Gypsum Board Finish Levels: Finish panels to levels indicated below and according to ASTM C840:
 - 1. Level 1: Embed tape at joints in ceiling plenum areas, concealed areas, and where indicated, unless a higher level of finish is required for fire-resistance-rated assemblies and sound-rated assemblies.
 - 2. Level 2: Embed tape and apply separate first coat of joint compound to tape, fasteners, and trim flanges where panels are substrate for tile and where indicated.
 - 3. Level 4: Embed tape and apply separate first, fill, and finish coats of joint compound to tape, fasteners, and trim flanges at panel surfaces that will be exposed to view, unless otherwise indicated.
 - 4. Level 5: Embed tape and apply separate first, fill, and finish coats of joint compound to tape, fasteners, and trim flanges, and apply skim coat of joint compound over entire surface where indicated on drawings.

- a. Primer and its application to surfaces are specified in Section 099123 "Interior Painting."

3.3 APPLICATION OF TEXTURE FINISHES

- A. Surface Preparation and Primer: Prepare and apply primer to gypsum panels and other surfaces receiving texture finishes. Apply primer to surfaces that are clean, dry, and smooth.
- B. Texture Finish Application: Mix and apply finish using powered spray equipment, to produce a uniform texture free of starved spots or other evidence of thin application or of application patterns.

3.4 PROTECTION

- A. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.
- B. Remove and replace panels that are wet, moisture damaged, and mold damaged.

END OF SECTION

SECTION 09 5123 - ACOUSTICAL TILE CEILINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Acoustical tiles for interior ceilings.
- B. Related Requirements:
- C. Products furnished, but not installed under this Section, include anchors, clips, and other ceiling attachment devices to be cast in concrete.

1.3 PREINSTALLATION MEETINGS

- A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each exposed product and for each color and texture specified, 6 inches (150 mm) in size.
- C. Samples for Initial Selection: For components with factory-applied finishes.
- D. Samples for Verification: For each component indicated and for each exposed finish required, prepared on Samples of sizes indicated below:
 - 1. Acoustical Tiles: Set of full-size Samples of each type, color, pattern, and texture.
 - 2. Exposed Moldings and Trim: Set of [6-inch- (150-mm-)] each type and color.

1.5 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:
 - 1. Ceiling suspension-system members.
 - 2. Structural members to which suspension systems will be attached.
 - 3. Method of attaching hangers to building structure.

- a. Furnish layouts for cast-in-place anchors, clips, and other ceiling attachment devices whose installation is specified in other Sections.
- 4. Carrying channels or other supplemental support for hanger-wire attachment where conditions do not permit installation of hanger wires at required spacing.
- 5. Size and location of initial access modules for acoustical tile.
- 6. Items penetrating finished ceiling and ceiling-mounted items including the following:
 - a. Lighting fixtures.
 - b. Diffusers.
 - c. Grilles.
 - d. Sprinklers.
 - e. Perimeter moldings.
- 7. Show operation of hinged and sliding components adjacent to acoustical tiles.
- 8. Minimum Drawing Scale: 1/8 inch = 1 foot (1:96)
- B. Qualification Data: For testing agency.
- C. Product Test Reports: For each acoustical tile ceiling, for tests performed by qualified testing agency.
- D. Evaluation Reports: For each acoustical tile ceiling suspension system and anchor and fastener type, from ICC-ES.
- E. Field quality-control reports.

1.6 CLOSEOUT SUBMITTALS

- A. Maintenance Data: For finishes to include in maintenance manuals.

1.7 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Acoustical Ceiling Units: Full-size tiles equal to 2 percent of quantity installed.
 - 2. Suspension-System Components: Quantity of each concealed grid and exposed component equal to 2 percent of quantity installed.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver acoustical tiles, suspension-system components, and accessories to Project site and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.
- B. Before installing acoustical tiles, permit them to reach room temperature and a stabilized moisture content.

1.9 FIELD CONDITIONS

- A. Environmental Limitations: Do not install acoustical tile ceilings until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.
 - 1. Pressurized Plenums: Operate ventilation system for not less than 48 hours before beginning acoustical tile ceiling installation.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Source Limitations:
 - 1. Suspended Acoustical Tile Ceilings: Obtain each type of acoustical ceiling tile and its suspension system from single source from single manufacturer.
 - 2. Directly Attached Acoustical Tile Ceilings: Obtain each type of acoustical ceiling tile from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

- A. Surface-Burning Characteristics: Comply with ASTM E84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
 - 1. Flame-Spread Index: Class A according to ASTM E1264.
 - 2. Smoke-Developed Index: 50 or less.

2.3 ACOUSTICAL TILES

- 1. Acceptable Manufacturers
 - a. Armstrong World Industries
 - b. CertainTeed Corp.
 - c. USG Corp.
 - d. Rockfon
- B. Acoustical Tile Standard: Provide manufacturer's standard tiles of configuration indicated that comply with ASTM E1264 classifications as designated by type, form, pattern, acoustical rating, and light reflectance unless otherwise indicated.
- C. Light Reflectance (LR): Match existing
- D. Ceiling Attenuation Class (CAC): Match existing.
- E. Noise Reduction Coefficient (NRC): Match existing
- F. Edge/Joint Detail: Match existing.
- G. Thickness: Match existing.

- H. Modular Size: 24 by 24 inches (610 by 610 mm)
- I. Antimicrobial Treatment: Manufacturer's standard broad spectrum, antimicrobial formulation that inhibits fungus, mold, mildew, and gram-positive and gram-negative bacteria and showing no mold, mildew, or bacterial growth when tested according to ASTM D3273, ASTM D3274, or ASTM G21 and evaluated according to ASTM D3274 or ASTM G21.

2.4 METAL SUSPENSION SYSTEM

- A. Metal Suspension-System Standard: Provide manufacturer's standard, direct-hung, metal suspension system and accessories of type, structural classification, and finish indicated that complies with applicable requirements in ASTM C635/C635M.
- B. 15/16" Prelude "XL"

2.5 ACCESSORIES

- A. Attachment Devices: Size for five times the design load indicated in ASTM C635/C635M, Table 1, "Direct Hung," unless otherwise indicated. Comply with seismic design requirements.
 - 1. Anchors in Concrete: Anchors of type and material indicated below, with holes or loops for attaching hangers of type indicated and with capability to sustain, without failure, a load equal to five times that imposed by ceiling construction, as determined by testing according to ASTM E488/E488M or ASTM E1512 as applicable, conducted by a qualified testing and inspecting agency.
 - a. Type: Postinstalled expansion anchors.
 - b. Corrosion Protection: Carbon-steel components zinc plated according to ASTM B633, Class SC 1 (mild) service condition.
 - c. Corrosion Protection: Stainless-steel components complying with ASTM F593 and ASTM F594, Group 1 Alloy 304 or 316.
 - 2. Power-Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hangers of type indicated, and with capability to sustain, without failure, a load equal to 10 times that imposed by ceiling construction, as determined by testing according to ASTM E1190, conducted by a qualified testing and inspecting agency.
- B. Wire Hangers, Braces, and Ties: Provide wires as follows:
 - 1. Zinc-Coated, Carbon-Steel Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper.
 - 2. Stainless-Steel Wire: ASTM A580/A580M, Type 304, nonmagnetic.
 - 3. Size: Wire diameter sufficient for its stress at three times hanger design load (ASTM C635/C635M, Table 1, "Direct Hung") will be less than yield stress of wire, but not less than 0.106-inch- (2.69-mm-) diameter wire.
- C. Hanger Rods: Mild steel, zinc coated or protected with rust-inhibitive paint.
- D. Flat Hangers: Mild steel, zinc coated or protected with rust-inhibitive paint.

- E. Angle Hangers: Angles with legs not less than 7/8-inch (22 mm) wide; formed with 0.04-inch- (1-mm-) thick, galvanized-steel sheet complying with ASTM A653/A653M, G90 (Z275) coating designation; with bolted connections and 5/16-inch- (8-mm-) diameter bolts.

2.6 METAL EDGE MOLDINGS AND TRIM

- A. Roll-Formed, Sheet-Metal Edge Moldings and Trim: Type and profile indicated or, if not indicated, manufacturer's standard moldings for edges and penetrations complying with seismic design requirements; formed from sheet metal of same material, finish, and color as that used for of suspension-system runners.
 - 1. For circular penetrations of ceiling, provide edge moldings fabricated to diameter required to fit penetration exactly.
 - 2. Finish: Painted to match color indicated by manufacturer's designation
- B. Extruded-Aluminum Edge Moldings and Trim: Where indicated, provide manufacturer's extruded-aluminum edge moldings and trim of profile indicated or referenced by manufacturer's designations, including splice plates, corner pieces, and attachment and other clips, complying with seismic design requirements.
 - 1. Clear Anodic Finish: AAMA 611, AA-M12C22A31, Class II, 0.010 mm or thicker.
 - 2. Baked-Enamel or Powder-Coat Finish: Minimum dry film thickness of 1.5 mils (0.04 mm). Comply with ASTM C635/C635M and coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

2.7 MISCELLANEOUS MATERIALS

- A. Acoustical Tile Adhesive: Type recommended in writing by acoustical tile manufacturer, bearing UL label for Class 0-25 flame spread.
- B. Staples: 5/16-inch- (8-mm-) long, divergent-point staples.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, including structural framing and substrates to which acoustical tile ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.
- B. Examine acoustical tiles before installation. Reject acoustical tiles that are wet, moisture damaged, or mold damaged.
- C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Testing Substrates: Before adhesively bonding tiles to wet-placed substrates such as cast-in-place concrete or plaster, test and verify that moisture level is below tile manufacturer's recommended limits.
- B. Measure each ceiling area and establish layout of acoustical tiles to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width tiles at borders unless otherwise indicated, and comply with layout shown on reflected ceiling plans.
- C. Layout openings for penetrations centered on the penetrating items.

3.3 INSTALLATION OF SUSPENDED ACOUSTICAL TILE CEILINGS

- A. Install suspended acoustical tile ceilings according to ASTM C636/C636M and manufacturer's written instructions.
- B. Suspend ceiling hangers from building's structural members and as follows:
 - 1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
 - 2. Splay hangers only where required to miss obstructions; offset resulting horizontal forces by bracing, countersplaying, or other equally effective means.
 - 3. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
 - 4. Secure wire hangers to ceiling suspension members and to supports above with a minimum of three tight turns. Connect hangers directly to structure or to inserts, eye screws, or other devices that are secure and appropriate for substrate and that will not deteriorate or otherwise fail due to age, corrosion, or elevated temperatures.
 - 5. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both the structure to which hangers are attached and the type of hanger involved. Install hangers in a manner that will not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
 - 6. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to cast-in-place hanger inserts, post installed mechanical or adhesive anchors, or power-actuated fasteners that extend through forms into concrete.
 - 7. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
 - 8. Do not attach hangers to steel deck tabs.
 - 9. Do not attach hangers to steel roof deck. Attach hangers to structural members.
 - 10. Space hangers not more than 48 inches (1200 mm) o.c. along each member supported directly from hangers unless otherwise indicated; provide hangers not more than 8 inches (200 mm) from ends of each member.
 - 11. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards.
- C. Secure bracing wires to ceiling suspension members and to supports with a minimum of four tight turns. Suspend bracing from building's structural members as required for hangers without attaching to permanent metal forms, steel deck, or steel deck tabs. Fasten bracing wires into concrete with cast-in-place or post installed anchors.

- D. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical tiles.
 - 1. Apply acoustical sealant in a continuous ribbon concealed on back of vertical legs of moldings before they are installed.
 - 2. Screw attach moldings to substrate at intervals not more than 16 inches (400 mm) o.c. and not more than 3 inches (75 mm) from ends. Miter corners accurately and connect securely.
 - 3. Do not use exposed fasteners, including pop rivets, on moldings and trim.
- E. Install suspension-system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.
- F. Arrange directionally patterned acoustical tiles as follows:
 - 1. As indicated on reflected ceiling plans.
- G. Install acoustical tiles in coordination with suspension system and exposed moldings and trim. Place splines or suspension-system flanges into kerfed edges of tiles so tile-to-tile joints are interlocked.
 - 1. Fit adjoining tiles to form flush, tight joints. Scribe and cut tiles for accurate fit at borders and around penetrations through ceiling.
 - 2. Hold tile field in compression by inserting leaf-type, spring-steel spacers between tiles and moldings, spaced 12 inches (305 mm) o.c.
 - 3. Protect lighting fixtures and air ducts according to requirements indicated for fire-resistance-rated assembly.

3.4 ERECTION TOLERANCES

- A. Suspended Ceilings: Install main and cross runners level to a tolerance of 1/8 inch (3 mm in 3.6 m), non-cumulative.
- B. Moldings and Trim: Install moldings and trim to substrate and level with ceiling suspension system to a tolerance of 1/8 inch in 12 feet (3 mm in 3.6 m) non-cumulative.

3.5 FIELD QUALITY CONTROL

- A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.
- B. Perform the following tests and inspections of completed installations of acoustical tile ceiling hangers and anchors and fasteners in successive stages and when installation of ceiling suspension systems on each floor has reached 20 percent completion, but no tiles have been installed. Do not proceed with installations of acoustical tile ceiling hangers for the next area until test results for previously completed installations of acoustical tile ceiling hangers show compliance with requirements.
 - 1. Within each test area, testing agency will select one of every 10 power-actuated fasteners and post installed anchors used to attach hangers to concrete and will test them for 200 lbf (890 N) of tension; it will also select one of every two post installed anchors used to attach bracing wires to concrete and will test them for 440 lbf (1957 N) of tension.

2. When testing discovers fasteners and anchors that do not comply with requirements, testing agency will test those anchors not previously tested until 20 pass consecutively and then will resume initial testing frequency.
- C. Acoustical tile ceiling hangers, anchors, and fasteners will be considered defective if they do not pass tests and inspections.
- D. Prepare test and inspection reports.

3.6 ADJUSTING

- A. Clean exposed surfaces of acoustical tile ceilings, including trim and edge moldings. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage.
- B. Remove and replace tiles and other ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

END OF SECTION

SECTION 09 6513 - RESILIENT BASE AND ACCESSORIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Rubber Wall base.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Samples: For each exposed product and for each color and texture specified, not less than 12 inches (300 mm) long.
- C. Samples for Initial Selection: For each type of product indicated.
- D. Samples for Verification: For each type of product indicated and for each color, texture, and pattern required in manufacturer's standard-size Samples, but not less than 12 inches (300 mm) long.
- E. Product Schedule: For resilient base and accessory products. Use same designations indicated on Drawings.

1.4 MAINTENANCE MATERIAL SUBMITTALS

- A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
 - 1. Furnish not less than 10 linear feet (3 linear m) or fraction thereof, of each type, color, pattern, and size of resilient product installed.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Store resilient products and installation materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than 50 deg F (10 deg C) or more than 90 deg F (32 deg C).

1.6 FIELD CONDITIONS

- A. Maintain ambient temperatures within range recommended by manufacturer, but not less than 70 deg F (21 deg C) or more than 95 deg F (35 deg C), in spaces to receive resilient products during the following periods:
 - 1. 48 hours before installation.
 - 2. During installation.
 - 3. 48 hours after installation.
- B. After installation and until Substantial Completion, maintain ambient temperatures within range recommended by manufacturer, but not less than 55 deg F (13 deg C) or more than 95 deg F (35 deg C).
- C. Install resilient products after other finishing operations, including painting, have been completed.

PART 2 - PRODUCTS

2.1 RUBBER WALL BASE

- A. Product Standard: ASTM F 1861, Type TS (rubber, vulcanized thermoset), Group I (solid, homogeneous).
 - 1. Style and Location:
 - a. Style: Cove
- B. Acceptable Manufacturers:
 - 1. Basis of Design: Armstrong
 - 2. Johnsonite
 - 3. Roppe
- C. Thickness: Match existing
- D. Height: 4 inches (102 mm)
- E. Lengths: Coils in manufacturer's standard length
- F. Outside Corners: Job formed or preformed.
- G. Inside Corners: Job formed or preformed.
- H. Colors: Match existing.

2.2 INSTALLATION MATERIALS

- A. Trowelable Leveling and Patching Compounds: Latex-modified, portland cement based or blended hydraulic-cement-based formulation provided or approved by resilient-product manufacturer for applications indicated.

- B. Adhesives: Water-resistant type recommended by resilient-product manufacturer for resilient products and substrate conditions indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
 - 1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resilient products.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
 - 1. Installation of resilient products indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Prepare substrates according to manufacturer's written instructions to ensure adhesion of resilient products.
- B. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound; remove bumps and ridges to produce a uniform and smooth substrate.
- C. Do not install resilient products until materials are the same temperature as space where they are to be installed.
 - 1. At least 48 hours in advance of installation, move resilient products and installation materials into spaces where they will be installed.
- D. Immediately before installation, sweep and vacuum clean substrates to be covered by resilient products.

3.3 RESILIENT BASE INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient base.
- B. Apply resilient base to walls, columns, pilasters, casework and cabinets in toe spaces, and other permanent fixtures in rooms and areas where base is required.
- C. Install resilient base in lengths as long as practical without gaps at seams and with tops of adjacent pieces aligned.
- D. Tightly adhere resilient base to substrate throughout length of each piece, with base in continuous contact with horizontal and vertical substrates.
- E. Do not stretch resilient base during installation.

F. Job-Formed Corners:

1. Outside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 3 inches (76 mm) in length.
 - a. Form without producing discoloration (whitening) at bends.
2. Inside Corners: Use straight pieces of maximum lengths possible and form with returns not less than 3 inches (76 mm) in length.
 - a. Cope corners to minimize open joints.

3.4 RESILIENT ACCESSORY INSTALLATION

- A. Comply with manufacturer's written instructions for installing resilient accessories.
- B. Resilient Molding Accessories: Butt to adjacent materials and tightly adhere to substrates throughout length of each piece. Install reducer strips at edges of floor covering that would otherwise be exposed.

3.5 CLEANING AND PROTECTION

- A. Comply with manufacturer's written instructions for cleaning and protecting resilient products.
- B. Perform the following operations immediately after completing resilient-product installation:
 1. Remove adhesive and other blemishes from surfaces.
 2. Sweep and vacuum horizontal surfaces thoroughly.
 3. Damp-mop horizontal surfaces to remove marks and soil.
- C. Protect resilient products from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.
- D. Cover resilient products subject to wear and foot traffic until Substantial Completion.

END OF SECTION

SECTION 09 9124 - INTERIOR PAINTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes surface preparation and the application of paint systems on the following interior substrates:
 - 1. Gypsum board.
 - 2. Concrete masonry units (CMU).

1.3 DEFINITIONS

- A. MPI Gloss Level 1: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D523.
- B. MPI Gloss Level 2: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D523.
- C. MPI Gloss Level 3: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D523.
- D. MPI Gloss Level 4: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D523.
- E. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D523.
- F. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D523.
- G. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D523.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product. Include preparation requirements and application instructions.
 - 1. Include printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
 - 2. Indicate VOC content.
- B. Samples for Initial Selection: For each type of topcoat product.
- C. Samples for Verification: For each type of paint system and in each color and gloss of topcoat.

1. Submit Samples on rigid backing, 8 inches (200 mm) square.
 2. Apply coats on Samples in steps to show each coat required for system.
 3. Label each coat of each Sample.
 4. Label each Sample for location and application area.
- D. Product List: Use same designations indicated on Drawings and in the Interior Painting Schedule to cross-reference paint systems specified in this Section. Include color designations.

1.5 QUALITY ASSURANCE

- A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
1. Architect will select one surface to represent surfaces and conditions for application of each paint system.
 - a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).
 - b. Other Items: Architect will designate items or areas required.
 2. Final approval of color selections will be based on mockups.
 - a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.
 3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
 4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F (7 deg C).
1. Maintain containers in clean condition, free of foreign materials and residue.
 2. Remove rags and waste from storage areas daily.

1.7 FIELD CONDITIONS

- A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F (10 and 35 deg C).
- B. Do not apply paints when relative humidity exceeds 85 percent; at temperatures of less than 5 deg F (3 deg C) above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Sherwin-Williams.
 2. PPG Architectural Finishes, Inc.
 3. Behr Paint Company

2.2 PAINT, GENERAL

- A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products List."
- B. Material Compatibility:
1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
- C. VOC Content of Field-Applied Interior Paints and Coatings: Provide products that comply with the the limits for VOC content indicated for each product in following paragraphs, exclusive of colorants added to a tint base, when calculated according to 40 CFR 59, Subpart D (EPA Method 24). Requirements do not apply to paints and coatings that are applied in a fabrication or finishing shop.
- D. Chemical Components of Field-Applied Interior Paints and Coatings: Provide topcoat paints and anti-corrosive and anti-rust paints applied to ferrous metals that comply with the following chemical restrictions; these requirements do not apply to paints and coatings that are applied in a fabrication or finishing shop:
1. Aromatic Compounds: Paints and coatings shall not contain more than 1.0 percent by weight of total aromatic compounds (hydrocarbon compounds containing one or more benzene rings).
 2. Restricted Components: Paints and coatings shall not contain any of the following:
 - a. Acrolein.
 - b. Acrylonitrile.
 - c. Antimony.
 - d. Benzene.
 - e. Butyl benzyl phthalate.
 - f. Cadmium.
 - g. Di (2-ethylhexyl) phthalate.
 - h. Di-n-butyl phthalate.
 - i. Di-n-octyl phthalate.
 - j. 1,2-dichlorobenzene.
 - k. Diethyl phthalate.
 - l. Dimethyl phthalate.
 - m. Ethylbenzene.
 - n. Formaldehyde.

- o. Hexavalent chromium.
- p. Isophorone.
- q. Lead.
- r. Mercury.
- s. Methyl ethyl ketone.
- t. Methyl isobutyl ketone.
- u. Methylene chloride.
- v. Naphthalene.
- w. Toluene (methylbenzene).
- x. 1,1,1-trichloroethane.
- y. Vinyl chloride.

E. Colors: Match existing.

2.3 BLOCK FILLERS

- A. Interior/Exterior Latex Block Filler: MPI #4.
1. MPI VOC Range (grams/liter): E3, less than 51 g/l.
 2. Available products include but are not limited to:
 - a. Benjamin Moore, Morecraft, Super Craft Latex Block Filler.
 - b. Cloverdale Paint, Hi-Performance, Int/Ext Acrylic Latex Block Filler.
 - c. Coronado Paint, Supercoat 5000, Commercial Latex Block Filler.
 - d. Farrel Calhoun, Int/Ext Acrylic Latex Masonry Block Filler.
 - e. Kelly Moore, Fill and Prime Acrylic Block Filler.
 - f. PPG, Speedhide, Int/Ext Masonry Block Filler.
 - g. Rodda Paint, Premium Classic, Int/Ext Latex Block Filler.
 - h. Sherwin Williams, PrepRite, Int/Ext Block Filler.

2.4 PRIMERS/SEALERS

- A. Interior Latex Primer/Sealer: MPI #50.
1. MPI VOC Range (grams/liter): E3, less than 51 g/l.
 2. Available products include but are not limited to:
 - a. Bennette Paint, Vinyl Primer Sealer.
 - b. Columbia Paint, Premium Pro, Basecoat Wallboard Primer Sealer.
 - c. Dunn-Edwards, Vinylastic, Interior Pigmented Sealer.
 - d. General Paint, Z Coat Low VOC Sealer.
 - e. ICI Paints (Canada), Glidden, Interior Latex Sealer.
 - f. Kelly Moore, Acry-Plex, Int. PVA Primer/Sealer Low Odor Formula.
 - g. Miller Paint, MPI#50 Primer Low Sheen.
 - h. Rodda Paint, Roseal Interior Latex Wallboard Sealer.
 - i. Sherwin-Williams, Quali-Kote, Interior Latex Primer.
 - j. Vista Paint, Carefree, Earth Coat Primer.
- B. Interior Alkyd Primer/Sealer: MPI #45.
- C. Interior, Institutional Low Odor/VOC Primer Sealer: MPI #149.
1. MPI VOC Range (grams/liter): E3, less than 11 g/l.
 2. Available products include but are not limited to:
 - a. Benjamin Moore, Eco Spec, Interior Latex Primer Sealer.
 - b. Cloverdale Paint, Horizon, Interior Low VOC Drywall Sealer.
 - c. Columbia Paint, Purecoat, Low Odor Interior Primer.
 - d. Dunn-Edwards, Ecoshield, Low-odor/zero-VOC Interior Latex Primer.
 - e. Frazee Paint, Envirokote Primer.

- f. General Paint, Z-coat, Drywall Sealer.
- g. Parker Paint, Klean Air Primer.
- h. Rodda Paint, Horizon, Interior Latex Sealer.

2.5 LATEX PAINTS

- A. Institutional Low-Odor/VOC Latex (Flat): MPI #143 (Gloss Level 1).
 - 1. MPI VOC Range (grams/liter): E3, less than 11 g/l.
 - 2. Available products include but are not limited to:
 - a. Benjamin Moore, Eco Spec, Int. Latex Flat.
 - b. Cloverdale Paint, Horizon, Interior Low VOC Flat.
 - c. Columbia Paint, Purecoat, Low Odor Acrylic Flat.
 - d. General Paint, Z-coat, Flat White.
 - e. ICI Paints, Dulux, Lifemaster Flat Interior Latex Enamel.
 - f. PPG, Pure Performance, Interior Latex Flat.
 - g. Sherwin Williams, Harmony, Interior Latex Flat.
- B. Institutional Low-Odor/VOC Latex (Low Sheen, used where "Eggshell" specified in Part 3): MPI #144 (Gloss Level 2).
 - 1. MPI VOC Range (grams/liter): E3, less than 11 g/l.
 - 2. Available products include but are not limited to:
 - a. Benjamin Moore, Eco Spec, Int. Latex Eggshell Enamel.
 - b. Cloverdale Paint, Horizon, Interior Low VOC Eggshell.
 - c. Columbia Paint, Purecoat, Low Odor Acrylic Eggshell.
 - d. General Paint, Z-coat, Eggshell White.
 - e. ICI Paints, Dulux, Lifemaster Eggshell Interior Latex Enamel.
 - f. PPG, Pure Performance, Interior Latex Eggshell.
 - g. Sherwin Williams, Harmony, Interior Latex Eggshell.
- C. Institutional Low-Odor/VOC Latex (Semigloss): MPI #147 (Gloss Level 5).
 - 1. MPI VOC Range (grams/liter): E3, less than 11 g/l.
 - 2. Available products include but are not limited to:
 - a. Benjamin Moore, Eco Spec, Int. Latex Semi-gloss Enamel.
 - b. Cloverdale Paint, Horizon, Interior Low VOC Semi-gloss.
 - c. Columbia Paint, Purecoat, Low Odor Acrylic Semi-gloss.
 - d. General Paint, Z-coat, Acrylic Semi-gloss Latex.
 - e. ICI Paints, Dulux, Interior Acrylic Semi-gloss.
 - f. PPG, Pure Performance, Interior Semi-gloss Latex.
 - g. Sherwin Williams, Harmony, Interior Latex Semi-gloss.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 - 1. Masonry (Clay and CMU): 12 percent.
 - 2. Gypsum Board: 12 percent.

- C. Gypsum Board Substrates: Verify that finishing compound is sanded smooth.
- D. Verify suitability of substrates, including surface conditions and compatibility, with existing finishes and primers.
- E. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and paint systems indicated.
- B. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.
 - 1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems indicated.
- C. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.3 INSTALLATION

- A. Apply paints according to manufacturer's written instructions and to recommendations in "MPI Manual."
 - 1. Use applicators and techniques suited for paint and substrate indicated.
 - 2. Paint surfaces behind movable equipment and furniture same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed equipment or furniture with prime coat only.
 - 3. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.
 - 4. Primers specified in painting schedules may be omitted on items that are factory primed or factory finished if acceptable to topcoat manufacturers.
- B. Tint each undercoat a lighter shade to facilitate identification of each coat if multiple coats of same material are to be applied. Tint undercoats to match color of topcoat, but provide sufficient difference in shade of undercoats to distinguish each separate coat.
- C. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.
- D. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

3.4 FIELD QUALITY CONTROL

- A. Dry-Film Thickness Testing: Owner may engage the services of a qualified testing and inspecting agency to inspect and test paint for dry-film thickness.
 - 1. Contractor shall touch up and restore painted surfaces damaged by testing.
 - 2. If test results show that dry-film thickness of applied paint does not comply with paint manufacturer's written recommendations, Contractor shall pay for testing and apply additional coats as needed to provide dry-film thickness that complies with paint manufacturer's written recommendations.

3.5 CLEANING AND PROTECTION

- A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.
- B. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.
- C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Architect, and leave in an undamaged condition.
- D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.6 INTERIOR PAINTING SCHEDULE

- A. Gypsum Board: Provide the following finish systems over interior gypsum board surfaces:
 - 1. Lower-Luster, Acrylic-Enamel Finish for walls : Two finish coats over a primer.
 - 1) Primer: Latex-based, interior primer applied at spreading rate recommended by the manufacturer.
 - 2) First and Second Coats: Low-luster (eggshell or satin: Match existing), acrylic-latex, interior enamel applied at spreading rate recommended by the manufacturer.

END OF SECTION

SECTION 09 9600 - HIGH PERFORMANCE COATINGS

PART 1 - GENERAL

1.1 SUMMARY

- A. Section includes surface preparation and the application of high-performance coating systems.
 - 1. Exterior Substrates:
 - a. Clay masonry.

1.2 DEFINITIONS

- A. MPI Gloss Level 5: 35 to 70 units at 60 degrees, according to ASTM D523.
- B. MPI Gloss Level 6: 70 to 85 units at 60 degrees, according to ASTM D523.
- C. MPI Gloss Level 7: More than 85 units at 60 degrees, according to ASTM D523.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product. Include preparation requirements and application instructions.
 - 1. Include printout of current "MPI Approved Products List" for each product category specified, with the proposed product highlighted.
- B. Samples: For each type of coating system and in each color and gloss of topcoat indicated.

1.4 QUALITY ASSURANCE

- A. Mockups: Apply mockups of each coating system indicated to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.
 - 1. Architect will select one surface to represent surfaces and conditions for application of each coating system.
 - a. Wall and Ceiling Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).
 - b. Other Items: Architect will designate items or areas required.
 - 2. Final approval of color selections will be based on mockups.
 - a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Architect at no added cost to Owner.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Sherwin-Williams
- B. Benjamin more
- C. PPG Paint
- D. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to products listed in the Exterior High-Performance Coating Schedule for the coating category indicated.

2.2 HIGH-PERFORMANCE COATINGS, GENERAL

- A. MPI Standards: Products shall comply with MPI standards indicated and shall be listed in its "MPI Approved Products Lists."
- B. Material Compatibility:
 - 1. Materials for use within each paint system shall be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer, based on testing and field experience.
 - 2. For each coat in a paint system, products shall be recommended in writing by topcoat manufacturers for use in paint system and on substrate indicated.
 - 3. Products shall be of same manufacturer for each coat in a coating system.
- C. Colors: Match existing .

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.
- B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:
 - 1. Concrete: 12 percent.
 - 2. Masonry (Clay and CMUs): 12 percent.
- C. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.
- D. Proceed with coating application only after unsatisfactory conditions have been corrected.
 - 1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

- A. Comply with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual" applicable to substrates and coating systems indicated.
- B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.
 - 1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.
- C. Clean substrates of substances that could impair bond of coatings, including dust, dirt, oil, grease, and incompatible paints and encapsulants.
 - 1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce coating systems indicated.

3.3 APPLICATION

- A. Apply high-performance coatings according to manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual."
- B. Apply coatings to produce surface films without cloudiness, spotting, holidays, laps, brush marks, runs, sags, ropiness, or other surface imperfections. Produce sharp glass lines and color breaks.

3.4 EXTERIOR HIGH-PERFORMANCE COATING SCHEDULE

- A. Clay Masonry Substrates:
 - 1. Epoxy System MPI EXT 4.1D:
 - a. Prime Coat: Epoxy, matching topcoat.
 - b. Intermediate Coat: Epoxy, matching topcoat.
 - c. Topcoat: Epoxy, gloss, MPI #77.
 - 2. Pigmented Polyurethane over Epoxy System MPI EXT 4.1J:
 - a. Prime Coat: Epoxy, matching intermediate coat.
 - b. Intermediate Coat: Epoxy, gloss, MPI #77.
 - c. Topcoat: Polyurethane, two component, pigmented, gloss (MPI Gloss Level 6), MPI #72.

END OF SECTION

SECTION 22 0101 - PLUMBING GENERAL PROVISIONS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. General provisions and requirements for all plumbing work.

1.2 RELATED SECTIONS

- A. Requirements of this section generally supplement requirements of Division 01.

1.3 REFERENCES

- A. NFPA 241: Safeguarding Construction, Alteration, and Demolition Operations.

1.4 SYSTEM DESCRIPTION

- A. The full set of Contract Documents applies to work of Division 22.
- B. Visit the site and study all aspects of the project and working conditions, as required by General and Supplementary Conditions, Bidding and Contracting Requirements, Drawings, and Specifications. Verify field dimensions.
- C. The work covered in technical sections includes the furnishing of all labor, equipment and materials, and the performance of all operations pertinent to the work described.
- D. Except as required otherwise in Division 01, promptly obtain and pay for, including all necessary signatures and paperwork, all permits, fees and inspections required for work of this division by authorities having jurisdiction, including any utility connection or extension charge. No payment will be made until a copy of the permit is forwarded to the Owner.
- E. Plumbing work of this project includes, as a brief general description, the following:
 - 1. Installation of foundation drains, and associated duplex sump pumps, to transfer ground water away from the existing underground structure.
- F. See Division 01 for requirements related to Owner's occupancy of the premises, limits on use of site, time restrictions on work, limits on utility outages or shutdowns, and phasing (sequencing) and scheduling.

1.5 PRODUCT OPTIONS

- A. Except as modified by provisions of Bidding and Contracting Requirements and Division 01, the following options apply to Division 22 specifications.
 - 1. Products specified by reference standards or by description only: Any product meeting those standards or description.
 - 2. Products specified by naming one or more manufacturers, or model name or catalog reference number: Products specified establish a standard of quality, options to be included, and performance.
 - a. Where other acceptable manufacturers are named, Contractor may provide products of those named manufacturers only, which meet the specifications.

- b. Where specification permits "equal" products, without naming other acceptable manufacturers, Contractor may use products of any manufacturer, which meet the specifications.
- 3. Products specified by naming one manufacturer and particular product, with no provision for other options: No options or substitutions allowed.
- B. General: Where Contractor is permitted to use a product other than the specified item and model named as the basis of design, Contractor is responsible for all coordination and additional costs as specified in the article "Substitutions," below for substitutions.

1.6 SUBSTITUTIONS

- A. Substitutions will be considered only as permitted or required by the Bidding and Contracting Requirements and Division 01. Except as modified by those requirements, the requirements below apply to Division 22 specifications.
- B. Substitutions may be considered when a product becomes unavailable through no fault of the Contractor.
- C. Document each request with complete data substantiating compliance of proposed substitution with contract documents.
- D. A request constitutes a representation that the Bidder or Contractor:
 - 1. Has investigated proposed product and determined that it meets or exceeds the quality level of the specified product.
 - 2. Will provide the same warranty for the substitution as for the specified product.
 - 3. Will coordinate installation and make changes to other work which may be required for the work to be complete with no additional cost to Owner.
 - 4. Waives claims for additional costs or time extension which may subsequently become apparent.
 - 5. Will reimburse Owner for review or redesign services associated with re-approval by authorities.
- E. Substitutions will not be considered when they are indicated or implied on shop drawing or product data submittals, without separate written request, or when acceptance will require revision to the Contract Documents.
- F. Substitution submittal procedure is specified in Bidding and Contracting Requirements and Division 01.

1.7 MATERIALS AND EQUIPMENT

- A. All materials and equipment shall be new and the best of their respective kinds, suitable for the conditions and duties imposed on them by the project, and of representative manufacturer. The description, characteristics and requirements of the materials to be used shall be in accordance with the specifications.
- B. All equipment, construction and installation must meet requirements of local, state and federal governing codes.
- C. Singular number: In cases where material, a device, or part of the equipment is referred to in the singular number in the specifications, it is intended that such reference shall apply to as many items of material, devices, or parts of the equipment as are required to complete the installation as shown on the drawings or required for proper operation of the system.

D. Terms have the following meanings:

1. Furnish: Supply item
2. Install: Mount and connect item
3. Provide: Furnish and install.

- E. All materials and equipment shall be installed and completed in a first class and workmanlike manner and in accordance with the best modern methods, practice and manufacturers' instructions. Any work which shall not present an orderly and neat or workmanlike appearance shall be removed and replaced with satisfactory work when so directed in writing by the Architect.
- F. The specifications and drawings are intended to define the minimum requirements, as to quality of materials, construction, finish and overall workmanship.
- G. General Conditions describe the correlation and intent of the Contract Documents. In case of discrepancies between the specifications and drawings, the specifications should be followed as to the general methods and principles and the drawings followed as to sizes, capacities and specifics for corresponding parts. If sizes are omitted, the Architect will determine sizes to be utilized.
- H. In all cases of doubt, uncertainty, or conflict as to the true meaning of the specifications or drawings, it is the responsibility of the Contractor to notify the Architect of said uncertainty, doubt, or conflict and obtain a decision as to the intent before starting any work which may be affected by this decision.

1.8 COORDINATION

- A. Should a situation develop during construction to prevent the proper installation of any equipment or item where shown on the drawings, call the situation to the attention of the Architect and await a written decision.
- B. Plan and coordinate all work to proceed in an orderly and continuous manner without undue delay, and in conformance with project schedule. Submit samples, shop drawings, schedules, insurance policies and certificates, and the like in time to avoid delays in actual construction. Coordinate plumbing work so that work of each trade is completed before other construction begins which would obstruct it.
- C. Coordinate trades to ensure that proper clearances between work of the various trades allow access to items which require operation and maintenance.
- D. Coordinate location and elevation of all piping, ductwork, light fixtures, equipment, and appurtenances in such a manner that the finished installation is as indicated on drawings. In the event difficulties are encountered which prevent this, it is the Contractor's responsibility to bring this to the attention of the Architect prior to initiation of work. Correct improperly coordinated installation at no additional cost.
- E. The Contractors' assistants shall include a competent foreman, who shall be on the premises at all times to check, layout, coordinate and superintend the installation of work. The foreman shall establish all grades and lines relative to the work before starting, and be responsible for the accuracy thereof.

1.9 SUBMITTALS

- A. Manufacturers' and subcontractors' lists:

1. As specified in Division 01, submit a complete list of proposed manufacturers for all equipment, materials and subcontractors used for the work of this division. Lists shall follow the sequence of the specifications. No considerations will be given for partial or incomplete lists. After review of lists, submit shop drawings and product data.

B. Shop drawings and product data:

1. Submit in accordance with the requirements of Division 01 or as established at the preconstruction conference, the required number of copies of shop drawings and product data for every item of equipment. Shop drawings or product data will not be considered until manufacturers' lists have been approved. Shop drawings and product data shall be submitted, as required by the General Conditions, with sufficient time for checking, return to Contractor, and resubmission as required before Contractor shall install any item.
2. Each item submitted shall be properly labeled, indicating the specific service for which the equipment or material is to be used, section and paragraph number of specification or drawing number to which it applies, Contractor's name and project name and number. Data submitted shall be specific and shall include product data and printed information in sufficient detail and scope to verify compliance with requirements of the contract documents. Clearly identify each item within the data. Data of a general nature will not be accepted. Each sheet must clearly show the project name and number.
3. The review of a shop drawing or product data shall not be considered as a guarantee of the measurements or building conditions or that the shop drawings or product data have been checked to see that item submitted properly fits the building conditions. This review shall not relieve the Contractor of the responsibility for furnishing material or performing work as required by the contract documents, for correctness of dimensions and quantities, or for proper coordination of details and interfaces among trades.
4. All exclusively electrical items furnished as items associated with plumbing items but not specifically described in the plumbing item submission, shall be submitted as a separate submittal but shall be clearly marked as associated with the plumbing item by identification specification paragraph.
5. Product data sheets shall be 8.5-inch by 11-inch cut sheets for operating and maintenance manual.

C. Submit at least three copies of the results of every test required under any section in this division.

D. Specialist shall submit a list of at least three projects similar to this project in type, size, and quality, which have been in place and operating satisfactorily for at least five years.

1. Include project name, address, name and phone number of owner's representative, and project type and size.

E. After the work is completed, submit all required certificates of approval from approved inspection agencies and authorities having jurisdiction over work of this division. Certificates of approval must be received by the Architect prior to final acceptance of the work.

1.10 SPECIALIST

- A. The term "Specialist" as used in the specification shall mean an individual or firm of established reputation (or, if newly organized, whose personnel have previously established a reputation in the same field,) which is regularly engaged in, and which maintains a regular force of workers skilled in either (as applicable) manufacturing or fabricating items required by the contract, installing items required by the contract, or otherwise performing work required by the contract. Where the specification requires installation by a specialist, the term shall also be deemed to mean the manufacturer of the item, an individual or firm licensed by the manufacturer, or an individual or firm who will perform the work under the manufacturer's direct supervision.

1.11 CONTRACT CLOSEOUT SUBMITTALS

A. Project record documents:

1. Maintain on site one set of the following record documents; record actual revisions to the work of this division:
 - a. Contract drawings.
 - b. Specifications
 - c. Addenda
 - d. Change orders and other modifications to the Contract
 - e. Reviewed shop drawings, product data, and samples
2. Maintain record documents separate from documents used for construction.
3. Record information concurrent with construction progress.
4. Specifications: Legibly mark and record in each section a description of actual products installed, including the following:
 - a. Manufacturer's name and product model and number
 - b. Product options, substitutions, or alternates utilized
 - c. Changes made by addenda and modifications
5. Record documents and shop drawings: Legibly mark each item to record actual construction, including:
 - a. Measured horizontal and vertical locations of underground utilities and appurtenances, referenced to permanent surface improvements.
 - b. Measured locations of internal utilities and appurtenances concealed in construction, referenced to visible and accessible features of the work.
 - c. Field changes of dimension and detail.
 - d. Details not on original Contract Drawings.
6. Submit documents as specified in Division 01.

B. Operation and maintenance data:

1. Submit sets prior to final inspection as specified in Division 01. Unless otherwise specified in Division 01, submit no fewer than three sets. In addition to requirements specified in Division 01, submit operating and maintenance manuals for the work of this division as specified below.
2. Lubrication charts: Prepare lubrication charts for each piece of mechanical equipment that requires grease or oil.
 - a. Include the following:
 - (1) Types of lubricants required
 - (2) Locations of lubrication points
 - (3) Frequency of lubrication.
3. Binders: Three-ring binders with vinyl-covered hard covers. Provide large enough binders, and sufficient quantity, that the required contents can be easily turned, removed, and reinserted.
4. Prepare binder covers with printed title "OPERATION AND MAINTENANCE INSTRUCTIONS," and title of project. Print on spine of binder "O & M INSTRUCTIONS." If more than one binder is required, print covers and spines with volume numbers. Include in the front of every binder an index to all binders.
5. Internally subdivide the binder contents with permanent page dividers, logically organized as described below; with tab titling clearly printed under reinforced laminated plastic tabs.

6. Contents: Prepare a Table of Contents for each volume, with each product or system description identified, typed on white paper.
7. Part 1: Directory, listing names, addresses, and telephone numbers of mechanical engineers; Contractor; mechanical subcontractors; and major mechanical equipment suppliers.
8. Part 2: Operation and maintenance instructions, arranged by specification section. For each category, identify names, addresses, and telephone numbers of subcontractors and suppliers. Identify the following:
 - a. Significant design criteria, including pump curves and similar performance charts.
 - b. List of plumbing equipment, including operating weight of each.
 - c. Parts list for each pump, including recommended spare parts list.
 - d. Operating instructions.
 - e. Maintenance instructions for plumbing equipment and systems.
 - f. Maintenance instructions for special finishes, including recommended cleaning methods and materials and special precautions identifying detrimental agents.
9. Part 3: Project documents and certificates, including the following:
 - a. Shop drawings and product data for plumbing systems.
 - b. Photocopies of certificates.
 - c. Photocopies of warranties and guarantees.
 - d. Test reports: Copies of the results of all tests required under all sections of specifications.
10. Submit one copy of completed volumes in final form 15 days prior to final inspection. This copy will be returned after final inspection, with Engineer comments. Revise content of documents as required prior to final submittal.
11. Submit final volumes revised, within ten days after final inspection.

1.12 REGULATORY REQUIREMENTS

- A. When these specifications call for materials or construction of a better quality or larger sizes than required by the following codes and standards, the provisions of the specifications shall take precedence.
- B. Provide, without extra charge, any additional materials and labor which may be required for compliance with these codes and standards even though the work is not mentioned in these specifications or shown on the contract drawings.
- C. Perform the work of this division in strict accordance with the following authorities. The latest revision of these codes accepted by the authority having jurisdiction as of the date of the contract documents shall apply.
 1. The plumbing, mechanical, electrical, building, fire, and safety codes of the state and county or city in which the work is being performed.
 2. The National Electric Code, NFPA 70 (NEC).
 3. The National Fire Protection Association Code. (NFPA).
 4. International Energy Conservation, Fire, Fuel Gas, Mechanical, and Plumbing Codes (ICC).

1.13 REFERENCE STANDARDS

- A. Perform the work of this division using the standards of the following organizations, as referred to in technical sections, as a minimum requirement for construction and testing. Unless specified otherwise in Bidding and Contract Documents or Division 01, the latest revision current as of the date of the contract documents shall apply. Products shall be certified by manufacturers to meet the requirements of referenced standards.

1. Federal Specifications (FS)
2. Military Specification (MS)
3. Military Standards (Mil. Std.)
4. Air Conditioning, Heating, and Refrigeration Institute (AHRI)
5. Air Movement and Control Association (AMCA)
6. Associated Air Balance Council (AABC)
7. American Association State Highway and Transportation Officials (AASHTO)
8. American National Standards Institute (ANSI)
9. American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE)
10. ASME International (ASME)
11. American Society for Testing and Materials (ASTM)
12. American Society of Sanitary Engineering (ASSE)
13. American Water Works Association (AWWA)
14. International Code Council (ICC)
15. Manufacturer's Standardization Society of the Valve and Fittings Industry Inc. (MSS)
16. National Electrical Code, NFPA 70 (NEC)
17. National Electrical Manufacturer's Association (NEMA)
18. National Fire Protection Association (NFPA)
19. National Fuel Gas Code, NFPA 54
20. National Sanitary Foundation (NSF)
21. National Standard Plumbing Code (NSPC)
22. The Occupational Safety and Health Act (OSHA)
23. Piping and Drainage Institute (PDI)
24. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA)
25. Underwriters Laboratory Inc. (UL)
26. Maryland Occupational Safety and Health Act (MOSHA)

1.14 TEMPORARY STORAGE

- A. Maintain upon premises, where directed, a storage area, and be responsible for all contents within these areas. Provide all security measures necessary for this area.
- B. Maintain the area and return it to its original condition at project completion.

1.15 PROTECTION

- A. Control dust resulting from construction work to prevent its spread beyond the immediate work area, and to avoid creation of a nuisance.
 1. Do not use water to control dust. Use drop cloths or other suitable barriers.
 2. In areas where dirt or dust is produced as a result of the work, sweep daily, or more often as required.
 3. Provide walk-off mats at entries and replace them at regular intervals.
 4. Construct dust partitions, where indicated on the drawings or as required.
 5. Protect areas occupied by Owner's personnel or equipment.
- B. Each trade and subcontractor is responsible for preventing damage and soiling of work performed by other trades or subcontractors. Each trade and subcontractor is responsible for providing temporary protection of its own work.
 1. Protect work from spills, splatters, drippings, adhesives, bitumens, mortars, paints, plasters, and damage from welding or burning.
 2. Protect finished work from damage, defacement, staining, or scratching.
 3. Protect finishes from cleaning agents, or grinding and finishing equipment.
 4. Protect adjacent and finished work from damage, using tape, masking, covers or coatings and protective enclosures.
 5. Coordinate installations and temporarily remove items to avoid damage from finishing work.

- C. Repair all damage or soiling to the complete satisfaction of the Architect; replace any materials or work damaged to such an extent that they cannot be restored to their original condition, all at no addition to the Contract Sum.
- D. Protect work stored in place and supplies stored in the building.
 - 1. Store materials and products, subject to damage from moisture, in dry locations. If necessary, protect in wraps or covers.
 - 2. Store plastics, other materials, and products subject to damage from heat or cold at manufacturer's recommended temperatures.
- E. Use of sidewalk or roadway areas outside of the property lines shall be with permission and approval of the project officer.

1.16 FIRE PROTECTION

- A. As a minimum, provide hand-carried, portable, UL-rated extinguishers with each work crew working inside the building.
- B. Select extinguishers in accordance with NFPA 10 and NFPA 241 for classification, extinguishing agent, and size required by location and class of fire exposure.

1.17 PROJECT CONDITIONS

- A. Drawings showing utilities in concealed locations are based on the best information available but are not represented as being precisely correct. Work of the contract includes digging, cutting, drilling, using nondestructive methods, and other methods of locating concealed utilities in the field, as well as patching and repairing as specified in "Cutting and Patching" below.
- B. If, in the course of the work, workers encounter a material they suspect to present some hazard:
 - 1. Promptly notify the Owner and Architect in writing.
 - 2. Do not perform any work which would disturb the suspected material until written instructions have been received.

1.18 WARRANTY

- A. All work and equipment provided as work of this division shall be fully warranted under the general project warranty. In addition, provide added special warranties specified in individual sections.
- B. During the correction period, the Contractor shall begin correcting any work found to be not in accordance with the requirements of the Contract Documents within 4-hours of receiving written notice from the Owner. Provide detailed schedule for completion of work within 24-hours of receiving written notice from the Owner and revise schedule based on any Owner comments generated. Except as otherwise required in General Conditions and Division 01, the correction period is one year after the date of substantial completion of the work. Work requiring correction shall promptly be repaired or completely replaced at no addition to the Contract Sum.
 - 1. Service reports for warranty work shall be provided to the Owner.
- C. Special warranties are warranties required by individual specification sections, incidental product warranties, manufacturers' standard warranties, installer or subcontractor service agreements, and other individual warranties in addition to the general project warranty.
- D. Provide copies of warranties as required for Operation and Maintenance Manual specified above, and by Division 01.

- E. For items of work delayed beyond date of substantial completion, provide updated submittal within ten days after acceptance, listing date of acceptance as start of warranty period.

PART 2 - PRODUCTS

Not used.

PART 3 - EXECUTION

3.1 CUTTING AND PATCHING

- A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.
- B. No cutting of any structure or finish shall be done until the condition requiring such cutting has been examined and approved by the Architect.
- C. New or existing surfaces disturbed as a result of such cutting or otherwise damaged shall be restored to match original work and all materials used for any patching or mending shall conform to the class of materials originally installed.
- D. Refinish surfaces to match adjacent finish. For continuous surfaces, refinish to nearest intersection or natural break. For an assembly, refinish entire unit.

3.2 TEMPORARY FACILITIES

- A. Temporary water facilities, electricity, telephone, toilet facilities, and temporary heat, shall be provided as specified in Division 01.

3.3 PROGRESS MEETINGS

- A. Progress meetings shall be held as specified in Division 01, and also when and if the Contractor or Architect finds them necessary or advantageous to progress of work.
- B. Contractor, those subcontractors and those material suppliers concerned with current progress or with the scheduling of future progress, Architect and Owner shall each be represented at these meetings by persons familiar with the details of work and authorized to conclude matters relating to work progress.

END OF SECTION

SECTION 22 0500 - COMMON WORK RESULTS FOR PLUMBING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Requirements applicable to more than one section of Division 22.
- B. Equipment installation requirements.
- C. Basic material and equipment required for the plumbing piping work.
- D. Identification of plumbing systems and equipment.
- E. Cleaning and painting.
- F. Operating instructions.
- G. Piping tests.

1.2 RELATED SECTIONS

- A. Project and special warranties: Division 01 and Section 22 0101.
- B. Operation and Maintenance Manuals: Division 01 and Section 22 0101.

1.3 REFERENCES

- A. American Society of Mechanical Engineers
 - 1. ASME A 13.1: Scheme for the Identification of Piping Systems
 - 2. ASME B 31.9: Building Services Piping
- B. American Society of Testing and Materials
 - 1. ASTM B 32: Standard Specification for Solder Metal
 - 2. ASTM B 88: Standard Specification for Seamless Copper Water Tube
 - 3. ASTM B 813: Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
 - 4. ASTM D 635: Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position
 - 5. ASTM E 84: Standard Test Method for Surface Burning Characteristics of Building Materials
 - 6. ASTM E 548: Standard Guide for General Criteria Used for Evaluating Laboratory Competence
 - 7. ASTM D 2564: Standard Specification for Solvent Cements for Poly (Vinyl Chloride) (PVC) Plastic Piping Systems
 - 8. ASTM F 656: Standard Specification for Primers for use in Solvent Cement Joints of Poly (Vinyl Chloride) (PVC) Plastic Pipe and Fittings
- C. American Welding Society
 - 1. AWS D1.1: Structural Welding - Steel
 - 2. AWS D10.9: Specification for Qualification of Welding Procedures and Welders for Piping and Tubing

D. NSF

1. NSF/ANSI 61: Drinking Water System Components - Health Affects

1.4 DEFINITIONS

- A. Project correction period: A period after Substantial Completion of the work during which the Contractor shall correct every part of the work found to be not in accordance with the requirements of the contract documents, promptly after receipt of written notice.
- B. Qualified testing agency: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspecting indicated, as documented according to ASTM E 548; and with additional qualifications specified in individual sections; and where required by authorities having jurisdiction, that is acceptable to authorities.
 1. NRTL: A nationally recognized testing laboratory according to 29 CFR 1910.7.
 2. NVLAP: A testing agency accredited according to NIST's National Voluntary Laboratory Accreditation Program.
- C. DN: Dimension Nominale, nominal pipe size in millimeters, in accordance with the metric system for construction, Systeme Internationale (SI).
- D. NPS: Nominal pipe size in inches, in accordance with standard U.S. designations for manufactured pipe. Pipe sizes do not change when projects are designed and built in metric units; each size has a consistent name (nominal dimension) in each system.

1.5 DESIGN REQUIREMENTS

- A. The drawings and system performances have been designed based on use of the particular manufacturer's products specified and scheduled on the drawings.
- B. Products of other manufacturers that are listed under the article "Acceptable Manufacturers," or permitted as "equal," are permitted provided:
 1. Product shall meet the specifications.
 2. Contractor shall make, without addition to the contract sum, all adjustments for deviations so that the final installation is complete and functions as the design basis product is intended.
- C. Do not propose products with dimensions or other characteristics different from the design basis product that render their use impractical, or cause functional fit, access, or connection problems.

1.6 SUBMITTALS

- A. Shop drawings:
 1. Schedule of welding procedures proposed for each piping system in the project.
 2. Shop drawings of backboards for piping specialties.
- B. Certifications: Proof of operator and testing agency personnel qualifications as required for welding and brazing in the article "Quality Assurance" below.
- C. Test reports: Field test results for each piping system as specified in Part 3 below.

1.7 QUALITY ASSURANCE

- A. Provide materials and perform work in accordance with the plumbing, mechanical, electrical, building, fire, health and safety, and other applicable codes and regulations of the state, county or city in which the work is performed.

- B. Welding procedures and operator qualifications for structural welding: AWS D1.1, Structural Welding Code Steel, electric arc process.
- C. Soldering procedures and operator qualifications for building systems piping:
 - 1. ASME B31.9, Building Services Piping.
 - 2. Copper Development Association "Copper Tube Handbook."
 - 3. Safe Drinking Water Act.
- D. UL label:
 - 1. Electrical control panels, equipment, materials and devices provided or installed as work of Division 23 shall be UL listed and shall bear a UL label.
 - 2. Equipment, including custom assemblies, shall be listed and labeled as an assembly.
 - 3. If a UL label is not available, the item shall be tested and labeled by a qualified nationally recognized testing agency, acceptable to authorities having jurisdiction, and in accordance with NFPA 70 (NEC).
 - 4. Provide testing, if required, without addition to the contract sum.
- E. VOC content: Field-applied adhesives and sealants, limits per South Coast Air Quality Management District (SCAQMD), Rule No. 1168.
- F. Products shall contain no urea-formaldehyde content.

PART 2 - PRODUCTS

2.1 GENERAL

- A. General piping techniques, testing, identification, painting, and operating instructions specified in this section apply to products specified in other sections of Division 22.
- B. Equipment that uses or processes date and time data to perform its function shall be warranted by the manufacturer to properly function and correctly use or process all time-related data for all dates and times which occur during a reasonable life expectancy of the equipment.

2.2 PIPING MATERIALS

- A. Soldering materials:
 - 1. Solder: Free of lead, antimony, and zinc and meeting the requirements of ASTM B 32. No solder containing lead is permitted.
 - a. Tin 95.5 percent, copper 4 percent, and silver 0.5 percent.
 - (1) Equal to "Silvabrite 100" manufactured by Engelhard Corporation.
 - b. Tin, copper, bismuth, and silver.
 - (1) Equal to "Oatey Silver" manufactured by Oatey.
 - 2. Flux:
 - a. Meeting the requirements of ASTM B 813 and NSF 61 certified.
 - b. Equal to Oatey H-2095.

B. Plastic PVC pipe joint materials:

1. Primer: ASTM F 656 and containing methyl ethyl ketone or acetone.
2. Solvent cement : ASTM D 2564.

2.3 IDENTIFICATION DEVICES AND MATERIALS

A. Stenciling materials:

1. Stencils: Manufactured standard stencils prepared for required applications, conforming to ASME A 13.1 for color and size of legend letters, including arrows showing direction of flow.
2. Paint: Exterior type enamel, colors conforming to ASME A 13.1, or black.

B. Equipment identification tags:

1. Laminated plastic with adhesive back, white core and black outer layers, which, when engraved, will produce white letters and numerals on a black background.
2. Tags installed on curved surfaces shall be aluminum or brass.

PART 3 - EXECUTION

3.1 INSTALLATION OF PRODUCTS AND EQUIPMENT

A. Manufacturers' instructions: Except as modified by drawings or specifications, install products and equipment in accordance with manufacturers' instructions and recommendations applicable to the project conditions.

1. Immediately notify Architect if a difference or discrepancy is found between manufacturers' instructions and the drawings or specifications.

B. The contract drawings are diagrammatic and do not indicate all fittings or offsets in pipe, or all specialties required. Provide required fittings, offsets, and specialties to coordinate the work.

C. Items which require access for operation or maintenance shall be easily accessible.

D. Provide installations compliant with NFPA 70.

3.2 EQUIPMENT INSTALLATION

A. Prepare for installation:

1. Coordinate work to assure that any required recesses for equipment are correct.
2. Examine areas and conditions where equipment will be installed for rough in locations, compliance with installation requirements and tolerances and other conditions affecting performance. Proceed with installation only after unsatisfactory conditions have been corrected.
3. Visually check equipment. Replace any damaged components prior to installation.
4. Remove temporary devices used for equipment transport.
5. Assemble equipment that is not assembled in the factory.

B. Install equipment:

1. Set equipment plumb and level.
2. Install such that labels and safety instructions are clearly visible.
3. Drawings show the general arrangement of piping, conduit and other elements. Coordinate elements to allow access doors to open fully, allow access to removable access panels or casing panels, maintain clearances in front of control panels and electrical components required by the NFPA 70, and maintain the manufacturer's recommended service clearances.

4. Install components inside of equipment enclosures such that they are serviceable when access panels are removed.
5. Install piping hangers and supports so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment. Support equipment independently from piping.
6. Support:
 - a. Floor-mounted equipment:
 - (1) Securely fasten in place with connections to wall and floor.
 - (2) Provide supplemental structure as required.
 - (3) Provide a concrete equipment foundation where indicated. Comply with requirements of Section 22 0529.
 - b. Structural frame-mounted equipment: Securely fasten to structural frame.
 - c. Coordinate supports so they do not interfere with access to other equipment that requires service.
7. Snugly fit equipment to architectural finishes in finished spaces unless otherwise indicated.
8. Connect associated piping, power, controls, and other associated systems.
9. Conceal associated piping and electrical connections in finished spaces unless otherwise indicated.
10. Install dielectric pipe fittings at connections where dissimilar metals are joined.
11. Coordinate with installation of building automation system controls.
12. Install equipment with a safety listing or certification as required to comply with their listing or certification requirements.

C. Install and wire items furnished with equipment:

1. Install control devices, accessories, and appurtenances furnished loose with equipment.
2. Wire devices furnished with equipment that are not factory-wired.
 - a. Comply with requirements in Division 26.
 - b. Comply with requirements in NFPA 70.
 - c. Comply with the manufacturer's installation requirements.

D. Prepare equipment for use:

1. Clean equipment and repair any damaged finishes. Remove paint splatters, dirt, and debris. Repair damaged finish to match original finish.
2. Lubricate equipment.
3. Adjust equipment packaged controls to put the system into operation as intended.
4. Start up and operate equipment in accordance with manufacturer's instructions.
5. Verify smooth rotation and proper direction of rotation of motors.
6. Test controls, unit operation, and safety devices. Replace any failed components and retest.
7. Provide equipment identification.

3.3 PIPE INSTALLATION

- A. Remove burrs resulting from cutting pipe or from any other operation.
- B. Install unions in the piping at each item of equipment to provide easy removal of the equipment, valve, or appliance.

C. Interface with other products:

1. Where pipe is provided through walls, provide finished, permanent, waterproof installation complete with inserts, sleeves, supports or hangers, seals, and other appurtenances as required. Do not pierce, cut, or notch any footing or other structural member.
2. Waterproofing and dampproofing of the building shall be unharmed by the installation of the work. Where pipe has to pierce waterproofing or dampproofing, including outside walls, the penetration shall be made watertight. Waterproofing damaged or destroyed shall be repaired or replaced with new waterproofing.

D. Thoroughly clean pipe and fittings before they are installed, and keep them clean until the acceptance of the completed work. Cap or plug the ends of the lines to prevent earth and other debris from entering during construction.

E. Copper tubing installation:

1. Cut pipe with a tubing cutter or fine-tooth saw. Cuts made with a saw shall be true and square, and the end shall be filed smooth with a fine-tooth file. Remove all marks and burrs with sandpaper.
2. Solder joints for copper tubing: Clean ends of tubing and inside of fitting ends thoroughly with emery cloth before applying flux.
3. Provide dielectric fittings between copper and steel piping to prevent electrolysis.
4. Follow the techniques for soldering and brazing pipe, fittings, and valves as recommended by the manufacturer.

F. Plastic piping installation:

1. Cut pipe true and square with a fine-tooth saw and file the end smooth with a fine-tooth file. Remove all saw marks and burrs with sandpaper.
2. Clean connecting surfaces of both pipe and fitting with primer.
3. Apply solvent cement liberally with clean brush, first to fitting and then to pipe (outer surface and end). Lap the solvent cement a minimum of 0.25 inch (6 mm) over depth of fitting.
4. Join pipe and fitting to full depth of fitting, giving fitting at least one-quarter turn on pipe to distribute cement.
5. Pipe and fitting shall show a small fillet or bead completely around pipe without any voids, or fitting shall be cut out and new fitting made up and installed. Allow a minimum of 48 hours drying time for each joint.

3.4 IDENTIFICATION

A. General: Do not apply identification until insulation and finish painting work is complete.

B. Equipment:

1. Stencil equipment with minimum two-inch (50-mm) -high letters or provide identification tags. Clearly identify function, equipment served, and area served.
2. Firmly fasten each identification tag to its appropriate piece of equipment with drive screws, sheet metal screws, or rivets. Do not interfere with operation of, or damage the item being marked.

C. Piping:

1. Mark by stenciling.
2. Mark to identify service with arrows showing direction of flow. Apply markings near building walls where pipes enter or leave an accessible space and in intermediate locations so that markings are no more than 30 feet (9 m) apart. They shall be readily visible to a person standing on the floor.
3. Fully identify all piping installed as work of the project.

4. Mark pipe with letters of height and with colors as required by OSHA and conforming to ASME A 13.1.
5. Identify every thermometer, gauge, and control device.
6. Provide valve tags for all valves except shutoff valves on individual fixtures or equipment where their function is obvious, or where the fixture or equipment is immediately adjacent. Numbers shall correspond to those shown on the Valve Chart. Attach tags to valve shaft.

3.5 CLEANING AND PAINTING

- A. Cleaning: Clean all piping and equipment.

3.6 OPERATING INSTRUCTIONS (DEMONSTRATION)

- A. Furnish the necessary technicians, skilled workers, and helpers to operate all the plumbing systems and equipment of the entire project for one 8-hour day.
- B. Where specified in technical sections, provide longer periods required for specialized equipment.
- C. Instruct the Owner or designated personnel in operation, maintenance, lubrication, and adjustment of all systems and equipment.
 1. Instructions by manufacturer's technical representative for each type of equipment shall include the performance of the recommended preventive maintenance procedures for that equipment.
- D. The Operating and Maintenance Manual shall be available at the time of the instructions, for use by instructors and Owner personnel.
- E. Schedule the general and specialized instruction periods for a time agreed upon by the Owner and Architect.

3.7 PIPING TESTS

- A. Hydrostatic testing:
 1. Notify Owner in writing at least 24 hours prior to the test.
 2. Test before pipes are concealed.
 3. Piping may be tested in sections as the work progresses.
 4. Provide fluid, pumps, valves, and gages required for testing.
 5. Where water is used as the test fluid, provide ambient temperature water and provide means to avoid freezing. Drain and dispose of test fluid when testing is concluded.
 6. Brace and support piping during the test, so that no movement, displacement, or damage results from the application of the test pressure.
 7. Replace piping or fittings found defective with new material.
 8. Storm drainage piping tests:
 - a. Before connection to site piping, cap or plug all new storm drainage piping.
 - b. Test following the methods of testing required by the plumbing code, and no less than the duration and pressures required in the Schedule of Piping Systems Tests.
 - c. Where pipes are in trenches, leave the trenches open until the completion of the test.
 9. Documentation of tests: Prepare a test report for each portion of piping tested, identified by service, material, location, and pipe size. Include these items:
 - a. Date of test.
 - b. Starting and completion times.
 - c. Initial test pressure.
 - d. Final test pressure.

- e. Problems or leaks detected.
- f. Corrective actions taken.
- g. Record of successful completion of testing.
- h. Name, title, and signature of person conducting test.

10. Piping Systems Test Schedule:

System	Test Pressure psig (kPa)	Duration	Allowable Drop	Medium
Pumped storm drainage	125 (860)	4 Hours	None	Water
Storm drainage piping	4.3 (30)	4 Hours	None	Water

END OF SECTION

SECTION 22 0501 - EXCAVATION AND FILL FOR PLUMBING WORK

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Trenching, backfilling, and compacting for plumbing work underground inside the building and extending five feet beyond exterior building walls, and outside the building as shown on drawings.

1.2 RELATED SECTIONS

- A. Cutting and patching: Division 01 and Section 22 0101.
- B. Piping:
 - 1. Storm drainage piping: Section 22 1413.

1.3 REFERENCES

- A. ASTM D 1557: Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbs/cu ft (2700 kN-m/cu m).

1.4 SUBMITTALS

- A. Shop drawings: At the same scale as the contract drawings, showing field verified locations of utilities, and proposed detailed trenching plan.
- B. Product data:
 - 1. Warning tape
- C. Certifications: Test reports showing that compaction meets specified requirements.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Backfill: Earth materials, free from perceptible amounts of wood, debris, or topsoil, free of frost at the time of placement, and not containing marl or other elements which tend to stay in a plastic state.
- B. Underground warning tape: Polyethylene 0.004 inch (0.102 mm) thick for metallic lines, and for non-metallic lines polyethylene both sides with metallic lining, six inches (152 mm) wide.
 - 1. Colors: In accordance with APWA and AASHTO standards.
 - 2. Markings: Repeated continuously along the entire length, legend appropriate for line being identified.

2.2 EQUIPMENT

- A. Mechanical tampers for compacting backfill: Capable of exerting a blow equal to 250 pounds per square foot (12 kPa) of area of the tamping face.

PART 3 - EXECUTION

3.1 PREPARATION

- A. The general locations of underground utilities are indicated on the drawings and are not to be assumed to be accurate or complete. Before beginning work, field check the area with the most accurate instruments available, such as Fisher Labs' Pipe and Cable Locators.

3.2 INSTALLATION

- A. Perform all excavating, trenching, sheeting, shoring, backfilling, and compacting required for the proper installation of the work.
- B. Where obstructions are encountered, obtain written approval and make necessary changes in line, grade or location.
- C. Protect existing utilities from damage during excavation and backfilling. Repair damaged new or existing work at no addition to the contract sum. Bracing, shoring and other protection of existing utilities is part of this work.
- D. Do not damage or remove existing shrubs or trees including their root systems, without prior notification to the Architect.
- E. Provide temporary roadways over trenches with railings and other safeguards, including amber blinker lamps or other warnings for night use.
- F. Note the depths of footings. In cases where piping is in close proximity to or below footings and where the natural earth under footings is disturbed, after the line is installed the voids shall be filled up to bottoms of such footings with solid concrete.

3.3 CUTTING

- A. Cut concrete and asphalt concrete with masonry saw prior to breaking it into smaller pieces for removal.
- B. Cut sidewalks perpendicular to the length at the closest existing joint that is a minimum of 24 inches back from either side of the top of the new trench.

3.4 TRENCHING

- A. Excavations outside the building shall generally follow the routes indicated on the drawings. Stockpile topsoil separately for later replacement. Excavations shall be of sufficient depths to provide, unless indicated otherwise on the drawings, a minimum cover as follows:
 - 1. Storm water: Elevations shown on drawings.
- B. Trenches shall be of necessary depth and width for the proper laying of pipe with a minimum of 8 inches (205 mm) on each side of the joint.
 - 1. The sides shall be as nearly vertical as practicable. Unless local regulations are more strict, trenches 4 ft. (1220 mm) and deeper shall have shored sides as required by OSHA trenching regulations.
 - 2. The bottoms of trenches shall be accurately graded to provide uniform bearing and support for each section of pipe on undisturbed soil at every point along its entire length, except for bell holes and for the proper sealing of the pipe joints.
 - 3. No greater length of trench shall be left open, in advance of the completed structure placed in it, than can be completed in that day's operation.

4. Except where rock is encountered, do not excavate below the depths required. Where rock excavation is required, excavate to a depth of at least 6 inches (150 mm) below the trench depth and fill the overdepth with compacted crusher run or bank run stone or sand. Unauthorized overdepths in excavation shall be backfilled with crushed stone, slag or gravel, thoroughly compacted.
 5. Whenever wet or otherwise unstable soil is encountered, it shall be removed to the depth and extent directed, and the trench backfilled to the proper grade with crushed stone, slag or gravel.
- C. Should springs be encountered within the work area, or soft soil conditions at the elevations required for load bearing, immediately notify the Architect and do not place any portion of the work on such surfaces until instructions are received.
 - D. Furnish and maintain pumps, flumes, gutters, and appurtenances if required to keep the excavations free from water. Water shall be directed to a point remote from building operations, shown on the approved shop drawing.
 - E. Excavation for manholes and similar structures shall be sufficient to leave a minimum of 12 inches (305 mm) and a maximum of 24 inches (610 mm) clearance on all sides. Fill over-depth excavation with concrete.

3.5 BACKFILL

- A. Place no backfill until the adjacent construction or the utility to be covered has been inspected, tested, and approved.
- B. Installing underground warning tape: Install in backfill above exterior buried lines not encased in concrete. Select legend and color appropriate for type of line. Install metallic lined tape for non-metallic lines. Install approximately 12 inches (305 mm) below grade.
- C. Plumbing systems backfill:
 1. Backfill and compact in six-inch (150-mm) layers up to spring line of the pipe. The installations shall then be inspected and tested.
 2. Following inspection, backfill in six-inch (150-mm) layers, each compacted, until the pipe has a cover of not less than one foot (305 mm). Place the remainder of the backfill material in the trench in eight-inch (200-mm) compacted layers.
 3. Excavations improperly backfilled shall be reopened, then refilled and compacted to the required grade and compaction, and smoothed off.
 4. Open trenches across roadways or other areas to be paved shall be backfilled as specified above, except that the entire depth of trench shall be backfilled in six-inch (150-mm) layers, and each layer shall be mechanically compacted.
 5. Completed work shall have uniform graded surface, in accordance with the surface and grade indicated on the drawings.

3.6 COMPACTION

- A. Test in accordance with the requirements of ASTM D 1557.
- B. Compact under slabs, roads, and sidewalks to a 95 percent density.
- C. Compact unpaved areas to a 90 percent density.
- D. Backfill and compact trench in unpaved areas to within 4 inches (102 mm) of existing grade. Furnish and install compacted select topsoil for the final layer to finish even with existing grade. Remove surplus earth and rake unpaved areas for final planting.

E. Take particular care in compaction of earth under joints of plumbing piping.

END OF SECTION

SECTION 22 0502 - SLEEVES AND PLATES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Sleeves.
- B. Sealing materials.
- C. Mechanical penetration seals.

1.2 SUBMITTALS

- A. Product data: For each type of product, include a description, characteristics, materials, composition, finishes, and performance. Include the following:
 - 1. Sleeves.
 - 2. Sealing materials.
 - 3. Mechanical penetration seals.

PART 2 - PRODUCTS

2.1 SLEEVES

- A. Sleeves without water stops for wall and floor penetrations:
 - 1. Steel sleeves: Schedule 40 black steel pipe, ASTM A 53.
 - 2. Copper sleeves: Type L, ASTM B 88 hard drawn.

2.2 SEALING MATERIALS

- A. Waterproof sealant:
 - 1. High-performance, moisture cured, 1-component, polyurethane-based, non-sag, elastomeric sealant. Use a primer for applications required by the manufacturer.
 - 2. Basis of design: Sika Corporation "Sikaflex - la."

2.3 MECHANICAL PENETRATION SEALS

- A. Mechanical penetration seals:
 - 1. Seals shall be modular mechanical type, consisting of interlocking synthetic links shaped to continuously fill the annular space between the pipe and wall opening. Bolt and nut fasteners for the seals shall be stainless steel for units used in penetrations below grade.
 - 2. Basis of design: PSI "Link-Seal Modular Seals" or Calpico Sealing Link "LINX".

PART 3 - EXECUTION

3.1 INSTALLING SLEEVES

- A. Provide sleeves for piping passing through walls.
- B. Sleeve materials:
 - 1. Provide steel sleeves for penetrations of cast iron piping.

2. Provide copper sleeves for penetrations of uninsulated copper tubing and piping.
3. A core drill may be provided in cast concrete walls in lieu of a sleeve.

C. Sleeve projection:

1. Sleeves through walls: Flush with finished surfaces.

3.2 INSTALLING SEALING MATERIALS

- A. Install sealant between sleeve interior and pipe:

3.3 INSTALLING MECHANICAL PENETRATION SEALS

- A. Penetrations in exterior masonry or concrete walls and foundations:

1. Mechanical penetration seal, at outside face of wall.

END OF SECTION

SECTION 22 0513 - COMMON MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Unless otherwise specified in a particular section or required for a particular application, motors shall conform to the following requirements, whether factory-installed or field-installed.

1.2 RELATED WORK SPECIFIED ELSEWHERE

- A. Pumps:
 - 1. Sump pumps: Section 22 1429.

1.3 REFERENCES

- A. NEMA MG 1: Motors and Generators.
- B. NEMA MG 11: Energy Management Guide for Selection and Use of Single-Phase Motors.
- C. UL 508: Industrial Control Equipment.

1.4 DEFINITIONS

- A. Energy efficient motor: Motor meeting the nominal and minimum efficiency levels listed for its horsepower and speed in Table 12-10 of NEMA MG 1.
- B. Nominal efficiency: Efficiency as defined in Table 12-8, Efficiency Levels, in NEMA MG 1, and identified on the motor nameplate.

1.5 SUBMITTALS

- A. Product data:
 - 1. Motors and drives not provided with equipment: Show nameplate data and ratings; characteristics; mounting arrangements; size and location of winding termination lugs, conduit entry, and grounding lugs, and coatings.
- B. Wiring diagrams required for the proper installation of plumbing equipment.
- C. Submit product data which verifies compliance with ASHRAE 90.1 or provide certified performance ratings by a qualified independent testing agency.

1.6 QUALITY ASSURANCE

- A. UL label: As specified in Section 22 0500, Common Work Results for Plumbing.
- B. Plumbing equipment shall meet the energy performance requirements of ASHRAE 90.1.

1.7 REGULATORY REQUIREMENTS

- A. Motors shall conform to the requirements of NEMA MG1 and applicable portions of the National Electric Code (NEC, NFPA 70).

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

A. Motors:

1. A.O. Smith
2. Baldor Electric Co.
3. Marathon
4. Rockwell
5. Siemens
6. Toshiba International

2.2 BASIC MOTOR REQUIREMENTS

- A. Capacity: Each motor shall have sufficient capacity and torque to start, accelerate, and operate the machine it drives without exceeding the motor nameplate rating at the speed specified, or at any speed and load which may be obtained by the drive actually furnished.
- B. Starting: Each automatically controlled motor shall be capable of starting as frequently as the control sequence may demand. Motors not automatically controlled shall be capable of making no fewer than 4 starts per hour.
- C. Loads: Belt-connected motors shall be equipped with shafts and bearings designed to withstand both the normal connected loads of the drive furnished, and momentary loads imposed during acceleration.
- D. Ratings: Motors shall be rated for continuous duty at 100 percent of rated capacity, and temperature rise shall be based on ambient temperature of 40 degrees C.
- E. Phase: Unless otherwise indicated, motors one-half horsepower and larger shall be polyphase and motors smaller than one-half horsepower shall be single-phase motors.
- F. Motor construction:
 - 1. ODP (open drip-proof) NEMA design B construction:
 - a. Motors for pumps, unless specified otherwise.
 - b. Motors outdoors but inside of a weather-tight enclosure, unless specified otherwise.
 - 2. TEFC severe duty:
 - a. Motors mounted outdoors and not inside of a weather-tight enclosure.
 - b. Where TEFC motors are specified.
- G. Efficiency: The term "energy efficient" is defined in the article "Definitions" in Part 1 above.
 - 1. Single-phase motors, alternating-current fractional horsepower, rated 1/20 to 1 horsepower, 250 volts or less: NEMA MG 11, types and efficiencies selected for their applications.
 - 2. Polyphase motors, medium alternating-current, squirrel-cage, 1 to 500 horsepower, 600 volts or less: NEMA MG 10, energy-efficient types selected for their application. Nominal full-load efficiencies shall meet or exceed ratings of Table 12-10 of NEMA MG 1.
 - 3. Motors for packaged hermetic and semi-hermetic refrigeration compressors need not comply with these efficiency requirements but they shall comply with the requirements indicated for power factor and power consumption.

2.3 SINGLE-PHASE MOTORS

- A. Permanent split-capacitor or split-phase type.
- B. Bearings: Sealed, pre-lubricated ball bearings.
- C. Thermal protection: Internal protection to automatically open power supply circuit to motor when winding temperature exceeds a safe value calibrated to temperature rating of motor insulation. Thermal-protection device shall automatically reset when motor temperature returns to normal.

2.4 ELECTRONICALLY COMMUTATED MOTOR (ECM)

- A. Brushless direct current (DC) variable speed motor supplied with alternating current, with a permanent magnet with near zero rotor losses, permanently-lubricated ball bearings, electronic commutation, designed for synchronous rotation, and at least 70% efficient at all operating speeds.
- B. As a minimum, the motor shall include the following features:
 - 1. Integrated controller / inverter that operates the wound stator and senses rotor position to electronically commutate the stator.
 - 2. Thermal overload protection.
 - 3. Built-in soft start and soft speed change ramps.
 - 4. Inductors to minimize harmonic distortion and line noise.
 - 5. Designed to overcome reverse rotation without affecting life expectancy.
 - 6. Include a pulse width modulating (PWM) controller for either manual or DDC controlled motor speed adjustment down to 25 percent of full speed. The manual PWM controller shall be field adjustable with a standard screwdriver. The remote PWM shall be capable of receiving a 0-10Vdc analog signal from a DDC controller provided by the controls contractor, as indicated or required by other sections of DIV 23 specifications.
 - 7. Software for motor control as indicated or described in other DIV 22 specifications.

2.5 TEFC (TOTALLY ENCLOSED FAN-COOLED) MOTORS

- A. Polyphase motors with the following additional requirements:
 - 1. TEFC construction for severe environment.
 - 2. Ventilating fans: Made of corrosion-resistant, non-sparking material.
 - 3. Conduit box: Heavy-wall cast construction, gasketed with a lead gasket between box and motor frame.
 - 4. Motor shaft shall be provided with an external slinger on the drive end.
 - 5. Rotor and stator air-gap surfaces coated to prevent corrosion.
 - 6. Finish: At least two coats of catalyzed epoxy enamel.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Mount direct-connected motors securely and in accurate alignment. The drive shall be free from both angular and parallel misalignment when both motor and driven machine are operating at normal temperatures.

3.2 OPERATING INSTRUCTIONS

- A. As specified in Section 22 0500, provide operating instructions.

END OF SECTION

SECTION 22 1413 - STORM DRAINAGE PIPING

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Storm water piping.
- B. Pumped storm water piping.
- C. Foundation drain.

1.2 RELATED SECTIONS

- A. Piping installation and testing: Section 22 0500.

1.3 REFERENCES

- A. Cast iron piping standards
 - 1. ASTM A 74: Standard Specification for Cast Iron Soil Pipe and Fittings
 - 2. ASTM A 888: Standard Specification for Hubless Cast Iron Soil Pipe and Fittings
 - 3. ASTM C 564: Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
 - 4. ASTM C1540: Standard Specification for Heavy-Duty Shielded Couplings Joining Hubless Cast Iron Soil Pipe and Fittings
 - 5. CISPI 301: Standard Specification for hubless cast iron soil pipe and fittings for sanitary and storm drain, waste, and vent piping applications
- B. Copper Type L piping standards
 - 1. ASME/ANSI B 16.18: Cast Copper Alloy Solder Joint Pressure Fittings
 - 2. ASME/ANSI B 16.22: Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
 - 3. ASTM B 88: Standard Specification for Seamless Copper Water Tube
- C. Perforated PVC piping standards
 - 1. ASTM D 1785: Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120

1.4 SUBMITTALS

- A. Product data: Each specified material and product.
- B. Submit a single manufacturer for each product. Submittals that include multiple manufacturers for a single product are not acceptable.

1.5 QUALITY ASSURANCE

- A. Prior to any new piping installation, verify the inverts of all piping to which new work is to be attached. Demonstrate to the satisfaction of the construction manager and/or Owner, that the connections to existing storm water pipes meet the intent of the contract.
- B. Pipe shall be certified by the manufacturer to meet referenced standards and shall bear a label, directly on the pipe, indicating compliance.

PART 2 - PRODUCTS

2.1 STORM WATER PIPE AND FITTINGS

A. Storm Water pipe and fittings:

1. Piping: NIH DRM Type D:
 - a. Cast iron hubless pipe, ASTM A74, ASTM A888, CISPI 301. Piping manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.
2. Fittings: NIH DRM Type IV:
 - a. Cast-iron hubless fittings, CISPI 301. Manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.
3. Joints: NIH DRM Type c:
 - a. Heavy-duty shielded couplings listed to ASTM C564, ASTM C1540 and either FM 1680 Class 1 or IAPMO IGC 237, with type 304 or 316 stainless steel shield and neoprene or approved equivalent virgin elastomeric gasket, no adhesives. Couplings shall be a minimum of 4-band type. Hubless style couplings are not acceptable for pipe size change (reducer) applications, use manufactured pipe fittings.

B. Pumped Storm Water pipe and fittings:

1. Piping: NIH DRM Type Q:
 - a. Seamless copper tube, ASTM B88, Type L hard drawn; except where copper is allowed the following additional requirements apply: Type K wall thickness is required for pipe size 100 mm (4 in.) and larger. Manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.
2. Fittings: NIH DRM Type XVIII:
 - a. Wrought copper solder cup type fittings, ANSI/ASME B16.22 or B16.18. Manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.
3. Joints: NIH DRM Type e:
 - a. ASTM B828 lead-free soldered joints with ASTM B32 Grade HB, Grade HN, or approved equal solder that is also listed in Section 1 of ASTM B32. ASTM B813 high temperature water soluble flux. Air cooled only, no quenching. System should be rinsed thoroughly as soon as possible after soldering to prevent ongoing flux activity. External surface flux residue shall also be removed.
4. Unions for copper pipe: Cast bronze or wrought copper with solder ends, ASME B16.18 or B16.22.
5. Dielectric Nipples:
 - a. Standard: IAPMO PS 66.
 - b. Electroplated steel nipple complying with ASTM F 1545.
 - c. Pressure Rating and Temperature: 300 psig (2070 kPa) at 225 deg F (107 deg C).
 - d. End Connections: Male threaded or grooved.
 - e. Lining: Inert and noncorrosive, propylene.

C. Foundation Drainage pipe and fittings:

1. Piping: NIH DRM Type QQ:

- a. Perforated PVC ASTM D1785 only. SDR 35 is not acceptable. Perforations shall be per ASTM C700 or ASTM F-758 and AASHTO M252 unless otherwise required by geotech or structural engineer. Piping manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.

2. Fittings: NIH DRM Type XVIII:

- b. Polyvinyl chloride, drainage pattern, ASTM D2665, ASTM D3311. Manufacturer to be ISO 9001 (or equivalent) and ISO 14001 or equivalent certified.

3. Joints: NIH DRM Type f:

- b. Solvent cemented ASTM 2564 NSF listed cement and ASTM F656 NSF listed primer to IAPMO installation standard 9-95. "Hot" cements or wet-type fast dry cements shall not be utilized. Solvent cement joints only in dry ambient conditions and cement type appropriate for ambient temperature and pipe size. No less than 24 hour cure time for pressurized systems, and no less than 4 hours cure time for non-pressurized systems, regardless of size or cement type.

2.2 FOUNDATION DRAIN PIPING ACCESSORIES

A. Filter fabric:

- 1. Nonwoven polypropylene material recommended by the manufacturer for use in filtration or drainage
- 2. Permittivity, ASTM D 4491: 90-115 gallons per minute per sq. ft. (6-7 L/s per 0.9 sq. m).
- 3. Puncture resistance, ASTM D 4833: 65 lbs (290 N).
- 4. Mullen burst, ASTM D 3786: 230 psi (1586 kPa).
- 5. Trapezoid tear strength, ASTM D 4533: 45 lbs.
- 6. Grab/tensile elongation, ASTM D 4632: 110 lbs/50 percent.
- 7. Basis of design: Carthage Mills FX-40NS, Amoco 4506, or Mirafi 160N.

- B. Pins for securing filter fabric: As recommended by the manufacturer, at least 15 inches (380 mm) long, 0.1875-inch (5-mm) diameter, with 1.5-inch-(38-mm) round washer attached to the top of the pin between staked ears.

- C. Stone filter material: Uniform gradation, AASHTO M 43 size 67 or 7.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Connect piping as shown on the drawings. Check elevations of connection points before installing new work.

3.2 INSTALLATION, GENERAL

- A. Use suitable tools and appliances for the safe and convenient handling and laying of pipe. Examine each section of pipe for defects. Do not lay any piece that is known to be defective. If any defective piece should be discovered after having been laid, remove and replace it at no change to the contract price.

- B. Install piping in accordance with the Pipe Installation Schedule at the end of this section, as indicated on the drawings, and in accordance with Section 22 0500, Common Work Results for Plumbing. Materials and work shall conform to local plumbing codes and health department regulations.
- C. Thoroughly clean all pipe and fittings before installing them, and keep them clean until the acceptance of the completed work. Cap or plug ends of lines to prevent debris from entering during construction.
- D. Make changes in direction of storm water piping with approved sanitary fittings, Y branches, 1/8 or 1/16 bends.
- E. Install all storm water piping at a 2 percent minimum downward slope in the direction of flow unless otherwise indicated.

3.3 INSTALLING CAST-IRON PIPING

- A. Install cast-iron soil piping according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook," Chapter IV, "Installation of Cast Iron Soil Pipe and Fittings." Install restraint assemblies at pipe couplings and fittings for piping NPS 5 (DN 125) and larger.

3.4 INSTALLING COPPER PIPING

- A. Install aboveground copper tubing according to Copper Development Association, Inc. (CDA) "Copper Tube Handbook."

3.5 INSTALLING FOUNDATION DRAINS

- A. Excavation for pipe laid against footings shall not be lower than the bottom of the footing.
- B. Installing stone filter material and geotextile filter fabric:
 - 1. Lay geotextile filter fabric, with material extending up as will be required to complete an envelope around stone filter material and piping.
 - 2. Install no less than 6 inches of stone filter material under pipe, and lay pipe.
 - 3. Fill with stone filter material no less than 6 inches (150 mm) on each side and 2 feet (610 mm) above piping, or to bottom of slab.
 - 4. Fold geotextile filter fabric over pipe and stone filter material to form a continuous envelope, overlapping no less than 2 inches. Top of geotextile envelope shall be directly under slab or between stone filter material and earth fill.
- C. Lay drain lines at elevations noted on drawings.
- D. Lay piping without rises or traps. Lay piping flat, or with uniform fall towards the points of discharge.
- E. Lay pipe with perforations downward.
- F. Verify joints are tight prior to backfilling. During backfilling assure that joints are not disturbed.

END OF SECTION

SECTION 22 1429 - SUMP PUMPS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Submersible sump pumps.
- B. Submersible sump pump controls.
- C. Metal framing systems.

1.2 RELATED SECTIONS

- A. UL label, demonstration, training, and equipment installation: Section 22 0500.
- B. Motors: Section 22 0513.
- C. Pumped storm water piping: Section 22 1413.
- D. Controls: On drawings.

1.3 SUBMITTALS

- A. Product data: Include certified performance curves and rated capacities of selected models; shipping, installed, and operating weights; furnished specialties; and accessories for each type and size of pump specified. Indicate pumps' operating point on curves.
- B. Shop drawings: Show layout and connections for pumps. Include setting drawings with templates, directions for installation of foundation and anchor bolts, and other anchorages.
 - 1. Wiring diagrams: Detail wiring for power, signal, and control systems and differentiate between manufacturer-installed and field-installed wiring.
- C. Maintenance data: For each pump specified, to include in maintenance manuals specified in Division 01 and Section 22 0101.

1.4 QUALITY ASSURANCE

- A. UL label: As specified in Section 22 0500, Common Work Results for Plumbing.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Specified units are the basis for design of the project. The following listed manufacturers also provide units of acceptable quality. If units by any of these manufacturers should be proposed, verify that they meet requirements specified in Division 01 and the article "Product Options" in Section 22 0101, and submit shop drawings as specified in the "Submittals" article above.
- B. Submersible sump pumps:
 - 1. Barnes, Crane Pumps and Systems
 - 2. Deming, Crane Pumps and Systems
 - 3. Federal Pump Corporation
 - 4. Goulds Pumps, ITT Industries

- 5. Pacific Pump Co.
- 6. Weil Pump Co.
- 7. Zoeller Pump Co.

2.2 GENERAL REQUIREMENTS

- A. Provide type, size, configuration, capacity, and current characteristics indicated on the drawings.
- B. Provide motors that meet requirements in Section 22 0513, Common Motor Requirements for Plumbing Equipment.

2.3 SUMPS

- A. Field-fabricated concrete sump manhole: Concrete manhole with cover as detailed on civil drawings.

2.4 SUBMERSIBLE SUMP PUMPS

- A. Submersible Sump Pump:
 - 1. Description: Factory-assembled and factory-tested fixed-position sump pump.
 - 2. Pump shaft seal:
 - a. Provide single-seal for pumps with motors 1-HP or less in size.
 - b. Provide double-seal for pumps with motors greater than 1-HP in size.
 - 3. Pump Type: Submersible, end-suction, single-stage, close-coupled, overhung-impeller, centrifugal sump pump.
 - 4. Casing: Cast iron, with strainer inlet and legs that elevate pump to permit flow into impeller.
 - 5. Support and discharge configuration:
 - a. Support legs and threaded top discharge for pumps without sump guide rails.
 - b. Cantilevered disconnect fitting support and horizontal flanged discharge for pumps with sump guide rails.
 - 6. Impeller: Dynamically balanced, bronze or stainless steel, keyed and secured to shaft.
 - 7. Shaft: Stainless steel.
 - 8. Bearings: Factory-sealed, grease-lubricated ball bearings.
 - 9. Seal: Carbon and ceramic mechanical seal.
 - 10. Inlet strainer: Brass or stainless steel.
 - 11. Motor: NEMA 6, hermetically sealed, air-filled, capacitor-start type; with built-in overload protection and lifting eye or lug, cast-iron shell, rated for continuous duty at 120 degrees.
 - 12. Waterproof power cable of length required and cable-sealing assembly for connection at pump.
 - 13. Basis-of-design product: Subject to compliance with requirements, provide Zoeller Pump Company Model 267, or comparable product manufactured by other listed manufacturer.

2.5 SUBMERSIBLE SUMP PUMP CONTROLS

- A. Duplex sump pump controls:
 - 1. Control panel: Provide a U.L. listed control panel with the following minimum features:
 - a. Enclosure: NEMA Type 4X.
 - b. Enclosure front:
 - (1) In areas accessible to the public: Dead front type panel with controls concealed behind a lockable door.

(2) In areas not accessible to the public: Standard front with HOA switches, lights, and reset buttons exposed.

- c. Power supply: Hard wired to control panel.
- d. Control power transformer.
- e. Circuit breaker disconnect for each controlled pump.
- f. Terminal strip connections for building automation system:
 - (1) Alarm status.
 - (2) Pump status for each controlled pump.
- g. Alarm light.
- h. Alarm horn or buzzer and alarm silence button.
- i. Thermal overload and reset button for each controlled pump.
- j. Pump run status sensing and indicator light for each controlled pump.
- k. HOA switch for each controlled pump.
- l. Pump starting:
 - (1) Single phase pump control panels: Motor contactor.
 - (2) Three phase pump control panels: Magnetic starter.

2. Level controls:

- a. Water level switch: PVC or polypropylene encased mechanical-float type switch, rated for operation in up to 140 Degree F (60 degree C) fluid.
- b. Mounting rod: Schedule 40 NPS 2 (DN 50) CPVC or other approved non-corrosive material with wall support brackets.
- c. Electric cables: SJOW oil and water resistant cable clad in CPE, of length required between switch and controller, clipped to mounting rod at required elevations by switch manufacturer provided plastic clip.
- d. Duplex level switch locations:
 - (1) High level alarm switch: Set below the lowest pipe inlet, and above the "Pump #2 on" switch.
 - (2) Pump #2 on switch: Set below the high level alarm switch, at a level that provides sufficient volume to minimize lag pump cycling.
 - (3) Pump #1 on switch: Set below the Pump #2 on switch, at a level that provides sufficient volume to minimize lead pump cycling.
 - (4) Pump off switch: Set at manufacturer recommended distance above the pump inlet.
- e. Duplex control sequence:
 - (1) On liquid level rise to the "Pump #1 on switch": lead pump shall be energized.
 - (2) On further liquid level rise to the "Pump #2 on switch": lag pump shall be energized.
 - (3) On further liquid level rise to the "High level alarm switch": alarm light and the alarm horn or buzzer shall be energized.
 - (4) On liquid level fall below the "Pump off switch": All pumps shall be deenergized.

2.6 METAL FRAMING SYSTEMS

A. Metal framing system for pedestal or wall support of control panels.

- 1. Description: Shop- or field-fabricated, pipe-support assembly made of channels, nuts, bolts, structural connections, accessories, fittings, and other manufactured components.
- 2. Standard: Comply with MFMA-4 for factory-fabricated components for field assembly.
- 3. Channels: Continuous slotted galvanized steel channel with in-turned lips, width selected for applicable load criteria.

4. Channel Nuts: Formed or stamped nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

- A. Install equipment as indicated on the drawings. Comply with requirements in Section 22 0500.

3.2 INSTALLING SUBMERSIBLE SUMP PUMPS

A. Piping:

1. Provide a union or flange, check valve, and shut-off valve in the discharge from each submersible sump pump. Arrange to facilitate pump, pump discharge pipe, and check valve replacement by closing the shut-off valve and disconnecting the pump and discharge piping via the union or flange.
 - a. Vertical discharge: Provide in an accessible location above sump cover.
 - b. Horizontal discharge below slab: Provide in manhole.
 - c. Duplex pumps: Provide separate valves and unions or flanges for each pump prior to combining pump discharge.

B. Inspection:

1. Inspect motors for proper direction of rotation.
2. Verify pumps rotate freely by hand. If bound or dragging, determine the cause and correct it.

3.3 INSTALLING SUBMERSIBLE SUMP PUMP POWER AND CONTROLS

A. Installation:

1. Mount control panel on wall of manhole.
2. Provide all control and power wiring and accessories for a complete and functional system.
3. Provide switches at required elevations, mounted to mounting rod with clips provided by the switch manufacturer and with wiring tied to the control rod within the sump.
4. Provide wire from switches to control panels.
5. Provide power wiring to control panels.
6. Provide power wiring from control panels to pumps.
7. Wiring pathway:
 - a. Where the sump is located in a mechanical space and an exposed wiring pathway is provided, route wiring through sump cover bushings, glands, or seals and then route in conduit to the control panel.
 - b. Where the wiring pathway is concealed below a slab, route wiring in conduit from liquid tight NEMA 4X junction box mounted in the sump, through a water tight penetration of the sump above the high water alarm, and to the control panel.
8. Provide interface between pump controls and building automation system.

B. Startup:

1. Verify that pump controls are correct for the application.
2. Open discharge valves.
3. Energize pumps.
4. Fill sumps with water and check operations of pumps, controls, and alarms.

3.4 INSTALLING METAL FRAMING SYSTEM

- A. Provide metal framing system anchored to wall of manhole to support sump pump control panel.

3.5 OPERATING INSTRUCTIONS

- A. As specified in Section 22 0500, provide operating instructions.

END OF SECTION

SECTION 26 0101 - ELECTRICAL GENERAL PROVISIONS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. General provisions and requirements for electrical work.

1.2 RELATED SECTIONS

- A. Requirements of this section generally supplement requirements of Division 01.
- B. Division 01 includes sections specifying requirements for commissioning and construction waste management.

1.3 REFERENCES

- A. NFPA 10: Portable Fire Extinguishers.
- B. NFPA 241: Safeguarding Construction, Alteration, and Demolition Operations.

1.4 SYSTEM DESCRIPTION

- A. The full set of Contract Documents applies to work of Division 26.
- B. Visit the site and study all aspects of the project and working conditions, as required by General and Supplementary Conditions, Bidding and Contracting Requirements, Drawings, and Specifications. Verify field dimensions.
- C. The work covered in technical sections includes the furnishing of all labor, equipment and materials, and the performance of all operations pertinent to the work described.
- D. Except as required otherwise in Division 01, promptly obtain and pay for, all necessary signatures and paperwork, all permits, fees and inspections required for work of this division by authorities having jurisdiction, including any utility connection or extension charge. No payment will be made until a copy of the permit is forwarded to the Government.
- E. Electrical work of this project includes, as a brief general description, the following:
 - 1. Renovation of the lighting, power, low-voltage and fire alarm systems in NIH Building 49 in the Level 1 lobby and adjacent corridors.
 - 2. The project includes commissioning under the direction of a Commissioning Agent (CxA).
- F. See Division 01 for requirements related to commissioning, Government's occupancy of the premises, limits on use of site, time restrictions on work, limits on utility outages or shutdowns, and phasing (sequencing) and scheduling.

1.5 PRODUCT OPTIONS

- A. Except as modified by provisions of Bidding and Contracting Requirements and Division 01, these options apply to Division 26 specifications.
- B. General: Where Contractor is permitted to use a product other than the specified item and model named as the basis of design, Contractor is responsible for all coordination and additional costs as specified in article "Substitutions" below for substitutions.

- C. Products specified by reference standards or by description only: Any product meeting those standards or description.
- D. Products specified by naming one or more manufacturers, or model name or catalog reference number: Products specified establish a standard of quality, options to be included, and performance, and shall not be construed as limiting competition. Contractor may use products of any manufacturer, which meet the specifications.
- E. Products specified by naming one manufacturer and particular product, with no provision for other options: No options or substitutions allowed.

1.6 SUBSTITUTIONS

- A. Substitutions will be considered only as permitted or required by the Bidding and Contracting Requirements and Division 01. Except as modified by those requirements, the requirements below apply to Division 26 specifications.
- B. Substitutions may be considered when a product becomes unavailable through no fault of the Contractor.
- C. Document each request with complete data substantiating compliance of proposed substitution with contract documents.
- D. A request constitutes a representation that the Bidder or Contractor:
 - 1. Has investigated proposed product and determined that it meets or exceeds the quality level of the specified product.
 - 2. Will provide the same warranty for the substitution as for the specified product.
 - 3. Will coordinate installation and make changes to other work which may be required for the work to be complete with no additional cost to the Government.
 - 4. Waives claims for additional costs or time extension which may subsequently become apparent.
 - 5. Will reimburse the Government for review or redesign services associated with re-approval by authorities.
- E. Substitutions will not be considered when they are indicated or implied on shop drawing or product data submittals, without separate written request, or when acceptance will require revision to the Contract Documents.
- F. Substitution submittal procedure is specified in Bidding and Contracting Requirements and Division 01.

1.7 MATERIALS AND EQUIPMENT

- A. All materials and equipment shall be new and the best of their respective kinds, suitable for the conditions and duties imposed on them by the project, and of representative manufacturer. The description, characteristics and requirements of the materials to be used shall be in accordance with the specifications.
- B. All equipment, construction and installation must meet requirements of local, state and federal governing codes.
- C. Singular number: In cases where material, a device, or part of the equipment is referred to in the singular number in the specifications, it is intended that such reference shall apply to as many items of material, devices, or parts of the equipment as are required to complete the installation as shown on the drawings or required for proper operation of the system.

D. Terms have the following meanings:

1. Furnish: Supply item
2. Install: Mount and connect item
3. Provide: Furnish and install

- E. All materials and equipment shall be installed and completed in a first class and workmanlike manner and in accordance with the best modern methods, practice and manufacturers' instructions. Any work which shall not present an orderly and neat or workmanlike appearance shall be removed and replaced with satisfactory work when so directed in writing by the Project Officer.
- F. The specifications and drawings are intended to define the minimum requirements, as to quality of materials, construction, finish and overall workmanship.
- G. General Conditions describe the correlation and intent of the Contract Documents. In case of discrepancies between the specifications and drawings, the specifications should be followed as to the general methods and principles and the drawings followed as to sizes, capacities and specifics for corresponding parts. If sizes are omitted, the Project Officer will determine sizes to be utilized.
- H. In all cases of doubt, uncertainty, or conflict as to the true meaning of the specifications or drawings, it is the responsibility of the Contractor to notify the Project Officer of said uncertainty, doubt, or conflict and obtain a decision as to the intent prior to initiating any work which may be affected by this decision.

1.8 COORDINATION

- A. Should a situation develop during construction to prevent the proper installation of any equipment or item where shown on the drawings, call the situation to the attention of the Project Officer and await a written decision.
- B. Plan and coordinate all work to proceed in an orderly and continuous manner without undue delay, and in conformance with the project schedule. Submit samples, shop drawings, schedules, insurance policies and certificates, and the like in time to avoid delays in actual construction. Coordinate electrical work so that work of each trade is completed before other construction begins which would obstruct it.
- C. Coordinate trades to ensure that proper clearances between work of the various trades allow access to items which require operation and maintenance.
- D. Coordinate location and elevation of all conduit, light fixtures, equipment, and appurtenances in such a manner that the finished installation is as indicated on drawings. In the event difficulties are encountered which prevent this, it is the Contractor's responsibility to bring this to the attention of the Project Officer prior to initiation of work. Correct improperly coordinated installation at no additional cost.
- E. The Contractors' assistants shall include a competent electrical foreman, who shall be on the premises at all times to check, layout, coordinate and superintend the installation of work. The foreman shall establish all basic requirements relative to the work before starting, and be responsible for the accuracy thereof.

1.9 SUBMITTALS

A. Manufacturers' and subcontractors' lists:

1. As specified in Division 01, submit a complete list of proposed manufacturers for all equipment, materials and subcontractors used for the work of this division. Lists shall follow the sequence of the specifications. No considerations will be given for partial or incomplete lists. After review of lists, submit shop drawings and product data.

B. Shop drawings and product data:

1. Submit in accordance with the requirements of Division 01 or as established at the preconstruction conference, the required number of copies of Shop Drawings and Product Data for every item of equipment. Shop drawings or product data will not be considered until Manufacturers' Lists have been approved. Shop drawings and product data shall be submitted, as required by the General Conditions, with sufficient time for checking, return to Contractor, and resubmission as required before Contractor shall install any item.
2. Each item submitted shall be properly labeled, indicating the specific service for which the equipment or material is to be used, section and paragraph number of specification or drawing number to which it applies, Contractor's name and project name and number. Data submitted shall be specific and shall include product data and printed information in sufficient detail and scope to verify compliance with requirements of the contract documents. Clearly identify each item within the data. Data of a general nature will not be accepted. Each sheet must clearly show the project name and number.
3. The review of a shop drawing or product data shall not be considered as a guarantee of the measurements or building conditions or that the shop drawings or product data have been checked to see that item submitted properly fits the building conditions. This review shall not relieve the Contractor of the responsibility for furnishing material or performing work as required by the contract documents, for correctness of dimensions and quantities, or for proper coordination of details and interfaces among trades.
4. All exclusively electrical items furnished as items associated with mechanical items but not specifically described in the mechanical item submission, shall be submitted as a separate submittal but shall be clearly marked as associated with the mechanical item by identified specification paragraph.
5. Product data sheets shall be 8.5-inches by 11-inches cut sheets for operating and maintenance manual.

C. Submit at least three copies of the results of every test required under any section in this division.

D. Specialist shall submit a list of at least three projects similar to this project in type, size, and quality, which have been in place and operating satisfactorily for at least five years.

1. Include project name, address, name and phone number of Government's representative, and project type and size.

E. After the work is completed, submit all required certificates of approval from approved inspection agencies and authorities having jurisdiction over work of this division. Certificates of approval must be received by the Project Officer prior to final acceptance of the work.

1.10 SPECIALIST

- A. The term "Specialist" as used in the specification shall mean an individual or firm of established reputation (or, if newly organized, whose personnel have previously established a reputation in the same field,) which is regularly engaged in, and which maintains a regular force of workers skilled in either (as applicable) manufacturing or fabricating items required by the contract, installing items required by the contract, or otherwise performing work required by the contract. Where the specification requires installation by a specialist, the term shall also be deemed to mean the manufacturer of the item, an individual or firm licensed by the manufacturer, or an individual or firm who will perform the work under the manufacturer's direct supervision.

1.11 CONTRACT CLOSEOUT SUBMITTALS

A. Project record documents:

1. Maintain on site one set of the following record documents; record actual revisions to the work of this division:
 - a. Contract Drawings.
 - b. Specifications.
 - c. Addenda.
 - d. Change Orders and other Modifications to the Contract.
 - e. Reviewed shop drawings, product data, and samples.
2. Maintain record documents separate from documents used for construction.
3. Record information concurrent with construction progress.
4. Specifications: Legibly mark and record in each section a description of actual products installed, including the following:
 - a. Manufacturer's name and product model and number.
 - b. Product options, substitutions, or alternates utilized.
 - c. Changes made by addenda and modifications.
5. Record documents and shop drawings: Legibly mark each item to record actual construction, including:
 - a. Measured locations of internal utilities and appurtenances concealed in construction, referenced to visible and accessible features of the work.
 - b. Field changes of dimension and detail.
 - c. Details not on original Contract Drawings.
6. Submit documents as specified in Division 01.

B. Operation and maintenance data:

1. Submit sets prior to final inspection as specified in Division 01. Unless otherwise specified in Division 01, submit no fewer than three sets. In addition to requirements specified in Division 01, submit operating and maintenance manuals for the work of this division as specified below.
2. Binders: Three-ring binders with vinyl-covered hard covers. Provide large enough binders, and sufficient quantity, that the required contents can be easily turned, removed, and reinserted.

3. Prepare binder covers with printed title "OPERATION AND MAINTENANCE INSTRUCTIONS," and title of project. Print on spine of binder "O & M INSTRUCTIONS." If more than one binder is required, print covers and spines with volume numbers. Include in the front of every binder an index to all binders.
4. Internally subdivide the binder contents with permanent page dividers, logically organized as described below; with tab titling clearly printed under reinforced laminated plastic tabs.
5. Contents: Prepare a Table of Contents for each volume, with each product or system description identified, typed on white paper.
6. Part 1: Directory, listing names, addresses, and telephone numbers of electrical engineers; contractor; electrical subcontractors; and major electrical equipment suppliers.
7. Part 2: Operation and maintenance instructions, arranged by specification section. For each category, identify names, addresses, and telephone numbers of subcontractors and suppliers. Identify the following:
 - a. Significant design criteria.
 - b. List of equipment.
 - c. Parts list for each component, including recommended spare parts list.
 - d. Operating instructions.
 - e. Maintenance instructions for equipment and systems.
 - f. Maintenance instructions for special finishes, including recommended cleaning methods and materials and special precautions identifying detrimental agents.
8. Part 3: Project documents and certificates, including the following:
 - a. Shop drawings and product data.
 - b. Photocopies of certificates.
 - c. Photocopies of warranties, guarantees, and bonds.
 - d. Test reports: Copies of the results of all tests required under all sections of specifications.
9. Submit one copy of completed volumes in final form 15 days prior to final inspection. This copy will be returned after final inspection, with Engineer comments. Revise content of documents as required prior to final submittal.
10. Submit final volumes revised, within ten days after final inspection.

1.12 REGULATORY REQUIREMENTS

- A. When these specifications call for materials or construction of a better quality or larger sizes than required by the following codes and standards, the provisions of the specifications shall take precedence.
- B. Provide, without extra charge, any additional materials and labor which may be required for compliance with these codes and standards even though the work is not mentioned in these specifications or shown on the contract drawings.
- C. Perform the work of this division in strict accordance with the following authorities. The latest revision of these codes accepted by the authority having jurisdiction as of the date of the contract documents shall apply.
 1. The electrical, building, fire, and safety codes of the state and county or city in which the work is being performed.
 2. The National Electric Code, NFPA 70 (NEC).
 3. The National Fire Protection Association Code (NFPA).
 4. International Building Code (IBC).
 5. International Energy Conservation, Fire, and Electrical Codes (ICC).

1.13 REFERENCE STANDARDS

- A. Perform the work of this division using the standards of the following organizations, as referred to in technical sections, as a minimum requirement for construction and testing. Unless specified otherwise in Bidding and Contract Documents or Division 01, the latest revision current as of the date of the contract documents shall apply.
 - 1. Factory Mutual (FM)
 - 2. Federal Specifications (FS)
 - 3. Military Standards (Mil. Std.)
 - 4. American National Standards Institute (ANSI)
 - 5. American Society for Testing and Materials (ASTM)
 - 6. International Code Council (ICC)
 - 7. Institute of Electrical and Electronics Engineers (IEEE)
 - 8. National Electrical Code (NEC) (NFPA 70)
 - 9. National Electrical Manufacturer's Association (NEMA)
 - 10. National Fire Protection Association (NFPA)
 - 11. The Occupational Safety and Health Act (OSHA)
 - 12. Underwriters Laboratory Inc. (UL)
 - 13. American Association of State Highway and Transportation Officials (AASHTO)
 - 14. American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE)
 - 15. Maryland Occupational Safety and Health Act (MOSHA)
 - 16. Illuminating Engineering Society of North America (IESNA)

1.14 TEMPORARY STORAGE

- A. Maintain upon premises, where directed, a storage area, and be responsible for all contents within these areas. Provide all security measures necessary for this area.
- B. Area shall be maintained and shall be returned to original condition at the completion of the project.
- C. Store electrical construction materials such as wire, raceways and boxes, devices, and equipment in buildings, enclosed trailers, or portable enclosed warehouses.
 - 1. Materials and products subject to damage from moisture: Store in dry locations. If necessary, protect with protective wraps or covers.
 - 2. Plastics and other materials and products subject to damage from heat or cold: Store at manufacturer's recommended temperatures.
 - 3. Plastics and other materials and products subject to damage from sunlight: Protect from sunlight.
- D. Electrical equipment such as motor controllers, panelboards and circuit breakers stored before installation and installed during construction: Provide clean, dry locations at manufacturer's recommended temperatures, and cover or wrap if required to protect from incidental damage.

1.15 PROTECTION

- A. Control dust resulting from construction work to prevent its spread beyond the immediate work area, and to avoid creation of a nuisance.
 - 1. Do not use water to control dust. Use drop cloths or other suitable barriers.
 - 2. In areas where dirt or dust is produced as a result of the work, sweep daily, or more often as required.
 - 3. Provide walk-off mats at entries and replace them at regular intervals.
 - 4. Construct dust partitions, where indicated on the drawings or as required.

5. Protect areas occupied by Government personnel or equipment.
 6. Seal off all return air registers and other mechanical systems to prevent dust from entering.
- B. Each trade and subcontractor is responsible for preventing damage and soiling of work performed by other trades or subcontractors. Each trade and subcontractor is responsible for providing temporary protection of its own work.
1. Protect work from spills, splatters, drippings, adhesives, bitumens, mortars, paints, plasters, and damage from welding or burning.
 2. Protect finished work from damage, defacement, staining, or scratching.
 3. Protect finishes from cleaning agents, or grinding and finishing equipment.
 4. Protect adjacent and finished work from damage, using tape, masking, covers or coatings and protective enclosures.
 5. Coordinate installations and temporarily remove items to avoid damage from finishing work.
- C. Repair all damage or soiling to the complete satisfaction of the Project Officer; replace any materials or work damaged to such an extent that they cannot be restored to their original condition, all at no addition to the Contract sum.
- D. Protect work stored in place and supplies stored in the building.
1. Store materials and products, subject to damage from moisture, in dry locations. If necessary, protect in wraps or covers.
 2. Store plastics, other materials, and products subject to damage from heat or cold at manufacturer's recommended temperatures.
- E. Protect electrical materials and products from weather events and accidents of construction.
- F. Use of sidewalk or roadway areas outside of the property lines shall be with permission and approval of the local authorities having jurisdiction.

1.16 FIRE PROTECTION

- A. As a minimum, provide hand-carried, portable, UL-rated extinguishers with each work crew working inside the building.
- B. Select extinguishers in accordance with NFPA 10 and NFPA 241 for classification, extinguishing agent, and size required by location and class of fire exposure.

1.17 PROJECT CONDITIONS

- A. Drawings showing utilities in concealed locations are based on the best information available but are not represented as being precisely correct. Work of the contract includes digging, cutting, drilling, using nondestructive methods, and other methods of locating concealed utilities in the field, as well as patching and repairing as specified in "Cutting and Patching" below.
- B. If, in the course of the work, workers encounter a material they suspect to present some hazard:
 1. Promptly notify the Project Officer in writing.
 2. Do not perform any work which would disturb the suspected material until written instructions have been received.

1.18 WARRANTY

- A. All work and equipment provided as work of this division shall be fully warranted under the general project warranty. In addition, provide added special warranties as specified in individual sections.
- B. During the correction period, the Contractor shall begin correcting any work found to be not in accordance with the requirements of the Contract Documents within 4-hours of receiving written notice from the Project Officer. Provide detailed schedule for completion of work within 24-hours of receiving written notice from the Project Officer and revise schedule based on any Government comments generated. Except as otherwise required in General Conditions and Division 01, the correction period is one year after the date of substantial completion of the work. Work requiring correction shall promptly be repaired or completely replaced at no addition to the Contract Sum.
 - 1. Service reports for warranty work shall be provided to the Government.
- C. When use of the permanent equipment has been permitted for temporary services during construction of the building, the warranty and correction periods shall nevertheless begin at the time of substantial completion, unless another date of acceptance has been agreed to by the Government.
- D. Special warranties are warranties required by individual specification sections, incidental product warranties, manufacturers' standard warranties, installer or subcontractor service agreements, and other individual warranties in addition to the general project warranty.
- E. Provide copies of warranties as required for Operation and Maintenance Manual specified above, and by Division 01.
- F. For items of work delayed beyond date of substantial completion, provide updated submittal within ten days after acceptance, listing date of acceptance as start of warranty period.

1.19 COMMISSIONING

- A. This project includes commissioning under the direction of a Commissioning Agent (CxA). Contractor's and subcontractors' responsibilities are described in Division 01.
- B. Cooperate with the CxA to accomplish the requirements of the Commissioning Plan during the construction and correction periods.

PART 2 - PRODUCTS

Not used.

PART 3 - EXECUTION

3.1 CUTTING AND PATCHING

- A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.
- B. Cut walls, floors, partitions, roofs, and other appurtenances for the passage or accommodation of conduits. Close superfluous openings and remove all debris caused by work of this division.
- C. No cutting of any structure or finish shall be done until the condition requiring such cutting has been examined and approved by the Project Officer.

- D. New or existing surfaces disturbed as a result of such cutting or otherwise damaged shall be restored to match original work and all materials used for any patching or mending shall conform to the class of materials originally installed.
- E. Refinish surfaces to match adjacent finish. For continuous surfaces, refinish to nearest intersection or natural break. For an assembly, refinish entire unit.

3.2 TEMPORARY FACILITIES

- A. Temporary water facilities, electricity, telephone, toilet facilities, and temporary heat, shall be provided as specified in Division 01.

3.3 PROGRESS MEETINGS

- A. Progress meetings shall be held as specified in Division 01, and also when and if the Contractor or Project Officer finds them necessary or advantageous to progress of work.
- B. Contractor, those subcontractors and those material suppliers concerned with current progress or with the scheduling of future progress, and Project Officer shall each be represented at these meetings by persons familiar with the details of work and authorized to conclude matters relating to work progress.

3.4 COMMISSIONING

- A. Comply with requirements of 'Commissioning' in Part 1 above.

END OF SECTION 26 0101

SECTION 26 0500 - COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Requirements applicable to work of more than one section of Division 26.
- B. Testing wiring systems.

1.2 RELATED SECTIONS

- A. Division 01 includes sections specifying requirements for commissioning and construction waste management.
- B. Operation and Maintenance Manuals: Division 01 and Section 26 0101.
- C. Painting: Division 09.

1.3 DEFINITIONS

- A. Project correction period: A period after Substantial Completion of the work during which the Contractor shall correct every part of the work found to be not in accordance with the requirements of the contract documents, promptly after receipt of written notice.
- B. Qualified testing agency: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with the experience and capability to conduct testing and inspecting indicated, as documented according to ASTM E 548; and with additional qualifications specified in individual sections; and where required by authorities having jurisdiction, that is acceptable to authorities.

1.4 DESIGN REQUIREMENTS

- A. The drawings and system performances have been designed on the basis of using the particular manufacturers' products specified and scheduled on the drawings.
- B. Products of other manufacturers that are listed under the article "Available Manufacturers," or permitted as "equal," are permitted provided:
 - 1. Product shall meet the specifications.
 - 2. Contractor shall make, without addition to the contract sum, all adjustments for deviations so that the final installation is complete and functions as the design basis product is intended.
- C. Do not propose products with dimensions or other characteristics different from the design basis product that make their use impractical or cause functional fit, access, or connection problems.
- D. The contract drawings are generally diagrammatic, and do not indicate all fittings or offsets in conduit or all pull boxes, access panels, or other specialties required.
 - 1. Install conduit exposed to view parallel with the lines of the building and as close to walls, columns, and ceilings as may be practical, maintaining adequate clearance for access at parts requiring servicing.
 - 2. Install conduit a sufficient distance from other work to permit a clearance of not less than 0.5 inch (15 mm) between its finished covering and adjacent work.
 - 3. No conduit shall be run below the head of a window or door.

4. Pull boxes and other appurtenances which require operation or maintenance shall be easily accessible. Do not cut or form handholes for operation or maintenance of appliances through walls or ceilings.

1.5 SUBMITTALS

- A. Test reports: Show that tests specified in Part 3 below demonstrate the specified results.

1.6 QUALITY ASSURANCE

- A. Provide materials and perform work in accordance with the electrical, building, fire, and safety codes and regulations of the state, county, or city in which the work is performed.
- B. Electrical control panels, equipment, materials and devices provided or installed as work of Division 26 shall bear UL label, or, if UL label is not available, the item shall be tested and labeled by a qualified testing agency, acceptable to authorities having jurisdiction, and in accordance with NFPA 70. Provide testing, if required, without addition to the contract sum.
- C. VOC content: Field-applied adhesives and sealants, limits per South Coast Air Quality Management District (SCAQMD), Rule No. 1168.
- D. Products shall contain no urea-formaldehyde content.

1.7 COMMISSIONING

- A. This project includes commissioning under the direction of a Commissioning Agent (CxA). Contractors' and subcontractors' responsibilities are described in Division 01.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Aircraft cable: 0.25-inch (6-mm) steel wire rope, galvanized, construction 7 by 19 strands, minimum 7000 lbs (31138 N) breaking strength.

2.2 DATE-SENSITIVE EQUIPMENT

- A. Date-sensitive equipment: Systems, equipment, or components which use or process date and time data in order to perform their functions.
- B. Each item of date-sensitive equipment used in the project shall be warranted by the manufacturer to properly function and correctly use or process all time-related data for all dates and times which occur during a reasonable life expectancy of the equipment.

PART 3 - EXECUTION

3.1 INSTALLATION OF PRODUCTS AND EQUIPMENT

- A. Manufacturers' instructions: Except as modified by drawings or specifications, install products and equipment in accordance with manufacturers' instructions and recommendations applicable to the project conditions.
 1. Immediately notify Project Officer if a difference or discrepancy is found between manufacturers' instructions and the drawings or specifications.
- B. Install plywood backing panels with finished face exposed.

3.2 TESTS

- A. During the progress of the work and after completion, test the branch circuits and distribution system, and the low-voltage alarm and signal systems.
- B. Results of the tests shall show that the wiring meets the requirements of this specification. Should any test indicate defect in materials or workmanship, immediately repair, or replace with new, the faulty installation, and retest the affected portions of the work.
- C. Furnish equipment and instruments necessary for testing.
- D. Tests shall demonstrate the following:
 - 1. Lighting, power, and control circuits are continuous and free from short circuits.
 - 2. Circuits are free from unspecified grounds.
 - 3. The resistance to ground of each non-grounded circuit is not less than one megohm.
 - 4. Circuits are properly connected in accordance with the applicable wiring diagrams.
 - 5. Circuits are operable. Demonstration shall include functioning of each control not less than ten times, and continuous operation of each lighting and power circuit for not less than 0.5 hour.
- E. Test circuit breakers larger than 100 amps at full voltage.
- F. Make voltage built-up tests with a voltage sufficient to determine that no short circuits exist.
- G. Immediately repair defects and retest until systems are operating correctly.
- H. Submit test reports.

3.3 OPERATING INSTRUCTIONS

- A. Furnish the necessary technicians, skilled workers, and helpers to operate the electrical systems and equipment of the entire project for one 8-hour day.
- B. Where specified in technical sections, provide longer periods required for specialized equipment.
- C. Instruct the Government's designated personnel in operation, maintenance, lubrication, and adjustment of systems and equipment.
- D. The Operating and Maintenance Manual shall be available at the time of the instructions for use by instructors and Government personnel.
- E. Schedule the general and specialized instruction periods for a time agreed upon by the Project Officer.

END OF SECTION

SECTION 26 0507 - FIRESTOPPING FOR ELECTRICAL WORK

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Through-penetration firestopping in fire-rated construction.
- B. Through-penetration smoke-stopping in smoke partitions.

1.2 RELATED SECTIONS

- A. Conduit: Section 26 0533.

1.3 REFERENCES

- A. Underwriters Laboratories
 - 1. UL Fire Resistance Directory
 - 2. UL 1479: Through Penetration Firestops.
- B. American Society for Testing and Materials Standards:
 - 1. ASTM E 814: Standard Test Method for Fire Tests of Through-Penetration Firestops.

1.4 DEFINITIONS

- A. Assembly: Particular arrangement of materials specific to given type of construction described in referenced documents.
- B. Barriers: Time-rated fire walls, smoke barrier walls, time-rated ceiling/floor assemblies and structural floors.
- C. Firestopping: Methods and materials applied in penetrations and unprotected openings to limit spread of heat, fire, gasses and smoke.
- D. Penetration: Opening or foreign materials passing through or into barrier or structural floor such that full thickness of rated materials is not obtained.
- E. Sleeve: Metal fabrication or pipe section extending through thickness of barrier and used to permanently guard penetration. Sleeves are described as part of penetrating system in other sections and may or may not be required.
- F. System: Specific products and applications, classified and numbered by the rating agency to close specific barrier penetrations.

1.5 SYSTEM DESCRIPTION

- A. Design requirements:
 - 1. Fire-rated construction: Maintain barrier and structural floor fire resistant ratings including resistance to cold smoke at all penetrations.
 - 2. Smoke barrier construction: Maintain barrier and structural floor resistance to cold smoke at all penetrations.

1.6 SUBMITTALS

- A. Product data: Manufacturer's specifications and technical data including the following:
 - 1. Detailed specification of construction and fabrication.
 - 2. Manufacturer's installation instructions.
- B. Shop drawings: Submit firestop assemblies and devices for all openings and through penetrations in fire-rated construction. Indicate dimensions, description of materials and finishes, general construction, specific modifications, component connections, anchorage methods, hardware, and installation procedures, plus the following specific requirements.
 - 1. Details of each proposed assembly identifying intended products and applicable rating agency classification.
 - 2. Manufacturer or manufacturer's representative shall provide qualified engineering judgments and drawings relating to conditions where rated assemblies do not exist.
- C. Quality control submittals:
 - 1. Statement of qualifications.
- D. Certifications: Letters or forms showing acceptance by local authorities for systems without acceptance by a rating agency.

1.7 QUALITY ASSURANCE

- A. Products and assemblies shall be tested and labeled by an independent, nationally recognized testing and labeling authority.
- B. Installer's qualification: Firm experienced in installation or application of systems similar in complexity to those required for this project, plus the following:
 - 1. Acceptable to or licensed by manufacturer, state, or local authority where applicable.
 - 2. At least 2 years' experience with systems.
 - 3. Successfully completed at least 5 projects of comparable scale, using these systems.
- C. Local and state regulatory requirements: Obtain acceptance for proposed assemblies not conforming to specific rating agency classifications or rated assemblies.
- D. Materials shall have been tested to provide fire rating at least equal to that of the construction in which they are to be installed.
- E. VOC content: Field-applied adhesives and sealants, limits per South Coast Air Quality Management District (SCAQMD), Rule No. 1168.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Packing and shipping:
 - 1. Deliver products in original unopened packaging with legible manufacturer's identification.
 - 2. Coordinate delivery with scheduled installation date, allow minimum storage at site.
- B. Storage and protection: Store materials in a clean, dry, ventilated location. Protect from soiling, abuse, moisture and freezing when required. Follow manufacturer's instructions.

1.9 PROJECT CONDITIONS

A. Existing conditions:

1. Verify existing conditions and substrates before starting work. Correct unsatisfactory conditions before proceeding.
2. Proceed with installation only after penetrations of the substrate and supporting brackets have been installed.

B. Environmental requirements:

1. Furnish adequate ventilation if using solvent.
2. Furnish forced-air ventilation during installation if required by manufacturer.
3. Keep flammable materials away from sparks or flame.
4. Provide masking and drop cloths to prevent contamination of adjacent surfaces by firestopping materials.
5. Comply with manufacturing recommendations for temperature and humidity conditions before, during and after installation of firestopping.

1.10 WARRANTY

- A. General project warranty and correction period, as required in general conditions and Division 01, requires repair or replacement of materials or systems which fail in joint adhesion, co-adhesion, abrasion resistance, weather resistance, extrusion resistance, migration resistance, stain resistance, or general durability or appear to deteriorate in any other manner not clearly specified by submitted manufacturer's data as an inherent quality of the material for the exposure indicated.

PART 2 - PRODUCTS

2.1 AVAILABLE MANUFACTURERS

- A. Manufacturers and products: Those listed in the UL Fire Resistance Directory for the UL System involved, or rated for the application by Warnock Hersey or by another acceptable rating agency.

2.2 THROUGH-PENETRATION FIRESTOPPING OF FIRE-RATED CONSTRUCTION

- A. Provide systems or devices listed and labeled by a rating agency, and conforming to the construction type, penetrant type, annular space requirements and fire rating involved in each separate instance. The system shall be symmetrical for wall applications. Systems or devices shall be asbestos-free.
1. Additional requirements: Firestopping shall withstand the passage of cold smoke either as an inherent property of the system, or by the use of a separate product included as a part of the rated system or device, and designed to perform this function.
 2. Additional requirements: Firestopping sealants shall be red in color to facilitate field verification of firestopping application.

2.3 SMOKE-STOPPING AT SMOKE PARTITIONS

- A. Through-penetration smoke-stopping: Any system complying with the requirements for through-penetration firestopping in fire-rated construction is acceptable, provided that the system includes the specified smoke seal or will provide a smoke seal. The length of time of the fire resistance may be disregarded.

2.4 ACCESSORIES

- A. Fill, void or cavity materials and forming materials: Classified for firestopping use, or included in a rated firestopping assembly, by a rating agency.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Verification of conditions: Examine areas and conditions under which work is to be performed and identify conditions detrimental to proper or timely completion.
 - 1. Verify barrier penetrations are properly sized and in suitable condition for application of materials.
 - 2. Do not proceed until unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean surfaces to be in contact with penetration seal materials, of dirt, grease, oil, loose materials, rust, or other substances that may affect proper fitting, adhesion, or the required fire resistance.

3.3 INSTALLATION

- A. Provide firestop devices or assemblies for every opening and penetration in floors or fire-rated construction.
- B. Install penetration seal materials in accordance with printed instructions of the rating agency and in accordance with manufacturer's instruction.
- C. Ensure an effective smoke barrier in each sealed penetration. Install smoke stopping as specified for firestopping.
- D. Protect materials from damage on surfaces subject to traffic.
- E. Where large openings are created in walls or floors to permit installation of conduits, cables, or other items, close unused portions of opening with firestopping material tested for the application.

3.4 FIELD QUALITY CONTROL

- A. Examine penetration seals to ensure proper installation before concealing or enclosing them.
- B. Keep areas of work accessible until inspection and acceptance by applicable authorities.
- C. Before substantial completion, patch and repair firestopping cut or penetrated by other construction work.

3.5 ADJUSTING AND CLEANING

- A. Clean up spills of liquid components.
- B. Neatly cut and trim materials as required.
- C. Remove equipment, materials and debris, leaving area in undamaged, clean condition.

END OF SECTION

SECTION 26 0519 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Copper building wire rated 600 V or less.
 - 2. Metal-clad cable, Type MC, rated 600 V or less.
 - 3. Connectors, splices, and terminations rated 600 V and less.

1.3 DEFINITIONS

- A. RoHS: Restriction of Hazardous Substances.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of product.
- B. Product Schedule: Indicate type, use, location, and termination locations.

1.5 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

PART 2 - PRODUCTS

2.1 COPPER BUILDING WIRE

- A. Description: Flexible, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.
- B. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Alpha Wire Company.
 - 2. American Bare Conductor.
 - 3. General Cable Technologies Corporation.
 - 4. Okonite Company.
 - 5. Southwire Company.
- C. Standards:
 - 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
 - 2. RoHS compliant.

3. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."
 - D. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.
 - E. Conductor Insulation:
 1. Type THHN and Type THWN-2: Comply with UL 83.
 2. Type THW or Type THW-2: Comply with NEMA WC-70/ICEA S-95-658 and UL 83.
- 2.2 METAL-CLAD CABLE, TYPE MC
- A. Description: A factory assembly of one or more current-carrying insulated conductors in an overall metallic sheath.
 - B. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 1. Alpha Wire Company.
 2. American Bare Conductor.
 3. General Cable Technologies Corporation.
 4. Okonite Company.
 5. Southwire Company.
 - C. Standards:
 1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
 2. Comply with UL 1569.
 - D. Circuits: Single circuit and multicircuit with color-coded conductors.
 - E. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.
 - F. Ground Conductor: Insulated, Green.
 - G. Conductor Insulation:
 1. Type THHN/THWN-2: Comply with UL 83.
 - H. Armor: Steel or aluminum, interlocked.
 - I. Jacket: PVC applied over armor.
- 2.3 CONNECTORS AND SPLICES
- A. Description: Factory-fabricated connectors, splices, and lugs of size, ampacity rating, material, type, and class for application and service indicated; listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.

- B. Jacketed Cable Connectors: For steel and aluminum jacketed cables, zinc die-cast with set screws, designed to connect conductors specified in this Section.
- C. Lugs: One piece, seamless, designed to terminate conductors specified in this Section.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

- A. Feeders: Copper; solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.
- B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

- A. Exposed Feeders: Type THHN/THWN-2, single conductors in raceway.
- B. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspace: Type THHN/THWN-2, single conductors in raceway.
- C. Exposed Branch Circuits: Type THHN/THWN-2, single conductors in raceway.
- D. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN/THWN-2, single conductors in raceway.
 - 1. Metal-clad cable, Type MC cables: Acceptable for branch circuits serving administrative and public spaces only (such as offices, lobbies, and corridors). MC cabling is not allowed in laboratory or laboratory support spaces. MC cabling is not allowed in mechanical, electrical, or telecommunications spaces.
 - a. MC cabling shall not be used for circuiting homeruns to panelboards.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

- A. Conceal cables in finished walls, ceilings, and floors unless otherwise indicated.
- B. Complete raceway installation between conductor and cable termination points according to Section 26 0533 "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables.
- C. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.
- D. Use pulling means; including fish tape, cable, rope, and basket-weave wire/cable grips that will not damage cables or raceway.
- E. Install exposed cables parallel and perpendicular to surfaces of exposed structural members, and follow surface contours where possible.
- F. Support cables according to Section 26 0529 "Hangers and Supports for Electrical Systems."

3.4 CONNECTIONS

- A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.
- B. Make splices, terminations, and taps that are compatible with conductor material.
- C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches (150 mm) of slack.

3.5 IDENTIFICATION

- A. Identify and color-code conductors and cables according to Section 26 0553 "Identification for Electrical Systems."
- B. Identify each spare conductor at each end with identity number and location of other end of conductor, and identify as spare conductor.

3.6 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 26 0544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.7 FIRESTOPPING

- A. Apply firestopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly according to Section 26 0507 "Firestopping for Electrical Work."

3.8 FIELD QUALITY CONTROL

- A. Perform tests and inspections with the assistance of a factory-authorized service representative, where needed.
 - 1. After installing conductors and cables and before electrical circuitry has been energized, test feeder conductors and branch circuit conductors feeding the following critical equipment and services for compliance with requirements:
 - a. Feeders or branch circuits serving emergency or optional standby loads.
 - 2. Perform visual and electrical tests in accordance with NETA ATS Standards for Acceptance Testing.
- B. Cables will be considered defective if they do not pass tests and inspections.
- C. Prepare test and inspection reports to record the following:
 - 1. Procedures used.
 - 2. Results that comply with requirements.
 - 3. Results that do not comply with requirements, and corrective action taken to achieve compliance with requirements.

END OF SECTION

SECTION 26 0521 - WIRING CONNECTIONS

PART 1 - GENERAL

1.1 SECTION INCLUDES

- A. Power and control wiring for equipment.

1.2 RELATED SECTIONS

- A. Equipment: Installed items requiring electricity, specified in other sections or shown on drawings.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Conduits, wires and cables, devices, and accessories as specified in other sections.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide power wiring from the motor starters to each motor and its manual controlling device. Interlock and control wiring related to the automatic temperature control system.
 - 1. Make flexible or liquid tight connections as specified in Section 26 0533, Conduits.
- B. Except where provided with equipment, furnish and install manual pushbutton stations and pilot lights, with wiring. Where stations and pilot lights are grouped at central locations, mount them under a common faceplate.
- C. Rough in and connect to equipment furnished under other sections and equipment furnished by Owner. Make connections as indicated on drawings with exact locations and details determined by approved shop drawings of the equipment.
 - 1. Under equipment sections, equipment will be set in position and the electrical devices and components furnished loose. Assemble, install, and wire under this section.
 - 2. Accomplish rough-in from walls with flush outlet boxes and from floors by means of conduit couplings finishing flush with finished floor.
- D. Certain equipment, as indicated, will be furnished with control panels and auxiliary control components. Mount the panels, furnish and install source wiring and disconnects, and completely connect controls and motors.
- E. Provide source wiring, connections, and disconnects for mechanical heating, ventilating, and air-conditioning (HVAC) equipment specified in Division 23. Refer to sections of Division 23 for equipment and controls.
 - 1. Mount starters where required, and provide proper size overload protection.
- F. Where a Division 23 section requires installation of equipment under supervision of equipment manufacturer's representative, coordinate electrical installation to cooperate with representative's requirements.
- G. Provide power sources for Owner-furnished equipment.

- H. Provide power and control wiring for fire protection system pumps, controllers, alarm panels and alarm bell. Mount bells and silence switches where indicated. Provide plastic nameplates under bell and switch.

END OF SECTION

SECTION 26 0526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes grounding and bonding systems and equipment.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product indicated.

1.4 INFORMATIONAL SUBMITTALS

- A. Coordination Drawings: Plans showing dimensioned locations of grounding features specified in "Field Quality Control" Article.
- B. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For grounding to include in emergency, operation, and maintenance manuals.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

- A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
- B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 CONDUCTORS

- A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
- B. Bare Copper Conductors:
 - 1. Solid Conductors: ASTM B 3.
 - 2. Stranded Conductors: ASTM B 8.
 - 3. Tinned Conductors: ASTM B 33.

2.3 CONNECTORS

- A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.

- B. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.
- C. Beam Clamps: Mechanical type, terminal, ground wire access from four directions, with dual, tin-plated or silicon bronze bolts.
- D. Cable-to-Cable Connectors: Compression type, copper or copper alloy.
- E. Cable Tray Ground Clamp: Mechanical type, zinc-plated malleable iron.
- F. Conduit Hubs: Mechanical type, terminal with threaded hub.
- G. Signal Reference Grid Clamp: Mechanical type, stamped-steel terminal with hex head screw.
- H. Straps: Solid copper, cast-bronze clamp or copper lugs. Rated for 600 A.
- I. Tower Ground Clamps: Mechanical type, copper or copper alloy, terminal one or two-piece clamp.

PART 3 - EXECUTION

3.1 APPLICATIONS

- A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger unless otherwise indicated.
- B. Conductor Terminations and Connections:
 - 1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
 - 2. Connections to Structural Steel: Welded connectors.

3.2 EQUIPMENT GROUNDING

- A. Install insulated equipment grounding conductors with all feeders and branch circuits.
- B. Air-Duct Equipment Circuits: Install insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping.

3.3 INSTALLATION

- A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.
- B. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.
 - 1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
 - 2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.

- C. Bonding Interior Metal Ducts: Bond metal air ducts to equipment grounding conductors of associated fans, blowers, electric heaters, and air cleaners. Install tinned bonding jumper to bond across flexible duct connections to achieve continuity.
- D. Connections: Make connections so possibility of galvanic action or electrolysis is minimized. Select connectors, connection hardware, conductors, and connection methods so metals in direct contact are galvanically compatible.
 - 1. Use electroplated or hot-tin-coated materials to ensure high conductivity and to make contact points closer in order of galvanic series.
 - 2. Make connections with clean, bare metal at points of contact.
 - 3. Make aluminum-to-steel connections with stainless-steel separators and mechanical clamps.
 - 4. Make aluminum-to-galvanized-steel connections with tin-plated copper jumpers and mechanical clamps.
 - 5. Coat and seal connections having dissimilar metals with inert material to prevent future penetration of moisture to contact surfaces.

3.4 FIELD QUALITY CONTROL

- A. Perform tests and inspections with the assistance of a factory-authorized service representative, where needed.
- B. Tests and Inspections:
 - 1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
 - 2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
 - 3. Test completed grounding system at each location where a maximum ground-resistance level is specified, and at service disconnect enclosure grounding terminal. Make tests at ground rods before any conductors are connected.
 - a. Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
 - b. Perform tests by fall-of-potential method according to IEEE 81.
- C. Grounding system will be considered defective if it does not pass tests and inspections.
- D. Prepare test and inspection reports.
- E. Report measured ground resistances that exceed the following values:
 - 1. Service with Capacity of 2500 kVA and Less: 5 ohms.
 - 2. Service with Capacity More Than 2500 kVA: 1 ohm.
- F. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Project Officer promptly and include recommendations to reduce ground resistance.

END OF SECTION

SECTION 26 0529 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

1. Steel slotted support systems.
2. Aluminum slotted support systems.
3. Conduit and cable support devices.
4. Mounting, anchoring, and attachment components, including mechanical expansion anchors, concrete inserts, clamps, through bolts, toggle bolts, and hanger rods.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for the following:
 - a. Slotted support systems, hardware, and accessories.
 - b. Clamps.
 - c. Hangers.
 - d. Sockets.
 - e. Eye nuts.
 - f. Fasteners.
 - g. Anchors.
 - h. Saddles.
 - i. Brackets.
2. Include rated capacities and furnished specialties and accessories.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

- A. Steel Slotted Support Systems: Preformed steel channels and angles with minimum 13/32-inch- (10-mm-) diameter holes at a maximum of 8 inches (200 mm) o.c. in at least one surface.
 1. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Allied Tube & Conduit
 - b. B-line; an Eaton business
 - c. ERICO International Corporation
 - d. Flex-Strut Inc.
 - e. Thomas & Betts Corporation
 - f. Unistrut; an Atkore International business

2. Standard: Comply with MFMA-4 factory-fabricated components for field assembly.
 3. Material for Channel, Fittings, and Accessories: Galvanized steel.
 4. Channel Width: Selected for applicable load criteria.
 5. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
 6. Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
- B. Aluminum Slotted Support Systems: Extruded-aluminum channels and angles with minimum 13/32-inch- (10-mm-) diameter holes at a maximum of 8 inches (200 mm) o.c. in at least one surface.
1. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Allied Tube & Conduit
 - b. B-line; an Eaton business
 - c. ERICO International Corporation
 - d. Flex-Strut Inc.
 - e. Thomas & Betts Corporation
 - f. Unistrut; an Atkore International business
 2. Standard: Comply with MFMA-4 factory-fabricated components for field assembly.
 3. Channel Material: 6063-T5 aluminum alloy.
 4. Fittings and Accessories Material: 5052-H32 aluminum alloy.
 5. Channel Width: Selected for applicable load criteria.
- C. Conduit and Cable Support Devices: Steel or stainless-steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.
- D. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:
1. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated or stainless steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
 2. Concrete Inserts: Steel or malleable-iron, slotted support system units are similar to MSS Type 18 units and comply with MFMA-4 or MSS SP-58.
 3. Clamps for Attachment to Steel Structural Elements: MSS SP-58 units are suitable for attached structural element.
 4. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A 325.
 5. Toggle Bolts: Stainless-steel springhead type.
 6. Hanger Rods: Threaded steel.

PART 3 - EXECUTION

3.1 APPLICATION

- A. Comply with the following standards for application and installation requirements of hangers and supports, except where requirements on Drawings or in this Section are stricter:
1. NECA 1.
 2. NECA 101
 3. NECA 102.
 4. NECA 105.

- B. Comply with requirements in Section 26 0507 "Firestopping for Electrical Work" for firestopping materials and installation for penetrations through fire-rated walls, ceilings, and assemblies.
- C. Comply with requirements for raceways and boxes specified in Section 26 0533 "Raceways and Boxes for Electrical Systems."
- D. Maximum Support Spacing and Minimum Hanger Rod Size for Raceways: Space supports for EMT, IMC, and RMC as scheduled in NECA 1, where its Table 1 lists maximum spacings that are less than those stated in NFPA 70. Minimum rod size shall be 1/4 inch (6 mm) in diameter.
- E. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.
 - 1. Secure raceways and cables to these supports with two-bolt conduit clamps.

3.2 SUPPORT INSTALLATION

- A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this article.
- B. Raceway Support Methods: In addition to methods described in NECA 1, metallic conduit may be supported by openings through structure members, according to NFPA 70.
- C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb (90 kg).
- D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:
 - 1. To Wood: Fasten with lag screws or through bolts.
 - 2. To New Concrete: Bolt to concrete inserts.
 - 3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
 - 4. To Existing Concrete: Expansion anchor fasteners.
 - 5. To Steel: Beam clamps (MSS SP-58, Type 19, 21, 23, 25, or 27), complying with MSS SP-69.
 - 6. To Light Steel: Sheet metal screws.
 - 7. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate.
- E. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

3.3 PAINTING

- A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.
 - 1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils (0.05 mm).

- B. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION

SECTION 26 0533 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Conduits and fittings.
 - 2. Boxes, enclosures, and cabinets.
- B. Related Requirements:
 - 1. Section 26 0507 "Firestopping for Electrical Work" for firestopping at conduit and box entrances.

1.3 DEFINITIONS

- A. GRC: Galvanized rigid steel conduit.
- B. IMC: Intermediate metal conduit.
- C. EMT: Electrical metallic tubing.
- D. PVC: Polyvinyl chloride.

1.4 ACTION SUBMITTALS

- A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

PART 2 - PRODUCTS

2.1 CONDUITS AND FITTINGS

- A. Metal Conduit:
 - 1. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. AFC Cable Systems; a part of Atkore International (FMC and LFMC)
 - b. Allied Tube & Conduit; a part of Atkore International
 - c. O-Z/Gedney; a brand of Emerson Industrial Automation
 - d. Republic Conduit
 - e. Wheatland Tube Company; a division of Zekelman Industries
 - 2. Listing and Labeling: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

3. GRC: Comply with ANSI C80.1 and UL 6.
4. IMC: Comply with ANSI C80.6 and UL 1242.
5. EMT: Comply with ANSI C80.3 and UL 797.
6. FMC: Comply with UL 1; zinc-coated steel.
7. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.

B. Non-metallic conduit, tubing and fittings:

1. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Allied Tube & Conduit; a Tyco International Ltd. Co.
 - b. Arnco Corp.
 - c. Beck Manufacturing.
 - d. CANTEX Inc.
 - e. Certainteed Corp.; Pipe and Plastics Group.
 - f. Lamson & Sessions; Carlon Electrical Products.
2. Non-metallic conduit: Polyvinyl chloride (PVC) Schedule 40, rated for use with 90-degree conductors, for exposed, underground, and encased applications, complying with NEMA Specification TC-2 and UL 651.
3. PVC conduit fittings, solvents and adhesives:
 - a. Fittings: Complying with NEMA TC 3 and UL 514.
 - b. Solvents and adhesives: As recommended by conduit manufacturer.

C. Metal Fittings:

1. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Appleton Electric Co.; a brand of Emerson Industrial Automation
 - b. Cooper Crouse-Hinds; an Eaton Corporation
 - c. Hubbell Incorporated
 - d. O-Z/Gedney; a brand of Emerson Industrial Automation
 - e. Thomas & Betts; a member of the ABB Group.
2. Comply with NEMA FB 1 and UL 514B.
3. Listing and Labeling: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
4. Fittings, General: Listed and labeled for type of conduit, location, and use.
5. Fittings for EMT:
 - a. Material: Steel.
 - b. Type: Compression.
6. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.

- D. Joint Compound for IMC or GRC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 BOXES, ENCLOSURES, AND CABINETS

- A. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - 1. Appleton Electric Co.; a brand of Emerson Industrial Automation
 - 2. Crouse-Hinds; an Eaton Corporation
 - 3. Hoffman; a division of Pentair
 - 4. Rayco; Hubbell Incorporated
 - 5. Steel City / Carlon; Thomas & Betts; a member of the ABB Group
 - 6. Square D; Schneider Electric
- B. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.
- C. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.
- D. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, aluminum, Type FD, with gasketed cover.
- E. Luminaire Outlet Boxes: Nonadjustable, designed for attachment of luminaire weighing 50 lb (23 kg). Outlet boxes designed for attachment of luminaires weighing more than 50 lb (23 kg) shall be listed and marked for the maximum allowable weight.
- F. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
- G. Cast-Metal Access, Pull, and Junction Boxes: Comply with NEMA FB 1 and UL 1773, cast aluminum with gasketed cover.
- H. Box extensions used to accommodate new building finishes shall be of same material as recessed box.
- I. Gangable boxes are allowed.
- J. Single Gang Mud Ring: Galvanized steel.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

- A. Indoors: Apply raceway products as specified below unless otherwise indicated:
 - 1. Exposed, Not Subject to Physical Damage: EMT.
 - 2. Exposed and Subject to Physical Damage: GRC.
 - 3. Concealed in Ceilings and Interior Walls and Partitions: EMT.
 - 4. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.

5. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 4 stainless steel in institutional and commercial kitchens and damp or wet locations.
- B. Outdoors: Apply raceway products as specified below unless otherwise indicated:
1. Direct-buried or concrete-encased: PVC.
- C. Minimum Raceway Size: 3/4-inch (21-mm) trade size.
- D. Raceway Fittings: Compatible with raceways and suitable for use and location.
1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
 2. EMT: Use compression, steel fittings. Comply with NEMA FB 2.10.
 3. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.
- E. Install nonferrous conduit or tubing for circuits operating above 60 Hz. Where aluminum raceways are installed for such circuits and pass through concrete, install in nonmetallic sleeve.
- F. Do not install aluminum conduits or non-metallic conduit unless indicated on drawings.
- G. Install surface raceways only where indicated on Drawings.
- 3.2 INSTALLATION
- A. Comply with requirements in Section 26 0529 "Hangers and Supports for Electrical Systems" for hangers and supports.
- B. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NECA 102 for aluminum conduits. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.
- C. Do not install raceways or electrical items on any "explosion-relief" walls or rotating equipment.
- D. Do not fasten conduits onto the bottom side of a metal deck roof.
- E. Keep raceways at least 6 inches (150 mm) away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.
- F. Complete raceway installation before starting conductor installation.
- G. Arrange stub-ups so curved portions of bends are not visible above finished slab.
- H. Install no more than the equivalent of three 90-degree bends in any conduit run except for control wiring conduits, for which fewer bends are allowed. Support within 12 inches (300 mm) of changes in direction.
- I. Make bends in raceway using large-radius preformed ells. Field bending shall be according to NFPA 70 minimum radii requirements. Use only equipment specifically designed for material and size involved.

- J. Conceal conduit within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.
- K. Support conduit within 12 inches (300 mm) of enclosures to which attached.
- L. Stub-Ups to Above Recessed Ceilings:
 - 1. Use EMT, IMC, or RMC for raceways.
 - 2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.
- M. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.
- N. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch (35mm) trade size and insulated throat metal bushings on 1-1/2-inch (41-mm) trade size and larger conduits terminated with locknuts.
- O. Install raceways square to the enclosure and terminate at enclosures with locknuts. Install locknuts hand tight plus 1/4 turn more.
- P. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure to assure a continuous ground path.
- Q. Cut conduit perpendicular to the length. For conduits 2-inch (53-mm) trade size and larger, use roll cutter or a guide to make cut straight and perpendicular to the length.
- R. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb (90-kg) tensile strength. Leave at least 12 inches (300 mm) of slack at each end of pull wire. Cap underground raceways designated as spare above grade alongside raceways in use.
- S. Install raceway sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings according to NFPA 70.
- T. Install devices to seal raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all raceways at the following points:
 - 1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
 - 2. Conduit extending into pressurized duct and equipment.
 - 3. Conduit extending into pressurized zones that are automatically controlled to maintain different pressure set points.
 - 4. Where otherwise required by NFPA 70.
- U. Comply with manufacturer's written instructions for solvent welding RNC and fittings.
- V. Expansion-Joint Fittings:

1. Install in each run of aboveground RNC that is located where environmental temperature change may exceed 30 deg F (17 deg C) and that has straight-run length that exceeds 25 feet (7.6 m). Install in each run of aboveground RMC and EMT conduit that is located where environmental temperature change may exceed 100 deg F (55 deg C) and that has straight-run length that exceeds 100 feet (30 m).
- W. Flexible Conduit Connections: Comply with NEMA RV 3. Use a maximum of 72 inches (1830 mm) of flexible conduit for recessed and semirecessed luminaires, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.
1. Use LFMC in damp or wet locations.
- X. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to top of box unless otherwise indicated.
- Y. Horizontally separate boxes mounted on opposite sides of walls so they are not in the same vertical channel.
- Z. Locate boxes so that cover or plate will not span different building finishes.
- AA. Support boxes from more than one side by spanning two framing members or mounting on brackets specifically designed for the purpose.
- BB. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.
- 3.3 INSTALLING UNDERGROUND CONDUIT WITHOUT CONCRETE ENCASEMENT
- A. Run conduit in straight lines except as necessary.
 - B. Trenches: At least 3-inch clearance on each side of the conduit.
 - C. Warning tape: Install in backfill approximately 12 inches below grade.
 - D. Under existing roads and paved areas not to be disturbed, jack rigid steel conduit into place.
- 3.4 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS
- A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 26 0544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."
- 3.5 FIRESTOPPING
- A. Install firestopping at penetrations of fire-rated floor and wall assemblies. Comply with requirements in Section 26 0507 "Firestopping for Electrical Work."
- 3.6 PROTECTION
- A. Protect coatings, finishes, and cabinets from damage and deterioration.
 1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.

2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION

SECTION 26 0544 - SLEEVES AND SLEEVE SEALS FOR ELECTRICAL RACEWAYS AND CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
2. Grout.
3. Silicone sealants.

B. Related Requirements:

1. Section 26 0507 "Firestopping for Electrical Work" for penetration firestopping installed in fire-resistance-rated walls, horizontal assemblies, and smoke barriers, with and without penetrating items.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Wall Sleeves:

1. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, plain ends.
2. Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.

- B. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch (0.6-mm) minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.

C. Sleeves for Rectangular Openings:

1. Material: Galvanized sheet steel.
2. Minimum Metal Thickness:
 - a. For sleeve cross-section rectangle perimeter less than 50 inches (1270 mm) and with no side larger than 16 inches (400 mm), thickness shall be 0.052 inch (1.3 mm).
 - b. For sleeve cross-section rectangle perimeter 50 inches (1270 mm) or more and one or more sides larger than 16 inches (400 mm), thickness shall be 0.138 inch (3.5 mm).

2.2 GROUT

- A. Description: Nonshrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.
- B. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
- C. Design Mix: 5000-psi (34.5-MPa), 28-day compressive strength.
- D. Packaging: Premixed and factory packaged.

2.3 SILICONE SEALANTS

- A. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.
 - 1. Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.
- B. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

- A. Comply with NECA 1.
- B. Comply with NEMA VE 2 for cable tray and cable penetrations.
- C. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:
 - 1. Interior Penetrations of Non-Fire-Rated Walls and Floors:
 - a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Comply with requirements in Division 07 for joint sealants.
 - b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.
 - 2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
 - 3. Size pipe sleeves to provide 1/4-inch (6.4-mm) minimum annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed.
 - 4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.
- D. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:
 - 1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.

2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.

END OF SECTION

SECTION 26 0553 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:

1. Color and legend requirements for raceways, conductors, and warning labels and signs.
2. Labels.
3. Tapes and stencils.
4. Signs.
5. Cable ties.
6. Paint for identification.
7. Fasteners for labels and signs.

1.3 ACTION SUBMITTALS

- A. Product Data: For each type of product.
 1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for electrical identification products.
- B. Identification Schedule: For each piece of electrical equipment and electrical system components to be an index of nomenclature for electrical equipment and system components used in identification signs and labels. Use same designations indicated on Drawings.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

- A. Comply with NFPA 70.
- B. Comply with 29 CFR 1910.144 and 29 CFR 1910.145.
- C. Comply with ANSI Z535.4 for safety signs and labels.
- D. Adhesive-attached labeling materials, including label stocks, laminating adhesives, and inks used by label printers, shall comply with UL 969.
- E. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

2.2 COLOR AND LEGEND REQUIREMENTS

- A. Raceways and Cables Carrying Circuits at 600 V or Less:
 1. Black letters on an orange field.
 2. Legend: Indicate voltage and system or service type.

- B. Color-Coding for Phase- and Voltage-Level Identification, 600 V or Less: Use colors listed below for ungrounded service, feeder, and branch-circuit conductors.
 - 1. Color shall be factory applied or field applied for sizes larger than No. 8 AWG.
 - 2. Colors for 208/120-V Circuits:
 - a. Phase A: Black.
 - b. Phase B: Red.
 - c. Phase C: Blue.
 - d. Neutral: White.
 - 3. Colors for 480/277-V Circuits:
 - a. Phase A: Brown.
 - b. Phase B: Orange.
 - c. Phase C: Yellow.
 - d. Neutral: Grey.
 - 4. Color for Equipment Grounds: Green.
- C. Warning Label Colors:
 - 1. Identify system voltage with black letters on an orange background.
- D. Warning labels and signs shall include, but are not limited to, the following legends:
 - 1. Workspace Clearance Warning: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES (915 MM)."
- E. Equipment Identification Labels:
 - 1. White letters on a black field.

2.3 LABELS

- A. Vinyl Wraparound Labels: Preprinted, flexible labels laminated with a clear, weather- and chemical-resistant coating and matching wraparound clear adhesive tape for securing label ends.
- B. Self-Adhesive Wraparound Labels: Preprinted, 3-mil- (0.08-mm-) thick, vinyl flexible label with acrylic pressure-sensitive adhesive.
 - 1. Self-Lamination: Clear; UV-, weather- and chemical-resistant; self-laminating, protective shield over the legend. Labels sized such that the clear shield overlaps the entire printed legend.
 - 2. Marker for Labels: Machine-printed, permanent, waterproof, black ink recommended by printer manufacturer.
- C. Self-Adhesive Labels: Vinyl, thermal, transfer-printed, 3-mil- (0.08-mm-) thick, multicolor, weather- and UV-resistant, pressure-sensitive adhesive labels, configured for intended use and location.
 - 1. Nominal Size:
 - a. 2-1/4 by 9 inches for raceways.
 - b. 3-1/2 by 5 inches for equipment.
 - c. As required by authorities having jurisdiction.

2.4 TAPES AND STENCILS

- A. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.
- B. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; not less than 3 mils (0.08 mm) thick by 1 to 2 inches (25 to 50 mm) wide; compounded for outdoor use.
- C. Tape and Stencil: 4-inch- (100-mm-) wide black stripes on 10-inch (250-mm) centers placed diagonally over orange background and are 12 inches (300 mm) wide. Stop stripes at legends.
- D. Floor Marking Tape: 2-inch- (50-mm-) wide, 5-mil (0.125-mm) pressure-sensitive vinyl tape, with black and white stripes and clear vinyl overlay.
- E. Stenciled Legend: In nonfading, waterproof, black ink or paint. Minimum letter height shall be 1 inch (25 mm).

2.5 SIGNS

- A. Baked-Enamel Signs:
 - 1. Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application.
 - 2. 1/4-inch (6.4-mm) grommets in corners for mounting.
 - 3. Nominal Size: 7 by 10 inches (180 by 250 mm).
- B. Metal-Backed Butyrate Signs:
 - 1. Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs, with 0.0396-inch (1-mm) galvanized-steel backing, punched and drilled for fasteners, and with colors, legend, and size required for application.
 - 2. 1/4-inch (6.4-mm) grommets in corners for mounting.
 - 3. Nominal Size: 10 by 14 inches (250 by 360 mm).
- C. Laminated Acrylic or Melamine Plastic Signs:
 - 1. Engraved legend.
 - 2. Thickness:
 - a. For signs up to 20 sq. in. (129 sq. cm), minimum 1/16 inch (1.6 mm) thick.
 - b. For signs larger than 20 sq. in. (129 sq. cm), 1/8 inch (3.2 mm) thick.
 - c. Engraved legend with white letters on black face.
 - d. Punched or drilled for mechanical fasteners with 1/4-inch (6.4-mm) grommets in corners for mounting.
 - e. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2.6 CABLE TIES

- A. General-Purpose Cable Ties: Fungus inert, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.
 - 1. Minimum Width: 3/16 inch (5 mm).
 - 2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 12,000 psi (82.7 MPa).
 - 3. Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
 - 4. Color: Black, except where used for color-coding.

- B. UV-Stabilized Cable Ties: Fungus inert, designed for continuous exposure to exterior sunlight, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.
 - 1. Minimum Width: 3/16 inch (5 mm).
 - 2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 12,000 psi (82.7 MPa).
 - 3. Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
 - 4. Color: Black.
- C. Plenum-Rated Cable Ties: Self-extinguishing, UV stabilized, one piece, and self-locking.
 - 1. Minimum Width: 3/16 inch (5 mm).
 - 2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 7000 psi (48.2 MPa).
 - 3. UL 94 Flame Rating: 94V-0.
 - 4. Temperature Range: Minus 50 to plus 284 deg F (Minus 46 to plus 140 deg C).
 - 5. Color: Black.

2.7 MISCELLANEOUS IDENTIFICATION PRODUCTS

- A. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Retain paint system applicable for surface material and location (exterior or interior).
- B. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Self-Adhesive Identification Products: Before applying electrical identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product.

3.2 INSTALLATION

- A. Verify and coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and operation and maintenance manual. Use consistent designations throughout Project.
- B. Install identifying devices before installing acoustical ceilings and similar concealment.
- C. Verify identity of each item before installing identification products.
- D. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and operation and maintenance manual.
- E. Apply identification devices to surfaces that require finish after completing finish work.
- F. Install signs with approved legend to facilitate proper identification, operation, and maintenance of electrical systems and connected items.
- G. System Identification for Raceways and Cables under 600 V: Identification shall completely encircle cable or conduit. Place identification of two-color markings in contact, side by side.
 - 1. Secure tight to surface of conductor, cable, or raceway.

- H. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, and signal connections.
- I. Elevated Components: Increase sizes of labels, signs, and letters to those appropriate for viewing from the floor.
- J. Accessible Fittings for Raceways: Identify the covers of each junction and pull box of the following systems with the wiring system legend, system voltage, panel source and circuit number. System legends shall be as follows:
 - 1. "EMERGENCY POWER."
 - 2. "POWER."
 - 3. "COMMUNICATIONS."
 - 4. "SECURITY."
 - 5. "FIRE ALARM."
- K. Vinyl Wraparound Labels:
 - 1. Secure tight to surface of raceway or cable at a location with high visibility and accessibility.
 - 2. Attach labels that are not self-adhesive type with clear vinyl tape, with adhesive appropriate to the location and substrate.
- L. Self-Adhesive Wraparound Labels: Secure tight to surface at a location with high visibility and accessibility.
- M. Self-Adhesive Labels:
 - 1. On each item, install unique designation label that is consistent with wiring diagrams, schedules, and operation and maintenance manual.
 - 2. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high label; where two lines of text are required, use labels 2 inches (50 mm) high.
- N. Marker Tapes: Secure tight to surface at a location with high visibility and accessibility.
- O. Self-Adhesive Vinyl Tape: Secure tight to surface at a location with high visibility and accessibility.
 - 1. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6 inches (150 mm) where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding.
- P. Tape and Stencil: Comply with requirements in painting Sections for surface preparation and paint application.
- Q. Floor Marking Tape: Apply stripes to finished surfaces following manufacturer's written instructions.
- R. Baked-Enamel Signs:
 - 1. Attach signs that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.
 - 2. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on minimum 1-1/2-inch- (38-mm-) high sign; where two lines of text are required, use signs minimum 2 inches (50 mm) high.

S. Metal-Backed Butyrate Signs:

1. Attach signs that are not self-adhesive type with stainless steel screws.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high sign; where two lines of text are required, use labels 2 inches (50 mm) high.

T. Laminated Acrylic or Melamine Plastic Signs:

1. Attach signs that are not self-adhesive type with stainless steel screws.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high sign; where two lines of text are required, use labels 2 inches (50 mm) high.

U. Cable Ties: General purpose, for attaching tags, except as listed below:

1. Outdoors: UV-stabilized nylon.
2. In Spaces Handling Environmental Air: Plenum rated.

3.3 IDENTIFICATION SCHEDULE

- A. Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment. Install access doors or panels to provide view of identifying devices.
- B. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, pull points, and locations of high visibility. Identify by system and circuit designation.
- C. Accessible Raceways and Metal-Clad Cables, 600 V or Less, for Service, Feeder, and Branch Circuits, More Than 30 A and 120 V to Ground: Identify with self-adhesive raceway labels.
 1. Locate identification at changes in direction, at penetrations of walls and floors, at 50-foot (15-m) maximum intervals in straight runs, and at 25-foot (7.6-m) maximum intervals in congested areas.
- D. Accessible Fittings for Raceways and Cables within Buildings: Identify the covers of each junction and pull box of the following systems with self-adhesive labels containing the wiring system legend and system voltage. System legends shall be as follows:
 1. "EMERGENCY POWER."
 2. "POWER."
- E. Control-Circuit Conductor Identification: For conductors and cables in pull and junction boxes, use self-adhesive labels with the conductor or cable designation, origin, and destination.
- F. Control-Circuit Conductor Termination Identification: For identification at terminations, provide self-adhesive labels with the conductor designation.
- G. Auxiliary Electrical Systems Conductor Identification: Self-adhesive vinyl tape that is uniform and consistent with system used by manufacturer for factory-installed connections.
 1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.

- H. Workspace Indication: Apply floor marking tape or tape and stencil to finished surfaces. Show working clearances in the direction of access to live parts. Workspace shall comply with NFPA 70 and 29 CFR 1926.403 unless otherwise indicated. Do not install at flush-mounted panelboards and similar equipment in finished spaces.
- I. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Self-adhesive labels.
 - 1. Apply to exterior of door, cover, or other access.
- J. Equipment Identification Labels:
 - 1. Indoor Equipment: Laminated acrylic or melamine plastic sign.
 - 2. Equipment to Be Labeled:
 - 3. Nameplates: Electrical equipment shall have nameplates with NIH Facility Numbers (FAC #). Coordinate exact NIH Facility Number with the Project Officer.
- K. Circuit Identification:
 - 1. Provide printed laminated tape or engraved or labels on cover plates of wiring devices, including receptacles, light switches, junction boxes and pull boxes, identifying the panelboard source and circuit number. In addition, provide phase sequence for the receptacles installed in the SMR.
 - 2. Receptacles for computers and printers shall be provided with engraved cover plates or nameplates identifying their specific use riveted to the cover plate.
 - 3. Identify wiring devices according to power source, i.e., normal power or emergency power.
 - 4. Wiring devices on emergency power shall be red in color.

END OF SECTION

SECTION 26 2816**ENCLOSED SWITCHES AND CIRCUIT BREAKERS****PART 1 - GENERAL****1.1 RELATED DOCUMENTS**

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Molded-case circuit breakers (MCCBs)

1.3 DEFINITIONS

- A. NC: Normally closed.
- B. NO: Normally open.
- C. SPDT: Single pole, double throw.

1.4 ACTION SUBMITTALS

- A. Product Data: For each type of enclosed switch, circuit breaker, accessory, and component indicated. Include nameplate ratings, dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.
 - 1. Enclosure types and details for types other than NEMA 250, Type 1.
 - 2. Current and voltage ratings.
 - 3. Short-circuit current ratings (interrupting and withstand, as appropriate).
 - 4. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices, accessories, and auxiliary components.
- B. Shop Drawings: For enclosed switches and circuit breakers.
 - 1. Include plans, elevations, sections, details, and attachments to other work.
 - 2. Include wiring diagrams for power, signal, and control wiring.

1.5 INFORMATIONAL SUBMITTALS

- A. Field quality-control reports.

1.6 CLOSEOUT SUBMITTALS

- A. Operation and Maintenance Data: For enclosed switches and circuit breakers to include in emergency, operation, and maintenance manuals.

1.7 WARRANTY

- A. Manufacturer's Warranty: Manufacturer and Installer agree to repair or replace components that fail in materials or workmanship within specified warranty period.

1. Warranty Period: One year from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS

- A. Source Limitations: Obtain enclosed switches and circuit breakers, overcurrent protective devices, components, and accessories, within same product category, from single manufacturer.
- B. Available Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
1. Eaton Corporation; Cutler-Hammer Products
 2. General Electric Company
 3. Siemens Energy & Automation, Inc.
 4. Square D Company (Schneider Electric)
- C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches and circuit breakers, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.
- D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
- E. Comply with NFPA 70.

2.2 MOLDED-CASE CIRCUIT BREAKERS

- A. Circuit breakers shall be constructed using glass-reinforced insulating material. Current carrying components shall be completely isolated from the handle and the accessory mounting area.
- B. Circuit breakers shall have a toggle operating mechanism with common tripping of all poles, which provides quick-make, quick-break contact action. The circuit-breaker handle shall be over center, be trip free, and reside in a tripped position between on and off to provide local trip indication. Circuit-breaker escutcheon shall be clearly marked on and off in addition to providing international I/O markings. Equip circuit breaker with a push-to-trip button, located on the face of the circuit breaker to mechanically operate the circuit-breaker tripping mechanism for maintenance and testing purposes.
- C. The maximum ampere rating and UL, IEC, or other certification standards with applicable voltage systems and corresponding interrupting ratings shall be clearly marked on face of circuit breaker. Circuit breakers shall be 100 percent rated.
- D. MCCBs shall be equipped with a device for locking in the isolated position.
- E. Lugs shall be suitable for 167 deg F (75 deg C) rated wire.
- F. Standard: Comply with UL 489 with interrupting capacity to comply with available fault currents.

- G. Thermal-Magnetic Circuit Breakers: Inverse time-current thermal element for low-level overloads and instantaneous magnetic trip element for short circuits. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
- H. Adjustable, Instantaneous-Trip Circuit Breakers: Magnetic trip element with front-mounted, field-adjustable trip setting.
- I. Electronic Trip Circuit Breakers: Field-replaceable rating plug, rms sensing, with the following field-adjustable settings:
 - 1. Instantaneous trip.
 - 2. Long- and short-time pickup levels.
 - 3. Long- and short-time time adjustments.
 - 4. Ground-fault pickup level, time delay, and I-squared t response.
- J. Features and Accessories:
 - 1. Standard frame sizes, trip ratings, and number of poles.
 - 2. Lugs: Mechanical type, suitable for number, size, trip ratings, and conductor material.
 - 3. Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and high-intensity discharge lighting circuits.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine elements and surfaces to receive enclosed switches and circuit breakers for compliance with installation tolerances and other conditions affecting performance of the Work.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
 - 1. Commencement of work shall indicate Installer's acceptance of the areas and conditions as satisfactory.

3.2 PREPARATION

- A. Interruption of Existing Electric Service: Do not interrupt electric service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electric service according to requirements indicated:
 - 1. Notify Project Officer no fewer than seven days in advance of proposed interruption of electric service.
 - 2. Indicate method of providing temporary electric service.
 - 3. Do not proceed with interruption of electric service without Project Officer's written permission.
 - 4. Comply with NFPA 70E.

3.3 ENCLOSURE ENVIRONMENTAL RATING APPLICATIONS

- A. Enclosed Switches and Circuit Breakers: Provide enclosures at installed locations with the following environmental ratings.
 - 1. Indoor, Dry and Clean Locations: NEMA 250, Type 1.

3.4 INSTALLATION

- A. Coordinate layout and installation of switches, circuit breakers, and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.
- B. Install individual wall-mounted switches and enclosed circuit breakers with tops at uniform height unless otherwise indicated.
- C. Temporary Lifting Provisions: Remove temporary lifting of eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.
- D. Install fuses in fusible devices.
- E. Comply with NFPA 70 and NECA 1.

3.5 IDENTIFICATION

- A. Comply with requirements in Section 26 0553 "Identification for Electrical Systems."
 - 1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
 - 2. Label each enclosure with engraved metal or laminated-plastic nameplate.

3.6 FIELD QUALITY CONTROL

- A. Perform tests and inspections.
- B. Tests and Inspections for Switches:
 - 1. Visual and Mechanical Inspection:
 - a. Inspect physical and mechanical condition.
 - b. Inspect anchorage, alignment, grounding, and clearances.
 - c. Verify that the unit is clean.
 - d. Verify blade alignment, blade penetration, travel stops, and mechanical operation.
 - e. Verify that fuse sizes and types match the Specifications and Drawings.
 - f. Verify that each fuse has adequate mechanical support and contact integrity.
 - g. Inspect bolted electrical connections for high resistance using one of the two following methods:
 - 1) Use a low-resistance ohmmeter.
 - a) Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.
 - 2) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or NETA ATS Table 100.12.

- a) Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use NETA ATS Table 100.12.
 - h. Verify that operation and sequencing of interlocking systems is as described in the Specifications and shown on the Drawings.
 - i. Verify correct phase barrier installation.
 - j. Verify lubrication of moving current-carrying parts and moving and sliding surfaces.
2. Electrical Tests:

- a. Perform resistance measurements through bolted connections with a low-resistance ohmmeter. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
- b. Measure contact resistance across each switchblade fuseholder. Drop values shall not exceed the high level of the manufacturer's published data. If manufacturer's published data are not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
- c. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with switch closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1 from the NETA ATS. Investigate values of insulation resistance less than those published in Table 100.1 or as recommended in manufacturer's published data.
- d. Measure fuse resistance. Investigate fuse-resistance values that deviate from each other by more than 15 percent.
- e. Perform ground fault test according to NETA ATS 7.14 "Ground Fault Protection Systems, Low-Voltage."

C. Tests and Inspections for Molded Case Circuit Breakers:

1. Visual and Mechanical Inspection:

- a. Verify that equipment nameplate data are as described in the Specifications and shown on the Drawings.
- b. Inspect physical and mechanical condition.
- c. Inspect anchorage, alignment, grounding, and clearances.
- d. Verify that the unit is clean.
- e. Operate the circuit breaker to ensure smooth operation.
- f. Inspect bolted electrical connections for high resistance using one of the two following methods:
 - 1) Use a low-resistance ohmmeter.
 - a) Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from those of similar bolted connections by more than 50 percent of the lowest value.
 - 2) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or NETA ATS Table 100.12.

- a) Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use NETA ATS Table 100.12.
 - g. Inspect operating mechanism, contacts, and chutes in unsealed units.
 - h. Perform adjustments for final protective device settings in accordance with the coordination study.
- 2. Electrical Tests:
 - a. Perform resistance measurements through bolted connections with a low-resistance ohmmeter. Compare bolted connection resistance values to values of similar connections. Investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - b. Perform insulation-resistance tests for one minute on each pole, phase-to-phase and phase-to-ground with circuit breaker closed, and across each open pole. Apply voltage in accordance with manufacturer's published data. In the absence of manufacturer's published data, use Table 100.1 from the NETA ATS. Investigate values of insulation resistance less than those published in Table 100.1 or as recommended in manufacturer's published data.
 - c. Perform a contact/pole resistance test. Drop values shall not exceed the high level of the manufacturer's published data. If manufacturer's published data are not available, investigate values that deviate from adjacent poles or similar switches by more than 50 percent of the lowest value.
 - d. Perform insulation resistance tests on all control wiring with respect to ground. Applied potential shall be 500-V dc for 300-V rated cable and 1000-V dc for 600-V rated cable. Test duration shall be one minute. For units with solid state components, follow manufacturer's recommendation. Insulation resistance values shall be no less than two megohms.
 - e. Determine the following by primary current injection:
 - 1) Long-time pickup and delay. Pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 2) Short-time pickup and delay. Short-time pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 3) Ground-fault pickup and time delay. Ground-fault pickup values shall be as specified. Trip characteristics shall not exceed manufacturer's published time-current characteristic tolerance band, including adjustment factors.
 - 4) Instantaneous pickup. Instantaneous pickup values shall be as specified and within manufacturer's published tolerances.
 - f. Verify correct operation of auxiliary features such as trip and pickup indicators; zone interlocking; electrical close and trip operation; trip-free, anti-pump function; and trip unit battery condition. Reset all trip logs and indicators. Investigate units that do not function as designed.
 - g. Verify operation of charging mechanism. Investigate units that do not function as designed.
- 3. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
- 4. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

- D. Enclosed switches and circuit breakers will be considered defective if they do not pass tests and inspections.
- E. Prepare test and inspection reports.
 - 1. Test procedures used.
 - 2. Include identification of each enclosed switch and circuit breaker tested and describe test results.
 - 3. List deficiencies detected, remedial action taken, and observations after remedial action.

3.7 ADJUSTING

- A. Adjust moving parts and operable components to function smoothly, and lubricate as recommended by manufacturer.

END OF SECTION 26 2816

SECTION 31 1000 - SITE CLEARING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Protection of existing trees.
 - 2. Clearing and grubbing.
 - 3. Removal of trees and other vegetation.
 - 4. Topsoil stripping.

1.3 DEFINITIONS

- A. Remove: Remove and legally dispose of items indicated. Removal includes digging out and off-site disposing of stumps and roots.
- B. Tree Protection Zone: The area surrounding individual trees or groups of trees to be protected during construction and defined by the drip line of individual trees or the perimeter drip line of groups of trees, unless otherwise indicated.
- C. Topsoil: Friable, clay loam surface soil, found in varying depths.

1.4 MATERIALS OWNERSHIP

- A. Except for stripped topsoil or other materials indicated to remain Owner's property, cleared materials shall become Contractor's property and shall be removed from Project site.

1.5 SUBMITTALS

- A. Photographs or videotape, sufficiently detailed, of existing conditions of trees, plantings and other improvements adjoining the construction that might be misconstrued as damage caused by the Work.

1.6 PROJECT CONDITIONS

- A. Traffic: Conduct site clearing operations to ensure minimum interference with roads, streets, walks, and other adjacent occupied or used facilities. Do not close or obstruct streets, walks or other occupied or used facilities without permission from authorities having jurisdiction.
- B. Protection of Existing Improvements: Provide protections necessary to prevent damage to existing improvements indicated to remain in place.
 - 1. Protect existing improvements on adjoining properties and on Owner's property.
 - 2. Restore existing improvements damaged by clearing operations to their original condition.

- C. The conditions existing at the time of inspection for bidding purposes will be maintained by the Owner to the extent practical. However, minor variations may occur due to natural occurrences prior to the start of clearing work.
- D. Do not commence site-clearing operations until erosion and sedimentation control measures are in place.

PART 2 - PRODUCTS

2.1 TREE PROTECTION FENCING

- A. Tree protection fencing shall conform to standard and specifications in the latest version of Montgomery County DPS & MDE design and construction standards, whichever is more stringent.

PART 3 – EXECUTION

3.1 PROTECTION OF EXISTING TREES AND VEGETATION

- A. Install tree protection fencing as indicated. Erect and maintain a temporary fence around the drip line of individual trees or around the perimeter drip line of groups of trees to remain.
 - 1. Do not store construction materials, debris, topsoil or other excavated material within the tree protection zone.
 - 2. Do not permit vehicles or other equipment within the tree protection zone.
 - 3. Maintain tree protection zones free of weeds and trash.
- B. Protect existing trees and other vegetation indicated to remain in place, against unnecessary cutting, breaking or skinning of roots, skinning or bruising of bark, smothering of trees by stockpiling construction materials or excavated materials within drip line, excess foot or vehicular traffic, or parking of vehicles within drip line.
- C. Provide protection for roots over 1-1/2 inch diameter that are cut during construction operations. Coat cut faces with emulsified asphalt, or other acceptable coating, formulated for use on damaged plant tissues. Temporarily cover exposed roots with wet burlap to prevent roots from drying out; cover with earth as soon as possible.
- D. Repair or replace trees and vegetation indicated to remain which are damaged by construction operations, in a manner acceptable to Architect.

3.2 SITE CLEARING

- A. General: Remove trees, shrubs, grass and other vegetation as required to permit installation of the Work. Cut minor roots and branches of trees indicated to remain in a clean and careful manner, where such roots and branches obstruct installation of the Work.
- B. Clearing and Grubbing: Clear site of trees, shrubs and other vegetation within the clearing limits indicated.
 - 1. Completely remove stumps, roots, and other debris.
 - 2. Use only hand methods for grubbing inside drip line of trees indicated to remain.
 - 3. Fill depressions caused by clearing and grubbing operations with satisfactory soil material, unless further excavation or earthwork is indicated. Place fill material in horizontal layers not

exceeding 6 inches loose depth, and thoroughly compact to a density equal to adjacent original ground.

- C. Selective Clearing: Clear areas designated as "Selective Clearing" of all ground covers, underbrush and trees less than 6 inches in diameter at breast height. Coordinate extent of material removed with Architect.
 - 1. Remove trees that appear to be dying or weakening for any reason and at any point during construction up to and including Substantial Completion at the Architect's direction.

3.3 TOPSOIL STRIPPING

- A. Remove heavy growths of grass from areas before stripping.
- B. Strip topsoil to whatever depths are encountered, but to a minimum of at least 4 inches.
- C. Strip topsoil in a manner to prevent intermingling with underlying subsoil or other material.
 - 1. Remove subsoil and non-soil materials from topsoil, including trash, debris, weeds, roots, and other waste materials.
- D. Where existing trees are indicated to remain, leave existing topsoil in place within drip lines to prevent damage to root system.
- E. Temporarily stockpile topsoil in storage piles in areas indicated or directed. Construct storage piles to provide free drainage of surface water. Cover storage piles, if required, to prevent wind erosion.
 - 1. Do not stockpile topsoil within tree protection zones.
- F. Dispose of unsuitable or excess topsoil in a legal manner off-site.

3.4 DISPOSAL OF WASTE MATERIALS

- A. Burning on Owner's Property: not permitted.
- B. Removal from Owner's Property: Remove waste materials generated by clearing operations from Owner's property and dispose of in a legal manner off-site.
 - 1. Remove waste materials and debris from the site in a manner to prevent spillage. Pavements and the area adjacent to the site shall remain free from mud, dirt and debris always.
 - 2. Clean up debris resulting from site clearing operations continuously with the progress of the work.

END OF SECTION 311000

SECTION 31 2000 – EARTHWORK

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.
- B. Refer to Section 01 2110 and the Bid Form for information concerning required allowances and unit prices.
- C. Refer to Section 31 1000 for topsoil stripping and Section 32 9200 for topsoil placement.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Excavation, filling, backfilling, and grading indicated and necessary for proper completion of the work.
 - 2. Excavating and backfilling of trenches.

1.3 SUBMITTALS

- A. SHA approved Job Mix for stone.
- B. Imported fill (if required): Submit location of borrow pit and a sample of the soil for approval to the Owner's Geotechnical Engineer a minimum of fourteen (14) working days prior to use
- C. Geotextile Fabric

1.4 DEFINITIONS

- A. Excavation: Removal of all material (except for rock) encountered to design subgrade elevations indicated for cut areas and to subsoil elevations in fill areas. Excavation also includes subsequent re-spreading, moisture conditioning, compaction, and grading of satisfactory materials removed.
- B. Unauthorized Excavation: Removal of materials beyond the limits indicated in the definition of "Excavation" without specific direction of Architect.
- C. Additional Excavation: Removal, disposal and replacement of materials beyond the limits indicated in the definition of "Excavation" at the direction of the Architect. Refer to Part 3 of this Section for requirements of Additional Excavation.
- D. Subgrade: The undisturbed earth (in cut) or the compacted soil layer (in fill) immediately below granular subbase, drainage fill, or topsoil materials.
- E. Subsoil: The undisturbed earth immediately below the existing topsoil layer.
- F. Structures: The area extending a minimum of ten (10) feet beyond the edge of foundations, slabs, curbs, underground tanks, piping or other man-made stationary features occurring above or below ground surface.
- G. Pavements: The area extending 10 feet beyond the exterior limits of paved areas and down to undisturbed soils at a one horizontal to one vertical slope. The area extending 3 feet beyond the exterior limits of walks and down to undisturbed soils at a one horizontal to one vertical slope
- H. Subbase Material: Artificially graded mixture of crushed gravel or crushed stone meeting SHA specifications. Material type is indicated on the drawings.

- I. Drainage/Porous Fill: Washed, evenly graded mixture of crushed stone, or crushed or uncrushed gravel meeting the requirements of SHA base Stone.
- J. Rock: Hard bed rock, boulders or similar material requiring the use of rock drills and/or explosives for removal. The criteria for classification of general excavation as rock is any material which cannot be dislodged by a Caterpillar D-8 Tractor, or equivalent, equipped with a single tooth hydraulically operated power ripper. The criteria for trench rock shall be that a Caterpillar 345 Backhoe, or equivalent, with a proper width bucket cannot remove the material.

1.5 ADDITIONAL WORK

- A. This Project is Unclassified.
- B. The risks of concealed, unknown, or unanticipated subsurface conditions from existing ground surface to the design subgrade elevations in cut areas and to subsoil elevations in fill areas shall be included in the Contract Amount and shall not be considered as grounds for additional costs to the Contract.
- C. The risks of concealed, unknown, or unanticipated subsurface conditions below the elevations stated above identified as additional excavation shall also be included in the Base Bid Contract amount and shall not be considered as grounds for additional cost to the Contract. The work includes the establishment of acceptable bearing conditions in both cut and fill situations.
- D. Rock Excavation: Any required rock excavation shall be included in the Base Bid Contract amount and shall not be considered as grounds for additional costs to the Contract.
- E. Time extensions will not be granted for Additional Excavation or Rock Excavation.

1.6 EARTHWORK BALANCE ADJUSTMENTS

- A. Adjustments of grades may be allowed with prior written approval of the Owner / Architect to accommodate shortfall or surplus of material that may occur. Should adjustments be allowed, maintenance of designed drainage patterns and required adjustments to drainage structures shall be a Contract responsibility. All grade revisions must be documented and approved by NIH. **No additional payment will be made for these adjustments.**
- B. The Owner / Architect has identified a material stockpile at the Southern Maryland Electric Cooperative site located within 1 mile of the project site that can be used as a source of fill material. See Geotechnical Report by ECS, Dated 11.25.2014 for reference. The soil tests conducted by ECS of existing stockpile materials classify as (SC), (CL-ML), and (CL) and may be suitable for reuse as structural fill. If utilized, the Contractor shall be responsible for operations as required to relocate the material, including all permitting and stabilization of the borrow site. The Contractor shall be responsible for adjusting the moisture content within ± 2 percentage of the optimum moisture content, prior to placing as fill on the CSM site. The planning of earthwork operations should recognize and account for these efforts and costs. These approved materials should be placed in accordance with the Fill Placement section of this specification and be observed and documented by ECS or a qualified soil technician to confirm that our report criteria has been met.

1.7 QUALITY ASSURANCE

- A. Codes and Standards: Perform excavation work in compliance with applicable requirements of authorities having jurisdiction.
- B. Environmental Compliance:
 - 1. Comply with the requirements of the latest edition of MDE design and construction standards during earthwork operations.
 - 2. Comply with the permit conditions for all work performed within wetlands.

- C. Testing and Inspection Service: Owner will employ and pay for an independent Geotechnical testing and inspection laboratory to perform soil testing and inspection service during earthwork operations. Cooperate with Owner's Geotechnical Engineer as required for testing and inspection of work. These services do not relieve the responsibility for compliance with Contract Document requirements.

1.8 PROJECT CONDITIONS

- A. Site Information: Data concerning subsurface materials or conditions, which are based on test borings, have been obtained by the Owner for his use in designing the project. This report is included in this project manual for information only.
 - 1. The accuracy or completeness of the data is not warranted or guaranteed by the Owner or the Architect/Engineer, and in no event shall be considered part of the Contract Documents. The Owner and Architect/Engineer expressly disclaim any responsibility for the data as being representative of the conditions and materials that may be encountered.
- B. Bidders and interested parties (prior to receipt of bids) are encouraged to conduct their own soil and subsurface investigations, examinations, tests, and exploratory borings to determine the nature of the soil conditions underlying the project site. Contact the Owner's office to make an appointment to enter the site for the purpose of conducting your own investigation prior to bid.
- C. Existing Utilities: Do not interrupt existing utilities serving facilities occupied by the Owner or others except when permitted under the following conditions and then only after arranging to provide acceptable temporary utility services.
 - 1. Notify Architect not less than 48 hours in advance of proposed utility interruptions.
 - 2. Do not proceed with utility interruptions without receiving Architect's written permission.
 - 3. Existing utilities across or along the line of work are indicated only in an approximate location. Locate all underground lines and structures. Call "Miss Utility" at 1-800-552-7001 prior to construction. If utilities are marked that are not shown on the plans, locate utility vertically and horizontally and provide information to architect. Repair and correct any damage to underground lines and structures.

1.9 SAFETY

- A. Protection of Persons and Property: Barricade open excavations occurring as part of this work and post with warning lights.
 - 1. Operate warning lights as recommended by authorities having jurisdiction and governing regulations and standards.
 - 2. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Work within the road right-of-way shall meet all requirements of the latest edition of the SHA Design and Construction Manuals.

PART 2 - PRODUCTS

2.1 SOIL MATERIALS

- A. Satisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CL, GC, SC, GW, GP, GM, SM, SW, and SP or as approved by the geotechnical engineer.
- B. Unsatisfactory soil materials are defined as those complying with ASTM D2487 soil classification groups CH, OL, OH, MH, ML and PT.
- C. Backfill and Fill Materials: Satisfactory soil materials free of clay, rock or gravel larger than 4 inches in any dimension (2 inches for material used in trench backfill), debris, waste, frozen materials, vegetation and other deleterious matter.
- D. Imported material for structural fill shall comply with ASTM D2487 soil classification groups CL, ML, SC, SM, SP, SW, GC, GM, GP, or GW or as approved by the geotechnical engineer.

2.2 ACCESSORIES

- A. Non-woven Geotextile Fabric (for drainage): Mirafi 140N, or equivalent.
- B. Woven Geotextile Fabric (for reinforcement): PROPEX 2002, or equivalent.

PART 3 – EXECUTION

3.1 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.
- B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, obstructions, and deleterious materials from ground surface is specified in Section 02230 "Site Clearing."
- C. Protect and maintain erosion and sedimentation controls during earthwork operations.

3.2 DEWATERING

- A. Prevent surface water and subsurface or groundwater from flowing into excavations and from flooding project site and surrounding area.
 - 1. Do not allow water to accumulate in excavations. Remove water to prevent softening of foundation bottoms, undercutting footings, and soil changes detrimental to stability of subgrade and foundations. Provide and maintain pumps, well points, sumps, suction and discharge lines, and other dewatering system components necessary to convey water away from excavations.
 - 2. Establish and maintain temporary drainage ditches and other diversions outside excavation limits to convey rain water and water removed from excavations to collecting or runoff areas. Do not use utility trench excavations as temporary drainage ditches.
- B. Should any springs or running water be encountered in the excavation, notify the Architect and provide discharge by trenches (or other acceptable means) and drain to an appropriate point of disposal. Provide temporary drainage facilities to minimize the flow of rainwater onto adjacent property. Repair any damage to property or to subgrade because of construction and/or dewatering (or lack thereof) operations at no additional cost to the Contract. If permanent provision must be made for disposal of water other than as indicated, the Contract price shall be adjusted.

3.3 EXCAVATION

- A. Excavation consists of removal, placement and disposal of material encountered when establishing required subgrade or finish grade elevations.
 - 1. Excavation includes removal and disposal of pavements and other obstructions visible on ground surface; underground structures, utilities and other items indicated to be demolished and removed; together with earth and other materials encountered that are not classified as rock or unauthorized excavation.
- B. Rock Excavation: If Rock is encountered the Owner's Geotechnical Engineer will verify that the material qualifies for classification as rock excavation.
 - 1. If rock is encountered in grading, remove to depths as follows:
 - a) Under surfaced areas, to 6" under the respective subgrade for such areas.
 - b) Under grass and planted areas - 12" minimum.
 - c) Under trenches – 6" below bottom of trench.
 - 2. Contractor shall employ a surveyor licensed in the State of Maryland to calculate the quantity of material removed as Rock Excavation. The quantity of rock calculated shall not exceed the volume determined by the payment limits. The Owner's Project Representative shall review the quantity calculated within 48 hours of receiving the survey notes.

3.4 EXCAVATION STABILITY

- A. General: Comply with local codes, ordinances, and requirements of agencies having jurisdiction.
- B. Slope sides of excavations to comply with local codes, ordinances, and requirements of agencies having jurisdiction. Shore and brace where sloping is not possible because of space restrictions or stability of material excavated. Maintain sides and slopes of excavations in safe condition until completion of backfilling.
- C. Shoring and Bracing: Provide materials for shoring and bracing, such as sheet piling, uprights, stringers, and cross braces, in good serviceable condition. Maintain shoring and bracing in excavations regardless of time period excavations will be open. Extend shoring and bracing as excavation progresses.

3.5 SUBGRADE INSPECTION

- A. Notify Architect when mass, trench and footing excavations have reached required subgrade. The Architect will arrange for an inspection of conditions by the Owner's Geotechnical Engineer. *Alternative procedures for arranging this review may be implemented at the Owner's written option.*
- B. If the Owner's Geotechnical Engineer determines that the subgrade bearing conditions are unacceptable, the Architect will authorize additional excavation until suitable bearing conditions are encountered.
- C. Proof-roll subgrade **below the building slabs and impervious pavements** with heavy pneumatic-tired equipment to identify soft pockets and areas of excess yielding. Do not proof-roll wet or saturated subgrades.
 - 1. Completely proof-roll subgrade in one direction, repeating proof-rolling in direction perpendicular to first direction. Limit vehicle speed to 3 mph.
 - 2. Proof-roll with a loaded 10-wheel, tandem-axle dump truck weighing not less than 15 tons.

3. Excavate soft spots, unsatisfactory soils, and areas of excessive pumping or rutting, as determined by Architect, and replace with compacted backfill or fill as directed.
- D. Under supervision of the Owner's Geotechnical Engineer, proof roll subgrade in cut areas below the building pad and pavement(s) with a loaded dump truck or other approved pneumatic tired vehicle. Should any unstable sub-soil be encountered below pavement or structures, break up the top eight inches of ground surface, pulverize, moisture-condition to optimum moisture content, and compact to percentage of maximum density as stated in Percentage of Maximum Density Requirements. Perform this work at no additional cost and/or time to the Contract.
- E. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by Architect, without additional compensation.

3.6 ADDITIONAL EXCAVATION

- A. Additional Excavation (Mass): Remove excavated materials and dispose of on-site as directed by the Architect. Replace this excavated material with satisfactory material placed and compacted according to the requirements of the "Placement and Compaction" section.
- B. Additional Excavation in Trenches: Remove excavated materials and dispose of on-site as directed by the Architect. Replace this excavated material with stone.
- C. The quantity of material removed as Additional Excavation (Mass, Trench or Footing) shall be calculated by a surveyor licensed in the State of Maryland and employed by the Contractor. The Owner's Project Representative shall review the quantity calculated within 48 hours of receiving the survey notes.
- D. Protect the subgrade during construction. During wet conditions, the subgrade soils may become saturated and soften, possibly resulting in damage to the subgrade if disturbed by equipment. Correct subgrade damaged in this manner. **No additional payment will be made to correct subgrade damaged in this manner.**

3.7 UNAUTHORIZED EXCAVATION

- A. Correct Unauthorized Excavation as follows:
 1. Elsewhere, backfill and compact unauthorized excavations as indicated for authorized excavations of same classification unless otherwise directed by Architect.

3.8 STORAGE OF EXCAVATED MATERIALS

- A. Temporarily stockpile excavated materials acceptable for use as backfill and fill. Place, grade, and shape stockpiles for proper drainage. Cover to prevent windblown dust.
 1. Stockpile excavated materials away from edge of excavations. Do not store within the drip line of trees to remain.

3.9 BACKFILL AND FILL

- A. Backfill excavations as promptly as work permits, but not until completion of the following:
 1. Acceptance by local authority having jurisdiction of construction below finished grade, including perimeter insulation.
 2. Review, approval, and recording of the locations of underground utilities.
 3. Removal of concrete formwork.

4. Removal of shoring and bracing (including backfilling of voids with satisfactory materials).
 5. Removal of trash and debris from excavation.
 6. Permanent or temporary horizontal bracing is in place on horizontally supported walls.
- B. Place backfill on subgrades free of mud, frost, snow or ice.
- C. Ground Surface Preparation: Remove vegetation, debris, obstructions, and deleterious materials from ground surface prior to placement of fills.
- D. Bench sloped surfaces steeper than 1 vertical to 4 horizontal so fill material will bond with existing material. Plow, scarify, bench or break up sloped surfaces flatter than 1 vertical to 4 horizontal so fill material will bond with existing material.
- E. Place soil material in layers to required subgrade elevations, for each area classification listed below, using materials indicated in Part 2 of this Section.
1. Under grassed areas, use satisfactory excavated or borrow material.
 2. Under walks, curbs use satisfactory excavated or borrow material.

3.10 SOIL MOISTURE CONTROL

- A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 2 percentage points of optimum moisture content.
1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
 2. Remove and replace or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 2 percentage points and is too wet to compact to specified dry unit weight.
- B. Moisture Control: Where subgrade or layer of soil material must be moisture conditioned before compaction, uniformly apply water to surface of subgrade or layer of soil material. Apply water in minimum quantity as necessary to prevent free water from appearing on surface during or subsequent to compaction operations. Maintain the moisture content of the structural fill materials to within 2 percentage points of the optimum moisture content until permanently covered.
- C. Remove and replace, or scarify and air dry, soil material that is too wet to permit compaction to required density.
1. Stockpile or spread soil material that has been removed because it is too wet to permit compaction. Assist drying by discing, harrowing, or pulverizing until moisture content is reduced to a satisfactory value.
 2. Work wet materials as directed by the Owner's Geotechnical Engineer. Base bids on working material daily for a maximum of five days of acceptable weather.
 3. No additional payment will be made for these operations.

3.11 COMPACTION OF SOIL BACKFILL AND FILLS

- A. Place backfill and fill materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.
- B. Before compaction, moisten or aerate each layer as necessary to provide optimum moisture content. Compact each layer to required percentage of maximum dry density or relative dry density

for each area classification. Do not place backfill or fill material on surfaces that are muddy, frozen, or contain frost or ice.

- C. Control soil and fill compaction, providing minimum percentage of density indicated for each area classification indicated below. Correct improperly compacted areas or lifts as directed by Architect if soil density tests indicate inadequate compaction.
- D. Percentage of Maximum Density Requirements: Compact soil to not less than the following percentages of maximum density at a moisture content within 2 percentage points of optimum in accordance with ASTM D698:
 - 1. Under structures compact each layer of backfill or fill material at 99 percent maximum density. This includes ground under future expansion areas.
 - 2. Under grass or unpaved areas, compact each layer of backfill or fill material at 95 percent maximum density.
- E. Seal all fill areas at the end of each working day, utilizing a smooth drum roller.

3.12 GRADING

- A. General: Rough grading of areas within the Project, including cut and fill sections and adjacent transition areas, shall be reasonably smooth, compacted and free from irregular surface changes. The degree of finish shall be that ordinarily obtainable from either blade-grader or motor patrol except as otherwise indicated. The finished subgrade surface from the grassed areas generally shall be not more than 0.2 feet (61 mm) above or below the final grade or approved cross section, with due allowance for topsoil.
- B. The tolerance for areas within 10 feet (3.0 m) of building perimeter, walks and all areas to be paved shall not exceed 0.10 (30.5 mm) feet above or below the established subgrade. Finish all ditches, swales and gutters to drain readily. Unless otherwise indicated, evenly slope the subgrade to provide drainage away from building walls in all directions at a grade not less than ¼ inch per foot. Provide rounding at top and bottom of cut and fill slopes and at other breaks in grade.
- C. Protection of Graded Areas: Protect newly graded areas and areas of cut, fill and design/subgrade elevations from the actions of the elements and from deterioration as a result of construction operations and weather conditions (frost, rains, snow, sleet, hail, etc.). Repair any settlement or washing that occurs prior to or after acceptance of the work. Fill to required subgrade levels any areas where settlement occurs. Protect trees to remain, and, at all areas of the Site where construction operations are in progress, provide protection for the safety of occupants of the existing facilities.
- D. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.
 - 1. Provide a smooth transition between adjacent existing grades and new grades.
 - 2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.
- E. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:
 - 1. Lawn or Unpaved Areas: Plus or minus **1 inch (25.4 mm)**.
 - 2. Walks: Plus or minus **1 inch (25.4 mm)**.
 - 3. Pavements: Plus or minus **1/2 inch (12.7 mm)**.

- F. Grading inside Building Lines: Finish subgrade to a tolerance of 1/2 inch (12.7 mm) when tested with a 10-foot (3.0 m) straightedge.

3.13 FIELD QUALITY CONTROL

- A. Quality Control Testing During Construction: Allow testing service to inspect and approve each subgrade and fill layer before further backfill or construction work is performed.
 - 1. If in the opinion of the Architect, based on testing service reports and inspection, subgrade or fills have been placed that are below required density, perform additional compaction and testing until required density is obtained.
- B. The Owner will engage, and pay for, the services of a Geotechnical Engineer whose function shall be to afford complete engineering control by testing of the conditions of all footing subgrades, the placement of all structural fills under structures, building pad and pavement areas, and all compaction where required, and to observe the proof rolling of the building pad and pavement areas.
- C. The Owner's Geotechnical Engineer will be present as deemed necessary during all phases of the Work requiring filling, compaction operations or testing. The Geotechnical Engineer will provide the Architect with written certification that fill and compaction was completed with accepted materials in accordance with the Documents, and give a professional opinion regarding shrinkage or settlement of fill and safe load bearing capacity of fill.
- D. Site Preparation and Proof rolling: The Owner's Geotechnical Engineer will determine if any additional excavation or in-place densification is necessary to prepare a subgrade for fill placement for slab or pavement support.
- E. Fill Placement and Compaction: The Owner's Geotechnical Engineer will witness all fill operations and take enough in-place density tests to verify that the indicated degree of fill compaction is achieved. The Owner's Geotechnical Engineer will observe and approve borrow materials used and shall determine if their existing moisture contents are suitable/acceptable.
- F. The Owner's Geotechnical Engineer will submit two (2) copies each of his reports, recommendations and/or opinions to the Architect/Engineer and the Owner. Pertinent information will be provided to the Contractor as required.

3.14 EROSION CONTROL:

- A. Provide erosion control methods in accordance with requirements of authorities having jurisdiction to include MDE and as indicated in the Contract Documents.

3.15 PROTECTION

- A. Repair and reestablish grades in settled, eroded, and rutted areas to indicated tolerances.
- B. Reconditioning Compacted Areas: Where subsequent construction operations or adverse weather disturbs completed compacted areas, scarify surface, reshape, and compact to required density prior to further construction.
- C. Settling: Where settling is measurable or observable at excavated areas during general project warranty period, remove surface (pavement, lawn, or other finish), add backfill material, compact, and replace surface treatment. Restore appearance, quality, and condition of surface or finish to match adjacent work and eliminate evidence of restoration to greatest extent possible.

- D. Protect excavation bottoms against freezing when atmospheric temperature is less than 35 degrees F.

3.16 DISPOSAL OF WASTE MATERIALS

- A. Removal from Owner's Property: Remove excess materials, including trash and debris, and dispose of it off Owner's property in a legal manner.
- B. Dispose of materials not acceptable for use as backfill or fill legally offsite.
- C. Do not remove topsoil from site until it has been demonstrated to the Owner's satisfaction that it is excess.

1.1 END OF SECTION 31 2000

SECTION 31 2500 - EROSION CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS:

- A. The provisions of the Contract Documents apply to the work of this Section.
- B. The Maryland Department of the Environment shall be responsible for the coordination and enforcement of the erosion and sediment control provisions of this Ordinance and for this project.
- C. The Maryland State Highway Administration, Office of Materials Technology, List of qualified producers and products.
- D. The Maryland Department of the Environment (MDE), Standards and Specifications for Soil Erosion and Sediment Control, latest edition.

1.2 SUMMARY

- A. This Section includes the installation, maintenance and removal of erosion control measures required for prevention of sediment leaving the project site.
- B. No person shall clear or grade land without obtaining an approval of an Erosion/Sediment Control and Stormwater Management Plan except as provided within MDE regulations.
- C. No person shall clear or grade land without obtaining a General or Individual Permit for Stormwater Associated with Construction Activity from MDE.
- D. Only individuals allowed by MD State law may perform work as required by the Grading and Sediment Control Ordinance.

1.3 SEDIMENT & EROSION CONTROL PLAN

- A. Prior to commencement of work, obtain a copy of the signed / approved Erosion/Sediment Control and Stormwater Management Plan and General or Individual Permit for Stormwater Associated with Construction Activity. These must be kept on site at all times.
- B. Post Sediment & Erosion Control Bond as may be required by MDE. The bond shall remain in full force and effect until all work and conditions encompassed under permit; these regulations and any other regulations governing this work, have been completed and approved by the Department and the Bond officially released.
- C. Schedule a pre-construction conference on-site with the Architect / Owner / as required. Hold this meeting prior to the start of any construction activities.
- D. No work approved in accordance with the provisions of the Grading and Sediment Control Ordinance shall proceed until the Department inspects the site and approves the work previously completed or notified the Permittee otherwise. Upon notification from the Permittee as required in Section 7.3 of this Article, said Department shall inspect the site and notify the Permittee of its approval or rejection within twenty-four (24) hours, exclusive of Saturdays, Sundays and holidays. If the inspector does not make an inspection within the specified time period, work may proceed without presumption of approval at the risk of the Permittee. The Department shall have the right to waive inspections, except the final inspection.
- E. The Contractor shall be responsible for implementation of the approved erosion and sediment control plan and the prevention of damage to any public utilities or services within the limits of grading and along any routes of travel of equipment.
- F. All applicable permits shall be extended by the Contractor from the date of issuance or as deter

mined by MDE. All requests for renewal and/or extension must be made in writing by the Contractor sixty (60) days prior to the date of expiration or as indicated on the permit. The requests for renewal and/or extension shall include a time table for the completion of the project.

- G. All modifications made to the approved grading plans shall be engineered and submitted, by the Contractor to the Department and reprocessed in the same manner as the original plan. Field modifications may be authorized by MDE, provided that written authorization is given to the person performing work pursuant to the Grading and Sediment Control Ordinance, with copies forwarded to the District.
- H. The Contractor is responsible for all permitting or use of off-site stockpile materials.
- I. If grading operations cease for a period of fourteen (14) consecutive calendar days permanent erosion and sediment control measures must be employed. Unless the MDE inspector determines otherwise.

1.4 MDE SEDIMENT CONTROL CONDITIONS

- A. During construction, the following requirements shall be met:
 - 1. The active site shall be inspected for compliance with the plan on the average once every 2 weeks or after a large storm event. Inspection reports to be written in accordance with the Grading and Sediment Control Ordinance.
 - 2. Correct any practice deficiencies in a timely manner.
 - 3. After commencing initial operations, the Permittee shall require inspections by the Department at the following stages in the development of the site, or of each subdivision thereof:
 - (1) Upon completion of installation of perimeter erosion and sediment controls, prior to proceeding with any other earth disturbance or grading. Other building or grading inspection approvals may be not be authorized until initial approval by the Department is made;
 - (2) Upon completion of stripping, the stockpiling of top soil, the disposal of all unsuitable materials, and preparation of this ground to receive fill, but prior to beginning any placement;
 - (3) Upon completion of rough grading, but prior to placing top soil, permanent drainage or other site development improvements;
 - (4) Upon completion of final grading, permanent drainage and erosion control facilities, but prior to any seeding, sodding or planting;
 - (5) Upon completion of installation of all vegetative measures and all other work in accordance with the Development Services Permit. The Department may make additional inspections as such Department might deem appropriate;
 - (6) Upon final stabilization before removal of sediment controls; and
 - (7) After final stabilization and after the removal of sediment controls.
- B. The Contractor shall maintain permanent files on their inspections. Upon completion of permitted work, the Department may require the following, to be produced and supplied by the Contractor, for their files and shall also require copies for the NIH:
 - (1) An "as-built" plan and survey by a Professional Land Surveyor or Professional Engineer at the same scale as the original plan and showing all improvements and final grades. (2) Certification by the Owner that all grading, drainage, erosion control measures, and facilities and vegetative measures have been completed in conformance with the approved plans and specifications. (3) A report summarizing the inspection reports, field and laboratory tests and location tests. (4) "As-built" plans for ESDs approved by the District. "As-built" plans shall be prepared per District requirements.

1.5 SUBMITTALS

- A. Responsible Personnel "Green Card" Certification for Contractor overseeing the installation and maintenance of erosion and sediment control practices.
- B. Copies of the weekly Erosion Control Measure inspection reports. *These may be submitted at the monthly progress meetings.*
- C. Material specifications for all E&S practices utilized for the project.

1.6 PAYMENT PROCEDURES FOR EROSION CONTROL MEASURES

- A. Establish a line item in the Schedule of Values for Erosion Control Maintenance. This line item shall represent a minimum of thirty percent (30%) of the total value of the erosion control for the project.
- B. Erosion control maintenance will be paid on a monthly basis, following the satisfactory installation and maintenance of the erosion control measures.

PART 2 - PRODUCTS

2.1 EROSION CONTROL PRODUCTS:

- A. Safety Fence
 - 1. Six foot (1.83 m) high chain link, temporary safety fence with post stabilization and gates complying with NIH standards.
 - 2. Post appropriate warning signs along the Safety Fence.
- B. Construction Entrance
 - 1. Heavy-duty stone aggregate and filter fabric construction entrance, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Detail, B-2.
 - 2. Reinforced concrete wash-rack, draining to a sediment trap.
 - 3. The water source for washing operations shall be the responsibility of the Contractor.
- C. Silt Fence
 - 1. Synthetic filter fabric, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Detail, E-1.
 - 2. Wooden stakes shall be 2" (50.8 mm) oak, a minimum length of five feet.
- D. Wire Reinforced Silt Fence
 - 1. Synthetic filter fabric, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Detail, E-3.
 - 2. Wooden stakes shall be 2" (50.8 mm) oak, a minimum length of five feet.
 - 3. Wire fence reinforcement shall be a minimum of 14-gauge and have a maximum mesh spacing of six inches.
- E. Storm Drain Inlet Protection
 - 1. Gravel Drop Inlet Sediment Filter, complying with the requirements Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and

Sediment Control Details E-9.

2. Gravel Curb Inlet Sediment Filter, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details E-9.
- F. Culvert Inlet Protection
1. Silt Fence Culvert Inlet Protection, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, G-1.
- G. Temporary Diversion Dike
1. Berm of compacted soil material, stabilized with vegetation, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section C.
- H. Temporary Fill Diversion
1. A channel with a berm on the lower side, placed at the top of a fill slope, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section C.
- I. Diversion
1. A channel with a berm on the lower side, stabilized with vegetation, complying with the requirements of the Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Detail, Section C.
- J. Temporary Sediment Trap
1. A temporary ponding area, formed with an earthen embankment and a stone outlet structure, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section G.
- K. Temporary Sediment Basin
1. A temporary pond with a controlled outfall release, complying with the requirements of the Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section G and to the details indicated on the Drawings.
 2. The sediment basin shall be reconfigured for use as a permanent stormwater management facility. Conversion of the basin from a temporary to a permanent facility is required. Refer to the narrative on the drawings.
- L. Temporary Slope Drain
1. A flexible pipe extending from the top to the bottom of a fill slope, complying with the requirements of the Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section C.
 2. Pipe shall be smooth lined polyethylene, complying with the requirements of ASTM F667 or AASHTO M294.
- M. Outlet Protection
1. A level area of riprap, placed over filter fabric, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Detail, Section D.

N. Riprap

1. Graded stone, placed over filter fabric, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section D.
2. The size of the stone required is indicated on the drawings.

O. Rock Check Dams

1. Temporary stone dams constructed across a swale or ditch, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section D.
2. Check dams shall be placed on filter fabric.

P. Dewatering Structure

1. A temporary filtering device used for dewatering operations, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section F.

Q. Temporary Seeding

1. Temporary vegetative cover for disturbed areas, complying with the requirements of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details, Section B.

R. Permanent Seeding

1. Refer to the "Lawns and Grasses" specification for permanent seeding requirements.

PART 3 - EXECUTION

3.1 INSTALLATION OF EROSION CONTROL MEASURES

- A. Install all erosion and sediment control measures per the requirements of the of Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control Details.
- B. Protect all points of construction ingress and egress to the site to prevent tracking of mud onto public streets. Provide temporary construction entrances at all points of access to the site.
- C. Clear only those areas necessary for installation of the perimeter erosion control measures. The balance of the site shall not be cleared or otherwise disturbed until the perimeter erosion control measures are installed, functional and approved by NIH.
- D. Follow the construction sequence and install erosion control measures as indicated on the Drawings and as directed by NIH.
- E. Install additional measures as necessary to prevent sediment from leaving the project site.

3.2 MAINTENANCE OF EROSION CONTROL MEASURES

- A. Maintain all erosion and sediment control measures per the requirements of the Maryland Department of the Environment (MDE), 2011 Maryland Standards and Specifications for Soil Erosion and Sediment Control.
- B. At a minimum, the following maintenance is required:
 1. Safety Fence

- a) Review fence regularly for damage. Repair any damage immediately.
 - b) Secure the fence at the end of each working day. Repair or replace all locking devices as necessary.
2. Construction Entrance
- a) Wash and rework stone and/or place additional stone as required preventing tracking of mud onto the roadways.
 - b) Clean out the sediment-trapping device for the wash rack.
 - c) Remove all materials spilled, dropped, washed or otherwise tracked onto roadways or into storm sewers immediately. Do not use water trucks to wash the roadways.
3. Silt Fence
- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Make any required repairs immediately. Give special attention to damage resulting from end-runs and undercutting.
 - c) Replace fabric that is decomposing or is otherwise ineffective.
 - d) Clean out accumulated sediment following every storm event. Do not allow sediment to accumulate higher than one-half the height of the barrier.
4. Wire Reinforced Silt Fence
- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Make any required repairs immediately. Give special attention to damage resulting from end-runs and undercutting.
 - c) Replace fabric that is decomposing or is otherwise ineffective.
 - d) Clean out accumulated sediment following every storm event. Do not allow sediment to accumulate higher than one-half the height of the barrier.
5. Storm Drain Inlet Protection
- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Remove and clean or replace stone filters that have been clogged with sediment. Make any required repairs immediately
 - c) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half the height of the measure.
6. Culvert Inlet Protection
- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Remove and clean or replace stone filters that have been clogged with sediment. Make any required repairs immediately
 - c) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half the height of the measure.
7. Temporary Diversion Dike
- a) Inspect immediately following each rainfall and at least daily during prolonged rainfall. Inspect at least once every two weeks, whether or not it has rained. Make any necessary repairs immediately.
 - b) Repair damages caused by construction activities by the end of each working day.

8. Temporary Fill Diversion
 - a) Review measure at the end of each working day to ensure its effective operation.
9. Diversion
 - a) Inspect diversion following every rainfall and at least once every two weeks.
 - b) Remove accumulated sediment and make repairs as necessary.
 - c) Re-seed as necessary to maintain vegetative cover.
10. Temporary Sediment Trap
 - a) Remove sediment and restore the trap to its original dimensions once the sediment accumulates to the cleanout level. Refer to the drawings for the appropriate cleanout level elevations.
 - b) Any pumping shall be discharged through an approved dewatering structure.
 - c) Remove and clean or replace stone choked with sediment.
 - d) Regularly check the structure to ensure that it is structurally sound. Immediately repair any damage discovered.
11. Temporary Sediment Basin
 - a) Remove sediment and restore the basin to its original dimensions once the sediment accumulates to the cleanout level. Refer to the drawings for the appropriate cleanout level elevations.
 - b) Any pumping shall be discharged through an approved dewatering structure.
 - c) Regularly inspect the principal spillway and outfall for proper function. Regularly inspect the emergency spillway to ensure that its lining is well established and erosion resistant. Immediately repair any damage discovered.
 - d) Regularly check the embankment to ensure that it is structurally sound. Immediately repair any damage discovered.
12. Temporary Slope Drain
 - a) Inspect the temporary slope drains weekly and following every storm event. Immediately make any necessary repairs to ensure a free flow through the pipe.
13. Outlet Protection
 - a) Inspect outlet protection following every storm event. Re-lay riprap as necessary to prevent concentrated flow from running across the outlet protection.
14. Riprap
 - a) Inspect riprap following every storm event. Re-lay riprap as necessary to prevent concentrated flow from running under or around the riprap.
 - b) Clean out accumulated sediment from the riprap.
15. Rock Check Dams
 - a) Inspect immediately following each rainfall and at least daily during prolonged rainfall.
 - b) Remove and clean or replace stone that has been clogged with sediment.
 - c) Inspect for evidence of by-pass flows. Make any required repairs immediately
 - d) Remove accumulated sediment as required. Do not allow sediment to accumulate higher than one-half of the height of the dam.

16. Dewatering Structure

- a) Repair or replace the filtering media to prevent sediment accumulation from affecting the filtering capacity of the structure.

17. Temporary Seeding

- a) Re-seed and mulch areas where cover is inadequate to protect against erosion until adequate cover is obtained.
- C. Remove accumulated sediment as required and at appropriate intervals to maintain the effective function of all erosion control measures.
 - D. Inspect, repair and remove accumulated sediment from erosion control measures following significant (greater than ½") rainfall events.
 - E. If erosion control measures become clogged, causing the impoundment of water, restore the measures immediately. Pooled water poses a potential drowning hazard and shall be relieved immediately by either pumping (through an approved dewatering structure) or by removal of the blockage.

3.3 REMOVAL OF EROSION CONTROL MEASURES

- A. Remove all temporary erosion control measures following the stabilization of the site. Do not remove erosion control measures until authorized by NIH.
- B. Topsoil, permanently seed and stabilize areas occupied by erosion control measures.

END OF SECTION 31 2500

SECTION 32 9200 - LAWNS AND GRASSES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:
 - 1. Fine grading and preparing lawn areas
 - 2. Topsoil Placement
 - 3. Soil amendments
 - 4. Fertilizers
 - 5. Seeding (Meadow, Ecolawn)
 - 6. Sodding (Ecolawn)
 - 7. Lawn Restoration

1.3 DEFINITIONS

- A. Finish Grade: Elevation of finished surface of planting soil.
- B. Lawns: All areas disturbed by construction and not otherwise covered by paving, buildings or other structures.

1.4 SUBMITTALS

- A. Certification by product manufacturer that the following products supplied comply with requirements:
 - 1. Grass Seed
 - a) Certification of seed from seed vendor for each seed mixture stating the botanical and common name and percentage by weight of each species and variety, and percentage of purity, germination, and weed seed. Include the year of production and date of packaging.
 - b) Blue Tag Certification tag for each bag of seed.
 - 2. Sod
 - a) Gold Tag Certification.
- B. Installers qualifications
 - 1. Provide a list, with references, of the past three projects of a similar magnitude.
- C. Topsoil Amendment Plan.
 - 1. Provide copy of topsoil testing report.
 - 2. List of amendments proposed for topsoil, including application rates.

1.5 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer, who has successfully completed meadow establishment projects similar in size and complexity to this project. The installer's primary business (defined as a minimum of 60% of total billings) shall be establishment of lawns/ meadows.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Seed: Deliver seed in original sealed, labeled, and undamaged containers.
- B. Sod: Harvest, deliver, store and handle sod according to the requirements of the American Sod Producers Association (ASPA) "Specifications for Turfgrass Sod Materials and Transplanting/Installing".

1.7 COORDINATION AND SCHEDULING

- A. Ecolawn Seed Planting Season: Sow ecolawn seed during normal planting seasons for type of lawn work required.
 - 1. Fall Planting Season: August 1 through October 15
 - 2. October 16 – July 31 is not approved for installation.
- B. Ecolawn Sod Planting Season: Sod during normal planting seasons for type of lawn work required.
 - 1. **Coordinate with Sod provider at the beginning of the project to ensure its availability.**
 - 2. Spring Planting Season: March 15 through May 15
 - 3. Summer Planting Season: May 15 through September 15 with approved temporary irrigation system.
 - 4. Fall Planting Season: September 15 through November 15
- C. Meadow Planting Season:
 - 1. March 1 through May 15
 - 2. May 16 – June 15 with supplemental watering to ensure plant establishment and annual nurse crop of Winter Oats. Apply Winter Oats at 1/3 its typical application rate 24 lbs. /acre. DO NOT USE CEREAL RYE.
 - 3. June 16-March 1 is not approved for installation.
- D. Weather Limitations: Proceed with planting only when existing and forecast weather conditions are suitable for work.
- E. Seed 60-90 days prior to pre-emergent application.
- F. Seeding Schedule
 - 1. If job completion schedule does not allow seeding within a normal planting season, provide interim temporary seeding necessary to stabilize site. Complete permanent seeding during the next planting season.

1.8 LIMITS OF SEEDING

- A. Prepare soil and seed all disturbed areas with mix appropriate to locations as indicated on Landscape Plans.
- B. Hydroseeding all slopes steeper than 4:1.

1.9 LIMITS OF SODDING

- A. Provide sod in the following areas:
 1. Areas between the sidewalk/ inner curb and the building foundation.
 2. Interior parking lot islands to outer perimeter as depicted in the "Interior Landscape Area" exhibit.

1.10 LIMITS OF LAWN RENOVATION

- A. All existing lawn areas disturbed by construction activities where partial lawn still exists.

PART 2 – PRODUCTS

2.1 TOPSOIL

- A. Topsoil: ASTM D 5268, pH range of 5.5 to 7, a minimum of 4 percent organic material content; free of stones 1" or larger in any dimension and other extraneous materials harmful to plant growth.
 1. Topsoil Source: Reuse surface soil stockpiled on-site. Verify suitability of stockpiled surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
 - a) Supplement with imported or manufactured topsoil from off-site sources when quantities are insufficient. Obtain topsoil displaced from naturally well-drained construction or mining sites where topsoil occurs at least 4 inches deep; do not obtain from agricultural land, bogs or marshes.
- B. Have topsoil tested by a certified soil testing laboratory to determine the type and quantity of soil amendments necessary. Add amendments to topsoil as necessary to meet these requirements.
 - a.

2.2 INORGANIC SOIL AMENDMENTS

- A. If the topsoil analysis indicates the need for inorganic soil amendments, the following standards apply:
- B. Lime: ASTM C 602, agricultural limestone containing a minimum 80 percent calcium carbonate equivalent and as follows:
 1. Class: Class O, with a minimum 95 percent passing through No. 8 (2.36-mm) sieve and a minimum 55 percent passing through No. 60 (0.25-mm) sieve.
 2. Provide lime in form of dolomitic limestone.
- C. Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, with a minimum 99 percent passing through No. 6 (3.35-mm) sieve and a maximum 10 percent passing through No. 40 (0.425-mm) sieve.
- D. Iron Sulfate: Granulated ferrous sulfate containing a minimum of 20 percent iron and 10 percent sulfur.
- E. Aluminum Sulfate: Commercial grade, unadulterated.
- F. Perlite: Horticultural perlite, soil amendment grade.

- G. Agricultural Gypsum: Finely ground, containing a minimum of 90 percent calcium sulfate.
- H. Sand: Clean, washed, natural or manufactured, free of toxic materials.
- I. Diatomaceous Earth: Calcined, diatomaceous earth, 90 percent silica, with approximately 140 percent water absorption capacity by weight.
- J. Zeolites: Mineral clinoptilolite with at least 60 percent water absorption by weight.

2.3 ORGANIC SOIL AMENDMENTS

- A. If the topsoil analysis indicates the need for organic soil amendments, the following standards apply:
- B. Compost: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 3/4-inch (19-mm) sieve; soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 percent of dry weight.
 - 2. Feedstock: Agricultural, food, or industrial residuals; biosolids; yard trimmings; or source-separated or compostable mixed solid waste.
 - 3. Peat: Finely divided or granular texture, with a pH range of 6 to 7.5, containing partially decomposed moss peat, native peat, or reed-sedge peat and having a water-absorbing capacity of 1100 to 2000 percent.
 - 4. Wood Derivatives: Decomposed, nitrogen-treated sawdust, ground bark, or wood waste; of uniform texture, free of chips, stones, sticks, soil, or toxic materials.
 - 5. Manure: Well-rotted, unleached, stable or cattle manure containing not more than 25 percent by volume of straw, sawdust, or other bedding materials; free of toxic substances, stones, sticks, soil, weed seed, and material harmful to plant growth.

2.4 HERBICIDES

- A. Selective Herbicides: EPA registered and approved, of type recommended by manufacturer for application.

2.5 FERTILIZER

- A. Commercial Fertilizer: Commercial-grade complete fertilizer of neutral character, consisting of fast- and slow-release nitrogen, 50 percent derived from natural organic sources of urea formaldehyde, phosphorous, and potassium in the following composition:
 - 1. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in topsoil analysis reports from a qualified soil-testing agency.
 - 2. Minimum Composition: No less than 1 lb (0.45 kg)/1000 sq. ft. (92.9 sq. m) of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.

2.6 ECOLAWN SEED

- A. Seed: All seed must be fresh, clean, and dry.
- B. Seed Species

Hardiness Zone: 7A Mix: Ecolawn- 80% Hard Fescue/14% Chewings Fescue/3% Sheep Fescue/3% Creeping Red Fescue Mix Rate: 150 lbs./acre					Fertilization Rate (0-10-10)			Lime Rate
Ecolawn Mix Species	Min. % Germination	Min. % Pure Seed	Seeding Rate	Seeding Depth	N	P2O5	K2O	As Needed
Festuca rubra ssp. commutata	97%+	100%	120lb/ac	1/8" to 1/4"	---	45lb/acre	45lb/acre	
Festuca rubra var. commutata	97%+	100%	21lb/ac	1/8" to 1/4"	---	45lb/acre	45lb/acre	
Festuca ovina	97%+	100%	4.5lb/ac	1/8" to 1/4"	---	45lb/acre	45lb/acre	
Festuca Rubra var. rubra	97%+	100%	4.5lb/ac	1/8" to 1/4"	---	45lb/acre	45lb/acre	

2.7 SOD

- A. Certified sod, complying with TPI's "Specifications for Turfgrass Sod Materials" in its "Guideline Specifications to Turfgrass Sodding." Comply with ASPA specifications for machine cut thickness, size, strength, moisture content, and mowed height and free of weeds and undesirable native grasses. Provide viable sod of uniform density, color, and texture, strongly rooted, and capable of vigorous growth and development when planted. Provide the following turfgrass species:

1. Ecolawn – 80% Predator Hard Fescue, 14% Chewing (Seven Seas), 3% Sheep Fescue, 3% Creeping Red Fescue

2.8 MEADOW SEED

- A. All meadow seed must be fresh, clean, and dry.
1. Meadow mix rates are calculated as Pure Live Seed (PLS). Do not use wet seed or seed that is moldy or otherwise damaged in transit or storage.
 2. All seed must meet the requirements of the Maryland State Seed Law. All seed shall be subject to retesting by a recognized seed laboratory. All Seed used must have been tested within the 6 months immediately preceding the date of sowing. Seed tags must be available upon request to the inspector to verify type of seed and seeding rate.

B. Meadow Seed Species

Hardiness Zone: 7A Mix: Meadow Mix Mix Rate: 26.5 lbs./acre					Fertilization Rate (0-10-10)			Lime Rate
Warm Season/Cool Season 3' Meadow Mix Species	Min. % Germination	Min. % Pure Seed	Seeding Rate	Seeding Depth	N	P2O5	K2O	
Andropogon virginicus	55%+	100%	2.5 lb./ac	1/4"-1/2"	--	45 lb./ acre	45 lb./ acre	As Needed
Schizachyrium scoparium 'Fort Indian Town Gap' -PA Ecotype	55%+	100%	4 lb./ac					
Festuca rubra ssp commutata	97%+	100%	10lb/ac					
Wildflowers to be incorporated into seed mix.								
Choose 1								
Aster cordifolius	50%	100%	2 lb./ac	1/4"-1/2"				
Aster oblongifolius	50%	100%	2 lb./ac					
Eupatorium coelestinum	50%	100%	2 lb./ac					
Choose 3 Genera								
Asclepias tuberosa	50%	100%	2 lb./ac	1/4"-1/2"				
Baptisia tinctoria	50%	100%	2 lb./ac					
Coreopsis grandiflora	50%	100%	2 lb./ac					
Coreopsis lanceolata	50%	100%	2 lb./ac					
Rudbeckia fulgida var. fulgida	50%	100%	2 lb./ac					
Rudbeckia triloba	50%	100%	2 lb./ac					
Solidago nemoralis	50%	100%	2 lb./ac					
Choose 1								
Liatris squarrosa	50%	100%	2 lb./ac	1/4"-1/2"				
Lupinus perennis	50%	100%	2 lb./ac					
Penstemon hirsutus	50%	100%	2 lb./ac					

2.1 MANUFACTURERS

- A. Ecolawn Mix including Sod
 - 1. Mix available from Woodward Turf Farms, Inc.
- B. Meadow Mix
 - 1. Mix available from Newsom Seed and Ernst Conservations Seed

2.2 MULCHES

- A. Paper Fiber Mulch: Provide 100% biodegradable, non-toxic, contaminate-free cellulose fiber, free of plant-growth or germination inhibitors; with maximum moisture content of 15 percent and a pH range of 4.5 to 6.5 meeting the following ASTM:
 - 1. ASTM D7367 for water holding capacity.
 - 2. ASTM D5338 for functional longevity and biodegradability.
- B. Compost Mulch: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 1-inch (25-mm) sieve; soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 percent of dry weight.

2.3 EROSION-CONTROL MATERIALS

- A. Erosion-Control Fiber Mesh: Biodegradable twisted jute or spun-coir mesh, a minimum of 0.92 lb/sq. yd. with 50 to 65 percent open area. Include manufacturer's recommended steel wire staples, 6 inches long.

2.4 MISCELLANEOUS PRODUCTS

- A. Tackifier Amendment: Provide a non-toxic, non-asphaltic, linear anionic copolymer of acrylamide.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive lawns and grass for compliance with requirements and for conditions affecting performance of the Work. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 TOPSOIL PLACEMENT FOR LAWNS

- A. Limit subgrade preparation to areas that will be planted in the immediate future.
- B. Loosen subgrade to a minimum depth of 4 inches. Remove stones, sticks and roots larger than 2 inches in any dimension from subgrade. Completely remove trash and other extraneous debris from subgrade.
- C. Have topsoil tested by a certified soil testing laboratory to determine the type and quantity of soil amendments necessary.

- D. Sift topsoil to remove stones and other objects larger than 1" in any dimension. Maximum object size for topsoil shall be achieved by sifting not by hand removal or raking following placement of topsoil.
- E. Mix soil amendments and fertilizers with topsoil at rates required by soil testing. Delay mixing fertilizer if planting does not follow placing of planting soil within 4 days. Either mix soil before spreading or apply soil amendments on surface of spread topsoil and mix thoroughly into top 4 inches of topsoil before planting.
- F. Mix lime with dry soil prior to mixing fertilizer.
- G. Spread topsoil to a minimum depth of six inches (6").

3.3 PREPARATION

- A. Protect structures, utilities, sidewalks, pavements, and other facilities, trees, shrubs, and plantings from damage caused by planting operations.
- B. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.
- C. Protect adjacent and adjoining areas and plant materials from hydro seed over spraying.
- D. Eradicate existing vegetation if present through either herbicide application and/or tillage
 - 1. If tilling, complete two tillage operations to an average depth of 2-4 inches with one to two weeks in between each operation.
 - 2. If applying herbicide, complete two applications of a non-selective herbicide which does not leave an active soil residual with one to two weeks between each application
 - a) Close mowing two weeks prior to spraying is recommended to stimulate weed growth.
- E. Ecolawn Seed Mix
 - 1. Graded areas must be maintained in a true and even grade as specified on the approved plan, and then loosened to a depth of 3 to 5 inches with a disc or other equipment.
 - 2. Incorporate amendments to a depth of 3 to 5 inches.
 - 3. Remove large clods and irregularities in the grade larger than 1 inch.
 - 4. Settle and compact the area with a water roller to sufficiently compress the soil.
- F. Meadow Mix
 - 1. Graded areas must be maintained in a true and even grade as specified on the approved plan, then scarified or otherwise loosened to a depth of 3 to 5 inches with a disc, by dragging with a heavy chain, or using other equipment.
 - 2. On slopes 3:1 or steeper, scarify the soil surface with a bulldozer, heavy chain or other equipment to loosen the soil 1 inch deep. Following the general contour when preparing the final surface.
 - 3. All rocks and other debris larger than 2 inch in diameter shall be removed
 - 4. Mix soil amendments into the top 3 to 5 inches of soil by disking or other suitable means.
 - 5. Track slopes 3:1 or flatter with tracked equipment leaving the soil in an irregular condition with ridges running parallel to the contour of the slope. Leave the top 1 to 3 inches of soil loose and friable. Seedbed loosening may be unnecessary on newly disturbed areas.

- G. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.
- H. Protect adjacent and adjoining areas from hydroseed over spraying.

3.4 SEEDING MEADOW

- A. Hand seed, drill or dry broadcast seed with a spreader or a seeding machine.
 - 1. Do not broadcast or drop seed when wind velocity exceeds 5 mph.
 - 2. Mix a flow-enhancing material (calcine clay-based kitty litter, sawdust, etc.) to aid in the uniformity of dry broadcasting.
 - 3. Dry broadcast seed in two directions, perpendicular to each other. Apply half the seeding rate in each direction.
- B. Rake or drag seed to the required depth and ensuring seed-to-soil contact, and water with fine spray in dry applications.
- C. No raking or dragging is necessary in wet applications.

3.5 MULCHING

- A. Apply fiber mulch to all seeded areas immediately after seeding.
- B. Wetland Meadow Mixes
 - 1. Spread fiber mulch according to manufacturer instructions.
 - 2. Apply tackifier per the manufacturer instructions.

3.6 HYDROSEEDING (OPTIONAL)

- A. Extra care must be taken when hydroseeding in order to ensure native wildflower germination.
 - 1. Hydroseeding
 - a) The application should incorporate wildflower seed into a low mulch slurry to ensure proper seed/soil contact and avoiding seed suspension in upper mulch layer (which prevents germination).
 - b) Follow seeding schedule when hydroseeding.
 - c) Add fiber mulch and tackifier per manufacturer instructions.

3.7 SODDING NEW LAWNS

- A. Lay sod within 24 hours of stripping. Do not lay sod if dormant or if ground is frozen.
- B. Lay sod to form a solid mass with tightly fitted joints. Butt ends and sides of sod; do not stretch or overlap. Stagger sod strips or pads to offset joints in adjacent courses. Avoid damage to subgrade or sod during installation. Tamp and roll lightly to ensure contact with subgrade, eliminate air pockets, and form a smooth surface. Work sifted soil or fine sand into minor cracks between pieces of sod; remove excess to avoid smothering sod and adjacent grass.

- C. Saturate sod with fine water spray within 2 hours of planting. During first week, water daily or more frequently as necessary to maintain moist soil to a minimum depth of 1-1/2 inches below sod.

3.8 LAWN RENOVATION

- A. Renovate existing lawn damaged by Contractor's operations, such as storage of materials or equipment and movement of vehicles.
 - 1. Reestablish lawn where settlement or washouts occur or where minor regrading is required.
- B. Remove sod and vegetation from diseased or unsatisfactory lawn areas; do not bury in soil.
- C. Remove topsoil containing foreign materials resulting from Contractor's operations, including oil drippings, fuel spills, stone, gravel, and other construction materials, and replace with new topsoil.
- D. Mow, de-thatch, core aerate, and rake existing lawn.
- E. Remove weeds before seeding. Where weeds are extensive, apply selective herbicides as required. Do not use pre-emergence herbicides.
- F. Remove waste and foreign materials, including weeds, soil cores, grass, vegetation, and turf, and legally dispose of them off Owner's property.
- G. Till stripped, bare, and compacted areas thoroughly to a soil depth of 6 inches.
- H. Apply soil amendments and initial fertilizers required for establishing new lawns and mix thoroughly into top 4 inches of existing soil. Provide new planting soil to fill low spots and meet finish grades.
- I. Apply seed and protect with mulch as required for new lawns.
- J. Water newly planted areas and keep moist until new lawn is established.

3.9 MAINTENANCE OF NEW LAWNS

- A. Begin maintenance of lawns and meadow immediately after each area is planted and continue until acceptable cover is established. Maintain seeded/ sodded areas until Substantial Completion. Maintain all areas as necessary to ensure a satisfactory cover is achieved at Substantial Completion.
- B. Watering: Contractor to provide a source of water for use in establishing meadow until acceptable.
 - 1. All areas
 - a) If a minimum ¼ inch of rainfall is not received within 10 days after seeding, irrigate with a minimum of ¼ inch of water.
 - b) Provide supplemental moisture if and when necessary to assure early survival and establishment of ecolawn and meadow.
- C. Establishment
 - 1. Contractor to replant bare areas with appropriate seed until establishment of acceptable cover.
 - 2. Contractor to control invasions by undesirable plants by pulling, mowing or spraying with a selective herbicide until establishment of acceptable meadow.
- D. First Year Maintenance (By Owner)
 - 1. Ecolawn

- a) When ecolawn reaches 6 to 8 inches tall, mow to a height of 4 inches. Repeat procedure to limit growth above 8 inches. Cease trimming by the middle of September.
 - b) Replant all areas with less than 40% ground cover by following the original seedbed preparations, amendment and seeding recommendations
 - c) Replant all areas with between 40% and 85% cover by over-seeding at half the original rate.
2. Meadow
- a) When Meadow reaches 18 to 24 inches tall, using a string trimmer, trim to a height of 8 inches. Repeat procedure to limit growth above 24 inches. Cease trimming by the middle of September.
 - b) Replant all areas with less than 40% ground cover by following the original seedbed preparations, amendment and seeding recommendations
 - c) Replant all areas with between 40% and 85% cover by over-seeding at half the original rate.
- E. Second Year/General Maintenance (By Owner)
- 1. Spray undesirable weeds with an approved aquatic selective herbicide as needed.

3.10 SATISFACTORY LAWN

- A. All areas shall be considered satisfactory/acceptable provided requirements, including maintenance, have been met and a healthy, uniform, close stand of desired vegetation is established and no bare spots exceeding 5 by 5 inches or any surface irregularities.
- B. Replant areas that do not meet requirements and continue maintenance until cover is satisfactory/acceptable by Owner.

3.11 CLEANUP AND PROTECTION

- A. Promptly remove soil and debris created by seeding/ sodding from sidewalks and paved areas. Clean wheels of vehicles before leaving site to avoid tracking soil onto surface of roads, walks, or other paved areas.
- B. Erect barricades and warning signs as required to protect newly planted areas from traffic. Maintain barricades throughout maintenance period until all eco-lawn and meadow areas are established.

END OF SECTION 32 9200

SECTION 32 9300 - EXTERIOR PLANTS

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY

- A. This Section includes the following:

- 1. Trees

1.3 SUBMITTALS

- A. Installers Qualifications: Provide a list, with references, of the past three projects of similar scope.
- B. Product Data: For each type of product indicated.
- C. Plant Material Certifications:
 - 1. Certificates of inspection as required by governmental authorities.
 - 2. Label data substantiating that plant materials comply with specified requirements.
- D. Planting Schedule:
 - 1. Typewritten planting schedule.
 - 2. Once accepted, revise dates only as approved in writing and submitted to Architect.
- E. Maintenance Schedules: Typewritten instructions recommending procedures for maintenance of landscape work for one full year. Submit prior to completion of project.

1.4 QUALITY ASSURANCE

- A. Installer Qualifications: Engage an experienced installer, who has successfully completed planting projects similar in size and complexity to this project. The installer's primary business (defined as a minimum of 60% of total billings) shall be exterior plant installation.
- B. Installer's Field Supervision: Installer to maintain an experienced full-time supervisor on the project site when exterior planting is in progress.
- C. Exterior Plant Materials:
 - 1. Provide plant materials of quantity, size, genus, species, and variety indicated on the Landscape Plan and in the Plant List for landscape work. All plants delivered shall be true to name and legible tagged with the names and sizes of materials. Plant names must comply with standardized plant names as adopted by the latest edition of the International Code of Botanical Nomenclature. Names of varieties not listed must conform generally with names accepted by the nursery trade.
 - 2. All plant materials and work shall comply with recommendations and requirements of ANSI Z60.1 "American Standard for Nursery Stock."
 - 3. Provide nursery-grown plants with healthy root systems developed by transplanting or root pruning. Provide well-shaped, fully branched, healthy, vigorous stock free of disease, insects, eggs or larvae.

4. Shape/Habit: The branch system shall be normal development and free from disfiguring knots, sun-scald, injuries, abrasions of the bark, dead or dry wood, broken terminal growth or other objectionable disfigurements.
5. Trees shall have reasonable straight stems and shall be well branched and symmetrical per their natural habits of growth.
6. Specimen Materials: Specimen shall mean an exceptionally heavy, symmetrical, tightly knit plant, so trained and favored in its development and appearance, as to be unquestionably superior in form, number of branches, compactness and symmetry.
7. Source: All plants shall be nursery grown at a minimum of two (2) years under climatic conditions similar to those in the locality of the project, for proper acclimatization to heat, humidity, and winter temperature extremes.
8. Do not make substitutions. If specified landscape material is not obtainable, submit proof of no availability to Architect, together with proposal for use of equivalent material.
9. Inspection of Plant Materials: The Architect may inspect plant materials either at place of growth or on site before planting for compliance with requirements for genus, species, variety, size, and quality. Architect retains right to further inspect trees for size and condition of balls and root systems, insects, injuries and latent defects, and to reject unsatisfactory or defective material at any time during progress of work.

D. Pre-installation Conference: Conduct conference with Owner/ Architect at Project site.

1.5 DELIVERY, STORAGE AND HANDLING

A. Packaged Materials:

1. Deliver packaged materials in containers showing weight, analysis, and name of manufacturer or grower.
2. Protect materials from deterioration during delivery, and while stored at site.

B. Exterior Plant Materials

1. Protect bark, branches, and root systems from sun scald, drying, sweating, whipping, and other handling and tying damage. Do not bend or bind-tie trees or shrubs in such a manner as to destroy their natural shape. Provide protective covering of exterior plants during delivery. Do not drop exterior plants during delivery or transport.
2. Deliver exterior plant materials after preparations for planting have been completed and plant immediately. If planting is delayed more than 6 hours after delivery, set plant materials in a vertical position in shade, untie, protect from weather and mechanical damage, and keep roots moist and free from frost. For extended storage beyond three days, obtain loose material such as wood chips or saw dust for the use of healing in the root balls of B&B material.
3. Do not remove container-grown stock from containers until planting time. Do not store containerized materials on pavement surfaces exposed to direct sunlight for more than 4 hours. Provide temporary shade as necessary. Water judiciously while in storage.
4. Balled and bur lapped material shall be freshly dug.
5. Handle planting stock by root ball strapping. Do not whip, drag or allow the material to spin while in transport.

1.6 PROJECT CONDITIONS

- A. Examine the subgrade, verify the elevations, and observe the conditions under which work is to be performed. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner acceptable to the installer.
- B. Determine location of underground utilities and perform work in a manner which will avoid possible damage. Hand excavate as required.
- C. When conditions detrimental to plant growth are encountered, such as rubble fill, adverse drainage conditions, or obstructions, notify Architect before planting.
- D. Provide all necessary safeguards for the protection of all planted areas until provisional inspection/acceptance is accomplished.
- E. Planting Restrictions: Plant during one of the following periods.
 - 1. Spring Planting: Unfrozen soil conditions March 1-June 1st.
 - 2. Fall Planting: September 1-November 1st or until frozen soil conditions prevent work.
 - 3. Summer Planting: June 1 – September 1 with approved temporary irrigation system.
- F. Coordination with Lawns: Install plant materials after finish grades are established and before planting lawns, unless otherwise acceptable to the Architect.
 - 1. When planting exterior plants after lawns, protect lawn areas and promptly repair damage caused by planting operations.

1.7 WARRANTY

- A. Warranty exterior plant materials for a period of **two** years after date of Final Completion against defects including death and unsatisfactory growth, except for defects resulting from neglect by Owner, abuse or damage by others, or unusual phenomena or incidents which are beyond Contractor's control.
 - 1. The Contractor shall provide written notice to the Architect of any practice which will affect the warranty if not remedied promptly. The Architect will render an opinion of the conflict if necessary.
 - 2. Make replacements of all dead plants or plants in impaired condition (more than 25% dead or dying) condition in early spring/fall following installation. Replacements of dead or rejected plants should again be made prior to the expiration of the warranty period.

1.8 MAINTENANCE

- A. The Owner is responsible for maintaining all exterior plant material throughout the warranty period according to the submitted Maintenance Schedule.
- B. Installer shall remove all stakes and guy wires 12 months after planting.

PART 2 – PRODUCTS

2.1 FERTILIZER

- A. Commercial Fertilizer: Commercial-grade complete fertilizer of neutral character, consisting of fast- and slow-release nitrogen, 50 percent derived from natural organic sources of urea formaldehyde, phosphorous, and potassium. Revise fertilizer mix to remedy deficiencies found in soil.
 - 1. Composition: 1 lb/1000 sq. ft. (0.45 kg/92.9 sq. m. of actual nitrogen, 4 percent phosphorous, and 2 percent potassium, by weight.
 - 2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from a qualified soil-testing agency.
- B. Slow-Release Fertilizer: Granular or pelleted fertilizer consisting of 50 percent water-insoluble nitrogen, phosphorus, and potassium. Revise fertilizer mix to remedy deficiencies found in soil.
 - 1. Composition: 20 percent nitrogen, 10 percent phosphorous, and 10 percent potassium, by weight.
 - 2. Composition: Nitrogen, phosphorous, and potassium in amounts recommended in soil reports from a qualified soil-testing agency.

2.2 MULCHES

- A. Organic Mulch: Six (6) month old well rotted double shredded native hardwood bark mulch not larger than 4" in length and 1/2" in width, free of woodchips and sawdust.

2.3 STAKES AND GUYS

- A. Stakes: Oak Stakes shall be used.
- B. Guys: Twelve (12) gauge galvanized wire and rubber hose shall be used. Arbor-tie is acceptable.

2.4 WATER

- A. Free of substances harmful to plant growth.

2.5 TOPSOIL

- A. Topsoil: ASTM D 5268, pH range of 5.5 to 7, a minimum of 4 percent organic material content. Topsoil shall be fertile, friable, natural topsoil of loamy character, without admixture of subsoil material, obtained from a well-drained arable site, reasonably free from clay, lumps, coarse sands, stones, plants, roots, sticks and other foreign materials.
- B. Topsoil Source:
 - 1. Reuse surface soil stockpiled on-site. Verify suitability of stockpiled surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.
 - 2. Import topsoil or manufactured topsoil from off-site sources. Obtain topsoil displaced from naturally well-drained sites where topsoil occurs at least 4 inches deep; do not obtain from agricultural land, bogs or marshes.
 - 3. Amend existing in-place surface soil to produce topsoil. Verify suitability of surface soil to produce topsoil. Clean surface soil of roots, plants, sod, stones, clay lumps, and other extraneous materials harmful to plant growth.

2.6 INORGANIC SOIL AMENDMENTS

- A. Lime: ASTM C 602, agricultural limestone containing a minimum 80 percent calcium carbonate equivalent and as follows:
 - 1. Class: Class T, with a minimum 99 percent passing through No. 8 (2.36-mm) sieve and a minimum 75 percent passing through No. 60 (0.25-mm) sieve.
 - 2. Class: Class O, with a minimum 95 percent passing through No. 8 (2.36-mm) sieve and a minimum 55 percent passing through No. 60 (0.25-mm) sieve.
 - 3. Provide lime in form of dolomitic limestone.
- B. Sulfur: Granular, biodegradable, containing a minimum of 90 percent sulfur, with a minimum 99 percent passing through No. 6 (3.35-mm) sieve and a maximum 10 percent passing through No. 40 (0.425-mm) sieve.
- C. Iron Sulfate: Granulated ferrous sulfate containing a minimum of 20 percent iron and 10 percent sulfur.
- D. Aluminum Sulfate: Commercial grade, unadulterated.
- E. Perlite: Horticultural perlite, soil amendment grade.
- F. Agricultural Gypsum: Finely ground, containing a minimum of 90 percent calcium sulfate.
- G. Sand: Clean, washed, natural or manufactured, free of toxic materials.
- H. Diatomaceous Earth: Calcined, diatomaceous earth, 90 percent silica, with approximately 140 percent water absorption capacity by weight.
- I. Zeolites: Mineral clinoptilolite with at least 60 percent water absorption by weight.

2.7 ORGANIC SOIL AMENDMENTS

- A. Compost: Well-composted, stable, and weed-free organic matter, pH range of 5.5 to 8; moisture content 35 to 55 percent by weight; 100 percent passing through 3/4-inch (19-mm) sieve; soluble salt content of 5 to 10 decisiemens/m; not exceeding 0.5 percent inert contaminants and free of substances toxic to plantings; and as follows:
 - 1. Organic Matter Content: 50 to 60 percent of dry weight.
 - 2. Feedstock: Agricultural, food, or industrial residuals; bio-solids; yard trimmings; or source-separated or compostable mixed solid waste.
- B. Sphagnum peat moss: Sphagnum peat moss shall be partially decomposed, finely divided or granular texture, with a pH range of 3.4 to 4.8.
- C. Peat: Finely divided or granular texture, with a pH range of 6 to 7.5, containing partially decomposed moss peat, native peat, or reed-sedge peat and having a water-absorbing capacity of 1100 to 2000 percent.
- D. Wood Derivatives: Decomposed, nitrogen-treated sawdust, ground bark, or wood waste; of uniform texture, free of chips, stones, sticks, soil, or toxic materials.
 - 1. In lieu of decomposed wood derivatives, mix partially decomposed wood derivatives with at least 0.15 lb (2.4 kg) of ammonium nitrate or 0.25 lb (4 kg) of ammonium sulfate per cubic foot (cubic meter) of loose sawdust or ground bark.
- E. Manure: Well-rotted, unleached, poultry, stable or cattle manure containing not more than 25 percent by volume of straw, sawdust, or other bedding materials; free of toxic substances, stones, sticks, soil, weed seed, and material harmful to plant growth.

2.8 MISCELLANEOUS PRODUCTS

- A. Antidesiccant: Water-insoluble emulsion, permeable moisture retarder, film forming, for trees and shrubs. Deliver in original, sealed, and fully labeled containers and mix according to manufacturer's written instructions.

PART 3 – EXECUTION

3.1 EXAMINATION

- A. Examine areas to receive exterior plants for compliance with requirements and conditions affecting installation and performance. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PLANTING BED ESTABLISHMENT (INTERIOR LANDSCAPE AREAS ONLY)

- A. Prior to placement of topsoil, Loosen subgrade of planting beds to a minimum depth of 2 ½ feet below top of curb. Remove stones, sticks and roots larger than 2 inches in any dimension and legally dispose of them off of Owner's property. Remove excess soil that would displace any portion thereof of 6" of topsoil using top of curb as finish grade. Amend with compost to produce a friable soil structure.
- B. Amendments and soil loosening shall not jeopardize the integrity of perimeter curbing/ pavement. Do not displace or remove underlying stone within the angle of repose.
- C. Apply fertilizer directly to subgrade.
- D. Spread topsoil, apply soil amendments and fertilizer on surface, and thoroughly blend planting soil mix.
 - 1. Mix lime with dry soil before mixing fertilizer.
 - 2. Delay mixing fertilizer with planting soil if planting will not proceed within a few days.
- E. Finish Grading: Grade planting beds to a smooth, uniform surface plane with loose, uniformly fine texture flush with pavement. Roll and rake, remove ridges, and fill depressions to meet finish grades.
- F. Restore planting beds if eroded or otherwise disturbed after finish grading and before planting.

3.3 PLANTING BED ESTABLISHMENT (EXCLUDING INTERIOR LANDSCAPE AREAS ONLY)

- A. Limit preparation to areas which will be immediately planted.
- B. If no topsoil is present, till and loosen subgrade to depth of 5 inches, rake and remove large clumps of grasses and weeds, and place 6" of topsoil on all areas.
- C. If topsoil has already been placed, add the appropriate amount of topsoil where needed to achieve a minimum of 6".
- D. Verify that the proper organic matter is present in the topsoil and amend as needed.
- E. Apply lime as required to adjust pH of topsoil. Distribute evenly by machine and incorporate thoroughly into topsoil.
- F. Apply fertilizer as required by the soil test. Thoroughly and evenly incorporate it into the soil to a depth of 3".

- G. Grade planting areas to smooth, free-draining and even surface with a loose, uniformly fine texture.
- H. Restore planting beds if eroded or otherwise disturbed after finish grading and before planting.

3.4 PREPARATION

- A. Tree save areas as indicated shall be tagged and approved by the Architect/ Engineer prior to any clearing and/or thinning.
- B. Protect structures, utilities, sidewalks, pavements, and other facilities, and lawns and existing exterior plants from damage caused by planting operations.
- C. Provide erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.
- D. Lay out individual tree and shrub locations and areas for multiple exterior plantings. Stake locations, outline areas, adjust locations when requested, and obtain Landscape Architect's acceptance of layout before planting. Make minor adjustments as required.
- E. Apply antidesiccant to evergreen trees using power spray to provide an adequate film over trunks, branches, stems, twigs, and foliage to protect during digging, handling, and transportation.
 - 1. If deciduous trees or shrubs are moved in full leaf, spray with antidesiccant at nursery before moving and again two weeks after planting.

3.5 TREE PLANTING

- A. Set all plant materials plumb and in center of pit or trench as per detail.
 - 1. Remove burlap and wire baskets from tops of root balls and partially from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
 - 2. Carefully remove root ball from container without damaging root ball or plant.
 - 3. Place planting soil mix around root ball in layers, tamping to settle mix and eliminate voids and air pockets. When pit is approximately two-thirds backfilled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.
 - 4. Container Plants: Spread roots without tangling or turning toward surface, and carefully work backfill around roots by hand. Puddle with water until backfill layers are completely saturated. Plumb before backfilling and maintain plumb while working backfill around roots and placing layers above roots. Tamp final layer of backfill. Remove injured roots by cutting cleanly, do not break.
 - 5. Dish top of backfill to allow for a mulch ring just beyond the extents of shrub/ tree canopy (dripline).
 - 6. Label at least one tree and one shrub of each variety and caliper with a securely attached, waterproof tag bearing legible designation of botanical name. Retain label in legible form for the length of the two year bond. Installer shall remove labels after the bond is released.
- B. Organic Mulching: Apply maximum of 1-inch average thickness of organic mulch extending 12 inches beyond edge of planting pit or trench. Do not place mulch within 3 inches of trunks or stems.

3.6 TREE PRUNING

- A. Prune, thin, and shape trees and shrubs as indicated.

3.7 GROUND COVER AND PLANT PLANTING

- A. Set out and space ground cover and plants as indicated in details.
- B. Water thoroughly after planting, taking care not to cover plant crowns with wet soil.

3.8 CLEANUP AND PROTECTION

- A. During exterior planting, keep adjacent pavement and construction clean and work area in an orderly condition.
- B. Protect exterior plants from damage due to landscape operations, operations by other contractors and trades, and others. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged exterior planting.

3.9 DISPOSAL

- A. Disposal: Remove surplus soil and waste material, including excess subsoil, unsuitable soil, trash, and debris, and legally dispose of them off Owner's property.

END OF SECTION 02930

SECTION 33 4100 - STORM DRAINAGE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS:

- A. The provisions of the Contract Documents apply to the work of this Section.

1.2 SUMMARY:

- A. This Section includes the roof drainage collection system, the storm sewerage system piping and appurtenances from a point 5 feet (1.52 m) outside the building to the point of disposal, and the outfall structures of the stormwater management basin.

1.3 SUBMITTALS

- A. Product data for:
 - 1. Polyethylene pipe
 - 2. Frames and covers.
 - 3. Grates
- B. Certification, signed by material producer and contractor, that standard precast and cast in place concrete storm drainage manholes and Drop Inlets comply with SHA standards and specifications.
- C. SHA approved job mix for bedding stone.
- D. Shop drawings for:
 - 1. Non-standard precast or cast-in-place concrete storm drainage manholes and Drop Inlets.
 - 2. Cleanouts
- E. Record drawings of installed storm drainage system.
- F. Detectable Marking Tape

1.4 QUALITY ASSURANCE

- A. Environmental Compliance: Comply with applicable portions of local environmental agency regulations pertaining to storm sewerage systems.
- B. Utility Compliance: Comply with state and local regulations and standards pertaining to storm sewerage systems.
- C. All materials shall be new and free of defects (i.e. pipe shall not have chipped spigots or bells).

1.5 PROJECT CONDITIONS

- A. Site Information: Perform site surveys, research public utility records, and verify existing utility locations. Verify that storm sewerage system piping may be installed in compliance with original design and referenced standards.
- B. Locate existing structures and piping to be closed and abandoned.
- C. Existing Utilities: Do not interrupt existing storm sewer serving facilities occupied by the Owner of others except when permitted under the following conditions and then only after arranging to provide acceptable temporary storm sewer services.

1. Notify Architect not less than 48 hours in advance of proposed storm sewer interruptions.
 2. Do not proceed with storm sewer interruptions without receiving Architect's written permission.
- D. Existing utilities across or along the line of work are indicated only in an approximate location. Locate all underground lines and structures. The Contractor is required to use a private utility locator per NIH standards. Call "Miss Utility" at 1-800-552-7001 prior to construction. If utilities are marked that are not shown on the plans, locate utility vertically and horizontally and provide information to architect.

1.6 SEQUENCING AND SCHEDULING

- A. Coordinate with interior building storm drainage piping.
- B. Coordinate with other utility work.

PART 2 - PRODUCTS

2.1 GENERAL

- A. All materials used for construction of the storm sewerage system shall comply with the requirements of the latest edition of the SHA Specifications.

2.2 PIPE AND FITTINGS

- A. Provide pipe and pipe fitting materials compatible with each other. Pipe materials are indicated on the drawings.
- B. Reinforced Concrete Pipe (RCP): Shall conform to the requirements of ASTM C76/AASHTO M170, Class III, unless otherwise indicated.
- C. PVC Storm Sewer Pipe: Shall conform to the requirements of ASTM D3034, SDR-35 with bell and spigot ends for gasketed joints with ASTM F 477 elastometric seals
 - a) Connections to the building downspouts shall be made with Schedule 40 PVC.

2.3 MANHOLES

- A. Precast Concrete Manholes: Comply with the requirements of SHA Specifications.
- B. Cast-in-Place Manholes: Comply with the requirements of SHA Specifications.
- C. Manhole Steps, Safety Slabs and Inlet Shaping: Comply with the requirements of SHA Specifications.
- D. Manhole Frames and Covers: Comply with the requirements of SHA Specifications.

2.4 CLEANOUTS

- A. Cast-iron ferrule and countersunk brass cleanout plug, with round cast-iron access frame and heavy-duty, secured, scoriated cast-iron cover.

2.5 DROP INLETS

- A. Precast Concrete Drop Inlets: Comply with the requirements of the SHA Specifications.
- B. Cast-in-Place Drop Inlets: Comply with the requirements of the SHA Specifications.
- C. Drop Inlet Steps, Safety Slabs and Inlet Shaping: Comply with the requirements the SHA Specifications.
- D. Drop Inlet Frames and Grates: Comply with the requirements of the SHA Specifications.
- E. Plastic Drain Basins: ADS 2812AG or approved equal.

2.6 CONCRETE AND REINFORCEMENT

- A. Concrete: Conform to the requirements of SHA Standard Class A3 concrete.
- B. Reinforcement: Steel conforming to the following:
 - 1. Fabric: ASTM A 185 welded wire fabric, plain.

Reinforcement Bars: ASTM A 615, Grade 60, deformed.

2.7 DETECTABLE MARKING TAPE

- A. Detectable marking tape shall be installed above all storm sewer pipes.
- B. Detectable marking tape shall be continuous.
- C. Plastic marking tape shall consist of one layer of aluminum foil laminated between two layers of inert plastic film. Tape shall be resistant to alkalis, acids and other destructive agents commonly found in the soil. The laminate shall be strong enough that the layers cannot be separated by hand.
- D. Tape shall be a minimum of 4 mils thick with a minimum tensile strength of 60 lbs. in the machine direction and 58 lbs. in the transverse direction per 6" wide strip. Tape color shall be APWA Color Coded for marking the particular utility line and shall be imprinted with a continuous warning message to indicate the type of utility being marked, the message normally being repeated every 16" to 36". Tape shall be inductively locatable and conductively traceable using a standard pipe and cable-locating device.

PART 3 - EXECUTION

3.1 GENERAL

- A. Install the storm sewerage system in accordance with the latest edition of the SHA Specifications.

3.2 PREPARATION OF FOUNDATION FOR BURIED STORM SEWERAGE SYSTEMS

- A. Grade trench bottom to provide a smooth, firm, stable, and rock-free foundation, throughout the length of the pipe.
- B. Remove unstable, soft, and unsuitable materials at the surface upon which pipes are to be laid and backfill with clean sand or pea gravel to indicated level.
- C. Install pipe bedding conforming to the requirements of the latest edition of the SHA Specifications.

3.3 PIPE INSTALLATION

- A. Install piping beginning at low point of systems, true to grades and alignment indicated with unbroken continuity of invert. Place bell ends of piping facing upstream. Install gaskets, seals, sleeves, and couplings in accordance with manufacturer's recommendations for use of lubricants, cements, and other installation requirements. Maintain swab or drag in line and pull past each joint as it is completed.
- B. Use proper size increasers, reducers, and couplings, where different size or material of pipes and fittings are connected. Reduction of the size of piping in the direction of flow is prohibited.
- C. Extend storm sewerage system piping to connect to building storm drains, of sizes and in locations indicated.
- D. Join and install concrete pipe and fittings per SHA specifications.
- E. Join and install PE pipe and fittings per manufacturer's recommendations.
- F. Join different types of pipe with standard manufactured couplings and fittings intended for that purpose.

3.4 MANHOLES

- A. General: Install manholes complete with accessories as indicated. Form continuous concrete or split pipe section channel and benches between inlets and outlet. Set tops of frames and covers flush with finish grade, unless otherwise indicated.
- B. Place precast concrete manhole sections as indicated and install in accordance with ASTM C 891.
- C. Construct cast-in-place manholes as indicated.
- D. Apply bituminous mastic coating at joints of sections.

3.5 CLEANOUTS

- A. Install cleanouts and extension from sewer pipe to cleanout at grade as indicated. Set cleanout frame and cover in concrete block 12 by 12 by 6 inches (0.30 by 0.30 by 0.15 m) deep or as indicated on the plans, except where location is in concrete paving. Set top of cleanout flush with finish grade.

3.6 DROP INLETS

- A. Construct drop inlets to sizes and shapes indicated.
- B. Set frames and grates to elevations indicated.

3.7 INLET SHAPING

- A. Construct inlet shaping conforming to SHA standards all drop inlets and manholes.

3.8 FIELD QUALITY CONTROL

- A. Cleaning: Clear interior of piping and structures of dirt and other superfluous material as work progresses. Maintain swab or drag in piping and pull past each joint as it is completed.
 - 1. In large, accessible piping, brushes and brooms may be used for cleaning.
 - 2. Place plugs in ends of uncompleted pipe at end of day or whenever work stops.
 - 3. Flush piping between manholes and drop inlets to remove collected debris. Flush pipes through an approved erosion and sediment control measure.

- B. Interior Inspection: Inspect piping to determine whether line displacement or other damage has occurred.
 - 1. Make inspections after pipe between manholes and manhole locations has been installed and approximately 2 feet (0.61 m) of backfill is in place, and again at completion of project.
 - 2. If inspection indicates poor alignment, debris, displaced pipe, infiltration, or other defects correct such defects and reinspect.

END OF SECTION 33 4100